SECANT OBJECTS AND HODGE CLASSES BEYOND THE WEIL LINES

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Abstract. We determine what closing the algebraic locus of the Weil classes requires of a single object at a single point: the semiregularity criterion is never met by an object whose Chern character is exactly a Weil class, for any CM field, and with powers of the polarisation added it becomes one number, computed exactly for quartic CM fields; we exclude the secant objects, sums and convolutions of line bundles and supports that the known constructions supply; and we prove that the Hodge conjecture is equivalent to two statements about algebraic loci, one on abelian varieties and one on the classes reached from them by correspondences, and that it holds for the very general complete intersection of diagonal quadrics of Vandermonde type in every dimension.

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This paper and its companion [I] together replace the earlier paper *Algebraic loci of Weil classes on abelian varieties* by the authors and Priyabrata Mandal (archive [BMB26]); every result of that paper appears in one of the two with its proof. The preprint *Relative secant cycles and Hodge classes* [BB26] contains errors and incomplete proofs and is replaced by them.

Every proof is given in the text. Programs that repeat much of the finite arithmetic, and a Lean 4 certificate of part of it, are archived as described in the statement on data availability at the end of the paper; no statement rests on them.

1. Introduction

Let X be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}^p_X\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) \coloneqq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every p. Beyond degree two the best understood Hodge classes are those of abelian varieties, and the first of them that divisors do not produce are the Weil classes, which Weil exhibited on abelian varieties with an action of an imaginary quadratic field whose two eigenspaces on the tangent space have the same dimension [Wei77]; the construction works for every CM field F. Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field, and, through a product argument of Schoen [Sch98, Proposition 10], on every abelian fourfold of Weil type [Mar25a, Mar25b]. His sheaves are built from secant sheaves on an abelian variety of half the dimension, and they extend along the family because they satisfy an equivariant form of semiregularity.

The companion paper [I] studies the set of members of a family on which a flat Hodge class is algebraic, its *algebraic locus*, and determines it for the Weil classes of every CM field and every dimension at once. Fix a CM field F of degree $2m$, an integer $n \geq 2$ and a discriminant δ. The abelian varieties of (F,n,δ)-Weil type are the members of one family over a connected period domain $\mathcal{D}_F$ of dimension mn^2, their Weil classes form a flat local system of rank $2m$ over it, and the algebraic locus

$$\Sigma_F = \{\, s \in \mathcal{D}_F : \text{the Weil classes of } A_s \text{ are algebraic} \,\}$$

contains an explicit point of complex multiplication, is dense, is the preimage of a countable union of Zariski closed subsets of the Shimura variety of the family, and is either all of $\mathcal{D}_F$ or meagre; so the following are equivalent: Σ_F is closed, it has an interior point, the Weil classes are algebraic on every member, and the Hodge conjecture holds for the very general member ([I, Theorem 1.1], which we call the main theorem throughout). The notation of that paper is kept here: the standard model of [I, Section 4], the integral basis ω_1, ω_2 of the Weil line of [I, Theorem 4.9], the base points of [I, Theorem 15.27] and [I, Theorem 17.5], and the strata of [I, Theorem 16.4].

This paper takes the main theorem further in three directions. It determines what closing Σ_F requires of a single object at a single point, and computes that requirement exactly; it examines the objects that the known constructions supply and shows, in each case, where they fail; and it shows that the whole Hodge conjecture is two statements about algebraic loci, one on abelian varieties and one on the classes that correspondences carry from them, proves each on the families where it can be proved, and proves the conjecture itself on a family of varieties of every dimension.

The paper also proves the Hodge conjecture itself on a family of varieties of every dimension. Let $X \subset \mathbb{P}^N$ be a complete intersection of c diagonal hypersurfaces $\sum_i a_{ki} x_i^d = 0$ of degree d. It is *of Vandermonde type* if $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl} \lambda_i^{l-1}$ for distinct numbers $\lambda_0, \ldots, \lambda_N$, nonzero numbers μ_i and an invertible matrix (g_{kl}); for $c \geq 2$ this says that the columns of (a_{ki}) are distinct points of a rational normal curve, and every smooth complete intersection of two diagonal hypersurfaces of the same degree is of this type. Such an X is isomorphic to a variety $X_{d,N-c}(\lambda)$ that depends only on λ (Theorem 6.55), and *very general* refers to λ. For $d = 2$ we call X a *Vandermonde quadric intersection.*

Theorem 1.1 (Diagonal complete intersections)**.** *Every rational Hodge class is algebraic on the very general Vandermonde quadric intersection, in every dimension and in every*

projective space. The same holds for the very general complete intersection of Vandermonde type of diagonal cubics or quartics of dimension at most seven, or of diagonal sextics of dimension at most five, and for every smooth complete intersection of two diagonal cubics, of two diagonal quartics or of two diagonal sextics in $\mathbb{P}^6$.

The first two statements are Theorem 6.66(iii) and the third is Corollary 6.56(ii). The variety X is a quotient of a power of a generalised Fermat curve, so its Hodge classes come from abelian subvarieties of the Jacobian of that curve. For quadrics the monodromy of the curve along a chain of vanishing cycles is the full symplectic group on every eigenspace, so every Hodge class comes from powers of polarisations and no theorem on abelian varieties of higher dimension is needed. For cubics, quartics and sextics we compute the monodromy by an induction on the number of branch points (Proposition 6.64), and the classes it leaves are divisor classes, Weil classes of abelian fourfolds of Weil type, and Weil classes of abelian sixfolds of split Weil type, all algebraic by the theorems of Lefschetz and Markman [Mar25a, Mar25b]. In odd dimension every Hodge class is a multiple of a power of the hyperplane class. So the bound seven for cubics and quartics is the point where abelian varieties of dimension eight appear, and the bound five for sextics comes from abelian sixfolds of non-split Weil type, which occur in dimension six.

1.1. **Closing the algebraic locus.** By [I, Theorem 1.1(iv)] the Hodge conjecture for the very general member of a Weil family is the closedness of Σ_F, and by (iii) it is enough to show that one stratum has an interior point. Two statements would do this, and the paper determines what each of them asks.

The first bounds the Hilbert data. If at every point of the orbit of the base point the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then finitely many strata cover a dense set and one of them is everything ([I, Theorem 16.9]). Read on an arbitrary representative this bound is equivalent to its conclusion ([I, Theorem 16.17]); read on the transported cycle, the one representative that can be named, it fails, because the transport multiplies the degree of a component by c^{2n} and divides it only by the order of a stabiliser ([I, Theorem 16.15]). So the bound restates the problem rather than reducing it. The bound is needed only on a small set: by the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21],[1] it suffices along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit ([I, Theorem 17.14]). Every member at which this paper exhibits algebraic Weil classes has endomorphisms beyond the field, so it lies in the locus of members with smaller Hodge group. That locus contains the base point, is dense, is stable under $G_F(\mathbb{Q})$ and is a countable union of closed algebraic strata, exactly as Σ_F is, and it is meagre ([I, Proposition 17.12], [I, Corollary 17.13]). The properties listed in [I, Theorem 1.1] therefore do not decide the dichotomy by themselves, and a proof that Σ_F is everything has to reach a member with full Hodge group.

The second is semiregularity. By the theorem of Buchweitz and Flenner, in the form of Pridham for perfect complexes [BF03, Pri24],[2] a perfect complex at one base point whose

[1] In the form used here: an irreducible closed subvariety of a Shimura variety of abelian type that contains a Zariski dense set of CM points is a special subvariety. Hence a sequence of CM points that leaves every proper special subvariety is Zariski dense, and so is every infinite subsequence of it.

[2] The semiregularity map of a perfect complex E on X sends $\mathrm{Ext}^2(E,E)$ to $\bigoplus_q H^{q+2}(X,\Omega_X^q)$ by the trace of the product with powers of the Atiyah class of E; it carries the obstruction to deforming E along a first order deformation of X to the obstruction to keeping $\mathrm{ch}(E)$ of Hodge type. If it is injective, E deforms, formally and then along the whole family, in every direction in which $\mathrm{ch}(E)$ stays a Hodge class. Bloch introduced it for local complete intersection subvarieties.

Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, and whose semiregularity map is injective, extends along the whole family and makes Σ_F everything (Theorem 2.3, [I, Theorem 17.7(v)]). The paper computes the injectivity condition exactly. The annihilator of the Weil class in $H^{0,2}$ is carried by the polarisation isomorphically onto the tangent space of the family ([I, Lemma 9.6], [I, Theorem 9.17]), so no object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully is semiregular ([I, Theorem 9.7]). When the Chern character is exactly a Weil class the criterion is never met, for any CM field and any $n \geq 2$: a semiregular object would extend to the very general Weil torus of the field, a complex torus that is not algebraic, on which every coherent sheaf has vanishing Chern character strictly between degree zero and the top degree (Proposition 3.4, Theorem 3.5, Theorem 5.4). With powers of the polarisation added the criterion becomes one number: for a Chern character $N\omega + \sum_k c_k \theta^k$ and $n \geq 3$ every object has $\dim \mathrm{Ext}^2(E,E) \geq (4+\rho)n^2 - 2n$, where $\rho \leq 3$ is the rank of the Hankel matrix of the numbers $k!\, c_k$, and an object attaining this bound meets the criterion (Theorem 3.9).[3] A flatness condition on the deformations of the object suffices in place of injectivity (Proposition 3.15). By descent and scalar extension the statement is needed only for the split families of the CM fields of degree at least four (Proposition 6.13), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point (Corollary 6.31).

For a quartic CM field and $n = 2$ we compute the rank exactly for every shape. It takes fifteen values; the least, 80, is taken only by the pure shape, which the Weil tori exclude, so an object meeting the criterion has $\dim \mathrm{Ext}^2(E,E) \geq 88$ and a polynomial part that is exponential or linear at each real place (Theorem 5.14, Corollary 5.15). The Hodge locus of every such character is determined: it is the Weil tori of the field only for the characters of θ^4 shape, which are therefore excluded at rank 88, and the polarised family itself for every other character of that rank (Proposition 5.17, Corollary 5.18).

1.2. **Secant objects.** The objects that a secant construction supplies are examined in Section 4. On the split member $X \times \widehat{X}$ of [I, Section 14] the secant plane is written down in every dimension, and a secant sheaf has rank a, $c_1 = bt$, $c_2 = \frac{ad+b^2}{2}t^2$ and Bogomolov discriminant $\mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d})\, t^2$ ([I, Theorem 14.5], [I, Theorem 14.13]). No object whose Chern character is an effective combination of exponentials of divisor classes contributes to the Weil line, by a positivity argument with the norm form (Theorem 4.5), and no object with a line bundle summand off the polarisation line meets the criterion, so no sum of line bundles does in any dimension ([I, Theorem 9.20], Corollary 4.21). Convolutions of line bundles, their iterated cones, are cut down by the maps between the pieces. Fewer than $2n$ pieces with nondegenerate differences never meet the criterion (Theorem 4.26). On E_0^{2n}, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, the signed Chern characters of $2 \cdot 4^{n-1}$ line bundles add up to a real Weil class (Proposition 4.27), and with three multiples of the polarisation no convolution of these pieces meets the criterion at $n = 3$ (Theorem 4.35), nor at $n = 4$ when the shifts of the pieces lie in six consecutive values (Corollary 4.48). The weakened criterion of [Mar25b, Question 11.4] reduces, for the transform of $\mathcal{F} \boxtimes \mathcal{F}$, to the single identity $\mathrm{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \mathrm{id}$ for the sheaf $\mathcal{F}$ on the fourfold (Theorem 4.56); a local computation at an isolated double point of the support decides it in the negative for Markman's candidate in dimension eight (Theorem 4.61), and at a quartic CM field a curve on which a complex is locally

[3] That is, ρ is the rank of the $(2n-1) \times 3$ matrix with entries μ_{i+j}, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$, where $\mu_k = k!\, c_k$ (Setup 3.8). A vector in its kernel is a linear recurrence of order at most two among the μ_k, and ρ does not change when the character is multiplied by $e^{t\theta}$.

free is tangent to at most one of the two eigenplanes of the real multiplication, which decides Markman's explicit pair [Mar25c, Example 11.2.7] in the same way (Theorem 5.28, Corollary 5.30).

The supports are constrained as well. A Cohen–Macaulay support carries no local obstruction, while two smooth branches of codimension two meeting transversally at a point carry one (Corollary 4.77). For a smooth support, a surface S in the fourfold, Grothendieck–Riemann–Roch fixes every numerical invariant, and the Bogomolov–Miyaoka–Yau inequality gives $d \leq b^2$ for the twist $b\Theta$, which leaves only $d \in \{1, 3, 5, 7\}$ for the twist 3Θ of Markman's candidate (Theorem 4.67, Corollary 4.68). Complete intersections are excluded for every b and d (Theorem 4.81), two-term complexes of powers of the polarisation at every rank, twist and discriminant (Theorem 4.89), and a Hilbert–Burch resolution of a secant support has rank at least two (Proposition 4.86); the secant class is an integer combination of classes of line bundles exactly when $d \equiv 15$ or $23 \pmod{24}$ (Theorem 4.83).

1.3. **Weil classes and the Hodge conjecture.** Sections 6 to 8 place the Weil classes inside the conjecture. Two statements carry all of it. The first, (F2), asks that every Hodge class on an abelian variety outside the subring generated by divisor classes and Weil classes of quotients be algebraic; multiplying by an exceptional class on the square of a Mumford fourfold moves every Hodge class of every abelian variety outside that subring, so (F2) is equivalent to the Hodge conjecture for all abelian varieties (Proposition 7.49). The second, (F3′), asks that every Hodge class on every smooth projective variety be algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences (Definition 6.34). The Hodge conjecture is equivalent to (F2) and (F3′) together (Proposition 6.35, Theorem 9.1).

We prove (F3′) for every variety whose cohomology is reached from abelian varieties by correspondences, a class closed under products, surjective images, blow-ups, projective bundles and Hilbert schemes of points on surfaces, which contains every Fermat hypersurface and every K3 surface whose Kuga–Satake correspondence is algebraic (Proposition 6.35, Proposition 6.42); for every variety of dimension at most three, and on a variety of dimension k in the degrees 0, 2, $2k-2$ and $2k$; for every fourfold and fivefold whose zero-cycles are supported in dimension at most three, among them every uniruled fourfold and every rationally connected fivefold (Proposition 6.43, Proposition 6.44); for every fourfold and fivefold dominated by a variety of that class (Proposition 6.46); for every smooth ample divisor of odd dimension in a variety of that class, and in even dimension outside the middle degree and for a very general member (Proposition 6.47); and, by a toric compactification whose boundary is again of the same kind, for every hypersurface of simplex type, so for every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, whose Hodge conjecture then follows from that of Fermat varieties of one degree (Theorem 6.52, Corollary 6.53). A complete intersection of diagonal hypersurfaces whose coefficients are of Vandermonde type, which holds for every smooth complete intersection of two of them, is a quotient of a power of a generalised Fermat curve, so it satisfies (F3′), and its Hodge conjecture reduces to abelian varieties attached to the characters of the curve; for two diagonal hypersurfaces of degree 3, 4 or 6 in $\mathbb{P}^6$ these have dimension at most five, and the Hodge conjecture holds, the exceptional classes being Weil classes of abelian fourfolds of Weil type (Theorem 6.55, Corollary 6.56). At a very general configuration the monodromy of the curve leaves only the Hodge classes that come from polarisations and from Weil classes; we compute that monodromy for cyclic covers of degree three, four and six by an induction on the number of branch points

(Proposition 6.64), so the Hodge conjecture holds for the very general such intersection of quadrics in every dimension, of cubics and quartics up to dimension seven, and of sextics up to dimension five, in every projective space (Theorem 6.66). On the powers, Hilbert schemes and moduli spaces of a K3 surface whose transcendental field is a CM field, and on the fourfolds of $\mathrm{K3}^{[2]}$ type with field $\mathbb{Q}$ or a CM field, the Hodge conjecture itself holds (Proposition 6.39, Proposition 6.40).

The first instance of (F2) is a pair of classes on the square of a Mumford fourfold, and Section 7 studies it in detail. The two classes are the real multiplication by a totally real cubic field on the primitive part of H^2, and they are algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is algebraic (Proposition 7.2, Theorem 7.4); they are algebraic at every CM point (Theorem 6.29); they are rigid, their Hodge locus being the compact Mumford curve (Theorem 7.21); and they are represented by algebraic cycles at every closed point of every reduction of the curve modulo a prime, so that their algebraicity on every member is a bound on the Hilbert data of those cycles (Proposition 7.42, Theorem 7.43). For an abelian scheme over a curve whose invariant classes are algebraic the Lefschetz standard conjecture holds on the total space (Theorem 6.21), and over a compact Shimura curve the total space of a Mumford family satisfies the Hodge conjecture (Proposition 6.26).

The structure that [I, Theorem 1.1] finds in one family is present in every family that carries (F2) or (F3′) (Section 8). Every Hodge class of every abelian variety lies in an abelian scheme with a member of CM type, and (F2) is equivalent to the statement that in every such scheme the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Proposition 8.4, Theorem 8.5); the base points of the Weil families are the seeds, through André's theorem on the Hodge classes of abelian varieties of CM type. (F3′) is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 8.7). On the squares of K3 surfaces with real multiplication, where (F3′) is first open, the algebraic locus of the real multiplication contains every CM point, is dense, is stable under rational Hodge isometries, and is either the whole family or meagre (Theorem 8.9). In each case the property still missing is the interior point (Theorem 8.10).

Finally, the implications of this paper, of its companion and of the literature form a finite rule set, and its consequences can be determined by hand. They give the minimal sets of statements from which the conjecture follows (Theorem 6.70, Remark 7.50): (F3′) together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, or with (F2), besides two sets that are each equivalent to the conjecture.[4] Section 9 collects the results in one theorem.

Every proof is written to be checked by hand. Where an argument rests on a finite calculation, the calculation is given where it is used, as an explicit identity, a determinant or a case analysis. The archive [BMB26] holds programs that repeat much of this arithmetic and a Lean 4 certificate of part of it; no statement of the paper rests on them.

2. The semiregularity criterion at one point

2.1. **The second criterion: a rank at one point.** The other way from a dense algebraic locus to an open one is to make the cycle move, and the classical tool for that

[4] The Lefschetz standard conjecture asks that the inverse of the hard Lefschetz isomorphism $L^{\dim X-k}\colon H^k(X,\mathbb{Q}) \to H^{2\dim X-k}(X,\mathbb{Q})$ be induced by an algebraic correspondence. The variational Hodge conjecture asks that a flat section of Hodge classes over the base of a smooth projective family that is algebraic at one fibre be algebraic at every fibre.

is semiregularity. The theorem below adds nothing to the theory of semiregularity itself. Its content is that, after [I, Theorem 4.14], [I, Theorem 15.27] and [I, Theorem 16.4], the semiregularity of a single object at a single point implies the Hodge conjecture for the whole family, with no further propagation argument. We begin with the object: it exists, and its Chern character is concentrated in one degree and is a flat Hodge class on the whole family.

Proposition 2.1 (The K-theory class at a base point)**.** *Let $s_0 \in \Sigma$ be a base point of* [I, Theorem 15.27] *and write $A_0 = A_{s_0}$. There are an integer $N \geq 1$ and a perfect complex E on A_0 with*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq n), \qquad \mathrm{ch}_n(E) = N\,\omega_1.$$

The tuple $v = \mathrm{ch}(E)$ extends to a flat section of $\bigoplus_k R^{2k} f_* \mathbb{Q}$ over $\mathcal{S}$ which is of type (k,k) at every point.

Proof. By [I, Theorem 15.27] there is $\xi \in \mathrm{CH}^n(A_0)_{\mathbb{Q}}$ with $\mathrm{cl}(\xi) = \omega_1$. For a smooth quasi-projective variety over a field the Chern character $\mathrm{ch}\colon K_0(A_0)_{\mathbb{Q}} \to \mathrm{CH}^*(A_0)_{\mathbb{Q}}$ is a ring isomorphism [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so there is a unique $u \in K_0(A_0)_{\mathbb{Q}}$ with $\mathrm{ch}(u) = \xi$, and its Chern character has no component outside degree n. Choose $N \geq 1$ such that Nu is the image of a class in $K_0(A_0)$; that class is a difference of classes of vector bundles, hence the class of a perfect complex E, and $\mathrm{ch}(E) = N\xi$. Passing to cohomology gives the displayed identities.

For the last sentence, $\mathrm{ch}(E)$ has a single nonzero component, equal to $N\omega_1$, and ω is a flat section of type (n,n) everywhere by [I, Lemma 16.2]. ∎

Theorem 2.2 (Buchweitz and Flenner [BF03]; Pridham [Pri24])**.** *Let X be a smooth projective complex variety, E a perfect complex on X, and $X \hookrightarrow \mathcal{X} \to \operatorname{Spec} R$ a deformation of X over an Artin local $\mathbb{C}$-algebra along which $\mathrm{ch}(E)$ stays of Hodge type. If the semiregularity map*

$$\sigma_E\colon \operatorname{Ext}^2_X(E,E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega^q_X), \qquad \sigma_E(\zeta) = \sum_q \tfrac{1}{q!}\,\mathrm{tr}\big(\zeta \circ \mathrm{At}(E)^q\big),$$

is injective, then E extends to a perfect complex on $\mathcal{X}$.

In this generality, for perfect complexes and for every obstruction annihilated by σ_E rather than only the curvilinear ones, the theorem is proved by Pridham [Pri24], who identifies σ_E with the tangent map of a generalised Abel–Jacobi map on the derived moduli of perfect complexes; for coherent sheaves see also [BLM23]. Perry [Per26] extends it to noncommutative varieties with group actions, which covers twisted derived categories and answers a question of Markman.

Theorem 2.3 (Semiregularity at one point closes the family)**.** *Let $n \geq 2$, let s_0 and E be as in Proposition 2.1, and suppose the semiregularity map σ_E of Theorem 2.2 is injective. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$ in that family.*

Proof. Let $\mathfrak{M}$ be the stack of perfect complexes with vanishing negative self-extensions on the fibres of $f\colon \mathcal{A} \to \mathcal{S}$. It is algebraic and locally of finite presentation over $\mathcal{S}$ [Lie06], and at a point $[E]$ over s_0 its deformation theory relative to $\mathcal{S}$ has tangent space $\operatorname{Ext}^1(E,E)$ and obstruction space $\operatorname{Ext}^2(E,E)$. If $\operatorname{Ext}^{<0}(E,E) \neq 0$, which that stack excludes,[5] use

[5] Lieblich's stack parametrises the complexes he calls universally gluable, those with $\operatorname{Ext}^{<0}(E,E) = 0$; without that condition a complex can have automorphisms of negative degree, and the natural moduli object is the derived stack of Toën and Vaquié.

instead the analytic Kuranishi germ of pairs from the proof of Theorem 3.5, which has the same tangent and obstruction spaces; the argument below then gives a nonempty analytic open set in place of a Zariski open one, which is all that [I, Corollary 16.5] needs.

By Proposition 2.1 the Chern character of E stays of Hodge type along every direction in $\mathcal{S}$, because it is a flat section of Hodge classes on the whole base. So Theorem 2.2 applies to every infinitesimal deformation of A_0 inside the family, and injectivity of σ_E kills every obstruction. The morphism $\mathfrak{M} \to \mathcal{S}$ is therefore formally smooth at $[E]$, and being locally of finite presentation it is smooth on an open neighbourhood of $[E]$, which we take connected. Smooth morphisms are open, so the image of that neighbourhood is a nonempty Zariski open subset $U \subseteq \mathcal{S}$.

For $s \in U$ pick a point of the neighbourhood over s, that is, a perfect complex E_s on A_s. Chern characters are locally constant in families, so $\mathrm{ch}(E_s)$ is the parallel transport of $\mathrm{ch}(E)$: it is $N\omega_1$ in degree n and zero elsewhere. Chern characters of perfect complexes are algebraic, so $N\omega_1$, and hence ω_1, is algebraic on A_s; the rest of the Weil line follows by [I, Lemma 16.3]. So $p^{-1}(U) \subseteq \Sigma$, which therefore contains a nonempty open set, and [I, Corollary 16.5] gives $\Sigma = \mathcal{D}_{n,n}$. Apply [I, Theorem 15.14(a)]. ∎

Remark 2.4 (The hypothesis is not met by the object above). Injectivity of σ_E is automatic for some objects and not for others, and its degree-zero component already shows both. For a line bundle L one has $\mathrm{Ext}^2(L,L) = H^2(\mathcal{O}_X)$ and $\sigma_0 = \mathrm{tr}$ is the identity on it, so σ_L is injective for every line bundle on every variety, and Theorem 2.2 returns the classical fact that a divisor class deforms wherever it stays of type $(1,1)$. The object of Proposition 2.1 has rank zero, so its trace component is not an isomorphism, and in general not injective, the source having no reason to be as small as $h^2(\mathcal{O}_{A_0}) = \binom{2n}{2}$. The hypothesis of Theorem 2.3 therefore rests on the components with $q \geq 1$, which is where semiregularity stops being formal and becomes a computation.

For the object of Proposition 2.1 the hypothesis fails: by [I, Proposition 9.10] it has $\dim \ker \sigma_E \geq n^2$ at every base point constructed in this paper (see also [I, Remark 9.13]). The hypothesis concerns a choice of perfect complex representing a class in K_0, and [I, Proposition 9.10] does not reach every representative. Theorem 2.5 and the results after it describe what a representative satisfying the hypothesis would have to look like, and Theorem 3.5 shows that there is none. The proof of Theorem 2.3 applies word for word to the polarised criterion (P2′) of Remark 3.6, and Remark 3.13 records which of the results below carry over to it.

2.2. **Consequences of the rank condition.** The hypothesis of Theorem 2.3 is a condition on a rank, which cannot be inspected until the object is written down. It nevertheless has consequences that hold for every representative at once, because the semiregularity map composed with the evaluation map of Hochschild cohomology is contraction into the Chern character (Setup 4.54), and the Chern character here is a Weil class. The annihilator of a Weil class in HT^2 is large, and injectivity forces the evaluation map to vanish on the whole of it. We compute the annihilator and draw the consequences. They are the exact analogue for (P2) of what Theorem 4.61 and Theorem 4.89 do for the secant route in Section 4, and they rest on the same local products.

Theorem 2.5 (Consequences of injectivity). *Let A be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, and write $H^1(A,\mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K, $V_\pm^{1,0}$, $V_\pm^{0,1}$ for their Hodge components and $T_\pm \subset H^0(T_A) = T_0A$ for the eigenspaces of K on the tangent space, each of dimension n. Let $\omega = a\alpha_+ + b\alpha_-$ be a class on the Weil line with $ab \neq 0$,*

for instance any nonzero rational one, and let E be a nonzero perfect complex on A with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.

(i) *The annihilator of $\mathrm{ch}(E)$ in $HT^2(A)$ is*

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\, E) = (V_+^{0,1} \otimes V_-^{0,1}) \ \oplus \ \mathrm{Hom}_K(H^{1,0}, H^{0,1}) \ \oplus \ (T_+ \otimes T_-),$$

of dimension $n^2 + 2n^2 + n^2 = 4n^2$: the mixed block of $H^{0,2}$, which is the annihilator of [I, Lemma 9.6]*; the K-linear first order deformations of A, which contain the tangent space of the polarised Weil family; and the mixed bivectors.*

(ii) *If σ_E is injective, then ev_E vanishes on this space. Explicitly: $z \cdot \mathrm{id}_E = 0$ for every $z \in V_+^{0,1} \otimes V_-^{0,1}$; E extends to first order along every K-linear deformation of A; and the translation classes satisfy $A_\theta A_{\theta'} = 0$ in $\mathrm{Ext}^2(E,E)$ for every $\theta \in T_+$ and $\theta' \in T_-$.*

(iii) *Under the same hypothesis the algebra map $H^*(\mathcal{O}_A) \to \mathrm{Ext}^*(E,E)$, $z \mapsto z \cdot \mathrm{id}_E$, kills the ideal generated by the mixed block, which is $\bigoplus_{a,b\geq 1} \bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$; so the image of $H^{0,k}(A)$ has dimension at most $2\binom{n}{k}$ for $1 \leq k \leq n$ and is zero for $k > n$, and every endomorphism f of E has $\mathrm{tr}(f) = 0$ in $H^0(\mathcal{O}_A) = \mathbb{C}$.*

(iv) *Under the same hypothesis, if at some point p of its support E is locally isomorphic to i_*F with F locally free of positive rank on a smooth closed subvariety $Z \ni p$ of codimension at least two, then $T_+ \subseteq T_pZ$ or $T_- \subseteq T_pZ$.*

Proof. (i) Write $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ and $\alpha_\pm$ for a generator of $\bigwedge^{2n} V_\pm = \bigwedge^n V_\pm^{1,0} \otimes \bigwedge^n V_\pm^{0,1}$. The three summands of HT^2 act as follows: $z \in H^{0,2} = \bigwedge^2 H^{0,1}$ by wedge product; $v \in H^1(T_A) = \mathrm{Hom}(H^{1,0}, H^{0,1})$ as the derivation that replaces one $(1,0)$ factor by its image; $\pi \in H^0(\bigwedge^2 T_A) = \bigwedge^2 (H^{1,0})^\vee$ by double contraction of $(1,0)$ factors. For z, the computation is [I, Lemma 9.6]. For v, decompose $\mathrm{Hom}(H^{1,0}, H^{0,1})$ into its four blocks by eigenspaces. An element of the block $\mathrm{Hom}(V_+^{1,0}, V_+^{0,1})$ replaces a factor of $\bigwedge^n V_+^{1,0}$ in α_+ by an element of $V_+^{0,1}$, which is already exhausted in α_+, so it kills α_+; it acts trivially on α_-, which has no $V_+^{1,0}$ factor. Likewise $\mathrm{Hom}(V_-^{1,0}, V_-^{0,1})$ kills ω. A nonzero element of the block $\mathrm{Hom}(V_+^{1,0}, V_-^{0,1})$ carries α_+ to a nonzero element of $\bigwedge^{n-1} V_+^{1,0} \otimes V_-^{0,1} \otimes \bigwedge^n V_+^{0,1}$ and kills α_-, and a nonzero element of $\mathrm{Hom}(V_-^{1,0}, V_+^{0,1})$ carries α_- to a nonzero element of a different summand of $H^{n-1,n+1}$ and kills α_+. Since $ab \neq 0$ and the two summands are independent, v kills ω if and only if its two mixed blocks vanish, that is, if and only if v is K-linear. The space of K-linear maps has dimension $2n^2$, and it contains the n^2-dimensional tangent space of the polarised family, which is its symmetric part. For π, a mixed bivector $\theta \wedge \theta'$ with $\theta \in T_+ = (V_+^{1,0})^\vee$ and $\theta' \in T_- = (V_-^{1,0})^\vee$ kills α_+ because θ' kills every factor of $\bigwedge^n V_+^{1,0}$, and kills α_- symmetrically; a nonzero bivector in $\bigwedge^2 T_+$ carries α_+ to a nonzero element of $\bigwedge^{n-2} V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$ and kills α_-, and symmetrically for $\bigwedge^2 T_-$, the two images lying in different summands; so the annihilator in $H^0(\bigwedge^2 T_A)$ is $T_+ \otimes T_-$. Since the images of the three summands of HT^2 lie in the three different Hodge components $H^{n,n+2}$, $H^{n-1,n+1}$ and $H^{n-2,n}$ of $H^*(A,\mathbb{C})$, the annihilator is the direct sum of the three annihilators computed, of dimensions n^2, $2n^2$ and n^2.

(ii) By the commutative square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ of Setup 4.54, σ_E vanishes on $\mathrm{ev}_E(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, E))$; if σ_E is injective, ev_E vanishes there. The three explicit statements are (15) on the three summands of (i): on $H^{0,2}$ the evaluation map is $z \mapsto z \cdot \mathrm{id}_E$; on $H^1(T_A)$ it is the obstruction map $v \mapsto v \lrcorner \mathrm{At}(E)$, whose vanishing on v is the extension of

E to first order along v [Tod09]; and on a bivector $\theta \wedge \theta'$ it is $\frac{1}{2}(\theta \wedge \theta') \lrcorner \mathrm{At}(E)^2 = A_\theta A_{\theta'}$, as in the proof of Theorem 4.61.

(iii) The map $z \mapsto z \cdot \mathrm{id}_E$ is a homomorphism of graded algebras from $H^*(\mathcal{O}_A) = \bigwedge^* H^{0,1}$ to the graded centre of the Yoneda algebra, so its kernel is an ideal, and by (ii) it contains the mixed block $V_+^{0,1} \otimes V_-^{0,1}$. The ideal generated by the mixed block in $\bigwedge^*(V_+^{0,1} \oplus V_-^{0,1}) = \bigwedge^* V_+^{0,1} \otimes \bigwedge^* V_-^{0,1}$ is the sum of the summands $\bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$ with $a, b \geq 1$, since each such summand consists of multiples of the mixed block, while the remaining summands, with $a = 0$ or $b = 0$, meet the ideal in zero; the quotient in degree $k \geq 1$ is $\bigwedge^k V_+^{0,1} \oplus \bigwedge^k V_-^{0,1}$, of dimension $2\binom{n}{k}$, and zero for $k > n$. In particular the top class of $H^{2n}(\mathcal{O}_A)$, which is mixed for $n \geq 1$, acts by zero. Serre duality on A, whose canonical bundle is trivial, is the perfect pairing $\mathrm{Ext}^{2n}(E, E) \times \mathrm{Hom}(E, E) \to H^{2n}(\mathcal{O}_A) \cong \mathbb{C}$, $(\xi, f) \mapsto \mathrm{tr}(\xi \circ f)$; for $\xi = w \cdot \mathrm{id}_E$ with w the top class this is $w \cdot \mathrm{tr}(f)$, where $\mathrm{tr}(f) \in H^0(\mathcal{O}_A) = \mathbb{C}$, so $w \cdot \mathrm{id}_E = 0$ forces $\mathrm{tr}(f) = 0$ for every f.

(iv) Let $\mathcal{O}$ be the local ring of A at p and choose coordinates with $Z = \{x_1 = \cdots = x_c = 0\}$, $c \geq 2$. Locally $E \cong (\mathcal{O}/J)^{\oplus r}$ with $J = (x_1, \ldots, x_c)$, a regular sequence, so $\mathrm{Ext}^*_{\mathcal{O}}(\mathcal{O}/J, \mathcal{O}/J) = \bigwedge^*_{\mathcal{O}/J} N$ as a graded algebra, where $N = \mathrm{Hom}(J/J^2, \mathcal{O}/J)$ is the normal module with basis the classes $\partial_1, \ldots, \partial_c$ dual to $x_1, \ldots, x_c$; the Ext^1 is computed on the Koszul resolution exactly as in the proof of Lemma 4.60, and the product is the wedge because the Yoneda algebra of a Koszul complex is the exterior algebra on the dual of the generators. For E of rank r the local Yoneda algebra is $\bigwedge^* N \otimes M_r(\mathcal{O}/J)$. By Lemma 4.59 the germ of A_θ at p is the jet class along θ; on the Koszul complex the jet class of a derivation D is the chain map $-D(d)$, and for $D = \theta = \sum_i c^i \partial_i$ its component in $\mathrm{Ext}^1 = N \otimes M_r(\mathcal{O}/J)$ is $\sum_{i \leq c} (c^i \bmod J)\, \partial_i \otimes \mathrm{id}$, the class of θ modulo T_Z. Multiplicativity of the germ map then gives the germ of $A_\theta A_{\theta'}$ as $(\theta \bmod T_Z) \wedge (\theta' \bmod T_Z) \otimes \mathrm{id}$ in $\bigwedge^2 N \otimes M_r(\mathcal{O}/J)$, whose value at p is $\bar{\theta}(p) \wedge \bar{\theta}'(p) \in \bigwedge^2 (T_p A / T_p Z)$. If $A_\theta A_{\theta'} = 0$ for all $\theta \in T_+$, $\theta' \in T_-$, then $\bar{a} \wedge \bar{b} = 0$ for all $\bar{a}$ in the image I_+ of T_+ and $\bar{b}$ in the image I_- of T_- in $N_p = T_p A / T_p Z$, and $I_+ + I_- = N_p$ since $T_+ \oplus T_- = T_p A$. If both $I_\pm$ were nonzero, every nonzero $\bar{b} \in I_-$ would be proportional to every nonzero $\bar{a} \in I_+$, so $I_+ = I_-$ would be a line and $N_p = I_+ + I_-$ of dimension one, against $c \geq 2$. Hence $I_+ = 0$ or $I_- = 0$, which is the claim. ∎

Corollary 2.6 (Locally free components are excluded)**.** *Let A be as in Theorem 2.5 and suppose that no proper abelian subvariety $B \subset A$ has $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$; equivalently, that A has no nonzero K-stable abelian subvariety on which K acts through a single embedding. This holds whenever A is simple, at every split member $X \times \widehat{X}$ with X simple of endomorphism algebra $\mathbb{Q}$, and at a general point of the quaternionic locus of* [I, Setup 15.24]*. Let E be a perfect complex on A with $\mathrm{ch}(E) = N\omega$ as above and σ_E injective. Then no irreducible component Z_0 of the support of E of codimension at least two has a dense open subset along which E is a locally free sheaf of positive rank on Z_0. In particular E is not a sheaf, and along every component of its support of codimension at least two it has, generically, either at least two nonvanishing cohomology sheaves or a cohomology sheaf that is not locally free on the reduced support.*

Proof. Suppose that Z_0 is such a component, with $U \subseteq Z_0^{\mathrm{sm}}$ dense open and $E|_U$ a locally free sheaf of positive rank on U. By Theorem 2.5(iv), $U = U_+ \cup U_-$ with $U_\pm = \{p \in U : T_\pm \subseteq T_p Z_0\}$, both closed in U; as U is irreducible, $U = U_+$ say. Then every constant vector field $\theta \in T_+$ is tangent to Z_0 along U, hence along Z_0^{sm}. The set $S = \{(p, t) \in Z_0 \times \mathbb{C} : p + \exp(t\theta) \in Z_0\}$ is a closed analytic subset of the irreducible space $Z_0 \times \mathbb{C}$, and it contains an open neighbourhood of $Z_0^{\mathrm{sm}} \times \{0\}$, because the local flow of the restriction of θ to the submanifold Z_0^{sm} exists on such a neighbourhood; so $S = Z_0 \times \mathbb{C}$,

and $Z_0 + \exp(t\theta) \subseteq Z_0$, hence $= Z_0$, for every t. Let $G = \{x \in A : Z_0 + x = Z_0\}$, a closed algebraic subgroup of A, and B its identity component, an abelian subvariety. The connected set $\exp(T_+)$ contains 0 and lies in G, hence in B, so $T_+ \subseteq T_0B$. By hypothesis $B = A$, so $Z_0 = A$, against the codimension. The last sentence of the statement is the negation of the condition just excluded. If E were a sheaf, its support would have codimension at least n, because the Chern character of a nonzero sheaf begins in the codimension of its support, with the support cycle, whose multiplicities are positive; this is excluded by Theorem 2.13(iii) below, whose proof uses nothing from the present one beyond the translation argument above.

For the equivalence in the hypothesis, let $\sqrt{-d}$ act on A through an order of K. If an abelian subvariety $B \neq A$ has $T_0B \supseteq T_+$, then $H_1(B,\mathbb{Q}) \cap \sqrt{-d}\,H_1(B,\mathbb{Q})$ is a K-stable sub-Hodge structure, hence $H_1(C,\mathbb{Q})$ for a K-stable abelian subvariety $C \subseteq B$, and $T_0C = T_0B \cap \sqrt{-d}\,T_0B$ still contains T_+, on which $\sqrt{-d}$ acts by a scalar. The K-stable complement C' of C for the polarisation is nonzero, since $C \subseteq B \neq A$, and T_0C' is a K-stable complement of $T_0C \supseteq T_+$, so $T_0C' \subseteq T_-$ and K acts on C' through a single embedding. Conversely, if a nonzero K-stable $C' \neq A$ has $T_0C' \subseteq T_-$, its K-stable complement B is proper with $T_0B \supseteq T_+$. For the three cases: a simple A has no nonzero proper abelian subvariety; for $X \times \widehat{X}$ with X simple and $\mathrm{End}^0(X) = \mathbb{Q}$ the sub-Hodge structures of $H_1(X) \otimes \mathbb{Q}^2$ are $H_1(X) \otimes W$ with $W \subseteq \mathbb{Q}^2$, and since K acts on $\mathbb{Q}^2$ through a quadratic field, W is K-stable only for $W = 0$ or $\mathbb{Q}^2$; and at a general quaternionic member A is either simple or, when the quaternion algebra of [I, Setup 15.24] splits, isogenous to $C \times C$ with C a general principally polarised abelian n-fold, whose sub-Hodge structures are again $H_1(C) \otimes W$ with the same conclusion. ∎

2.3. **The Hochschild action and the two branches.** Theorem 2.5 computes the annihilator summand by summand and reads off what the vanishing of the evaluation map on it means for E. That description hides a structure which is easily brought out and gives more: the annihilator is the mixed part of a splitting of $HH^1(A)$ into two halves, one for each of the two conjugate pieces α_+ and α_- of the Weil class, and the same splitting describes the annihilator in every degree. The results of this subsection concern the Hochschild cohomology of A alone, and their consequences for E follow formally.

Setup 2.7 (Hochschild cohomology in these terms)**.** For an abelian variety A the Hochschild–Kostant–Rosenberg isomorphism identifies

$$HH^*(A) \;\cong\; HT^*(A) = \bigoplus_{p,q} H^p(\textstyle\bigwedge^q T_A) = \textstyle\bigwedge^*(H^{0,1}(A) \oplus H^0(T_A)),$$

as graded rings, the grading being $p+q$: the Todd class of A is trivial, so the twist by $\mathrm{td}^{1/2}$ in [Cal05], [CRV12] is the identity and the plain identification is already multiplicative. Write $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$, of dimension $4n$, so that $HH^*(A)$ is the exterior algebra on it, of dimension 2^{4n}. It acts on $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ by wedging with the classes of $H^{0,1}$ and contracting the classes of $H^{1,0}$ against $H^0(T_A)$; the two families of operators involve disjoint generators, so they pairwise anticommute and the action is an action of the exterior algebra. In degree two this is the operator $x\lrcorner$ of Setup 4.54, and $\mathrm{ev}_E \colon HH^*(A) \to \mathrm{Ext}^*(E,E)$ is a homomorphism of graded rings into the graded centre, with $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner \mathrm{ch}(E)$ [BF08, Section 6], [Cal05], [CRV12].

Theorem 2.8 (The annihilator is the mixed part of a splitting)**.** *Let A be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $\omega = a\alpha_+ + b\alpha_-$ lie on the Weil line with $ab \neq 0$. Put*

$$P = \mathrm{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \qquad Q = \mathrm{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

(i) $\dim P = \dim Q = 2n$ *and* $HH^1(A) = P \oplus Q$*; in particular* $\mathrm{Ann}_{HH^1}(\omega) = 0$.

(ii) $\mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$*, of dimension* $4n^2$*. This is the space computed in Theorem 2.5(i), written as the mixed part of the splitting.*

(iii) *Contraction with* α_- *is injective on* $\bigwedge^* P$*, and contraction with* α_+ *is injective on* $\bigwedge^* Q$*; the two images meet in exactly one line, in degree* $2n$*.*

(iv) *For every* $k \geq 1$*,*

$$\mathrm{Ann}_{HH^k}(\omega) = \Big(\bigoplus_{\substack{i+j=k \\ i,j \geq 1}} \bigwedge^i P \wedge \bigwedge^j Q \Big) \ \oplus \ \epsilon_k,$$

where $\epsilon_k = 0$ *for* $k \neq 2n$ *and* ϵ_{2n} *is the line of* (iii)*. So* $\dim \mathrm{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$ *for* $k \geq 1$*, while* $\mathrm{Ann}_{HH^0}(\omega) = 0$*, and the annihilator is the ideal generated by its degree-two part together with that one line.*

Proof. Choose bases $x_1, \dots, x_n$ of $V_+^{1,0}$ and $y_1, \dots, y_n$ of $V_-^{1,0}$, and bases $\overline{x}_i$ of $V_+^{0,1}$ and $\overline{y}_j$ of $V_-^{0,1}$ (the bar is notation only: complex conjugation carries $V_+^{1,0}$ to $V_-^{0,1}$), so that $\alpha_+ = x_1 \cdots x_n \overline{x}_1 \cdots \overline{x}_n$ and $\alpha_- = y_1 \cdots y_n \overline{y}_1 \cdots \overline{y}_n$, and write $\partial_{x_i}, \partial_{y_j}$ for the dual basis of $H^0(T_A) = T_+ \oplus T_-$.

(i) Of the $4n$ basis operators, $\overline{x}_i \wedge$ kills α_+ because $\overline{x}_i$ already occurs in it, and $\partial_{y_j} \lrcorner$ kills α_+ because no y occurs in it; so $P \supseteq V_+^{0,1} \oplus T_-$. Conversely, the classes $\overline{y}_j \wedge \alpha_+$ are linearly independent, as are the classes $\partial_{x_i} \lrcorner \alpha_+$, and the two families lie in different summands of H^*, so no nonzero combination of the operators $\overline{y}_j \wedge$ and $\partial_{x_i} \lrcorner$ kills α_+: $P = V_+^{0,1} \oplus T_-$, of dimension $2n$. The computation for Q is the same with the roles of the two eigenspaces exchanged, and $P \cap Q = 0$ with $\dim P + \dim Q = 4n$, so $HH^1 = P \oplus Q$. A class of degree one that kills ω kills α_+ and α_- separately, the two images lying in different summands and $ab \neq 0$, so it lies in $P \cap Q = 0$.

(iii) A monomial $\overline{x}_I \partial_{y_J} \in \bigwedge^* P$ sends α_- to $\pm\, \overline{x}_I \, y_{[n]\setminus J} \, \overline{y}_{[n]}$, and these are linearly independent for distinct pairs (I, J), which is the injectivity; the statement for Q is symmetric. For the intersection, the image of $\bigwedge^* P$ is spanned by monomials with no factor x and all n factors $\overline{y}$, and the image of $\bigwedge^* Q$, by the same computation, by monomials with no factor y and all n factors $\overline{x}$. Both images are spanned by monomials, so their intersection is spanned by the monomials common to both, and the only such monomial is $\overline{x}_{[n]} \overline{y}_{[n]}$. It is the image of $\pi_P = \overline{x}_{[n]} \partial_{y_{[n]}}$ and of $\pi_Q = \overline{y}_{[n]} \partial_{x_{[n]}}$, both of degree $2n$. So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose $\bigwedge^k HH^1 = \bigoplus_{i+j=k} \bigwedge^i P \wedge \bigwedge^j Q$. A summand with $i, j \geq 1$ annihilates α_+, because it has a factor in P and the action is an action of the algebra, and annihilates α_- for the same reason with Q; so it lies in the annihilator of ω. The sum of these summands over all k is the ideal generated by $P \wedge Q$, since each of their monomials has, after reordering, a factor $p \wedge q$ with $p \in P$ and $q \in Q$. On the remaining two summands, $\xi_P \in \bigwedge^k P$ has $\xi_P \lrcorner \omega = b\, \xi_P \lrcorner \alpha_-$ and $\xi_Q \in \bigwedge^k Q$ has $\xi_Q \lrcorner \omega = a\, \xi_Q \lrcorner \alpha_+$, so a combination $\xi_P + \xi_Q$ annihilates ω if and only if $b\, \xi_P \lrcorner \alpha_- = -a\, \xi_Q \lrcorner \alpha_+$. By (iii) both contractions are injective and their images meet only in degree $2n$, in one line; so this happens only for $k = 2n$, and then only on the line spanned by $b^{-1}\pi_P - a^{-1}\pi_Q$, with π_P and π_Q scaled so that $\pi_P \lrcorner \alpha_- = \pi_Q \lrcorner \alpha_+$. In degree zero the annihilator is zero because $\omega \neq 0$. Taking $k = 2$ and $n \geq 2$, so that $2 \neq 2n$, gives (ii), the dimension being $\dim P \cdot \dim Q = 4n^2$; the general count is $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$ plus the exceptional line. ∎

Corollary 2.9 (Consequences for the Hochschild action)**.** *Let A and ω be as above and let E be a nonzero perfect complex on A with $\mathrm{ch}(E) = N\omega$, $N \neq 0$.*

(i) *Unconditionally,* $\dim \mathrm{Ext}^2(E,E) \geq 2n(2n-1)$.

(ii) *If σ_E is injective then ev_E vanishes on $P \wedge Q$, hence on the ideal it generates, and the ring homomorphism $\mathrm{ev}_E \colon HH^*(A) \to \mathrm{Ext}^*(E,E)$ factors through*

$$R \;=\; \textstyle\bigwedge^* P \;\oplus_{\mathbb{C}}\; \bigwedge^* Q, \qquad \dim_{\mathbb{C}} R = 2^{2n+1} - 1,$$

the two exterior algebras glued along their scalars, inside $HH^*(A)$ of dimension 2^{4n}. At $n = 2$ that is 31 out of 256.

(iii) *Granting the compatibility $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ in degree k [Cal05], [CRV12], the induced map on R is injective away from the line of Theorem 2.8(iii), so*

$$\dim \mathrm{Ext}^k(E,E) \;\geq\; 2\binom{2n}{k} \quad (1 \leq k < 2n), \qquad \sum_{k \geq 1} \dim \mathrm{Ext}^k(E,E) \;\geq\; 2^{2n+1} - 3.$$

Proof. (i) By Setup 2.7, $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^2}(\mathrm{ch}\, E)$ in degree two, since $\sigma_E(\mathrm{ev}_E(\xi)) = \xi \lrcorner \mathrm{ch}(E)$ and the right side is nonzero off the annihilator. Hence $\dim \mathrm{Ext}^2(E,E) \geq \dim HH^2(A) - \dim \mathrm{Ann}_{HH^2}(\omega) = \binom{4n}{2} - 4n^2 = 2n(2n-1)$ by Theorem 2.8(ii). No hypothesis on σ_E is used.

(ii) ev_E vanishes on the annihilator by Theorem 2.5(ii), its kernel is an ideal because it is a ring map, and the ideal generated by $P \wedge Q$ in $\bigwedge^*(P \oplus Q) = \bigwedge^* P \otimes \bigwedge^* Q$ is the sum of the summands with $i, j \geq 1$, whose complement is $\bigwedge^* P \oplus_{\mathbb{C}} \bigwedge^* Q$ of dimension $2 \cdot 2^{2n} - 1$.

(iii) With the compatibility in degree k, the argument of (i) gives $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^k}(\omega)$, and by Theorem 2.8(iv) that annihilator meets $\bigwedge^k P \oplus \bigwedge^k Q$ only in the exceptional line. So the map induced on R has kernel contained in that one line and its image has dimension at least $\dim R - 1 = 2^{2n+1} - 2$. One of those dimensions sits in degree zero, where the image is $\mathbb{C} \cdot \mathrm{id}_E$, nonzero because E is; the remaining $2^{2n+1} - 3$ lie in positive degrees, which is the second inequality. For $1 \leq k < 2n$ the exceptional line has no component in degree k, so $\bigwedge^k P \oplus \bigwedge^k Q$, of dimension $2\binom{2n}{k}$, maps injectively, which is the first. ∎

Remark 2.10 (The two branches). The ring R of Corollary 2.9(ii) is the ring of functions on two odd affine spaces of dimension $2n$ crossing at the origin. In these terms, if E satisfies the criterion, the deformations of $D^b(A)$ along which its Hochschild action survives form two branches, P and Q, which meet only at the trivial deformation and are the annihilators of the two conjugate pieces α_+ and α_- of the Weil class. The criterion says that E sees the two halves of the class separately and their interaction not at all.

The splitting $HH^1 = P \oplus Q$ is not the splitting into eigenspaces of K, which are $V_+^{0,1} \oplus T_+$ and $V_-^{0,1} \oplus T_-$; P and Q cut across them. It is the splitting into the annihilators of α_+ and of α_-, and it exists because each of the two pieces of the Weil class is a top form on an eigenspace. Nor is the conclusion a contradiction in itself. R is a perfectly good ring, and the bound of (iii) is consistent with the value of $\chi(E,E)$ computed in [I, Proposition 9.11], since $\sum_k (-1)^k 2\binom{2n}{k} = 0$. The corollary says how large such an E must be and how little of the Hochschild cohomology it may see, but nothing in the Hochschild computation forbids an E of the required shape. The exclusion comes from a different direction, deformation to non-algebraic Weil tori, in Theorem 3.5.

2.4. **The numerical form of the criterion and the support.** The same square gives two further results. The first converts the rank of σ_E into a single dimension, the form in which a candidate can be tested. The second locates the support of E, and it excludes the first place one would look: a cycle representing the Weil class.

Theorem 2.11 (The criterion as a dimension)**.** *Let A, ω and E be as in Corollary 2.9, so that* $\operatorname{ch}(E) = N\omega$ *with* $N \neq 0$ *and no other component.*

(i) *If* $\dim \operatorname{Ext}^2(E,E) = 2n(2n-1)$*, the least value that Corollary 2.9(i) allows, then* $\operatorname{ev}_E \colon HH^2(A) \to \operatorname{Ext}^2(E,E)$ *is surjective with kernel exactly* $P \wedge Q$*, the map* σ_E *is injective, and every class of* $\operatorname{Ext}^2(E,E)$ *is a sum of products* $\operatorname{ev}_E(p)\operatorname{ev}_E(p') + \operatorname{ev}_E(q)\operatorname{ev}_E(q')$ *with* $p, p' \in P$ *and* $q, q' \in Q$*, hence lies in the graded centre of* $\operatorname{Ext}^*(E,E)$*.*

(ii) *Conversely, if* σ_E *is injective, then* $\operatorname{ev}_E(HH^2(A))$ *has dimension exactly* $2n(2n-1)$*, and* σ_E *carries it isomorphically onto* $HH^2(A) \lrcorner \omega$*.*

(iii) *Consequently, if at one base point* s_0 *of a family some perfect complex* E *with* $\operatorname{ch}(E) = N\omega_1$ *has* $\dim \operatorname{Ext}^2(E,E) = 2n(2n-1)$*, then* $\Sigma = \mathcal{D}_{n,n}$*, and the Hodge conjecture holds in every degree for every member of that family with* $\operatorname{Hg}(A) = \operatorname{SU}(V,H)$*.*

Proof. (i) The square of Setup 2.7 gives $\ker \operatorname{ev}_E \subseteq \operatorname{Ann}_{HH^2}(\omega) = P \wedge Q$ in degree two, the equality being Theorem 2.8(ii). Hence $\operatorname{rank} \operatorname{ev}_E \geq \binom{4n}{2} - 4n^2 = 2n(2n-1) = \dim \operatorname{Ext}^2(E,E)$, so ev_E is onto and its kernel, of dimension $4n^2$ and contained in $P \wedge Q$, is all of $P \wedge Q$. Every class of $\operatorname{Ext}^2(E,E)$ is then $\operatorname{ev}_E(\xi)$ for some ξ, and $\sigma_E(\operatorname{ev}_E(\xi)) = N\,\xi \lrcorner \omega$ vanishes only for $\xi \in P \wedge Q = \ker \operatorname{ev}_E$; so σ_E is injective. Since $HH^2(A) = \bigwedge^2 P \oplus (P \wedge Q) \oplus \bigwedge^2 Q$ and ev_E is a ring map into the graded centre, the image is $\operatorname{ev}_E(\bigwedge^2 P) + \operatorname{ev}_E(\bigwedge^2 Q)$, which is the description of the classes.

(ii) By Theorem 2.5(ii) the kernel of ev_E in degree two contains $P \wedge Q$, and by the argument of (i) it is contained in it, so it equals $P \wedge Q$ and the image has dimension $2n(2n-1)$. The composite $\sigma_E \circ \operatorname{ev}_E$ is contraction into $\operatorname{ch}(E) = N\omega$, whose kernel is exactly $P \wedge Q = \ker \operatorname{ev}_E$, so σ_E maps the image isomorphically onto $HH^2(A) \lrcorner \omega$.

(iii) By (i) the map σ_E is injective, and Theorem 2.3 applies. ∎

Theorem 2.11(iii) is a sufficient condition for $\mathbf{W}(K,n,\delta)$ that mentions no map at all: one object, at one point, and one number. At $n = 2$ the number is 12 and at $n = 3$ it is 30. By Theorem 3.5 no complex attains it, but the first two parts of the theorem hold for the polarised criterion (P2′) with the number $r(\gamma)$ of Theorem 3.9(iv) in place of $2n(2n-1)$, and that is the form in which the polarised criterion can be tested.

The dimension of $\operatorname{Ext}^2(E,E)$ is computable in a way the rank of σ_E is not. For a complex built from sheaves whose cohomology is known, the local-to-global spectral sequence $E_2^{p,q} = H^p(A, \mathcal{E}xt^q(E,E)) \Rightarrow \operatorname{Ext}^{p+q}(E,E)$ reduces the count to the cohomology of the local $\mathcal{E}xt$ sheaves, and on an abelian variety the cohomology of a sheaf supported on an abelian subvariety, or of a homogeneous bundle, is a question of linear algebra. The rank of σ_E, by contrast, requires the trace of the exponential of the Atiyah class on every class of Ext^2, a computation with the object and not only with its invariants. Part (ii) says that, on the central part of Ext^2, nothing is lost by reading the criterion this way. Whether an injective σ_E forces the whole of $\operatorname{Ext}^2(E,E)$ to be central is, by Serre duality, a statement about $\operatorname{Ext}^{2n-2}$, which the square in degree two does not reach; nothing below uses it.

The second result rests on a local fact about complexes supported at a point, which we state and prove first.

Lemma 2.12 (Two jet classes on a complex of finite length)**.** *Let* $R = \mathbb{C}[[x_1, \dots, x_c]]$ *with* $c \geq 2$*, let* M *be a bounded complex of finitely generated free* R*-modules whose cohomology has finite length, and put* $\chi(M) = \sum_i (-1)^i \ell(H^i(M))$*. For* $a \in \mathbb{C}^c$ *let* $J(a) \in \operatorname{Ext}^1_R(M,M)$ *be the jet class of* M *along the constant vector field* $\sum_i a_i \partial_i$*, represented by the chain*

map $-\sum_i a_i\partial_i(d)$ *as in the proof of Theorem 2.5(iv). If* $\chi(M)\neq 0$ *and* a, b *are linearly independent, then* $J(a)J(b)\neq 0$ *in* $\mathrm{Ext}^2_R(M,M)$.

Proof. Let T be an abelian variety of dimension c and $t\in T$ a point. Linear coordinates on the universal cover identify the completed local ring $\widehat{\mathcal{O}}_{T,t}$ with R in such a way that the translation invariant vector fields become the constant fields $\sum_i a_i\partial_i$: the exponential map from $T_tT=\mathbb{C}^c$ is a local biholomorphism carrying the constant vector fields to the translation invariant ones, and its germ at the origin identifies $\widehat{\mathcal{O}}_{T,t}$ with $\mathbb{C}[[x_1,\dots,x_c]]$ compatibly with derivations. A complex of R-modules with finite length cohomology is the same thing as a complex of $\mathcal{O}_{T,t}$-modules with finite length cohomology, since modules of finite length over either ring are modules over $\mathcal{O}_{T,t}/\mathfrak{m}^N$ for N large, so M defines a perfect complex M_T on T supported at t. Its local $\mathcal{E}xt$ sheaves are skyscrapers at t, so the local-to-global spectral sequence gives $\mathrm{Ext}^*_T(M_T,M_T)=\mathrm{Ext}^*_R(M,M)$, and by Lemma 4.59 the translation classes A_a, A_b of M_T restrict to $J(a)$, $J(b)$. Apply the square of Setup 4.54 on T to the bivector $a\wedge b$:

$$\sigma_{M_T}(A_aA_b)=(a\wedge b)\lrcorner\,\mathrm{ch}(M_T).$$

A sheaf of length ℓ supported at a point has Chern character $\ell\cdot[\mathrm{pt}]$, and Chern characters are additive over the cohomology of a complex, so $\mathrm{ch}(M_T)=\chi(M)\,[\mathrm{pt}]$. The class of a point is a nonzero multiple of $dz_1\cdots dz_c\,d\overline{z}_1\cdots d\overline{z}_c$, and contracting it with $a\wedge b$ removes two holomorphic factors and leaves a nonzero class of type $(c-2,c)$, because a and b are independent. So the right side is nonzero when $\chi(M)\neq 0$, hence so is A_aA_b, and so is its restriction $J(a)J(b)$, the restriction map being an isomorphism. ∎

The hypothesis on $\chi(M)$ cannot be dropped. The Euler characteristic is the only additive invariant here: the Grothendieck group of complexes with finite length cohomology over a regular local ring is $\mathbb{Z}$, generated by the residue field, since every module of finite length has a composition series whose factors are all isomorphic to it. The lemma says that the product of two independent jet classes detects this generator, and on a complex of Euler characteristic zero the product can vanish. Let K be the Koszul resolution of the residue field k of $\mathbb{Q}[x_1,x_2]$, on which the jet class of a derivation D is the chain map $-D(d_K)$, and let $u=\partial_2(d_K)\partial_1(d_K)\colon K\to K[2]$ represent $J(\partial_2)J(\partial_1)$. The entries of d_K are $\pm x_1,\pm x_2$, so u has constant entries. Let C be the cone of u, with $C^m=K^{m+1}\oplus K^{m+2}$ and $d_C(a,b)=(-d_Ka,\,u(a)+d_Kb)$. As u induces zero on cohomology, C has two cohomology modules, each isomorphic to k, so $\chi(C)=0$. Since $\partial_i(u)=0$, the matrix $\partial_i(d_C)$ is block diagonal, and the product $J(\partial_2)J(\partial_1)$ on C is $(a,b)\mapsto(u(a),u(b))$, which equals d_CH+Hd_C for $H(a,b)=(b,0)$: it is null homotopic, the gluing has killed it. On the split complex $k\oplus k[1]$, whose Euler characteristic is also zero, the same product is $u\oplus u[1]$, which is not null homotopic because $u\neq 0$ in $\mathrm{Ext}^2(k,k)$ by the lemma applied to k.

Theorem 2.13 (The support of an object meeting the criterion)**.** *Let* A *be an abelian* $2n$*-fold of* $(K,-1,n)$*-Weil type with* $n\geq 2$*, let* $\omega=a\alpha_+ + b\alpha_-$ *with* $ab\neq 0$*, and let* E *be a perfect complex on* A *with* $\mathrm{ch}(E)=N\omega$*,* $N\neq 0$*, no other component, and* σ_E *injective. For an irreducible component* Z *of the support of* E *write*

$$m_Z(E)=\sum_i(-1)^i\,\ell_{\mathcal{O}_{A,Z}}(\mathcal{H}^i(E)_Z)$$

for the multiplicity of Z *in the cycle of* E.

(i) *If* Z *has codimension at least two and* $m_Z(E)\neq 0$*, then* Z *is a union of cosets of an abelian subvariety* $B\subsetneq A$ *with* $T_0B\supseteq T_+$ *or* $T_0B\supseteq T_-$.

(ii) *If A has no proper abelian subvariety B with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, which by Corollary 2.6 holds when A is simple, at every split member $X \times \widehat{X}$ with X simple and $\mathrm{End}^0(X) = \mathbb{Q}$, and at a general point of the quaternionic locus, then $m_Z(E) = 0$ for every component Z of codimension at least two, and the cycle of E is a divisor.*

(iii) *At every member of the family, whatever its endomorphisms, the support of E has an irreducible component of codimension less than n.*

In particular no perfect complex whose support has codimension at least n, at any member of any Weil family, satisfies the criterion of Theorem 2.3. This includes every complex assembled from the structure sheaves of the components of a cycle representing ω, the base cycle of [I, Theorem 15.27] *among them.*

Proof. (i) Let $c \geq 2$ be the codimension of Z and let $U \subseteq Z$ be the dense open set of smooth points of Z_{red} that lie on no other component of the support. Fix $\theta \in T_+$ and $\theta' \in T_-$; by Theorem 2.5(ii), $A_\theta A_{\theta'} = 0$ in $\mathrm{Ext}^2(E,E)$. Suppose that at some $p \in U$ the images $\bar\theta, \bar\theta'$ of θ, θ' in $N_p = T_pA/T_pZ$ are linearly independent, and complete them by images of constant fields $\nu_3, \dots, \nu_c$ to a basis of N_p. In linear coordinates on the universal cover the constant fields are coordinate derivations, and the germ S at p of the translate of the span of $\theta, \theta', \nu_3, \dots, \nu_c$ is a smooth germ of dimension c, transverse to Z at p, to which θ and θ' are tangent. Let $j\colon S \to A$ be the inclusion and $M = Lj^*E$ on $\widehat{\mathcal{O}}_{S,p} \cong \mathbb{C}[[x_1, \dots, x_c]]$. Derived restriction is a ring map $\mathrm{Ext}^*(E,E) \to \mathrm{Ext}^*(M,M)$, and it carries the germ of A_θ to the jet class of M along $\theta|_S$, because the Atiyah class is functorial for pullback and θ is tangent to S [BF03, Section 3]. So $J(\theta)J(\theta') = 0$ on M. Near p the support of E is Z, and $Z \cap S = \{p\}$, so M has finite length cohomology; its Euler characteristic is the intersection multiplicity of the cycle of E with S at p, which, S being transverse to Z at a smooth point lying on no other component, is $m_Z(E)$, by Serre's formula for intersection multiplicities applied to each cohomology sheaf and summed with signs [Ser00, Chapter V].[6] Since θ and θ' are independent directions on S, Lemma 2.12 applied to M gives $m_Z(E) = 0$.

So if $m_Z(E) \neq 0$, then at every $p \in U$ and for all $\theta \in T_+$, $\theta' \in T_-$ the images satisfy $\bar\theta \wedge \bar\theta' = 0$ in $\bigwedge^2 N_p$. As in the proof of Theorem 2.5(iv), the images $I_\pm$ of $T_\pm$ in N_p span N_p, which has dimension $c \geq 2$, so one of them is zero: $T_+ \subseteq T_pZ$ or $T_- \subseteq T_pZ$. The two conditions cut out closed subsets of the irreducible set U whose union is U, so one of them, say the first, holds on all of U, and the argument of Corollary 2.6 shows that Z is stable under translation by the identity component B of its stabiliser, an abelian subvariety with $T_0B \supseteq T_+$. Since $Z \neq A$, $B \neq A$.

(ii) follows from (i), since no such B exists; if A itself is a component of the support, its multiplicity is the rank of E, which is zero, so the cycle of E is a divisor.

(iii) Suppose every component of the support has codimension at least n. Then $\mathrm{ch}_k(E) = 0$ for $k < n$ and $\mathrm{ch}_n(E) = \sum_Z m_Z(E)[Z]$, the sum over the components of codimension exactly n; this is Riemann–Roch without denominators in its simplest case [Ful98, Chapter 15]. By (i), each Z with $m_Z(E) \neq 0$ is $\pi^{-1}(\pi(Z))$ for the quotient $\pi\colon A \to A/B$ by an abelian subvariety B with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, so $[Z] = \pi^*[\pi(Z)]$ lies in the subring of $H^*(A,\mathbb{C})$ generated by $\pi^*H^1(A/B,\mathbb{C})$, the space of classes of degree one that vanish on T_0B and on its conjugate. When $T_0B \supseteq T_+$, a $(1,0)$-form vanishing on

[6] Serre's formula expresses $\sum_i (-1)^i \ell(\mathrm{Tor}_i(F, \mathcal{O}_S))$, for a module F and a smooth germ S meeting its support properly at p, as a sum over the components Z' of the support of $\ell(F_{Z'})$ times the intersection multiplicity of Z' and S at p. For S transverse to a component smooth at p the intersection multiplicity is one. For a complex the Tor spectral sequence gives the same formula with alternating sums over the cohomology, which define $m_Z(E)$.

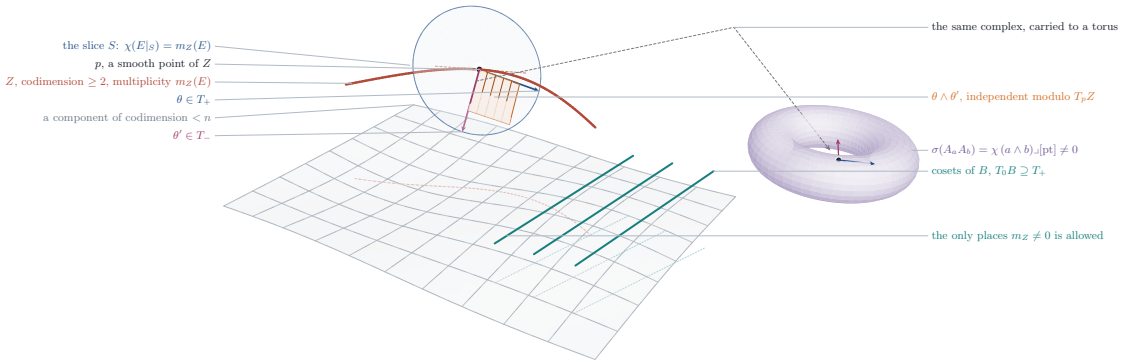


FIGURE 1. The proof of Theorem 2.13. The grey sheet stands for a component of the support of codimension less than n, which the theorem forces to exist; the clay curve Z for a component of codimension at least two. At a smooth point p of Z the directions $\theta \in T_+$ and $\theta' \in T_-$ span the slice S, transverse to Z, on which E restricts to a complex of finite length with Euler characteristic $m_Z(E)$; the amber parallelogram is the bivector $\theta \wedge \theta'$, independent modulo T_pZ. Carried to a torus of the same dimension, the product of the two jet classes is seen by the semiregularity map as $m_Z(E)$ times the class of a point contracted with the bivector, which is not zero, so $m_Z(E) = 0$ or the bivector degenerates. The teal lines are cosets of an abelian subvariety B tangent to T_+, the only components of codimension at least two on which a nonzero multiplicity survives. The scene is schematic in the dimensions and exact in the incidences.

T_+ lies in $V_-^{1,0}$, and a $(0,1)$-form vanishing on the conjugate of T_+ lies in $V_+^{0,1}$, because complex conjugation exchanges the two eigenvalues of K;[7] so the subring lies in $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$. When $T_0B \supseteq T_-$ it lies in $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$. Choose bases of the four pieces. Both exterior algebras are spanned by monomials in these bases, so their sum is too, and α_+, a single monomial in n generators of $V_+^{1,0}$ and n of $V_+^{0,1}$, is not among the monomials spanning it; nor is α_-. As $ab \neq 0$, the class $N\omega$ has nonzero coordinates on both and is not in the sum, which contradicts $\mathrm{ch}_n(E) = N\omega$. So some component has codimension less than n. The last assertion follows, since a complex assembled from structure sheaves of the components of a cycle of codimension n is supported on that cycle. ∎

Figure 1 draws the slice and the bivector of the proof.

2.5. **Indecomposability and endomorphisms.** The numerical criterion of Theorem 2.11 fixes one number, $\dim \mathrm{Ext}^2(E,E)$. Together with Serre duality, the lower bounds of Corollary 2.9 in the other degrees and the Hodge–Riemann relations for the Weil class, it fixes a good deal more: the object that carries the class can be taken indecomposable, and it has at least three linearly independent endomorphisms at $n = 2$ and, granting the compatibility of Corollary 2.9(iii), at $n = 3$, the first case in which the criterion is needed at a nontrivial discriminant.

[7] The field K acts on $H_1(A,\mathbb{Q})$ by rational endomorphisms, so on the complexification its action commutes with complex conjugation, which carries the eigenspace of $\sqrt{-d}$ for one embedding of K to that for the other. On the tangent space the conjugate of T_+ is the eigenspace in $T^{0,1}$ for the second embedding; it pairs with $V_-^{0,1}$, not with $V_+^{0,1}$.

Theorem 2.14 (The shape of an object meeting the criterion)**.** *Let A, ω and E be as in Theorem 2.11, so that* $\mathrm{ch}(E) = N\omega$ *with* $N \neq 0$ *and no other component, and suppose that* $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$*. Write* $e_k = \dim \mathrm{Ext}^k(E,E)$*.*

(i) $\chi(E,E) = \sum_k (-1)^k e_k = (-1)^n N^2 \int_A \omega^2$*, and this number is positive.*

(ii) $E \cong E_0 \oplus E'$ *with* E_0 *indecomposable,* $\mathrm{ch}(E_0) = \mathrm{ch}(E)$*,* $\dim \mathrm{Ext}^2(E_0,E_0) = 2n(2n-1)$*,* $\mathrm{ch}(E') = 0$ *and* $\mathrm{Ext}^2(E_0,E') = \mathrm{Ext}^2(E',E_0) = \mathrm{Ext}^2(E',E') = 0$*. So E_0 meets the criterion as well.*

(iii) *If E is indecomposable, then* $\mathrm{End}(E) = \mathbb{C}\cdot \mathrm{id} \oplus J$ *with J the nilpotent radical, every $f \in J$ annihilates* $\mathrm{Ext}^2(E,E)$ *and* $\mathrm{Ext}^{2n-2}(E,E)$ *under the Yoneda product, and* $\mathrm{ev}_E(\bigwedge^{2n} P) = \mathrm{ev}_E(\bigwedge^{2n} Q)$ *is a single line in* $\mathrm{Ext}^{2n}(E,E)$*. If moreover $e_1 = 4n$, then J annihilates* $\mathrm{Ext}^1(E,E)$ *as well.*

(iv) *If $n = 2$ or $n = 3$, then*

$$\dim \mathrm{End}(E) \;\geq\; 2 + \tfrac{1}{2}\,\chi(E,E) \;>\; 2,$$

so E has at least three linearly independent endomorphisms, and at $n = 3$ also $e_1 \leq e_0 + 9$. In particular no simple object meets the criterion: no stable sheaf, no complex with $\mathrm{End} = \mathbb{C}$*, and no image of one under an autoequivalence of $D^b(A)$; and an indecomposable object that meets it has a nonzero nilpotent endomorphism. The smallest profiles $(e_0,\dots,e_{2n})$ that the constraints allow are $(3,8,12,8,3)$ at $n = 2$, with $\chi = 2$, and $(3,12,30,40,30,12,3)$ and $(3,12,30,41,30,12,3)$ at $n = 3$, with $\chi = 2$ and $\chi = 1$.*

Parts (i) and (ii), and part (iv) at $n = 2$, use the compatibility $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 2.7 in degree two only, as Theorem 2.11 does. Part (iii), and part (iv) at $n = 3$, use it in the further degrees in which Corollary 2.9(iii) grants it: $2n-2$ and $2n$ for (iii), and three for (iv).

Proof. (i) The Todd class of A is trivial, so the Riemann–Roch theorem gives $\chi(E,E) = \int_A \mathrm{ch}(E)^\vee \mathrm{ch}(E)$, and $\mathrm{ch}(E)^\vee = (-1)^n N\omega$ since ω has degree $2n$. The polarisation pairs V_+ only with V_- (see [I, Figure 9]), so its class θ lies in $V_+ \otimes V_-$ and $\theta \wedge \alpha_\pm = 0$, because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So ω is a nonzero rational primitive class of type (n,n) in the middle degree, and the Hodge–Riemann relations give $(-1)^{n(2n-1)} \int_A \omega^2 > 0$, where $(-1)^{n(2n-1)} = (-1)^n$.

(ii) The category $D^b(A)$ has finite dimensional morphism spaces and split idempotents, so $E = \bigoplus_i E_i$ with each E_i indecomposable. The evaluation map and the semiregularity map are compatible with direct sums, and by Theorem 2.11(i) the kernel of ev_E in degree two is $P \wedge Q$; so $\mathrm{ev}_{E_i}(P \wedge Q) = 0$ and, composing with σ_{E_i}, $(P \wedge Q)\lrcorner\,\mathrm{ch}(E_i) = 0$ for every i. We claim that the only classes of Hodge type killed by $P \wedge Q$ are those of $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$. Write $H^*(A,\mathbb{C}) = \bigwedge^*(A_+ \oplus B_+ \oplus A_- \oplus B_-)$ with $A_\pm = V_\pm^{1,0}$ and $B_\pm = V_\pm^{0,1}$, and grade it by the four degrees (i,j,k,l). As in Setup 2.7, P acts by wedging with B_+ and contracting A_-, and Q by wedging with B_- and contracting A_+, so $P \wedge Q$ is spanned by four kinds of operators, each of which shifts the four degrees by a fixed amount: wedging with $b_+ \wedge b_-$ kills a component of degree (i,j,k,l) for all choices exactly when $j = n$ or $l = n$; wedging with b_+ after contracting a_+, exactly when $i = 0$ or $j = n$; contracting a_- and wedging with b_-, exactly when $k = 0$ or $l = n$; and contracting a_- and a_+, exactly when $i = 0$ or $k = 0$. Each of these is a statement about a tensor product of maps that are injective unless the degree in question is extreme. A class of type (p,p) has $i+k = j+l = p$ in every component. If $j = n$, then $l = n$ would force $i = k = n$, against the fourth condition, so $l < n$ and the third condition gives $k = 0$; hence $i = n + l$, so $i = n$ and $l = 0$: the component lies

in $\bigwedge^n A_+ \otimes \bigwedge^n B_+ = \mathbb{C}\alpha_+$. If $j < n$, the first two conditions give $l = n$ and $i = 0$, hence $k = n$ and $j = 0$: the component lies in $\mathbb{C}\alpha_-$. This proves the claim. Since $\mathrm{ch}(E_i)$ is a rational class of Hodge type killed by $P \wedge Q$, it lies on the rational Weil plane, and when it is nonzero Corollary 2.9(i) gives $\dim \mathrm{Ext}^2(E_i, E_i) \geq 2n(2n-1)$, a nonzero rational Weil class having both coefficients a and b nonzero. As $\mathrm{Ext}^2(E,E) = \bigoplus_{i,j} \mathrm{Ext}^2(E_i, E_j)$ has dimension $2n(2n-1)$ and $\sum_i \mathrm{ch}(E_i) = N\omega \neq 0$, exactly one summand E_0 has nonzero Chern character and every other block of Ext^2 vanishes.

(iii) An indecomposable object of a category with finite dimensional morphism spaces and split idempotents has a local endomorphism ring, so $\mathrm{End}(E) = \mathbb{C} \cdot \mathrm{id} \oplus J$ with J nilpotent. By Theorem 2.11(i), ev_E maps $\bigwedge^2 P \oplus \bigwedge^2 Q$ isomorphically onto $\mathrm{Ext}^2(E,E)$. By Serre duality $e_{2n-2} = e_2$, and contraction with ω is injective on $\bigwedge^{2n-2} P \oplus \bigwedge^{2n-2} Q$ by Theorem 2.8(iv), since $2n - 2 \neq 2n$; so, with the compatibility in degree $2n-2$, ev_E maps that space isomorphically onto $\mathrm{Ext}^{2n-2}(E,E)$. In degree $2n$ the semiregularity map is the trace, since $H^{q+2n}(\Omega^q_A) = 0$ for $q > 0$, so $\mathrm{tr}\,\mathrm{ev}_E(\xi)$ is the $H^{0,2n}$ component of $\xi \lrcorner \mathrm{ch}(E)$; for the generators π_P, π_Q of $\bigwedge^{2n} P$ and $\bigwedge^{2n} Q$ it is Nc_P and Nc_Q with $c_P c_Q \neq 0$. Indeed π_P wedges with a basis of B_+ and contracts a basis of A_-, so it kills α_+, whose factor $\bigwedge^n B_+$ is already full, and carries α_- to $\pm$ the generator $\bigwedge^n B_+ \wedge \bigwedge^n B_-$ of $H^{0,2n}$; so $c_P = \pm b$ and likewise $c_Q = \pm a$, both nonzero. For $f \in \mathrm{End}(E)$ and $x \in \bigwedge^2 P \oplus \bigwedge^2 Q$ write $f \cdot \mathrm{ev}_E(x) = \mathrm{ev}_E(\varphi_f(x))$; then $f \mapsto \varphi_f$ is a homomorphism of algebras. Let $x \in \bigwedge^2 P$ and $y \in \bigwedge^{2n-2} P$, and write $x \wedge y = \beta(x,y)\pi_P$. Since ev_E is a ring map that kills the ideal generated by $P \wedge Q$,

$$\beta(x,y)\,\mathrm{tr}(f\,\mathrm{ev}_E(\pi_P)) = \mathrm{tr}(f\,\mathrm{ev}_E(x)\,\mathrm{ev}_E(y)) = \mathrm{tr}\,\mathrm{ev}_E(\varphi_f(x) \wedge y) = Nc_P\,\beta(\varphi_f(x)_P, y),$$

and the pairing β is perfect, so $\varphi_f(x)_P = \mu_f x$ with $\mu_f = \mathrm{tr}(f\,\mathrm{ev}_E(\pi_P))/Nc_P$. The same computation with $y \in \bigwedge^{2n-2} Q$, where $x \wedge y$ lies in the ideal generated by $P \wedge Q$, gives $\varphi_f(x)_Q = 0$. So f acts on $\mathrm{ev}_E(\bigwedge^2 P)$ by a scalar μ_f, and in the same way on $\mathrm{ev}_E(\bigwedge^2 Q)$ by a scalar ν_f; μ and ν are characters of $\mathrm{End}(E)$ and vanish on J. So J kills $\mathrm{Ext}^2(E,E)$, on both sides because ev_E lands in the graded centre, and the same argument with the roles of the two degrees exchanged shows that it kills $\mathrm{Ext}^{2n-2}(E,E)$. In particular $\mathrm{tr}(f\,\mathrm{ev}_E(\pi_P)) = \mu_f Nc_P = 0$ for $f \in J$, so $\mathrm{ev}_E(\pi_P)$ and, likewise, $\mathrm{ev}_E(\pi_Q)$ lie in the annihilator of J under the perfect pairing of $\mathrm{End}(E)$ with $\mathrm{Ext}^{2n}(E,E)$. That annihilator is a line, J having codimension one, and both elements are nonzero, their traces being Nc_P and Nc_Q; so they span the same line. Finally, if $e_1 = 4n$ then $\mathrm{Ext}^1(E,E) = \mathrm{ev}_E(HH^1)$, because ev_E is injective on HH^1, as shown in (iv). For $f \in J$ and $h \in HH^1$ write $f\,\mathrm{ev}_E(h) = \mathrm{ev}_E(h')$; then $\mathrm{ev}_E(h' \wedge h'') = f\,\mathrm{ev}_E(h \wedge h'') = 0$ for every $h'' \in HH^1$, so $h' \wedge HH^1 \subseteq P \wedge Q$, which forces $h' = 0$ since $\dim P = \dim Q \geq 2$.

(iv) By Serre duality $e_k = e_{2n-k}$. The map ev_E is injective on HH^1 using degree two alone: if $\mathrm{ev}_E(h) = 0$ then $\mathrm{ev}_E(h \wedge h') = 0$ for every h', so $h \wedge HH^1 \subseteq \mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$, and $h = 0$ as in (iii). So $e_1 \geq 4n$, and Corollary 2.9(iii) gives $e_3 \geq 2\binom{6}{3} = 40$ at $n = 3$. At $n = 2$ the Euler characteristic is $2e_0 - 2e_1 + 12$, and at $n = 3$ it is $2e_0 - 2e_1 + 60 - e_3$. So $2e_0 = \chi + 2e_1 - 12 \geq \chi + 4$ at $n = 2$, and $2e_0 = \chi + 2e_1 + e_3 - 60 \geq \chi + 4$ at $n = 3$, where also $e_1 = e_0 + 30 - \frac{1}{2}(\chi + e_3) \leq e_0 + 9$. With (i) this gives the bound. An autoequivalence preserves End, and a stable sheaf is simple. An indecomposable object with $\dim \mathrm{End} \geq 3$ has $J \neq 0$. For the smallest profiles take $e_0 = 3$. At $n = 2$, χ is even and positive, so $2e_1 = 18 - \chi \leq 16$, and $e_1 \geq 8$ forces $e_1 = 8$ and $\chi = 2$. At $n = 3$, $2e_1 + e_3 = 66 - \chi \leq 65$ with $e_1 \geq 12$ and $e_3 \geq 40$ forces $e_1 = 12$ and $e_3 \in \{40, 41\}$, with $\chi = 2$ and $\chi = 1$. ∎

Remark 2.15 (Objects excluded by the shape). Theorem 2.14 gives necessary conditions only, and their effect is to exclude the objects one would construct first. Line bundles, semi-homogeneous bundles, skyscraper sheaves, stable sheaves and their Fourier–Mukai transforms are simple, and direct sums of them are excluded by (ii) together with Theorem 6.32. At $n = 3$, granting the compatibility of Corollary 2.9(iii), the shape allowed is an indecomposable object with a local endomorphism algebra of dimension at least three whose radical kills Ext^2 and Ext^4 and pairs to zero with the line $\mathrm{ev}_E(\bigwedge^6 P)$. The smallest conceivable one has $\mathrm{Hom}(E, E)$ of dimension three, $\mathrm{Ext}^1(E, E)$ equal to the image of $HH^1(A)$, of dimension twelve and killed by the radical, and $\mathrm{Ext}^3(E, E)$ of dimension forty or forty-one. Iterated self-extensions and thickenings are the natural objects with nilpotent endomorphisms, but an extension of a simple object F by itself has $\mathrm{ch} = 2\,\mathrm{ch}(F)$, so F would already have to carry the class, and the question returns one level down. For $n \geq 4$ write $e_k = 2\binom{2n}{k} + \delta_k$ for $1 \leq k \leq 2n-1$, with $\delta_k \geq 0$ and $\delta_2 = \delta_{2n-2} = 0$. Since $\sum_{k=1}^{2n-1}(-1)^k\binom{2n}{k} = -2$, the Euler characteristic is $2e_0 - 4 + \sum_k (-1)^k\delta_k$, and an excess δ_k in an even degree $4 \leq k \leq 2n-4$ can make it positive with $e_0 = 1$; so the bound on End disappears. By Theorem 3.5 none of these shapes is realised.

3. The semiregularity criterion in its pure and polarised forms

3.1. **Weil tori and the pure form of the criterion.** The object E of [I, Section 16] has Chern character exactly $N\omega$. For such an object, injectivity of σ_E would make E extend along every K-linear deformation of A (Theorem 2.5(ii)). These deformations reach non-algebraic tori, on which the Weil class is a Hodge class but not a Chern class of any coherent sheaf. This settles the pure form of the criterion, negatively, in every dimension.

Setup 3.1 (Weil tori)**.** Let A be an abelian $2n$-fold of $(K, -1, n)$-Weil type, $V = H^1(A, \mathbb{Q})$ with its action of K, and $V \otimes \mathbb{C} = V_+ \oplus V_-$ the two eigenspaces, each of dimension $2n$. For complementary n-dimensional subspaces $X, Y \subset V_+$ the subspace $X \oplus \overline{Y}$ of $V \otimes \mathbb{C}$ is the space of holomorphic 1-forms of a complex structure on the real torus underlying A; call the resulting complex torus $T_{X,Y}$. On $T_{X,Y}$ the field K acts with signature (n, n), so $T_{X,Y}$ is a Weil torus in the sense of [Voi02b], and every complex structure on that real torus commuting with K and of signature (n, n) arises in this way. The pairs (X, Y) form a connected complex manifold S of dimension $2n^2$, containing the point of A; the polarised Weil family is the submanifold of dimension n^2 on which the polarisation of A is of type $(1, 1)$, and the tangent space of S is the space $\mathrm{Hom}_K(H^{1,0}, H^{0,1})$ of K-linear first order deformations in Theorem 2.5(i). Since α_+ spans $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$ and X is holomorphic while Y is antiholomorphic, the Weil classes are of type (n, n) on every $T_{X,Y}$. We say that a property holds for a very general member of S if it holds outside a countable union of proper analytic subsets.

Lemma 3.2 (The Hodge classes of a very general Weil torus)**.** *For a very general member T of S, $\mathrm{Hdg}^k(T) = 0$ for $1 \leq k \leq 2n-1$, $k \neq n$, and $\mathrm{Hdg}^n(T)$ is the rational Weil plane.*

Proof. There are countably many rational classes, and the Hodge locus of each is an analytic subset of the connected manifold S; so a rational class that is of Hodge type at a very general member is of Hodge type on all of S. Let $G = \mathrm{Res}_{K/\mathbb{Q}}\,\mathrm{GL}_{2n}$ be the group of K-linear automorphisms of V. Its real points act transitively on S, since $\mathrm{GL}(V_+)$ acts transitively on pairs of complementary subspaces. The Mumford–Tate group M of a very general member lies in G, since K acts by endomorphisms of every member, and it contains the Mumford–Tate group of every member, whose Hodge classes include those

of the very general one. So the real Lie algebra of M contains the complex structure J_s of every member, which is the derivative of its Hodge cocharacter, and therefore contains the $\mathrm{Ad}\,G(\mathbb{R})$-span of one J_s. This span is an ideal of $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$ containing the non-central element J_s, which acts on V_+ by i on X and by $-i$ on Y; hence it contains $\mathfrak{sl}_{2n}(\mathbb{C})$, and M contains $G^{\mathrm{der}} = \mathrm{Res}_{K/\mathbb{Q}}\,\mathrm{SL}_{2n}$. Over $\mathbb{C}$ this is $\mathrm{SL}(V_+) \times \mathrm{SL}(V_-)$ acting on $\bigwedge^*(V_+ \oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$, whose invariants are spanned by 1, α_+, α_- and $\alpha_+\alpha_-$. Hodge classes are invariant under M, which gives the statement. ∎

Lemma 3.3 (A very general Weil torus has no subvarieties)**.** *A very general member T of S has no analytic subsets other than finite sets and T itself.*

Proof. Let $Z \subset T$ be an irreducible analytic subset of codimension c with $0 < c < 2n$. For every Kähler class κ the class $[Z]$ satisfies $\int_T [Z]\,\kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By Lemma 3.2 this forces $c = n$ and $[Z]$ into the Weil plane. Take for κ_0 the class of a positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \overline{Y}$. Its blocks lie in $X \otimes \overline{X}$ and $\overline{Y} \otimes Y$, both inside $V_+ \otimes V_-$, so every power of κ_0 has a factor from V_+ and a factor from V_- in each term, and $\alpha_\pm \wedge \kappa_0^n = 0$ because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So $\int_T [Z]\,\kappa_0^n = 0$, which is impossible. ∎

Proposition 3.4 (Coherent sheaves on a very general Weil torus)**.** *Let $n \geq 2$ and let T be a very general member of S. Every coherent analytic sheaf F on T has $\mathrm{ch}_k(F) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \leq k \leq 2n-1$. For $n = 2$ this is Voisin's theorem that every coherent sheaf on a general Weil torus of dimension four has $c_2(F) = 0$* [Voi02b, Theorem 10]*, whose proof we follow.*

Proof. Chern classes of coherent sheaves on T exist in rational Deligne cohomology, extend those of bundles and are additive [Gri10], so they are Hodge classes. By Lemma 3.3 the torsion of F, the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf G. Every coherent sheaf on T has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$ (Lemma 3.2); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of G therefore has stable torsion-free factors Q_j, stable for every Kähler class, and $\mathrm{ch}(G) = \sum_j \mathrm{ch}(Q_j)$. By the theorem of Bando and Siu [BS94] the stable reflexive sheaf Q_j^{**} carries an admissible Hermite–Einstein metric for every Kähler class κ and satisfies $\int_T (2rc_2 - (r-1)c_1^2)(Q_j^{**})\,\kappa^{2n-2} \geq 0$, with equality exactly when Q_j^{**} is a projectively flat vector bundle. Here $c_1 = 0$ and $c_2(Q_j^{**}) \in \mathrm{Hdg}^2(T)$. For $n \geq 3$ that group is zero; for $n = 2$ it is the Weil plane, and then $\int_T c_2(Q_j^{**})\,\kappa_0^2 = 0$ for the class κ_0 of the proof of Lemma 3.3. In either case equality holds for some κ, so Q_j^{**} is a projectively flat bundle with $c_1 = 0$, hence flat, and its rational Chern classes of positive degree vanish. Since Q_j and Q_j^{**} differ only on a finite set, $\mathrm{ch}_k(Q_j) = 0$ for $1 \leq k \leq 2n-1$, and the same holds for G and for F. ∎

Theorem 3.5 (The pure form of the criterion is never met)**.** *Let $n \geq 2$, let A be an abelian $2n$-fold of $(K, -1, n)$-Weil type, ω a nonzero rational Weil class, and E a perfect complex on A with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component. Then σ_E is not injective. In particular $\dim \mathrm{Ext}^2(E, E) > 2n(2n-1)$, the hypotheses of Theorem 2.3 and Theorem 2.11(iii) are satisfied by no object at any member of any Weil family of an*

imaginary quadratic field, and the propagation statement (P2) *of Remark 3.7, in the form stated there, is false.*

Proof. Suppose that σ_E is injective. Since A is projective, E is isomorphic to a bounded complex $E^\bullet$ of holomorphic vector bundles. The class $N\omega$ is of Hodge type on every member of S (Setup 3.1), so by the theorem of Buchweitz and Flenner in Pridham's form (Theorem 2.2, [BF03, Pri24]), every obstruction to extending E along an Artinian deformation of A inside S vanishes, exactly as in the proof of Theorem 2.3. The deformations of the complex structure inside S together with those of the holomorphic structures and differentials of $E^\bullet$ are governed by a differential graded Lie algebra of Dolbeault type with finite dimensional cohomology, and Kuranishi's method [Kur65], in the form of [GM88, Sections 2–3], gives an analytic germ (M, m_0) with a morphism $\pi\colon (M, m_0) \to (S, s_0)$ and a complex of holomorphic vector bundles on the pulled back family of tori deforming $E^\bullet$, whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that π is formally smooth, hence smooth, so its image contains a neighbourhood U of s_0. Thus every member T_s with $s \in U$ carries a bounded complex of holomorphic vector bundles $E_s^\bullet$ whose Chern character is the parallel transport of $N\omega$. Choose $s \in U$ very general. By Proposition 3.4, $\mathrm{ch}_n(E_s) = \sum_i (-1)^i \mathrm{ch}_n(E_s^i) = 0$, while it is the transport of $N\omega$, which is not zero. This contradiction proves that σ_E is not injective. The argument uses only that $\mathrm{ch}(E)$ is of Hodge type on every member of S, so it applies verbatim when $\mathrm{ch}(E) - N\omega$ has components in degrees 0 and $4n$ as well. The remaining statements follow, since $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ by Corollary 2.9(i), and equality forces σ_E to be injective by Theorem 2.11(i). ∎

Remark 3.6 (The polarised criterion). Theorem 3.5 turns Theorem 2.5(ii) from a constraint into a contradiction: an object that extends along every K-linear deformation of A reaches non-algebraic tori, where the Weil class is a Hodge class but not a Chern class. The results that describe such an object, Theorem 2.5, Corollary 2.9(ii)–(iii), Theorem 2.11, Theorem 2.13, Theorem 6.32 and Theorem 2.14, remain true as implications, but they are vacuous: no object satisfies their hypotheses. They record what the pure form asks of an object, and Remark 3.13 says which of them extend to the polarised form defined next.

The obstruction does not reach a Chern character with a component that is of Hodge type on the polarised family and not beyond it, unless a twist removes that component. Call (P2′) the statement that at one base point of each Weil family some perfect complex E with

$$\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\, \theta^k, \qquad N \neq 0,\ c_k \in \mathbb{Q},$$

θ the polarisation, has injective semiregularity map. The proof of Theorem 2.3 applies to it word for word, with the deformation taken over the polarised family, along which $\mathrm{ch}(E)$ stays of Hodge type: at a member A_s it produces $\mathrm{ch}_n(E_s) = N\omega_1 + c_n\theta^n$ algebraic, and θ^n is algebraic, so ω_1 is. The form above is already the most general one allowed by the subring generated by θ and the Weil classes: a Weil class is primitive, so $\theta\omega_1 = 0$ and that subring is spanned by the powers of θ and the Weil plane. Theorem 3.5 excludes directly a Chern character that remains of Hodge type on all of S, which by Lemma 3.2 is a multiple of the Weil class plus terms in degrees 0 and $4n$, and after a line bundle twist the characters $c\, e^{t\theta} + N\omega_1 + c'\theta^{2n}$ with $t\theta$ integral (Theorem 3.9(v)). For every other polarised character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, the component of the Hodge locus of $\mathrm{ch}(E)$ through A is the polarised family itself, whose members are abelian varieties, so nothing forces E onto non-algebraic tori and the argument cannot be run on E (Proposition 3.11);

the one exception is a ratio at $n = 2$ at which the locus is five-dimensional (Remark 3.12). The semiregular objects in Markman's proofs for abelian fourfolds and for the sixfolds of trivial discriminant [Mar25a, Mar25b] are not of the pure form: their Chern characters are not multiples of a Weil class, and they are deformed only along families on which those characters stay of Hodge type, and nothing below depends on whether they have exactly the form above. From here on, in Corollary 6.71, Theorem 6.70 and the frontier of Section 6, (P2) is read in this polarised form. The annihilator computations of [I, Section 16] assume that $\mathrm{ch}(E)$ is a pure Weil class; the next subsection carries them out for a Chern character with components in every even degree (Theorem 3.9, Proposition 3.10, Proposition 3.11).

The same obstruction does not reach the Mumford target of Section 7.2. The exceptional classes there are rigid: their Hodge locus among all complex tori is the Mumford curve (Theorem 7.21, Proposition 7.24), so a semiregular complex with Chern character $N\omega$ on the square of a Mumford fourfold would only have to deform along the curve, whose members are all algebraic.

Remark 3.7 (Two finite statements). After [I, Theorem 16.9] and Theorem 2.3 the Hodge conjecture for abelian varieties of Weil type follows from either of the following, each of which is a statement about explicit finite data and neither of which mentions a general point. The first is settled by [I, Theorem 16.17], and we record it because it is the form in which the question first presents itself. The second is false as stated here (Theorem 3.5), and its polarised form, (P2′) of Remark 3.6, is computed in closed form below.

(P1) *A bound.* At every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class ω is represented by a cycle whose Hilbert data comes from one finite set. [I, Theorem 16.15] shows that the transport of the base cycle does not satisfy this, for any cycle whose components have finite stabiliser, and [I, Theorem 16.17] shows that, read on an arbitrary representative, the statement is equivalent to $\mathbf{W}(K, n, \delta)$ itself. It is therefore a reformulation of the conjecture for the family, and we record it as such rather than as a criterion.

(P2) *A rank.* The semiregularity map of a perfect complex with the Chern character of Proposition 2.1 at one base point is injective. In this form the statement is false (Theorem 3.5); its polarised form, in Remark 3.6, allows the Chern character to carry powers of the polarisation, and what follows records what the pure form asks. The source of the map is $\mathrm{Ext}^2(E, E)$ for a single object, and its target has dimension $\sum_q h^{q,q+2}(A_0) = \binom{4n}{2n-2}$, computed in [I, Proposition 9.2]. So the target is large, which is the direction that helps, and the question is the rank of the map. Full rank would force the following (Theorem 2.5): the evaluation map of Hochschild cohomology must vanish on the $4n^2$-dimensional annihilator of the Weil class, so E must be killed by the mixed block of $H^{0,2}$, must extend along every K-linear deformation, must have all products of transverse translation classes zero, and must have no endomorphism of nonzero trace; and at a base point with no abelian subvariety tangent to an eigenspace it can be a locally free sheaf on no smooth germ of its support of codimension at least two (Corollary 2.6). The representative that the construction supplies, a virtual sum of line bundles, fails at the first of these ([I, Proposition 9.10]), and any representative that is a sheaf fails at the last. Theorem 2.8 identifies that annihilator as the mixed part of a splitting $HH^1 = P \oplus Q$ into the annihilators of the two conjugate pieces of the class, and Corollary 2.9 draws the consequence: the whole Hochschild action of E factors through a ring of dimension $2^{2n+1} - 1$ inside one of dimension 2^{4n}, while $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ for every representative, semiregular or not. Theorem 2.11 turns the rank condition into that number: equality suffices. Theorem 2.13 locates the support: at every member

it has a component of codimension less than n, so no complex carried by a cycle representing ω can serve, and at a general member every component of codimension at least two has multiplicity zero. Theorem 2.14 fixes its shape: an indecomposable summand carries the class and meets the criterion alone, and at $n = 2$, and at $n = 3$ granting the compatibility of Corollary 2.9(iii), it has at least three independent endomorphisms, so no simple object serves.

Neither statement is the variational Hodge conjecture: (P1) says nothing about deformations, and (P2) is an infinitesimal statement at one point, the global propagation being supplied by [I, Theorem 16.4], which is what [I, Remark B.11] identifies as missing. Nor is either implied by the results of [I, Section 15.5]: the base points and the cycles are constructed there, and (P1) and (P2) are questions about those same cycles that the construction does not answer. In the forms stated here both are settled: (P1) is equivalent to its conclusion, and (P2) is false. Its polarised form in Remark 3.6 is the criterion that the rest of this section computes.

3.2. **The polarised criterion in closed form.** The annihilator computations of [I, Section 16] assume that $\mathrm{ch}(E)$ is a pure Weil class, as Remark 3.6 notes. This subsection carries them out for a Chern character with components in every even degree. In degree two the answer is complete and holds for every n; in the other degrees it is a closed formula, with one correction term in the middle degree. The same computation gives the Hodge locus of the character, and with it the list of the obstructions of this paper that apply to polarised characters (Proposition 3.11, Remark 3.13). None of this constructs an object or decides whether one exists. It gives, for each shape of Chern character, the number that $\dim \mathrm{Ext}^2(E,E)$ has to reach and the exact kernel that ev_E must then have, which is the form in which a candidate can be tested.

Setup 3.8 (Polarised characters)**.** Let A be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, with polarisation θ, and let $\omega = u\alpha_+ + \overline{u}\,\alpha_-$ be a nonzero real class on the Weil line, in the notation of Theorem 2.5. For complex numbers $c_0, \dots, c_{2n}$ put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \qquad \gamma = \omega + p(\theta), \qquad \mu_m = m!\, c_m,$$

and let $\rho \in \{0,1,2,3\}$ be the rank of the $(2n-1) \times 3$ Hankel matrix $H = [\mu_{j+i}]$, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$. Write $r(\gamma)$ for the rank of contraction $HT^2(A) \to H^*(A,\mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. For $a < n \leq b$, in the basis of Theorem 2.8 adapted to an eigenbasis e_j of V_+ and f_j of V_- with e_j, f_j of type $(1,0)$ for $j < n$ and $j \geq n$ respectively, write $z_{ab} = w(e_b)w(f_a) \in V_+^{0,1} \otimes V_-^{0,1}$, $v^1_{ab} = w(e_b)c(e_a)$ and $v^2_{ab} = w(f_a)c(f_b)$ in $H^1(T_A)$, and $\pi_{ab} = c(e_a)c(f_b) \in T_+ \otimes T_-$, where w is wedge product with a $(0,1)$ class and c contraction of a $(1,0)$ class.

The matrix H is a catalecticant of the moment sequence $\mu_m = m!\, c_m$. A nonzero vector (x,λ,w) in its kernel is a linear recurrence $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ of order at most two, so $\rho \leq 2$ says that the moments satisfy one, and the shapes of Theorem 3.9(ii) are its solutions, read off from the roots of $wT^2 + \lambda T + x$. For a Chern character, $\gamma = N\omega_1 + \sum_k c_k\theta^k$ with rational c_k, and u is determined by $N\omega_1$. Since $\alpha_\pm^2 = 0$, $\int_A \omega^2 = 2|u|^2 \int_A \alpha_+\alpha_-$; scaling the eigenbases so that $\int_A \alpha_+\alpha_- = (-1)^n \int_A \theta^{2n}/(2n)!$, which is possible because the sign of $\int_A \alpha_+\alpha_-$ is $(-1)^n$ by the Hodge–Riemann relations, one has $|u|^2 = (-1)^n \frac{(2n)!}{2} \int_A \omega^2 / \int_A \theta^{2n}$. The rank ρ does not change when γ is multiplied by $e^{t\theta}$, which acts on the moments by a

binomial transform, and $\omega e^{t\theta} = \omega$ because a Weil class is primitive; so ρ is an invariant of γ up to twist.

Theorem 3.9 (The polarised criterion as a number)**.** *In Setup 3.8:*

(i) *For $n \geq 3$,*

$$\mathrm{Ann}_{HT^2}(\gamma) = \bigoplus_{a<n\leq b} \{x\, z_{ab} + y_1 v^1_{ab} + y_2 v^2_{ab} + w\, \pi_{ab} \ : \ (x,\ i(y_1 - y_2),\ w) \in \ker H\},$$

so that $\dim \mathrm{Ann}_{HT^2}(\gamma) = n^2(4-\rho)$ and $r(\gamma) = (4+\rho)n^2 - 2n$, whatever $u \neq 0$ is.

(ii) *$\rho = 3$ for a general p. If $\rho = 3$, and either $n \geq 3$ or $n = 2$ and $e = 0$ in the notation of (iii), then $\mathrm{Ann}_{HT^2}(\gamma)$ is exactly the tangent space $T_{[A]}\mathcal{D}_{n,n}$ of the polarised Weil family, spanned by the n^2 vectors $v^1_{ab} + v^2_{ab}$, inside $H^1(T_A)$, with no component in $H^{0,2}$ or in $H^0(\bigwedge^2 T_A)$; and $r(\gamma) = \binom{4n}{2} - n^2 = 7n^2 - 2n$. Moreover $\rho \leq 2$ exactly when p is one of $Ae^{s\theta} + Be^{t\theta}$, $(A + B\theta)e^{t\theta}$, $Ae^{t\theta} + B\theta^{2n}$, $A\theta^{2n-1} + B\theta^{2n}$, and $\rho \leq 1$ exactly when $p \in \mathbb{C}e^{t\theta} \cup \mathbb{C}\theta^{2n}$; the pure case $\rho = 0$ gives back $r = 2n(2n-1)$ of Corollary 2.9(i).*

(iii) *For $n = 2$, $\dim \mathrm{Ann}_{HT^2}(\gamma) = 4(4-\rho) + e$ with $e = \dim\ker((HK)^2 - |u|^2) \leq 2$ and K the antidiagonal matrix with entries $1, -2, 1$; so $r(\gamma) \in \{12, 16, 18, 20, 22, 23, 24\}$, and $e > 0$ exactly when $X(X-I)^2 = 4J^2$, where $X = |u|^2$ and I, J are the two invariants of the binary quartic $\sum_m \binom{4}{m}\mu_m x^{4-m}y^m$.*

(iv) *Let E be a perfect complex on A with $\mathrm{ch}(E) = \gamma$ and $N \neq 0$. Then $\dim \mathrm{Ext}^2(E,E) \geq r(\gamma)$ with no hypothesis on σ_E. If equality holds, then ev_E is onto with kernel $\mathrm{Ann}_{HT^2}(\gamma)$ and σ_E is injective, so that (P2′) holds at that point and, by Remark 3.6 and Theorem 2.3, the Hodge conjecture holds for every member of the family with full Hodge group. Conversely, if σ_E is injective then $\ker \mathrm{ev}_E = \mathrm{Ann}_{HT^2}(\gamma)$.*

(v) *If $p \in \mathbb{C}e^{t\theta} + \mathbb{C}\theta^{2n}$ with $t\theta$ the class of a line bundle L, a set of shapes with $\rho \leq 2$ that contains $\mathbb{C}\theta^{2n}$, then no perfect complex E with $\mathrm{ch}(E) = \gamma$ has injective σ_E.*

Proof. (i) Let the torus $(\mathbb{C}^\times)^{2n}$ act by $e_j \mapsto \tau_j e_j$, $f_j \mapsto \tau_j^{-1} f_j$. It preserves the Hodge decomposition and θ, and scales $\alpha_\pm$ by $(\prod_j \tau_j)^{\pm 1}$; contraction is equivariant. A weight of HT^2 has at most two nonzero coordinates, so for $n \geq 3$ two such weights never differ by a nonzero multiple of $(1, \dots, 1)$, and the images of one weight component in the θ-part and in the two α-parts land in distinct weight spaces. Hence $\mathrm{Ann}(\gamma) = \mathrm{Ann}(p(\theta)) \cap \mathrm{Ann}(\alpha_+) \cap \mathrm{Ann}(\alpha_-)$, and the last two meet in $P \wedge Q$ by Theorem 2.8(ii). The space $P \wedge Q$ is the sum over $a < n \leq b$ of the four-dimensional weight spaces spanned by $z_{ab}, v^1_{ab}, v^2_{ab}, \pi_{ab}$, and $\mathrm{Ann}(p(\theta))$ is torus-stable, so the computation splits pair by pair. Put $m = e_b \wedge f_a$ and $\theta' = \theta - t_a - t_b$, where $t_j = is_j e_j \wedge f_j$ with $s_j = \pm 1$ the sign that makes $\theta = \sum_j t_j$ positive. Since the t_j are even with $t_j^2 = 0$, direct evaluation gives

$$z \lrcorner \theta^k = m\, \theta'^k, \quad v^1 \lrcorner \theta^k = ik\, m\, \theta'^{k-1}, \quad v^2 \lrcorner \theta^k = -ik\, m\, \theta'^{k-1}, \quad \pi \lrcorner \theta^k = k(k-1)\, m\, \theta'^{k-2},$$

so for $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$ and $\lambda = i(y_1 - y_2)$,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!}(x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2}).$$

The classes $m \wedge \theta'^j$, $0 \leq j \leq 2n-2$, are nonzero and of distinct degrees, so ξ kills $p(\theta)$ exactly when $(x, \lambda, w) \in \ker H$. The map $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$ is onto with kernel spanned by $v^1 + v^2$, which gives $1 + (3 - \rho)$ dimensions for each of the n^2 pairs. (ii) The vector $v^1_{ab} + v^2_{ab}$ is K-linear and kills θ, and these n^2 vectors span the tangent space of

the polarised family by [I, Proposition 9.14]; when $\rho = 3$ they are all of the annihilator. If $\rho \le 2$, a nonzero $(x, \lambda, w) \in \ker H$ turns the equations $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ into a linear recurrence of full rank on $\mu_0, \dots, \mu_{2n}$, whose two-dimensional space of solutions is spanned by the listed shapes according to the roots of $wT^2 + \lambda T + x$; $\rho \le 1$ is the intersection of two such spaces. (iii) At $n = 2$ the argument of (i) fails for exactly one pair of weights, $\epsilon_2 + \epsilon_3$ in $\bigwedge^2 P$ and $-\epsilon_0 - \epsilon_1$ in $\bigwedge^2 Q$, whose difference is $(1, 1, 1, 1)$; these give one coupled block of size eight, whose kernel is isomorphic to $\ker(X - T)$ with T conjugate on its image to $(HK)^2$, and the characteristic polynomial of HK is $y^3 - Iy - 2J$, which is traceless, so $e \le 2$. (iv) is the argument of Theorem 2.11(i)–(ii) with $\mathrm{Ann}_{HT^2}(\gamma)$ in place of $P \wedge Q$: $\sigma_E \circ \mathrm{ev}_E$ is contraction into γ, so $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\gamma)$. (v) If $\gamma - \omega$ has components in degrees 0 and $4n$ only, the proof of Theorem 3.5 applies verbatim. If $p = c\,e^{t\theta} + c'\theta^{2n}$ with $t\theta = c_1(L)$, then $\mathrm{ch}(E \otimes L^{-1}) = \gamma e^{-t\theta} = c + \omega + c'\theta^{2n}$, because $\omega\theta = 0$ and $\theta^{2n+1} = 0$, and $\sigma_{E\otimes L^{-1}} = e^{-c_1(L)} \cdot \sigma_E$ is injective exactly when σ_E is, so the first case applies to $E \otimes L^{-1}$. ∎

For a general shape, then, the polarised criterion asks of one complex that $\dim \mathrm{Ext}^2(E, E)$ be 24 at $n = 2$, 57 at $n = 3$ and 104 at $n = 4$, against 12, 30 and 56 for the pure form, and injectivity then forces nothing beyond first-order extension along the polarised family, which is what the criterion is for. The larger number comes from the components $c_k\theta^k$: every direction of HT^2 other than the n^2 directions of the family moves γ off the Hodge classes, so an object meeting the bound has ev_E injective on a complement of $T_{[A]}\mathcal{D}_{n,n}$. Figure 2 plots the number against n for the four Hankel ranks.

Proposition 3.10 (The Hochschild profile of a polarised character)**.** *In Setup 3.8, let ρ_k be the rank of contraction $HT^k(A) \to H^*(A, \mathbb{C})$ into γ, put $C_l = l!\,c_l$, let r' be the rank of the $(n+1) \times (n+1)$ Hankel matrix $H' = [C_{p+q}]$, and $d = \dim \ker((K'H')^2 - |u|^2)$ with $K' = \mathrm{antidiag}((-1)^a \binom{n}{a})$.*

(i) *$\rho_0 = \rho_{2n} = 1$, $\rho_k = 0$ for $k > 2n$, and for $1 \le k \le 2n-1$, $k \ne n$,*

$$\rho_k = 2\binom{2n}{k} + M_k \min(k+1,\, 2n+1-k,\, r'), \qquad M_k = \sum_{\substack{i+j=k \\ 1\le i,j\le n-1}} \binom{n}{i}\binom{n}{j};$$

in the middle degree the same formula holds with d subtracted. In particular $\rho_k = \rho_{2n-k}$, and the general profiles are $(1, 8, 24, 8, 1)$ at $n = 2$ and $(1, 12, 57, 112, 57, 12, 1)$ at $n = 3$.

(ii) *$\chi(E, E) = V\,Q(c) + W$ for a perfect complex with $\mathrm{ch}(E) = \gamma$, where $V = \int_A \theta^{2n} > 0$, $W = (-1)^n \int_A \omega^2 > 0$ and $Q(c) = \sum_k (-1)^k c_k c_{2n-k}$; and $\chi(E, E)$ is even.*

(iii) *Granting the compatibility of Corollary 2.9(iii) in every degree, $\dim \mathrm{Ext}^k(E, E) \ge \rho_k$. At $n = 2$, at the points with $d = 1$ one has $\rho_2 = 23$, which is odd, so $\dim \mathrm{Ext}^2(E, E) = r(\gamma)$ cannot hold there, since $\chi(E, E)$ is even and Serre duality gives $\chi(E, E) \equiv \dim \mathrm{Ext}^2(E, E) \bmod 2$.*

Proof. (i) Split $HT^1 = Q \oplus P$ as in Theorem 2.8, so that $HT^k = \bigoplus_{i+j=k} \bigwedge^i Q \otimes \bigwedge^j P$ and Q, P act on the two tensor factors of $H^*(A) = H_+ \otimes H_-$ separately. A mixed monomial kills ω, and on $p(\theta)$ the block of mixed monomials with fixed index sets I, J has entries $C_{s+t+m+m'}$ depending only on $s + t$ and $m + m'$, so its rank is that of a catalecticant of the binary form $\sum_k c_k x^k y^{2n-k}/(2n-k)!$, which is $\min(k+1, 2n+1-k, r')$ by the classical Hilbert function of an apolar ideal. The pure sources $\bigwedge^k Q$ and $\bigwedge^k P$ kill α_- and α_+ respectively and contract the other class injectively (Theorem 2.8(iii)). To see

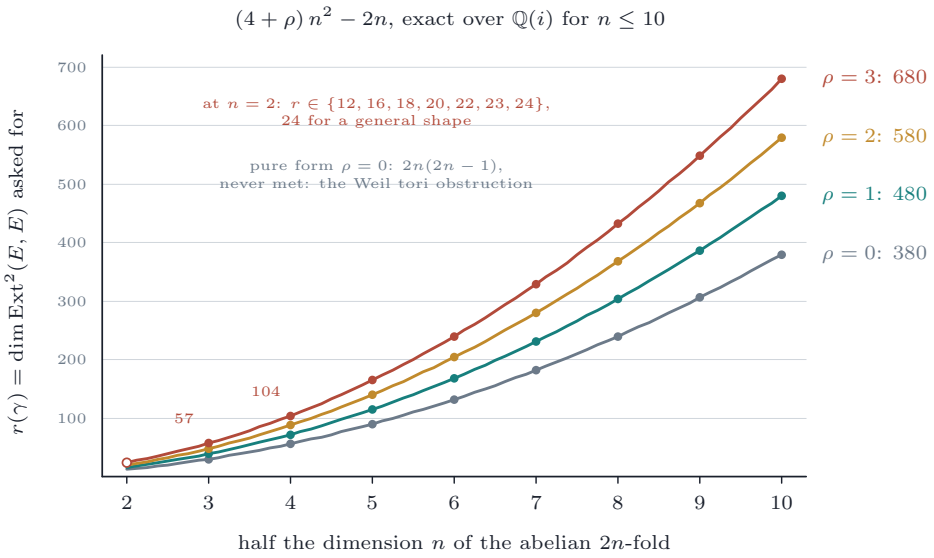


FIGURE 2. The dimension $r(\gamma) = (4+\rho)n^2 - 2n$ that $\mathrm{Ext}^2(E,E)$ must have for a complex with Chern character $\gamma = N\omega_1 + \sum_k c_k\theta^k$ to meet the polarised criterion, for the four Hankel ranks ρ of Theorem 3.9; the dots mark the values at $n = 2, \dots, 10$, and at $n = 10$ the space HT^2 has dimension 780. The pure form, $\rho = 0$, is the bottom curve $2n(2n-1)$, which no complex attains (Theorem 3.5); a general shape lies on the top curve $7n^2 - 2n$. At $n = 2$ the weight argument fails for one pair of weights and seven values occur.

where their images go, use the torus of the proof of Theorem 3.9(i) and put $I_Q = \{j < n\}$, $I_P = \{j \ge n\}$. The basis vectors $w(f_j), c(e_j)$ of Q have weight $-\epsilon_j$ for $j \in I_Q$, the vectors $w(e_j), c(f_j)$ of P have weight ϵ_j for $j \in I_P$, θ has weight 0 and $\alpha_\pm$ has weight $\pm\sum_j \epsilon_j$. If $\xi \in \bigwedge^k Q$ is a monomial, $k < 2n$, and another monomial η of HT^k sends $p(\theta)$ or α_- to the weight of $\xi \lrcorner \alpha_+$, which is $+1$ at every $j \in I_P$, then it is not α_-, which would need two factors from P at every $j \in I_P$; so η has one factor from P for each $j \in I_P$, ξ has weight at most -1 at every $j \in I_Q$, and $k \ge n$. For $k = n$ this forces ξ and η to lie in the spans $D_Q \subset \bigwedge^n Q$ and $D_P \subset \bigwedge^n P$ of the 2^n monomials with one factor for each index. The same holds with P and Q exchanged. A mixed monomial kills ω, so for $k < n$ the matrix of contraction is block triangular, the pure blocks injective onto targets of their own, and ρ_k is $2\binom{2n}{k}$ plus the rank of the mixed blocks; for $k > n$ the formula follows from Lemma 7.33, which also gives $\rho_k = \rho_{2n-k}$.

For $k = n$, a pure monomial outside $D_Q \cup D_P$ repeats an index, so it kills 1 and t_j at that index and hence $p(\theta)$, and keeps a target of its own; no mixed monomial reaches the weights $\sum_{j\in I_P}\epsilon_j$ and $-\sum_{j\in I_Q}\epsilon_j$ of the images of $D_Q \oplus D_P$, since it would need n factors from one side. So ρ_n is the value of the formula minus the dimension of the kernel of the block $\beta\colon D_Q \oplus D_P \to T_1 \oplus T_2$, T_1 and T_2 being the targets of those two weights. For $R \subseteq I_Q$ and $J \subseteq I_P$ let $X_R = \prod_{j\in I_Q} z_j$ with $z_j = w(f_j)$ for $j \in R$ and $c(e_j)$ otherwise, and $Y_J = \prod_{j\in I_P} z_j$ with $z_j = c(f_j)$ for $j \in J$ and $w(e_j)$ otherwise, the factor of largest index acting first, and write $\alpha_+ = a\, e_0 \cdots e_{2n-1}$, $\alpha_- = b\, f_0 \cdots f_{2n-1}$. Since $\theta^{2n}/(2n)! = \prod_j t_j = \prod_j s_j\, e_0 \cdots e_{2n-1} f_0 \cdots f_{2n-1}$, the normalisation of Setup 3.8 gives

$ab = (-1)^n \prod_j s_j$. As the t_j are even with $t_j^2 = 0$, $p(\theta) = \sum_S \mu_{|S|} \prod_{j\in S} t_j$. Each operator acts on one factor of $H^*(A) = \bigotimes_j \bigwedge^*\langle e_j, f_j\rangle$ with the Koszul sign of the factors before it, and in each of the four products below the parities of those factors do not depend on R or J; the signs are $+1$ on $p(\theta)$, $\epsilon_1 = (-1)^{n(n-1)/2}$ for X_R on α_+ and $\epsilon_2 = (-1)^{n(3n-1)/2}$ for Y_J on α_-. With $\tau_R = e_n \cdots e_{2n-1} \prod_{j\in R} t_j$ and $\tau'_{R'} = f_0 \cdots f_{n-1} \prod_{j\in R'} t_j$, and using $c(f_j)t_j = -is_j e_j$, $c(e_j)t_j = is_j f_j$ and $e_j f_j = -is_j t_j$, one finds

$$X_R \lrcorner \alpha_+ = \epsilon_1 a \prod_{j\in R} (is_j)\, \tau_R, \qquad Y_J \lrcorner p(\theta) = \sum_R \mu_{|J|+|R|} \prod_{j\in J} (-is_j)\, \tau_R,$$

$$Y_J \lrcorner \alpha_- = \epsilon_2 b \prod_{j\in I_P\setminus J} (-is_j)\, \tau'_{I_P\setminus J}, \qquad X_R \lrcorner p(\theta) = \sum_{R'} \mu_{n-|R|+|R'|} \prod_{j\in I_Q\setminus R} (is_j)\, \tau'_{R'}.$$

For $x = \sum_R x_R X_R$ and $y = \sum_J y_J Y_J$ put $\tilde{x}_R = \prod_{j\in R}(is_j)\, x_R$ and $\tilde{y}_J = \prod_{j\in J}(-is_j)\, y_J$, and let $\Pi_P = \prod_{j\in I_P}(-is_j)$, $\Pi_Q = \prod_{j\in I_Q}(is_j)$. Then $\beta(x,y) = 0$ says

$$\epsilon_1 a\, u\, \tilde{x}_R = -\sum_J \mu_{|J|+|R|}\, \tilde{y}_J, \qquad \epsilon_2 b\, \overline{u}\, \Pi_P (-1)^{|J|} \tilde{y}_J = -\Pi_Q \sum_R \mu_{2n-|R|-|J|} (-1)^{|R|} \tilde{x}_R$$

for all R and J. So $\tilde{x}_R = g_{|R|}$ and $(-1)^{|J|}\tilde{y}_J = h_{|J|}$ depend only on $|R|$ and $|J|$, and with $D = \operatorname{diag}((-1)^c \binom{n}{c})$ and the exchange matrix J_0, so that $DJ_0 = K'$ and $J_0 D J_0 = (-1)^n D$, the conditions read $\epsilon_1 a\, u\, g = -H' D h$ and $\epsilon_2 b\, \overline{u}\, \Pi_P h = -\Pi_Q J_0 H' J_0 D g$. Eliminating h gives $(H'K')^2 g = \lambda g$ with $\lambda = (-1)^n \epsilon_1\epsilon_2\, ab\, |u|^2\, \Pi_P/\Pi_Q$, and $\epsilon_1\epsilon_2 = (-1)^n$, $ab = (-1)^n \prod_j s_j$, $\Pi_P/\Pi_Q = (-1)^n \prod_j s_j$, so $\lambda = |u|^2$. Conversely an eigenvector g determines h, x and y. Hence $\dim\ker\beta = \dim\ker((H'K')^2 - |u|^2) = d$, as $H'K'$ and $K'H'$ are conjugate.

(ii) $\operatorname{ch}(E)^\vee$ changes the sign of the components of odd codegree, $\omega\theta = 0$ kills the cross terms, and Hodge–Riemann gives $W > 0$; parity holds because the Chern character of a perfect complex on a complex torus is integral, and the intersection form on $\bigwedge^{2n} H^1(A,\mathbb{Z})$ pairs each monomial only with its complement, so $\int x^2$ is even. (iii) is the argument of Corollary 2.9(iii). ∎

The profile also shows what the polarised criterion leaves unconstrained. For the pure form, Theorem 2.14(iv) derives $\dim \operatorname{End}(E) \geq 3$ at $n = 2$ and $n = 3$ from $\chi(E,E) > 0$. For a polarised character, $\chi(E,E) = VQ(c) + W$ can have either sign, and when it is small enough the Euler characteristic forces nothing. At $n = 2$, in the model of [I, Setup 4.1] with $d = 1$, where $\int \eta^4 = 24$ and $\int \omega_1^2 = 8$ (see the proof of Proposition 3.11), the character $\omega_1 + p(\eta)$ with $c = (2,1,0,1,-1)$ has $Q(c) = 2c_0c_4 - 2c_1c_3 + c_2^2 = -6$, so $\chi(E,E) = -144 + 8 = -136$, and a simple object is not excluded.

Proposition 3.11 (The Hodge locus of a polarised character)**.** (i) *Let T be a complex torus, let x be a real class of type (k,k) on T, and for $v \in \operatorname{Hom}(H^{1,0}, H^{0,1}) = H^1(T_T)$ let J_v be the complex structure on the underlying real torus with $H^{1,0}_v = (1+v)H^{1,0}$. Then x is of type (k,k) for J_v if and only if $v \lrcorner x = 0$. So in this chart the Hodge locus of x is the linear space $\operatorname{Ann}_{H^1(T_T)}(x)$, and it coincides with its first-order locus.*

(ii) *Let J be any complex structure on the real torus underlying A. If θ^k is of type (k,k) for J for some $1 \leq k \leq 2n-1$, then θ is of type $(1,1)$ for J.*

(iii) *In Setup 3.8 with $n \geq 3$, the germ at A of the Hodge locus of γ among all complex structures is the polarised Weil family $\mathcal{D}_{n,n}$, of dimension n^2, if $c_k \neq 0$ for some $1 \leq k \leq 2n-1$, and the family of all K-linear deformations, of dimension $2n^2$, otherwise. In the first case the connected component of the Hodge locus through A is $\mathcal{D}_{n,n}$, and all its members are abelian varieties polarised by θ.*

(iv) *For $n=2$ the same holds, with one exception: if $c_1=c_3=0$ and $c_2^2\int_A\theta^4=\frac{3}{4}\int_A\omega^2$, the germ has dimension 5. In the model of* [I, Setup 4.1] *this is the character $N\omega_1\pm\frac{d}{2}N\eta^2+c_0+c_4\eta^4$; with rational data the ratio is attained exactly in the families of trivial discriminant.*

Proof. (i) Write D_v for the derivation of $\bigwedge^* H^1$ extending v. Since $v^2=0$ on H^1, $\bigwedge^*(1+v)=\exp(D_v)$, so the Hodge filtration of J_v is $F_v^p=\exp(D_v)F^p$. Now $\exp(-D_v)x=\sum_m(-1)^m D_v^m x/m!$ with $D_v^m x$ of type $(k-m,k+m)$, so $x\in F_v^k$ exactly when $D_v x=v\lrcorner x=0$; and a real class lies in $F_v^k\cap\overline{F_v^k}=H_v^{k,k}$ as soon as it lies in F_v^k. (ii) Let $t\mapsto t_J$ be the circle acting on $H^{p,q}$ by t^{p-q}. It acts on $\bigwedge^* H^1(A,\mathbb{R})$ by algebra automorphisms and fixes θ^k, so the path $t\mapsto t_J\theta$ lies in $\{y\in\bigwedge^2 H^1(A,\mathbb{R}):y^k=\theta^k\}$. At a nondegenerate y the differential $h\mapsto k\,y^{k-1}h$ of $y\mapsto y^k$ is injective on $\bigwedge^2$ when $2+(k-1)\le 2n$, by the linear hard Lefschetz theorem for the symplectic form y on a space of dimension $4n$: for a nondegenerate $y\in\bigwedge^2 U^\vee$ on a real space U of dimension $2m$, multiplication by y^j is injective from $\bigwedge^p$ to $\bigwedge^{p+2j}$ whenever $p+j\le m$, because y generates an $\mathfrak{sl}_2$-action on $\bigwedge^* U^\vee$ in which $\bigwedge^p$ has weight $p-m$; here $m=2n$, $p=2$ and $j=k-1$. So near θ that set is discrete, the path is constant, and θ is of type $(1,1)$. The bound $k\le 2n-1$ is sharp, since θ^{2n} is a multiple of the volume form for every complex structure. (iii) By (i), applied to each homogeneous component, the Hodge locus of γ in the chart is $\mathrm{Ann}_{H^1(T_A)}(\gamma)$, the part of $\mathrm{Ann}_{HT^2}(\gamma)$ in $H^1(T_A)$. In the description of Theorem 3.9(i) that part is $x=w=0$ with $(0,\lambda,0)\in\ker H$; the vector $(0,1,0)$ lies in $\ker H$ exactly when $\mu_1=\cdots=\mu_{2n-1}=0$, so the part is spanned by the n^2 vectors $v^1_{ab}+v^2_{ab}$, which span $T_{[A]}\mathcal{D}_{n,n}$, when some c_k with $1\le k\le 2n-1$ is nonzero, and by all v^1_{ab},v^2_{ab}, the K-linear deformations, otherwise. A germ of submanifold of dimension n^2 inside a linear space of the same dimension is that space. On $\mathcal{D}_{n,n}$ the class θ stays of type $(1,1)$ and nondegenerate, so the signature of the hermitian form it defines is constant and every member is polarised by θ. The same argument at every member shows that $\mathcal{D}_{n,n}$ is open in the Hodge locus; it is closed in the space of complex structures, since K-linearity is a closed condition and a limit of positive hermitian forms that stays nondegenerate stays positive; and it is connected. (iv) At $n=2$ the weight argument of Theorem 3.9(i) fails for one pair of weights. The other weights contribute as in (iii): the K-linear deformations, of dimension 8, when $c_1=c_2=c_3=0$, and the tangent space of the polarised family, of dimension 4, otherwise. The remaining pair of weights couples $\xi'\in\mathrm{Hom}(V_+^{1,0},V_-^{0,1})$ with $\xi''\in\mathrm{Hom}(V_-^{1,0},V_+^{0,1})$. Choose bases $x_{\pm,k}$ of $V_\pm^{1,0}$ and $y_{\pm,k}$ of $V_\pm^{0,1}$, $k=1,2$, with $\theta=\sum_k(x_{+,k}y_{-,k}+x_{-,k}y_{+,k})$, put $\beta_\pm=x_{\pm,1}x_{\pm,2}y_{\pm,1}y_{\pm,2}$, a generator of $\bigwedge^4 V_\pm$, and write $\omega=u_+\beta_+ + u_-\beta_-$. The component of $(\xi'+\xi'')\lrcorner\gamma$ of degree 3 in V_- and 1 in V_+ is $u_-\,\xi''\lrcorner\beta_- + c_2\,\xi'\lrcorner\theta^2$, because $c_1=c_3=0$ and $\xi'\lrcorner\theta^4=4\theta^3(\xi'\lrcorner\theta)=0$; the component of degree 3 in V_+ is the same with $+$ and $-$ exchanged. Here $\xi'\lrcorner\theta$ lies in the line spanned by $y_{-,1}y_{-,2}$, so $\xi'\lrcorner\theta^2=2\theta\,(\xi'\lrcorner\theta)$ is a multiple of $\theta\,y_{-,1}y_{-,2}$. The map $\xi''\mapsto\xi''\lrcorner\beta_-$ is injective, and $\theta\,y_{-,1}y_{-,2}=\xi''_0\lrcorner\beta_-$ for $\xi''_0(x_{-,1})=-y_{+,2}$, $\xi''_0(x_{-,2})=y_{+,1}$, which has $\xi''_0\lrcorner\theta=2y_{+,1}y_{+,2}$. With ξ'_0 defined in the same way, a solution has $\xi'=a\xi'_0$ and $\xi''=b\xi''_0$ with $u_+a+4c_2b=0$ and $4c_2a+u_-b=0$. So the annihilator meets this pair in a line when $16c_2^2=u_+u_-$ and in zero otherwise; since $\theta^4=24\,\beta_+\beta_-$ and $\omega^2=2u_+u_-\,\beta_+\beta_-$, the condition is $c_2^2\int\theta^4=\frac{3}{4}\int\omega^2$. In the model, where $\int\theta^4$ and $\int\omega^2$ are 24 and $2|u|^2$ times the same volume, the ratio reads $|u|=4|c_2|$. In the model of [I, Setup 4.1], $\eta^4=4!\prod_j\epsilon_j x_j y_j$ gives $\int\eta^4=24$; in ω_1^2 only the products $w^{(T)}w^{(T^c)}$ survive, and each of the eight with $|T|$ even contributes d^2, so $\int\omega_1^2=8d^2$; likewise $\int\omega_2^2=8d$, and $\omega_1\omega_2=0$ because T and T^c have the same parity. This gives $c_2=\pm dN/2$. For a general rational Weil class $\omega=x\omega_1+y\omega_2$ and hermitian

form $\mathrm{diag}(-a_1,-a_2,a_3,a_4)$ one has $\int\omega^2 = 8d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})$ and $\int\theta^4 = 24\,a_1a_2a_3a_4$, so the ratio asks $(c_2/N)^2 = d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})/(4a_1a_2a_3a_4)$, a rational square exactly when $a_1a_2a_3a_4$ is a norm from K. ∎

The argument of Theorem 3.5 therefore does not extend to the polarised form. A semiregular complex deforms along the Hodge locus of its Chern character. For the pure form that locus is the whole $2n^2$-dimensional family of K-linear tori, whose very general member is not algebraic; for a polarised character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, it is the polarised family itself, all of whose members are abelian varieties, and the argument through Weil tori has nothing to act on. This concerns the argument run on E itself: an autoequivalence can still remove a component, and a line bundle twist is how the argument reaches the polarised characters of Theorem 3.9(v). The only other case it can reach is the exceptional ratio of (iv), where the Hodge locus leaves the polarised family.

Remark 3.12 (The exceptional ratio at $n=2$). At $n=2$, with $c_1=c_3=0$ and $c_2^2\int\theta^4 = \frac{3}{4}\int\omega^2$, put $\gamma_2=\omega+c_2\theta^2$, and let $\mathfrak{g}$ be its stabiliser in $\mathfrak{gl}(H^1(A,\mathbb{R}))$. In the notation of the proof of Proposition 3.11(iv), put $(e_1,\dots,e_4)=(x_{+,1},x_{+,2},y_{+,1},y_{+,2})$ and $(f_1,\dots,f_4)=(y_{-,1},y_{-,2},-x_{-,1},-x_{-,2})$; then $\theta=\sum_j e_jf_j$, $\omega = u_+e_{1234}+u_-f_{1234}$ with $e_{1234}=e_1\wedge\cdots\wedge e_4$, and the ratio is $16c_2^2=u_+u_-$. Passing to the basis se_j, $s^{-1}f_j$ with $s^8=u_+/u_-$ fixes θ and makes the two coefficients of ω equal, and passing from f_j to $\sqrt{-1}f_j$, if necessary, changes the sign of θ^2 and fixes f_{1234}. So over $\mathbb{C}$ the class γ_2 is a nonzero multiple of

$$\tfrac{1}{2}(e_{1234}+f_{1234}) - \tfrac{1}{8}\Big(\textstyle\sum_j e_jf_j\Big)^2,$$

which is the Cayley form $\frac{1}{2}\omega_0^2+\Re\Omega_0$ of $\mathbb{C}^4$, with $\omega_0=\frac{\sqrt{-1}}{2}\sum dz_j\wedge d\bar z_j$ and $\Omega_0=dz_1\wedge\cdots\wedge dz_4$, written in the basis $e_j=dz_j$, $f_j=d\bar z_j$ [HL82]. The stabiliser of the Cayley form in $\mathrm{GL}(8,\mathbb{R})$ is Spin(7), acting on $\mathbb{R}^8$ by its spin representation, and its invariants in $\bigwedge^*\mathbb{R}^8$ are spanned by 1, the form and the volume form [Bry87]. A stabiliser in $\mathfrak{gl}$ is cut out by linear equations, so it commutes with extension of scalars, and $\mathfrak{g}\otimes\mathbb{C}\cong\mathfrak{so}(7,\mathbb{C})$ acts on $H^1(A,\mathbb{C})$ by the spin representation, irreducibly, with invariants in $\bigwedge^*H^1$ spanned by 1, γ_2 and the class of a point. At an ordinary ratio the stabiliser is the Lie algebra $\mathfrak{su}(2,2)$ of the Hodge group, of dimension 15, which also fixes γ_2 here.

The real form is $\mathfrak{so}(4,3)$. The spin representation of $\mathfrak{so}(7,\mathbb{C})$ carries an invariant symmetric form, unique up to a scalar by irreducibility, so $\mathfrak{g}$ preserves a real symmetric form B on $H^1(A,\mathbb{R})$. On $H^1(A,\mathbb{C})=V_+\oplus V_-$ the subalgebra $\mathfrak{su}(2,2)\otimes\mathbb{C}=\mathfrak{sl}(V_+)$ acts on V_+ and dually on V_-, and its only invariant bilinear forms pair V_+ with V_-; so B is a multiple of the real part of the hermitian form of θ, of signature $(4,4)$, and $\mathfrak{g}\subseteq\mathfrak{so}(4,4)$. The real forms of $\mathfrak{so}(7,\mathbb{C})$ are the $\mathfrak{so}(p,q)$ with $p+q=7$, and the maximal compact subalgebra $\mathfrak{so}(p)\oplus\mathfrak{so}(q)$ lies in a maximal compact subalgebra of $\mathfrak{so}(4,4)$, which is $\mathfrak{so}(4)\oplus\mathfrak{so}(4)\cong\mathfrak{su}(2)^4$. A simple subalgebra of $\mathfrak{su}(2)^4$ projects injectively to a factor, so it has dimension 3; hence $p,q\leq 4$, and $\{p,q\}=\{4,3\}$. So γ_2 is the invariant 4-form of Spin(4,3), a split form of the Cayley form.

The Hodge decomposition of $\mathfrak{g}$ has dimensions 5, 11, 5. Indeed, the complex structure J of A fixes the class γ_2 of type $(2,2)$ and lies in the image of the Hodge group, which is connected and contained in the group of $\mathfrak{g}$; as $J^2=-1$ is the nontrivial central element of Spin(4,3), the image of J in $\mathrm{SO}(7,\mathbb{C})$ is an involution with $2k$ eigenvalues -1 and k odd, since a lift of such an involution squares to $(-1)^k$. The eigenvalue -1 of $\mathrm{Ad}\,J$ on $\mathfrak{g}\otimes\mathbb{C}$ then has multiplicity $2k(7-2k)$, which is 10 or 6, and it contains the (-1)-eigenspace of $\mathfrak{sl}(V_+)$, of dimension 8; so $k=1$, and $\mathfrak{g}^{-1,1}\oplus\mathfrak{g}^{1,-1}$ has dimension 10.

The Hodge locus of γ near A is the orbit of A under the connected real group G of $\mathfrak{g}$, since that orbit has dimension $\dim \mathfrak{g}^{-1,1} = 5$ and lies in the locus, which near A is smooth of dimension 5 (Proposition 3.11(i) and (iv)). The Hodge group of a very general member T has Lie algebra an ideal of $\mathfrak{g}$, hence all of it, so the Hodge classes of T are 1, γ_2 and the class of a point, and $\mathrm{NS}(T) = 0$. The locus therefore contains tori that are not algebraic, and the proof of Theorem 3.5 can be started there. It breaks down at its last step: Proposition 3.4 needs a Kähler class on which the degree-four Chern character integrates to zero, and here

$$\int_A \gamma_2\, \kappa_h^2 = \frac{|c_2|}{6}\Big(\operatorname{sign}(c_2)\, \sigma_2(h) + 4\operatorname{Re}\big(\phi \det h_{01,23}\big)\Big) \int_A \theta^4, \qquad |\phi| = 1,$$

for the Kähler class $\kappa_h = i\sum h_{jk} z_j \wedge \overline{z}_k$ of a positive hermitian h in a unitary frame for θ, and the right-hand side has the sign of c_2. Indeed, writing $h = \sum_j \lambda_j w_j w_j^*$ with (w_j) orthonormal, Cauchy–Binet gives $\sigma_2(h) \pm 4\operatorname{Re}(\phi \det h_{01,23}) = \sum_{j<k} \lambda_j\lambda_k\big(1 \pm 4\operatorname{Re}(\phi\, \xi_{01}\overline{\xi}_{23})\big)$ with $\xi = w_j \wedge w_k$ of norm one. The Plücker relation and the inequality of the means give $|\xi_{01}\xi_{23}| \le \frac{1}{4}$, so each term is nonnegative, and by Parseval the brackets sum to 6 over $j<k$, so the sum is at least $6\min_j \lambda_j^2 > 0$. Since G fixes γ_2 and carries Kähler cones to Kähler cones, the same holds on every member of the locus. So at this ratio the argument excludes at most a torsion-free sheaf with $c_2 > 0$, which Bogomolov's inequality already forbids to be θ-semistable, and it excludes no complex. With $R = c_2^2 \int \theta^4 / \int \omega^2$, the ratio that equals $\frac{3}{4}$ here, Figure 3 draws the infimum of the ratio as a function of R. Since Markman's theorem settles every Weil fourfold, this remark concerns the shape of (P2′), not the conjecture.

Remark 3.13 (The obstructions for a polarised character). The obstruction theorems proved from [I, Section 9] on for a Chern character on the Weil line divide as follows for a polarised character. The following hold unchanged, with $r(\gamma)$ in place of $2n(2n-1)$ where a number occurs: [I, Theorem 9.20] and Corollary 4.21, whose only hypothesis is that $\mathrm{ch}(E)$ stay of Hodge type to first order along $\mathcal{D}_{n,n}$; Theorem 4.5 and Corollary 4.6, since a polarised character is a flat Hodge class in every degree; Lemma 2.12; and the first two parts of Theorem 2.11, which become Theorem 3.9(iv). The following hold only when $c_0 = \cdots = c_{2n-2} = 0$, so that $H^{0,2}$ still kills γ: [I, Theorem 9.7], [I, Proposition 9.10], [I, Corollary 9.18], [I, Theorem 9.19] and Theorem 2.5(iii). The following hold only when $\ker H$ contains a vector with $w \neq 0$, which needs $\rho \le 2$, the vector giving a compensated class of $P \wedge Q$ with a component in $T_+ \otimes T_-$: Theorem 2.5(iv), Corollary 2.6 and Theorem 2.13. The following do not extend: Theorem 3.5, outside Theorem 3.9(v); the positivity of $\chi(E,E)$ in [I, Proposition 9.11] and with it Theorem 2.14(i)–(iii), by Proposition 3.10(ii); and Theorem 6.32. So for a general shape, $\rho = 3$, what these results exclude is a line bundle summand off the polarisation line and a Chern character effective in the exponential basis, and nothing else; Proposition 3.14 below extends the first to summands with a scalar Atiyah class. An object with no such summand, whose Chern character is not effective, and with $\dim \mathrm{Ext}^2(E,E) = 7n^2 - 2n$, is not excluded by anything proved here, and none is known; by Lemma 3.16 below it may be taken indecomposable.

Proposition 3.14 (Summands with a scalar Atiyah class)**.** *Let A be of $(K,-1,n)$-Weil type with $n \ge 2$, let $T = T_{[A]}\mathcal{D}_{n,n}$, and let E be a perfect complex on A with $v \lrcorner \mathrm{ch}(E) = 0$ for every $v \in T$. Suppose E has a direct summand $F[p]$ with $\mathrm{At}(F) = \delta \cdot \mathrm{id}_F$, where δ is a real class of type $(1,1)$ that is not a real multiple of θ, and $\mathrm{rk}\, F \neq 0$. Then there is $v \in T$ with $\mathrm{ev}_E(v) \neq 0$ and $\sigma_E(\mathrm{ev}_E(v)) = 0$. In particular no finite direct sum of shifted bundles*

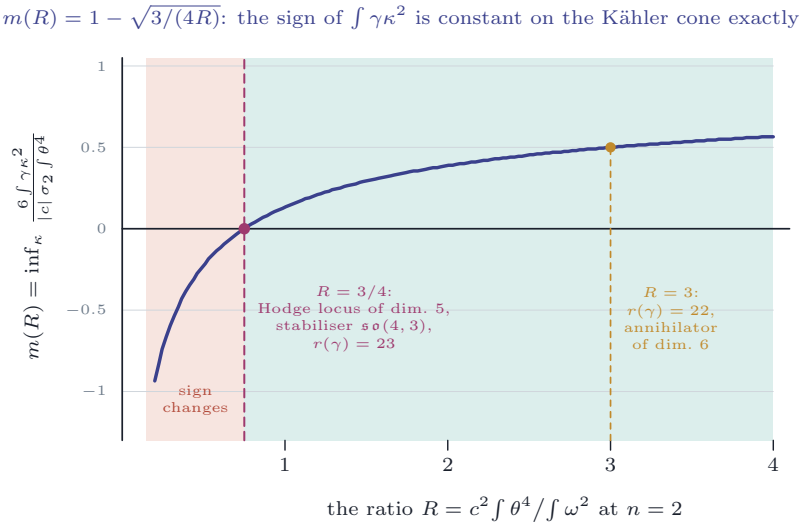


FIGURE 3. The exceptional ratio of Remark 3.12. By the identity displayed there, which is linear in γ, the infimum over the Kähler cone of $6\int\gamma\kappa^2/(|c|\,\sigma_2(h)\int\theta^4)$ is $m(R) = 1 - \sqrt{3/(4R)}$, approached at the boundary of the cone by a positive hermitian form of rank two whose Plücker coordinates satisfy $|\xi_{01}\xi_{23}| = \frac{1}{4}$. It is negative exactly for $R < 3/4$, and $R = 3/4$ is also where the Hodge locus of γ jumps from dimension 4 to 5 and its stabiliser becomes $\mathfrak{so}(4,3)$.

π_*L, π an isogeny, among them the simple semi-homogeneous bundles, has a polarised character γ with $N \neq 0$ and injective semiregularity map.

Proof. As in [I, Theorem 9.20], $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$ on $H^1(T_A)$ and the Atiyah class of a direct sum is block diagonal, so the component of $\mathrm{ev}_E(v)$ in $\mathrm{Ext}^2(F,F)$ is $(v\lrcorner\delta)\cdot\mathrm{id}_F$; the map $z \mapsto z\cdot\mathrm{id}_F$ on $H^{0,2}$ is injective because $\mathrm{tr}(z\cdot\mathrm{id}_F) = \mathrm{rk}(F)\,z$, and some $v \in T$ has $v\lrcorner\delta \neq 0$ by [I, Proposition 9.15(i)]. On the other hand $\sigma_E(\mathrm{ev}_E(v)) = v\lrcorner\mathrm{ch}(E) = 0$. For the last sentence, the proof of Lemma 4.91(i) gives $\mathrm{At}(\pi_*L) = \delta\cdot\mathrm{id}$ with $\pi^*\delta = c_1(L)$ for any line bundle L, and $\mathrm{ch}(\pi_*L) = \mathrm{rk}(\pi_*L)\,e^{\delta}$; if every δ were a multiple of θ the Chern character of the sum would lie in $\mathbb{Q}[\theta]$, which does not contain ω because $\omega\theta = 0$ and $\theta^{n+1} \neq 0$. ∎

Theorem 2.3 asks that σ_E be injective, but its proof uses only that the relative deformations of E dominate the family, and Pridham's form of Theorem 2.2, in which σ_E annihilates every obstruction, gives this under a weaker hypothesis. That hypothesis allows an object that is obstructed to first order, which is exactly the way in which [I, Theorem 9.20], Corollary 4.21 and Proposition 3.14 exclude an object.

Proposition 3.15 (A flatness form of the criterion)**.** *Let $n \geq 2$, $s_0 \in \mathcal{D}_{n,n}$, and let E be a bounded complex of vector bundles on A_{s_0} with $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ and $N \neq 0$. Let K_E be the base of a miniversal deformation of E on the fixed variety A_{s_0}, put $e = \dim\mathrm{Ext}^1(E,E)$ and $k = \dim\ker\sigma_E$. By Theorem 2.2 in Pridham's form, σ_E annihilates every obstruction, both on A_{s_0} and along the Artinian deformations of A_{s_0} in $\mathcal{D}_{n,n}$.*

(i) $\dim K_E \geq e - k$.

(ii) *If* $\dim K_E = e - k$, *then* $\Sigma = \mathcal{D}_{n,n}$, *and the Hodge conjecture holds in every degree for every member of the family with* $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

(iii) $\dim K_E = e - k$ *as soon as the cone* $\{\xi \in \mathrm{Ext}^1(E, E) : \xi \circ \xi = 0\}$ *has codimension* k.

For $k = 0$ *this is Theorem 2.3.*

Proof. Let R be the completed local ring of $\mathcal{D}_{n,n}$ at s_0, regular of dimension n^2. By hypothesis $\ker \sigma_E$ is a complete obstruction space for the deformations of the pair (A_s, E) over Artinian R-algebras, whose relative tangent space is $\mathrm{Ext}^1(E, E)$; so this functor has a hull P/J with $P = R[[x_1, \dots, x_e]]$ and J generated by at most k elements [FM98], and its fibre over the closed point of R is a hull of the absolute functor, of dimension $\dim K_E$. Krull's height theorem gives $\dim P/J \geq n^2 + e - k$, and the same count on the fibre gives (i). If $\dim K_E = e - k$, the dimension inequality for a local homomorphism [Mat86, Theorem 15.1] gives $\dim P/J \leq n^2 + e - k$, so equality holds, J has height k and is generated by a regular sequence, P/J is Cohen–Macaulay, and it is flat over R by miracle flatness [Mat86, Theorem 23.1]: a local homomorphism from a regular local ring to a Cohen–Macaulay local ring is flat exactly when the dimension of the target is the dimension of the source plus that of the fibre, and here both sides equal $n^2 + e - k$. The analytic Kuranishi germ of pairs, as in the proof of Theorem 3.5, has P/J as its completion, so it is flat over the germ of $\mathcal{D}_{n,n}$ at s_0 and therefore open onto a neighbourhood U of s_0. Every A_s with $s \in U$ then carries a bounded complex of vector bundles, algebraic by GAGA, whose Chern character is the parallel transport of $\mathrm{ch}(E)$, so $N\omega_{1,s}$ differs from an algebraic class by a polynomial in θ and is algebraic; [I, Corollary 16.5] gives (ii). For (iii), the quadratic part of the Kuranishi map of E is $\xi \mapsto \xi \circ \xi$, so the tangent cone of K_E lies in the cone of the hypothesis, and $\dim K_E \leq e - k$. Every step uses only deformations along $\mathcal{D}_{n,n}$, which are projective, whereas the proof of Theorem 3.5 needs non-projective ones. ∎

No object is known that meets (iii) with $k > 0$. Among direct sums the flatness form fails for a large class (Remark 3.29 below); for indecomposable objects nothing in this paper excludes it, and the proposition says only that injectivity of σ_E is more than propagation needs.

3.3. **Candidate objects for the polarised criterion.** The polarised criterion asks for one perfect complex. The results below say what such a complex can be made of, for K imaginary quadratic: a direct sum meets the criterion only through one indecomposable summand (Lemma 3.16); no sum of objects whose Ext algebra is an exterior algebra meets it (Corollary 3.17); the Orlov products of secant objects meet its twisted form, with $\dim \mathrm{Ext}^2$ equal to the number of Theorem 3.9, at $n = 2$ and $n = 3$ (Theorem 3.19), and cannot meet it at $n = 4$ when the factors have no negative self-extensions (Proposition 3.20); line bundles on smooth divisors, which give the certificates, give no factor from $n = 4$ on, nor do bundles restricted from the abelian variety at $n = 5$ (Proposition 3.23); and at $n = 4$ a combination of natural objects needs at least eight of them, on one conic (Theorem 3.27). None of this gives a new case of the Hodge conjecture: the certificates are for twisted objects, in families where Markman's theorem already holds.

Lemma 3.16 (The summands of an object meeting the criterion)**.** *Let* A *be a member of a* $(K, -1, n)$*-Weil family,* K *imaginary quadratic and* $n \geq 2$, *let* $T = T_{[A]}\mathcal{D}_{n,n} \subseteq H^1(T_A)$, *and let* $E = \bigoplus_i F_i[k_i]$ *be a finite direct sum of perfect complexes with* $v \lrcorner \mathrm{ch}(E) = 0$ *for every* $v \in T$, *as holds for every polarised character* $N\omega_1 + \sum_k c_k \theta^k$. *Suppose that* σ_E *is*

injective on $\mathrm{ev}_E(T)$*; this holds if* σ_E *is injective, and under the weakened criterion of* [Mar25b, Question 11.4].

(i) *For every* i *and every* $v \in T$, $\mathrm{ev}_{F_i}(v) = 0$ *and* $v \lrcorner \mathrm{ch}(F_i) = 0$*, and every* $\mathrm{ch}(F_i)$ *lies in* $\bigoplus_k \mathbb{Q}\theta^k \oplus W(A)$.

(ii) *If* $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ *with* $N \neq 0$*, some* F_i *has* $\mathrm{ch}(F_i) = \omega' + \sum_k c'_k\theta^k$ *with* ω' *a nonzero rational Weil class, and then* $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i) \geq 2n(2n-1)$*. If* σ_E *is injective, so is every* σ_{F_i}*, and that* F_i *meets the polarised criterion on its own.*

So the polarised criterion is a statement about indecomposable objects.

Proof. For $v \in T$, $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E) = 0$ by the commutative square of Setup 4.54, so $\mathrm{ev}_E(v) = 0$. The Atiyah class is natural, so $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$ is block diagonal with the blocks $\pm\,\mathrm{ev}_{F_i}(v)$, which therefore vanish, and $v \lrcorner \mathrm{ch}(F_i) = \sigma_{F_i}(\mathrm{ev}_{F_i}(v)) = 0$. By Proposition 3.11(i) every homogeneous component of $\mathrm{ch}(F_i)$, a rational class of type (k,k), stays of Hodge type on the germ of $\mathcal{D}_{n,n}$ at A, which in the chart of that proposition is the linear space T; so it is a Hodge class at a member with Hodge group $\mathrm{SU}(V,H)$, and [I, Theorem 4.14] puts it in $\mathbb{Q}\theta^k$, or in $\mathbb{Q}\theta^n \oplus W$ for $k = n$. That is (i). For (ii), the Weil parts of the $\pm\,\mathrm{ch}(F_i)$ add up to $N\omega_1 \neq 0$, so one of them is nonzero; Theorem 3.9 gives $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i)$ with no hypothesis on σ, and $r \geq (4+0)n^2 - 2n = 2n(2n-1)$ for $n \geq 3$, while at $n = 2$ the seven values of Figure 2 are at least 12. Injectivity passes to the summands by Lemma 7.38. By the Krull–Schmidt property of $D^b(A)$ the F_i may be taken indecomposable. ∎

The Weil part ω' of the summand need not be a multiple of ω_1; propagating any nonzero rational Weil class propagates the whole Weil plane, through the algebraic action of K. The lemma is the Weil-family analogue of Corollary 7.41(ii).

Corollary 3.17 (Sums of objects with exterior Ext algebras)**.** *In Lemma 3.16, let every* F_i *have* $\dim \mathrm{Ext}^2(F_i, F_i) < 2n(2n-1)$*. This holds when* $\mathrm{Ext}^*(F_i, F_i) \cong \bigwedge^* \mathbb{C}^{2n}$*, and so for line bundles, for the sheaves* $\mathcal{O}_B \otimes L$ *with* $B \subseteq A$ *an abelian subvariety, for simple semi-homogeneous bundles, for skyscraper sheaves, and for the images of all of these under autoequivalences of* $D^b(A)$*. Then no such sum with* $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ *and* $N \neq 0$ *has* σ_E *injective on* $\mathrm{ev}_E(T)$*: it fails the polarised criterion and the weakened criterion, some* $v \in T$ *obstructs it to first order, and* $\dim \mathrm{Ext}^2(E, E) > r(\mathrm{ch}\, E)$.

Proof. The first assertion is Lemma 3.16(ii), since $\binom{2n}{2} = n(2n-1) < 2n(2n-1)$, and $\dim \mathrm{Ext}^2(E, E) \geq r(\mathrm{ch}\, E)$ with equality only if σ_E is injective (Theorem 3.9). For $\mathcal{O}_B$ choose a complementary abelian subvariety B'; the isogeny $\mu\colon B \times B' \to A$ is étale with group $G = B \cap B'$, and $\mu^*\mathcal{O}_B = \bigoplus_{g \in G} \mathcal{O}_{B \times \{g\}}$ with G permuting the summands simply transitively, so $\mathrm{Ext}_A(\mathcal{O}_B, \mathcal{O}_B) = \mathrm{Ext}_{B \times B'}(\mathcal{O}_{B \times 0}, \mathcal{O}_{B \times 0}) = \bigwedge^*(\mathbb{C}^{\dim B} \oplus \mathbb{C}^{\dim B'})$. For a simple semi-homogeneous bundle V, $V \otimes V^\vee$ is a sum of line bundles in Pic^0 of which $\mathcal{O}$ occurs once [Muk78], so $\mathrm{Ext}^*(V, V) = H^*(\mathcal{O}_A)$. Equivalences preserve Ext algebras. ∎

By Lemma 3.16 the objects that Remark 3.13 allows may be taken indecomposable. The corollary recovers, for polarised characters, the part of Theorem 6.32 that concerns objects with exterior Ext algebras, and it extends Proposition 3.14, whose summand has a scalar Atiyah class, to the subtori, the points and the Fourier–Mukai images. In the language of pure spinors, a summand whose character is a rational multiple of the pure spinor of a rational maximal isotropic subspace is excluded unless that character lies in $\mathbb{Q}[\theta]$, because the rational maximal isotropic subspaces whose pure spinors are flat over $\mathcal{D}_{n,n}$ are those with pure spinor 1, the class of a point or $e^{c\theta}$.

The secant objects of [I, Section 14] have their own Hochschild profile, which Theorem 4.56(i) and (ii) compute in degrees one and two.

Proposition 3.18 (The Hochschild profile of a secant class)**.** *Let X be a principally polarised abelian n-fold, $n \geq 2$, with polarisation θ, let $d \geq 1$, and let $v = a\,u_\theta + b\,v_\theta$ with $(a,b) \in \mathbb{Q}^2 \setminus \{0\}$, in the notation of* [I, (30)]*. The rank of contraction $HT^k(X) \to H^*(X,\mathbb{C})$, $\xi \mapsto \xi \lrcorner v$, is 1 for $k = 0$ and $k = n$, $2\binom{n}{k}$ for $1 \leq k \leq n-1$, and 0 for $k > n$. A perfect complex F on X with $\operatorname{ch}(F) = v$ has $\dim \operatorname{Ext}^k(F,F) \geq 2\binom{n}{k}$ for $k = 1, 2$, and*

$$\chi(F,F) = \begin{cases} (-1)^{n/2}\, 2^{n-1} d^{\,n/2-1}(a^2 d + b^2) & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

Proof. Put $s = \sqrt{-d}$ and $w = \tfrac{1}{2}(a - bs/d)$. By [I, Theorem 14.5(i)], $v = w\,e^{s\theta} + \overline{w}\,e^{-s\theta}$, and $w \neq 0$. By Lemma 4.55(i), $\xi \lrcorner e^{\pm s\theta} = y_\pm e^{\pm s\theta}$, where $y_\pm \in H^{0,k}(X)$ is the component of $e^{\pm s\theta}\xi$ in $\bigwedge^k H^{0,1}$. The map $\xi \mapsto (y_+, y_-)$ is onto $H^{0,k} \oplus H^{0,k}$ for $k \geq 1$: $\xi = z \in \bigwedge^k H^{0,1}$ gives (z,z), and $\xi = z' \otimes \partial$ with $z' \in \bigwedge^{k-1} H^{0,1}$ gives $\pm s\, z' \wedge (\partial \lrcorner \theta)$, where $\partial \lrcorner \theta$ runs over $H^{0,1}$. If $w\,y_+ e^{s\theta} + \overline{w}\,y_- e^{-s\theta} = 0$, the components of degrees k and $k+2$ give $wy_+ + \overline{w}y_- = 0$ and $(wy_+ - \overline{w}y_-) \wedge \theta = 0$, and for $k < n$ wedge product with $\theta = \sum_a p_a \wedge q_a$ is injective on $\bigwedge^k H^{0,1}$, so $y_\pm = 0$. So the rank is $2\binom{n}{k}$ for $1 \leq k < n$. For $k = n$, $y \wedge \theta = 0$ for $y \in H^{0,n}$ and the image is the line spanned by $wy_+ + \overline{w}y_-$; for $k > n$, $H^{0,k} = 0$; and for $k = 0$ the rank is one. The bounds on Ext^1 and Ext^2 follow, as in Theorem 4.56(i) and (ii), from the compatibility of the characteristic morphism with contraction into the Chern character. Finally the Todd class is trivial, so $\chi(F,F) = \int_X \operatorname{ch}(F)^\vee \operatorname{ch}(F)$ with $\operatorname{ch}(F)^\vee = w\,e^{-s\theta} + \overline{w}\,e^{s\theta}$, and $\int_X (e^{2s\theta} + e^{-2s\theta}) = (2s)^n(1 + (-1)^n)$ because $\int_X \theta^n = n!$; with $|w|^2 = (a^2 d + b^2)/(4d)$ this is the formula. ∎

At $n = 4$ and $(a,b) = (1,3)$, the class of Proposition 4.52, this gives $\chi(\mathcal{F},\mathcal{F}) = 8d(d+9)$, the rank recorded there.

Theorem 3.19 (Twisted certificates at $n = 2$ and $n = 3$)**.** *Let $d \geq 1$, $K = \mathbb{Q}(\sqrt{-d})$, let Φ be Orlov's equivalence of Proposition 4.52(v), and for a perfect complex F on a principally polarised X put $E = \Phi(F \boxtimes F^\vee)$ on $A = X \times \widehat{X}$, with the K-action of* [I, (29)]*, the Poincaré class ℓ and the polarisation $\eta = d\beta + \widehat{\beta}$ of the split member (*[I, Notation 15.1]*).*

(i) *Let X be the Jacobian of a curve C of genus two, $C \subset X$ by an Abel–Jacobi map, and $F = \mathcal{O}_C(p)$ for a point $p \in C$. Then $\operatorname{ch}(F) = v_\theta$, $\dim \operatorname{Ext}^k(F,F) = 1, 4, 1$, and $\dim \operatorname{Ext}^k(E,E) = 1, 8, 18, 8, 1$ for $k = 0, \dots, 4$, which are the ranks of contraction from $HT^k(A)$ into $\operatorname{ch}(E)$. So $\dim \operatorname{Ext}^2(E,E) = 18 = r(\operatorname{ch} E)$ and σ_E is injective.*

(ii) *Let X be the Jacobian of a non-hyperelliptic curve C of genus three, $i\colon C^{(2)} \to X$ the embedding of the theta divisor, x the class of $\{p+q : q \in C\}$ and θ the restriction of the polarisation, and let $d = k(k+1)$ with $k \geq 1$ and $F = i_*\mathcal{O}_{C^{(2)}}((2k+1)x - k\theta)$. Then $\operatorname{ch}(F) = v_\theta$, $\dim \operatorname{Ext}^k(F,F) = 1, 6, 6, 1$, and $\dim \operatorname{Ext}^k(E,E) = 1, 12, 48, 74, 48, 12, 1$. So $\dim \operatorname{Ext}^2(E,E) = 48 = r(\operatorname{ch} E)$ and σ_E is injective.*

(iii) *In both cases $\operatorname{ch}(E)$ is not killed by $T = T_{[A]}\mathcal{D}_{n,n}$, and no line bundle twist of E has a character killed by T; the twisted character $\kappa = \operatorname{ch}(E)e^{\ell/2}$ is killed by T, has the form $N'\omega' + \sum_k c_k \eta^k$ with ω' a nonzero rational Weil class, and has the same contraction ranks as $\operatorname{ch}(E)$. A class $B \in \mathrm{NS}(A)_\mathbb{Q}$ with $\operatorname{ch}(E)e^B$ killed by T is congruent to $\ell/2$ modulo $\mathbb{Q}\eta$. At $n = 2$, κ has $c_1 = c_3 = 0$ and lies at the exceptional ratio of Remark 3.12, and its annihilator in $H^1(T_A)$ has dimension 5; at $n = 3$ that annihilator has dimension $9 = n^2$.*

Proof. (i) By Grothendieck–Riemann–Roch $\mathrm{ch}(i_*\mathcal{O}_C(p)) = [C] + (1+1-2)[\mathrm{pt}] = \theta = v_\theta$, since $v_\theta = \theta$ when $n = 2$. For the curve, $\mathrm{Ext}^1(\mathcal{O}_C, \mathcal{O}_C) = H^1(\mathcal{O}_C) \oplus H^0(N_C)$ with $N_C = K_C$ by adjunction, of dimension $2+2$, and $\mathrm{Ext}^0 = \mathrm{Ext}^2 = \mathbb{C}$ by Serre duality; twisting by $\mathcal{O}_C(p)$ changes nothing. The Ext algebra of $F^\vee$ is that of F, Φ is an equivalence, and by the Künneth formula $\mathrm{Ext}^*(E,E) = \mathrm{Ext}^*(F,F) \otimes \mathrm{Ext}^*(F^\vee, F^\vee)$, which gives $1, 8, 18, 8, 1$. By Proposition 3.18 and the Künneth decomposition $HT^2(X \times X) = HT^2 \otimes 1 \oplus HT^1 \otimes HT^1 \oplus 1 \otimes HT^2$ the rank of contraction into $\mathrm{ch}(F) \boxtimes \mathrm{ch}(F^\vee)$ is $1 + 4 \cdot 4 + 1 = 18$, and it is transported by Φ as in the proof of Theorem 4.56(iv). The other degrees are the same count. Equality in degree two forces σ_E injective, as in Theorem 3.9. (ii) On $C^{(2)}$ one has $x^2 = 1$, $x\theta = 3$, $\theta^2 = 6$ and $K_{C^{(2)}} = \theta$, $i_*x = \frac{1}{2}\Theta^2$ and $i_*\theta = \Theta^2$, so Riemann–Roch gives $\mathrm{ch}_2(F) = 0$ and $\mathrm{ch}_3(F) = -k(k+1)[\mathrm{pt}]$, that is $\mathrm{ch}(F) = \Theta - d[\mathrm{pt}] = v_\theta$. The local-to-global spectral sequence, with $h^p(\mathcal{O}_{C^{(2)}}) = 1, 3, 3$ and $h^p(K_{C^{(2)}}) = 3, 3, 1$, gives $\dim \mathrm{Ext}^1, \dim \mathrm{Ext}^2 \le 6$, and Proposition 3.18 gives ≥ 6; $\mathrm{Ext}^0 = \mathbb{C}$ and Serre duality gives $\mathrm{Ext}^3 = \mathbb{C}$. The rest is as in (i), with $r(\mathrm{ch}\, E) = 6n^2 - 2n = 48$ by Theorem 4.56(iv).

(iii) By Proposition 3.21 below, whose proof does not use this theorem, κ lies in $\mathbb{Q}[\eta] \oplus W(A)$ and is killed by T; both summands are defined over $\mathbb{Q}$, so its Weil part $N'\omega'$ is rational, and $r(\kappa) = r(\mathrm{ch}\, E)$ is Lemma 4.55(i). The proof of Proposition 3.21 also gives κ explicitly. At $n = 2$, $\mathrm{ch}(F) = \theta = \frac{1}{2s}(e^{s\theta} - e^{-s\theta})$ and $\mathrm{ch}(F^\vee) = -\theta$, so, with $t = s/2d$,

$$\kappa = -(e^{t\eta} + e^{-t\eta}) - \tfrac{1}{4d}(\gamma^2 - d\ell^2) = -2 + \tfrac{1}{4d}\eta^2 - \tfrac{1}{192d^2}\eta^4 - \tfrac{1}{4d}(\gamma^2 - d\ell^2).$$

Hence $c_1 = c_3 = 0$ and $c_2 = 1/4d$. Write $\gamma - s\ell = a_1 + a_2$ and $\gamma + s\ell = b_1 + b_2$ for the decompositions over the two factors of $X = E_1 \times E_2$, as in the proof of Proposition 3.21. Then $a_i^2 = b_i^2 = 0$, so the Weil part is $\omega = -\frac{1}{4d}(a_1a_2 + b_1b_2)$, and $a_ib_i = \gamma_i^2 + d\ell_i^2 = -4d\,\beta_i\widehat{\beta}_i$; so $\int \omega^2 = \frac{2}{16d^2}\int a_1b_1a_2b_2 = 2$, while $\int \eta^4 = 6\int \eta_1^2\eta_2^2 = 24d^2$. Thus $c_2^2 \int \eta^4 = \frac{3}{2} = \frac{3}{4}\int \omega^2$, the exceptional ratio of Proposition 3.11(iv), where the annihilator has dimension 5. At $n = 3$, $\mathrm{ch}(F) = v_\theta = \frac{1}{2s}(e^{s\theta} - e^{-s\theta})$ and $\mathrm{ch}(F^\vee) = -v_\theta$, and the same computation gives the η-part $-2s(e^{t\eta} - e^{-t\eta}) = 2\eta + \cdots$; so $c_1 = 2 \neq 0$ and the annihilator is $T_{[A]}\mathcal{D}_{3,3}$, of dimension 9, by Proposition 3.11(iii).

For B, put $B' = B - \frac{1}{2}\ell$, so that $\mathrm{ch}(E)e^B = \kappa e^{B'}$. Contraction with $v \in T$ is a derivation and kills κ, so $v \lrcorner (\kappa e^{B'}) = \kappa\,(v \lrcorner B')\,e^{B'}$. Multiplication by κ is injective on $H^2(A, \mathbb{C})$: at $n = 2$ the degree zero part of κ is -2, and at $n = 3$ its lowest part is 2η, and multiplication by η is injective from H^2 to H^4 on the sixfold A. So T kills $\kappa e^{B'}$ exactly when $v \lrcorner B' = 0$ for all $v \in T$. By Proposition 3.11(i) the real class B' of type $(1,1)$ then stays of type $(1,1)$ on the germ of $\mathcal{D}_{n,n}$ at A, hence on all of $\mathcal{D}_{n,n}$, and at a very general member $\mathrm{NS} \otimes \mathbb{Q} = \mathbb{Q}\eta$ ([I, Lemma 7.5]); so $B' \in \mathbb{Q}\eta$. The classes $x_j\xi_j$ appear in $\frac{1}{2}\ell + c\eta$ with coefficient $\frac{1}{2}$ for every c, so no integral class is congruent to $\frac{1}{2}\ell$ modulo $\mathbb{Q}\eta$, on any X: no line bundle twist of E, and in particular not E itself, has a character killed by T. ∎

In both cases the numerical form of the criterion holds with equality, so the injectivity of σ_E comes out of a dimension count and does not need [Mar25b, Question 11.4]. The certificates concern the twisted object only: the character κ stays of Hodge type along $\mathcal{D}_{n,n}$, while $\mathrm{ch}(E)$ does so only on a locus of dimension 3 at $n = 2$ and 6 at $n = 3$, so propagation needs the extension of Theorem 2.2 to twisted objects, which Perry proves [Per26], and an argument extracting untwisted cycles, as in [Mar25b, Section 2]. The families concerned, of trivial discriminant with $n = 2$ and $n = 3$, are Markman's.

Proposition 3.20 (No Orlov product at $n = 4$)**.** *Let X be a principally polarised abelian fourfold, $d \geq 1$, and F_1, F_2 perfect complexes on X with characters in the secant plane $P_\theta \setminus \{0\}$ and $\operatorname{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves. Then $E = \Phi(F_1 \boxtimes F_2^\vee)$ has $\dim \operatorname{Ext}^2(E, E) > 88 = r(\operatorname{ch} E)$.*

Proof. By the Künneth formula $\dim \operatorname{Ext}^2(E, E) = e_0 e_2' + e_1 e_1' + e_2 e_0'$ with $e_k = \dim \operatorname{Ext}^k(F_1, F_1)$ and e_k' likewise for F_2, and by Proposition 3.18 $e_0 \geq 1$, $e_1 \geq 8$, $e_2 \geq 12$; so the sum is at least $12 + 64 + 12 = 88$, which is $r(\operatorname{ch} E)$ by Theorem 4.56(iv). Equality would force $e_0, e_1, e_2 = 1, 8, 12$. Since $e_k = 0$ for $k < 0$ and Serre duality gives $e_k = e_{4-k}$, this would give $\chi(F_1, F_1) = 1 - 8 + 12 - 8 + 1 = -2$, while Proposition 3.18 gives $\chi(F_1, F_1) = 8d(a^2 d + b^2) > 0$. ∎

Without $\operatorname{Ext}^{<0} = 0$ the count changes, because negative Ext groups of the F_i contribute products $e_{-j} e_{2+j}'$ to $\operatorname{Ext}^2(E, E)$ and change the Euler characteristic. The next two results treat the template in every dimension: the twisted character is always flat, and the count says exactly which secant objects would give equality.

Proposition 3.21 (Flatness in every dimension)**.** *Let (X, θ) be a principally polarised abelian n-fold, $n \geq 1$, let $d \geq 1$ and $s = \sqrt{-d}$, and put $\Psi(Z) = \operatorname{ch}(\Phi(Z)) e^{\ell/2}$ for classes Z on $X \times X$, extended linearly, with $\gamma = d\beta - \widehat{\beta}$ as in* [I, Theorem 15.2] *and $\eta = d\beta + \widehat{\beta}$. For $\sigma = \pm 1$, $\Psi(e^{\sigma s\theta} \boxtimes e^{\sigma s\theta})$ is a nonzero multiple of $e^{\sigma(s/2d)\eta}$, and $\Psi(e^{\sigma s\theta} \boxtimes e^{-\sigma s\theta})$ is a nonzero multiple of the Weil class $(\gamma - \sigma s\ell)^n$. Consequently, for perfect complexes F_1, F_2 with $\operatorname{ch}(F_i) \in P_\theta$, the class $\kappa = \operatorname{ch}(\Phi(F_1 \boxtimes F_2^\vee)) e^{\ell/2}$ lies in $\mathbb{Q}[\eta] \oplus W(A)$, is killed by $T = T_{[A]}\mathcal{D}_{n,n}$, and has Hankel rank $\rho \leq 2$; for $F_1 = F_2 \neq 0$ its Weil part is nonzero.*

Proof. A symplectic basis of $H^1(X, \mathbb{Z})$ for θ identifies $H^*(X, \mathbb{Q})$, with θ, with the tensor product of n copies of the cohomology of an elliptic curve with its class θ_i, $\theta = \sum_i \theta_i$. The action of Φ on cohomology is the correspondence $(1 \times \Phi_P) \circ \mu_*$ with $\mu(a, b) = (a - b, b)$, and μ, the Poincaré class $\ell = \sum_i \ell_i$, the classes η and γ, and the exponentials $e^{\pm s\theta} = \bigotimes_i e^{\pm s\theta_i}$ all decompose as tensor products over i of classes of even degree, so no signs arise. Each identity is therefore the tensor product of n copies of the identity for $n = 1$, up to a nonzero scalar, where for the Weil class one uses $(\gamma - \sigma s\ell)^n = n! \prod_i (\gamma_i - \sigma s\ell_i)$, valid because the factors commute and $(\gamma_i - \sigma s\ell_i)^2 = 0$: by [I, Theorem 15.2(i)] at $n = 1$ the class $\gamma_i - \sigma s\ell_i$ spans the top exterior power of a two-dimensional eigenspace.

It remains to treat $n = 1$. Let y_1, y_2 be a symplectic basis of $H^1(X)$, z_1, z_2 its copy on the second factor of $X \times X$ and ξ_1, ξ_2 the dual basis of $H^1(\widehat{X})$, so that $\theta = \beta = y_1 y_2$, $\widehat{\beta} = \xi_1 \xi_2$, $\ell = y_1 \xi_1 + y_2 \xi_2$ and $\ell^2 = -2\beta\widehat{\beta}$. On cohomology Φ sends w to $\int_z \mu^* w \cdot e^{z_1 \xi_1 + z_2 \xi_2}$, the integral being over the second factor; with the opposite sign of the Poincaré class, ℓ is replaced by $-\ell$ in this proposition and in Theorem 3.19. For $a, b \in \{\pm 1\}$,

$$\mu^*(e^{as\theta} \boxtimes e^{bs\theta}) = (1 + as(y_1 + z_1)(y_2 + z_2))(1 + bs\, z_1 z_2),$$
$$e^{z_1 \xi_1 + z_2 \xi_2} = 1 + z_1 \xi_1 + z_2 \xi_2 - z_1 z_2 \xi_1 \xi_2,$$

and collecting the coefficient of $z_1 z_2$ in the product gives

$$\operatorname{ch} \Phi(e^{as\theta} \boxtimes e^{bs\theta}) = (a + b)s - ab\, d\, \beta - \widehat{\beta} - as\, \ell - as\, \beta\widehat{\beta}.$$

Multiplying by $e^{\ell/2} = 1 + \frac{1}{2}\ell - \frac{1}{4}\beta\widehat{\beta}$, and using $\beta\ell = \widehat{\beta}\ell = 0$, gives $\Psi = 2as - d\beta - \widehat{\beta} - \frac{as}{2}\beta\widehat{\beta} = 2as\, e^{(as/2d)\eta}$ when $a = b$, and $\Psi = d\beta - \widehat{\beta} - as\ell = \gamma - as\ell$ when $a = -b$. So for general n the multiples are $(2\sigma s)^n$ and $1/n!$.

For the consequence, write $\mathrm{ch}(F_1) = w_1 e^{s\theta} + \overline{w}_1 e^{-s\theta}$ and $\mathrm{ch}(F_2^\vee) = w_2 e^{-s\theta} + \overline{w}_2 e^{s\theta}$ as in the proof of Proposition 3.18; the terms $e^{\pm s\theta} \boxtimes e^{\pm s\theta}$ give multiples of $e^{\pm(s/2d)\eta}$ and the mixed terms give the Weil part $w_1 w_2 c_+ \alpha_+ + \overline{w}_1 \overline{w}_2 c_- \alpha_-$, with $c_\pm \neq 0$. The classes of $\mathbb{Q}[\eta] \oplus W(A)$ are Hodge classes on every member of the family, so T kills κ, and the η-part lies in the span of the two exponentials $e^{\pm(s/2d)\eta}$, which gives $\rho \leq 2$. ∎

Proposition 3.22 (The equality criterion for the Orlov template)**.** *Let (X, t) be a polarised abelian n-fold, $n \geq 2$, $d \geq 1$, and F_1, F_2 perfect complexes with characters in $P_t \setminus \{0\}$; put $e_k = \dim \mathrm{Ext}^k(F_1, F_1)$ and $e'_k = \dim \mathrm{Ext}^k(F_2, F_2)$ for all $k \in \mathbb{Z}$, and $E = \Phi(F_1 \boxtimes F_2^\vee)$.*

(i) *$\dim \mathrm{Ext}^2(E, E) = \sum_{k \in \mathbb{Z}} e_k e'_{2-k} \geq R_n$, where $R_n = 6n^2 - 2n$ for $n \geq 3$ and $R_2 = 18$, and $R_n = r(\mathrm{ch}\, E) = r(\kappa)$.*

(ii) *Equality holds if and only if $(e_0, e_1, e_2) = (e'_0, e'_1, e'_2) = (1, 2n, r_2)$, with $r_2 = n(n-1)$ for $n \geq 3$ and $r_2 = 1$ for $n = 2$, and $e_{-j} e'_{2+j} = e_{2+j} e'_{-j} = 0$ for every $j \geq 1$.*

(iii) *For $F_1 = F_2 = F$ and $n \geq 3$ the last condition says that the set $S = \{j \geq 1 : \mathrm{Ext}^{-j}(F, F) \neq 0\}$ lies in $[n-1, \infty)$ and that $j \in S$ implies $j - (n-2) \notin S$; for $n = 2$ it says $\mathrm{Ext}^{<0}(F, F) = 0$.*

(iv) *Equality implies that σ_E is injective.*

Proof. (i) Φ is an equivalence and the Künneth formula holds in all degrees, negative ones included, and dualising is an anti-equivalence, so $\mathrm{Ext}^*(E, E) = \mathrm{Ext}^*(F_1, F_1) \otimes \mathrm{Ext}^*(F_2, F_2)$. By Proposition 3.18, $e_0 \geq 1$, $e_1 \geq 2n$ and $e_2 \geq r_2$, so the three terms with $k = 0, 1, 2$ already give R_n; the equality $R_n = r(\mathrm{ch}\, E)$ is the Künneth count of Theorem 4.56(iv) and Theorem 3.19, and $r(\kappa) = r(\mathrm{ch}\, E)$ by Lemma 4.55(i). (ii) Every term is nonnegative, and $e_0 e'_2 \geq e'_2 \geq r_2 > 0$ with equality only if $e_0 = 1$ and $e'_2 = r_2$, and likewise for the other two terms; every other product must vanish. (iii) Serre duality gives $e_k = e_{n-k}$. For $1 \leq j \leq n-2$, e_{2+j} is at least the rank of contraction from $HT^{2+j}(X)$ into $\mathrm{ch}(F)$, which is positive by Proposition 3.18, because the characteristic morphism is compatible with contraction into the Chern character in every degree [CRV12]; so $e_{-j} = 0$. For $j \geq n-1$, $e_{2+j} = e_{-(j-(n-2))}$. (iv) The composite $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\mathrm{ch}(E)$, of rank $R_n \leq \mathrm{rk}\, \sigma_E \leq \dim \mathrm{Ext}^2(E, E)$, so equality forces σ_E to be injective. ∎

For $n = 2$ and $n = 3$ the certificates of Theorem 3.19 attain the equality; for $n = 4$ it is excluded by Proposition 3.20 when $\mathrm{Ext}^{<0} = 0$, with negative Ext groups in degrees at most $-(n-1)$ the Euler characteristic leaves room in every dimension, and from $n = 5$ on it leaves room without them (Proposition 3.23(i)). If a secant object F attaining the equality existed for one d and unboundedly many n, then, since κ is flat in every dimension by Proposition 3.21, a twisted form of Theorem 2.3 together with Proposition 6.13 would make the Weil classes of every family of $\mathbb{Q}(\sqrt{-d})$ algebraic. That is the secant route of (C1) and (C2) in Section 6.5, with the question of [Mar25b] replaced by a dimension count, and it lies off every minimal route (Theorem 6.70(iv)).

Both certificates are sheaves on theta divisors, a curve of genus two and the surface $C^{(2)}$. The next result explains why this construction stops at $n = 3$ and what a continuation would need.

Proposition 3.23 (Sheaves on divisors in the Orlov template)**.** *Let (X, θ) be a principally polarised abelian n-fold and $d \geq 1$; put $S(x) = \sum_{j \geq 0} (-d)^j x^{2j} / (2j+1)!$, so that $v_\theta = \theta S(\theta)$, and $T_b(x) = bx/(1 - e^{-bx})$ for an integer $b \geq 1$. Write $\beta_0 = \beta_n = 1$, $\beta_k = 2\binom{n}{k}$ for $0 < k < n$ and $\beta_k = 0$ otherwise; these are the Betti numbers of the connected sum of two real n-tori.*

(i) *Let $n \geq 5$ and let F be a perfect complex with $\operatorname{ch}(F) \in P_\theta \setminus \{0\}$. There are integers e_k, zero for $k \notin [0,n]$, with $e_k = e_{n-k}$, $(e_0, e_1, e_2) = (1, 2n, n(n-1))$, $e_k \geq \beta_k$ for all k, and $\sum_k (-1)^k e_k = \chi(F,F)$. So Serre duality, the contraction bounds $e_k \geq \beta_k$ (Proposition 3.18 and [CRV12]) and the Euler characteristic allow equality in Proposition 3.22 with $\operatorname{Ext}^{<0}(F,F) = 0$. For odd n one may take $e_k = \beta_k$, and at $n = 5$ this is the only choice; for even n the sequence (β_k) has Euler characteristic $-2 \neq \chi(F,F)$, so some e_k with $3 \leq k \leq n-3$ must exceed β_k.*

(ii) *Let $n \geq 3$, let $i\colon D \hookrightarrow X$ be a smooth effective divisor, and let G be a vector bundle of rank $r \geq 1$ on D with $\operatorname{ch}(G) \in i^*H^*(X,\mathbb{Q})$. If $\operatorname{ch}(i_*G) \in P_\theta$, then $[D] = b\theta$ for an integer $b \geq 1$ and $\operatorname{ch}(G) = r\, i^*\big(S(\theta)\, T_b(\theta)\big)$; in particular $c_1(G) = \frac{rb}{2}\,\theta|_D$, $\operatorname{ch}_2(G) = r\big(\frac{b^2}{12} - \frac{d}{6}\big)\theta^2|_D$ and $r \geq 2$. If $n \geq 4$, every line bundle on D satisfies the hypothesis, so no line bundle L on a smooth divisor has $\operatorname{ch}(i_*L) \in P_\theta$, and rb is even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{d+1}{3}\,\theta^2|_D$, which is integral only for $d \equiv 2 \pmod 3$.*

(iii) *Let $n \geq 2$, let $i\colon D \hookrightarrow X$ be a smooth divisor with $[D] = b\theta$, V a vector bundle on X, $G = V|_D$ and $F = i_*G$. Then*

$$\dim \operatorname{Ext}^k(F,F) = h^k(\operatorname{End} G) + h^{k-1}\big(\operatorname{End} G \otimes \mathcal{O}_D(D)\big)$$

for every k. For $b = 1$ this is $\beta_k + h^k(\operatorname{End}_0 G) + h^{k-1}(\operatorname{End}_0 G \otimes \mathcal{O}_D(D))$, so for a line bundle V the sheaf F has exactly the profile required in Proposition 3.22(ii). For $b \geq 2$, $\dim \operatorname{Ext}^1(F,F) \geq b^n + n - 1 > 2n$.

(iv) *At $n = 5$, for no smooth divisor D and no vector bundle V on X is $i_*(V|_D)$ a factor of an Orlov product attaining the equality of Proposition 3.22.*

Proof. (i) The rank a of F is an integer, and $c_1(F) = b\theta$ with θ primitive, so $b \in \mathbb{Z}$; for even n the number $c = 2^{n-1}d^{n/2-1}(a^2 d + b^2)$ is therefore a positive integer divisible by 2^{n-1}, and $\chi(F,F) = (-1)^{n/2}c$ by Proposition 3.18, while $\chi(F,F) = 0$ for odd n. The conditions fix e_0, e_1, e_2 and, by symmetry, e_{n-2}, e_{n-1}, e_n, with the values β_k; for $n = 5$ these are all the e_k. Since $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$, the sequence (β_k) has Euler characteristic 0 for odd n and -2 for even n. For odd n take $e_k = \beta_k$. For even $n \geq 6$ take $e_k = \beta_k$ for $k \neq n/2$ and $e_{n/2} = \beta_{n/2} + c + 2(-1)^{n/2} \geq \beta_{n/2}$, which is possible because $c \geq 2^{n-1} > 2$ and $3 \leq n/2 \leq n-3$; its Euler characteristic is $-2 + (-1)^{n/2}(c + 2(-1)^{n/2}) = \chi(F,F)$.

(ii) The rank of i_*G is zero, so $\operatorname{ch}(i_*G) = b' v_\theta$ for some $b' \in \mathbb{Q}$, and comparing degree two gives $r[D] = b'\theta$. As $[D]$ is integral and nonzero and θ is primitive, $[D] = b\theta$ with b a positive integer, and D is ample. The tangent bundle of X is trivial and the normal bundle of D is $\mathcal{O}_D(D)$, so $\operatorname{td}(D) = i^*T_b(\theta)^{-1}$; writing $\operatorname{ch}(G) = i^*y$, Grothendieck–Riemann–Roch and the projection formula give

$$\operatorname{ch}(i_*G) = i_*\big(i^*(y\, T_b(\theta)^{-1})\big) = y\, T_b(\theta)^{-1}\, b\theta = y\,(1 - e^{-b\theta}).$$

So $\operatorname{ch}(i_*G) = rb\,\theta S(\theta)$ says $b\theta\, z\, T_b(\theta)^{-1} = 0$ for $z = y - r\,S(\theta)T_b(\theta)$, that is $\theta z = 0$. Let z_k be the component of z in $H^{2k}(X,\mathbb{Q})$. If $2k \leq n-1$, multiplication by θ is injective on $H^{2k}(X,\mathbb{Q})$ by the hard Lefschetz theorem, so $z_k = 0$. If $2k \geq n$, the Gysin map i_* is injective on $H^{2k}(D,\mathbb{Q})$, being dual to the restriction $H^{2n-2-2k}(X,\mathbb{Q}) \to H^{2n-2-2k}(D,\mathbb{Q})$, which is onto by the Lefschetz hyperplane theorem, and $i_*i^*z_k = b\theta z_k = 0$; so $i^*z_k = 0$. Hence $\operatorname{ch}(G) = r\, i^*(S(\theta)T_b(\theta))$, and the expansion $S(x)T_b(x) = 1 + \frac{b}{2}x + \big(\frac{b^2}{12} - \frac{d}{6}\big)x^2 + O(x^3)$ gives c_1 and ch_2. If $r = 1$ then $\operatorname{ch}_2(G) = \frac{1}{2}c_1(G)^2$, so $\big(\frac{b^2}{24} + \frac{d}{6}\big)\theta^2|_D = 0$, which is false: $\int_D (\theta|_D)^{n-1} = b\,n! \neq 0$ and $n - 1 \geq 2$. If $n \geq 4$, the Lefschetz hyperplane theorem

gives $H^2(X,\mathbb{Z}) \cong H^2(D,\mathbb{Z})$, so every line bundle L on D has $\operatorname{ch}(L) = i^*e^\lambda$ for some $\lambda \in H^2(X,\mathbb{Z})$, and $\theta|_D$ is primitive, so the integrality of $c_1(G)$ makes rb even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{1}{2}c_1^2 - \operatorname{ch}_2 = \frac{d+1}{3}\theta^2|_D = \frac{2(d+1)}{3}\, i^*(\frac{1}{2}\theta^2)$. The class $\frac{1}{2}\theta^2$ is integral and primitive. The affine variety $X \setminus D$ has the homotopy type of a CW complex of dimension n, so by Lefschetz duality $H^4(X,D;\mathbb{Z}) \cong H_{2n-4}(X \setminus D;\mathbb{Z}) = 0$ and $H^5(X,D;\mathbb{Z}) \cong H_{2n-5}(X \setminus D;\mathbb{Z})$ is torsion free; hence $i^*\colon H^4(X,\mathbb{Z}) \to H^4(D,\mathbb{Z})$ is injective with torsion-free cokernel, so $i^*(\frac{1}{2}\theta^2)$ is primitive modulo torsion, and $c_2(G)$ is integral only if 3 divides $2(d+1)$.

(iii) Let s be the section of $\mathcal{O}_X(D)$ vanishing on D. The sheaf F has the locally free resolution $0 \to V(-D) \xrightarrow{s} V \to F \to 0$, and applying $\operatorname{Hom}(-,F)$ gives a long exact sequence with $\operatorname{Ext}^k(V,F) = H^k(\operatorname{End} G)$ and $\operatorname{Ext}^k(V(-D),F) = H^k(\operatorname{End} G \otimes \mathcal{O}_D(D))$, in which the map between these two groups is multiplication by $s|_D = 0$. So it splits into short exact sequences $0 \to H^{k-1}(\operatorname{End} G \otimes \mathcal{O}_D(D)) \to \operatorname{Ext}^k(F,F) \to H^k(\operatorname{End} G) \to 0$. Write $\operatorname{End} G = \mathcal{O}_D \oplus \operatorname{End}_0 G$. On an abelian variety $H^k(\mathcal{O}_X(D)) = 0$ for $k > 0$ and $H^k(\mathcal{O}_X(-D)) = 0$ for $k < n$, and $h^0(\mathcal{O}_X(D)) = b^n$ by Riemann–Roch. From $0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$ one gets $h^k(\mathcal{O}_D) = \binom{n}{k}$ for $k \leq n-2$, and also for $k = n-1$ when $b = 1$, because $H^n(\mathcal{O}_D) = 0$ makes the map $H^n(\mathcal{O}_X(-D)) \to H^n(\mathcal{O}_X)$ between lines onto. From $0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to \mathcal{O}_D(D) \to 0$ one gets $h^0(\mathcal{O}_D(D)) = b^n - 1 + n$ and $h^k(\mathcal{O}_D(D)) = \binom{n}{k+1}$ for $k \geq 1$. For $b = 1$ the two contributions of $\mathcal{O}_D$ add up to β_k in every degree, which gives the formula; for a line bundle V, $\operatorname{End}_0 G = 0$, and F is a sheaf, so $\operatorname{Ext}^{<0}(F,F) = 0$. For $b \geq 2$, $\dim \operatorname{Ext}^1(F,F) \geq h^0(\mathcal{O}_D(D)) = b^n + n - 1$, and $b^n > n+1$.

(iv) Such a product needs $\operatorname{ch}(F) \in P_\theta$ for $F = i_*(V|_D)$, so $[D] = b\theta$ by (ii), and its factors have $(e_0,e_1,e_2) = (1,10,20)$ by Proposition 3.22(ii); by (iii) $b = 1$. Then Serre duality gives $e_k = \beta_k$ for every k, so by (iii) $H^*(\operatorname{End}_0 G) = 0$ and $\chi(D, \operatorname{End}_0 G) = 0$. On the other hand (ii) gives $\operatorname{ch}(G) = r\, i^*(S(\theta)T_1(\theta))$, so $\operatorname{ch}(\operatorname{End} G) = \operatorname{ch}(G)\operatorname{ch}(G)^\vee = r^2 i^*\big(S(\theta)^2 T_1(\theta) T_1(-\theta)\big)$, and Hirzebruch–Riemann–Roch on D gives

$$\chi(D, \operatorname{End} G) = r^2 \int_X S(\theta)^2 \, \frac{\theta^2}{e^\theta - 1} = \frac{r^2(32d^2 - 20d - 1)}{6},$$

while $\chi(D,\mathcal{O}_D) = \int_X (1 - e^{-\theta}) = 1$. So $6\chi(D, \operatorname{End}_0 G) = r^2(32d^2 - 20d - 1) - 6 \geq 5$, because $32d^2 - 20d - 1 \geq 11$ for $d \geq 1$. ∎

So from $n = 5$ on the numbers allow a secant object attaining the equality, the line bundles on theta divisors have the Ext groups it requires in every dimension, and what fails is their character. The line bundle of Theorem 3.19(ii) has $c_1 = (2k+1)x - k\theta$, which by Proposition 3.23(ii), valid at $n = 3$, is not restricted from the Jacobian; from $n = 4$ on the Lefschetz theorem forbids such classes. At $n = 5$, by (ii) and (iv), a vector bundle G on a smooth divisor with i_*G a factor attaining the equality is neither a line bundle nor restricted from X; if G lives on a theta divisor and has Chern character restricted from X, its rank is even, and rank two needs $d \equiv 2 \pmod 3$. Bundles of this kind that are not restricted from X, sheaves on singular divisors, sheaves that are not locally free on their support, and objects of positive rank remain; we know no object of these kinds with a secant character and a small Ext algebra.

The singular divisors closest to the two certificates are the theta divisors of Jacobians: both certificates are line bundles on the symmetric product $C^{(g-1)}$, pushed forward along the Abel–Jacobi map, for $g = 2, 3$. From $g = 4$ on the theta divisor is singular and the Lefschetz argument of Proposition 3.23(ii) no longer applies, since $C^{(g-1)}$ carries the class

x, which does not come from the Jacobian. The character excludes this continuation all the same.

Proposition 3.24 (The Jacobian continuation stops at genus three)**.** *Let C be a smooth projective curve of genus $g \geq 2$, $X = J(C)$ with its principal polarisation Θ, and $u\colon C^{(g-1)} \to X$ the Abel–Jacobi map, whose image is a translate of the theta divisor. Let x be the class of the divisor $p + C^{(g-2)}$ and $\theta = u^*\Theta$, let L be a line bundle on $C^{(g-1)}$ with $c_1(L) = \alpha x + \beta\theta$, and put $\gamma = \beta - \frac{1}{2}$. Then the perfect complex Ru_*L has*

$$\operatorname{ch}(Ru_*L) = \Theta + \Big(\frac{\alpha}{2} + \gamma\Big)\Theta^2 + c_3\Theta^3 + c_4\Theta^4 + \cdots,$$

and when $\alpha = -2\gamma$, that is $\alpha + 2\beta = 1$,

$$c_3 = -\frac{4\gamma^2 - 1}{24}, \qquad c_4 = -\frac{\gamma(4\gamma^2 - 1)}{72}.$$

Consequently $\operatorname{ch}(Ru_*L) \in P_\Theta$ for some $d \geq 1$ exactly when $g = 2$ and $\alpha + 2\beta = 1$, or $g = 3$, $\alpha + 2\beta = 1$ and $d = \beta(\beta - 1)$; these are the sheaves of Theorem 3.19(i) and (ii). For $g \geq 4$ no line bundle on $C^{(g-1)}$ with c_1 in the span of x and θ has a secant character.

Proof. The Todd class of X is 1, so by Grothendieck–Riemann–Roch $\operatorname{ch}(Ru_*L) = u_*(e^{\alpha x + \beta\theta}\operatorname{td}(C^{(g-1)}))$. By Macdonald [Mac62], [ACGH85, Chapter VII, Section 5], the total Chern class of $C^{(k)}$ is $(1+x)^{k-g+1}e^{-\theta/(1+x)}$; for $k = g-1$ the first factor is 1. In Macdonald's description of the cohomology ring $\theta = \sum_{i=1}^{g}\sigma_i$ with $\sigma_i^2 = 0$, so $e^{-\theta/(1+x)} = \prod_i (1 + x - \sigma_i)/(1+x)$ and, with $\operatorname{Td}(y) = y/(1-e^{-y})$ and $q = \operatorname{Td}'/\operatorname{Td}$,

$$\operatorname{td}(C^{(g-1)}) = \prod_i \frac{\operatorname{Td}(x - \sigma_i)}{\operatorname{Td}(x)} = \prod_i (1 - \sigma_i q(x)) = e^{-\theta q(x)}.$$

Since $\log \operatorname{Td}(y) = \frac{y}{2} - \frac{y^2}{24} + O(y^4)$, we have $-q(x) = -\frac{1}{2} + \frac{x}{12} + O(x^3)$. The class x^a is that of the image of $C^{(g-1-a)}$ under $D \mapsto D + p_1 + \cdots + p_a$, which u maps birationally onto a translate of W_{g-1-a}, of class $\Theta^{a+1}/(a+1)!$ by Poincaré's formula [ACGH85, Chapter I, Section 5]; with $\theta = u^*\Theta$ and the projection formula, $u_*(x^a\theta^b) = \Theta^{a+b+1}/(a+1)!$. So the coefficient c_k of Θ^k is $\sum_{a+b=k-1} m_{ab}/(a+1)!$, where m_{ab} is the coefficient of $x^a\theta^b$ in $e^{\alpha x}e^{\theta(\gamma + x/12)}$ for $k \leq 4$ (the term $O(x^3)$ first contributes to Θ^5), namely $m_{ab} = \sum_j \frac{\alpha^{a-j}}{(a-j)!}\frac{\gamma^{b-j}}{(b-j)!}\frac{1}{12^j j!}$. This gives $c_1 = 1$, $c_2 = \frac{\alpha}{2} + \gamma$,

$$c_3 = \frac{\alpha^2}{12} + \frac{\alpha\gamma}{2} + \frac{1}{24} + \frac{\gamma^2}{2}, \qquad c_4 = \frac{\alpha^3}{144} + \frac{\alpha^2\gamma}{12} + \frac{\alpha}{72} + \frac{\alpha\gamma^2}{4} + \frac{\gamma}{24} + \frac{\gamma^3}{6},$$

and at $\alpha = -2\gamma$ these are the stated values.

The rank of Ru_*L is 0, so $\operatorname{ch}(Ru_*L) \in P_\Theta$ means $\operatorname{ch}(Ru_*L) = v_\Theta = \Theta - \frac{d}{6}\Theta^3 + \frac{d^2}{120}\Theta^5 - \cdots$, the coefficient of Θ being 1. For $g = 2$ this is $c_2 = 0$. For $g \geq 3$ it also asks $c_3 = -d/6$, that is $d = \gamma^2 - \frac{1}{4} = \beta(\beta - 1)$, which settles $g = 3$. For $g \geq 4$ it asks in addition $c_4 = 0$; as $4\gamma^2 - 1 = 4d$, this is $\gamma d = 0$, so $d = 0$ or $\gamma = 0$, where $d = -\frac{1}{4}$. At $g = 3$, $L = \mathcal{O}((2k+1)x - k\theta)$ has $\beta = -k$ and $d = k(k+1)$, and at $g = 2$, $C^{(1)} = C$ and $L = \mathcal{O}_C(p)$ has $\alpha = 1$, $\beta = 0$. ∎

So the continuation of the two certificates must leave the line bundles on $C^{(g-1)}$: a bundle of higher rank, a sheaf on the theta divisor of a principally polarised abelian variety that is not a Jacobian, or a complex of positive rank.

Remark 3.25 (Untwisting). The template $E = \Phi(F_1 \boxtimes F_2^\vee)$ does not give an untwisted object meeting the polarised criterion by the obvious operations. No line bundle twist makes $\mathrm{ch}(E)$ flat (Theorem 3.19(iii)). At $n = 2$ the object $S^2(E[1]) \otimes P$, for P a line bundle with $c_1(P) = \ell$, is untwisted with flat character, of rank 3 and at the exceptional ratio after a twist by a multiple of η, but its number is $r = 23$, which is odd, while Serre duality and the evenness of the Mukai pairing on an abelian variety make $\dim \mathrm{Ext}^2 \equiv \chi(E, E) \equiv 0 \pmod 2$; so equality is impossible. The object $\bigwedge^2(E[1]) \otimes P$ has no Weil part. For $E \otimes V$ with V a simple semi-homogeneous bundle of slope $\ell/2$ [Muk78], the character $\mathrm{rk}(V)\kappa$ is flat, but $\sigma_{E\otimes V}$ kills the summands $\mathrm{Ext}^2(E, E \otimes P_\xi)$, $\xi \neq 0$, of $\mathrm{Ext}^2(E \otimes V, E \otimes V)$ coming from the decomposition of $V \otimes V^\vee$, and at $n = 2$ these are nonzero; this last point uses the decomposition of $\mathrm{End}\, V$ and the way Φ intertwines $\otimes P_\xi$ with translations, which we recall without proof. None of this shows that the polarised criterion must be met in twisted form; it shows that this template meets it only there.

At $n = 4$ the natural objects of the split member can be described completely, and the description turns the criterion into a statement about points on a quadric.

Setup 3.26 (The quadric of Lagrangians)**.** Let X be a very general principally polarised abelian fourfold, $U = H^1(X, \mathbb{Q})$, and $A = X \times \widehat{X}$ with the K-action of [I, (29)], so that $H^1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^2$ and the rational Hodge ring $\mathcal{H}_A$ of A is the ring of $\mathrm{Sp}(U)$-invariants in $\bigwedge^*(U \otimes \mathbb{Q}^2)$, of dimension 55, with $1, 3, 6, 10, 15, 10, 6, 3, 1$ in degrees $0, 2, \dots, 16$. The symmetric pairing on $H^1(A, \mathbb{Q}) \oplus H_1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^4$ is the product of the symplectic form of U and a symplectic form on $\mathbb{Q}^4$, so the maximal isotropic subspaces stable under $\mathrm{Sp}(U)$ are the $U \otimes L$ with L a Lagrangian plane in $\mathbb{Q}^4$. Let $Q \subset \mathbb{P}^4$ be the Lagrangian Grassmannian $\mathrm{LG}(2, 4)$, a smooth quadric threefold in its Plücker embedding. Call a perfect complex on A *natural* if it is a line bundle, a sheaf $\mathcal{O}_B \otimes L$ on an abelian subvariety, a point, or the image of one of these under an autoequivalence. Then the character of a natural F is a multiple of the pure spinor of $U \otimes L_F$ for a rational point $L_F \in Q$, and there is a linear isomorphism $\Phi_4 \colon H^0(Q, \mathcal{O}(4))^\vee \to \mathcal{H}_A$ with $\mathrm{ch}(F) = \Phi_4(\nu_4(L_F))$ up to a rational factor, ν_4 the fourth Veronese map. Under it the powers of η span the image of a conic $C_\eta \subset Q$, and the Weil plane is the rational part of the line spanned by $\nu_4(L_+)$ and $\nu_4(L_-)$ for a conjugate pair of K-points $L_\pm$ of Q; the line $l_W = L_+L_-$ meets Q only in $L_\pm$, and the plane of C_η is its polar plane.

The pure spinor description of the characters of natural objects is classical, and we recall it without proof. The abelian subvarieties $B_{(p,q)} = \{(px, q\varphi x)\}$, with $\varphi \colon X \to \widehat{X}$ the polarisation, form a conic of natural objects, and on it the two points $L_\pm$ have coordinates $(p : q) = (1 : \mp\sqrt{-d})$: by [I, Theorem 14.2(iv)] the eigenspaces $W_\pm$ are the graphs of $\mp\sqrt{-d}\,\beta$. So the Weil classes are the values of the octic $(p, q) \mapsto [B_{(p,q)}]$ at those points.

Theorem 3.27 (Minimal support at $n = 4$)**.** *In Setup 3.26, let $F_1, \dots, F_s$ be natural objects whose points L_{F_i} are pairwise distinct and not on C_η, and suppose that $\sum_i m_i \,\mathrm{ch}(F_i) + P = N\omega_1 + \sum_k c_k \eta^k$ with $m_i \in \mathbb{Q}$, $P \in \mathbb{Q}[\eta]$ and $N \neq 0$. Then $s \geq 8$. If $s = 8$, the eight points lie on one smooth conic $\Pi \cap Q$ with Π a plane containing l_W, the conic is disjoint from C_η, and $\sum_i m_i \,\mathrm{ch}(F_i) = N\omega_1$ exactly. Conversely any eight rational points of such a conic carry such a relation with every $m_i \neq 0$.*

Proof. Through Φ_4 the relation reads $\sum_i m_i \nu_4(L_{F_i}) + P' = c_+\nu_4(L_+) + c_-\nu_4(L_-)$ in $H^0(Q, \mathcal{O}(4))^\vee$, with P' in the span of $\nu_4(C_\eta)$ and $c_\pm$ conjugate and nonzero because $N \neq 0$. A quartic f vanishing on C_η, at L_- and at every L_{F_i} gives $c_+ f(L_+) = 0$; so it suffices to produce such an f with $f(L_+) \neq 0$ unless the configuration is as stated. Let h_- be the

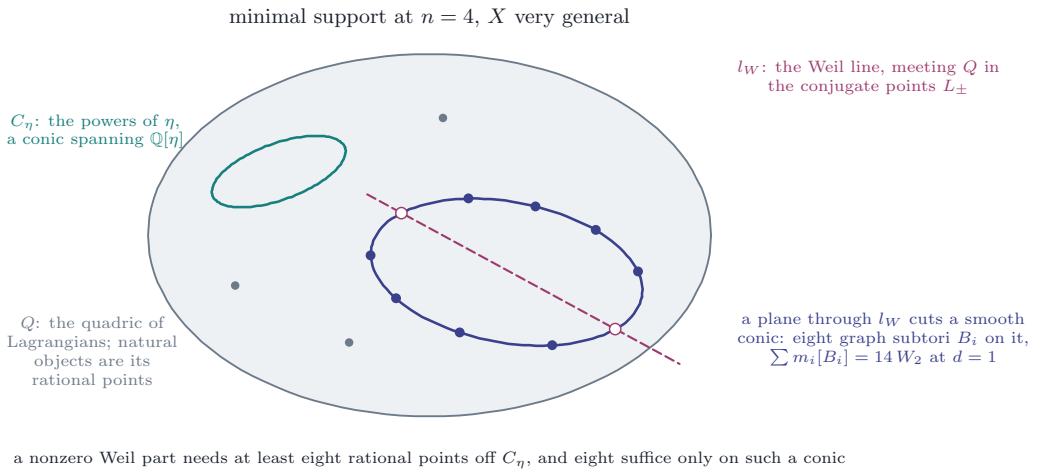


FIGURE 4. Theorem 3.27. The quadric Q of Lagrangians; the rational points are the Chern characters of the natural objects on the split member at $n = 4$, up to the Veronese map ν_4. The powers of the polarisation fill the conic C_η (teal), and the Weil line is spanned by the conjugate points $L_\pm$ (open circles) in which the line l_W (dashed) meets Q. A combination of natural characters with a nonzero Weil part needs at least eight rational points off C_η, and with eight they lie on one conic cut out by a plane through l_W (indigo), such as the eight graph subvarieties B_i of Example 3.28; the grey points are natural objects off any such conic, which no relation of this size can use.

linear form $B(L_-, \cdot)$, with B the bilinear form of the quadric Q; it vanishes on the polar plane of l_W, hence on C_η, and at L_-, but not at L_+, since $l_W \not\subset Q$. If the points split into at most three groups G_k with L_+ outside the span of each, linear forms l_k vanishing on G_k and not at L_+ give the quartic $h_- l_1 l_2 l_3$, which kills everything but c_+, so $c_+ = 0$. A rational line through L_+ also passes through L_-, so it is l_W, which has no rational point on Q; hence groups of one or two points never contain L_+ in their span, and a triple does only if it lies in a plane through l_W. For $s \leq 6$ split into pairs. For $s = 7$, either some triple is not in a plane through l_W, or all seven lie in one such plane Π, hence on $C'' = \Pi \cap Q$. That conic is smooth, since a degenerate one is a pair of conjugate lines with at most one rational point, and it misses C_η, since Π meets the plane of C_η in one point z and $z \in Q$ would put the lines $zL_\pm$ in Q. The quartics vanishing on C_η restrict onto all binary octics on $C'' \cong \mathbb{P}^1$, and nine distinct points impose independent conditions on octics, so $L_\pm$ and the seven points are independent. For $s = 8$, if no plane through l_W holds seven of the points, choose two triples not in such planes and a pair; if exactly seven lie in Π, modify the quartic by a multiple of one vanishing on Π and at the eighth point; if all eight lie on C'', the ten points $L_\pm, L_{F_1}, \ldots, L_{F_8}$ of the rational normal curve $\nu_4(C'')$ of degree eight satisfy exactly one relation, with every coefficient nonzero, and it is rational by Galois symmetry. Its part along the span of $\nu_4(C_\eta)$ vanishes, because quartics restrict onto the sum of the octics on the two disjoint conics, so the relation is pure. ∎

Figure 4 draws the quadric of Lagrangians, the conic C_η and a conic through eight natural characters.

Example 3.28 (Fourteen times a Weil class)**.** Let $d = 1$ and write $(\gamma - \sqrt{-1}\,\ell)^4 = W_1 + \sqrt{-1}\,W_2$ with W_1, W_2 rational, as in [I, Theorem 15.2(iii)]. The eight subvarieties $B_i = B_{(p_i,q_i)}$ with

$$(p_i, q_i) = (1,-3),\ (1,-2),\ (1,2),\ (1,3),\ (2,-1),\ (2,1),\ (3,-1),\ (3,1)$$

and $m = (-1, 16, -16, 1, -16, 16, 1, -1)$ satisfy

$$\textstyle\sum_i m_i [B_i] = 14\, W_2,$$

with the sign of φ for which $B_{(p,q)}$ has conormal forms $p\xi_j - q x_{n+j}$ and $p\xi_{n+j} + q x_j$; with the opposite sign the right side is $-14W_2$. Among the 12870 sets of eight subvarieties with $|p|, |q| \le 3$ this one has the least $\sum_i m_i^2$, namely 1028, and the least $\sum_i |m_i|$, namely 68. In general the coefficients of the relation of Theorem 3.27 for eight subvarieties are proportional to $\big((q_i^2 + d p_i^2)\prod_{j\neq i}(p_i q_j - p_j q_i)\big)^{-1}$. The sum $E = \bigoplus_i \mathcal{O}_{B_i}^{\oplus |m_i|}[k_i] \oplus \mathcal{O}(\eta) \oplus \mathcal{O}(2\eta) \oplus \mathcal{O}(3\eta)$, with $k_i = 0$ for $m_i > 0$ and 1 otherwise, has $\mathrm{ch}(E) = 14W_2 + \sum_k c_k \eta^k$ with $k!\, c_k = 1 + 2^k + 3^k$, so $\rho = 3$ and $r = 104$, while $\dim \mathrm{Ext}^2(E, E) = 28868$; by Corollary 3.17 it is far from the criterion, and every direct sum of natural objects with $N \neq 0$ has $\dim \mathrm{Ext}^2 \ge 8 \cdot 28 = 224$.

Remark 3.29 (The flatness form for direct sums). Lemma 3.16 does not apply to the flatness form of Proposition 3.15, but a direct sum fails that form too when its cross-extensions cannot interact. Let $E = \bigoplus_i (\mathcal{O}_{B_i} \otimes L_i)^{\oplus m_i}[k_i]$ with the B_i pairwise distinct abelian subvarieties of dimension n meeting pairwise in finite sets, such as the $B_{(p,q)}$. The extensions between different summands are concentrated in degree n, so an arrow of Ext^1 between summands raises the shift by $n-1$, and a chain of $a \ge 1$ arrows lands in $\mathrm{Ext}^{2+a(n-1)} \neq \mathrm{Ext}^2$. Every term of the Maurer–Cartan equation that involves a cross-extension therefore lands in a zero group, the base K_E is the product of the varieties of commuting $2n$-tuples of $m_i \times m_i$ matrices with the space of cross-extensions, and a count of dimensions, using the simultaneously diagonalisable component, gives $\dim K_E > e - k$; so (ii) and (iii) of Proposition 3.15 fail. We give this in outline only. Indecomposable objects, among them iterated extensions of at least eight natural blocks, and sums whose cross-extensions compose to nonzero classes are not covered.

4. Secant objects on the split member

4.1. **Sums of line bundles and the positivity obstruction.** Theorem 2.3 asks for one perfect complex at one base point. The cheapest candidate is a sum of line bundles on a split member, because there the Weil classes are already polynomials in β, $\widehat{\beta}$ and ℓ by [I, Theorem 15.2], so the Chern character one wants is manifestly in reach. This subsection settles that candidate. It does not exist, and the obstruction is not semiregularity: it is the Chern character condition, and it fails because the norm form of K is positive definite.

The norm rules out three different constructions in this paper, by three different mechanisms, and they should be kept apart. [I, Theorem 5.6] uses the norm *character* of $K^\times$ and an equivariance hypothesis. [I, Proposition 15.18] uses the fact that the part R of H^2 on which $K^\times$ acts through τ^2 and $\overline{\tau}^2$ has no Hodge classes at a general point, which is a statement about the Hodge group. Theorem 4.5 below uses neither: it uses that $\mathrm{N}_{K/\mathbb{Q}}$ is a positive definite quadratic form on K and that the multiplicities of a filtration are positive integers. The three are independent, and only the third is available at a special point, which is exactly where the object of Proposition 2.1 has to live.

Lemma 4.1 (The semiregularity map of a split object)**.** *Let Y be a smooth projective complex variety and let*

$$E = \bigoplus_{i=1}^{s} L_i[p_i]^{\oplus m_i}, \qquad \lambda_i = c_1(L_i),$$

with the L_i line bundles, the λ_i pairwise distinct and $p_i \in \mathbb{Z}$. Then for $\zeta = (\zeta_{ij}) \in \operatorname{Ext}^2(E,E)$,

$$\sigma_E(\zeta)_q = \frac{1}{q!} \sum_{i=1}^{s} (-1)^{p_i} \operatorname{tr}(\zeta_{ii})\, \lambda_i^q, \tag{1}$$

so σ_E annihilates $\operatorname{Ext}^2(L_i[p_i], L_j[p_j])$ for $i \neq j$ and the traceless part of each diagonal block. Consequently σ_E is injective if and only if all three of the following hold:

(i) *$m_i = 1$ for every i;*
(ii) *$H^{\,2+p_i-p_j}\big(Y, L_j L_i^{-1}\big) = 0$ for all $i \neq j$;*
(iii) *the map*

$$\Phi\colon H^2(\mathcal{O}_Y)^{\oplus s} \longrightarrow H^*(Y,\mathbb{C}), \qquad (\zeta_i) \longmapsto \sum_i e^{\lambda_i}\zeta_i,$$

is injective.

Proof. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of a line bundle is its first Chern class, so $\operatorname{At}(E)$ is diagonal with entries λ_i and $\operatorname{At}(E)^q$ is diagonal with entries λ_i^q. The matrix product $\zeta \circ \operatorname{At}(E)^q$ therefore has (i,j) entry $\zeta_{ij}\lambda_i^q$, and the trace keeps only the diagonal; a shift by p multiplies the trace by $(-1)^p$. That is (1), and it exhibits the two spaces the map kills.

For the equivalence, the off-diagonal part of $\operatorname{Ext}^2(E,E)$ is $\bigoplus_{i\neq j} \operatorname{Ext}^2(L_i[p_i], L_j[p_j]) = \bigoplus_{i\neq j} H^{2+p_i-p_j}(L_jL_i^{-1})$, which must vanish, giving (ii); the diagonal block at i is $H^2(\mathcal{O}_Y)\otimes \operatorname{End}(\mathbb{C}^{m_i})$ and only its trace survives, so $m_i = 1$, giving (i). What is left is $\bigoplus_i H^2(\mathcal{O}_Y)$, and by (1) an element (ζ_i) of it dies exactly when $\sum_i \lambda_i^q \zeta_i = 0$ for every $q \geq 0$, which is (iii) after absorbing the signs $(-1)^{p_i}$ into the ζ_i and the factorials into the grading. ∎

Condition (ii) is cheap, because the shifts p_i are at our disposal: only their parities are fixed by the Chern character, and a line bundle whose first Chern class is nondegenerate of index k has cohomology in degree k alone, so one may choose the p_i to avoid every index that occurs. Condition (iii) is a rank condition, and condition (i) is free. What is not free is the Chern character itself, and that is the subject of the rest of this subsection.

Notation 4.2 (Coordinates on the divisor classes). Let $A_0 = X \times \widehat{X}$ be a split member, $n \geq 2$, and keep $\gamma = d\beta - \widehat{\beta}$ from [I, Notation 15.1]. Write

$$\theta^{\pm} = \gamma \mp \sqrt{-d}\,\ell \in \textstyle\bigwedge^2 V_{\pm}, \qquad R_0 = \mathbb{Q}\gamma \oplus \mathbb{Q}\ell \subset R.$$

Every $\theta \in R_0$ is $\theta = u\theta^+ + \overline{u}\,\theta^-$ for a unique $u \in K$, and in coordinates $u = g + h\sqrt{-d}$ corresponds to $\theta = 2g\gamma + 2dh\,\ell$. By [I, Proposition 15.19(iv)] the group $\mathrm{NS}(A_0)_{\mathbb{Q}}$ at a split member with X general is $\mathbb{Q}\eta \oplus R_0$, so every divisor class λ is $e\eta + \theta$ with $e \in \mathbb{Q}$ and $\theta \in R_0$, and we write $u(\lambda) \in K$ for the coordinate of its R_0-part.

Lemma 4.3 (A basis in degree four)**.** *For $n \geq 2$ the six products $\beta^2, \widehat{\beta}^2, \ell^2, \beta\widehat{\beta}, \beta\ell, \widehat{\beta}\ell$ are linearly independent in $H^4(A_0,\mathbb{Q})$. Equivalently the six classes η^2, $\eta\theta^+$, $\eta\theta^-$, $(\theta^+)^2$, $(\theta^-)^2$, $\theta^+\theta^-$ are a basis of the degree four part of the subring generated by the divisor classes.*

Proof. Grade $H^*(A_0,\mathbb{Q}) = \bigwedge^* U \otimes \bigwedge^* U^\vee$ by bidegree. Then β has bidegree $(2,0)$, $\widehat{\beta}$ has $(0,2)$ and ℓ has $(1,1)$, so the six products have bidegrees $(4,0)$, $(0,4)$, $(2,2)$, $(2,2)$, $(3,1)$, $(1,3)$. All are distinct except for the pair $\beta\widehat{\beta}$ and ℓ^2, and those two are independent in $\bigwedge^2 U \otimes \bigwedge^2 U^\vee$ because $\beta\widehat{\beta} = \beta \otimes \widehat{\beta}$ is decomposable there while $\ell^2 = -2\sum_{i<j}(x_i x_j)\otimes(\xi_i\xi_j)$ is diagonal in the induced bases, of tensor rank $\binom{2n}{2} > 1$. The second description follows because the change of basis between the two sets of six classes is invertible over K: $\eta = d\beta + \widehat{\beta}$ and $\gamma = d\beta - \widehat{\beta}$ span the same plane as β and $\widehat{\beta}$ since $\det\left(\begin{smallmatrix} d & 1 \\ d & -1 \end{smallmatrix}\right) = -2d \neq 0$, and $\theta^{\pm}$ span the same plane as γ and ℓ over $\mathbb{C}$. ∎

The next lemma says the same thing at every abelian variety of Weil type, not only at a split member, and it says it without coordinates. It is the fact on which the positivity argument turns.

Lemma 4.4 ($\theta^+\theta^-$ is never proportional to η^2)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $0 \neq \theta^{\pm} \in \bigwedge^2 V_{\pm}$. Then $\theta^+\theta^-$ is not a scalar multiple of η^2 in $H^4(A,\mathbb{C})$.*

Proof. Both classes lie in the summand $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$ of $H^4(A,\mathbb{C})$, the isotypic component for the character N^2, so the question is one of tensor rank there. The polarisation class η lies in $V_+ \otimes V_-$ and is the hermitian form of [I, Proposition 3.7], hence nondegenerate: in dual bases $\eta = \sum_{a=1}^{2n} v_a \wedge w_a$ with v_a a basis of V_+ and w_a a basis of V_-. Squaring,

$$\eta^2 = \sum_{a,b}(v_a \wedge w_a)(v_b \wedge w_b) = -2\sum_{a<b}(v_a v_b)\otimes(w_a w_b),$$

which is diagonal in the induced bases of $\bigwedge^2 V_+$ and $\bigwedge^2 V_-$ and therefore has tensor rank exactly $\binom{2n}{2}$. The class $\theta^+\theta^-$ is, up to sign, $\theta^+ \otimes \theta^-$, of tensor rank one. For $n \geq 2$ one has $\binom{2n}{2} \geq 6 > 1$, so the two cannot be proportional. ∎

Theorem 4.5 (The positivity obstruction)**.** *Let (A,ι,η) be of $(K,-1,n)$-Weil type with $n \geq 2$, and suppose that the only divisor classes on which $K^\times$ acts through the norm character are the multiples of η, that is*

$$\mathrm{NS}(A)_{\mathbb{Q}} \cap (V_+ \otimes V_-) = \mathbb{Q}\,\eta. \tag{2}$$

Let E be a perfect complex on A whose Chern character is effective in the exponential basis, meaning

$$\mathrm{ch}(E) = \sum_{i=1}^{s} m_i\, e^{\lambda_i}, \qquad m_i \in \mathbb{Q}_{>0}, \quad \lambda_i \in \mathrm{NS}(A)_{\mathbb{Q}}, \tag{3}$$

and write $\lambda_i = e_i\eta + \theta_i$ with θ_i in the part R of $H^2(A,\mathbb{Q})$ carrying the characters τ^2 and $\overline{\tau}^2$, and $u_i \in K$ for the θ^+-coordinate of θ_i. If $\mathrm{ch}_2(E)$ is a flat Hodge class, then

$$\sum_{i=1}^{s} m_i\, \mathrm{N}_{K/\mathbb{Q}}(u_i) = 0, \tag{4}$$

and since $\mathrm{N}_{K/\mathbb{Q}}$ is positive definite and every m_i is positive, $u_i = 0$ for every i. Hence every λ_i is a rational multiple of η, and $\mathrm{ch}_k(E) \in \mathbb{Q}\,\eta^k$ for every k. In particular $\mathrm{ch}_n(E)$ has no component on the Weil line.

Proof. From (3), $\mathrm{ch}_2(E) = \frac{1}{2}\sum_i m_i\lambda_i^2$, and hypothesis (2) is what makes the decomposition $\lambda_i = e_i\eta + \theta_i$ available with e_i a scalar. Decompose $H^4(A,\mathbb{C})$ into $K^\times$-isotypic pieces and

keep the one for N^2, namely $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$. Writing $\theta_i = u_i\theta^+ + \overline{u}_i\theta^-$ for a fixed choice of generators $\theta^\pm$ of $\bigwedge^2 V_\pm$, one has

$$\lambda_i^2 = e_i^2\eta^2 + 2e_i\eta\theta_i + u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}_{K/\mathbb{Q}}(u_i)\,\theta^+\theta^-,$$

and of these five terms only the first and the last are in the N^2 piece: $\eta\theta_i$ carries $\mathrm{N}\tau^2$ and $\mathrm{N}\overline{\tau}^2$, and $(\theta^\pm)^2$ carry τ^4 and $\overline{\tau}^4$, all four distinct from N^2. So the N^2-part of $\mathrm{ch}_2(E)$ is

$$\tfrac{1}{2}\Big(\sum_i m_i e_i^2\Big)\eta^2 + \Big(\sum_i m_i \mathrm{N}_{K/\mathbb{Q}}(u_i)\Big)\theta^+\theta^-.$$

Now compare with the flat Hodge classes of degree four. For $n \geq 3$ they are $\mathbb{Q}\eta^2$ by [I, Theorem 4.14]. For $n = 2$ they are $\mathbb{Q}\eta^2 \oplus W(A)$, and $W(A)$ spans the summands for τ^4 and $\overline{\tau}^4$, which are not N^2. In both cases the N^2-part of a flat Hodge class of degree four is a multiple of η^2. By Lemma 4.4 the class $\theta^+\theta^-$ is not such a multiple, so its coefficient must vanish, which is (4).

Finally $\mathrm{N}_{K/\mathbb{Q}}(g + h\sqrt{-d}) = g^2 + dh^2$ with $d > 0$, so $\mathrm{N}_{K/\mathbb{Q}}(u) \geq 0$ with equality only at $u = 0$, and a sum of nonnegative terms with positive weights vanishes only when every term does. Then $\mathrm{ch}_k(E) = \frac{1}{k!}(\sum_i m_i e_i^k)\eta^k$, and η^k is not on the Weil line by [I, Theorem 4.14]. ∎

Corollary 4.6 (Objects filtered by line bundles)**.** *Let A satisfy* (2)*, which holds at every split member of* [I, Theorem 15.2] *and at every quaternionic base point of* [I, Theorem 15.25]*. Then Theorem 2.3 cannot be applied at A with any of the following:*

(a) *a coherent sheaf admitting a filtration by line bundles, in particular any direct sum of line bundles;*
(b) *a simple semi-homogeneous vector bundle in the sense of Mukai* [Muk78]*, since such a bundle of rank r has $\mathrm{ch}(E) = r\,e^{c_1(E)/r}$;*
(c) *any iterated extension of the objects in (a) and (b).*

In each case the failure is not of the semiregularity hypothesis but of the Chern character hypothesis of Proposition 2.1: no such E has a Chern character which is a flat section of Hodge classes with a nonzero Weil component.

Proof. First, (2) holds at the points named. At a split member $\mathrm{NS}(A)_\mathbb{Q}$ has rank three with basis $\beta, \widehat{\beta}, \ell$ by [I, Proposition 15.19(iv)], and of these η spans the norm-character part while γ and ℓ span R. At a quaternionic point $\mathrm{NS}(A)_\mathbb{Q}$ is the space of Rosati-symmetric elements of $B = (-d, b)_\mathbb{Q}$, which is spanned by 1, j and ij; the first gives η and the other two are semilinear over K, hence lie in R. In both cases the norm-character part of $\mathrm{NS}(A)_\mathbb{Q}$ is exactly $\mathbb{Q}\eta$.

Each of (a), (b) and (c) has a Chern character of the shape (3) with positive multiplicities, by additivity of ch along a filtration and by Mukai's computation of the Chern character of a simple semi-homogeneous bundle. The conclusion of Theorem 4.5 then contradicts the requirement that $\mathrm{ch}_n(E)$ have a nonzero component on the Weil line. ∎

Remark 4.7 (Scope of the theorem, and the objects of Markman). At a very general member of the family $\mathrm{NS}(A)_\mathbb{Q} = \mathbb{Q}\eta$, so Theorem 4.5 is true for the empty reason that there are no θ_i to begin with. Its content is at the special members, which is exactly where Theorem 2.3 needs its object to live and where the base points of [I, Theorem 15.27] are. That is the opposite of the usual situation in this paper, where the special members are the easy ones.

The theorem does not touch the objects of Markman [Mar25a, Mar25b]. Those are sheaves whose class in K_0 is not a positive combination of exponentials of divisor classes, so (3) fails for them at the first step; and they live at very general points, where, as just

noted, the statement is vacuous. What the theorem does is remove the class of objects one would reach for first, and it removes it for a reason which is visible before any cohomology is computed.

Remark 4.8 (Signed sums of line bundles). Theorem 4.5 uses positivity of the multiplicities and nothing else, so it says exactly where to go next: the object must be a complex whose class in K_0 is not effective, that is one in which the line bundles appear with multiplicities of both signs. Such objects exist. Writing $E = F \oplus G[1]$ with F and G sums of line bundles, the class in K_0 is $[F]-[G]$ and (4) becomes $\sum_{L_i \subset F} \mathrm{N}(u_i) = \sum_{L_j \subset G} \mathrm{N}(u_j)$, which is satisfiable; and Lemma 4.1 still applies verbatim, with the shifts absorbing condition (ii).

The object of Theorem 2.3, if it exists at a split member, is not a sum of line bundles in any arrangement of signs and shifts: Corollary 4.21 below proves this in every dimension. It must have Ext^2 that the trace does not see, that is, it must be indecomposable in the derived category. This description of the remaining construction is sharper than the one in Remark 3.7.

4.2. **An explicit object and the Prouhet–Tarry–Escott problem.** Proposition 2.1 produces the object of Theorem 2.3 by an existence argument: the Chern character is an isomorphism onto the Chow ring with rational coefficients, so a K-theory class with the right character exists. That is enough for the theorem and tells one nothing about the object. This subsection writes it down. The answer turns the question into a Diophantine one, and the Diophantine problem has a classical name.

Setup 4.9 (Objects supported on R)**.** Let A be of $(K,-1,n)$-Weil type, $n \geq 2$, satisfying (2), and keep $\theta^{\pm}$ from Notation 4.2. For $u \in K$ with $\theta(u) = u\theta^+ + \overline{u}\,\theta^- \in \mathrm{NS}(A)_{\mathbb{Q}}$, and for multiplicities $m_i \in \mathbb{Z}$, consider

$$v = \sum_{i=1}^{s} m_i\, e^{\theta(u_i)}, \qquad T_{q,r} = \sum_{i=1}^{s} m_i\, u_i^q\, \overline{u}_i^r \quad (q,r \geq 0).$$

Proposition 4.10 (The Chern character in terms of the u_i)**.** *In Setup 4.9,*

$$v_k = \sum_{\substack{q+r=k \\ 0 \leq q,r \leq n}} \frac{T_{q,r}}{q!\,r!}\,(\theta^+)^q(\theta^-)^r, \tag{5}$$

because $(\theta^{\pm})^q = 0$ for $q > n$. Consequently v is a flat section of Hodge classes with a nonzero Weil component if

$$T_{q,r} = 0 \ \text{ for all } 0 \leq q,r \leq n \text{ with } (q,r) \notin \{(0,0),(n,0),(0,n)\}, \qquad T_{n,0} \neq 0, \tag{6}$$

and then $v_0 = T_{0,0} = \sum_i m_i$, $v_k = 0$ for $0 < k < 2n$ with $k \neq n$, and

$$v_n = \frac{1}{n!}\Big(T_{n,0}(\theta^+)^n + \overline{T}_{n,0}(\theta^-)^n\Big),$$

a nonzero rational multiple of a generator of the Weil line, by [I, Theorem 15.2].

Proof. Expand $\theta(u_i)^k$ by the binomial theorem, divide by $k!$ and collect; the vanishing of $(\theta^{\pm})^q$ for $q > n$ is because $\theta^{\pm} \in \bigwedge^2 V_{\pm}$ and $\dim_{\mathbb{C}} V_{\pm} = 2n$. Under (6) the only surviving terms of (5) are the ones listed, and $(\theta^+)^n$ spans $\bigwedge^{2n} V_+$, which is the Weil line over $\mathbb{C}$. ∎

So the search for the object of Theorem 2.3 is the search for a finitely supported integer-valued function on $\mathcal{O}_K$ that annihilates every monomial $u^q\overline{u}^{\,r}$ with $q,r \leq n$ except three of them. Two things follow at once, and they pull in opposite directions.

Lemma 4.11 (Equal norms force the Prouhet–Tarry–Escott condition)**.** *Suppose all the u_i in Setup 4.9 have the same norm $N = \mathrm{N}_{K/\mathbb{Q}}(u_i) > 0$. Then (6) holds if and only if*

$$\sum_i m_i u_i^k = 0 \quad \text{for } k = 0, 1, \dots, n-1, \qquad \sum_i m_i u_i^n \neq 0. \tag{7}$$

With $m_i = \pm 1$ this says that the u_i split into two multisets on the circle $\mathrm{N} = N$ whose power sums agree through degree $n-1$ and differ at degree n: a Prouhet–Tarry–Escott pair of degree $n-1$, constrained to a norm circle of $\mathcal{O}_K$.

Proof. For $q \geq r$ one has $u^q \overline{u}^r = N^r u^{q-r}$, so $T_{q,r} = N^r \sum_i m_i u_i^{q-r}$, and the pairs (q, r) with $q, r \leq n$ realise every difference $k = q - r$ from 0 to n. The excluded pairs $(0, 0)$ and $(n, 0)$ are the ones with $r = 0$ and $k = 0$ or $k = n$; but $k = 0$ also occurs at $(1, 1), \dots, (n, n)$, which are not excluded, so $\sum_i m_i = 0$ is still required. The remaining differences $k = 1, \dots, n-1$ each occur at some non-excluded pair, and $k = n$ occurs only at $(n, 0)$. ∎

Theorem 4.12 (Equal norms cannot be semiregular)**.** *Let $E = \bigoplus_i L_i[p_i]$ realise (7) with all norms equal. Then $\mathrm{ch}_0(E) = 0$, and the semiregularity map σ_E has a kernel of dimension at least n^2.*

Proof. By Lemma 4.11 the case $k = 0$ of (7) gives $\sum_i m_i = 0$, which is $\mathrm{ch}_0(E)$. For the kernel, take $\zeta_i = m_i \zeta$ for a single $\zeta \in H^2(\mathcal{O}_A)$; then by Lemma 4.1(iii)

$$\Phi\big((m_i\zeta)_i\big) = \Big(\sum_i m_i e^{\lambda_i}\Big)\zeta = \mathrm{ch}(E)\,\zeta,$$

which by Proposition 4.10 is a multiple of $\omega\,\zeta$ with ω a generator of the Weil line. Now $W(A)_{\mathbb{C}}$ is spanned by $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, and the Weil type condition says $\dim V_\pm^{0,1} = n$; writing $H^2(\mathcal{O}_A) = \bigwedge^2(V_+^{0,1} \oplus V_-^{0,1})$, the product of a generator of $\bigwedge^{2n} V_+$ with ζ kills the components of ζ in $\bigwedge^2 V_+^{0,1}$ and in $V_+^{0,1} \otimes V_-^{0,1}$, and likewise with the roles exchanged. So $\omega\,\zeta = 0$ for every ζ in $V_+^{0,1} \otimes V_-^{0,1}$, a space of dimension n^2, and all of those lie in the kernel of Φ. ∎

Corollary 4.13 (At least $n+1$ norms)**.** *If E as in Setup 4.9 satisfies (6) and $\mathrm{ch}_0(E) \neq 0$, then the u_i take at least $n+1$ distinct norms.*

Proof. Let $N_1, \dots, N_\nu$ be the distinct norms and M_j the sum of the m_i over the level $\mathrm{N}(u_i) = N_j$. The conditions $T_{r,r} = 0$ for $r = 1, \dots, n$ read $\sum_j M_j N_j^r = 0$ for $r = 1, \dots, n$. The matrix $(N_j^r)_{1 \leq r \leq n,\, 1 \leq j \leq \nu}$ has rank $\min(n, \nu)$ because the N_j are distinct and nonzero, so $\nu \leq n$ forces every $M_j = 0$ and hence $\mathrm{ch}_0(E) = \sum_j M_j = 0$. ∎

The smallest solution. At $n = 3$ and $K = \mathbb{Q}(\sqrt{-3})$ the Prouhet–Tarry–Escott condition of Lemma 4.11 has a particularly simple solution: the norm circle $\mathrm{N} = 1$ carries the six roots of unity of K, and the two cosets of the cube roots of -1 and of $+1$ are a Prouhet–Tarry–Escott pair of degree two. Explicitly, with ζ_6 a primitive sixth root of unity, the cube roots of -1 are $\zeta_6, \zeta_6^3, \zeta_6^5$ and those of $+1$ are $1, \zeta_6^2, \zeta_6^4$; each triple sums to zero and has zero sum of squares, while the sums of cubes are -3 and $+3$.

Proposition 4.14 (An explicit object at $n = 3$)**.** *Let $K = \mathbb{Q}(\sqrt{-3})$, $n = 3$, and let $A_0 = X \times \widehat{X}$ be a split member. In the basis $\beta, \widehat{\beta}, \ell$ of $\mathrm{NS}(A_0)_{\mathbb{Q}}$ put*

$$\lambda_1 = (3, -1, 3), \quad \lambda_2 = (-6, 2, 0), \quad \lambda_3 = (3, -1, -3),$$

and $\lambda_{3+i} = -\lambda_i$ for $i = 1, 2, 3$, with $m_i = +1$ for $i \le 3$ and $m_i = -1$ for $i \ge 4$. Then the six classes are distinct and integral, the perfect complex

$$E = (L_1 \oplus L_2 \oplus L_3) \ \oplus \ (L_4 \oplus L_5 \oplus L_6)[3]$$

has

$$\mathrm{ch}_k(E) = 0 \quad (k \neq 3), \qquad \mathrm{ch}_3(E) = 12\,\omega_1,$$

and every difference $\lambda_j - \lambda_i$ with $i \neq j$ is nondegenerate of index exactly 3, so that conditions (i) and (ii) of Lemma 4.1 hold for this shift. Condition (iii) fails: the map Φ has rank 75 instead of 90, with kernel of dimension $15 = \dim H^2(\mathcal{O}_{A_0})$.

Proof. With $d = 3$, the class $\theta(u) = 2g\gamma + 6h\ell$ of $u = g + h\sqrt{-3}$ has coordinates $(6g, -2g, 6h)$, so $\lambda_1, \lambda_2, \lambda_3$ are the images of the cube roots $\zeta_6, \zeta_6^3, \zeta_6^5$ of -1 and $\lambda_4, \lambda_5, \lambda_6$ those of the cube roots of $+1$; the six classes are distinct and integral. All six have norm 1, and $\sum_i m_i u_i^k$ is 0 for $k = 0, 1, 2$ and $-3 - 3 = -6$ for $k = 3$, so (7) holds at $n = 3$. By Lemma 4.11 and Proposition 4.10, $\mathrm{ch}_k(E) = 0$ for $k \neq 3$ and $\mathrm{ch}_3(E) = -((\theta^+)^3 + (\theta^-)^3)$, and by [I, Example 15.3] at $n = 3$, $\omega_1 \pm \sqrt{-3}\,\omega_2 = -\frac{1}{6}(\theta^{\mp})^3$; so $\mathrm{ch}_3(E) = 12\,\omega_1$.

Every difference is $\lambda_j - \lambda_i = \theta(w)$ with $w = u_j - u_i \neq 0$, and $\theta(w)$ is nondegenerate of index n for every $w \in K^\times$. Indeed $\theta(1) = 2\gamma$ and $\theta(\sqrt{-d}) = 2d\ell$ are of type $(1,1)$, hence so are $\theta^{\pm}$, so $\theta^+ \in V_+^{1,0} \otimes V_+^{0,1}$, and $V_+^{0,1}$ is the conjugate of $V_-^{1,0}$. In the splitting of the tangent space dual to $H^{1,0} = V_+^{1,0} \oplus V_-^{1,0}$, the hermitian form of $\theta(w)$ therefore has the block form $\left(\begin{smallmatrix} 0 & wP \\ \bar{w}P^* & 0 \end{smallmatrix}\right)$, with P the block of 2γ, which is invertible because γ is nondegenerate. Its eigenvalues are plus and minus the singular values of wP, so it has signature (n, n), and a line bundle with this class has cohomology only in degree n [BL04, Chapter 3]. For the shift, condition (ii) asks that $2 + p_i - p_j \neq 3$ whenever $i \neq j$, that is $p_i - p_j \neq 1$; within each of the two blocks the difference is 0, and between them it is ± 3.

For the rank, the u_i are the six sixth roots of unity, each once, and $\overline{u}_i = u_i^{-1}$. Absorb the signs into the ζ_i as in Lemma 4.1(iii) and expand $e^{\lambda_i} = e^{u_i\theta^+} e^{\overline{u}_i\theta^-}$:

$$\Phi(\zeta) = \sum_{a,b=0}^{3} \frac{(\theta^+)^a (\theta^-)^b}{a!\,b!}\, Z_{a-b}, \qquad Z_k = \sum_i u_i^k \zeta_i,$$

with k read modulo 6. The $Z_0, \dots, Z_5$ determine the ζ_i, by a Vandermonde matrix. Split $H^{0,2}(A_0) = \bigwedge^2 V_+^{0,1} \oplus (V_+^{0,1} \otimes V_-^{0,1}) \oplus \bigwedge^2 V_-^{0,1}$, write $Z_k = Z_k^+ + Z_k^0 + Z_k^-$ accordingly, and grade $H^*(A_0) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$ by the Hodge types of the two factors. The term $(\theta^+)^a(\theta^-)^b Z_k^0$ has types $(a, a+1)$ and $(b, b+1)$, the term with Z_k^+ has types $(a, a+2)$ and (b, b), and the term with Z_k^- has types (a, a) and $(b, b+2)$. These are different for different (a, b) and different pieces, so every term vanishes separately. Since $\theta^{\pm}$ pairs $V_{\pm}^{1,0}$ nondegenerately with $V_{\pm}^{0,1}$, both of dimension 3, multiplication by $(\theta^{\pm})^a$ is injective on $\bigwedge^j V_{\pm}^{0,1}$ when $a + j \le 3$ and zero otherwise; in bases with $\theta^+ = \sum_i e_i f_i$ it sends f_J to a signed sum of the $e_I f_{I \cup J}$ with $I \cap J = \emptyset$, $|I| = a$, and different J give disjoint supports. Hence $Z_k^0 = 0$ unless no $a, b \le 2$ have $a - b \equiv k$, which leaves $k = 3$; $Z_k^+ = 0$ unless no $a \le 1$, $b \le 3$ have $a - b \equiv k$, which leaves $k = 2$; and likewise $Z_k^- = 0$ unless $k = 4$. So $\ker \Phi$ is $V_+^{0,1} \otimes V_-^{0,1}$ at $k = 3$ together with $\bigwedge^2 V_+^{0,1}$ at $k = 2$ and $\bigwedge^2 V_-^{0,1}$ at $k = 4$, of dimension $9 + 3 + 3 = 15 = \dim H^2(\mathcal{O}_{A_0})$, and Φ has rank $90 - 15 = 75$. The first summand is the kernel of Theorem 4.12. ∎

Remark 4.15 (What the construction settles). Three things are settled. The object of Proposition 2.1 is not an abstraction: at $n = 3$ over $\mathbb{Q}(\sqrt{-3})$ it is six line bundles and a

shift, and its Chern character is exactly 12 times a Weil class with every other graded piece zero. The construction is completely general, and Lemma 4.11 identifies the constraint as a Prouhet–Tarry–Escott problem on a norm circle, a question of a kind that can be attacked directly. And Corollary 4.13 says what any semiregular solution must look like: the u_i must take at least $n+1$ distinct norms, because otherwise the rank is zero and the argument of Theorem 4.12 produces a kernel of dimension n^2.

This leaves the existence of a solution with $n+1$ norms. The level sums are then forced up to scale (Proposition 4.20), and Corollary 4.21 shows that no solution could serve, because a line bundle summand off the polarisation line is obstructed along the family whatever the norms.

The obstruction classes of the explicit object. [I, Corollary 9.18] and [I, Remark 9.21] say that the kernel of σ_E contains a copy of the tangent space of the Weil family, and that this is a statement about where the semiregularity map is blind, not a statement that the object is obstructed. For the explicit object the obstruction classes can be written down, and they are not zero. The computation is elementary and we record it in general.

Recall the conventions of [I, Section 9.3]: a first-order deformation of A is $v \in \operatorname{Hom}(H^{1,0}, H^{0,1}) = H^1(A, T_A)$, extended by zero on $H^{0,1}$ and acting on $H^*(A,\mathbb{C})$ as a derivation $v\lrcorner$; a class ξ of type (p,p) stays of type (p,p) to first order along v exactly when $v\lrcorner\xi = 0$. For a line bundle L with $\lambda = c_1(L)$ the Atiyah class is $\lambda \in H^1(\Omega^1_A)$, its contraction with v is $v\lrcorner\lambda \in H^2(\mathcal{O}_A) = H^{0,2}(A)$, and this is the obstruction to deforming L along v; a shift does not change it. For $E = \bigoplus_i L_i[p_i]$ the Atiyah class is diagonal with entries λ_i, as in the proof of Lemma 4.1, so the obstruction to deforming E along v is the diagonal element

$$\operatorname{ob}_E(v) = \big(v\lrcorner\lambda_i\big)_i \in \bigoplus_i H^{0,2}(A) \subset \operatorname{Ext}^2(E,E). \tag{8}$$

Proposition 4.16 (The obstruction map of a split object)**.** *Let $A_0 = X \times \widehat{X}$ be a split member with β principal and X general, so that $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \widehat{\beta}, \ell\rangle$ by* [I, Proposition 15.19(iv)]*. Let $T = T_{[A_0]}\mathcal{D}_{n,n}$ be the space of polarised first-order deformations commuting with K, of dimension n^2 by* [I, Proposition 9.14]*, and put*

$$T_{\mathrm{spl}} = \big\{\, v \in \operatorname{Hom}(H^{1,0}, H^{0,1}) \;:\; v\lrcorner\beta = v\lrcorner\widehat{\beta} = v\lrcorner\ell = 0 \,\big\}.$$

Let $E = \bigoplus_{i=1}^s L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and let $\operatorname{ob}_E\colon T \to \bigoplus_i H^{0,2}(A_0)$ be the map (8) *restricted to T.*

(i) *T_{spl} is the tangent space to the split locus: it has dimension $n(n+1)/2$, it is contained in T, and it consists of the deformations of $X \times \widehat{X}$ induced by the polarised deformations of (X,β).*

(ii) *If the classes λ_i together with η span $\mathrm{NS}(A_0)_{\mathbb{Q}}$, then $\ker \operatorname{ob}_E = T_{\mathrm{spl}}$ and $\operatorname{rank} \operatorname{ob}_E = n(n-1)/2$, the codimension of the split locus in $\mathcal{D}_{n,n}$.*

(iii) *If $\operatorname{ch}_k(E) \in \mathbb{Q}\eta^k$ for $k \neq n$ and $\operatorname{ch}_n(E) \in \mathbb{Q}\eta^n \oplus W(A_0)$, then $\sigma_E \circ \operatorname{ob}_E = 0$ on T: for every $v \in T$,*

$$\sigma_E\big(\operatorname{ob}_E(v)\big) = v\lrcorner \operatorname{ch}(E) = 0. \tag{9}$$

(iv) *For the object of Proposition 4.14, at $n = 3$ and $K = \mathbb{Q}(\sqrt{-3})$: $\ker \operatorname{ob}_E = T_{\mathrm{spl}}$ has dimension 6 and ob_E has rank 3; the image of ob_E lies in $\ker \sigma_E$; the subspace $\Delta = \{z \cdot \mathrm{id}_E : z \in \operatorname{Ann}_{H^{0,2}}(\omega_1)\}$ of* [I, Corollary 9.18] *has dimension 9 and meets the image of ob_E only in zero, so that $\Delta \oplus \operatorname{im} \operatorname{ob}_E$ is a twelve-dimensional subspace of the fifteen-dimensional kernel of σ_E; and $\operatorname{ob}_E(v)$ has components $v\lrcorner\lambda_i$ on L_i*

and $-v\lrcorner\,\lambda_i$ on $L_i^{-1}[3]$, so it lies on the line of vectors of the shape $(z,\dots,z)$ only when it vanishes. In particular E has a nonzero first-order obstruction along every $v \in T$ not tangent to the split locus, and no element $z\cdot\mathrm{id}_E$ of $\mathrm{Ext}^2(E,E)$ cancels it.

Proof. (i) Let $P = H^{1,0}(X) \subset H^1(X,\mathbb{C}) = U_{\mathbb{C}}^\vee$ with basis $p_1,\dots,p_n$, so that $p_1,\dots,p_n,\overline{p}_1,\dots,\overline{p}_n$ is a basis of $U_{\mathbb{C}}^\vee$, and let $q_a,\overline{q}_a$ be the dual basis of $U_{\mathbb{C}} = H^1(\widehat{X},\mathbb{C})$; the dual of a conjugate basis is the conjugate of the dual basis, so the notation is consistent. The Poincaré class is the identity element of $U^\vee\otimes U$,

$$\ell = \sum_{a=1}^{n}\bigl(p_a\wedge q_a + \overline{p}_a\wedge\overline{q}_a\bigr),$$

and it is of type $(1,1)$, which forces $H^{1,0}(\widehat{X}) = \langle\overline{q}_a\rangle$ and $H^{0,1}(\widehat{X}) = \langle q_a\rangle$: the Hodge structure of $\widehat{X}$ is the dual one. Since β is of type $(1,1)$ and nondegenerate, $\beta = \sum_{a,b} h_{ab}\,p_a\wedge\overline{p}_b$ with $h\in\mathrm{GL}_n(\mathbb{C})$; and since β is principal with Gram matrix B in a symplectic basis of U and $\widehat{\beta}$ has Gram matrix $-B^{-1} = B$ in the dual basis of $U^\vee$, the Gram matrix of $\widehat{\beta}$ in the basis $p_a,\overline{p}_a$ is minus the inverse of the Gram matrix $\left(\begin{smallmatrix}0 & h\\ -h^\top & 0\end{smallmatrix}\right)$ of β in the basis $q_a,\overline{q}_a$, which gives $\widehat{\beta} = -\sum_{a,b}(h^{-1})_{ab}\,\overline{q}_a\wedge q_b$.

Now $H^{1,0}(A_0) = \langle p_a\rangle\oplus\langle\overline{q}_a\rangle$ and $H^{0,1}(A_0) = \langle\overline{p}_a\rangle\oplus\langle q_a\rangle$, so a deformation v is four $n\times n$ matrices,

$$v(p_a) = \sum_c (A_{ca}\,\overline{p}_c + C_{ca}\,q_c), \qquad v(\overline{q}_a) = \sum_c (B_{ca}\,\overline{p}_c + D_{ca}\,q_c),$$

and v kills $\overline{p}_a$ and q_a. Contracting,

$$v\lrcorner\,\ell = \sum_{a,c}\Bigl(A_{ca}\,\overline{p}_c\wedge q_a + C_{ca}\,q_c\wedge q_a + B_{ca}\,\overline{p}_a\wedge\overline{p}_c + D_{ca}\,\overline{p}_a\wedge q_c\Bigr),$$

and reading off the three kinds of monomials, $v\lrcorner\,\ell = 0$ if and only if $D = -A^\top$ and B, C are symmetric. Next

$$v\lrcorner\,\beta = \sum_{a,b,c} h_{ab}\Bigl(A_{ca}\,\overline{p}_c\wedge\overline{p}_b + C_{ca}\,q_c\wedge\overline{p}_b\Bigr),$$

so $v\lrcorner\,\beta = 0$ if and only if $Ch = 0$, that is $C = 0$, and Ah is symmetric. Likewise $v\lrcorner\,\widehat{\beta} = 0$ if and only if $B = 0$ and Dh^{-1} is symmetric. Given $D = -A^\top$, the last condition reads $A^\top h^{-1} = h^{-\top}A$, which after multiplication by $h^\top$ on the left and by h on the right is $h^\top A^\top = Ah$, the symmetry of Ah again. Hence

$$T_{\mathrm{spl}} = \bigl\{(A,B,C,D) = (A,0,0,-A^\top)\ :\ Ah \text{ symmetric}\bigr\},$$

of dimension $n(n+1)/2$. It is the tangent space to the split locus: the same computation on X alone shows that the polarised first-order deformations of (X,β) are the matrices A with Ah symmetric, the space $T_{[X]}\mathcal{A}_n$ of dimension $n(n+1)/2$; the deformation of $\widehat{X}$ induced by A is the dual one, $-A^\top$; and the map $A\mapsto(A,0,0,-A^\top)$ is injective. Finally $T_{\mathrm{spl}}\subseteq T$: for $v\in T_{\mathrm{spl}}$, $v\lrcorner\,\eta = 0$ because $\eta\in\langle\beta,\widehat{\beta}\rangle$, so v is polarised, and $v\lrcorner\,\omega_1 = v\lrcorner\,\omega_2 = 0$ because ω_1 and ω_2 are polynomials in $\beta,\widehat{\beta},\ell$ by [I, Theorem 15.2] and $v\lrcorner$ is a derivation; by [I, Proposition 9.14] such a v commutes with K.

(ii) Every $v\in T$ has $v\lrcorner\,\eta = 0$. If the λ_i and η span $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle\beta,\widehat{\beta},\ell\rangle$, then $v\lrcorner\,\lambda_i = 0$ for all i is equivalent, for $v\in T$, to $v\lrcorner\,\beta = v\lrcorner\,\widehat{\beta} = v\lrcorner\,\ell = 0$, so $\ker\mathrm{ob}_E = T\cap T_{\mathrm{spl}} = T_{\mathrm{spl}}$ by (i), and the rank is $n^2 - n(n+1)/2 = n(n-1)/2$, which is the codimension of [I, Proposition 15.19(iii)].

(iii) By (1), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_{ii}\, e^{\lambda_i}$ for a diagonal ζ. Since $v \lrcorner$ is a derivation and $v \lrcorner \lambda_i$ has even degree, $v \lrcorner \lambda_i^q = q\, \lambda_i^{q-1} (v \lrcorner \lambda_i)$ and so $v \lrcorner e^{\lambda_i} = (v \lrcorner \lambda_i)\, e^{\lambda_i}$. Therefore

$$\sigma_E(\mathrm{ob}_E(v)) = \sum_i (-1)^{p_i} (v \lrcorner \lambda_i)\, e^{\lambda_i} = v \lrcorner \sum_i (-1)^{p_i} e^{\lambda_i} = v \lrcorner \mathrm{ch}(E).$$

For $v \in T$ one has $v \lrcorner \eta^k = k \eta^{k-1} (v \lrcorner \eta) = 0$, and $v \lrcorner \omega = 0$ for every $\omega \in W(A_0)$ by [I, Proposition 9.14]; under the hypothesis on $\mathrm{ch}(E)$ this gives (9).

(iv) The three classes $\lambda_1 = \gamma + 3\ell$, $\lambda_2 = -2\gamma$, $\lambda_3 = \gamma - 3\ell$ span $\langle \gamma, \ell \rangle$, and $\langle \eta, \gamma \rangle = \langle \beta, \widehat{\beta} \rangle$ by the determinant in the proof of Lemma 4.3, so (ii) applies and gives the kernel and the rank; Proposition 4.14 gives the hypothesis of (iii), so the image lies in $\ker \sigma_E$. The subspace Δ has dimension $\dim \mathrm{Ann}_{H^{0,2}}(\omega_1) = n^2 = 9$ by [I, Lemma 9.6]. The components of $\mathrm{ob}_E(v)$ on L_i and on $L_i^{-1}[3]$ are $v \lrcorner \lambda_i$ and $v \lrcorner (-\lambda_i) = -v \lrcorner \lambda_i$, so an element of the image of the shape $(z, \dots, z)$ has $z = -z$, hence is zero; this is $\Delta \cap \mathrm{im}\, \mathrm{ob}_E = 0$, and with $\dim \ker \sigma_E = 15$ from Proposition 4.14 the dimension count follows. ∎

Remark 4.17 (Meaning of the obstruction classes). Proposition 4.16 sharpens [I, Remark 9.21] in one respect and exposes a limit of the criterion in another. First, for the explicit object the obstruction classes along the family are nonzero, and they are nonzero exactly in the $n(n-1)/2$ directions normal to the split locus, which are the directions in which the Weil class has to be transported. Second, those classes lie in the kernel of σ_E by (9), and (9) holds for every object whose Chern character is a flat section of Hodge classes along the family, so the semiregularity map cannot detect the obstruction of any such object. The obstruction is therefore nonzero, and the criterion cannot see it. For the explicit object the situation is the worst possible: the twelve dimensions $\Delta \oplus \mathrm{im}\, \mathrm{ob}_E$ of the kernel are accounted for by the two mechanisms of [I, Corollary 9.18] and (8), the remaining three lie on the even side of [I, Corollary 9.8] as well, and none of the fifteen is an artefact. The classes of the shape $z \cdot \mathrm{id}_E$ are the ones a change of the twisting class would contribute, one and the same z to every rank one summand; the last assertion of (iv) says that no such change removes the obstruction. So the explicit object does not move off the split locus to first order, either as a complex or as a twisted complex, and the question of [I, Question 15.13] for it is not one of verifying a hypothesis but of replacing the object.

The positive content is a recipe. By (ii), a split object whose classes span $\mathrm{NS}(A_0)_{\mathbb{Q}}$ with η has obstruction map of rank $n(n-1)/2$ against the family, and the spanning hypothesis is more than is needed: a single class $\theta \in \mathrm{NS}(A_0)_{\mathbb{Q}}$ not proportional to η already has $\{v \in T : v \lrcorner \theta = 0\} = T_{\mathrm{spl}}$. Indeed, every $v \in T$ kills η and commutes with K, so it is a pair (v_+, v_-) with $v_\pm \in \mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$. Since $\theta^\pm$ is of type $(1,1)$ in $\bigwedge^2 V_\pm$, the R_0-part $u\theta^+ + \overline{u}\,\theta^-$ of θ, with $u \neq 0$, gives $v \lrcorner \theta = u\, v_+ \lrcorner \theta^+ + \overline{u}\, v_- \lrcorner \theta^-$, whose two terms lie in the different summands $\bigwedge^2 V_+^{0,1}$ and $\bigwedge^2 V_-^{0,1}$ of $H^{0,2}(A_0)$. So the kernel of $v \mapsto v \lrcorner \theta$ on T does not depend on $u \neq 0$; it is therefore the common kernel for γ and ℓ, the cases $u = 1$ and $u = \sqrt{-d}$, which is T_{spl} by (ii). So every split object built from line bundles other than powers of η is obstructed in exactly the normal directions to the split locus, and Theorem 4.5 says that an object built from powers of η carries no Weil class. What survives is the statement of [I, Remark 9.9]: the object must have positive rank.

The evaluation map of the explicit object. [I, Theorem 9.19] places the copy Δ of the tangent space inside the image of the evaluation map. For a split object the whole of ev_E can be written down, and for the explicit object the entire kernel of σ_E turns out to lie in its image. We identify $HH^2(A_0)$ with

$$HT^2(A_0) = H^{0,2}(A_0) \;\oplus\; \mathrm{Hom}(H^{1,0}, H^{0,1}) \;\oplus\; \textstyle\bigwedge^2 (H^{1,0})^\vee \tag{10}$$

by the Hochschild–Kostant–Rosenberg isomorphism, the three summands being $H^2(\mathcal{O})$, $H^1(T)$ and $H^0(\bigwedge^2 T)$ for the trivial tangent bundle. An element $x = (z, v, \pi)$ acts on $H^*(A_0, \mathbb{C})$ by $z\wedge$, by the derivation $v\lrcorner$ of [I, Section 9.3], and by the double contraction $\pi\lrcorner$, which removes two $(1,0)$ slots; we write $x\lrcorner\xi$ for the sum of the three.

The double contraction is a second order operator. For a class λ of type $(1,1)$, which has a single $(1,0)$ slot, $\pi\lrcorner\lambda = 0$ and $\pi\lrcorner\lambda^k = \binom{k}{2}\lambda^{k-2}\,(\pi\lrcorner\lambda^2)$, because $\pi\lrcorner$ is a composite of two interior products, each of which is an antiderivation, and the only nonzero terms are those in which the two interior products hit two different factors. Summing over k with the factorials gives $\pi\lrcorner e^{\lambda} = \frac{1}{2}(\pi\lrcorner\lambda^2)\,e^{\lambda}$, exactly parallel to $v\lrcorner e^{\lambda} = (v\lrcorner\lambda)\,e^{\lambda}$ for the first order operator $v\lrcorner$. This is the whole content of the compatibility (12) below: the semiregularity map of a line bundle is multiplication by e^{λ}, and every summand of HT^2 acts on e^{λ} as multiplication by a class in $H^{0,2}$ which is precisely the value of ev_L. The same identity, for arbitrary E, is the commutative square of Buchweitz and Flenner [BF03, Corollary 4.3] and its Hochschild form in [Mar25b, Lemma 11.3]; on an abelian variety the Todd class is trivial and no correction to the HKR map is needed [Cal05, BF08].

Proposition 4.18 (The evaluation map of a split object)**.** *Let A_0 be a split member with X general and let $E = \bigoplus_{i=1}^{s} L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and all $\mathrm{Ext}^2(L_i[p_i], L_j[p_j])$, $i \neq j$, zero, so that $\mathrm{Ext}^2(E,E) = \bigoplus_i H^{0,2}(A_0)$.*

(i) *Under (10), ev_E is diagonal:*

$$\mathrm{ev}_E(z,v,\pi)_i = z + v\lrcorner\lambda_i + \tfrac{1}{2}\,\pi\lrcorner\lambda_i^2 \;\in\; H^{0,2}(A_0) = \mathrm{Ext}^2\big(L_i[p_i], L_i[p_i]\big). \tag{11}$$

(ii) *For every $x \in HT^2(A_0)$,*

$$\sigma_E\big(\mathrm{ev}_E(x)\big) = x\lrcorner\,\mathrm{ch}(E), \tag{12}$$

and consequently $\ker\sigma_E \cap \mathrm{im}\,\mathrm{ev}_E = \mathrm{ev}_E\big(\{x : x\lrcorner\,\mathrm{ch}(E) = 0\}\big)$. The weakened hypothesis of [Mar25b, Question 11.4] *holds for E if and only if ev_E kills every x that preserves $\mathrm{ch}(E)$ to first order.*

(iii) *$\dim HT^2(A_0) = 2\binom{2n}{2} + 4n^2$ and $\dim\mathrm{Ext}^2(E,E) = s\binom{2n}{2}$; for $s \geq 5$ and every $n \geq 2$ the second exceeds the first, so ev_E is not surjective and* [Mar25b, Lemma 11.3] *cannot apply to a sum of five or more line bundles.*

(iv) *For the object of Proposition 4.14, with $\lambda_i = u_i\theta^+ + \overline{u}_i\theta^-$ as in Notation 4.2: the space $\{x : x\lrcorner\,\mathrm{ch}(E) = 0\}$ is $\mathrm{Ann}_{H^{0,2}}(\omega_1) \oplus \{v : v\lrcorner\omega_1 = 0\} \oplus \{\pi : \pi\lrcorner\omega_1 = 0\}$, of dimensions 9, 18 and 9; the first and the third summands are both carried by ev_E onto Δ; the second is carried onto*

$$\Sigma = \big\{\,(u_i a + \overline{u}_i b)_{i=1}^{6} \;:\; a \in \textstyle\bigwedge^2 V_+^{0,1},\; b \in \bigwedge^2 V_-^{0,1}\,\big\}, \qquad \dim\Sigma = n(n-1) = 6,$$

which contains the image of the obstruction map of Proposition 4.16; $\Delta \cap \Sigma = 0$; and

$$\ker\sigma_E = \Delta \oplus \Sigma \subseteq \mathrm{im}\,\mathrm{ev}_E\,.$$

So every direction in which σ_E is blind is the value of ev_E on a Hochschild class that preserves $\mathrm{ch}(E)$ to first order, and the weakened hypothesis fails for E by all fifteen dimensions. The rank of ev_E is 45.

Proof. (i) The evaluation map is the action of $HH^*(A_0)$ on $\mathrm{Ext}^*(E,E)$, which under HKR is contraction of a polyvector in $H^q(\bigwedge^p T)$ with $\frac{1}{p!}\mathrm{At}(E)^p$ [BF08]; for $p = 0$ this is $z \cdot \mathrm{id}_E$ as in the proof of [I, Theorem 9.19], for $p = 1$ it is $v\lrcorner\,\mathrm{At}(E)$, the map (8), and for $p = 2$ it is $\frac{1}{2}\pi\lrcorner\,\mathrm{At}(E)^2$. The Atiyah class of E is diagonal with entries λ_i, so all three are diagonal with the entries displayed.

(ii) By (1), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_i e^{\lambda_i}$ for diagonal ζ. The three terms of (11) are exactly the factors by which $z\wedge$, $v\lrcorner$ and $\pi\lrcorner$ multiply e^{λ_i}: the first trivially, the second because $v\lrcorner$ is a derivation and $v\lrcorner\lambda_i$ has even degree, the third by the formula for $\pi\lrcorner e^{\lambda}$ above. Hence $\sigma_E(\mathrm{ev}_E(x)) = \sum_i (-1)^{p_i}\, x\lrcorner e^{\lambda_i} = x\lrcorner \operatorname{ch}(E)$. The description of $\ker\sigma_E \cap \operatorname{im}\mathrm{ev}_E$ follows: $\mathrm{ev}_E(x)$ lies in $\ker\sigma_E$ if and only if $x\lrcorner \operatorname{ch}(E) = 0$.

(iii) The three summands of (10) have dimensions $\binom{2n}{2}$, $4n^2$ and $\binom{2n}{2}$. The inequality $s\binom{2n}{2} > 2\binom{2n}{2} + 4n^2$ is $(s-2)(2n-1) > 4n$, and $4n/(2n-1) \le 8/3$ for $n \ge 2$, so it holds for $s \ge 5$.

(iv) Since $\operatorname{ch}(E) = 12\omega_1$ is of type $(3,3)$, the three summands of $x\lrcorner \operatorname{ch}(E)$ have types $(3,5)$, $(2,4)$ and $(1,3)$, so the kernel is the direct sum of the three kernels. The first is $\mathrm{Ann}_{H^{0,2}}(\omega_1)$, of dimension $n^2 = 9$ by [I, Lemma 9.6]. The second is $\mathrm{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \mathrm{Hom}(V_-^{1,0}, V_-^{0,1})$, of dimension $2n^2 = 18$, by the proof of [I, Proposition 9.14], whose argument for (i)⇔(iii) does not use the polarisation. For the third, decompose $\bigwedge^2 (H^{1,0})^\vee$ as $\bigwedge^2 (V_+^{1,0})^\vee \oplus (V_+^{1,0})^\vee \otimes (V_-^{1,0})^\vee \oplus \bigwedge^2 (V_-^{1,0})^\vee$; here ω_1 is a combination of $(\theta^+)^3$ and $(\theta^-)^3$ with both coefficients nonzero, $(\theta^\pm)^3$ spans $\bigwedge^3 V_\pm^{1,0} \otimes \bigwedge^3 V_\pm^{0,1}$, the outer blocks of π act injectively on the respective factor $\bigwedge^3 V_\pm^{1,0}$ and land in different summands of $H^{1,3}$, and the middle block kills both; so the third kernel is the middle block, of dimension $n^2 = 9$.

At a split point $\theta^\pm$ are of type $(1,1)$, so $\theta^\pm \in V_\pm^{1,0} \otimes V_\pm^{0,1}$, and they are nondegenerate pairings because $(\theta^\pm)^3 \ne 0$. Write $\lambda_i^2 = u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}(u_i)\,\theta^+\theta^-$. A middle block π kills $(\theta^\pm)^2$, which have no $V_\mp^{1,0}$ slot, so $\frac{1}{2}\pi\lrcorner\lambda_i^2 = \mathrm{N}(u_i)\,\pi\lrcorner(\theta^+\theta^-)$; the six u_i of Proposition 4.14 are $\pm\zeta_6, \pm 1, \pm\zeta_6^{-1}$, all of norm one, so this value is independent of i and $\mathrm{ev}_E(\pi) = z_\pi \cdot \mathrm{id}_E$ with $z_\pi = \pi\lrcorner(\theta^+\theta^-)$. The map $\pi \mapsto z_\pi$ from the middle block to $V_+^{0,1} \otimes V_-^{0,1}$ is the tensor product of the two nondegenerate pairings, hence an isomorphism onto the middle block of $H^{0,2}$, which is $\mathrm{Ann}_{H^{0,2}}(\omega_1)$ by [I, Lemma 9.6]. So ev_E carries the third kernel onto Δ, as it carries the first by [I, Theorem 9.19].

For $v = (v_+, v_-)$ in the second kernel, $v\lrcorner\lambda_i = u_i\, v_+\lrcorner\theta^+ + \overline{u}_i\, v_-\lrcorner\theta^-$ with $a = v_+\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $b = v_-\lrcorner\theta^- \in \bigwedge^2 V_-^{0,1}$; as $v_\pm$ ranges over $\mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$, a and b range over the whole of $\bigwedge^2 V_\pm^{0,1}$ because the pairings $\theta^\pm$ are nondegenerate, which is the surjectivity of $\mathrm{alt}(Vc)$ in [I, (26)]. So the image is Σ. The map $(a,b) \mapsto (u_i a + \overline{u}_i b)_i$ is injective, because $(u_1, \overline{u}_1)$ and $(u_2, \overline{u}_2)$ are linearly independent in $\mathbb{C}^2$, $u_1/\overline{u}_1 = \zeta_6^2$ differing from $u_2/\overline{u}_2 = 1$; hence $\dim\Sigma = 2\binom{n}{2} = n(n-1) = 6$. The obstruction map of Proposition 4.16 is the restriction to the polarised K-commuting v, so its image lies in Σ. An element of Σ of the shape $(z, \dots, z)$ has $u_i a + \overline{u}_i b = z = -z$ on comparing i with $3+i$, so $z = 0$ and then $a = b = 0$: $\Delta \cap \Sigma = 0$. By (ii), $\Delta \oplus \Sigma \subseteq \ker\sigma_E \cap \operatorname{im}\mathrm{ev}_E$, and its dimension $9 + 6 = 15$ equals $\dim\ker\sigma_E$ by Proposition 4.14, so the three spaces coincide.

For the rank, each term of (11) is a scalar depending on i times a class independent of i. Write $\mathbf{u}^k = (u_i^k)_i \in \mathbb{C}^6$ and $M = V_+^{0,1} \otimes V_-^{0,1}$. The class z contributes $\mathbf{u}^0 \otimes z$. Split v into its four blocks $v_{\varepsilon\varepsilon'} \colon V_\varepsilon^{1,0} \to V_{\varepsilon'}^{0,1}$; then $v\lrcorner\theta^+$ is the sum of $v_{++}\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $v_{+-}\lrcorner\theta^+ \in M$, the first ranging over $\bigwedge^2 V_+^{0,1}$ as above and the second over M because θ^+ is a nondegenerate pairing, and symmetrically for $v\lrcorner\theta^-$ with the blocks v_{--} and v_{-+}. As $\overline{u}_i = u_i^{-1}$, the values on the second summand of (10) are therefore $\mathbf{u}^1 \otimes (a+c) + \mathbf{u}^{-1} \otimes (b+c')$ with a, b, c, c' ranging independently over $\bigwedge^2 V_+^{0,1}$, $\bigwedge^2 V_-^{0,1}$, M, M. On the third summand, the outer block π_{++} kills $(\theta^-)^2$ and $\theta^+\theta^-$, and $\pi_{++} \mapsto \frac{1}{2}\pi_{++}\lrcorner(\theta^+)^2$ is an isomorphism onto $\bigwedge^2 V_+^{0,1}$ (in bases with $\theta^+ = \sum_j e_j f_j$ it sends $e_j^\vee \wedge e_k^\vee$ to $\pm f_j f_k$); likewise for π_{--}, and

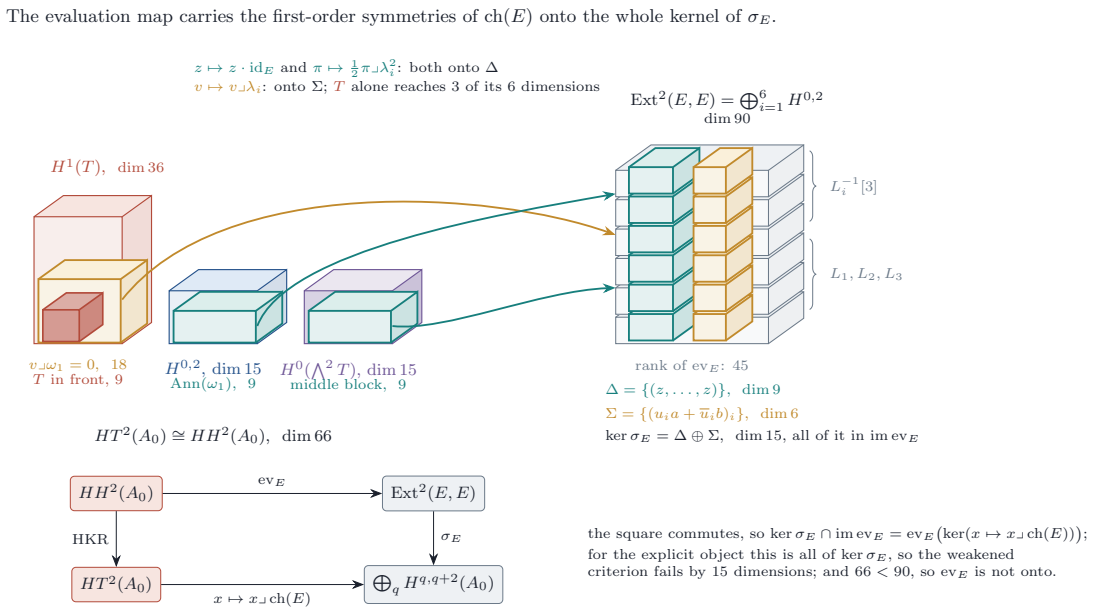


FIGURE 5. Proposition 4.18 for the explicit object at $n = 3$. Left: the three summands of $HT^2(A_0)$, drawn with heights proportional to their dimensions, and inside each the part that preserves $\mathrm{ch}(E)$ to first order: the annihilator of ω_1, the deformations killing ω_1 with the polarised K-commuting tangent space T in front, and the middle block of bivectors. Right: $\mathrm{Ext}^2(E,E)$ as six stacked copies of $H^{0,2}$, one per summand of E, and inside it the kernel of σ_E: the column Δ, one and the same class in every layer, and the column Σ, whose entries depend on the summand through u_i. The three arrows are the evaluation map on the three kernels; the two teal ones both land on Δ, the ochre one on Σ, and together they cover the kernel. Below, the commutative square that makes this a statement about the weakened criterion of [Mar25b, Question 11.4].

the middle block contributes $\mathbf{u}^0 \otimes z_\pi$. So

$$\operatorname{im} \mathrm{ev}_E = \mathbf{u}^0 \otimes H^{0,2} + \mathbf{u}^{\pm 1} \otimes \left(\textstyle\bigwedge^2 V_\pm^{0,1} \oplus M\right) + \mathbf{u}^{\pm 2} \otimes \textstyle\bigwedge^2 V_\pm^{0,1}.$$

The u_i are the six sixth roots of unity, so the vectors $\mathbf{u}^k$ for $k = 0, \pm 1, \pm 2$ are distinct characters of the cyclic group they form and are linearly independent. The sum is direct, and the rank is $15 + 2 \cdot 12 + 2 \cdot 3 = 45$. ∎

Figure 5 draws the three kernels and their images in $\mathrm{Ext}^2(E,E)$.

Remark 4.19 (Question 11.4 for split objects). For a split object the two criteria stand or fall together, and for the explicit object both fall completely. [I, Theorem 9.19] makes the first n^2 dimensions of the failure uniform over every concentrated object in every dimension; Proposition 4.18(iv) accounts for the remaining $n(n-1)$ at $n = 3$ by the deformations of the two halves $V_\pm$ of the Hodge structure separately, and the equality $\ker \sigma_E = \Delta \oplus \Sigma$ says that the two mechanisms exhaust the kernel: nothing in it is accidental. None of this touches the case for which the question was asked. A sheaf of positive rank has $z \cdot \mathrm{ch}(E) \neq 0$ for every $z \neq 0$, so its Δ is zero, and whether ev_E kills the deformations that preserve $\kappa(E)$

is a statement about the sheaf and not about its Chern character. Markman's candidate in [Mar25b, Section 12], a rank one complex on a principally polarised abelian fourfold with Chern character on the secant $\langle e^{\pm\sqrt{-d}\Theta}\rangle$, is exactly such a sheaf, and he records that the verification remains to be done. Theorem 4.56 reduces that verification to one identity on the fourfold, and Theorem 4.61 shows that the identity fails: the candidate does not satisfy the weakened criterion either.

Solutions of nonzero rank. We return to the Diophantine problem of Setup 4.9. By Corollary 4.13 a solution with $\mathrm{ch}_0(E) \neq 0$ needs at least $n+1$ distinct norms, and multiplicity one bounds each level sum by the number of elements of $\mathcal{O}_K$ of that norm.

Proposition 4.20 (The level sums)**.** *Keep Setup 4.9 and suppose E satisfies (6) with $\mathrm{ch}_0(E) \neq 0$, so that the u_i take $\nu \geq n+1$ distinct norms by Corollary 4.13. Write $N_1 < \cdots < N_\nu$ for those norms, M_j for the sum of the m_i at level N_j, and $r_K(N) = \#\{u \in \mathcal{O}_K : \mathrm{N}_{K/\mathbb{Q}}(u) = N\}$. If $\nu = n+1$ then, with $P = \prod_k N_k$, $S = \mathrm{ch}_0(E)$ and $D_j = N_j \prod_{k\neq j}(N_j - N_k)$,*

$$M_j = (-1)^n \frac{S\,P}{D_j}, \tag{13}$$

the least $S > 0$ making every M_j an integer is $S_{\min} = \mathrm{lcm}_j\big(|D_j|/\gcd(|D_j|, P)\big)$, and multiplicity one, Lemma 4.1(i), forces $|M_j| \leq r_K(N_j)$ for every j.

Proof. The conditions $T_{r,r} = 0$ for $r = 1, \ldots, n$ say that the linear functional $f \mapsto \sum_j M_j f(N_j)$ kills $x, x^2, \ldots, x^n$, hence sends every polynomial f of degree at most n to $S\,f(0)$. Applying it to $f(x) = \prod_{k\neq j}(x - N_k)$, which has degree n, gives

$$M_j \prod_{k\neq j}(N_j - N_k) = S\,(-1)^n \prod_{k\neq j} N_k,$$

and multiplying by N_j turns this into (13). The value of $S_{\min}$ is the definition of a least common multiple, and the bound on $|M_j|$ holds because each u of norm N_j contributes at most one term $m_i = \pm 1$ to M_j. ∎

So the level sums are determined by the norms up to the scale S, and they compete with the supply $r_K(N_j)$ of elements at each level. That competition is arithmetic. A second mechanism, in the cohomology of A, closes the split route for every n and every set of norms at once.

Corollary 4.21 (Sums of line bundles fail both criteria)**.** *Let A be of $(K, -1, n)$-Weil type with $n \geq 2$, and let $E = \bigoplus_i L_i[p_i]^{\oplus m_i}$ be a sum of shifted line bundles, twisted by a common class or not, whose (twisted) Chern character remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component. Then σ_E is not injective on the image of ev_E: E is neither semiregular nor semiregular in the weakened sense of* [Mar25b, Question 11.4]*, and it is obstructed to first order along some direction of the Weil family. In particular every object of Setup 4.9 satisfying (6), for every $n \geq 2$, every field K, every rank and every set of norms, fails both criteria.*

Proof. Write λ_i for the twisted first Chern classes of the summands. If every λ_i were a real multiple of η, it would be a rational multiple, both classes being rational, and the twisted Chern character $\sum_i m_i e^{\lambda_i}$ would be a polynomial in η; its component in degree $2n$ would then lie in $\mathbb{Q}\eta^n$, which meets the Weil line in zero by [I, Proposition 5.1], against the hypothesis. So some λ_i is not a real multiple of η, and [I, Theorem 9.20] applies to the summand $L_i[p_i]$. For an object of Setup 4.9 the classes are $\lambda_i = \theta(u_i) \in R_0$, and

$R_0 \cap \mathbb{Q}\eta = 0$; no twist is needed because $c_1(E) = \theta(T_{1,0}) = 0$ by (6), as $(1,0)$ is not an excluded pair for $n \geq 2$; and (6) gives a flat Chern character with a nonzero Weil component by Proposition 4.10. ∎

Remark 4.22 (What the corollary asks of an object). By Corollary 4.21 there is no sum of line bundles, at any member of any Weil family in any dimension, that could run the semiregularity argument in either form. The reason is [I, Proposition 9.15(i)]: the only divisor class that stays of type $(1,1)$ across the whole family is the polarisation, so every line bundle summand off that line is obstructed in some direction of the family, and the semiregularity map, whose composite with the obstruction is contraction with the flat class $\mathrm{ch}(E)$, cannot see it. The object the argument needs is one with no rank one summand off the polarisation line, of positive rank, indecomposable enough that its Atiyah class contracted with every direction of T vanishes. Markman's reflexive sheaves at $n = 3$ are of that kind, and nothing here says that their analogues at $n \geq 4$ do not exist.

4.3. **Convolutions of line bundles.** After Corollary 4.21 the next objects built from line bundles are their convolutions, the iterated cones of shifted line bundles. Their Chern character is the same signed sum of exponentials as for a direct sum, but their Ext groups are cut down by the maps between the pieces, so the argument of [I, Theorem 9.20] no longer applies as it stands. This subsection shows that fewer than $2n$ pieces with nondegenerate differences never serve (Theorem 4.26); writes down, at a CM member for $K = \mathbb{Q}(i)$, a signed sum of $2 \cdot 4^{n-1}$ line bundles that is exactly a real Weil class (Proposition 4.27); and determines, for the convolutions of those pieces with three multiples of the polarisation at $n = 3$, which products in the twisted differential can act on the 525 diagonal classes of degree two, of which at least 468 must disappear before the polarised criterion can hold (Theorem 4.31). Nothing enters the diagonal, cup products remove at most 276 of these classes, and everything else would have to be removed by fourfold products along a path through the three multiples and one of the Weil pieces: one family of them for some placements of the multiples, and two families that exclude each other for the others (Proposition 4.32); one order of the pieces gives a convolution that solves the Maurer–Cartan equation (Proposition 4.33). Off the diagonal, the classes between two pieces that differ in every coordinate are reached by no product (Lemma 4.34), and no convolution of these pieces meets the polarised criterion (Theorem 4.35). At $n = 4$ the same classes, now between pieces that differ in three coordinates, are reached by no product in any convolution, and they exclude every convolution whose 128 pieces L_ζ occupy two shifts (Theorem 4.37) or three consecutive ones (Theorem 4.38); with the groups between pieces whose shifts differ by two or three (Lemma 4.39) they exclude every convolution whose pieces have their shifts within five consecutive values (Corollary 4.44); the diagonal classes at a lonely shift (Lemma 4.45) and the interleaved shifts (Proposition 4.47) extend this to six (Corollary 4.48).

Setup 4.23 (Convolutions)**.** Let A be an abelian variety of dimension g with a flat Kähler metric, and let $L_1, \dots, L_m$ be line bundles on A with hermitian metrics of constant curvature, hermitian forms $H_1, \dots, H_m$ and pairwise distinct first Chern classes. For integers $d_1, \dots, d_m$ put $P_a = L_a[-d_a]$. A *convolution* of $P_1, \dots, P_m$ is an object E given

by a one-sided twisted complex[8]

$$\Big(\bigoplus_a \mathcal{A}^{0,*}(L_a)[-d_a],\ \bar\partial+\delta\Big), \qquad \delta=(\delta_{ab})_{a<b}, \quad \delta_{ab}\in\mathcal{A}^{0,1+d_a-d_b}\big(L_a^{-1}\otimes L_b\big),$$

with $\bar\partial\delta+\delta\circ\delta=0$, so that $\mathrm{ch}(E)=\sum_a(-1)^{d_a}e^{c_1(L_a)}$. Write $\mathcal{A}(a,b)=\mathcal{A}^{0,*}(L_a^{-1}\otimes L_b)$ and $\mathcal{H}(a,b)$ for its harmonic forms, so that $\mathcal{H}^q(a,b)\cong H^q(A,L_a^{-1}\otimes L_b)$, and write π, G and $h=\bar\partial^*G$ for the harmonic projection, the Green operator and the homotopy, with $1-\pi=\bar\partial h+h\bar\partial$ and $\pi h=h\pi=h^2=0$. A form of degree q in $\mathcal{A}(a,b)$ is a morphism $P_a\to P_b$ of degree $w=q-d_a+d_b$; we call q its Ext degree and, when $w=1$, call $q-1$ its *excess*. When H_b-H_a is nondegenerate, $\mathcal{H}(a,b)$ is concentrated in the single degree $n(H_b-H_a)$, the number of negative eigenvalues [BL04, Chapter 3].

Lemma 4.24 (The harmonic model)**.** *In Setup 4.23:*

(i) *E is isomorphic to a convolution of the same pieces with $h\delta=0$, and then $\delta=\delta^H-h(\delta\circ\delta)$ with $\delta^H=\pi\delta$ harmonic. Substituting repeatedly, every δ_{ab} is a finite sum of planar binary trees whose leaves are components δ^H_{cd} with $c<d$, whose internal vertices compose, and whose internal edges carry $-h$.*

(ii) *$\mathrm{Ext}^*(E,E)$ is the cohomology of $\mathcal{H}_E=\bigoplus_{a,b}\mathcal{H}(a,b)$, graded by w, with the differential $d_E=\sum_{k\ge0}\pi\,Q(-hQ)^k$, where $Q=[\delta,-]$.*

(iii) *For $x\in\mathcal{H}(a,b)$, the component of $d_E(x)$ in $\mathcal{H}(c,e)$ is a sum of terms $\pm\pi\,T$, one for each planar binary tree T with $k\ge2$ leaves, internal vertices composing, $-h$ on the internal edges other than the root, whose leaves read in order are the components along a path $c=a_0\to a_1\to\cdots\to a_k=e$ in which one step is $x\colon a\to b$ and the others are components $\delta^H_{a_ja_{j+1}}$ with $a_j<a_{j+1}$. The* Ext *degree of such a term is the sum of the* Ext *degrees of its leaves minus $k-2$.*

The terms with k leaves in (iii) are the Massey products of length k of the pieces;[9] (iii) is all that is used of them below.

Proof. (i) Suppose $h\delta_{cd}=0$ for $d-c<r$. Conjugating by $1+g$, with g of degree 0 supported at distance r, replaces δ_{ab} for $b-a=r$ by $X_{ab}+\bar\partial g_{ab}$, where X_{ab} depends only on δ, and leaves the components at smaller distance unchanged. With $g_{ab}=-hX_{ab}$ the new component is $X_{ab}-\bar\partial hX_{ab}$, and $h\bar\partial h=h(1-\pi-h\bar\partial)=h$ gives $h(X_{ab}-\bar\partial hX_{ab})=0$. Induction on r gives $h\delta=0$, and then $\delta=\pi\delta+\bar\partial h\delta+h\bar\partial\delta=\delta^H-h(\delta\circ\delta)$ by the Maurer–Cartan equation. The component $(\delta\circ\delta)_{ab}$ involves only components at distance less than $b-a$, so the substitution ends. (ii) The endomorphism complex of E is $\bigoplus_{a,b}\mathcal{A}(a,b)$ with the differential $\bar\partial+Q$. Blockwise, the inclusion of harmonic forms, π and $-h$ form a strong deformation retraction of $(\bigoplus\mathcal{A}(a,b),\bar\partial)$ onto $(\mathcal{H}_E,0)$, and Q raises $b-a$, so hQ is nilpotent. The basic perturbation lemma gives a homotopy equivalence of $(\bigoplus\mathcal{A}(a,b),\bar\partial+Q)$ with $(\mathcal{H}_E,d_E)$, with the chain maps $\iota_E=\sum_k(-hQ)^k$ on $\mathcal{H}_E$ and $\pi_E=\pi\sum_k(-Qh)^k$. (iii) Each Q in d_E composes with one component of δ on the left or on the right; writing that component as in (i) gives the trees, the edges from (i) and those from d_E being exactly the

[8]Every object obtained from $P_1,\dots,P_m$ by iterated cones in this order is of this form, because the Dolbeault complexes form a differential graded enhancement of the full subcategory of line bundles. All the objects of this subsection are given in this way.

[9]In the language of A_∞-categories they are the higher products m_k of the minimal model of the line bundles, evaluated on the Maurer–Cartan element δ^H of the twisted complex; for $k=2$ they are cup products.

internal edges other than the root. Composition adds degrees, h lowers them by one, and a binary tree with k leaves has $k-2$ internal edges. ∎

Lemma 4.25 (The index of a sum)**.** *For hermitian forms X, Y on $\mathbb{C}^g$, $n(X+Y) \le n(X)+n(Y)$. If $H_1+\cdots+H_k=0$, then $\sum_j n(H_j) \ge g - \dim\ker H_i$ for every i; in particular $\sum_j n(H_j) \ge g$ when one H_i is nondegenerate.*

Proof. A subspace on which $X+Y$ is negative definite meets the subspaces on which $X \ge 0$ and $Y \ge 0$, of codimensions $n(X)$ and $n(Y)$, only in zero, so its dimension is at most $n(X)+n(Y)$. For the second statement, $-H_i = \sum_{j\ne i} H_j$ gives $g - n(H_i) - \dim\ker H_i = n(-H_i) \le \sum_{j\ne i} n(H_j)$. ∎

Theorem 4.26 (Fewer than $2n$ pieces)**.** *Let A be of $(K,-1,n)$-Weil type with $n \ge 2$, and let E be a convolution of $m < 2n$ shifted line bundles whose differences $H_b - H_a$ are all nondegenerate. Suppose that $\mathrm{ch}(E)$ remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component, as every polarised character $N\omega_1 + p(\theta)$ with $N \ne 0$ does. Then there is $v \in T_{[A]}\mathcal{D}_{n,n}$ with*

$$\mathrm{ev}_E(v) \ne 0 \qquad \text{and} \qquad \sigma_E(\mathrm{ev}_E(v)) = 0,$$

so σ_E is not injective, and for a polarised character $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\, E)$ by Theorem 3.9(iv).

Proof. First, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero component in the diagonal classes $\mathcal{H}^2(c,c) = H^2(A,\mathcal{O}_A)$. Such a component is a sum of terms of Lemma 4.24(iii) whose k leaves have degree one and run around a closed path $c \to \cdots \to c$; the pieces before the step x increase from c and those after it increase to c, so the path passes through k distinct pieces and $k \le m$. Its leaves have Ext degrees $q_j = n(H_{a_{j+1}} - H_{a_j})$, and the output has Ext degree $\sum_j q_j - (k-2) = 2$, so $\sum_j q_j = k < 2n$. The k differences are nondegenerate and add up to zero, so Lemma 4.25 gives $\sum_j q_j \ge g = 2n$, a contradiction. Hence taking diagonal components gives a well defined map $\pi_D\colon \mathrm{Ext}^2(E,E) \to \bigoplus_a H^2(A,\mathcal{O}_A)$.

Next, $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$, as in the proof of [I, Theorem 9.20]. In the twisted complex the Atiyah class is represented by the commutator of the Chern $(1,0)$-connection $\bigoplus_a \nabla_a$ with $\bar\partial + \delta$; its diagonal entries are the curvatures F_a, constant forms representing $c_1(L_a)$, and its other entries $\nabla\delta_{ab}$ lie above the diagonal. So $\mathrm{ev}_E(v)$ is represented by an upper triangular z with $z_{aa} = v \lrcorner F_a$, and in $\mathcal{H}_E$ by $\pi_E(z)$. The operators Q and h preserve upper triangular forms and Q makes them strictly upper triangular, so the diagonal of $\pi_E(z)$ is that of $\pi(z)$, and $\pi_D(\mathrm{ev}_E(v)) = (v \lrcorner c_1(L_a))_a$.

If every $c_1(L_a)$ were a real multiple of the polarisation, it would be a rational one and $\mathrm{ch}(E)$ a polynomial in it, whose component in degree $2n$ meets the Weil line in zero by [I, Proposition 5.1]. So some $c_1(L_a)$ is not, and by [I, Proposition 9.15(i)] some $v \in T_{[A]}\mathcal{D}_{n,n}$ has $v \lrcorner c_1(L_a) \ne 0$; then $\pi_D(\mathrm{ev}_E(v)) \ne 0$, while $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E) = 0$. ∎

[I, Theorem 9.20] is the case $\delta = 0$, with any number of pieces. A convolution reaches its diagonal classes once it has $2n$ pieces, when a cycle through all of them can have total index $2n$, the equality case of Lemma 4.25. A convolution meeting the polarised criterion must therefore have at least $2n$ pieces, or pieces with degenerate differences. The next construction has both.

Proposition 4.27 (A pure Weil class from line bundles)**.** *Let $E_0 = \mathbb{C}/\mathbb{Z}[i]$, $A = E_0^{2n}$ with coordinates $z_1,\dots,z_{2n}$ and $n \ge 2$, let $K = \mathbb{Q}(i)$ act by i on $z_1,\dots,z_n$ and by $-i$ on*

$z_{n+1},\dots,z_{2n}$, and let θ be the product principal polarisation; then A is of $(\mathbb{Q}(i),-1,n)$-Weil type, a CM member as in [I, Remark 17.6]. *Write B_j for the surface with coordinates (z_j,z_{n+j}), so that $A=B_1\times\cdots\times B_n$. For $p\in\mathbb{Z}$ and $\zeta\in\mu_4^n$ let L_ζ be a line bundle whose hermitian form is $\left(\begin{smallmatrix} p & \zeta_j \\ \overline{\zeta}_j & p\end{smallmatrix}\right)$ on B_j and zero between different surfaces, and put $\alpha=\prod_j \frac{i}{2}\,dz_j\wedge d\bar z_{n+j}$.*

(i) *α and $\overline{\alpha}$ span the Weil classes of A, and*

$$\sum_{\prod\zeta=1} e^{c_1(L_\zeta)} - \sum_{\prod\zeta=-1} e^{c_1(L_\zeta)} = 2\cdot 4^{n-1}\,(\alpha+\overline{\alpha}).$$

(ii) *For distinct integers t_1,t_2,t_3 and signs σ_i, the character $\gamma=2\cdot4^{n-1}(\alpha+\overline{\alpha})+\sum_i\sigma_i e^{t_i\theta}$ has Hankel rank 3, so $r(\gamma)=7n^2-2n$. A convolution with this character of the $2\cdot4^{n-1}$ bundles L_ζ, with $(-1)^d=\prod\zeta$, and of three line bundles M_i with $c_1(M_i)=t_i\theta$ has $2\cdot4^{n-1}+3$ pieces and $(2\cdot4^{n-1}+3)\,n(2n-1)$ diagonal classes of degree two: 525 against $r=57$ at $n=3$, and 3668 against 104 at $n=4$.*[10]

Proof. The hermitian forms have imaginary parts integral on $\mathbb{Z}[i]^{2n}$, so the line bundles exist. On B_j write c_1 of the restriction as $p(a_j+b_j)+\zeta_j c_j+\overline{\zeta}_j c_j'$, where $a_j=\frac{i}{2}dz_j\wedge d\bar z_j$, $b_j=\frac{i}{2}dz_{n+j}\wedge d\bar z_{n+j}$, $c_j=\frac{i}{2}dz_j\wedge d\bar z_{n+j}$ and $c_j'=\frac{i}{2}dz_{n+j}\wedge d\bar z_j$. Since $c_j^2=c_j'^2=0$,

$$e^{c_1(L_\zeta)}=\prod_j f_j(\zeta_j),\qquad f_j(z)=e^{p(a_j+b_j)}(1+zc_j+\overline{z}c_j'+c_jc_j').$$

For $u\in\mu_4$, $\frac{1}{2}(u+\overline{u})$ is u when $u=\pm1$ and 0 when $u=\pm i$, so the left side of (i) is $\frac{1}{2}(\prod_j\sum_z zf_j(z)+\prod_j\sum_z\overline{z}f_j(z))$. Now $\sum_{z\in\mu_4}z=\sum z^2=0$ and $\sum|z|^2=4$, so $\sum_z zf_j(z)=4e^{p(a_j+b_j)}c_j'=4c_j'$, because a_jc_j' and b_jc_j' repeat a differential, and likewise $\sum_z\overline{z}f_j(z)=4c_j$. This gives $2\cdot4^{n-1}(\prod c_j'+\prod c_j)=2\cdot4^{n-1}(\overline{\alpha}+\alpha)$. The differentials dz_j, $j\le n$, and $d\bar z_{n+j}$ span the i-eigenspace of K on $H^1(A,\mathbb{C})$, and α spans its top exterior power, which is a Weil class, and $\overline{\alpha}$ the conjugate one. (ii) The moments $\mu_m=\sum_i\sigma_i t_i^m$ make the Hankel matrix of Setup 3.8 a product of Vandermonde matrices of rank 3, and Theorem 3.9(ii) gives r. The counts are immediate. ∎

The pieces L_ζ and $L_{\zeta'}$ have a degenerate difference as soon as ζ and ζ' share a coordinate, so Theorem 4.26 does not apply to them, and there are more than $2n$ of them. The rest of this subsection computes what the Massey products can do to the diagonal classes of such a convolution. The pieces are external products over the surfaces B_j, and the first tool is a bound on how far a product can lower the degree on each surface.

Lemma 4.28 (Degree drop on a product)**.** *Let $A=B_1\times\cdots\times B_n$ with a product metric, and let every L_a be an external product with a product metric. Decompose the leaves of a term of Lemma 4.24(iii) with k leaves into tensor products of harmonic forms on the factors. Then the output has degree s_j-d_j on B_j, where s_j is the sum of the degrees of the leaves on B_j and $d_j\ge0$, $\sum_j d_j=k-2$, and*

$$d_j\le\max(0,\,u_j-2),$$

where u_j is the number of leaves whose component on B_j is not a multiple of the unit section of $\mathcal{O}_{B_j}$ (such a unit occurs only where source and target agree on B_j).

[10] At $n=2$ the same construction has $8+3=11$ pieces and 66 diagonal classes against $r=24$.

Proof. The Laplacian is $\Delta = \sum_j \Delta_j$, so $h = \sum_j h_j$ with $h_j = \bar\partial_j^* G = G\bar\partial_j^*$, which lowers the degree on B_j by one and keeps the others; expand every internal edge accordingly, and let d_j be the number of edges carrying h_j. Let V_j, U_j and W_j be the forms whose component on B_j is Δ_j-harmonic, a multiple of the unit, or in the image of $\bar\partial_j^*$. The operators G and h_i, $i \neq j$, preserve all three, h_j kills V_j and W_j, π kills W_j, and multiplication by an element of U_j preserves V_j, U_j and W_j. For an edge e carrying h_j let $N(e)$ be the set of leaves below e whose component on B_j is not a unit. If $|N(e)| \leq 1$ the value below e lies in V_j and the term vanishes. If e lies below the next such edge e' with $N(e) = N(e')$, everything composed in between lies in U_j, the value at e' lies in W_j, and the term vanishes; if $N(e)$ is the set of all u_j such leaves for the highest such edge, the value at the root lies in W_j and π kills it. So in a nonzero term the sets $N(e)$ are distinct, of size at least two, different from the whole set, and pairwise nested or disjoint. Together with the whole set they form a family of that kind, which has at most $u_j - 1$ members, since its members are the internal vertices of a forest with u_j leaves in which every internal vertex has at least two children. Hence $d_j \leq u_j - 2$. ∎

Proposition 4.29 (Nothing enters the diagonal)**.** *Keep Proposition 4.27 with $n \geq 3$, and let t_1, t_2, t_3 be distinct with $|t_i - p| \geq 2$. For every convolution E of the pieces L_ζ and M_i, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero diagonal component. Consequently*

$$\dim \operatorname{Ext}^2(E,E) \;\geq\; \dim \ker\bigl(d_E|_{D^2}\bigr), \qquad D^2 = \bigoplus_a H^2(A, \mathcal{O}_A).$$

Proof. As in the proof of Theorem 4.26, a term that reaches the diagonal runs around a closed path of distinct pieces whose k steps have degree one and total excess 0. The step from L_ζ to $L_{\zeta'}$ has Ext degree 1 on each surface where $\zeta_j \neq \zeta'_j$, the difference there having eigenvalues $\pm|\zeta'_j - \zeta_j|$, and at least 0 elsewhere. If $\prod \zeta = \prod \zeta'$ the two differ on at least two surfaces; otherwise the step has even Ext degree, the shifts having opposite parities. Either way its excess is at least 1. The difference between L_ζ and M_i has eigenvalues $t_i - p \pm 1$ on every surface, so it is definite: the step has Ext degree 0 and excess -1 if it goes up the value ϕ, where $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$, and Ext degree $2n$ and excess $2n - 1$ if it goes down; the same holds between the M_i. A maximal run of the cycle through the M_i starts and ends at the value p, or is the whole cycle, so it has at least one step down and at most three steps up; its excess is at least $2n - 1 - 3 \geq 2$. So every cycle has excess at least 1. Twisting by line bundles of degree zero leaves the forms unchanged and can only make some $\mathcal{H}(a,b)$ vanish. For the inequality, the diagonal components of cocycles of degree two vanish on coboundaries, so $\dim \operatorname{Ext}^2(E,E)$ is at least the dimension of their image, which contains the kernel of d_E on D^2. ∎

At $n = 2$ the bound on a run is 0 and a cycle through the three multiples can reach the diagonal; this is why the analysis below starts at $n = 3$.

Proposition 4.30 (Cup products on the diagonal)**.** *Keep Proposition 4.29 with $n = 3$, so that D^2 has dimension $35 \cdot 15 = 525$. The terms of d_E on D^2 with two leaves, the cup products, vanish on the 45 diagonal classes of the M_i, and on the 480 diagonal classes of the L_ζ they have a kernel of dimension at least 204. So their kernel on D^2 has dimension at least 249.*

Proof. For $y = (y_a) \in D^2$ the cup products in $\mathcal{H}(a,b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, since y has even degree. On a surface where the pieces differ, δ^H_{ab} has degree 1, the only degree of the

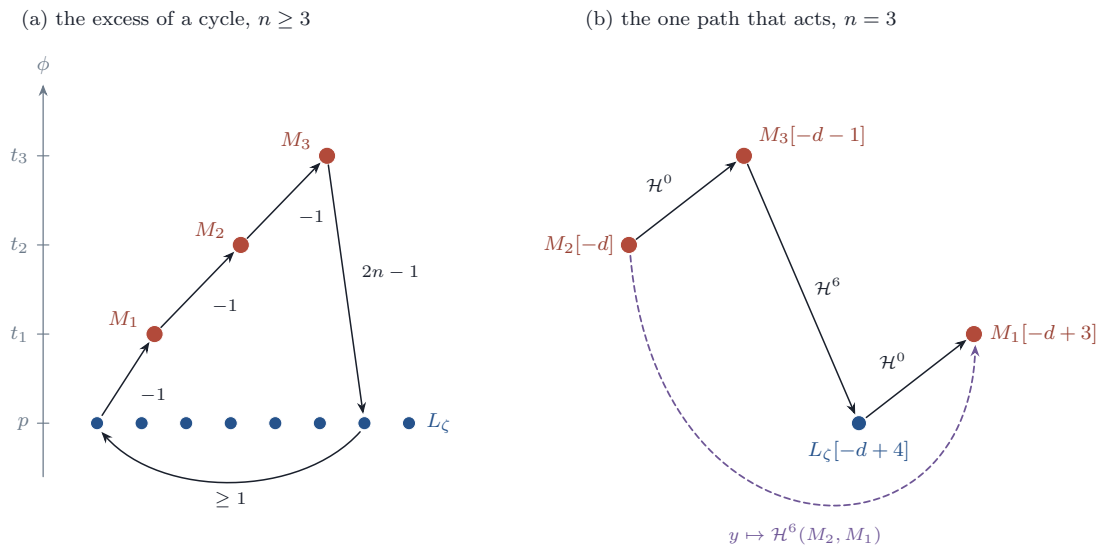


FIGURE 6. The pieces of Proposition 4.27 at their values ϕ, with $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$. (a) A step of degree one between pieces with a definite difference has Ext degree 0 going up and $2n$ going down, so excess -1 or $2n-1$; a step between two L_ζ has excess at least 1. The cycle drawn runs up through all three M_i and down once, the worst case, and still has excess $2n-3 > 0$, so for $n \geq 3$ no term reaches the diagonal (Proposition 4.29). (b) The one path whose products act on the diagonal at $n = 3$ when the t_i lie above p (Theorem 4.31(i)), drawn at the same heights and in the order of the convolution, with the Ext degrees of its steps and the shifts they force; the dashed arrow is the value of the product with a diagonal class y.

cohomology there, and the product with a component of y of positive degree lands in a degree where the cohomology vanishes. The pieces M_i differ from every other piece on every surface, which gives the first statement. Write y_a by its type $\tau \in \{0,1,2\}^3$, $|\tau| = 2$, the degrees on the three surfaces: three types $(2,0,0)$ of dimension one and three types $(1,1,0)$ of dimension four. The degree count on each surface and the parity of Ext degrees show that L_ζ and $L_{\zeta'}$ can carry a nonzero cup product on the type τ only if either ζ' differs from ζ by a sign in one coordinate outside the support of τ, or ζ' is ζ multiplied by (i,i) or $(-i,-i)$ in two coordinates and τ is 2 on the third. Let Γ_τ be the graph on the 32 pieces with these edges. If each component y^τ is constant on the connected components of Γ_τ, every cup product vanishes. For $\tau = (2,0,0)$ the coordinate ζ_1 is constant along the edges, and the eight pieces with a given ζ_1 are connected, so Γ_τ has 4 components; for $\tau = (1,1,0)$ the edges change the sign of ζ_3 only, and Γ_τ has 16 components. The kernel therefore contains a space of dimension $3\cdot 4\cdot 1 + 3\cdot 16\cdot 4 = 12 + 192 = 204$. ∎

Theorem 4.31 (The diagonal on E_0^6)**.** *Let $n = 3$ and let E be a convolution of the 32 bundles L_ζ and of M_1, M_2, M_3 as in Proposition 4.27, with $p+2 \leq t_1 < t_2 < t_3$*

and signs σ_i, so that $\mathrm{ch}(E) = 32(\alpha+\overline{\alpha}) + \sum_i \sigma_i e^{t_i\theta}$ *and the polarised criterion asks for* $\dim \mathrm{Ext}^2(E,E) = 57$.

(i) *The Massey products of length at least three vanish on D^2, except those of length four along a path $M_2 \to M_3 \to L_\zeta \to M_1$ taken in this order in the convolution, which occur only if $\sigma_1 = \sigma_3 = -\sigma_2$ and take values in $\mathcal{H}^6(M_2, M_1) \cong H^6\big(A, \mathcal{O}_A(-(t_2 - t_1)\theta)\big)$, of dimension $(t_2 - t_1)^6$.*

(ii) $\dim \mathrm{Ext}^2(E,E) \geq 249 - \min\big(249, (t_2-t_1)^6\big)$, *and* $\dim \mathrm{Ext}^2(E,E) \geq 249$ *unless the order and the signs are as in (i). In particular E does not meet the polarised criterion when $t_2 - t_1 \leq 2$, since $249 - 64 = 185 > 57$, nor when the signs or the order differ from (i).*

(iii) *If E meets the polarised criterion, then the length four products of (i) have rank at least $249 - 57 = 192$ on the kernel of the cup products, and $t_2 - t_1 \geq 3$.*

The same holds for $t_1 < t_2 < t_3 \leq p-2$ with E replaced by its dual.

Proof. (i) By Lemma 4.24(iii) a term with $k \geq 3$ leaves acting on $y \in H^2(\mathcal{O}_A)$ at a piece c runs along a chain $s = a_0 \to \cdots \to a_{k-1} = e$ of distinct pieces, increasing in the order of the convolution, whose $k-1$ steps are components of degree one, with y inserted at c. Its value lies in $\mathcal{H}(s,e)$, in Ext degree $2 + \sum_j q_j - (k-2) = 3 + \varepsilon$, where ε is the sum of the excesses of the steps; so $-3 \leq \varepsilon \leq 3$, and $\mathcal{H}(s,e)$ must be nonzero in degree $3+\varepsilon$. By the proof of Proposition 4.29 a step between two L_ζ has excess at least 1, with equality only between pieces of opposite parities, and a step between pieces at different values of ϕ has excess -1 and Ext degree 0 going up, excess 5 and Ext degree 6 going down. Let U and D be the numbers of steps up and down and λ the total excess of the steps between two L_ζ, so that $\varepsilon = 5D - U + \lambda$. For two distinct L_ζ, $L_{\zeta'}$ differing in $m \geq 1$ coordinates, $\mathcal{H}(L_\zeta, L_{\zeta'})$ vanishes above degree $6-m$; for pieces at different values it sits in degree 0 if the target is higher and in degree 6 if it is lower.

Two observations remove chains. First, a chain of total excess -2 has no step down, and so consists of steps between L_ζ followed by at most three steps up. Second, let $a \to b \to c$ be two consecutive steps of Ext degree 0, sections x_{ab} and x_{bc} of line bundles. The Maurer–Cartan equation $\pi(\delta \circ \delta) = 0$ of the proof of Proposition 4.33(i) has, in $\mathcal{H}^0(a,c)$, one term for each chain from a to c of total excess -2. If $a \to b \to c$ is the only such chain, the equation reads $x_{ab}x_{bc} = 0$, a product of holomorphic sections being holomorphic and so harmonic; since A is connected, x_{ab} or x_{bc} vanishes, and every term along a chain through both steps is zero.

Here every step up ends at one of M_1, M_2, M_3, so $U \leq 3$, and $D \leq 1$. If $D = 0$, the chain is $r \geq 0$ pieces L_ζ followed by multiples in increasing order. When it ends at an L_ζ, $k \geq 3$ gives $\lambda \geq 2$, so s and e differ in one coordinate and $\lambda = 2$; then the two steps change the parity twice, whereas s and e have opposite parities. When it ends at a multiple, e lies above s, so $\varepsilon = -3$, which forces $U = 3$, $\lambda = 0$ and the chain $L_\zeta \to M_1 \to M_2 \to M_3$; by the first observation $L_\zeta \to M_1 \to M_2$ is the only chain of excess -2 from L_ζ to M_2, and the second removes it. If $D = 1$, then $U \geq 2 + \lambda$, so (U, λ) is $(2,0)$, $(3,0)$ or $(3,1)$ and ε is 3, 2 or 3. For $\varepsilon = 3$ the piece s is a multiple and e lies below it. With $(3,1)$ the three steps up would end at three multiples other than s, which do not exist. With $(2,0)$ the chain has three steps, and a step down lands at an L_ζ, since a fourth multiple does not exist. It is $M_1 \to M_2 \to M_3 \to L_\zeta$, removed by the second observation at $M_1 \to M_2 \to M_3$; or $M_a \to M_b \to L_\zeta \to M_c$ with $t_a < t_b$ and $t_c < t_a$, that is $M_2 \to M_3 \to L_\zeta \to M_1$; or $M_a \to L_\zeta \to M_b \to M_c$ with $t_b < t_c < t_a$, that is $M_3 \to L_\zeta \to M_1 \to M_2$, removed at $L_\zeta \to M_1 \to M_2$. For $\varepsilon = 2$, with $(U, \lambda) = (3,0)$, the pieces s and e are L_ζ differing in one

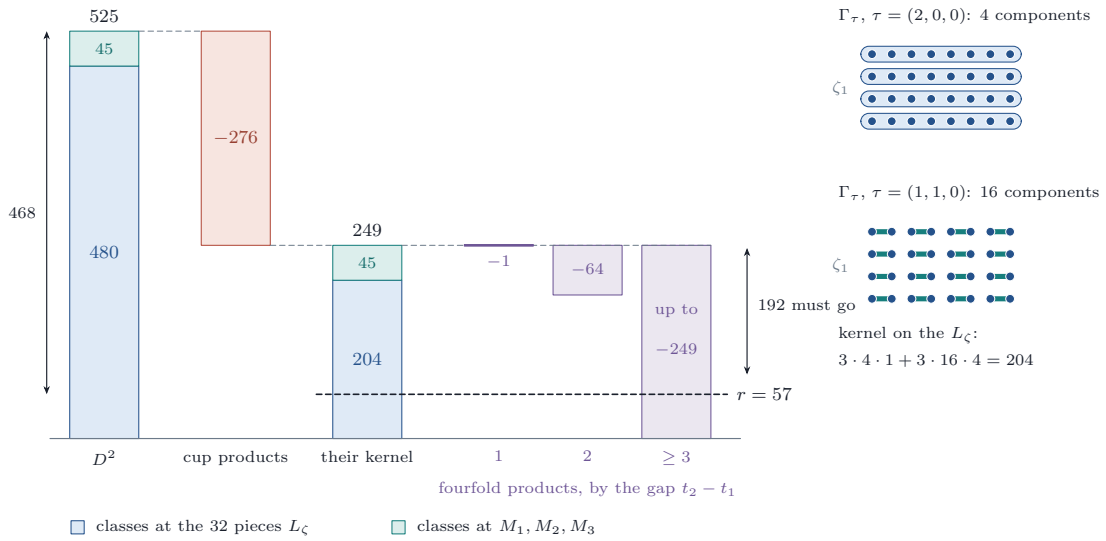


FIGURE 7. The 525 diagonal classes of degree two at $n = 3$ (Proposition 4.30 and Theorem 4.31). The cup products remove 276 of them for general components, and at most that many in any case, leaving 204 at the pieces L_ζ and the 45 at the M_i; when the t_i lie on one side of p, the fourfold products of the surviving path remove at most $(t_2 - t_1)^6$ more, which is 1 or 64 for the gaps 1 and 2 and no constraint from the third gap on. The criterion asks for 57, so 468 classes must disappear in all and 192 after the cup products. On the right, the graphs Γ_τ on the 32 pieces, one row for each value of ζ_1: for $\tau = (2, 0, 0)$ each row is a component, for $\tau = (1, 1, 0)$ the components are the pairs differing in the sign of ζ_3; a class constant on the components of every Γ_τ has vanishing cup products.

coordinate; the last step is the step down, preceded by three steps up from s, so the chain is $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, removed at $L_\zeta \to M_1 \to M_2$.

Only $M_2 \to M_3 \to L_\zeta \to M_1$ remains. Its steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_L = d_{M_3} - 5$ and $d_{M_1} = d_L + 1$; this gives the signs, and the value lies in $\mathcal{H}^{3+d_{M_2}-d_{M_1}}(M_2, M_1) = \mathcal{H}^6(M_2, M_1)$, whose dimension is $|\chi(\mathcal{O}_A((t_1 - t_2)\theta))| = (t_2 - t_1)^6$. (ii) By Proposition 4.29 it suffices to bound the kernel of d_E on D^2. By Proposition 4.30 the cup products have a kernel Y of dimension at least 249, and on Y the differential is the sum of the products of (i), of rank at most $(t_2 - t_1)^6$. (iii) follows from (ii). For the dual, $E \mapsto E^\vee$ preserves Ext^2 and sends the configuration with p, t_i to the one with $-p$, $-t_i$, the set of the L_ζ being stable under $\zeta \mapsto -\zeta$. ∎

The placements of the t_i on both sides of p are not covered by Theorem 4.31. There two families of fourfold products occur instead of one, and no convolution carries both.

Proposition 4.32 (Placements on both sides of p)**.** *Keep Theorem 4.31, but let $t_1 \le p - 2$ and $p + 2 \le t_2 < t_3$.*

(i) *The Massey products of length at least three vanish on D^2 unless $\sigma_1 = \sigma_2 = -\sigma_3$. In that case the only ones that act have length four and run along one of the paths*

(a) $M_2 \to M_3 \to M_1 \to L_\zeta$, (b) $M_3 \to M_1 \to L_\zeta \to M_2$,

taken in this order in the convolution, with $\prod\zeta=\sigma_3$. *The products along (a) take values in* $\mathcal{H}^6(M_2,L_\zeta)$, *of dimension* $\big((t_2-p)^2-1\big)^3$, *and those along (b) in* $\mathcal{H}^6(M_3,M_2)$, *of dimension* $(t_3-t_2)^6$. *A convolution carries at most one of the two kinds, since (a) puts* M_2 *before* M_3 *and (b) puts* M_3 *before* M_2.

(ii) *If* E *meets the polarised criterion, then* $\sigma_1=\sigma_2=-\sigma_3$, *and the products along (a) or those along (b) have rank at least* 192 *on the kernel of the cup products. For (b) this needs* $t_3-t_2\geq 3$. *The sixteen targets of (a) have dimension* $16\big((t_2-p)^2-1\big)^3\geq 432$, *so for (a) the count gives no condition.*

For $t_1<t_2\leq p-2$ *and* $t_3\geq p+2$ *the same holds for* $E^\vee[1]$, *and the condition on the signs becomes* $\sigma_2=\sigma_3=-\sigma_1$.

Proof. (i) We use the notation and the two observations of the proof of Theorem 4.31(i). Now $\phi(M_1)=t_1<p<t_2<t_3$; a step up ends at M_2, at M_3, or at an L_ζ coming from M_1, so again $U\leq 3$ and $D\leq 1$. If $D=0$, the values do not decrease along the chain. When e lies above s, $\varepsilon=-3$ forces $U=3$, $\lambda=0$ and the chain $M_1\to L_\zeta\to M_2\to M_3$, removed at $L_\zeta\to M_2\to M_3$, the only chain of excess -2 from L_ζ to M_3. When e and s have the same value, they are two L_ζ joined by steps between L_ζ, which is excluded as before. If $D=1$, (U,λ) is $(2,0)$, $(3,0)$ or $(3,1)$. For $\varepsilon=3$, e lies below s. With $(3,1)$ the steps up end at M_2, M_3 and an L_ζ following M_1; then s is an L_ζ, so $e=M_1$, which is followed by a step: impossible. With $(2,0)$ the chain has three steps and no two consecutive L_ζ. As M_1 is the lowest piece, it is never the end of a step up, and a step up into an L_ζ comes from M_1; so two consecutive steps up end at M_2 or M_3. The chain cannot start at M_1. If it starts at an L_ζ, it ends at M_1 with the step down, after two steps up: $L_\zeta\to M_2\to M_3\to M_1$, removed at $L_\zeta\to M_2\to M_3$. If it starts at M_3, the first step goes down and the last two go up, ending at M_2; the middle piece is then an L_ζ entered from M_1, and the chain is $M_3\to M_1\to L_\zeta\to M_2$, path (b). If it starts at M_2, then $e\in\{L_\zeta,M_1\}$, so the last two steps are not both up; the first step goes up to M_3, the second down, and the third up into $e=L_\zeta$ from M_1: the chain is $M_2\to M_3\to M_1\to L_\zeta$, path (a). For $\varepsilon=2$, $(U,\lambda)=(3,0)$ and s, e are L_ζ differing in one coordinate. The last step enters e from M_1, since a step down into e would follow three steps up from s, one more than the values allow; M_1 is entered by the step down, after two steps up from s: $L_\zeta\to M_2\to M_3\to M_1\to L_{\zeta'}$, removed at $L_\zeta\to M_2\to M_3$. Along (a) the steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3}=d_{M_2}+1$, $d_{M_1}=d_{M_3}-5$ and $d_L=d_{M_1}+1$. Along (b) they have Ext degrees 6, 0 and 0, which forces $d_{M_1}=d_{M_3}-5$, $d_L=d_{M_1}+1$ and $d_{M_2}=d_L+1$. This gives the signs and puts the values in $\mathcal{H}^{3+d_{M_2}-d_L}(M_2,L_\zeta)=\mathcal{H}^6(M_2,L_\zeta)$ and in $\mathcal{H}^{3+d_{M_3}-d_{M_2}}(M_3,M_2)=\mathcal{H}^6(M_3,M_2)$. On B_j the line bundle $L_\zeta\otimes M_2^{-1}$ has the hermitian form $\left(\begin{smallmatrix}p-t_2 & \zeta_j\\ \overline{\zeta}_j & p-t_2\end{smallmatrix}\right)$, with eigenvalues $p-t_2\pm 1<0$ and determinant $(t_2-p)^2-1$, which gives the first dimension. The second is that of Theorem 4.31(i) with t_3, t_2 in place of t_2, t_1. (ii) Propositions 4.29 and 4.30 do not depend on the placement of the t_i, so as in Theorem 4.31(ii) $\dim\mathrm{Ext}^2(E,E)\geq 249-\rho$, where ρ is the rank on the kernel of the cup products of the products of (i) that E carries, and $\rho=0$ unless $\sigma_1=\sigma_2=-\sigma_3$. The criterion asks for 57, so $\rho\geq 192$, and $(t_3-t_2)^6\geq 192$ exactly when $t_3-t_2\geq 3$. For the last statement, $E\mapsto E^\vee[1]$ preserves Ext^2, sends p and the t_i to $-p$ and the $-t_i$, reverses the order of the pieces and changes every sign. ∎

The Maurer–Cartan equation can be solved for one order of the pieces.

Proposition 4.33 (An explicit convolution)**.** *Keep Theorem 4.31 with* $p=0$ *and* $t=(2,5,6)$, *take the pieces to be external products* $L_\zeta=N_{1,\zeta_1}\boxtimes N_{2,\zeta_2}\boxtimes N_{3,\zeta_3}$ *and* $M_i=$

$M_{i,1} \boxtimes M_{i,2} \boxtimes M_{i,3}$ of line bundles on the B_j, and order them as M_2, M_3, the sixteen L_ζ with $\prod \zeta = 1$, M_1, the sixteen L_ζ with $\prod \zeta = -1$, with the shifts 0, 1, −4, −3, −5.

(i) *The components that the degrees allow are $x_1 \in \mathcal{H}^0(M_2, M_3)$, $x_2^\zeta \in \mathcal{H}^6(M_3, L_\zeta)$, $x_3^\zeta \in \mathcal{H}^0(L_\zeta, M_1)$, components in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ for the $48 + 96$ pairs with $\prod \zeta = 1 = -\prod \zeta'$ that differ in one or two coordinates, and components in $\mathcal{H}^6(M_2, L_{\zeta'})$, $\prod \zeta' = -1$, which we take to be zero. For harmonic components the Maurer–Cartan equation is exactly*

$$x_1 x_2^\zeta = 0 \text{ in } H^6(A, M_2^{-1} \otimes L_\zeta) \text{ for every } \zeta, \qquad \sum_\zeta x_2^\zeta x_3^\zeta = 0 \text{ in } H^6(A, M_3^{-1} \otimes M_1),$$

and every solution is the harmonic part of a convolution E with $\operatorname{ch}(E) = 32(\alpha + \overline{\alpha}) + e^{5\theta} - e^{6\theta} - e^{2\theta}$.

(ii) *There are solutions with every $x_2^\zeta \neq 0$ and $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$.*

(iii) *For every solution $\dim \operatorname{Ext}^2(E, E) \geq 2560$, so E does not meet the polarised criterion.*

Proof. (i) A component from P_a to P_b has Ext degree $1 + d_a - d_b$, and the forms give the cohomology of each pair: $\mathcal{H}(M_2, M_3)$ and $\mathcal{H}(L_\zeta, M_1)$ sit in degree 0, $\mathcal{H}(M_3, L_\zeta)$, $\mathcal{H}(M_2, L_{\zeta'})$ and $\mathcal{H}(M_2, M_1)$ in degree 6, and $\mathcal{H}(L_\zeta, L_{\zeta'})$ in degree 2 exactly for the pairs listed; no other pair has cohomology in the degree a component needs. Put $\delta = \delta^H - h(\delta \circ \delta)$ recursively, as in Lemma 4.24(i), and $\Phi = \bar{\partial}\delta + \delta \circ \delta$. Then $\Phi = \pi(\delta \circ \delta) + h\bar{\partial}(\delta \circ \delta)$ and $\bar{\partial}(\delta \circ \delta) = \Phi \circ \delta - \delta \circ \Phi$, so by induction on $b - a$ the twisted complex exists exactly when $\pi(\delta \circ \delta) = 0$. The components of $\pi(\delta \circ \delta)$ come from paths of two or more components; those that land in a nonzero group are $x_1 x_2^\zeta$ in $\mathcal{H}^6(M_2, L_\zeta)$ and $\sum_\zeta x_2^\zeta x_3^\zeta$ in $\mathcal{H}^6(M_3, M_1)$, while the product along $M_2 \to M_3 \to L_\zeta \to M_1$ would lie in $\mathcal{H}^5(M_2, M_1) = 0$ and x_2^ζ composed with a component in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ in degree eight. The signs $(-1)^d$ are $+, -, +, -, -$, which gives the Chern character by Proposition 4.27(i). (ii) Take x_1 and the x_3^ζ to be external products of nonzero sections on the B_j. On each B_j the products with their factors map $H^2(B_j, M_3^{-1} L_\zeta)$, of dimension 35, to spaces of dimensions 24 and 16, so they have kernels of dimensions at least 11 and 19. A sum of external products whose factor on one surface lies in the first kernel and whose factor on another lies in the second is a nonzero x_2^ζ with $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$, which solves (i).[11] (iii) The sixteen L_ζ with $\prod \zeta = 1$ have shift −4 and the others −5, so Lemma 4.34 below applies to every pair of them that differ in all three coordinates: there are 112 such pairs, and their groups $\mathcal{H}^3$ have dimensions adding up to $16 \cdot 160 = 2560$. ∎

The classes counted in (iii) are the ones no product can reach. Two pieces L_ζ, $L_{\zeta'}$ that differ in every coordinate have a difference that is indefinite on every surface, so $\mathcal{H}(L_\zeta, L_{\zeta'})$ is $\bigotimes_j H^1(B_j, \cdot)$, concentrated in degree three, of dimension $D(\zeta, \zeta') = \prod_j |\zeta_j - \zeta'_j|^2$; no component can join them, and if their shifts differ by one their classes have degree two.

Lemma 4.34 (Pieces that differ everywhere)**.** *Keep Proposition 4.29 with $n = 3$. Suppose that for some integer r either $p + 2 \leq t_1 < t_2 < t_3$ and M_1, M_2, M_3 have the shifts $r + 1$, $r + 4$, $r + 5$, or $t_1 \leq p - 2 < p + 2 \leq t_2 < t_3$ and they have the shifts $r - 1$, $r + 3$, $r + 4$ or $r - 1$, $r + 1$, $r + 4$; these are the shifts forced by the paths of Theorem 4.31(i) and of Proposition 4.32(i), (a) and (b), with r the shift of the L_ζ on the path. Let L_ζ, $L_{\zeta'}$ differ*

[11] Such solutions are far from unique: the equations of (i) are linear in each of x_1, x_2^ζ, x_3^ζ separately, and the components on the 144 pairs do not enter them at all.

in all three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. *Then* d_E *vanishes on* $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, *and no element of degree one has a differential with a nonzero component there. Consequently*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; \dim \ker(d_E|_{D^2}) + \sum D(\zeta,\zeta'),$$

the sum running over the pairs of this kind.

Proof. Only the shifts are used, and the order of the pieces only removes components, so we ignore it. A component from P_a to P_b of Ext degree q has $d_b = d_a + 1 - q$. Between distinct L_ζ the degree q is at least 2: there is no $\mathcal{H}^0$, and $\mathcal{H}^1$ occurs only for pieces differing in one coordinate, which have opposite parities, while $q = 1$ would need equal shifts. So a component between two L_ζ lowers the shift by at least one, and by exactly one only in degree two. For a multiple M_i let μ_i and ν_i be the shifts of an L_ζ that a component from M_i reaches and from which a component reaches M_i: $\mu_i = d_{M_i} - 5$ and $\nu_i = d_{M_i} - 1$ when $t_i > p$, and $\mu_i = d_{M_i} + 1$, $\nu_i = d_{M_i} + 5$ when $t_i < p$. Between the multiples the degrees allow only $M_2 \to M_3$ in the first case, $M_2 \to M_3 \to M_1$ in the second, and $M_3 \to M_1$ in the third. Let λ_i be the largest μ_k and κ_i the least ν_k over the multiples M_k reachable from M_i, respectively reaching M_i, through these steps, M_i included. The values, in the order M_1, M_2, M_3 and relative to r, are

	μ			ν			λ			κ		
$r+1,\ r+4,\ r+5$	-4	-1	0	0	3	4	-4	0	0	0	3	3
$r-1,\ r+3,\ r+4$	0	-2	-1	4	2	3	0	0	0	2	2	2
$r-1,\ r+1,\ r+4$	0	-4	-1	4	0	3	0	-4	0	3	0	3

and in all three cases, for all i and k,

$$\lambda_i \leq r \leq \kappa_k, \qquad \kappa_i \geq \lambda_i + 2, \qquad \lambda_i \leq \mu_i + 2, \qquad \nu_i \leq \kappa_i + 2, \tag{14}$$

while $\mu_i \leq \lambda_i$ and $\nu_i \geq \kappa_i$ by definition. A run $L \to M_k \to \cdots \to M_i \to L'$ through multiples goes from the shift $\nu_k \geq \kappa_k$ to $\mu_i \leq \lambda_k \leq \kappa_k - 2$, so every chain of components from one L to another lowers the shift, by at least two if it passes through a multiple and by exactly one only if it is a single component of degree two. In particular a chain from an L of shift x into M_i has $x \geq \kappa_i$, and a chain from M_i to an L of shift y has $y \leq \lambda_i$.

By Lemma 4.24(iii), a term of d_E on a class u of Ext degree q_u in $\mathcal{H}(a',b')$ runs along a chain $c \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to e$, and since the degrees of the components telescope, its value lies in $\mathcal{H}(c,e)$ in Ext degree

$$q_u + 1 + (d_c - d_{a'}) + (d_{b'} - d_e).$$

Write $a = L_\zeta$ and $b = L_{\zeta'}$, so that $d_a = d_b + 1$.

Nothing reaches $\mathcal{H}^3(a,b)$. Let u have degree one, so $q_u = 1 + d_{a'} - d_{b'}$, with chains from a to a' and from b' to b and $(a',b') \neq (a,b)$. If a' and b' are both pieces L, then $d_{a'} \leq d_a$ and $d_{b'} \geq d_b$, not both with equality, so $q_u \leq 1$; $q_u = 0$ is impossible, and $q_u = 1$ needs either $a' = b'$, and then a chain from a to b lowering the shift by one, a single component in $\mathcal{H}^2(a,b) = 0$, or a', b' differing in one coordinate, of opposite parities, against $d_{a'} = d_{b'}$. If $a' = M_i$ and b' is an L, then u lies in the degree of a component, so $d_{b'} = \mu_i$, while $d_{b'} \geq d_b = d_a - 1 \geq \kappa_i - 1 \geq \lambda_i + 1 > \mu_i$. If a' is an L and $b' = M_k$, then $d_{a'} = \nu_k$ and $d_a - d_b \geq \nu_k - \lambda_k \geq \kappa_k - \lambda_k \geq 2$. If $a' = M_i$ and $b' = M_k$, then $d_a - d_b \geq \kappa_i - \lambda_k$; for $i = k$ this is at least 2, and for $i \neq k$ the class u lies in the degree of a step $M_i \to M_k$, so $\lambda_k \leq \lambda_i$ and again $d_a - d_b \geq 2$. Each case contradicts $d_a - d_b = 1$.

d_E vanishes on $\mathcal{H}^3(a,b)$. Here u is the class itself and a term lands in $\mathcal{H}(c,e)$, $(c,e) \neq (a,b)$, in degree $3 + d_c - d_e$, with chains from c to a and from b to e. If c and e are both pieces L, then $d_c - d_e \geq 2$; $c = e$ is impossible, and otherwise a class of degree at least 5

needs c, e differing in one coordinate, in degree 5, so $d_c - d_e = 2$ against their opposite parities. If $c = M_i$ and e is an L, the degree is that of a component when $d_e = \mu_i + 2$, but $d_e \le d_b < d_a \le \lambda_i \le \mu_i + 2$. If c is an L and $e = M_k$, it is that of a component when $d_c = \nu_k - 2$, but $d_c \ge d_a > d_b \ge \kappa_k \ge \nu_k - 2$. If $c = M_i$ and $e = M_k$, then $d_a \le \lambda_i \le r \le \kappa_k \le d_b$, against $d_a > d_b$.

For the inequality: a coboundary has no component on D^2 (Proposition 4.29) and none on these groups, so the cocycles of $\ker(d_E|_{D^2})$ and of these groups span a space that meets the coboundaries only in zero. Twisting the pieces by line bundles of degree zero does not change $D(\zeta, \zeta')$, the forms being nondegenerate on every surface, and can only remove components. ∎

Theorem 4.35 (No convolution of these pieces meets the criterion)**.** *Keep Proposition 4.29 with $n = 3$: let E be any convolution of the 32 bundles L_ζ and of M_1, M_2, M_3, with $t_1 < t_2 < t_3$ and $|t_i - p| \ge 2$, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero. Then $\dim \mathrm{Ext}^2(E, E) \ge 69$. In particular no such convolution meets the polarised criterion, which asks for 57.*

Proof. Replacing E by $E^\vee$ or $E^\vee[1]$, which preserves Ext^2, we may assume that the t_i are as in Theorem 4.31 or in Proposition 4.32. If no fourfold product of those results acts, $\dim \mathrm{Ext}^2(E, E) \ge 249$. Otherwise the path fixes the shifts of the M_i as in Lemma 4.34, with r the shift of every L_ζ on a path; let P be the sixteen pieces of the parity of r, Q the other sixteen, and m the number of pieces on paths.

The fourfold products act on D^2 only through the classes at the M_i and at the m pieces on paths, so their rank is at most $45 + 15m$, and by Proposition 4.30 $\dim \ker(d_E|_{D^2}) \ge 249 - 45 - 15m = 204 - 15m$. If $m \le 9$ this is at least 69.

Let $m \ge 10$, so that at least ten pieces of P have the shift r. Every $W \in Q$ has 3, 6 and 7 partners in P differing from it in one, two and three coordinates, and for the seven $D = 16$ six times and $D = 64$ once.[12] If $d_W = r \pm 1$, at most six pieces of P lack the shift r, so one of the seven has it, and Lemma 4.34 gives at least 16 classes. If $d_W \ne r \pm 1$, consider for each type τ of the form $(1, 1, 0)$ the partner $W^\tau \in P$ obtained by changing the sign of the coordinate where τ vanishes. By the proof of Proposition 4.30, W^τ is the only piece with which W can carry a nonzero cup product on the type τ, and conversely; a cup product between them needs a component of degree two, hence shifts differing by one. If they do, the classes of type τ equal at W and W^τ and zero elsewhere have vanishing cup products, and W^τ, of shift $d_W \pm 1 \ne r$, lies on no path; if they do not, the classes of type τ at W alone have vanishing cup products. Either way these 4 classes lie in $\ker(d_E|_{D^2})$, 12 for each such W, and those of different W are independent, since $W \mapsto W^\tau$ is injective. Adding over Q, Lemma 4.34 gives $\dim \mathrm{Ext}^2(E, E) \ge 16 \cdot 12 = 192$. ∎

At $n = 4$ there are 128 pieces L_ζ, and the classes that no product reaches are again those between pieces that differ in three coordinates and have adjacent shifts. The shifts along a chain are now controlled by counting alone.

Lemma 4.36 (Shifts along chains at $n = 4$)**.** *Keep Proposition 4.27 with $n = 4$, take the L_ζ to be external products $N_{1,\zeta_1} \boxtimes \cdots \boxtimes N_{4,\zeta_4}$ of line bundles on the B_j, let t_1, t_2, t_3 be distinct with $|t_i - p| \ge 2$, and let E be a convolution of the L_ζ and of M_1, M_2, M_3 with shifts of the given parities. Call a component of δ^H, or an element of degree one of $\mathcal{H}_E$ in*

[12] The ratios ζ_j'/ζ_j of two pieces of opposite parities have product -1; when all three differ from 1 they are $(-1, -1, -1)$, with $D = 4^3$, or a permutation of $(i, -i, -1)$, with $D = 2 \cdot 2 \cdot 4$.

one group $\mathcal{H}(a,b)$, a step from P_a to P_b; a step of Ext *degree q has $d_b = d_a + 1 - q$. Put $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$.*

(i) *A step between distinct L_ζ has* Ext *degree at least two, so it lowers the shift by at least one, and by exactly one only in* Ext *degree two. A step in $\mathcal{H}^1(a,a) = H^1(A, \mathcal{O}_A)$ keeps the shift.*

(ii) *A step between an L_ζ and a multiple, or between two multiples, has* Ext *degree* 0 *if it goes up the value ϕ and* 8 *if it goes down, so it raises the shift by one or lowers it by seven.*

(iii) *A sequence of steps from an L_ζ to an $L_{\zeta'}$ through multiples, distinct before one marked step and distinct after it, where the marked step may lie in $\mathcal{H}^1(a,a)$ at a multiple, lowers the shift by at least four. Consequently a sequence of steps from one L_ζ to another, at most one of them not a component, lowers the shift by at least one, and by exactly one only if it consists of steps between pieces L_ζ, one of* Ext *degree two and the others in $\mathcal{H}^1(a,a)$.*

Proof. (i) For $\zeta \neq \zeta'$ the group $\mathcal{H}(L_\zeta, L_{\zeta'})$ is the tensor product over the surfaces of H^1 of a line bundle with a nondegenerate form, on the $k \geq 1$ surfaces where ζ and ζ' differ, and of $H^*(B_j, \mathcal{O}_{B_j})$ on the others, the factors of L_ζ and $L_{\zeta'}$ on those being equal. So its degrees are at least k, and degree one needs $k = 1$. Then the ratio $\zeta'_j/\zeta_j \neq 1$ in the one coordinate is the ratio of $\prod \zeta'$ and $\prod \zeta$, which is -1, and the shifts have opposite parities, while a step of Ext degree one keeps the shift. (ii) The forms of $M_i \otimes L_\zeta^{-1}$ have the eigenvalues $t_i - p \pm 1$ on every surface, which have the sign of $t_i - p$; the form of $M_k \otimes M_i^{-1}$ is $(t_k - t_i)\,\mathrm{Id}$. A definite form has cohomology in degree 0 if it is positive and $2n = 8$ if it is negative [BL04, Chapter 3].[13] (iii) Let U and D be the numbers of steps up and down outside $\mathcal{H}^1(a,a)$; by (ii) the shift changes by $U - 7D$. The values start and end at p and pass through the values of multiples, so $D \geq 1$. If $D = 1$, the values increase strictly before the step down and strictly after it, so those before it lie above p and those after it below p, each multiple occurs once, and $U \leq 3$: the drop $7 - U$ is at least four.[14] If $D \geq 2$, at most six multiples occur, counted with repetition, so there are at most seven steps outside $\mathcal{H}^1(a,a)$, $U \leq 7 - D$, and the drop $7D - U \geq 8D - 7$ is at least nine. For the consequence, split the sequence at the pieces L_ζ it visits: each part through multiples lowers the shift by at least four, and by (i) each other part lowers it by at least one, unless it lies in $\mathcal{H}^1(a,a)$. ∎

Theorem 4.37 (Two shifts on E_0^8)**.** *Keep Lemma 4.36, and let L_ζ and $L_{\zeta'}$ differ in exactly three coordinates and have the shifts $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then $\mathcal{H}(L_\zeta, L_{\zeta'})$ has its classes of degree two in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$, which is* 16 *or* 64*, and:*

(i) *for every convolution, no element of degree one has a differential with a nonzero component in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$;*

(ii) *if the L_ζ occupy two consecutive shifts, those with $\prod \zeta = \epsilon$ the shift s and the others $s - 1$, for a sign ϵ and an integer s, then d_E vanishes on these groups, and for every*

[13] This is Mumford's index theorem: a line bundle on an abelian variety whose hermitian form is nondegenerate with i negative eigenvalues has cohomology only in degree i. On $B_j = E_0 \times E_0$ the form of $M_i \otimes L_\zeta^{-1}$ is $\left(\begin{smallmatrix} t_i - p & -\zeta_j \\ -\bar{\zeta}_j & t_i - p \end{smallmatrix}\right)$, with eigenvalues $t_i - p \pm 1$, and $|t_i - p| \geq 2$ makes both of the sign of $t_i - p$; over the four surfaces the index is 0 or 8. The hypothesis $|t_i - p| \geq 2$ is what keeps the steps between a piece and a multiple out of the middle degrees.

[14] The bound is attained: with p below the three values t_i, the run $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, up three times and down once, changes the shift by $3 - 7 = -4$. With p in the middle, $t_1 < p < t_2 < t_3$, the run up through M_2 and M_3, down to M_1 and up to $L_{\zeta'}$ changes it by $3 - 7 = -4$ as well.

order of the pieces, every choice of the shifts of the M_i and every placement of the t_i,

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64 \cdot 640 = 40960.$$

In particular E does not meet the polarised criterion, which asks for $\dim \mathrm{Ext}^2(E,E) = 104$.

Proof. Write $W = L_\zeta$ and $V = L_{\zeta'}$. By the proof of Lemma 4.36(i), $\mathcal{H}(W,V)$ has degrees 3, 4 and 5, the last two from the surface where ζ and ζ' agree, and its classes of degree $w = 2$ have Ext degree $2 + d_W - d_V = 3$; the dimension of $\mathcal{H}^3(W,V)$ is the product over the three other surfaces of the determinants $|\zeta_j - \zeta'_j|^2$, as in Lemma 4.34.

(i) As in the proof of Lemma 4.34, a term of the differential of an element u of degree one in $\mathcal{H}(a',b')$ with a component in $\mathcal{H}(W,V)$ runs along a chain of components $W \to \cdots \to a'$, the step u, and a chain $b' \to \cdots \to V$, with $(a',b') \neq (W,V)$. This is a sequence of steps from W to V, one of them u, lowering the shift by one. By Lemma 4.36(iii) it consists of steps between pieces L_ζ, one of Ext degree two and at most one, u, in $\mathcal{H}^1(a,a)$. If u is not in $\mathcal{H}^1(a,a)$, it is the step of degree two and both chains are empty, against $(a',b') \neq (W,V)$. If u is in $\mathcal{H}^1(a,a)$, the rest is a single component $W \to V$ of Ext degree two, in $\mathcal{H}^2(W,V) = 0$.

(ii) A term of d_E on $x \in \mathcal{H}^3(W,V)$ runs along chains of components $c \to \cdots \to W$ and $V \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3+1+(d_c - d_W)+(d_V - d_e) = 3 + d_c - d_e$. No piece $L_{\zeta''}$ other than W lies on the first chain: by Lemma 4.36(iii) it would have the shift at least $d_W + 1 = s + 1$. Likewise only multiples follow V on the second chain, and each chain has at most three pieces besides W or V. Let U and D be the numbers of steps up and down along both chains, so that $U + D \leq 6$; by Lemma 4.36(ii), $d_W = d_c + U_1 - 7D_1$ and $d_e = d_V + U_2 - 7D_2$, and the Ext degree is

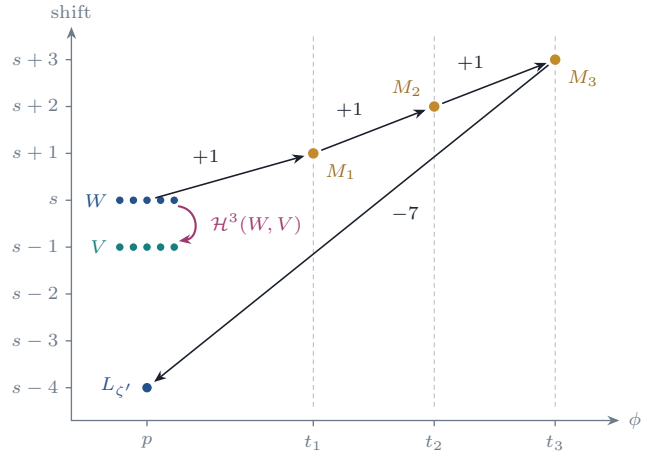
$$3 + d_c - d_e = 4 - U + 7D.$$

If $c = e$, this degree is 3, so $U = 7D + 1$, and $D \geq 1$ because the values return to $\phi(c) \neq p$; then $U \geq 8$, which is impossible. If $c \neq e$, the group $\mathcal{H}(c,e)$ sits in degree 0 if $\phi(e) > \phi(c)$ and in degree 8 if $\phi(e) < \phi(c)$, by Lemma 4.36(ii). Degree 0 needs $U = 4 + 7D \geq 4$ with $D = 0$, so the values increase strictly, the multiples are distinct and $U \leq 3$. Degree 8 needs $7D = U + 4$, so $D = 1$ and $U = 3$; the values increase strictly before the step down, from $\phi(c)$, and after it, up to $\phi(e) < \phi(c)$, so the five values along the path, with W and V counted once, are distinct, and four of them are values of multiples, of which there are three. So no term reaches a nonzero group, and d_E vanishes on $\mathcal{H}^3(W,V)$.

By (i) and (ii) the classes of these groups are cocycles of degree two, and no nonzero combination of them is a coboundary, since every coboundary has zero components there. Each piece of the shift s has 28 partners of the other parity differing from it in exactly three coordinates,[15] all of the shift $s-1$, and the 28 groups have dimensions adding up to 640; there are 64 pieces of the shift s. ∎

Figure 8 draws the shifts along a run. The hypothesis of (ii) is the arrangement of the explicit convolution of Proposition 4.33 at $n = 3$, where the sixteen pieces of each parity share a shift. The pieces must not be twisted by line bundles of degree zero here, unlike in Theorem 4.35: a twist that is nontrivial on the surface where ζ and ζ' agree removes $\mathcal{H}(L_\zeta, L_{\zeta'})$.

[15] The ratios ζ'_j/ζ_j in the three coordinates have product -1, so they are $(-1,-1,-1)$, with $D = 64$, or a permutation of $(i,-i,-1)$, with $D = 16$; with four places for the coordinate where they agree this gives $4 \cdot 7 = 28$ partners and $4 \cdot (64 + 6 \cdot 16) = 640$ classes.



Two shifts on E_0^8

W: the 64 pieces with $\prod\zeta=\epsilon$, at shift s;
V: the other 64, at shift $s-1$;
a step up in ϕ has Ext degree 0: shift $+1$;
a step down has Ext degree 8: shift -7;
a run through the multiples drops ≥ 4,
so no product reaches $\mathcal{H}^3(W,V)$,
and no term of d_E leaves it;
each top piece has 28 partners V, with
$D=16$ or 64, 640 classes in all:
$\dim \mathrm{Ext}^2(E,E)\geq 64\cdot 640=40960$,
against 104 for the polarised criterion;
three consecutive: ≥ 22528; more stay open

FIGURE 8. Lemma 4.36 and Theorem 4.37, in the placement $p<t_1<t_2<t_3$. Horizontal: the value ϕ of a piece; vertical: its shift. The pieces L_ζ with $\prod\zeta=\epsilon$ sit at the shift s and the others at $s-1$. A run through the multiples, up three times with Ext degree 0 and down once with Ext degree 8, ends four shifts lower, the least drop of Lemma 4.36(iii), so it never joins a top piece W to a partner V one shift lower; the group $\mathcal{H}^3(W,V)$ survives in $\mathrm{Ext}^2(E,E)$.

With a third shift, a component of Ext degree two can follow or precede a class of $\mathcal{H}^3(L_\zeta,L_{\zeta'})$, so the differential no longer vanishes on these classes. When the three shifts are consecutive, such terms reach only the groups between the highest and the lowest shift, which are too small to remove the classes.

Theorem 4.38 (Three consecutive shifts on E_0^8)**.** *Keep Lemma 4.36, and suppose that the shifts of the L_ζ lie in $\{s,s-1,s-2\}$ for an integer s. By their parities, for a sign ϵ, the pieces with $\prod\zeta=\epsilon$ have the shift $s-1$ and the others the shift s or $s-2$; let T and B be the sets of pieces of the shifts s and $s-2$. Then, for every order of the pieces, every choice of the shifts of the M_i and every placement of the t_i:*

(i) *every two pieces L_ζ, $L_{\zeta'}$ that differ in exactly three coordinates have shifts differing by one, and the groups $\mathcal{H}^3(L_\zeta,L_{\zeta'})$ of Theorem 4.37, taken from the piece of the larger shift, have dimensions adding up to* 40960*;*

(ii) *d_E maps the sum of these groups into $\bigoplus_{c\in T,\,e\in B}\mathcal{H}^5(c,e)$, and on them it consists of cup products with a single component of* Ext *degree two;*

(iii)
$$\dim \mathrm{Ext}^2(E,E)\ \geq\ 40960-\sum_{c\in T,\,e\in B}\dim\mathcal{H}^5(c,e)\ \geq\ 40960-18432=22528.$$

In particular E does not meet the polarised criterion. When T or B is empty this is Theorem 4.37(ii).

Proof. (i) The ratios ζ'_j/ζ_j in the three coordinates have product -1, so the two pieces have opposite parities: one has the shift $s-1$ and the other s or $s-2$. Each of the 64

pieces of the shift $s-1$ has 28 such partners, whose groups have dimensions adding up to 640, as in the proof of Theorem 4.37; this gives $64 \cdot 640 = 40960$.

(ii) Let $x \in \mathcal{H}^3(a,b)$ with $d_a = d_b + 1$. As in the proof of Theorem 4.37(ii), a term of $d_E(x)$ runs along chains of components $c \to \cdots \to a$ and $b \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3 + d_c - d_e$. The shifts of the pieces L_ζ differ by at most two, so by Lemma 4.36(iii) no part of a chain between two pieces L_ζ passes through a multiple. Hence a chain $c \to \cdots \to a$ that starts at a multiple M_i is a run $M_i = m_0 \to \cdots \to m_r \to L_1$ of distinct multiples, $r \le 2$, followed by pieces L_ζ only; let U_1 and D_1 be the numbers of its steps up and down the value ϕ before L_1, so that $d_{L_1} = d_{M_i} + U_1 - 7D_1$ by Lemma 4.36(ii) and $U_1 + D_1 = r + 1 \le 3$. Likewise a chain $b \to \cdots \to e$ ending at a multiple M_k ends with a run from a piece L_2 with U_2 steps up and D_2 down, $U_2 + D_2 \le 3$ and $d_{M_k} = d_{L_2} + U_2 - 7D_2$. Put $L_1 = c$ if c is a piece L_ζ and $L_2 = e$ if e is one, and $\delta = d_{L_1} - d_{L_2}$; then $1 = d_a - d_b \le \delta \le 2$.

If c and e are both pieces L_ζ, each nonempty chain lowers the shift by at least one, so $3 + d_c - d_e \ge 5$, while $\delta \le 2$ gives at most 5. So the degree is 5, $c \in T$, $e \in B$, and the two chains together consist of one component, of Ext degree two by Lemma 4.36(i): the term is a cup product of x with that component.

If $c = M_i$ and e is a piece L_ζ, the degree is $3 + \delta - U_1 + 7D_1$, and $\mathcal{H}(c,e)$ sits in degree 0 or 8 by Lemma 4.36(ii). Degree 0 needs $U_1 \ge 3 + \delta \ge 4$, and degree 8 needs $7D_1 - U_1 = 5 - \delta \in \{3,4\}$, so $D_1 \ge 1$ and $U_1 \ge 3$; both contradict $U_1 + D_1 \le 3$. The case of a piece c and a multiple e is the same with the second run.

If $c = M_i$ and $e = M_k$, the degree is $3 + \delta - U + 7D$ with $U = U_1 + U_2 \le 6$ and $D = D_1 + D_2$. For $i = k$ it is 3, so $U - 7D = \delta \ge 1$; $D = 0$ is impossible, since the values would increase from t_i to p along the first run and from p to t_i along the second, and $D \ge 1$ gives $U \ge 8$. For $i \ne k$, $\mathcal{H}(c,e)$ sits in degree 0 if $t_k > t_i$ and 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + 7D$, so $D = 0$ and $U \ge 4$; but then the multiples of the first run have values below p and those of the second values above p, so they are distinct and $U \le 3$. Degree 8 needs $7D - U = 5 - \delta$, so $D = 1$ and $U \ge 3$: the runs have at least four steps, so some multiple lies on both. One run increases strictly. If it is the second, its values lie in $(p, t_k]$, while those of the first lie at or above $t_i > t_k$ before its step down and below p after it. If it is the first, its values lie in $[t_i, p)$, while those of the second lie above p before its step down and at or below $t_k < t_i$ after it. Either way no multiple lies on both runs, a contradiction.

(iii) By Theorem 4.37(i), which holds for every convolution, no coboundary has a component in these groups, so the kernel of d_E on their sum injects into $\mathrm{Ext}^2(E,E)$, and by (ii) its dimension is at least 40960 minus the dimension of the targets. For pieces c, e of the same parity, $\mathcal{H}^5(c,e)$ is nonzero only when they differ in two or three coordinates, and each piece has such partners with groups $\mathcal{H}^5$ of dimensions adding up to 960.[16] So the sum in (iii) is at most $\min(|T|,|B|) \cdot 960 \le 32 \cdot 960 = 30720$, which already gives $\dim \mathrm{Ext}^2(E,E) \ge 10240$. The dimension of $\mathcal{H}^5(c,e)$ depends only on the ratio e/c, so the sum is the weight of the cut (T,B) of a weighted Cayley graph on the group of ratios of two pieces of one parity, of order 64 and degree 960. For the vector $x = \pm 1$ of the cut, the weight is $\frac{1}{4}(64 \cdot 960 - x^{\mathsf{T}} A x) \le 16\,(960 - \lambda)$, with A the weighted adjacency matrix and λ its least eigenvalue. Its eigenvalues are the character sums $\sum_g \dim \mathcal{H}^5(1,g)\,\chi(g)$. A character

[16] The ratios $e_j/c_j \ne 1$ have product 1. In two coordinates they are $(-1,-1)$, with $D = 16$, or $(i,-i)$, $(-i,i)$, with $D = 4$; then $\mathcal{H}^5 = \mathcal{H}^1 \otimes \mathcal{H}^1 \otimes \mathcal{H}^3(B_j \times B_k, \mathcal{O})$ has dimension $4D$, and the six pairs of coordinates give $6 \cdot 64 + 12 \cdot 16 = 576$. In three coordinates they are a permutation of $(i,i,-1)$ or of $(-i,-i,-1)$, with $D = 16$, and $\mathcal{H}^5 = (\mathcal{H}^1)^{\otimes 3} \otimes H^2(B_l, \mathcal{O})$ has dimension 16; the four triples give $4 \cdot 6 \cdot 16 = 384$. In four coordinates $\mathcal{H}(c,e)$ sits in degree 4.

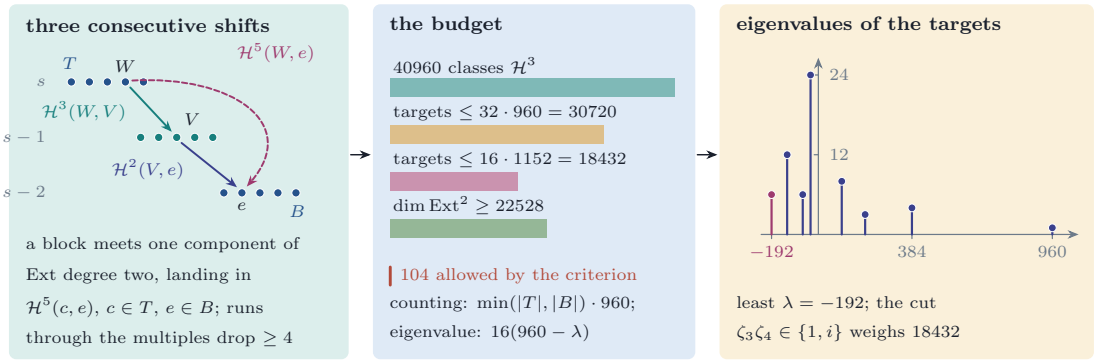


FIGURE 9. Theorem 4.38. Left: the pieces L_ζ at three consecutive shifts, those of one parity at $s-1$ and the others split between T at s and B at $s-2$; a group $\mathcal{H}^3$ between two pieces that differ in three coordinates (teal) can meet a component of Ext degree two (indigo) only in a group $\mathcal{H}^5(c,e)$ with $c \in T$ and $e \in B$ (magenta). Middle: the 40960 classes against the targets, at most 30720 by counting and 18432 by the least eigenvalue, against the 104 that the criterion allows. Right: the eigenvalues of the Cayley graph of the targets.

is $\chi(g) = \prod_j g_j^{k_j}$ with $k \in (\mathbb{Z}/4)^4$, up to adding $(1,1,1,1)$; put $\omega_j = i^{k_j}$, $W = \sum_j \omega_j$, $\varrho = \sum_j \omega_j^2$ and $\Omega = \prod_j \omega_j$. By the list in the footnote, the ratios that move the pair of coordinates $\{j,l\}$ contribute $64\,\omega_j^2\omega_l^2 + 32\,\Re(\omega_j\overline{\omega}_l)$, and those that fix only the coordinate o contribute $32\,\Re(\Omega\overline{\omega}_o(W-\omega_o))$. Summing,

$$\chi\big(\textstyle\sum_g \dim \mathcal{H}^5(1,g)\, g\big) = 32\varrho^2 + 16|W|^2 - 192 + 32\,\Re(\Omega)\big(|W|^2 - 4\big).$$

If $\Re\Omega = 0$, the sum of the k_j is odd, so an odd number of them are odd, $|\varrho| = 2$, and the value is at least -64. If $\Re\Omega = 1$, an even number are odd: with none or four, $|\varrho| = 4$ and the value is at least 192; with two, $\varrho = 0$, and $W = 0$ would force the k_j to be $0,1,2,3$, whose sum is not divisible by 4, so $|W|^2 \geq 4$ and the value is at least -128. If $\Re\Omega = -1$, the value is $32\varrho^2 - 16|W|^2 - 64$, and $|W|^2 \leq 8 + 2\varrho^2$: when all k_j have the same parity, $|W| \leq 4 = |\varrho|$, and when two are odd, $\varrho = 0$ and W is a sum of two of ± 1 and two of $\pm i$, so $|W|^2 \leq 8$. Hence $\lambda \geq -192$,[17] and the sum is at most $16 \cdot 1152 = 18432$. ∎

Figure 9 draws the three shifts and the budget.

The groups of Theorem 4.37 join two pieces whose shifts differ by one. All groups between pieces whose shifts differ by three, and thirteen kinds between pieces whose shifts differ by two, are also left alone by the differential, and with them the argument covers every arrangement of the shifts within five consecutive values. For pieces a and b call $d_a - d_b$ the *spread* of $\mathcal{H}(a,b)$; its classes of degree two have Ext degree $2 + d_a - d_b$. For $\zeta, \zeta' \in \mu_4^4$ call $\zeta'/\zeta = (\zeta_j'/\zeta_j)_j$ their *ratio*; it *moves* the coordinates where it differs from 1, and it changes the parity, $\prod \zeta' = -\prod \zeta$, exactly when the product of its entries is -1. As before, $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta_j'} |\zeta_j - \zeta_j'|^2$.

[17] The eight values are 960, 384, 192, 96, −32, −64, −128 and −192. The bound 18432 is attained by the cut $\zeta_3\zeta_4 \in \{1,i\}$ against $\zeta_3\zeta_4 \in \{-1,-i\}$, so 22528 is the best bound that this count gives.

Lemma 4.39 (Groups of spread two and three)**.** *Keep Lemma 4.36, and let E be any convolution of the L_ζ and the M_i, in any order, with any shifts of the given parities and any placement of the t_i.*

(i) *For a class x of degree two in $\mathcal{H}(L_\zeta, L_{\zeta'})$ of positive spread, every term of $d_E(x)$ has all its leaves between pieces $L_{\zeta''}$.*

(ii) *Let $\prod\zeta = \prod\zeta'$, let ζ'/ζ move all four coordinates, one of them by -1, and let $d_{L_\zeta} = d_{L_{\zeta'}} + 2$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ form $\mathcal{H}^4(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') \in \{64, 256\}$; no element of degree one has a differential with a nonzero component in it, and d_E vanishes on it.*

(iii) *Let $\prod\zeta = -\prod\zeta'$ and $d_{L_\zeta} = d_{L_{\zeta'}} + 3$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ lie in $\mathcal{H}^5(L_\zeta, L_{\zeta'})$, and d_E vanishes on them.*

(iv) *Let ζ and ζ' differ in exactly three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Every term of d_E on $\mathcal{H}^3(L_\zeta, L_{\zeta'})$ runs along chains of components $c \to \cdots \to L_\zeta$ and $L_{\zeta'} \to \cdots \to e$ between pieces $L_{\zeta''}$ that together lower the shift by one or two.*

Proof. Write $a = L_\zeta$, $b = L_{\zeta'}$ and $\delta = d_a - d_b$. As in the proof of Theorem 4.37(ii), a term of d_E on a class of degree two in $\mathcal{H}(a, b)$ runs along chains of components $c \to \cdots \to a$ and $b \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c, e)$ in Ext degree $3 + d_c - d_e$, which is 3 if $c = e$.

(i) Let L_1 be the first piece $L_{\zeta''}$ of the first chain and L_2 the last of the second, and put $X = (d_{L_1} - d_a) + (d_b - d_{L_2})$. By Lemma 4.36(iii), $X \geq 0$, and $X \geq 4$ if a multiple lies between L_1 and a or between b and L_2. Before L_1 the first chain is a run of distinct multiples, with U_1 steps up and D_1 steps down the value ϕ and $U_1 + D_1 \leq 3$; after L_2 the second chain is a run with U_2 and D_2 steps, $U_2 + D_2 \leq 3$. By Lemma 4.36(ii) the term has Ext degree $3 + \delta + X - U + 7D$, where $U = U_1 + U_2$ and $D = D_1 + D_2$. If c and e are pieces, this is $3 + \delta + X$; a multiple on a chain makes it at least 8, but $\mathcal{H}^8(c, e) \neq 0$ only for $c = e$, where the degree is 3. If $c = M_i$ and e is a piece, $\mathcal{H}(c, e)$ sits in degree 0 if $t_i < p$ and in degree 8 if $t_i > p$. Degree 0 needs $U_1 \geq 3 + \delta + X \geq 4$. Degree 8 needs a step down on the run from t_i to p, so $D_1 \geq 1$, and then $U_1 = 7D_1 - 5 + \delta + X \geq 3$. Both contradict $U_1 + D_1 \leq 3$, and a piece c with a multiple e is the same. Let $c = M_i$ and $e = M_k$. For $i = k$ the degree is 3, so $U - 7D = \delta + X \geq 1$; $D \geq 1$ would give $U \geq 8$, and $D = 0$ would make the values increase from t_i to p and then from p to t_i. For $i \neq k$, $\mathcal{H}(c, e)$ sits in degree 0 if $t_k > t_i$ and in degree 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + X + 7D$, so $D = 0$ and $U \geq 4$; but then the first run lies below p and the second above, so the multiples are distinct and $U \leq 3$. Degree 8 needs $D \geq 1$, since with $D = 0$ the values would satisfy $t_i < p < t_k$; then $7D - U = 5 - \delta - X \leq 4$ gives $D = 1$ and $U \geq 3$, so the runs have at least four steps and share a multiple, which the last paragraph of the proof of Theorem 4.38(ii) excludes.

(ii) By the proof of Lemma 4.36(i), $\mathcal{H}(a, b)$ is the tensor product of four groups H^1 and sits in degree 4, of dimension $D(\zeta,\zeta')$; the ratio is $(-1,-1,-1,-1)$, with $D = 256$, or a permutation of $(i,-i,-1,-1)$, with $D = 64$.[18] A coboundary with a component in $\mathcal{H}(a, b)$ is a sum of terms of Lemma 4.24(iii) along a path $a \to \cdots \to b$ of morphisms of degree one, one of them an element u and the others components, that lowers the shift by two. By Lemma 4.36(iii) no multiple lies on it, and by Lemma 4.36(i) each step between distinct

[18] A ratio that moves every coordinate has entries in $\{i, -1, -i\}$ and product 1. With a -1 among them it is $(-1,-1,-1,-1)$ or has exactly two entries -1 and the entries i and $-i$, twelve ratios; together thirteen, and $256 + 12 \cdot 64 = 1024$. The other eight ratios have all four entries in $\{i, -i\}$ and $D = 16$; (ii) makes no statement about them.

pieces lowers the shift by at least one. So the path consists of a component of Ext degree three from a to b, which lies in $\mathcal{H}^3(a,b)=0$, and a class of $\mathcal{H}^1(a,a)$ or $\mathcal{H}^1(b,b)$; or of two steps $a\to c\to b$ of Ext degree two, possibly with a class of $H^1(A,\mathcal{O}_A)$ at one of the three pieces. The output has degree one on every surface. Without the class of H^1 the two steps have degrees adding up to one on every surface, so they move disjoint pairs of coordinates and have degree zero elsewhere; a step of Ext degree two changes the parity, so its two moved entries have product -1 and are (i,i) or $(-i,-i)$, and ζ'/ζ has no entry -1. With the class of H^1, Lemma 4.28 gives a drop of one on a surface where all three leaves have positive degree, and the output has degree at least two there.

Now let a term of d_E on $\mathcal{H}^4(a,b)$ carry the components $y_1,\dots,y_m$, $m\ge 1$, between pieces by (i), and land in $\mathcal{H}(c,e)$; let K be the set of coordinates fixed by the ratio of c and e, and $X\ge m$ the drop of the chains. The output has Ext degree $5+X$. Let n_j be the number of the y_i of positive degree on B_j; the class x has degree one there, so Lemma 4.28 bounds the drop on B_j by $\max(0,n_j-1)$. For $j\notin K$ the output has degree 1 on B_j, so the degrees of the y_i there add up to at most $\max(0,n_j-1)$, which forces $n_j=0$. For $j\in K$ the y_i undo the entry $\zeta'_j/\zeta_j\neq 1$, so $n_j\ge 1$, and the output has degree at least $1+n_j-(n_j-1)=2$ there, hence exactly 2. So $5+X=(4-|K|)+2|K|$, that is $X=|K|-1$. The components change the parity X times modulo two, and together their ratio is ζ/ζ' on K and 1 elsewhere, so

$$\prod_{j\in K}\zeta'_j/\zeta_j=(-1)^{|K|-1}.$$

For the ratio $(-1,-1,-1,-1)$ the left side is $(-1)^{|K|}$. For a permutation of $(i,-i,-1,-1)$ the left side is real only if K contains both or neither of the coordinates moved by $\pm i$, and then it is again $(-1)^{|K|}$. So no term is nonzero.

(iii) The classes of degree two have Ext degree 5. By (i) a term of d_E on them lands in Ext degree $6+X$, with $X\ge 1$ the drop of the chains between pieces. For $X\ge 2$ this is at least 8, which needs $c=e$ and degree 3. For $X=1$ the pieces c, e have shifts differing by four, hence the same parity, so their ratio moves at least two coordinates and $\mathcal{H}(c,e)$ has no degree above 6.

(iv) By (i) a term lands in Ext degree $4+X$, with $X\ge 1$ the drop of the chains between pieces. For $X\ge 4$ the degree is at least 8, which needs $c=e$. For $X=3$ the degree is 7, which needs a ratio moving one coordinate, hence opposite parities, while $d_c-d_e=4$ is even. ∎

The groups of (ii) are reached by no coboundary and left by no term in every convolution, so they survive in $\mathrm{Ext}^2(E,E)$ as soon as the shifts give them the spread two. That happens along every edge of a Cayley graph that crosses between two shift values of one parity, and a cut of that graph is never light.

Lemma 4.40 (Cuts of a Cayley graph)**.** *Let G be a finite abelian group, $S\subset G\setminus\{0\}$ a generating set and $w\colon S\to\mathbb{R}_{>0}$ a weight, extended by 0 to G. For every subset $T\subset G$ with $T\neq\emptyset,G$,*

$$\sum_{c\in T,\ e\notin T} w(e-c)\ \ge\ \min_{H}\,|H|\cdot w(S\setminus H),$$

the minimum over the proper subgroups H of G. For the group G_0 of the 64 ratios of pieces of one parity, with S the thirteen ratios of Lemma 4.39(ii) and $w=D$, the minimum is 1024, attained only at $H=0$, so every such cut weighs at least 1024.

Proof. The left side is $\sum_{s\in S} w(s)\,|(T+s)\setminus T|$. Let H be the stabiliser of T, a proper subgroup since $T\neq\emptyset, G$. Then T and $T+s$ are unions of cosets of H, and for $s\notin H$ the set $(T+s)\setminus T$ is nonempty, since $|T+s|=|T|$, so it has at least $|H|$ elements. This gives the inequality. In G_0, written additively as the quadruples in $(\mathbb{Z}/4)^4$ of sum 0, the ratios of type $(i,-i,-1,-1)$ generate, since the differences of two of them give every e_j-e_k. The weights are 256 once and 64 otherwise, and $|S\cap H|\le|H|-1$. So $|H|\,w(S\setminus H)$ is 1024 for $H=0$, at least $|H|\,(1024-256-64(|H|-2))$, which is 1536, 2560 and 3072, for $|H|=2,4,8$, and at least $64|H|\ge 1024$ for $|H|\ge 16$, because some generator lies outside H.[19] ∎

Theorem 4.41 (One parity at two or three values)**.** *Keep Lemma 4.36, and suppose that the shifts of the pieces L_ζ with $\prod\zeta=\epsilon$ take exactly two values s and $s-2$, or exactly three values $s+2$, s and $s-2$, for a sign ϵ and an integer s. Then, for every order of the pieces, all shifts of the other pieces, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E)\ \ge\ 1024.$$

In particular E does not meet the polarised criterion.

Proof. Let T be the set of pieces of parity ϵ at the shift s; it is neither empty nor all of them. Two pieces of parity ϵ, one in T and the other not, have shifts differing by exactly two. When their ratio is one of the thirteen of Lemma 4.39(ii), the group $\mathcal{H}^4$ from the piece of the larger shift to the other is reached by no coboundary and killed by no term, so its classes inject into $\mathrm{Ext}^2(E,E)$, and those of different pairs lie in different summands $\mathcal{H}(a,b)$. Their dimensions add up to the weight of the cut $(T,\complement T)$ of the Cayley graph of Lemma 4.40, since the pieces of parity ϵ form a coset of G_0; so they add up to at least 1024. ∎

Theorem 4.42 (Shifts three apart)**.** *Keep Lemma 4.36, and suppose that the pieces L_ζ with $\prod\zeta=\epsilon$ all have the shift s and the others have shifts in $\{s-3,s+3\}$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E,E)\ \ge\ 64\cdot 1072-128\cdot 8=67584.$$

Proof. Every piece W of parity $-\epsilon$ and every piece V of parity ϵ have shifts differing by three. By Lemma 4.39(iii) the classes of degree two of the group from the higher to the lower of the two lie in $\mathcal{H}^5$, and d_E vanishes on them. For each W these groups have dimensions adding up to 1072.[20] A coboundary with a component in such a group runs along a path of morphisms of degree one that lowers the shift by three; no multiple lies on it, by Lemma 4.36(iii), and a step between distinct pieces lowers the shift by at least one, while no piece has a shift strictly between those of W and V. So the path is the component between W and V together with a class of $H^1(A,\mathcal{O}_A)$ at one of them, and the components of all coboundaries in these groups come from $\bigoplus_a \mathcal{H}^1(a,a)$ over the 128 pieces, of dimension $128\cdot 8=1024$. The $64\cdot 1072=68608$ classes are cocycles, and at most 1024 dimensions of them are coboundaries. ∎

[19] The orders of the subgroups of $G_0\cong(\mathbb{Z}/4)^3$ are powers of two. A cut at one vertex weighs exactly 1024, so the bound is sharp.

[20] Let the ratio move k coordinates. For $k=1$ it is -1 in one coordinate, $D=4$, and $\mathcal{H}^5$ is H^1 tensored with the part of degree four of $H^*(\mathcal{O})$ on the other three surfaces, of dimension $\binom{6}{4}=15$: four partners with 60 each. For $k=2$ it is (i,i) or $(-i,-i)$, $D=4$, and the part of degree three on the other two surfaces has dimension $\binom{4}{3}=4$: twelve partners with 16 each. For $k=3$ the part of degree two on the last surface has dimension one, and the 28 partners give 640 as in Theorem 4.37. For $k=4$ the group sits in degree 4. So $240+192+640=1072$.

When one parity sits at a single shift, the groups $\mathcal{H}^3$ of Theorem 4.37 survive next to it unless pieces of the other parity crowd it from both sides.

Proposition 4.43 (A gap of four)**.** *Keep Lemma 4.36, let the pieces L_ζ with $\prod\zeta=\epsilon$ all have the shift y, and let T be the set of pieces of parity $-\epsilon$ at the shift $y+1$. If T is not empty and no piece has the shift $y-1$ or $y+3$, then for every order, every choice of the shifts of the M_i and every placement of the t_i, the groups $\mathcal{H}^3(W,V)$ with $W\in T$ and V differing from W in exactly three coordinates are reached by no coboundary and killed by no term, and*

$$\dim\operatorname{Ext}^2(E,E)\ \geq\ 640\,|T|\ \geq\ 640.$$

The same holds with $y+1$, $y-1$, $y+3$ replaced by $y-1$, $y+1$, $y-3$ and the groups $\mathcal{H}^3(V,W)$.

Proof. By Theorem 4.37(i), which holds for every convolution, no coboundary reaches these groups. By Lemma 4.39(iv) a term of d_E on $\mathcal{H}^3(W,V)$ runs along chains of components between pieces into W and out of V that lower the shift by one or two in all; the first would pass through a piece at the shift $y+2$ or $y+3$, the second through one at $y-1$ or $y-2$. The shifts $y\pm2$ have parity ϵ, whose pieces are all at y, and $y-1$, $y+3$ are excluded. Each $W\in T$ has 28 partners V, all at the shift y, with groups of dimensions adding up to 640. The second statement has the same proof, with the chains into V and out of W in place of those into W and out of V. ∎

Corollary 4.44 (Five consecutive shifts)**.** *Keep Lemma 4.36. If the shifts of the* 128 *pieces L_ζ lie in five consecutive integers, then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim\operatorname{Ext}^2(E,E)\geq640$; no such convolution meets the polarised criterion.*

Proof. Let the values be $s,\dots,s-4$, with one parity at values in $\{s,s-2,s-4\}$ and the other at values in $\{s-1,s-3\}$. If the second parity takes both its values, or the first takes two values two apart or all three, Theorem 4.41 applies. Otherwise the second parity sits at one value y and the first at one value or at $\{s,s-4\}$. One value each, at an odd distance at most three: Theorem 4.37 or Theorem 4.42. The first at $\{s,s-4\}$: for $y=s-1$ the pieces at $y+1=s$ are not empty and $y-1$, $y+3$ are empty, and for $y=s-3$ the pieces at $y-1=s-4$ are not empty and $y+1$, $y-3$ are empty, so Proposition 4.43 applies. The least of the bounds is 640. ∎

Within six consecutive values two arrangements are left, up to exchanging the parities: each parity at one shift, five apart, and the parities at $\{s,s-4\}$ and $\{s-1,s-5\}$. The first is settled by the diagonal, whose classes at a shift with no other piece nearby can be reached only through all three multiples, and the second by the groups of Theorem 4.37 once more.

Lemma 4.45 (The diagonal at a lonely shift)**.** *Keep Lemma 4.36, and let the pieces L_ζ with $\prod\zeta=\epsilon$ all have the shift s and the others the shift $s-g$, for a sign ϵ and an odd integer $g\geq5$. Let $y\in\mathcal{H}^2(a,a)$ be a diagonal class at a piece a, and let a term of $d_E(y)$ in Lemma 4.24(iii) be nonzero. Then a is the only piece on its path, the path passes through M_1, M_2 and M_3, and the shifts of a, M_1, M_2, M_3 are pairwise incongruent modulo* 4.

Proof. Write the path as $c\to\cdots\to a\to\cdots\to e$, with the chain of components into a and the chain out of a not both empty, so that $c\neq e$, since the components of a convolution go up the order of the pieces. Let L_1 and L_2 be the first and the last piece on it and $X=d_{L_1}-d_{L_2}$; before L_1 and after L_2 the path is a run of distinct multiples, with U steps

up and D steps down the value ϕ in all. As in the proof of Lemma 4.39(i), now with $\delta = 0$, the term has Ext degree $3 + X - U + 7D$ by Lemma 4.36(ii). A sequence of components from one piece to another lowers the shift by Lemma 4.36(iii), and the pieces have the shifts s and $s - g$; so $X \in \{0, g\}$, and $X = 0$ exactly when a is the only piece on the path.

If c and e are pieces, then $U = D = 0$ and $X = g$, and $\mathcal{H}^{3+g}(c, e) = 0$ since $3 + g \geq 8$ and $c \neq e$. If c is a multiple and e a piece, then $\mathcal{H}(c, e)$ sits in degree 0 or 8. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$: for $X = 0$ this gives $D = 1$ and $U = 2$; for $X = g$ it gives $D = 0$ and $U = g - 5$, which is at least one because the run before L_1 is not empty and at most three, so $g = 7$; but then the path lowers the shift by seven from L_1 to e, by a single component, of Ext degree eight between distinct pieces, or by a run through multiples, and neither is possible.[21] A piece c with a multiple e is the same with the runs exchanged. If c and e are multiples, both runs are nonempty and together pass through at most three multiples, so $U + D \leq 3$ and $U + D \geq 2$. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$; $D = 0$ gives $U = X - 5 \geq 2$, so $X = g = 7$, which is excluded as before; $D = 1$ gives $U = 2 + X \leq 2$, so $X = 0$ and $U = 2$; $D \geq 2$ gives $U \geq 9$.

So in every case $X = 0$ and $U + D = 3$: the path consists of a and the three multiples, with three steps. A step up raises the shift by one and a step down lowers it by seven, so every step adds one modulo 4, and the four nodes of the path have the shifts $d_c, d_c + 1, d_c + 2, d_c + 3$ modulo 4 in the order of the path.[22] ∎

Proposition 4.46 (Single shifts)**.** *Keep Lemma 4.36, and suppose that the pieces L_ζ with $\prod \zeta = \epsilon$ all have the shift s and the others all have the shift $s - g$, for a sign ϵ and an odd integer g. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i,*

$$\dim \mathrm{Ext}^2(E, E) \;\geq\; 64 \cdot 28 = 1792.$$

Proof. Exchanging the parities, we may take $g > 0$. For $g = 1$ and $g = 3$ this is Theorem 4.37 and Theorem 4.42. Let $g \geq 5$. If a term of d_E were nonzero on a diagonal class at a piece of shift s and another on a diagonal class at a piece of shift $s - g$, then by Lemma 4.45 the shifts of the three multiples would be pairwise incongruent modulo 4 and miss both the residue of s and that of $s - g$, which differ, since g is odd. So at one of the two shifts d_E vanishes on the 28 diagonal classes of each of the 64 pieces there, and by Proposition 4.29, which holds for $n = 4$, these 1792 classes inject into $\mathrm{Ext}^2(E, E)$.[23] ∎

Proposition 4.47 (Interleaved shifts)**.** *Keep Lemma 4.36, and suppose that the shifts of the pieces L_ζ with $\prod \zeta = \epsilon$ take exactly the two values s and $s - 4$, and those of the others exactly the two values $s - 1$ and $s - 5$. Then, for every order, every choice of the shifts of the M_i and every placement of the t_i, $\dim \mathrm{Ext}^2(E, E) \geq 640$.*

[21] Between distinct pieces $\mathcal{H}^8 = 0$. A run from a piece through $m \leq 3$ distinct multiples to a piece has $m + 1$ steps, at least one up and one down, and lowers the shift by $7D' - U'$ with $U' + D' = m + 1$: by 6, 5, 4 for $D' = 1$, by 13, 12 for $D' = 2$ and by 20 for $D' = 3$, never by seven.

[22] With the values $t_1 < t_2 < t_3$ and p, the path with $U = 3$ visits the four nodes in the order of their values; a path with $U = 2$ and $D = 1$ ending in degree eight visits the largest j values in increasing order and then the others, $j = 1, 2, 3$. So each placement has exactly four such paths, and for each the shifts of the multiples are determined by d_a.

[23] The proof does not look at the classes between the two parities. For $g = 5$ these are $64 \cdot 16 = 1024$ classes of degree two in groups $\mathcal{H}^7$, on which d_E vanishes by Lemma 4.39(i); but the coboundaries of the loops in $\mathcal{H}^1(a, a)$, and, when two multiples have suitable shifts, the products along runs $W \to M_i \to M_j \to V$, reach them, and counting alone gives nothing there.

Proof. Let A and A' be the pieces of parity ϵ at the shifts s and $s-4$, and B and B' those of parity $-\epsilon$ at $s-1$ and $s-5$; all four are nonempty. Let W and V differ in exactly three coordinates, with $d_W = d_V + 1$, that is $W \in A$ and $V \in B$, or $W \in A'$ and $V \in B'$. By Theorem 4.37(i) no coboundary reaches $\mathcal{H}^3(W, V)$, and by Lemma 4.39(iv) a term of d_E on it passes through a piece at the shift $d_W + 1$ or $d_W + 2$ or at $d_V - 1$ or $d_V - 2$. These are $s+1$, $s+2$, $s-2$, $s-3$ for $W \in A$ and $s-3$, $s-2$, $s-6$, $s-7$ for $W \in A'$, and no piece has these shifts. So the groups inject into $\mathrm{Ext}^2(E, E)$.

Let Γ be the group of the ratios of two pieces, of order 128, which acts simply transitively on the pieces, and $S_3 \subset \Gamma$ the 28 ratios that move exactly three coordinates and change the parity, with the weight D, which adds up to 640 (Theorem 4.37). With $X = A \cup B'$ the pairs above are exactly the edges of the Cayley graph of (Γ, S_3) between X and its complement $A' \cup B$; so the dimensions add up to the weight of that cut, and X is neither empty nor everything. Written additively, as the quadruples in $(\mathbb{Z}/4)^4$ of even sum, the differences of two ratios of type $(i, -i, -1, 1)$ give every $e_j - e_k$, which generate the quadruples of sum 0 modulo 4, and the elements of S_3 have sum 2 modulo 4; so S_3 generates Γ. In Lemma 4.40, $H = 0$ gives 640; for $|H| = 2, 4, 8, 16$ at most $|H| - 1$ elements of S_3 lie in H, at most four of them of weight 64, so $|H|\, w(S_3 \setminus H)$ is at least $2 \cdot 576$, $4 \cdot 448$, $8 \cdot 336$ and $16 \cdot 208$; for $|H| = 32$ some $r \in S_3$ lies outside H, and so does $-r$, which gives weight at least 32, or 64 if $r = -r$; and for $|H| = 64$ the weight outside H is at least 16. So every such cut weighs at least 640.[24] ∎

Corollary 4.48 (Six consecutive shifts)**.** *Keep Lemma 4.36. If the shifts of the* 128 *pieces L_ζ lie in six consecutive integers, then, for every order, every choice of the shifts of the M_i and every placement of the t_i,* $\dim \mathrm{Ext}^2(E, E) \geq 640$*; no such convolution meets the polarised criterion.*

Proof. By Corollary 4.44 we may assume that the shifts take both values s and $s-5$ of the window $s, \dots, s-5$; let the parity at s have its values in $\{s, s-2, s-4\}$ and the other in $\{s-1, s-3, s-5\}$. If one of them takes two values two apart, or three, Theorem 4.41 applies. Otherwise the first takes $\{s\}$ or $\{s, s-4\}$ and the second $\{s-5\}$ or $\{s-1, s-5\}$. For $\{s\}$ and $\{s-5\}$ this is Proposition 4.46, and for $\{s, s-4\}$ and $\{s-1, s-5\}$ it is Proposition 4.47. For $\{s\}$ and $\{s-1, s-5\}$, Proposition 4.43 applies in its second form with $y = s$: the shift $s-1$ is taken and $s+1$, $s-3$ are not. For $\{s, s-4\}$ and $\{s-5\}$ it applies in its first form with $y = s-5$: the shift $s-4$ is taken and $s-6$, $s-2$ are not. The least of the bounds is 640. ∎

For the arrangements outside these results the diagonal still carries a large part of the obstruction.

Proposition 4.49 (Cup products on the diagonal at $n = 4$)**.** *Keep Lemma 4.36, so that D^2 has dimension* $131 \cdot 28 = 3668$*. For every convolution the cup products vanish on the* 84 *diagonal classes of the M_i, and on the* 3584 *diagonal classes of the L_ζ they have a kernel of dimension at least* 400*. So their kernel on D^2 has dimension at least* 484*, and a convolution meeting the polarised criterion needs products of length at least three that remove at least* 380 *diagonal classes.*

Proof. As in the proof of Proposition 4.30, the cup products in $\mathcal{H}(a, b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, they vanish on the classes of the M_i, and for L_ζ, $L_{\zeta'}$ and a type τ of degree two on the four surfaces they vanish on the part of type τ unless ζ'/ζ fixes the coordinates in the

[24] The subgroups of $\Gamma \cong (\mathbb{Z}/4)^3 \times \mathbb{Z}/2$ have orders that are powers of two. A cut at a single vertex weighs exactly 640, so the bound is sharp.

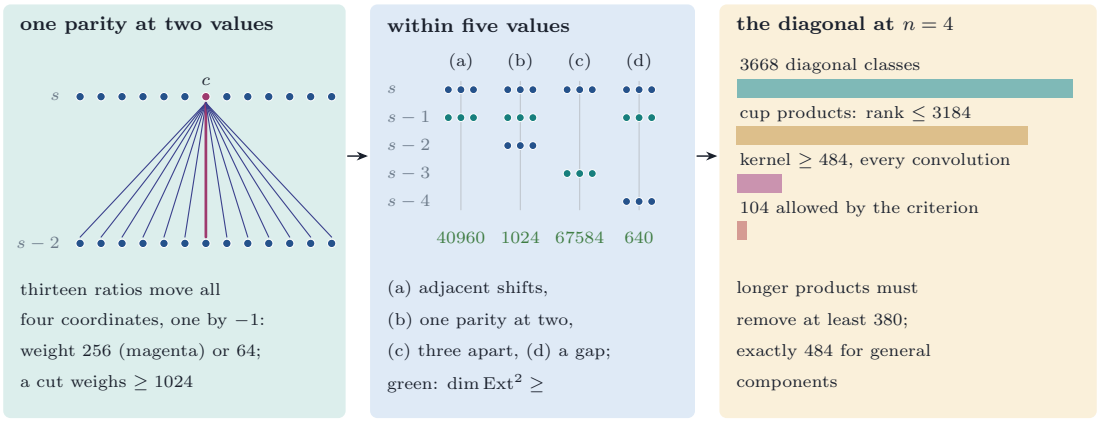


FIGURE 10. The shift arrangements of the pieces L_ζ on E_0^8. Left: one parity at two values two apart; a piece c is joined to the thirteen pieces whose ratio moves every coordinate, one by -1, and the groups $\mathcal{H}^4$ of the edges crossing between the values survive, weighing at least 1024 (Lemma 4.39(ii), Lemma 4.40). Middle: the four kinds of arrangement within five consecutive values, with the result that excludes each and its bound on $\dim \mathrm{Ext}^2$. Right: the diagonal at $n = 4$; the cup products leave at least 484 of the 3668 classes, and the criterion allows 104 (Proposition 4.49).

support of τ. So a class whose part of type τ depends only on the coordinates of ζ in the support of τ has vanishing cup products. The four types $(2,0,0,0)$ have dimension one and four values of the coordinate, the six types $(1,1,0,0)$ have dimension four and sixteen values of the two coordinates: $4 \cdot 4 + 6 \cdot 16 \cdot 4 = 400$. By Proposition 4.29, which holds for $n = 4$, the kernel of d_E on D^2 injects into $\mathrm{Ext}^2(E,E)$, and it is at most 104 only if the longer products remove 380 dimensions of these 484. ∎

Figure 10 and Figure 11 draw the arrangements and the bounds.

Remark 4.50 (What the route gives). At $n = 3$ the route through convolutions of these pieces is decided. The diagonal obstruction of Theorem 4.31 leaves room only for the fourfold products along one path, and the explicit convolution of Proposition 4.33 solves the Maurer–Cartan equation. The obstruction that decides the route lies off the diagonal, in classes that no product of any length can reach: those between pieces that differ in every coordinate and have adjacent shifts (Lemma 4.34), and, at pieces whose shifts are not adjacent to the shift of the paths, the classes of type $(1,1,0)$. No order and no choice of shifts or twists avoids both, and Theorem 4.35 excludes every convolution of the 35 pieces; in the explicit convolution the first kind alone gives 2560 classes against the 57 allowed. In each placement exactly two of the eight sign patterns carry products, and the two are exchanged by a symmetry of the construction.[25] The member E_0^6 with this Weil structure has trivial discriminant, where the Weil classes are algebraic on every member by Markman's theorem [Mar25a, Mar25b], so a convolution meeting the criterion here would have given a second proof of a known case rather than a new one. At $n = 4$ the

[25]The automorphism of A that is -1 on the coordinate z_{n+1} and the identity on the others changes α into $-\alpha$, fixes θ and permutes the L_ζ, changing ζ_1 into $-\zeta_1$; composed with the shift $[1]$ it changes every σ_i into $-\sigma_i$ and keeps everything else.

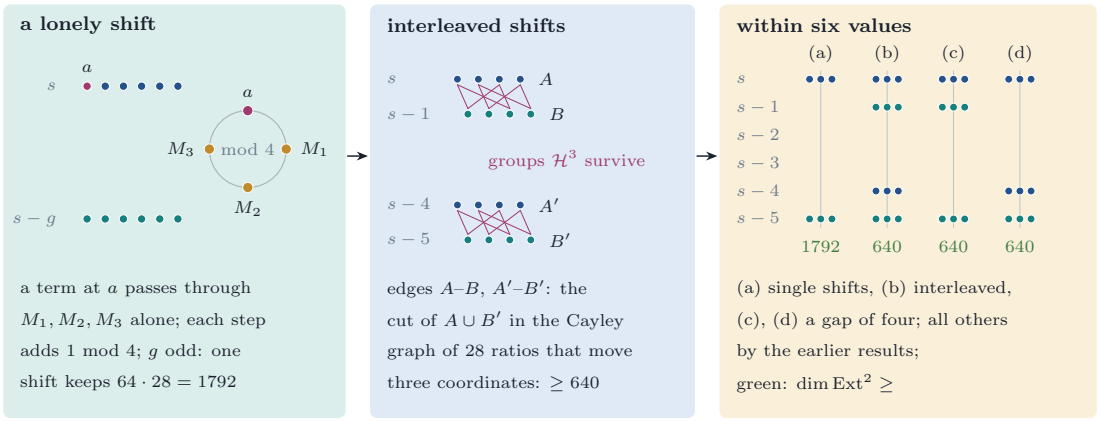


FIGURE 11. The arrangements within six consecutive values on E_0^8. Left: the parities at single shifts s and $s-g$, g odd and at least five; a term on a diagonal class at a piece a runs through M_1, M_2, M_3 and no other piece, and since each step adds one to the shift modulo 4 the four nodes have distinct residues, so the multiples serve at most one of the two shifts and the 1792 diagonal classes at the other survive (Lemma 4.45, Proposition 4.46). Middle: the interleaved shifts; the groups $\mathcal{H}^3$ between A and B and between A' and B' survive and form a cut of the Cayley graph of the 28 ratios that move three coordinates, of weight at least 640 (Proposition 4.47). Right: the four arrangements left within six values by the results for five values, each with its bound on $\dim \mathrm{Ext}^2$ (Corollary 4.48).

classes between pieces that differ in three coordinates and have adjacent shifts are again reached by nothing, in every convolution, and when the pieces L_ζ occupy two shifts, as in the explicit convolution, no term of d_E leaves them, so that $\dim \mathrm{Ext}^2(E,E) \geq 40960$ against the 104 allowed (Theorem 4.37). With three consecutive shifts the differential reaches these classes, but only through cup products into the groups between the highest and the lowest shift, and at least 22528 of them survive (Theorem 4.38). The groups of spread three, and those of spread two whose ratio moves every coordinate, one of them by -1, are untouched as well, in every convolution (Lemma 4.39): the latter cross every cut between two shift values of one parity, and weigh at least 1024 there (Theorem 4.41); with one parity at a single shift, those of spread three, or of spread one beside a gap, survive in their turn (Theorem 4.42, Proposition 4.43). Together they exclude every convolution whose pieces L_ζ have their shifts within five consecutive values (Corollary 4.44). The two arrangements left within six values fall as well: with each parity at a single shift, at any odd distance, the diagonal classes at one of the two shifts can be reached only through all three multiples, and the multiples serve at most one of them (Lemma 4.45, Proposition 4.46); the interleaved shifts leave the groups of Theorem 4.37 on a cut (Proposition 4.47). So every convolution whose shifts lie in six consecutive values fails (Corollary 4.48). For wider arrangements the cup products still leave 484 diagonal classes, so products of length at least three would have to remove 380 (Proposition 4.49), and products of length three and four among the L_ζ pass every test of Theorem 4.31. Whether some convolution of the 131 pieces whose shifts do not lie in six consecutive values, and whose parities do not each sit at one shift, meets the criterion at $n = 4$ is open. Markman's theorem does not reach the

eightfolds of Weil type, so such a convolution would give a new case. Figure 6 shows the excess count and the surviving path, and Figure 7 the budget of diagonal classes.

Markman's candidate in dimension eight. Above $n = 3$ there is one concrete candidate of this kind. Markman proposes it in [Mar25b, Section 12] and notes that it remains to be checked against the weakened criterion. Its Chern character is a finite computation in the ring $\mathbb{Q}[\Theta]/(\Theta^5)$. We carry the computation out: it places the candidate at a rational point of the secant plane of [I, Definition 14.4] and reduces what has to be checked to a single condition, which Theorem 4.61 decides in the negative.

Setup 4.51 (Markman's object)**.** Let (X, Θ) be a principally polarised abelian fourfold, $d \geq 3$ odd and squarefree, $K = \mathbb{Q}(\sqrt{-d})$ and $N = (d+9)/2$. Let D_i, $i \in \mathbb{Z}/N\mathbb{Z}$, be N translates of Θ in general position: any two, three or four of them meet transversally, in a surface, a curve and $\Theta^4 = 24$ points, and any five have empty intersection. Put $Z_i = D_i \cap D_{i+1}$, $Z = \bigcup_i Z_i$, let $C_i = Z_i \cap Z_{i+1} = D_i \cap D_{i+1} \cap D_{i+2}$ be the curves along which consecutive components meet, and let $\nu\colon \widetilde{Z} \to Z$ be the partial normalisation of Z at $m = (d+9)(2d-1)$ of its isolated points of self-intersection, the points of $Z_i \cap Z_j$ for i, j at cyclic distance at least three. Put

$$\mathcal{F} = [\mathcal{O}_X \longrightarrow \nu_*\mathcal{O}_{\widetilde{Z}}] \otimes \mathcal{O}_X(3\Theta),$$

the complex placed in degrees 0 and 1.

Proposition 4.52 (Markman's candidate on the secant plane)**.** *In Setup 4.51, with $t = \Theta$ and u_t, v_t as in* [I, (30)] *for $n = 4$:*

(i) $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$, *the isolated self-intersection points of Z number* $12N(N-5) = (d+9)(3d-3)$, *and* $\chi(\mathcal{O}_{\widetilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(27-d)$;

(ii) $\operatorname{ch}(\nu_*\mathcal{O}_{\widetilde{Z}}) = N\Theta^2 - 2N\Theta^3 + \chi(\mathcal{O}_{\widetilde{Z}})\,[\mathrm{pt}]$;

(iii) $\operatorname{ch}(\mathcal{F}) = u_t + 3v_t$, *that is*

$$\operatorname{ch}(\mathcal{F}) = 1 + 3\Theta - \tfrac{d}{2}\Theta^2 - \tfrac{d}{2}\Theta^3 + \tfrac{d^2}{24}\Theta^4,$$

the rational point $(a, b) = (1, 3)$ of the secant plane P_Θ; with the twist $\mathcal{O}_X(\Theta)$ in place of $\mathcal{O}_X(3\Theta)$ the degree two component would instead be $-\frac{d+8}{2}\Theta^2$, so the twist in [Mar25b, Section 12] *must be read as 3Θ;*

(iv) $\mathrm{N}_{K/\mathbb{Q}}(3 + \sqrt{-d}) = d + 9 = 2N$, *with N the number of divisors, and by* [I, Theorem 14.7] *the Bogomolov discriminant of $\mathcal{F}$ is $\Delta(\mathcal{F}) = (d+9)\Theta^2$;*

(v) *the objects $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ and $\Phi(\mathcal{F} \boxtimes \mathcal{F}^\vee)$ on $X \times \widehat{X}$ have ranks $\int_X \operatorname{ch}(\mathcal{F})^2 = 8d(d-9)$ and $\int_X \operatorname{ch}(\mathcal{F})\operatorname{ch}(\mathcal{F}^\vee) = 8d(d+9)$, up to sign, both nonzero for squarefree d; the same formula gives $-8q$ for the pair of secant sheaves of* [Mar25b, Section 11] *on a threefold, which is his rank $8q$ up to the shift.*

Consequently $\mathcal{F} \boxtimes \mathcal{F}$ satisfies conditions (1), (2b) and (2c) of Markman's strategy [Mar25b, Section 4]*: the transform has nonzero rank, its normalised Chern character is invariant under the special Mumford–Tate group by* [Mar25b, Corollary 9.2]*, and its Weil component is nonzero by* [Mar25b, Proposition 10.2]*, the relevant coefficient being* $\mathrm{N}(c)$ *with* $c = \frac{1}{2}(1+3/\sqrt{-d}) \neq 0$. *What remains is condition (2a) in its weakened form: for $E = \Phi(\mathcal{F}\boxtimes\mathcal{F})$, the map ev_E must kill every class in $H^1(T) \oplus H^0(\bigwedge^2 T)$ that preserves $\kappa(E)$ to first order.*

Proof. On a principally polarised abelian fourfold $\chi(\mathcal{O}_X(k\Theta)) = k^4$ and $\Theta^4 = 24$. The Koszul complex of r transversal translates of Θ gives ch of the structure sheaf of their intersection as $(1 - e^{-\Theta})^r$, so $\operatorname{ch}(\mathcal{O}_{Z_i}) = \Theta^2 - \Theta^3 + \frac{7}{12}\Theta^4$, $\operatorname{ch}(\mathcal{O}_{C_i}) = \Theta^3 - \frac{3}{2}\Theta^4$ and ch of

four divisors is Θ^4. Integrating gives $\chi(\mathcal{O}_{Z_i}) = 14$, $\chi(\mathcal{O}_{C_i}) = -36$ and 24 points; the same numbers are the Koszul sums $0 - 2 \cdot 1 + 16$, $0 - 3 + 48 - 81$ and $0 - 4 + 96 - 324 + 256$ of the values $\chi(\mathcal{O}_X(-k\Theta)) = k^4$.

(i), (ii). At every point Z is locally a union of coordinate subspaces of $\mathbb{C}^4$: along C_i two surfaces $\{x_1 = x_2 = 0\} \cup \{x_2 = x_3 = 0\}$, at a point of $C_i \cap C_{i+1}$ three of them, at an isolated point $\{x_1 = x_2 = 0\} \cup \{x_3 = x_4 = 0\}$. The ideals of such an arrangement are monomial and distribute over sums, so the Mayer–Vietoris complex

$$0 \to \mathcal{O}_Z \to \bigoplus_i \mathcal{O}_{Z_i} \to \bigoplus_{i<j} \mathcal{O}_{Z_i \cap Z_j} \to \bigoplus_{i<j<k} \mathcal{O}_{Z_i \cap Z_j \cap Z_k} \to \cdots$$

with scheme-theoretic intersections is exact. Pairs at cyclic distance one meet in C_i, N of them; pairs at distance at least two meet in the 24 points of four divisors, $N(N-3)/2$ of them; the only triples with nonempty intersection are the consecutive ones $Z_i \cap Z_{i+1} \cap Z_{i+2} = C_i \cap C_{i+1}$, 24 points each, N of them; and no four components meet, since that would involve at least five divisors. Hence

$$\operatorname{ch}(\mathcal{O}_Z) = N \operatorname{ch}(\mathcal{O}_{Z_i}) - N \operatorname{ch}(\mathcal{O}_{C_i}) - \tfrac{N(N-3)}{2}\Theta^4 + N\Theta^4,$$

whose degree two part is $N\Theta^2$, degree three part $-N\Theta^3 - N\Theta^3 = -2N\Theta^3$, and degree four part $\chi(\mathcal{O}_Z)[\mathrm{pt}]$ with $\chi(\mathcal{O}_Z) = 14N + 36N - 12N(N-3) + 24N = 110N - 12N^2$. The isolated singular points are those of pairs at distance at least three, of which there are $N(N-5)/2$, giving $12N(N-5) = 3(d+9)(d-1)$ points; the pairs at distance two lie on the double curves and are not isolated. At an isolated point the two branches meet transversally, and separating them gives $0 \to \mathcal{O}_Z \to \nu_*\mathcal{O}_{\widetilde{Z}} \to k(p) \to 0$, so each of the m normalisations raises χ by one and leaves the lower degrees of ch unchanged. With $N = (d+9)/2$, $\chi(\mathcal{O}_{\widetilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(55 - 3(d+9) + 2d - 1) = (d+9)(27-d)$.

(iii) $\operatorname{ch}(\mathcal{F}) = e^{3\Theta}(1 - \operatorname{ch}(\nu_*\mathcal{O}_{\widetilde{Z}}))$. The degree two part is $(\frac{9}{2} - N)\Theta^2 = -\frac{d}{2}\Theta^2$, which is where $N = (d+9)/2$ enters; with the twist $k\Theta$ it would be $(\frac{k^2}{2} - N)\Theta^2$, equal to $-\frac{d}{2}\Theta^2$ exactly for $k^2 = 9$, and $k = -3$ is excluded by the degree three part. The degree three part is $(\frac{27}{6} - 3N + 2N)\Theta^3 = (\frac{9}{2} - N)\Theta^3 = -\frac{d}{2}\Theta^3$, and the degree four part is $(\frac{81}{24} - \frac{9N}{2} + 6N - \frac{\chi(\mathcal{O}_{\widetilde{Z}})}{24})\Theta^4 = \frac{81 + 36N - \chi(\mathcal{O}_{\widetilde{Z}})}{24}\Theta^4 = \frac{d^2}{24}\Theta^4$ by (i). Comparing with [I, (30)] at $n = 4$, $u_t = 1 - \frac{d}{2}t^2 + \frac{d^2}{24}t^4$ and $v_t = t - \frac{d}{6}t^3$, gives $\operatorname{ch}(\mathcal{F}) = u_t + 3v_t$.

(iv) This follows from $\mathrm{N}(3 + \sqrt{-d}) = 9 + d$ and [I, (31)] with $(a, b) = (1, 3)$.

(v) Orlov's equivalence is $\Phi = (\mathrm{id} \times \Phi_{\mathcal{P}})^{-1} \circ \mu^*$ with $\mu(x, y) = (x + y, y)$, so the restriction of $\Phi(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$ to $\{x\} \times \widehat{X}$ is the Fourier–Mukai transform of $\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2$, whose derived fibre at $L \in \widehat{X}$ is $R\Gamma(X, \tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L^{\pm 1})$; the rank of a perfect complex is the Euler characteristic of its derived fibre, and $\chi(\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L) = \int_X \operatorname{ch}(\mathcal{F}_1) \operatorname{ch}(\mathcal{F}_2)$ by Riemann–Roch, translations and $L \in \mathrm{Pic}^0$ acting trivially on cohomology. With $\operatorname{ch}(\mathcal{F}^\vee) = u_t - 3v_t$, the involution τ changing the sign of the odd degrees, $\int u_t^2 = 8d^2$, $\int u_t v_t = 0$ and $\int v_t^2 = -8d$ give the two values. On a threefold $\int (u_t + v_t)^2 = 2 \int u_t v_t = -8q$.

For the final assertion, condition (2b) is the $\mathrm{Spin}(V_{\mathbb{Q}})_B$-invariance of $\kappa(E)$, which [Mar25b, Corollary 9.2] derives from $\operatorname{ch}(\mathcal{F}) \in B$ and the nonvanishing of the rank. Condition (2c) is [Mar25b, Proposition 10.2], which for K imaginary quadratic reduces to the nonvanishing of the component of $\operatorname{ch}(\mathcal{F}) \otimes \operatorname{ch}(\mathcal{F})$ in $\ell_W \otimes \ell_{\overline{W}}$, and writing $\operatorname{ch}(\mathcal{F}) = c\,\ell_W + \bar{c}\,\ell_{\overline{W}}$ with $\ell_W = e^{\sqrt{-d}\Theta}$ gives that coefficient as $c\bar{c}$. ∎

Remark 4.53 (The remaining condition). Proposition 4.52 does not verify Markman's candidate. It verifies every statement about the candidate that concerns classes, and

it shows that the one remaining statement concerns the object. For $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ on the eightfold $X \times \widehat{X}$ of Weil type, with $\kappa(E) = \operatorname{ch}(E) e^{-c_1(E)/\operatorname{rk} E}$, that statement reads: every $x \in H^1(T) \oplus H^0(\bigwedge^2 T)$ with $x \lrcorner \kappa(E) = 0$ satisfies $\operatorname{ev}_E(x) = 0$ in $\operatorname{Ext}^2(E,E)$. The $H^2(\mathcal{O})$ summand imposes nothing, because $\operatorname{rk} E \neq 0$ makes $z \cdot \kappa(E) \neq 0$ for $z \neq 0$. By [I, Theorem 9.20] the condition fails as soon as E has a rank one direct summand whose class is not a multiple of the polarisation, so the first question about E is whether it is indecomposable. Markman's object at $n = 3$ is reflexive of rank $8q$ away from a divisor, and he proves the condition by descending to a quotient on which σ_E is injective on an invariant subspace [Mar25b, Proposition 11.5]. Whether the same can be done for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ is a question about the Ext groups of one object on one eightfold, not about any Hodge class. Theorem 4.56 below moves the question from the eightfold to the fourfold and from E to $\mathcal{F}$ itself: the condition is equivalent to the single identity (19) in $\operatorname{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2)$, the other half of it holding by construction. Theorem 4.61 shows that the identity fails, so the answer is negative.

Reduction to the fourfold. The condition of Remark 4.53 concerns an object on an eightfold. It is equivalent to a condition on the factor $\mathcal{F}$ on the fourfold, and there it splits into two conditions, one of which holds for Markman's object because the object is constructed uniformly over the moduli of principally polarised fourfolds. The reduction is not special to that object. It works for every secant object on every abelian n-fold with $n \geq 3$, and its ingredients are three computations in the Hodge algebra of a single abelian variety, which we carry out by hand. We first fix conventions.

Setup 4.54 (The weakened criterion for one object)**.** Let Y be an abelian variety and F a nonzero perfect complex on Y. Write $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$ as in (10), with $x = (z, v, \pi)$ acting on $H^*(Y, \mathbb{C})$ by $x \lrcorner$, and likewise $HT^1(Y) = H^{0,1}(Y) \oplus H^0(T_Y)$, with (z, θ) acting by $z\wedge$ and by the interior product $\theta \lrcorner$. Normalise the Atiyah class $\operatorname{At}(F) \in \operatorname{Ext}^1(F, F \otimes \Omega_Y^1)$ so that $\operatorname{ch}(F) = \operatorname{tr} \exp \operatorname{At}(F)$; for a line bundle L it is then $c_1(L)$, as in (8).[26] The evaluation map of [I, Theorem 9.19] is, under HKR,

$$\operatorname{ev}_F(z, v, \pi) = z \cdot \mathrm{id}_F + v \lrcorner \operatorname{At}(F) + \tfrac{1}{2}\, \pi \lrcorner \operatorname{At}(F)^2 \ \in\ \operatorname{Ext}^2(F, F), \tag{15}$$

the contraction of the polyvector with the exponential Atiyah class [Hua21, Theorem A], [BF08]; its restriction to $H^1(T_Y)$ is the obstruction map, whose kernel is the space of first order deformations of Y to which F extends [Tod09], [Mar25a, Section 8.3]. Put

$$\operatorname{Ann}_{HT^2}(\operatorname{ch} F) = \{x \in HT^2(Y) : x \lrcorner \operatorname{ch}(F) = 0\},$$

the Hochschild classes preserving $\operatorname{ch}(F)$ to first order. We say that F *satisfies the weakened criterion* if ev_F vanishes on $\operatorname{Ann}_{HT^2}(\operatorname{ch} F)$. By the commutative square $\sigma_F \circ \operatorname{ev}_F = (\,\cdot\,) \lrcorner \operatorname{ch}(F)$ of [BF08, Proposition 6.2.1 and Corollary 6.3.2], in the form quoted in [Mar25b, Lemma 11.3],

$$\operatorname{ev}_F(\operatorname{Ann}_{HT^2}(\operatorname{ch} F)) = \ker \sigma_F \cap \operatorname{im} \operatorname{ev}_F,$$

so the weakened criterion is exactly the hypothesis of [Mar25b, Question 11.4], that the restriction of σ_F to the image of ev_F is injective. For a split object this is Proposition 4.18(ii).

Lemma 4.55 (The B-field transform)**.** *Let $B \in H^{1,1}(Y)$. For $x = (z, v, \pi) \in HT^2(Y)$ put*

$$e^B x = \left(z + v \lrcorner B + \tfrac{1}{2}\, \pi \lrcorner B^2,\ v + \pi \lrcorner B,\ \pi\right), \tag{16}$$

[26] This is the convention of [Tod09] and of [Mar25a, footnote 14]. Buchweitz and Flenner use the opposite sign, which replaces $\operatorname{At}(F)$ by $-\operatorname{At}(F)$; this leaves $\operatorname{At}(F)^2$ unchanged and changes only the sign of the middle term of ev_F, which does not affect the vanishing statements below.

where $\pi \lrcorner B \in \mathrm{Hom}(H^{1,0}, H^{0,1}) = H^1(T_Y)$ is the contraction of one slot of π with B: for $\pi = \partial_a \wedge \partial_b$ it sends p_b to $\partial_a \lrcorner B$, p_a to $-\partial_b \lrcorner B$ and the other basis vectors to zero. Then e^B is a linear automorphism of $HT^2(Y)$ with inverse e^{-B}, and:

(i) *$x \lrcorner (w\, e^B) = \big((e^B x) \lrcorner w\big)\, e^B$ for every even class $w \in H^{\mathrm{ev}}(Y, \mathbb{C})$;*

(ii) *for a line bundle L with $c_1(L) = B$, $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$ and $\mathrm{Ann}_{HT^2}(\mathrm{ch}(F \otimes L)) = e^{-B}\,\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$; hence F satisfies the weakened criterion if and only if $F \otimes L$ does.*

The computation in (ii) is formal: it uses only that the twisted object has Atiyah class $\mathrm{At}(F) + B \cdot \mathrm{id}_F$ and Chern character $\mathrm{ch}(F)e^B$. It therefore applies verbatim to the twisted sheaves of [Mar25b, Section 2]*, with B rational. In particular, for F of nonzero rank r, the criterion for the twisted object $F \otimes L^{-1}$ with class $\kappa(F) = \mathrm{ch}(F)e^{-c_1(F)/r}$ is the same condition as the criterion for F with class $\mathrm{ch}(F)$.*

Proof. The computation takes place in a graded algebra on which the interior products $\iota_a = \partial_a \lrcorner$ act as odd derivations. It uses only that w is even; that B is even and central with $\iota_a\iota_b B = 0$, which holds for B of type $(1,1)$ and ∂_a, ∂_b of type $(1,0)$; and that the odd scalar classes $\iota_a B$ anticommute with every odd element. In $H^*(Y, \mathbb{C})$ these properties are clear. They also hold in $\bigoplus_{p,q} \mathrm{Ext}^p(F, F \otimes \Omega_Y^q) = \mathrm{Ext}^*(F,F) \otimes \bigwedge^* H^{1,0}$, on which the polyvectors act through the second factor by odd derivations, because the classes coming from $H^*(Y, \mathbb{C})$ are graded central in the Yoneda algebra. First, $\iota_a(B^k) = k(\iota_a B)B^{k-1}$, since $\iota_a B$ is odd and B is even, so $\iota_a e^B = (\iota_a B)e^B$. For w even this gives $v \lrcorner (we^B) = (v \lrcorner w)e^B + w \sum_a v(p_a)(\iota_a B)e^B = (v \lrcorner w + (v \lrcorner B)w)e^B$, and $z(we^B) = (zw)e^B$. For the bivector, with $\pi \lrcorner = \iota_a\iota_b$,

$$\iota_b(we^B) = (\iota_b w)e^B + w(\iota_b B)e^B,$$

and applying ι_a, using that $\iota_b w$ and $\iota_b B$ are odd,

$$\iota_a\iota_b(we^B) = \big[\iota_a\iota_b w + (\iota_a B)\iota_b w - (\iota_b B)\iota_a w + w\big((\iota_a\iota_b B) - (\iota_b B)(\iota_a B)\big)\big]e^B.$$

The middle two terms are $(\pi \lrcorner B) \lrcorner w$ for the element $\pi \lrcorner B$ of $\mathrm{Hom}(H^{1,0}, H^{0,1})$ displayed in the statement, acting as a derivation, and the last bracket is $\frac{1}{2}\iota_a\iota_b(B^2) = \iota_a((\iota_b B)B)$. Summing the three actions gives (i). Applying (i) with B replaced by $-B$ and then by B, and using $(\pi \lrcorner B) \lrcorner B = 2(\iota_a B)(\iota_b B) = \pi \lrcorner B^2$ when $\iota_a\iota_b B = 0$, shows $e^{-B}e^B = \mathrm{id}$.

(ii) The Atiyah class of $F \otimes L$ is $\mathrm{At}(F) + B \cdot \mathrm{id}_F$, the Atiyah class being additive on tensor products [BF03, Section 3], and the two summands commute, so $\exp \mathrm{At}(F \otimes L) = \exp \mathrm{At}(F) \cdot e^B$. By (i) in the algebra $\mathrm{Ext}^*(F,F) \otimes \bigwedge^* H^{1,0}$, with $w = \exp \mathrm{At}(F)$,

$$x \lrcorner \exp \mathrm{At}(F \otimes L) = \big((e^B x) \lrcorner \exp \mathrm{At}(F)\big)e^B,$$

and the right hand side has the same component in $\mathrm{Ext}^2(F,F) \otimes \bigwedge^0$ as $(e^B x) \lrcorner \exp \mathrm{At}(F)$, because $e^B - 1 = B + \frac{1}{2}B^2 + \cdots$ raises the exterior degree. This is $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$. By (i) in $H^*(Y, \mathbb{C})$ with $w = \mathrm{ch}(F)$, $x \lrcorner \mathrm{ch}(F \otimes L) = 0$ if and only if $(e^B x) \lrcorner \mathrm{ch}(F) = 0$, which is the statement on the annihilators. Hence $\mathrm{ev}_{F\otimes L}(\mathrm{Ann}_{HT^2}(\mathrm{ch}(F \otimes L))) = \mathrm{ev}_F(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F))$. ∎

Theorem 4.56 (Reduction to the factor)**.** *Let (X, t) be a polarised abelian variety of dimension $n \geq 3$, let $d \geq 1$, and let F be a perfect complex on X with $\mathrm{ch}(F) = a\, u_t + b\, v_t$ for some $(a, b) \in \mathbb{Q}^2 \setminus \{0\}$, in the notation of* [I, (30)]*. Then:*

(i) *No nonzero class in $HT^1(X)$ preserves $\mathrm{ch}(F)$: the map $HT^1(X) \to H^*(X, \mathbb{C})$, $y \mapsto y \lrcorner \mathrm{ch}(F)$, is injective. Its two lowest components form the matrix $\left(\begin{smallmatrix} a & b \\ b & -ad \end{smallmatrix}\right)$ of determinant $-(a^2 d + b^2) = -\mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d}) \neq 0$.*

(ii) *The classes in* $HT^2(X)$ *preserving* $\mathrm{ch}(F)$ *to first order are*

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F) = \{(0,v,0) : v \lrcorner t = 0\} \;\oplus\; \{(\tfrac{d}{2}\,\pi \lrcorner t^2,\, 0,\, \pi) : \pi \in H^0(\textstyle\bigwedge^2 T_X)\}, \tag{17}$$

the polarised first order deformations of (X,t)*, of dimension* $n(n+1)/2$*, and the Poisson directions each compensated by a class in* $H^{0,2}$*, of dimension* $n(n-1)/2$*; the total is* $n^2 = \dim \mathcal{D}_{n,n}$*, and the space does not depend on* (a,b)*.*

(iii) F *satisfies the weakened criterion if and only if*

(a) $v \lrcorner \mathrm{At}(F) = 0$ *in* $\mathrm{Ext}^2(F,F)$ *for every* $v \in H^1(T_X)$ *with* $v \lrcorner t = 0$*, that is,* F *extends to first order along every polarised deformation of* (X,t)*; and*

(b) $\mathrm{At}(F)^2 = -d\,t^2 \cdot \mathrm{id}_F$ *in* $\mathrm{Ext}^2(F, F \otimes \Omega_X^2)$*.*

The trace of (b) is automatic: $\operatorname{tr} \mathrm{At}(F)^2 = 2\,\mathrm{ch}_2(F) = -ad\,t^2 = \operatorname{tr}(-d\,t^2 \cdot \mathrm{id}_F)$*, so (b) is the vanishing of the trace free part of the square of the Atiyah class.*

(iv) *Let* F_1, F_2 *be two such complexes,* $G = F_1 \boxtimes F_2$ *on* $X \times X$ *and* $E = \Phi(G)$ *on* $X \times \widehat{X}$*, with* Φ *Orlov's equivalence as in Proposition 4.52(v). Then* $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G) = \mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_1) \otimes 1 \oplus 1 \otimes \mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_2)$*, of dimension* $2n^2$*, and*

$$E \text{ satisfies the weakened criterion} \iff G \text{ does} \iff F_1 \text{ and } F_2 \text{ do.}$$

Proof. Choose a basis $p_1, \dots, p_n$ of $H^{1,0}(X)$ and a basis $q_1, \dots, q_n$ of $H^{0,1}(X)$ with $t = \sum_a p_a \wedge q_a$, which is possible because t is a positive class of type $(1,1)$; let ∂_a be the dual basis of T, so that $\iota_a p_b = \delta_{ab}$ and $\iota_a q_b = 0$. Write $\mathrm{ch}(F) = \sum_k c_k t^k$ with $c_k\, k! = a(-d)^{k/2}$ for k even and $c_k\, k! = b(-d)^{(k-1)/2}$ for k odd, so that

$$c_{k+1}\,(k+1)! = -d\,c_{k-1}\,(k-1)! \qquad (k \geq 1). \tag{18}$$

Two facts about the Hodge algebra are used throughout: for $v \in H^1(T)$ acting as a derivation, $v \lrcorner t^k = k\,(v \lrcorner t)\,t^{k-1}$; and for a bivector π of type $(2,0)$ in the tangent directions, $\pi \lrcorner t = 0$ because t has a single $(1,0)$ slot in each term, so $\pi \lrcorner t^k = \binom{k}{2}(\pi \lrcorner t^2)\,t^{k-2}$ by the footnote to (10). Finally, multiplication by t is injective on $H^{0,1}$ when $n \geq 2$ and on $H^{0,2}$ when $n \geq 3$: $q_a \wedge t$ contains the monomial $p_c q_c q_a$ for any $c \neq a$, and $q_a q_b \wedge t$ contains $p_c q_c q_a q_b$ for any $c \notin \{a,b\}$, and distinct monomials are independent.

(i) For $y = (z,\theta)$ the class $y \lrcorner \mathrm{ch}(F)$ has components of type $(k-1,k)$ equal to $c_{k-1} z\, t^{k-1} + c_k\, \theta \lrcorner t^k = (c_{k-1} z + k\, c_k\, (\theta \lrcorner t))t^{k-1}$. For $k = 1$ this is $az + b\,(\theta \lrcorner t)$ and for $k = 2$ it is $(bz - ad\,(\theta \lrcorner t))t$, by (18). Both z and $\theta \lrcorner t = \sum_a \theta^a q_a$ lie in $H^{0,1}$, on which t acts injectively, so $y \lrcorner \mathrm{ch}(F) = 0$ forces $(z, \theta \lrcorner t)$ into the kernel of the displayed matrix, whose determinant is $-a^2 d - b^2 < 0$. Hence $z = 0$ and $\theta \lrcorner t = 0$, and the latter gives $\theta = 0$ because t is nondegenerate.

(ii) For $x = (z, v, \pi)$ the component of $x \lrcorner \mathrm{ch}(F)$ of type $(k-1, k+1)$ is

$$c_{k-1} z\, t^{k-1} + c_k\, v \lrcorner t^k + c_{k+1}\, \pi \lrcorner t^{k+1} = \Big(c_{k-1} z + k\, c_k\, W + \tbinom{k+1}{2} c_{k+1}\, P\Big) t^{k-1},$$

with $W = v \lrcorner t$ and $P = \pi \lrcorner t^2$, both in $H^{0,2}$. By (18), $\binom{k+1}{2} c_{k+1} = -\frac{d}{2} c_{k-1}$, so the bracket is $c_{k-1} Z + k\, c_k W$ with $Z = z - \frac{d}{2} P$. For $k = 1$ and $k = 2$ this gives $aZ + bW$ and $(bZ - adW)\,t$, and t is injective on $H^{0,2}$ because $n \geq 3$; the same determinant forces $Z = W = 0$. Conversely $Z = W = 0$ kills every component. So $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F)$ is the set of (z, v, π) with $v \lrcorner t = 0$ and $z = \frac{d}{2}\,\pi \lrcorner t^2$, which is (17); the two summands are independent because the first has $\pi = 0$ and the second is the graph of a linear map from the π component. A $v \in \mathrm{Hom}(H^{1,0}, H^{0,1})$ has $v \lrcorner t = \sum_a v(p_a) \wedge q_a$, which vanishes exactly when the matrix of v is symmetric; this is the tangent space of the polarised deformations, of dimension $n(n+1)/2$, as in Proposition 4.16(i). The bivectors contribute $\dim \bigwedge^2 T = n(n-1)/2$. Nothing in the description involves (a,b).

(iii) By (ii) and (15), ev_F vanishes on $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F)$ if and only if it vanishes on both summands. On the first it is $v \mapsto v \lrcorner \mathrm{At}(F)$, which is (a). On the second it is

$$\tfrac{d}{2}(\pi \lrcorner t^2)\cdot \mathrm{id}_F + \tfrac{1}{2}\,\pi \lrcorner \mathrm{At}(F)^2 = \tfrac{1}{2}\,\pi \lrcorner (\mathrm{At}(F)^2 + d\,t^2 \cdot \mathrm{id}_F),$$

since $\pi \lrcorner (t^2 \cdot \mathrm{id}_F) = (\pi \lrcorner t^2)\cdot \mathrm{id}_F$. Now $\Omega_X^2 = \mathcal{O}_X \otimes \bigwedge^2 H^{1,0}$, so $\mathrm{Ext}^2(F, F\otimes\Omega_X^2) = \mathrm{Ext}^2(F,F)\otimes\bigwedge^2 H^{1,0}$, and the contractions with all $\pi \in \bigwedge^2 (H^{1,0})^\vee$ vanish on an element of this space if and only if the element is zero, the pairing being perfect. This is (b). For the trace, $\mathrm{ch}(F) = \mathrm{tr}\exp \mathrm{At}(F)$ gives $\mathrm{ch}_2(F) = \frac{1}{2}\,\mathrm{tr}\,\mathrm{At}(F)^2$, and $\mathrm{ch}_2(F) = c_2 t^2 = -\frac{ad}{2}t^2$, while $\mathrm{tr}(\mathrm{id}_F) = \mathrm{rk}\,F = c_0 = a$.

(iv) The Hochschild cohomology of a product is the tensor product, $HT^2(X\times X) = HT^2\otimes HT^0 \oplus HT^1\otimes HT^1 \oplus HT^0\otimes HT^2$, the Atiyah class of G is $\mathrm{At}(F_1)\boxtimes 1 + 1\boxtimes \mathrm{At}(F_2)$ [BF03, Proposition 3.14], [Mar25a, (8.4.1)], so $\exp \mathrm{At}(G) = \exp \mathrm{At}(F_1)\boxtimes \exp \mathrm{At}(F_2)$, and $\mathrm{Ext}^2(G,G)$ is the corresponding Künneth sum of $\mathrm{Ext}^i(F_1,F_1)\otimes\mathrm{Ext}^j(F_2,F_2)$. A polyvector on a factor contracts only the forms of that factor, so for $x_1 \in HT^i(X)$ and $x_2 \in HT^j(X)$ with $i+j=2$,

$$(x_1\otimes x_2)\lrcorner\,\mathrm{ch}(G) = \pm(x_1\lrcorner\,\mathrm{ch}\,F_1)\otimes(x_2\lrcorner\,\mathrm{ch}\,F_2), \qquad \mathrm{ev}_G(x_1\otimes x_2) = \pm\,\mathrm{ev}_{F_1}(x_1)\otimes \mathrm{ev}_{F_2}(x_2),$$

with $\mathrm{ev}(1) = \mathrm{id}$ on HT^0 and Koszul signs that play no role; this is the decomposition [Mar25a, (8.4.2), (8.4.3)]. Let $x = x_{20}+x_{11}+x_{02}$ be a class in $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)$ according to the three summands. Since $\mathrm{ch}(F_i)$ is even, $x_{11}\lrcorner\,\mathrm{ch}(G)$ lies in $H^{\mathrm{odd}}\otimes H^{\mathrm{odd}}$ while the other two terms lie in $H^{\mathrm{ev}}\otimes H^{\mathrm{ev}}$, so $x_{11}\lrcorner\,\mathrm{ch}(G)=0$ separately; and $x_{11}\lrcorner\,\mathrm{ch}(G)$ is the image of x_{11} under the tensor product of the two maps of (i), which is injective, so $x_{11}=0$. For the even part write $x_{20} = x_1\otimes 1$ and $x_{02} = 1\otimes x_2$; then $(x_1\lrcorner\,\mathrm{ch}\,F_1)\otimes\mathrm{ch}(F_2) + \mathrm{ch}(F_1)\otimes(x_2\lrcorner\,\mathrm{ch}\,F_2) = 0$. Applying a linear form on the second factor taking the value 1 on $\mathrm{ch}(F_2)$ shows $x_1\lrcorner\,\mathrm{ch}(F_1) = \lambda\,\mathrm{ch}(F_1)$ for a scalar λ, and then $x_2\lrcorner\,\mathrm{ch}(F_2) = -\lambda\,\mathrm{ch}(F_2)$. But every component of $x_1\lrcorner\,\mathrm{ch}(F_1)$ has Hodge type $(p,p+2)$ while $\mathrm{ch}(F_1)\neq 0$ has type (p,p); so $\lambda = 0$ and $x_i \in \mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_i)$. This proves the description of $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)$ and its dimension $2n^2$ by (ii). On it $\mathrm{ev}_G(x_1\otimes 1 + 1\otimes x_2) = \pm\,\mathrm{ev}_{F_1}(x_1)\otimes\mathrm{id}_{F_2} \pm \mathrm{id}_{F_1}\otimes\mathrm{ev}_{F_2}(x_2)$, whose two terms lie in different Künneth summands and vanish exactly when $\mathrm{ev}_{F_i}(x_i)=0$, as $\mathrm{id}_{F_i}\neq 0$. So G satisfies the criterion if and only if F_1 and F_2 do.

For the transport under Φ we use the commutative diagram of [Mar25a, Lemma 8.3.10]: an equivalence $\Phi\colon D^b(A)\to D^b(B)$ of derived categories of abelian varieties induces isomorphisms $\Phi^{HT}\colon HT^2(A)\to HT^2(B)$ and $\Phi^H\colon H^*(A,\mathbb{C})\to H^*(B,\mathbb{C})$ with $\Phi^H(x\lrcorner\,\mathrm{ch}\,G) = \Phi^{HT}(x)\lrcorner\,\mathrm{ch}\,\Phi(G)$ [CRV12], and $\Phi\circ\mathrm{ev}_G = \mathrm{ev}_{\Phi(G)}\circ\Phi^{HT}$ on $\mathrm{Hom}(G,G[2])\to\mathrm{Hom}(\Phi G,\Phi G[2])$ [Hua21, Theorem A]; on abelian varieties the Todd class is trivial, so there is no Duflo correction [Mar25a, Remark 8.3.11]. The first square gives $\Phi^{HT}(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)) = \mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)$ and the second $\mathrm{ev}_E(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)) = \Phi(\mathrm{ev}_G(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)))$, with Φ an isomorphism on Ext^2. So E satisfies the criterion if and only if G does. ∎

Figure 12 draws the Künneth stack and the two conditions at $n=4$.

Corollary 4.57 (The remaining condition on the fourfold)**.** *In Setup 4.51 let $E = \Phi(\mathcal{F}\boxtimes\mathcal{F})$. Then E satisfies the weakened criterion of* [Mar25b, Question 11.4] *if and only if*

$$\mathrm{At}(\mathcal{F})^2 = -d\,\Theta^2\cdot\mathrm{id}_{\mathcal{F}} \qquad \text{in } \mathrm{Ext}^2(\mathcal{F},\mathcal{F}\otimes\Omega_X^2). \tag{19}$$

Equivalently, writing $\mathrm{At}(\mathcal{F}) = \sum_a A_a\otimes p_a$ with $A_a\in\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$ in a basis p_a of $H^{1,0}(X)$, and $\Theta = \sum_a p_a\wedge q_a$ with $q_a\in H^{0,1}(X)$, all the commutators $A_aA_b - A_bA_a\in\mathrm{Ext}^2(\mathcal{F},\mathcal{F})$ are the scalars $\pm 2d\,(q_a\wedge q_b)\cdot\mathrm{id}_{\mathcal{F}}$ prescribed by Θ^2, the sign being fixed by the Koszul rule.

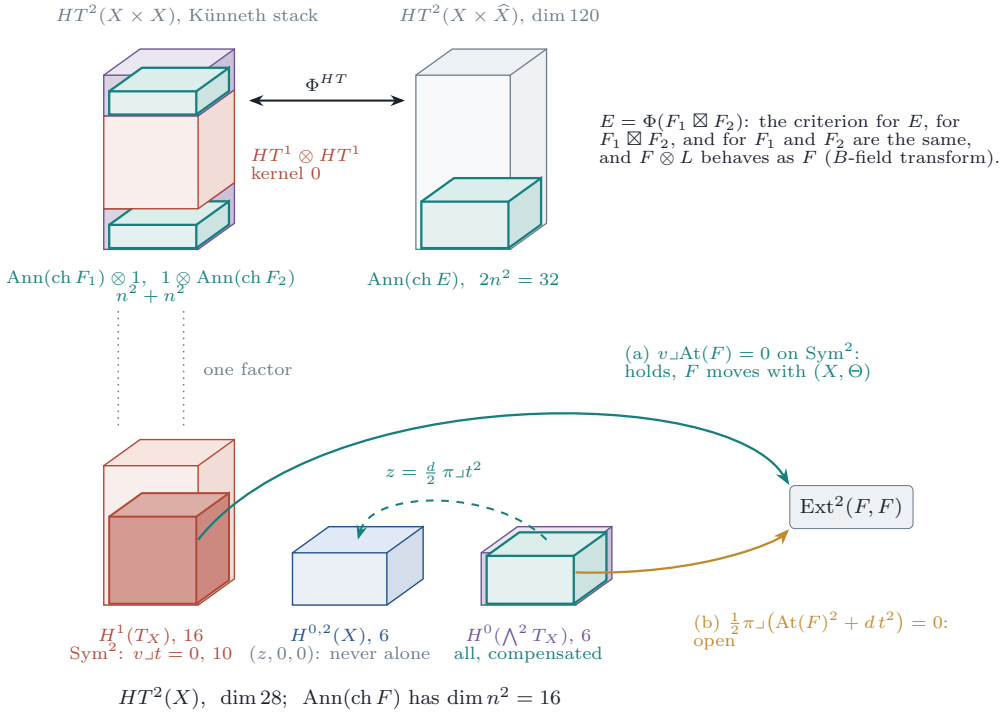


FIGURE 12. Theorem 4.56 for $n = 4$. Top: the Künneth stack of $HT^2(X \times X)$, whose middle summand $HT^1 \otimes HT^1$ meets the kernel in zero by (i), so that the classes preserving $\mathrm{ch}(F_1 \boxtimes F_2)$ are two copies of the kernel on one factor; across Orlov's isomorphism, HT^2 of the eightfold with the $2n^2$ classes preserving $\mathrm{ch}(E)$. Bottom: $HT^2(X)$ as three slabs with heights proportional to their dimensions, and inside them the n^2 classes of (17): the polarised deformations in front of $H^1(T_X)$, and the whole of $H^0(\bigwedge^2 T_X)$, each bivector tilted into $H^{0,2}$ by its compensation $\frac{d}{2}\, \pi \lrcorner t^2$; nothing in $H^{0,2}$ alone preserves $\mathrm{ch}(F)$. On the right, the two conditions that the evaluation map imposes on the two parts, (a) closed by the construction of $\mathcal{F}$ and (b) the identity (19).

Proof. By Proposition 4.52(iii), $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$, so Theorem 4.56 applies with $n = 4$, $(a, b) = (1, 3)$ and $F_1 = F_2 = \mathcal{F}$, and E satisfies the criterion if and only if $\mathcal{F}$ satisfies (a) and (b). It remains to show that (a) holds. The data of Setup 4.51 are a principally polarised fourfold (X, Θ) and N points $x_i \in X$ with $D_i = \tau_{x_i}^* \Theta$. The conditions imposed on the D_i, that the intersections of two, three and four of them be smooth of the expected codimension and that five of them be disjoint, are open in the parameter space of such data over the local universal deformation S of (X, Θ), by the openness of smoothness for a proper flat family and the properness of the projection. The isolated points of Z, being transversal intersections of four divisors, form a finite étale cover of the open set where the conditions hold, so the m points to be normalised can be followed locally; the partial normalisation separates two smooth branches at each of them and commutes with base change. The construction therefore produces a perfect complex $\mathcal{F}_S$ on the universal family over a neighbourhood of the base point in S, whose fibre at the base point is $\mathcal{F}$ and whose base has Kodaira–Spencer image the whole space of polarised deformations, of dimension $10 = n(n+1)/2$. The obstruction $v \lrcorner \mathrm{At}(\mathcal{F})$ to extending $\mathcal{F}$ along v vanishes for every v along which $\mathcal{F}$ extends [Tod09], and this is (a).

For the reformulation, $\Omega_X^1 = \mathcal{O}_X \otimes H^{1,0}$ gives $\mathrm{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$, and $\mathrm{At}(\mathcal{F})^2 = -\sum_{a<b}(A_a A_b - A_b A_a) \otimes p_a \wedge p_b$ by the Koszul rule, while $\Theta^2 = -2\sum_{a<b}(p_a \wedge p_b)(q_a \wedge q_b)$ for $\Theta = \sum_a p_a \wedge q_a$. Comparing coefficients of $p_a \wedge p_b$ gives the commutator relations. ∎

Remark 4.58 (Scope of the reduction). The reduction accomplishes three things. First, it removes the eightfold: the condition of Remark 4.53, on an object of rank $8d(d-9)$ on $X \times \widehat{X}$ of Weil type, is equivalent to one identity (19) for a rank one complex on the fourfold, in the Yoneda algebra $\mathrm{Ext}^*(\mathcal{F}, \mathcal{F})$. The dimension of the space of classes that has to be killed drops from $2n^2$ to n^2 and, by (a), to $n(n-1)/2 = 6$. At $n = 3$ the count $n^2 = 9$ is the nine-dimensional kernel of [Mar25a, Corollaries 8.3.12 and 8.5.2], which (17) writes out for every secant object in every dimension. Second, the half of that space made of geometric deformations imposes nothing, because Markman's construction is uniform over the moduli of principally polarised fourfolds. Third, by Lemma 4.55 the two forms of the criterion, for E with $\mathrm{ch}(E)$ and for the twisted object with $\kappa(E)$, coincide, so nothing is lost by working with the untwisted object.

What remains is the identity itself, which is a Heisenberg relation: the components of the Atiyah class in the n tangent directions must commute up to the scalars prescribed by Θ^2, and the Chern character forces only the trace of the relation. The identity says exactly that $\mathcal{F}$ deforms, as an object of the deformed category, along the Poisson directions of X compensated by the gerbe classes $\frac{d}{2}\pi \lrcorner \Theta^2$; these are the noncommutative first order deformations of X with a gerby correction, in the language of [Mar25a, Section 8.5]. It holds in the one case where the method is known to work. For Markman's sheaves $\mathcal{F}_1 = I_{\bigcup C_i}(\Theta)$ on the Jacobian threefold of [Mar25b, Section 11.1], with $\mathrm{ch} = u_\Theta + v_\Theta$ and $d = q$, the kernel of the obstruction map equals the annihilator of the Chern character [Mar25a, Corollary 8.3.12], which is stronger than the weakened criterion; so by Theorem 4.56(iii) these sheaves satisfy $\mathrm{At}(\mathcal{F}_1)^2 = -q\,\Theta^2 \cdot \mathrm{id}$ on the threefold. For the rank one complex of Setup 4.51 on the fourfold, the identity is a computation of Yoneda products in $\mathrm{Ext}^*(\mathcal{F}, \mathcal{F})$, and the computation can be made local. Below we carry it out at one point of the support and find that the identity fails.

Failure at an isolated double point. The identity (19) can be tested locally. The evaluation map is an algebra homomorphism [Mar25a, (8.3.3)], so on a bivector $\theta \wedge \theta'$ it is the Yoneda

product of the classes

$$A_\theta = \mathrm{ev}_{\mathcal{F}}(\theta) = \theta \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^1(\mathcal{F}, \mathcal{F}), \qquad \theta \in H^0(T_X),$$

which are the first order deformations of $\mathcal{F}$ along the translations of X. The product of two global classes restricts to the product of their germs. At a point where the support of $\mathcal{F}$ has two smooth branches of codimension two meeting transversally, the germs multiply nontrivially; since the compensating classes $z \cdot \mathrm{id}_{\mathcal{F}}$ have zero germs, the identity fails. Markman's construction has such points and cannot avoid them.

Lemma 4.59 (Localisation)**.** *Let F be a perfect complex on a smooth projective variety Y, let $\mathcal{E}xt^m(F,F)$ be the sheaf associated with $U \mapsto \mathrm{Hom}_{D(U)}(F|_U, F|_U[m])$, and let $e\colon \mathrm{Ext}^m(F,F) \to H^0(Y, \mathcal{E}xt^m(F,F))$ send a morphism $F \to F[m]$ to its germs. Then e is compatible with Yoneda products; $e(z \cdot \mathrm{id}_F) = 0$ for $z \in H^m(\mathcal{O}_Y)$, $m \geq 1$; and for $\theta \in H^0(T_Y)$ the germ of A_θ at p is the class of the jet extension $0 \to F_p \to J_\theta(F_p) \to F_p \to 0$ of the germ F_p along the derivation θ of $\mathcal{O}_{Y,p}$, which is $\mathcal{O}_{Y,p}$-linear in θ.*

Proof. Composition of morphisms commutes with restriction to open sets; this is the first assertion. For the second, $z \cdot \mathrm{id}_F$ restricted to an affine open U is $z|_U \cdot \mathrm{id}_{F|_U}$, and $z|_U \in H^m(U, \mathcal{O}_U) = 0$. For the third, the Atiyah class of F is the class of the first jet sequence $0 \to F \otimes \Omega^1_Y \to J^1(F) \to F \to 0$ [BF03, Section 3], and its restriction to U is the jet sequence of $F|_U$. Contracting with θ is pushing out along $\mathrm{id} \otimes \theta\colon F \otimes \Omega^1 \to F$, which produces the extension $J_\theta(F)$ whose underlying sheaf is $F \oplus F$ with $f \cdot (s, s') = (fs + \theta(f)s', fs')$. Linearity in θ is linearity of the contraction. ∎

Lemma 4.60 (Two branches through a point)**.** *Let $\mathcal{O}$ be the local ring of $\mathbb{C}^4$ at the origin or its completion $\mathbb{C}[[x_1, \dots, x_4]]$; let $P_1 = \{x_1 = x_2 = 0\}$ and $P_2 = \{x_3 = x_4 = 0\}$, two smooth surfaces meeting transversally at the origin; put $N_1 = \langle \partial_1, \partial_2 \rangle$ and $N_2 = \langle \partial_3, \partial_4 \rangle$, the normal spaces of P_1 and of P_2 at the origin, so that $\bigwedge^2 T_0 = \bigwedge^2 N_1 \oplus N_1 \otimes N_2 \oplus \bigwedge^2 N_2$. For a germ of vector field θ and a perfect complex F of $\mathcal{O}$-modules write $a_\theta \in \mathrm{Ext}^1_{\mathcal{O}}(F,F)$ for the class of the jet extension along θ.*

(i) *For $F = I = I_{P_1} \cap I_{P_2}$, the ideal of the union, $\mathrm{Ext}^2_{\mathcal{O}}(I,I) \cong N_1 \otimes N_2$, a vector space of dimension four on which the maximal ideal acts trivially, and $a_\theta a_{\theta'}$ is, up to a fixed nonzero scalar, the component of $\theta(0) \wedge \theta'(0)$ in $N_1 \otimes N_2$. In particular $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in \bigwedge^2 N_1 \oplus \bigwedge^2 N_2$.*

(ii) *For $F = K = [\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, the complex in degrees 0 and 1 whose cohomology is I in degree 0 and the residue field k in degree 1, $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in N_1 \otimes N_2$.*

In both cases the bilinear map $(\theta, \theta') \mapsto a_\theta a_{\theta'}$ is nonzero, and it depends only on $\theta(0) \wedge \theta'(0)$.

Proof. Write $A = \mathbb{C}[x_1, x_2]$, $B = \mathbb{C}[x_3, x_4]$, $R = A \otimes_{\mathbb{C}} B$, with maximal ideals $\mathfrak{m}_A$, $\mathfrak{m}_B$ at the origin, and $k = \mathbb{C}$. The ideal of the union is $I = \mathfrak{m}_A R \cap \mathfrak{m}_B R = \mathfrak{m}_A \otimes_{\mathbb{C}} \mathfrak{m}_B$, because the two ideals are generated by disjoint sets of variables. Everything below is homogeneous, the Ext groups in degree two are of finite length, and finite length modules and jet classes are unchanged by the flat base change from R to $\mathcal{O}$; so we compute over R. For finitely generated modules M, M' over A and N, N' over B, the tensor product of free resolutions resolves $M \otimes_{\mathbb{C}} N$ and the Künneth formula gives

$$\mathrm{Ext}^n_R(M \otimes N, M' \otimes N') = \bigoplus_{i+j=n} \mathrm{Ext}^i_A(M, M') \otimes_{\mathbb{C}} \mathrm{Ext}^j_B(N, N'), \tag{20}$$

compatibly with Yoneda products up to the Koszul sign, since a product of chain maps on the tensor product of resolutions is the tensor product of the products.

The basic case is a single plane. The Koszul complex $0 \to A \xrightarrow{(-x_2,x_1)^t} A^2 \xrightarrow{(x_1,x_2)} \mathfrak{m}_A \to 0$ is a free resolution, so $\mathfrak{m}_A$ has projective dimension one and $\operatorname{Ext}^i_A(\mathfrak{m}_A,-)=0$ for $i\geq 2$; and $\operatorname{Ext}^1_A(\mathfrak{m}_A,\mathfrak{m}_A)=\operatorname{coker}\big(\operatorname{Hom}(A^2,\mathfrak{m}_A)\to\operatorname{Hom}(A,\mathfrak{m}_A)\big)=\mathfrak{m}_A/\mathfrak{m}_A^2$, a two-dimensional space killed by $\mathfrak{m}_A$. On a complex of free modules with differential d the jet extension along a derivation D is the complex with terms $P_i\oplus P_i$ and differential $\left(\begin{smallmatrix} d & -D(d)\\ 0 & d\end{smallmatrix}\right)$, $D(d)$ the entrywise derivative, because the twisted module $J_D(P_i)$ is free on the elements $(e,0)$ and $(0,e)$ and $d(0,e)=\sum_l d_{le}(0,e_l)-\sum_l D(d_{le})(e_l,0)$; so the jet class is represented by the degree -1 chain map $-D(d)$. For $D=\partial_i$ on the Koszul complex this is $e\mapsto \delta_{i2}f_1-\delta_{i1}f_2$ on the generator e of the last term, whose image in $\mathfrak{m}_A/\mathfrak{m}_A^2$ is $\delta_{i2}x_1-\delta_{i1}x_2$:

$$\alpha_1 := a_{\partial_1} = -[x_2], \qquad \alpha_2 := a_{\partial_2} = [x_1],$$

a basis of $\operatorname{Ext}^1_A(\mathfrak{m}_A,\mathfrak{m}_A)$, which we identify with N_1 by $\partial_i\mapsto\alpha_i$. Similarly β_3,β_4 form a basis of $\operatorname{Ext}^1_B(\mathfrak{m}_B,\mathfrak{m}_B)$, identified with N_2.

(i) By (20) with $M=M'=\mathfrak{m}_A$, $N=N'=\mathfrak{m}_B$,

$$\operatorname{Ext}^2_R(I,I)=\operatorname{Ext}^1_A\otimes\operatorname{Ext}^1_B=N_1\otimes N_2,$$

the other two summands vanishing by the projective dimension, and it is killed by $\mathfrak{m}_A$ and $\mathfrak{m}_B$. The derivation ∂_1 acts on the factor A, so the jet extension of $I=\mathfrak{m}_A\otimes\mathfrak{m}_B$ along it is $J_{\partial_1}(\mathfrak{m}_A)\otimes\mathfrak{m}_B$ and $a_{\partial_1}=\alpha_1\otimes 1$; likewise $a_{\partial_2}=\alpha_2\otimes 1$, $a_{\partial_3}=1\otimes\beta_3$, $a_{\partial_4}=1\otimes\beta_4$. Hence $a_{\partial_i}a_{\partial_j}=\pm\alpha_i\otimes\beta_j$ for $i\leq 2<j$, the four of them a basis of $N_1\otimes N_2$, while $a_{\partial_1}a_{\partial_2}=\alpha_1\alpha_2\otimes 1=0$ and $a_{\partial_3}a_{\partial_4}=0$ because $\operatorname{Ext}^2_A(\mathfrak{m}_A,\mathfrak{m}_A)=0$. For germs $\theta=\sum_i c^i\partial_i$ and $\theta'=\sum_j c'^j\partial_j$ the linearity of Lemma 4.59 and the triviality of the module structure give $a_\theta a_{\theta'}=\sum_{i,j}c^i(0)c'^j(0)\,a_{\partial_i}a_{\partial_j}$, the image of $\theta(0)\wedge\theta'(0)$ under the projection of $\bigwedge^2 T_0$ onto $N_1\otimes N_2$ followed by the isomorphism $\partial_i\otimes\partial_j\mapsto\pm\alpha_i\otimes\beta_j$.

(ii) Let $Q=\mathcal{O}_{P_1}\oplus\mathcal{O}_{P_2}$. The map $\mathcal{O}\to Q$ has kernel I and cokernel $\mathcal{O}/(I_{P_1}+I_{P_2})=k$, so K sits in an exact triangle $I\xrightarrow{\iota}K\xrightarrow{\pi}k[-1]\xrightarrow{\varepsilon}I[1]$. The jet extension along θ is an exact functor, natural in its argument, so the jet classes commute with ι and π; and for the residue field a^k_θ depends only on $\theta(0)$, because $\operatorname{Ext}^1(k,k)$ is killed by $\mathfrak{m}$ and the class is linear in θ. Therefore

$$\pi[2]\circ a^K_\theta a^K_{\theta'}=a^k_\theta a^k_{\theta'}\circ\pi=\pi^*(a^k_{\theta(0)}a^k_{\theta'(0)}),\qquad \pi^*\colon \operatorname{Ext}^2(k,k)\to\operatorname{Ext}^1(K,k),$$

and it suffices to show that the right hand side is nonzero. The long exact sequence of the triangle for $\operatorname{Hom}(-,k)$ reads

$$\operatorname{Hom}(I,k)\xrightarrow{\varepsilon^*}\operatorname{Ext}^2(k,k)\xrightarrow{\pi^*}\operatorname{Ext}^1(K,k),$$

so $\ker\pi^*=\operatorname{im}\varepsilon^*$. Now $k=k_A\otimes k_B$ and, by (20), $\operatorname{Ext}^*(k,k)$ is the tensor product of $\operatorname{Ext}^*_A(k,k)$ and $\operatorname{Ext}^*_B(k,k)$. On the Koszul resolution $A\to A^2\to A$ of k_A the jet chain maps of ∂_1,∂_2 are $-D(d)$ as above, and their composite $A\to A$ is the scalar -1, whose class in $\operatorname{Ext}^2_A(k,k)=\operatorname{Hom}(A,k)=k$ is nonzero; the degree one classes γ_1,γ_2 of ∂_1,∂_2 are the basis $-e_1^*,-e_2^*$ of $\operatorname{Ext}^1_A(k,k)=\operatorname{Hom}(A^2,k)$, and their product spans $\operatorname{Ext}^2_A(k,k)$. With the analogous classes γ_3,γ_4 over B, the products $\gamma_1\gamma_2\otimes 1$, $\gamma_i\otimes\gamma_j$ $(i\leq 2<j)$ and $1\otimes\gamma_3\gamma_4$ form a basis of $\operatorname{Ext}^2(k,k)=k\oplus(k^2\otimes k^2)\oplus k$, so $\partial_i\wedge\partial_j\mapsto a^k_{\partial_i}a^k_{\partial_j}$ identifies $\bigwedge^2(N_1\oplus N_2)$ with $\operatorname{Ext}^2(k,k)$, and $a^k_{\theta(0)}a^k_{\theta'(0)}$ is the image of $\theta(0)\wedge\theta'(0)$. The class ε is the Künneth product $\varepsilon_A\otimes\varepsilon_B$ of the classes of $0\to\mathfrak{m}_A\to A\to k\to 0$ and its B-analogue, because K is the two-step truncation of the tensor product of the complexes $[A\to k_A]$ and $[B\to k_B]$ and ε is the inclusion of the omitted term $k_A\otimes k_B$. Hence ε^* maps $\operatorname{Hom}(I,k)=\operatorname{Hom}(\mathfrak{m}_A,k)\otimes\operatorname{Hom}(\mathfrak{m}_B,k)$ into $\operatorname{Ext}^1_A(k,k)\otimes\operatorname{Ext}^1_B(k,k)=N_1\otimes N_2$, and onto it, since $\operatorname{Hom}(\mathfrak{m}_A,k)\to\operatorname{Ext}^1_A(k,k)$ is an isomorphism: it is the connecting

homomorphism of the sequence, injective because $\mathrm{Hom}(A,k)\to\mathrm{Hom}(\mathfrak{m}_A,k)$ is zero and surjective because $\mathrm{Ext}^1_A(A,k)=0$. So $\ker\pi^*=N_1\otimes N_2$ and $\pi^*(\theta(0)\wedge\theta'(0))\neq 0$ exactly when $\theta(0)\wedge\theta'(0)\notin N_1\otimes N_2$. ∎

Theorem 4.61 (Markman's candidate does not satisfy the weakened criterion)**.** *In Setup 4.51, for every odd squarefree $d\geq 3$, every choice of the translates D_i in general position and every choice of the m points to be normalised:*

(i) *the identity (19) fails: there is a constant bivector $\pi\in H^0(\bigwedge^2 T_X)$ with $\frac{1}{2}\pi\lrcorner(\mathrm{At}(\mathcal{F})^2+d\,\Theta^2\cdot\mathrm{id}_{\mathcal{F}})\neq 0$ in $\mathrm{Ext}^2(\mathcal{F},\mathcal{F})$;*

(ii) *$\mathcal{F}$ does not satisfy the weakened criterion: the class $x_{\mathcal{F}}=(\frac{d}{2}\,\pi\lrcorner\Theta^2,0,\pi)$ has $x_{\mathcal{F}}\lrcorner\,\mathrm{ch}(\mathcal{F})=0$ and $\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})\neq 0$;*

(iii) *for every perfect complex F_2 on X with Chern character on the secant plane, in particular for $F_2=\mathcal{F}$ and $F_2=\mathcal{F}^\vee$, the object $E=\Phi(\mathcal{F}\boxtimes F_2)$ on the eightfold $X\times\widehat{X}$ of Weil type does not satisfy the weakened criterion of* [Mar25b, Question 11.4]*: the class $x=\Phi^{HT}(x_{\mathcal{F}}\otimes 1)$ satisfies $x\lrcorner\,\mathrm{ch}(E)=0$ and $\mathrm{ev}_E(x)\neq 0$, so σ_E is not injective on the image of ev_E.*

Proof. (i) The surfaces Z_i and Z_j at cyclic distance at least three meet in the 24 points of $D_i\cap D_{i+1}\cap D_j\cap D_{j+1}$, and there are $12N(N-5)=(d+9)(3d-3)$ such points, of which $m=(d+9)(2d-1)$ are normalised; the remaining $(d+9)(d-2)\geq 12$ are not. Let p be one of the latter. The four divisors through p meet transversally, so their local equations $x_1,\dots,x_4$ form a regular system of parameters of $\mathcal{O}_{X,p}$ and identify its completion with $\mathbb{C}[[x_1,\dots,x_4]]$, with $D_i=\{x_1=0\}$, $D_{i+1}=\{x_2=0\}$, $D_j=\{x_3=0\}$ and $D_{j+1}=\{x_4=0\}$; hence $Z_i=P_1$ and $Z_j=P_2$ in the notation of Lemma 4.60. The completion is flat, so a class whose completed germ is nonzero is nonzero. No other D_k passes through p, since five divisors have empty intersection, so no other component of Z does. Near p the map ν is an isomorphism, $\nu_*\mathcal{O}_{\widetilde{Z}}=\mathcal{O}_Z$, and $\mathcal{F}$ is the ideal sheaf $I_Z=I_{P_1}\cap I_{P_2}$ twisted by the locally trivial $\mathcal{O}_X(3\Theta)$: the germ $\mathcal{F}_p$ is the module I of Lemma 4.60(i).

Let $\theta,\theta'\in H^0(T_X)$ be constant vector fields whose values at p are ∂_1 and ∂_3; they exist because the constant fields span every tangent space. Put $\pi=\theta\wedge\theta'$. Since $\mathrm{ev}_{\mathcal{F}}$ is an algebra homomorphism, $\mathrm{ev}_{\mathcal{F}}(\pi)=\frac{1}{2}\pi\lrcorner\,\mathrm{At}(\mathcal{F})^2=A_\theta A_{\theta'}$. By Lemma 4.59 its germ at p is $a_\theta a_{\theta'}$, which is nonzero by Lemma 4.60(i), because $\partial_1\wedge\partial_3$ has nonzero component in $N_1\otimes N_2$. The germ of $\frac{d}{2}(\pi\lrcorner\Theta^2)\cdot\mathrm{id}_{\mathcal{F}}$ at p is zero by Lemma 4.59. So the germ at p of $\frac{1}{2}\pi\lrcorner(\mathrm{At}(\mathcal{F})^2+d\Theta^2\cdot\mathrm{id}_{\mathcal{F}})$ is $a_\theta a_{\theta'}\neq 0$, and the class itself is nonzero.

(ii) By Theorem 4.56(ii) the class $x_{\mathcal{F}}$ preserves $\mathrm{ch}(\mathcal{F})$, and by (15) $\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})$ is the class of (i).

(iii) Both $\mathcal{F}$ and $\mathcal{F}^\vee$ are admissible as F_2, since $\mathrm{ch}(\mathcal{F}^\vee)=u_\Theta-3v_\Theta$ lies on the secant plane. By Theorem 4.56(iv), E satisfies the criterion only if both factors do, and $\mathcal{F}$ does not. The class x of the statement is the transport of $x_{\mathcal{F}}\otimes 1\in\mathrm{Ann}_{HT^2}(\mathrm{ch}(\mathcal{F}\boxtimes F_2))$, on which ev is $\pm\,\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})\otimes\mathrm{id}_{F_2}\neq 0$; since Φ is an isomorphism on Ext^2, $\mathrm{ev}_E(x)\neq 0$. The last assertion follows from the commutative square of Setup 4.54: $\sigma_E(\mathrm{ev}_E(x))=x\lrcorner\,\mathrm{ch}(E)=0$ with $\mathrm{ev}_E(x)\neq 0$. ∎

Figure 13 draws the local computation at the double point.

Remark 4.62 (Normalised points). At a normalised point the germ of $\mathcal{F}$ is the complex K of Lemma 4.60(ii), and the same argument, with $\theta(p)=\partial_1$ and $\theta'(p)=\partial_2$, produces a nonzero germ. The failure therefore does not depend on which points are separated. Separating all of them would not help even if the construction allowed it: normalising every

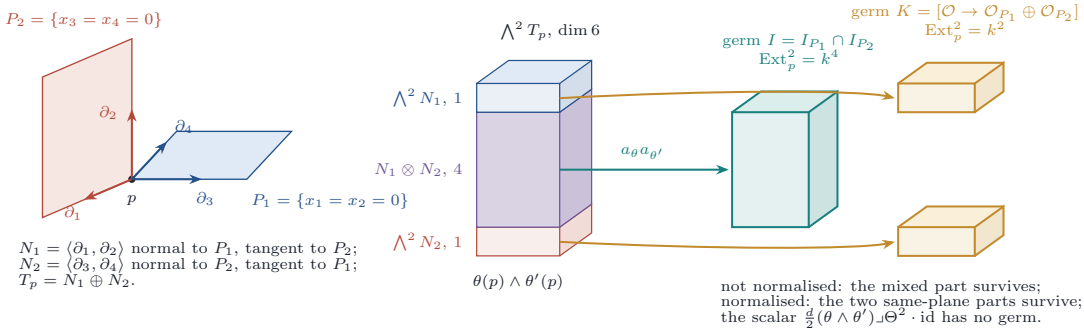


FIGURE 13. Lemma 4.60 and Theorem 4.61. Left: two smooth surfaces of a fourfold meeting transversally at p, with the normal spaces N_1 and N_2, each tangent to the other branch. Centre: the bivectors at p, split into $\bigwedge^2 N_1$, $N_1 \otimes N_2$ and $\bigwedge^2 N_2$, with heights proportional to the dimensions. Right: the local Ext^2 of the germ of $\mathcal{F}$ at p in the two cases, and the part of the bivectors that the products of the translation classes carry onto it: the mixed part when the point is not normalised, the two same-plane parts when it is. The compensating scalar has no germ, so in either case the identity (19) fails at p.

isolated point changes $\chi(\mathcal{O}_{\widetilde{Z}})$ to $110N - 12N^2 + 12N(N-5) = 50N$, while the secant plane requires $(d+9)(27-d) = 72N - 4N^2$, and the two agree only for $N = 11/2$. The two local computations are complementary: at a point of type (i) the obstruction lies in $N_1 \otimes N_2$, at a point of type (ii) in $\bigwedge^2 N_1 \oplus \bigwedge^2 N_2$, and together these exhaust $\bigwedge^2 T_p$.

Corollary 4.63 (A local obstruction in every dimension)**.** *Let (X, t) be a polarised abelian variety of dimension $n \geq 4$ and F a perfect complex on X with $\mathrm{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$. Suppose that at some point p the germ F_p is isomorphic, in the derived category of $\mathcal{O}_{X,p}$-modules, to the ideal of the union of two smooth germs of codimension two meeting transversally along a germ of codimension four, or, when $n = 4$, to the complex $[\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$ of two such germs. Then F does not satisfy the weakened criterion, and neither does $\Phi(F \boxtimes F_2)$ on $X \times \widehat{X}$ for any secant F_2. In particular, in every dimension $n \geq 4$, the union Z of the intersections $D_i \cap D_{i+1}$ of consecutive translates of a polarisation in general position contains, for every pair (i, j) of nonconsecutive indices, a subvariety $Z_i \cap Z_j$ of codimension four along which Z has two transversal branches; any secant object of the form $[\mathcal{O}_X \to \nu_* \mathcal{O}_{\widetilde{Z}}] \otimes L$ in which ν is an isomorphism at a general point of one such $Z_i \cap Z_j$ fails the criterion.*

Proof. In formal coordinates $x_1, \ldots, x_n$ at p with the two branches $\{x_1 = x_2 = 0\}$ and $\{x_3 = x_4 = 0\}$, which exist because the four local equations of the branches form part of a regular system of parameters, the ideal is $I \otimes_{\mathbb{C}} \mathbb{C}[x_5, \ldots, x_n]$ with I as in Lemma 4.60, and (20) with the third factor gives $\mathrm{Ext}^2(I_p, I_p) = N_1 \otimes N_2 \otimes \mathcal{O}_{W,p}$, where W is the intersection

of the branches. The jet classes along $\partial_5, \dots, \partial_n$ vanish, since these derivations act on a free factor, and the products of the classes along $\partial_1, \dots, \partial_4$ are as in the lemma. Reducing modulo the maximal ideal of $\mathcal{O}_{W,p}$ shows that $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ has a nonzero component in $N_1 \otimes N_2$. When $n = 4$ and F_p is the complex $[\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, Lemma 4.60(ii) applies directly. The rest of the argument is that of Theorem 4.61, using Theorem 4.56 for $n \geq 3$. For the last assertion, $Z_i \cap Z_j = D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$ is a transversal intersection of four divisors, of codimension four, and at a general point of it no fifth divisor passes, so Z has exactly the two branches Z_i and Z_j there. ∎

Remark 4.64 (Scope of the local obstruction). Theorem 4.61 answers the question posed in [Mar25b, Section 12]: the secant objects proposed there do not satisfy the weakened criterion of [Mar25b, Question 11.4], for any discriminant. The reason is local and geometric. That question asks whether the semiregularity map may be required to be injective only on the image of the evaluation map. For a secant object this image contains, by Theorem 4.56, the products $A_\theta A_{\theta'}$ of the translation classes shifted by the scalars $\frac{d}{2}(\theta \wedge \theta')\lrcorner\,\Theta^2$, and the semiregularity map kills them all. For the criterion to hold these shifted products must vanish, and they cannot: at an isolated double point of the support, the translation classes of the two branches multiply to a nonzero local class, while a scalar has no local class at all. The fourth ingredient of [I, Remark 14.16] can accordingly be stated more sharply. A secant object of positive rank on a fourfold that satisfies the weakened criterion must have, at every point, germs whose translation classes multiply to zero in the local Ext^2. By Lemma 4.60 this excludes every support with two transversal branches of codimension two through a point, whether or not the branches are separated, and by Corollary 4.63 it excludes the chain of consecutive intersections of translates in every dimension. An object supported on a smooth surface, or on surfaces meeting only along curves, is not excluded by this argument, but no such object with Chern character on the secant plane is known.

4.4. **Smooth supports and four discriminants.** Theorem 4.61 and Corollary 4.63 exclude every secant object whose support has two transversal branches of codimension two through a point. This leaves supports that are smooth or that meet themselves only along curves, and Remark 4.64 ends with the observation that no such object is known. In this subsection we explain why the local argument cannot reach the smooth case, and then determine what a smooth support can be. The answer is narrow: on the fourfold the invariants of the support are forced, and the discriminant is confined to a finite list.

We begin with the fact that at a smooth point there is nothing local left to compute.

Proposition 4.65 (No local obstruction at a smooth point)**.** *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$ and let $S \subset Y$ be smooth of codimension two. Then*

$$\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y, \qquad \mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S) = N_{S/Y}, \qquad \mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) = 0 \quad (q \geq 2),$$

and the same holds for $\mathcal{I}_S \otimes L$ for any line bundle L. Consequently, for $F = \mathcal{I}_S \otimes L$ both sides of (19) have vanishing germs in $\mathcal{E}xt^2$, and Lemma 4.59 decides nothing.

Proof. Twisting by L changes neither side, so take $F = \mathcal{I}_S$. The sheaf $\mathcal{I}_S$ is torsion free of rank one, and its dual is $\mathcal{O}_Y$ because S has codimension two; hence every local endomorphism of $\mathcal{I}_S$ is multiplication by a function, and $\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y$. Apply $\mathcal{H}om(-, \mathcal{I}_S)$ to

$$0 \to \mathcal{I}_S \to \mathcal{O}_Y \to \mathcal{O}_S \to 0. \tag{21}$$

Since $\mathcal{E}xt^q(\mathcal{O}_Y, \mathcal{I}_S) = 0$ for $q \geq 1$, the long exact sequence gives

$$\mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) \cong \mathcal{E}xt^{q+1}(\mathcal{O}_S, \mathcal{I}_S) \qquad (q \geq 1). \tag{22}$$

Now apply $\mathcal{H}om(\mathcal{O}_S,-)$ to (21). Because S is smooth of codimension two, hence a local complete intersection, the Koszul resolution gives $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_Y)=0$ for $k\neq 2$ and $\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)=\omega_S\otimes\omega_Y^{-1}=\omega_S$, while $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_S)=\bigwedge^k N_{S/Y}$ for $0\le k\le 2$ and zero otherwise. Computing both from the same Koszul complex shows that the map

$$\rho\colon \mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)=\omega_S\longrightarrow \mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S)=\textstyle\bigwedge^2 N_{S/Y}=\omega_S$$

induced by $\mathcal{O}_Y\to\mathcal{O}_S$ is the identity: in local coordinates in which $S=\{x_1=x_2=0\}$, both groups are computed as $M/(x_1,x_2)M$ for $M=\mathcal{O}_Y$ and $M=\mathcal{O}_S$, and the map is reduction modulo (x_1,x_2), which is the identity on $\mathcal{O}_S$.

Consider the long exact sequence of $\mathcal{H}om(\mathcal{O}_S,-)$ in degrees two and three. Since ρ is surjective and $\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{O}_Y)=0$, the segment

$$\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)\xrightarrow{\ \rho\ }\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S)\to\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{I}_S)\to\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{O}_Y)=0$$

gives $\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{I}_S)=0$, so $\mathcal{E}xt^2(\mathcal{I}_S,\mathcal{I}_S)=0$ by (22). For $k\ge 3$ both $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_Y)$ and $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_S)$ vanish, so $\mathcal{E}xt^{k+1}(\mathcal{O}_S,\mathcal{I}_S)=0$ and $\mathcal{E}xt^k(\mathcal{I}_S,\mathcal{I}_S)=0$. In degrees one and two, since ρ is injective, the segment

$$0=\mathcal{E}xt^1(\mathcal{O}_S,\mathcal{O}_Y)\to N_{S/Y}\to\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{I}_S)\to\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)\xrightarrow{\ \rho\ }\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S),$$

in which $N_{S/Y}=\mathcal{E}xt^1(\mathcal{O}_S,\mathcal{O}_S)$, gives $\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{I}_S)\cong N_{S/Y}$, and this is $\mathcal{E}xt^1(\mathcal{I}_S,\mathcal{I}_S)$ by (22).

For the last assertion, the right side of (19) has vanishing germs by Lemma 4.59, and the left side lies in $\mathrm{Ext}^2(F,F\otimes\Omega_Y^2)$, whose sheaf of germs in the $\mathcal{E}xt^2$ direction is $\mathcal{E}xt^2(F,F)\otimes\Omega_Y^2=0$. ∎

Remark 4.66 (The global form of the criterion). Proposition 4.65 is the precise reason why the argument of Theorem 4.61 stops at a smooth support, and it also shows what takes its place. Since $\mathcal{E}xt^2=0$, the local-to-global spectral sequence for $\mathrm{Ext}^2(\mathcal{I}_S,\mathcal{I}_S)$ has only two surviving rows, $H^1(S,N_{S/Y})$ and $H^2(Y,\mathcal{O}_Y)$, and they sit in different steps of the filtration: the first is the quotient that records the geometry of S, the second the subgroup of scalars. A product $A_\theta A_{\theta'}$ of translation classes therefore has a well defined image in $H^1(S,N_{S/Y})$, while the scalar on the right of (19) has image zero there. The criterion thus splits into two conditions of different kinds: a vanishing in $H^1(S,N_{S/Y})$, which is a statement about the normal bundle and not about any point, and an identity in $H^2(Y,\mathcal{O}_Y)$. At a double point the first condition already fails locally. At a smooth support it is a global question, and what is decided here is which surfaces S are available at all.

The invariants of a smooth support are far from free. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch converts the Chern character of the object into the complete set of numerical invariants of the surface, and three classical inequalities then constrain them.

Theorem 4.67 (The invariants of a smooth support)**.** *Let (X,Θ) be a principally polarised abelian fourfold, let $d>0$, let $b\in\mathbb{Z}_{>0}$, and let $\mathcal{F}=\mathcal{I}_S(b\Theta)$ be a coherent sheaf with*

$$\mathrm{ch}(\mathcal{F})=u_\Theta+b\,v_\Theta$$

and $S\subset X$ smooth. Write $N=\frac{1}{2}\mathrm{N}_{K/\mathbb{Q}}(b+\sqrt{-d})=\frac{1}{2}(b^2+d)$, $\iota\colon S\hookrightarrow X$ and $\lambda=\iota^*\Theta$. Then S is a surface and

$$\begin{aligned}
&[S]=N\Theta^2, && \iota_*K_S=\tfrac{4bN}{3}\,\Theta^3, && \chi(\mathcal{O}_S)=4N(2b^2-N),\\
&K_S^2=12N(4b^2-N), && e(S)=12N(4b^2-3N), && K_S\cdot\lambda=32bN,
\end{aligned}\tag{23}$$

with $\lambda^2 = 24N$ and $[S]^2 = 24N^2$. Moreover K_S is globally generated, hence nef; S is minimal of general type; and

$$\tfrac{4}{9}b^2 \;\le\; N \;\le\; b^2, \qquad \textit{equivalently} \qquad -\tfrac{1}{9}b^2 \;\le\; d \;\le\; b^2. \tag{24}$$

The lower bound is the Hodge index theorem and is implied by $d > 0$; the upper bound is the Bogomolov–Miyaoka–Yau inequality. In terms of b and d the five invariants read

$$\chi(\mathcal{O}_S) = (b^2+d)(3b^2-d), \quad K_S^2 = 3(b^2+d)(7b^2-d), \quad e(S) = 3(b^2+d)(5b^2-3d), \tag{25}$$

and the defect in the Bogomolov–Miyaoka–Yau inequality is

$$3e(S) - K_S^2 = 24\,(b^4 - d^2), \tag{26}$$

so that the inequality amounts exactly to $d \le b^2$. Finally, K_S is never numerically proportional to λ.

Proof. Since $\mathrm{ch}_0(\mathcal{F}) = 1$ and $c_1(\mathcal{F}) = b\Theta$, the sheaf $\mathcal{F}(-b\Theta) = \mathcal{I}_S$ has $\mathrm{ch}_0 = 1$, $\mathrm{ch}_1 = 0$, so S has codimension at least two, and

$$\mathrm{ch}(\mathcal{O}_S) = 1 - \mathrm{ch}(\mathcal{I}_S) = 1 - \mathrm{ch}(\mathcal{F})\, e^{-b\Theta}.$$

Put $z = b + \sqrt{-d}$, so that $z\overline{z} = 2N$. Since $u_\Theta = \Re e^{\sqrt{-d}\Theta}$ and $\sqrt{d}\, v_\Theta = \Im e^{\sqrt{-d}\Theta}$, we have $\mathrm{ch}(\mathcal{F}) = \Im(z\, e^{\sqrt{-d}\Theta})/\sqrt{d}$, hence $\mathrm{ch}(\mathcal{F})e^{-b\Theta} = \Im(z\, e^{-\overline{z}\Theta})/\sqrt{d}$, whose coefficient of Θ^k is $(-1)^k\Im(z\overline{z}^k)/(\sqrt{d}\,k!) = (-1)^{k+1}\,2N\Im(z^{k-1})/(\sqrt{d}\,k!)$. With $\Im z = \sqrt{d}$, $\Im z^2 = 2b\sqrt{d}$ and $\Im z^3 = (3b^2-d)\sqrt{d}$ this gives

$$\mathrm{ch}_2(\mathcal{O}_S) = \tfrac{b^2+d}{2}\Theta^2 = N\Theta^2, \quad \mathrm{ch}_3(\mathcal{O}_S) = -\tfrac{2bN}{3}\Theta^3, \quad \mathrm{ch}_4(\mathcal{O}_S) = \tfrac{3b^4+2db^2-d^2}{24}\Theta^4.$$

The first identity says S has codimension exactly two, so it is a surface, with class $N\Theta^2$. Because T_X is trivial, $\mathrm{td}(T_X) = 1$, and Grothendieck–Riemann–Roch for ι reads $\mathrm{ch}(\iota_*\mathcal{O}_S) = \iota_*\,\mathrm{td}(T_S)$. Comparing degrees three and four,

$$\mathrm{ch}_3(\mathcal{O}_S) = \iota_*\tfrac{c_1(T_S)}{2} = -\tfrac{1}{2}\iota_*K_S, \qquad \mathrm{ch}_4(\mathcal{O}_S) = \iota_*\tfrac{c_1^2+c_2}{12} = \chi(\mathcal{O}_S)\,[\mathrm{pt}],$$

the second by Noether's formula. With $\Theta^4 = 24\,[\mathrm{pt}]$ these give $\iota_*K_S = \frac{4bN}{3}\Theta^3$ and, after substituting $d = 2N - b^2$, $\chi(\mathcal{O}_S) = 3b^4 + 2db^2 - d^2 = 4N(2b^2 - N)$. Then $K_S \cdot \lambda = \int_X \iota_*K_S \cdot \Theta = \frac{4bN}{3}\cdot 24 = 32bN$ and $\lambda^2 = \int_X [S]\cdot\Theta^2 = 24N$.

For the last two invariants, the normal bundle sequence and the triviality of T_X give $c(T_S)c(N_{S/X}) = 1$, so $c_1(N_{S/X}) = K_S$ and $c_2(N_{S/X}) = K_S^2 - e(S)$; the self-intersection formula $[S]^2 = \int_S c_2(N_{S/X})$ then reads

$$24N^2 = K_S^2 - e(S),$$

while Noether's formula reads $12\chi(\mathcal{O}_S) = K_S^2 + e(S)$. Solving the two gives the stated K_S^2 and $e(S)$.

The conormal sequence presents Ω^1_S as a quotient of $\Omega^1_X|_S = \mathcal{O}_S^{\oplus 4}$, so Ω^1_S is globally generated, and therefore so is $K_S = \det\Omega^1_S$, a quotient of $\bigwedge^2\mathcal{O}_S^{\oplus 4}$. A globally generated line bundle is nef. A globally generated rank two bundle on a surface has $c_2 \ge 0$, the class c_2 being represented by the vanishing scheme of a general section, which is empty or finite; hence

$$e(S) = c_2(\Omega^1_S) = 12N(4b^2 - 3N) \;\ge\; 0,$$

so $N \le \frac{4}{3}b^2 < 4b^2$ and therefore $K_S^2 = 12N(4b^2 - N) > 0$. A nef divisor of positive self-intersection is big, so $\kappa(S) = 2$ and, K_S being nef, S is minimal of general type. The Bogomolov–Miyaoka–Yau inequality $c_1^2 \le 3c_2$ therefore applies and reads $K_S^2 \le 3e(S)$.

Substituting $N = \frac{1}{2}(b^2 + d)$ in (23) gives (25), hence $3e(S) - K_S^2 = 24(b^4 - d^2)$, and the inequality is $d \le b^2$, that is $N \le b^2$.

For the lower bound, λ is ample on S, so the Hodge index theorem gives $(K_S \cdot \lambda)^2 \ge K_S^2 \, \lambda^2$, that is $(32bN)^2 \ge 12N(4b^2 - N) \cdot 24N$, which simplifies to $9N \ge 4b^2$. Since $N = \frac{1}{2}(b^2 + d)$ with $d > 0$, this holds automatically. Equality in the Hodge index theorem occurs exactly when K_S is numerically proportional to λ, hence exactly when $9N = 4b^2$, and this never happens, because $9N = \frac{9}{2}(b^2 + d) > 4b^2$ for $d > 0$. ■

Figure 14 draws the Bogomolov–Miyaoka–Yau defect over the quarter plane of (b, d).

Corollary 4.68 (Four discriminants on the fourfold)**.** *Let (X, Θ) be a principally polarised abelian fourfold and $d > 0$ squarefree. A sheaf $\mathcal{I}_S(3\Theta)$ with Chern character $u_\Theta + 3v_\Theta$, the value of Proposition 4.52(iii), and with smooth support S exists only if $d \le 9$, that is $d \in \{1, 2, 3, 5, 6, 7\}$; if in addition $N = (9 + d)/2$ is an integer, as in Setup 4.51, only*

$$d \in \{1,\ 3,\ 5,\ 7\}, \qquad N \in \{5,\ 6,\ 7,\ 8\}.$$

For these values the invariants are forced to be those of Table 1. In particular, for $d > 9$ no such sheaf has smooth support.

Proof. This is immediate from (24) at $b = 3$ and the arithmetic of (23), and the table evaluates (23) at $b = 3$. The value $d = 9$ is the equality case of the Bogomolov–Miyaoka–Yau inequality, where $K_S^2 = 3e(S) = 2916$ and S would have to be a quotient of the ball; it is excluded because 9 is not squarefree. ■

d	N	$\chi(\mathcal{O}_S)$	K_S^2	$e(S)$	$K_S \cdot \lambda$	λ^2
1	5	260	1860	1260	480	120
3	6	288	2160	1296	576	144
5	7	308	2436	1260	672	168
7	8	320	2688	1152	768	192

TABLE 1. The only invariants a smooth support can have on an abelian fourfold, by Theorem 4.67 and Corollary 4.68, at $b = 3$ and with N an integer. Every entry is determined by $\mathrm{ch}(\mathcal{F})$ alone. The slope $K_S^2/\chi(\mathcal{O}_S)$ runs from 93/13 at $d = 1$ to 42/5 at $d = 7$, and reaches the Bogomolov–Miyaoka–Yau ceiling of 9 only at the excluded value $d = 9$.

Remark 4.69 (Status of the smooth case)*.* The local argument does not apply here, for a structural reason given by Proposition 4.65: the sheaf $\mathcal{E}xt^2$ of a smooth ideal vanishes, so both sides of (19) are locally zero, and the comparison of germs that decides Theorem 4.61 has nothing to compare.

The surface is rigidly constrained. At $b = 3$ its class, its canonical class, its Euler characteristic, its Euler number and its two intersection numbers with the polarisation are all determined by d, and they are compatible with existence only for $d \le 9$. This is the content of Corollary 4.68, and it restricts the construction: for all but finitely many imaginary quadratic fields the secant construction of Setup 4.51 with smooth support is not available, whatever the geometry of the surface.

What remains is a single existence question. For $d \in \{1, 3, 5, 7\}$, does a principally polarised abelian fourfold carry a smooth surface S with the invariants of Table 1, and if

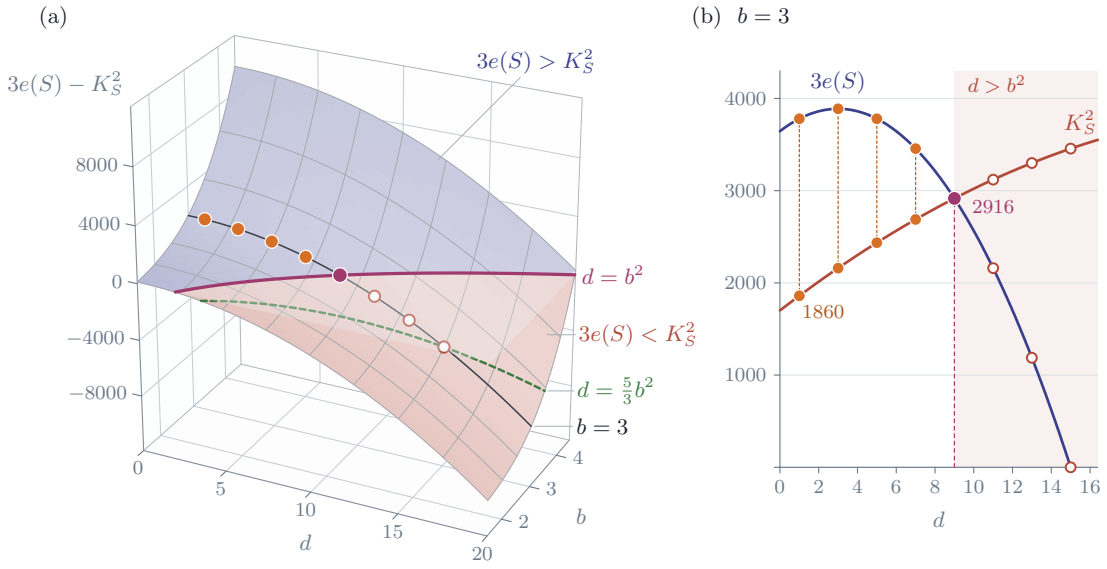


FIGURE 14. Theorem 4.67 over the quarter plane of (b, d). The height of the surface is the defect $3e(S) - K_S^2 = 24(b^4 - d^2)$ in the Bogomolov–Miyaoka–Yau inequality for the surface the theorem attaches to the pair, compressed by a signed cube root so that four orders of magnitude fit one frame; the compression is odd and monotone, so the zero locus and the sign are the true ones. The surface is drawn indigo where the defect is positive and clay where it is negative, and it crosses zero along $d = b^2$, which is therefore the whole content of the inequality. The weaker wall $d = \frac{5}{3}b^2$, where $e(S)$ itself changes sign, is drawn dashed and lies above it. On the line $b = 3$, which is the value Proposition 4.52(iii) forces on the fourfold, the four squarefree discriminants $d = 1, 3, 5, 7$ with N an integer sit above the wall and are marked with their drop lines; $d = 9$ sits exactly on it, the equality case, and $d = 11, 13, 15$ sit below and are excluded.

so, does the image of $A_\theta A_{\theta'}$ in $H^1(S, N_{S/X})$ vanish for all $\theta, \theta' \in H^0(T_X)$? The first half is a question about surfaces in abelian fourfolds with prescribed Chern numbers, of the kind that is answered by a construction or by a classification argument; the second is a question about one cohomology group of one normal bundle. Neither is settled here. The simplest surfaces of the right class are ruled out by one entry of the table: a complete intersection of divisors in $|a\Theta|$ and $|b'\Theta|$ has class $ab'\Theta^2$, so $ab' = N$, and canonical class $(a + b')\lambda$ by adjunction, so $K_S \cdot \lambda = 24(a + b')N$; since $ab' = N \geq 5$ forces $a + b' \geq 5$, this exceeds the required $96N$. The surface, if it exists, is not cut out by two theta divisors, nor by any two divisors algebraically equivalent to multiples of Θ; nor, by Corollary 4.84 below, is its ideal sheaf presented by any two bundles whose classes lie in the subring generated by line bundles. It must also move with X to first order in every polarised direction, in the exact sense of Proposition 4.73, while the injectivity that Lemma 4.88 asks of it holds automatically when X is simple (Lemma 4.70).

A positive answer to both, for a surface that moves in this way, together with an affirmative answer to [Mar25b, Question 11.4], would produce Weil classes on abelian eightfolds for at most four discriminants; the other discriminants, every $n \geq 5$ and every

field of degree larger than two would still have to be reached through Proposition 6.13 and the propagation statement.

Two of the conditions on a smooth support are decided by the table, and the third, which Theorem 4.56(iii)(a) imposes on every candidate, has an exact form. The first is the injectivity of $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ that Lemma 4.88 below requires of every support.

Lemma 4.70 (The injectivity condition for a smooth support)**.** *Let X be a simple complex abelian fourfold, let $V = H^0(T_X)$, and let $S \subset X$ be a smooth surface. The following are equivalent:*

(i) *the restriction $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ is injective;*
(ii) *no nonzero translation-invariant holomorphic 2-form on X restricts to zero on S;*
(iii) *S is not Lagrangian for any translation-invariant holomorphic symplectic form on X;*
(iv) *the Gauss image of S in $\mathrm{Gr}(2, V) \subset \mathbb{P}(\bigwedge^2 V)$ spans $\mathbb{P}(\bigwedge^2 V)$.*

If $e(S) \neq [S]^2$, they hold. In particular they hold for every smooth support of Table 1, where $e(S) = 12N(36 - 3N)$ and $[S]^2 = 24N^2$ agree only for $5N = 36$.

Proof. The restriction $\iota^*\colon H^2(X, \mathbb{C}) \to H^2(S, \mathbb{C})$ is a morphism of Hodge structures defined over $\mathbb{Q}$, so it commutes with complex conjugation, which interchanges the $(0, 2)$ and $(2, 0)$ parts; hence ι^* is injective on $H^{0,2}(X) = H^2(\mathcal{O}_X)$ if and only if it is injective on $H^{2,0}(X) = \bigwedge^2 V^\vee$, the translation-invariant 2-forms. A holomorphic 2-form on a compact surface is zero as a class in $H^{2,0}(S) = H^0(K_S)$ if and only if it is zero as a form; this is (i)⇔(ii). The restriction of $p \wedge q \in \bigwedge^2 V^\vee$ to S is the pull-back of the Plücker coordinate $p \wedge q$ along the Gauss map $\gamma\colon S \to \mathrm{Gr}(2, V)$, $s \mapsto T_s S$, as a section of $K_S = \gamma^*\mathcal{O}(1)$, so a 2-form vanishes on S if and only if the corresponding hyperplane contains $\gamma(S)$; this is (ii)⇔(iv).

A nonzero 2-form on $V \cong \mathbb{C}^4$ has rank two or four. If ω has rank four, $\omega|_S = 0$ says that every tangent plane of S is ω-isotropic, that is, S is Lagrangian for ω. If $\omega = \alpha \wedge \beta$ has rank two, with $\alpha, \beta \in V^\vee$ independent, then $\alpha|_S$ and $\beta|_S$ are independent holomorphic 1-forms on S, because S generates the simple X and a nonzero invariant 1-form restricts nontrivially to a generating subvariety, and $\alpha|_S \wedge \beta|_S = 0$. By the theorem of Castelnuovo and de Franchis [BHPV04] there is a fibration $f\colon S \to C$ onto a curve of genus at least two with $\alpha|_S$ and $\beta|_S$ pulled back from C. A fibre F of f is a curve in X on which α and β vanish, so all its tangent lines lie in the fixed plane $B = \ker\alpha \cap \ker\beta \subset V$. The abelian subvariety generated by $F - F$ is the image of a sum map from a power of F, whose differential at a general point is spanned by tangent vectors of F; so its tangent space lies in B, its dimension is at most two, and it is nonzero because F is not a point. This contradicts the simplicity of X. Hence only the rank-four case occurs, which is (ii)⇔(iii).

Finally, if S is Lagrangian for a symplectic ω, then ω identifies T_S with the annihilator of T_S in $\Omega^1_X|_S$, which is the conormal bundle, so $N_{S/X} \cong \Omega^1_S$ and $c_2(N_{S/X}) = e(S)$; since T_X is trivial, $c_2(N_{S/X}) = [S]^2$ by the self-intersection formula. So $e(S) \neq [S]^2$ excludes (iii). For the table, Theorem 4.67 gives $[S]^2 = 24N^2$ and $e(S) = 12N(36 - 3N)$ at $b = 3$, and $24N^2 = 12N(36 - 3N)$ reads $5N = 36$. For general b the two agree only when $N = \frac{4}{5}b^2$, that is $d = \frac{3}{5}b^2$, whose first integral instance is $b = 5$, $d = 15$. ∎

Remark 4.71 (The class $K_S - 4\lambda$). Theorem 4.67 records $K_S \cdot \lambda = 96N$ and $\lambda^2 = 24N$ at $b = 3$, so $K_S \cdot \lambda = 4\lambda^2$, and that K_S is never numerically proportional to λ. The sharper statement is that

$$\delta = K_S - 4\lambda \in \ker\bigl(\iota_*\colon H^2(S, \mathbb{Q}) \to H^6(X, \mathbb{Q})\bigr), \qquad \delta^2 = -12N(N - 4) < 0,$$

because $\iota_*K_S = 4N\Theta^3$ and $\iota_*\lambda = \Theta \cup [S] = N\Theta^3$. So δ is a nonzero algebraic class of type $(1,1)$, orthogonal by the projection formula to the whole image of $H^2(X,\mathbb{Q})$ in $H^2(S,\mathbb{Q})$, and of negative square: the Picard number of S is at least two, and the kernel of the Gysin map on $H^2(S,\mathbb{Q})$, a sub-Hodge structure, contains an algebraic class. For general b the class is $K_S - \frac{4b}{3}\lambda$, with square $\frac{4N}{3}(4b^2 - 9N)$, and its vanishing is the equality case $9N = 4b^2$ of the Hodge index bound in Theorem 4.67. This excludes none of the four cases; it describes them.

The third condition is the one that Theorem 4.56(iii)(a) imposes on every candidate and that the construction of Setup 4.51 meets by its uniformity over the moduli of principally polarised fourfolds (Corollary 4.57): the object must extend to first order along every polarised deformation of (X,Θ). For $\mathcal{F} = \mathcal{I}_S(3\Theta)$ with S smooth this is a condition on S. If S extends along a deformation X_v of X that keeps Θ of type $(1,1)$, then so does $\mathcal{F}$. Conversely, the obstruction $v \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^2(\mathcal{F},\mathcal{F})$ has, under the edge map to $H^1(S,\mathcal{E}xt^1(\mathcal{F},\mathcal{F})) = H^1(S,N_{S/X})$ of Remark 4.66, the image of v under $H^1(T_X) \to H^1(\iota^*T_X) \to H^1(N_{S/X})$, by Lemma 4.59 applied on the overlaps of a cover; and that image is the obstruction $\mathrm{ob}_S(v)$ to extending the subscheme S along X_v. So condition (a) for $\mathcal{F}$ is the statement that S extends to first order along every polarised deformation of X, and Bloch's semiregularity map [Blo72] puts it in a form that names the object it concerns. For $S \subset X$ smooth of codimension two with normal bundle $N = N_{S/X}$, let

$$\pi\colon H^1(S,N) \xrightarrow{\cong} H^1(S,N^\vee \otimes K_S) \xrightarrow{j} H^1(S,\iota^*\Omega^1_X \otimes K_S) \xrightarrow{\iota_*} H^3(X,\Omega^1_X)$$

be the composite of $N \cong N^\vee \otimes \bigwedge^2 N$, of the conormal inclusion $N^\vee \to \iota^*\Omega^1_X$ tensored with K_S, and of the Gysin map, with $K_X \cong \mathcal{O}_X$. This is the semiregularity map of Bloch for codimension two, and it satisfies $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S]$ for every $v \in H^1(X,T_X)$.

Lemma 4.72 (Duality)**.** *Under Serre duality $H^1(X,T_X) \times H^3(X,\Omega^1_X) \to H^4(\mathcal{O}_X) = \mathbb{C}$,*

$$\{v \in H^1(X,T_X) : S \text{ extends to first order along } v\} = \pi\big(H^1(S,N)\big)^\perp.$$

Proof. S extends along v if and only if $\mathrm{ob}_S(v) = 0$, and ob_S is the composite of the restriction $r\colon H^1(X,T_X) \to H^1(S,\iota^*T_X)$ with the map induced by $\iota^*T_X \to N$. The Serre dual of r is the Gysin map $\iota_*\colon H^1(S,\iota^*\Omega^1_X \otimes K_S) \to H^3(X,\Omega^1_X \otimes K_X)$, and the Serre dual of $H^1(\iota^*T_X) \to H^1(N)$ is the map $H^1(N^\vee \otimes K_S) \to H^1(\iota^*\Omega^1_X \otimes K_S)$ induced by the conormal inclusion. So the dual of ob_S is $\iota_* \circ j$, which is π after the identification $N \cong N^\vee \otimes K_S$, and the kernel of a linear map is the annihilator of the image of its dual. ∎

Proposition 4.73 (The first-order condition is a rank condition)**.** *Let $S \subset X$ be a smooth surface with $[S] = N\Theta^2$, $N \neq 0$. Then $\pi(H^1(S,N)) \supseteq \Theta \cup H^{0,2}(X)$, a subspace of dimension six, and the following are equivalent:*

(i) *S extends to first order along every polarised deformation of (X,Θ), that is, $\mathcal{I}_S(3\Theta)$ satisfies condition (a) of Theorem 4.56(iii);*
(ii) *$\pi(H^1(S,N)) = \Theta \cup H^{0,2}(X)$;*
(iii) *$\operatorname{rank}\pi = 6$, the least value possible.*

In particular a smooth secant support is as far from semiregular in Bloch's sense as a surface with this class can be: $\dim\ker\pi = h^1(N) - 6$, with $h^1(N) = 2h^0(N) - 2\chi(\mathcal{O}_S) + 24N^2 \geq 8 - 2\chi(\mathcal{O}_S) + 24N^2$, which is at least 88, 296, 568 and 904 for $N = 5,6,7,8$.

Proof. The polarised deformations form $P = \{v : v \lrcorner \Theta = 0\}$, of dimension ten in $H^1(T_X) = V \otimes H^{0,1}(X)$. For $z \in H^{0,2}(X)$ the pairing is $\langle v, \Theta \cup z\rangle = (v \lrcorner \Theta) \cup z$, so $\Theta \cup H^{0,2}(X) \subseteq P^\perp$;

the map $z \mapsto \Theta \cup z$ is injective by hard Lefschetz, so this subspace has dimension six, and $\dim P^{\perp} = 16 - 10 = 6$, so $P^{\perp} = \Theta \cup H^{0,2}(X)$. By Lemma 4.72, (i) says $P \subseteq \pi(H^1(N))^{\perp}$, that is $\pi(H^1(N)) \subseteq P^{\perp}$. On the other hand $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S] = 2N\,\Theta \cup (v \lrcorner \Theta)$, and $v \mapsto v \lrcorner \Theta$ is surjective onto $H^{0,2}(X)$, because Θ identifies V with $\overline{V}^{\vee}$ and the contraction becomes the exterior product $V \otimes \overline{V}^{\vee} \to \bigwedge^2 \overline{V}^{\vee}$. So $\pi(H^1(N)) \supseteq \Theta \cup H^{0,2}(X)$ always, and (i), (ii), (iii) are equivalent. For $h^1(N)$, Riemann–Roch with $c_1(N) = K_S$ and $c_2(N) = [S]^2 = 24N^2$ gives $\chi(N) = 2\chi(\mathcal{O}_S) - 24N^2$, Serre duality with $N^{\vee} \otimes K_S \cong N$ gives $h^2(N) = h^0(N)$, and $h^0(N) \geq 4$ because the four translation vector fields map injectively to $H^0(N)$: a translation vector field tangent to S everywhere would restrict to a nowhere vanishing vector field on S, so that $e(S) = c_2(T_S) = 0$, while a secant support has $e(S) = 12N(36 - 3N) \neq 0$ for $N = 5, 6, 7, 8$ by Theorem 4.67. ∎

Remark 4.74 (What the rank condition asks). Proposition 4.73 does not decide the condition; it converts it into the statement that the kernel of Bloch's map has codimension exactly six, as it has for the complete intersections $\Theta \cap \Theta_x$, which do move with X. The kernel of π is $j^{-1}(V^{\vee} \otimes \ker(H^1(K_S) \to H^3(\mathcal{O}_X)))$, and for $q(S) = 4$ it is the image in $H^1(N^{\vee} \otimes K_S)$ of $H^0(\Omega^1_S \otimes K_S)$ modulo $V^{\vee} \otimes H^0(K_S)$. A count with $\chi(\Omega^1_S \otimes K_S) = 2\chi(\mathcal{O}_S) + 24N^2$ and $h^1(T_S) \leq h^0(N) + 12$ shows the rank condition to be consistent with every inequality available, so no exclusion follows at this level; what would decide it is $\ker \pi$ for an actual surface, and no candidate surface is known.

4.5. **Depth and the local obstruction.** By Proposition 4.65, at a smooth point of the support there is nothing local to compare. This is a special case of a much more general statement, and the general statement shows exactly which supports Theorem 4.61 and Corollary 4.63 exclude and which lie beyond their reach. The dividing line is neither smoothness nor the local complete intersection property, but depth.

Theorem 4.75 (Cohen–Macaulay supports carry no local obstruction)**.** *Let Y be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$, let $Z \subset Y$ be a closed subscheme of codimension two, and let $p \in Z$. If the local ring $\mathcal{O}_{Z,p}$ is Cohen–Macaulay, then*

$$\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0 \qquad \text{for every } q \geq 2,$$

and the same holds for $\mathcal{I}_Z \otimes L$ for every line bundle L. Consequently both sides of (19) have vanishing germ at p, and Lemma 4.59 decides nothing there.

Proof. Twisting by L changes no $\mathcal{E}xt$ sheaf, so take $F = \mathcal{I}_Z$. Write $A = \mathcal{O}_{Y,p}$, a regular local ring of dimension m, and $I = \mathcal{I}_{Z,p}$. By hypothesis A/I is Cohen–Macaulay of dimension $m - 2$, so $\operatorname{depth} A/I = m - 2$, and the Auslander–Buchsbaum formula gives

$$\operatorname{pd}_A(A/I) = m - \operatorname{depth}_A(A/I) = 2.$$

From the exact sequence $0 \to I \to A \to A/I \to 0$ with A free it follows that $\operatorname{pd}_A I = 1$. A module of projective dimension one has a free resolution of length one, so $\operatorname{Ext}^q_A(I, M) = 0$ for every A-module M and every $q \geq 2$; taking $M = I$ gives the vanishing, since $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = \operatorname{Ext}^q_A(I, I)$. The last assertion follows from Lemma 4.59: the right side of (19) is $z \cdot \mathrm{id}_F$ with $z \in H^2(\mathcal{O}_Y)$ and has vanishing germs, and the left side lies in a sheaf whose $\mathcal{E}xt^2$ factor is zero. ∎

Remark 4.76 (Sharpness of the hypothesis). Two observations show that Theorem 4.75 is the right statement.

First, Cohen–Macaulayness is strictly weaker than the more familiar conditions that could be imposed instead. Among codimension two germs in $\mathbb{C}^4$, the smooth germ

(x_1, x_2), two planes meeting along a line, (x_1, x_2x_3), three planes meeting along a line, $(x_1, x_2x_3x_4)$, the double plane (x_1^2, x_2) and the quadric cone $(x_1, x_2^2 - x_3x_4)$ are complete intersections. The fat plane $(x_1, x_2)^2$ and the cone over the twisted cubic are the ideals of 2×2 minors of $\left(\begin{smallmatrix} x_1 & x_2 & 0 \\ 0 & x_1 & x_2 \end{smallmatrix}\right)$ and of $\left(\begin{smallmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{smallmatrix}\right)$, of the expected codimension two, so they are Cohen–Macaulay by the Hilbert–Burch theorem, although neither is a local complete intersection, both needing three generators. By Theorem 4.75, $\mathcal{E}xt^2(\mathcal{I}_Z, \mathcal{I}_Z)$ vanishes at all seven. Reducedness, irreducibility and the complete intersection property are therefore all irrelevant; what matters is the projective dimension, which by Auslander–Buchsbaum is a matter of depth.

Second, the vanishing fails at the germs that the construction of Setup 4.51 produces. Two planes meeting at a point and three planes meeting pairwise at a point are reduced surface germs that become disconnected when the point is removed, so they have depth 1: at least 1 because they are reduced, and at most 1 by the connectedness theorem recalled below. A plane with an embedded point has depth 0. By Auslander–Buchsbaum the projective dimensions of $\mathcal{O}_{Z,p}$ are 3, 3 and 4, and the proof of Corollary 4.77(i) below gives $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p \neq 0$ for $q = 2$, 2 and 3. By Hartshorne's connectedness theorem, a scheme whose local ring at p is Cohen–Macaulay is connected in codimension one at p, so a support with two components through p meeting only at p is never Cohen–Macaulay there. This is precisely the configuration analysed in Lemma 4.60.

Corollary 4.77 (The local dichotomy)**.** *Let (X, t) be a polarised abelian variety of dimension $n \geq 4$ and F a perfect complex with $\mathrm{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$, whose germ at each point is the germ of $\mathcal{I}_Z \otimes L$ for a subscheme Z of codimension two, and let $p \in Z$.*

(i) *If Z is not Cohen–Macaulay at p, then $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p \neq 0$ for $q = \operatorname{pd} \mathcal{I}_{Z,p} \geq 2$, so the comparison of germs of Lemma 4.59 has room at p. If moreover Z has at p two smooth branches of codimension two meeting transversally, as in Lemma 4.60 and Corollary 4.63, then F does not satisfy the weakened criterion, nor does $\Phi(F \boxtimes F_2)$ for any secant F_2.*

(ii) *If Z is Cohen–Macaulay, then there is no local obstruction at any point. Condition (b) of Theorem 4.56(iii) is then a single global condition: it holds if and only if the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ vanishes for all $\theta, \theta' \in H^0(T_X)$, the scalar on the right of (19) being forced by the trace. Condition (a), the extension of F to first order along the polarised deformations, is not affected by the depth and remains to be checked.*

In case (ii), $\mathcal{I}_Z$ has projective dimension one at every point, so it is locally the ideal of maximal minors of a Hilbert–Burch matrix; if Z is moreover smooth, then $\mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z) = N_{Z/X}$ and the invariants of Z are those of Theorem 4.67.

Proof. (i) Put $A = \mathcal{O}_{X,p}$, of dimension n, and $I = \mathcal{I}_{Z,p}$. Since A/I is not Cohen–Macaulay, $\operatorname{depth} A/I < \dim A/I \leq n - 2$, so $\operatorname{pd}_A(A/I) \geq 3$ by the Auslander–Buchsbaum formula and $q = \operatorname{pd}_A I \geq 2$. In a minimal free resolution $F_\bullet \to I$ the entries of the differentials lie in the maximal ideal $\mathfrak{m}$, so $\operatorname{Ext}^q_A(I, I)$, the cokernel of $\operatorname{Hom}(F_{q-1}, I) \to \operatorname{Hom}(F_q, I)$, surjects onto $\operatorname{Hom}(F_q, I) \otimes A/\mathfrak{m} \neq 0$ by Nakayama's lemma. At two transversal branches the conclusion is Theorem 4.61 and Corollary 4.63. (ii) Theorem 4.75 gives $\mathcal{E}xt^q = 0$ for $q \geq 2$ at every point, so the local-to-global spectral sequence for $\operatorname{Ext}^2(\mathcal{I}_Z, \mathcal{I}_Z)$ has only the two rows $H^1(\mathcal{E}xt^1)$ and $H^2(\mathcal{O}_X)$, and these lie in different steps of the filtration, the second being the subgroup of scalars. The right side of (19) lies in that subgroup, so its image in the quotient $H^1(\mathcal{E}xt^1)$ is zero, and (b) forces the image of the left side there to vanish. Conversely, if that image vanishes, then $\mathrm{At}(F)^2 = z \cdot \mathrm{id}_F$ for some

$z \in H^2(\mathcal{O}_X) \otimes \bigwedge^2 H^{1,0}$, the map $z \mapsto z \cdot \mathrm{id}_F$ being injective because its composite with the trace is multiplication by $\operatorname{rk} F = 1$; taking traces, $z = \operatorname{tr} \mathrm{At}(F)^2 = -d\,t^2$ by the last assertion of Theorem 4.56(iii), which is (b). The statement on projective dimension was established in the proof of Theorem 4.75, and the Hilbert–Burch theorem is the structure theorem for ideals of projective dimension one over a regular local ring; the last clause is Proposition 4.65 together with Theorem 4.67. ∎

Remark 4.78 (Normalisation and depth). Corollary 4.77 explains the tension recorded in Remark 4.62. The partial normalisation performed in Setup 4.51 separates the two branches at an isolated double point, and separating them is exactly what would make the local ring Cohen–Macaulay there; the normalisation is thus an attempt to reach case (ii). Remark 4.62 shows that the attempt cannot succeed: normalising every isolated point moves $\chi(\mathcal{O}_{\widetilde{Z}})$ to $50N$, while the secant plane demands $(d+9)(27-d) = 72N - 4N^2$, and the two agree only at $N = 11/2$. Repairing the depth destroys the Chern character, and keeping the Chern character keeps a point where the depth fails. This is the precise form of the obstruction, and it explains why a construction assembled from intersections of translates of a polarisation cannot be made to work: by Corollary 4.63 such a chain always has a point of the bad type, and by Remark 4.62 it cannot be separated without leaving the plane.

4.6. **Hilbert–Burch resolutions.** Corollary 4.77 leaves one class of supports and, beside the extension along the polarised deformations, one condition on them. The class consists of the Cohen–Macaulay subschemes of codimension two, and for these the proof of Theorem 4.75 gives more than the vanishing it was used for: the ideal sheaf has projective dimension one at every point. This is the Hilbert–Burch shape, whose invariants are polynomials in the twists.

Setup 4.79 (A resolution of the support)**.** Let (X, Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $Z \subset X$ be Cohen–Macaulay of codimension two, and suppose $\mathcal{I}_Z$ admits a global resolution

$$0 \longrightarrow E_1 \xrightarrow{\varphi} E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0, \qquad \operatorname{rk} E_0 = r+1, \quad \operatorname{rk} E_1 = r, \tag{27}$$

with $E_0 = \bigoplus_{i=1}^{r+1} \mathcal{O}(e_i\Theta)$ and $E_1 = \bigoplus_{j=1}^{r} \mathcal{O}(f_j\Theta)$ sums of line bundles. Put $p_i = e_i + b$ and $q_j = f_j + b$.

Lemma 4.80 (The resolution in moments)**.** *In Setup 4.79,* $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\,v_\Theta$ *holds if and only if*

$$\sum_{i=1}^{r+1} p_i^k - \sum_{j=1}^{r} q_j^k = c_k, \qquad k = 0,1,2,3,4, \qquad (c_k) = (1,\ b,\ -d,\ -db,\ d^2). \tag{28}$$

The equation at $k = 0$ is the rank count and is automatic. In addition, φ can be injective only if $\max_i e_i \geq \max_j f_j$, and $\mathcal{O}_X$ embeds in $E_0^\vee$ only if $\min_i e_i \leq 0$.

Proof. From (27), $\operatorname{ch}(\mathcal{I}_Z) = \operatorname{ch}(E_0) - \operatorname{ch}(E_1)$, so $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = \sum_i e^{p_i\Theta} - \sum_j e^{q_j\Theta}$. Comparing coefficients of $\Theta^k/k!$ with those of $u_\Theta + bv_\Theta$, which are the c_k displayed, gives (28). For the last two statements, recall that on an abelian variety $\mathrm{Hom}(\mathcal{O}(f\Theta), \mathcal{O}(e\Theta)) = H^0(\mathcal{O}((e-f)\Theta))$ vanishes unless $e \geq f$. If φ is injective, each summand of E_1 must map nonzero to E_0, which requires $\max_i e_i \geq f_j$ for every j. Dualising (27) and using $\mathcal{I}_Z^\vee = \mathcal{O}_X$ gives an injection $\mathcal{O}_X \hookrightarrow E_0^\vee$, which requires $-e_i \geq 0$ for some i. ∎

The case $r = 1$ consists of the complete intersections, and it is settled for every b and every d by an argument that needs no search.

Theorem 4.81 (No complete intersection supports a secant object)**.** *Let (X,Θ) be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $d>0$ and $b\geq 1$. No complete intersection $Z = D_1 \cap D_2$ of two divisors satisfies $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\, v_\Theta$.*

Proof. Such a Z is Setup 4.79 with $r=1$, the resolution being the Koszul complex of the two equations. Put $w(x) = x^2 + d$, which is strictly positive on $\mathbb{R}$ because $d>0$, and read (28) in the combinations $c_{k+2} + d\,c_k$ for $k = 0,1,2$. Those three numbers are

$$c_2 + dc_0 = -d + d = 0, \qquad c_3 + dc_1 = -db + db = 0, \qquad c_4 + dc_2 = d^2 - d^2 = 0,$$

so that

$$\sum_i w(p_i)\, p_i^k = \sum_j w(q_j)\, q_j^k, \qquad k = 0,1,2.$$

The two sides are the moments of order 0, 1 and 2 of the positive measures $\nu_P = \sum_i w(p_i)\delta_{p_i}$ and $\nu_Q = \sum_j w(q_j)\delta_{q_j}$, so those measures have the same mass, the same mean and the same variance. At $r=1$ the measure ν_Q has one atom, hence variance zero, so ν_P has variance zero as well, and a positive measure with two atoms has variance zero only if the atoms coincide. Hence $p_1 = p_2$, and equality of the means forces $q_1 = p_1$. The resolution (27) then cancels, and $\mathcal{I}_Z \cong \mathcal{O}(e_1\Theta)$ would be a line bundle, which the ideal sheaf of a subscheme of codimension two is not.

In terms of the divisors, the same computation shows that their degrees a_1 and a_2 would have to satisfy $a_1 + a_2 = \frac{4}{3}b$ and $(b-a_1)(b-a_2) = \frac{1}{2}(\frac{1}{3}b^2 + d)$; the quadratic with roots $b-a_1$ and $b-a_2$ then has discriminant $-\frac{2}{9}b^2 - 2d < 0$, so the degrees are not even real. ∎

For $r \geq 2$ the moment conditions alone no longer decide the question. The Chern characters of the bundles of Setup 4.79, and of every bundle built from line bundles, lie in one lattice, and the secant plane meets that lattice only along a congruence. Write a class $\xi \in \mathbb{Q}[\Theta]/(\Theta^5)$ as $\xi = \sum_{k=0}^4 c_k(\xi)\,\Theta^k/k!$, as in Lemma 4.80, so that $c_k(e^{j\Theta}) = j^k$, and let $\mathcal{L}_\Theta$ be the subgroup generated by the classes $e^{j\Theta} = \mathrm{ch}\,\mathcal{O}_X(j\Theta)$, $j\in\mathbb{Z}$. It is a subring, since $e^{a\Theta}e^{b\Theta} = e^{(a+b)\Theta}$, and it contains ch of every bundle obtained from line bundles algebraically equivalent to multiples of Θ by direct sums, tensor products, duals, exterior and symmetric powers, kernels of surjections and cokernels of injections between such bundles, and extensions; in particular the kernel bundles $M_{m\Theta} = \ker\big(H^0(\mathcal{O}(m\Theta))\otimes\mathcal{O}_X \to \mathcal{O}_X(m\Theta)\big)$, of class $m^4 - e^{m\Theta}$, and their duals and twists.

Lemma 4.82 (Binomial moments)**.** *Define $m_k(\xi)\in\mathbb{Q}$, $0\leq k\leq 4$, by*

$$c_i(\xi) = \sum_{k=0}^{i} S(i,k)\, k!\, m_k(\xi), \tag{29}$$

where $S(i,k)$ are the Stirling numbers of the second kind; explicitly

$$m_0 = c_0, \qquad m_1 = c_1, \qquad m_2 = \tfrac{1}{2}(c_2 - c_1),$$
$$m_3 = \tfrac{1}{6}(c_3 - 3c_2 + 2c_1), \qquad m_4 = \tfrac{1}{24}(c_4 - 6c_3 + 11c_2 - 6c_1).$$

Then $\xi\in\mathcal{L}_\Theta$ if and only if $m_k(\xi)\in\mathbb{Z}$ for $k = 0,\dots,4$.

Proof. For $\xi = \sum_j a_j e^{j\Theta}$ one has $c_i(\xi) = \sum_j a_j j^i$, and $j^i = \sum_k S(i,k)\,k!\binom{j}{k}$, so (29) holds with $m_k(\xi) = \sum_j a_j\binom{j}{k}\in\mathbb{Z}$. Conversely, the five vectors $\big(\binom{j}{0},\dots,\binom{j}{4}\big)$ for $j=0,\dots,4$ form a unitriangular matrix, so every integer vector $(m_0,\dots,m_4)$ is an integer combination of them, and the corresponding $\sum_{j=0}^4 a_j e^{j\Theta}$ lies in $\mathcal{L}_\Theta$ with the prescribed moments. Since (29) is invertible over $\mathbb{Q}$, the moments determine ξ. ∎

Equivalently, sending $e^{j\Theta}$ to x^j identifies $\mathbb{Q}[\Theta]/(\Theta^5)$ with $\mathbb{Q}[y]/(y^5)$, $y = x-1$, the class $e^{j\Theta}$ going to $(1+y)^j$; then $m_k(\xi)$ is the coefficient of y^k and $\mathcal{L}_\Theta$ is $\mathbb{Z}[y]/(y^5)$.

Theorem 4.83 (The secant plane and the lattice of line bundles)**.** *For $(a,b) \in \mathbb{Z}^2$ and $d \geq 1$,*

$$m(a\,u_\Theta + b\,v_\Theta) = \Big(a,\ b,\ -\frac{ad+b}{2},\ \frac{3ad+2b-bd}{6},\ \frac{ad^2-11ad+6bd-6b}{24}\Big).$$

Consequently $a\,u_\Theta + b\,v_\Theta \in \mathcal{L}_\Theta$ if and only if

$$ad \equiv b \pmod 2, \qquad 3ad+2b-bd \equiv 0 \pmod 6,$$
$$ad^2-11ad+6bd-6b \equiv 0 \pmod{24}.$$

For $a=1$ and $b=3$ the moments are $(1,3,-\frac{d+3}{2},1,\frac{(d+9)(d-2)}{24})$, and $u_\Theta + 3v_\Theta \in \mathcal{L}_\Theta$ if and only if d is odd and $24 \mid (d+9)(d-2)$, that is, if and only if

$$d \equiv 15 \text{ or } 23 \pmod{24}.$$

Proof. In the variable $y = x-1$ put $L = \log(1+y) = y - \frac12 y^2 + \frac13 y^3 - \frac14 y^4$ modulo y^5. Under $e^{j\Theta} \mapsto x^j$ the class Θ goes to L, hence $u_\Theta \mapsto 1 - \frac{d}{2}L^2 + \frac{d^2}{24}L^4$ and $v_\Theta \mapsto L - \frac{d}{6}L^3$. With $L^2 = y^2 - y^3 + \frac{11}{12}y^4$, $L^3 = y^3 - \frac32 y^4$ and $L^4 = y^4$ modulo y^5, the coefficients of $1, y, \dots, y^4$ in $a\,u_\Theta + b\,v_\Theta$ are the five displayed expressions, and the congruences are the integrality of the last three. For $b=3$ the second reads $3d+6-3d=6$, so $m_3=1$, and the third reads $d^2+7d-18 = (d+9)(d-2)$. Modulo 8 the condition is $8 \mid (d+1)(d-2)$ with d odd, hence $d \equiv 7 \pmod 8$; modulo 3 it is $3 \mid d(d-2)$, hence $d \equiv 0,2 \pmod 3$; together $d \equiv 15, 23 \pmod{24}$. ∎

Corollary 4.84 (Split resolutions, for every rank and every twist)**.** *Let $Z \subset X$ be Cohen–Macaulay of codimension two with a resolution $0 \to E_1 \to E_0 \to \mathcal{I}_Z \to 0$ in which $\operatorname{ch} E_0$ and $\operatorname{ch} E_1$ both lie in $\mathcal{L}_\Theta$: for instance Setup 4.79 with any rank and any twists, or E_0 and E_1 built from such line bundles and from the bundles $M_{m\Theta}$ by the operations listed above. If $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = u_\Theta + 3v_\Theta$, then $d \equiv 15$ or $23 \pmod{24}$. In particular this never happens for the four discriminants of Corollary 4.68.*

Proof. $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = e^{3\Theta}(\operatorname{ch} E_0 - \operatorname{ch} E_1) \in \mathcal{L}_\Theta$, and Theorem 4.83 applies. ∎

Remark 4.85 (What the congruence measures)*.* The fractional part of m_4 is the invariant that separates the secant plane from the line bundles. For $d \in \{1,3,5,7\}$ it is $-\frac{5}{12}, \frac12, \frac34, \frac13$ modulo $\mathbb{Z}$, and any object with Chern character $u_\Theta + 3v_\Theta$ must carry a class outside $\mathcal{L}_\Theta$ with exactly that defect. Two checks. A point has $m = (0,0,0,0,1/24)$, and the partial normalisation of Setup 4.51 at $(d+9)(2d-1)$ isolated points shifts m_4 by $(d+9)(2d-1)/24$; since $(d+9)(d-2) + (d+9)(2d-1) = 3(d+9)(d-1) \equiv 0 \pmod{24}$ for odd d, the unnormalised chain, which lies in $\mathcal{L}_\Theta$, is moved onto the plane by exactly the right defect, as it must be. For a smooth support the class of the structure sheaf has $m(\mathcal{O}_S) = \big(0,0,N,-3N,N(83-2N)/12\big)$, and $N(83-2N)/12$ is not an integer for $N \in \{5,6,7,8\}$; so a smooth secant support is a surface whose structure sheaf is not in the line-bundle subring, unlike every complete intersection of translates of Θ and every degeneracy locus of a map between bundles in $\mathcal{L}_\Theta$. The Fourier–Mukai transforms of line bundles and the push-forwards of line bundles along isogenies do have classes outside $\mathcal{L}_\Theta$, which is why the semi-homogeneous bundles need the separate argument of Theorem 4.92.

The lattice constrains the classes of the two terms of a resolution; their ranks constrain the Chern classes, because a vector bundle of rank s has $c_k = 0$ for $k > s$, and on a fourfold this bites for $s \leq 3$.

Proposition 4.86 (Resolutions of rank one and two)**.** *Let (X,Θ) be a principally polarised abelian fourfold, let $b\geq 1$ and $d>0$, and let $Z\subset X$ be of codimension two with a resolution*

$$0\longrightarrow E_1\longrightarrow E_0\longrightarrow \mathcal{I}_Z\longrightarrow 0$$

by vector bundles of ranks r and $r+1$, and with $\operatorname{ch}(\mathcal{I}_Z(b\Theta))=u_\Theta+b\,v_\Theta$.

(i) *$r\geq 2$, whatever the bundles. Equivalently, Z is not the zero locus of a section of a vector bundle of rank two.*

(ii) *If $r=2$ and the Chern classes of E_1 are polynomials in Θ, as they are when $\operatorname{Hdg}^2(X)=\mathbb{Q}\Theta$ and $\operatorname{Hdg}^4(X)=\mathbb{Q}\Theta^2$, then, writing $c_1(E_1(b\Theta))=a\Theta$ with $a\in\mathbb{Z}$,*

$$3d-b^2-4ab\equiv 0 \pmod{12}.$$

At $b=3$ this says $d\equiv 3 \pmod 4$.

In particular, at the discriminants $d=1$ and $d=5$ of Corollary 4.68, a resolution whose Chern classes are polynomials in Θ has $r\geq 3$.

Proof. Put $E=E_1(b\Theta)$, $G=E_0(b\Theta)$ and $F=\mathcal{I}_Z(b\Theta)$, so that $0\to E\to G\to F\to 0$ is exact and $c(G)=c(E)\,c(F)$. Newton's identities applied to $\operatorname{ch}(F)=1+b\Theta-\frac{d}{2}\Theta^2-\frac{db}{6}\Theta^3+\frac{d^2}{24}\Theta^4$ give, with $N=\frac12(b^2+d)$,

$$c(F)=1+b\Theta+N\Theta^2+\tfrac{bN}{3}\Theta^3+\tfrac{N(b^2-3d)}{12}\Theta^4.$$

Since E and G are vector bundles, $c_k(E)=0$ for $k>r$ and $c_k(G)=0$ for $k>r+1$.

(i) Let $r=1$ and $\ell=c_1(E)$, so that $c(G)=(1+\ell)\,c(F)$. The rank two bundle G has $c_3(G)=\frac{bN}{3}\Theta^3+N\ell\,\Theta^2=0$, and cup product with Θ^2 is an isomorphism $H^2(X,\mathbb{Q})\to H^6(X,\mathbb{Q})$ by hard Lefschetz, so $\ell=-\frac{b}{3}\Theta$. Then

$$c_4(G)=c_4(F)+\ell\,c_3(F)=N\Big(\frac{b^2-3d}{12}-\frac{b^2}{9}\Big)\Theta^4=-\frac{N(b^2+9d)}{36}\,\Theta^4,$$

which is not zero, since $N>0$ and $d>0$. A section of a rank two bundle V vanishing in codimension two exactly on Z gives the Koszul resolution $0\to\det V^\vee\to V^\vee\to\mathcal{I}_Z\to 0$, which has $r=1$; conversely, at $r=1$ the map $E_1\to E_0$ is a section of $E_0\otimes E_1^\vee$ with zero locus Z.

(ii) Let $r=2$, and write $c_1(E)=a\Theta$ and $c_2(E)=\frac{m}{2}\Theta^2$ with $a,m\in\mathbb{Z}$, since Θ and $\frac12\Theta^2$ are primitive integral classes. Now $c_3(E)=c_4(E)=0$, and the rank three bundle G has

$$c_4(G)=c_4(F)+c_1(E)\,c_3(F)+c_2(E)\,c_2(F)=0.$$

Integrating, with $\int_X\Theta^4=24$, and dividing by $2N$ gives $6m=3d-b^2-4ab$. By Riemann–Roch with trivial Todd class, $\chi(E)=\int_X\operatorname{ch}_4(E)=a^4-2a^2m+\frac12 m^2$ is an integer, so m is even, which is the congruence. At $b=3$ it reads $3(d-3)\equiv 12a\equiv 0$ modulo 12, that is, $d\equiv 3 \pmod 4$. ∎

Remark 4.87 (What the ranks exclude). Part (i) extends Theorem 4.81 from two divisors to the zero locus of a section of any rank two bundle, stable or not, so the construction of Serre produces no secant support; it holds for every X, with no condition on $\operatorname{NS}(X)$. Part (ii) holds for arbitrary bundles whose Chern classes are polynomials in Θ, not only for sums of line bundles, and at $b=3$ it excludes $r=2$ whenever $d\not\equiv 3 \pmod 4$. The integrality of the Chern classes of E_0 is automatic, since $c(E_0(b\Theta))=c(E_1(b\Theta))\,c(F)$ and $c(F)$ is integral, so it is the vanishing above the rank, and not integrality, that constrains. Neither condition restricts $r\geq 3$: a bundle of rank four has no vanishing Chern class on a fourfold, and at $r=3$ the only condition, $c_4(E)=0$, involves E alone.

Corollary 4.84 says where a split resolution can have the right Chern character. A stronger statement holds: no object with a split resolution satisfies the criterion, at any rank and for any discriminant (Theorem 4.89). It rests on a necessary condition that we isolate, because it applies to every object and can be tested on any candidate before any global computation is attempted.

Lemma 4.88 (Morphisms to a line bundle are killed by $H^2(\mathcal{O})$)**.** *Let (X,t) be a polarised abelian variety of dimension $n \geq 3$, let F be a perfect complex with $\operatorname{ch}(F) = au_t + bv_t$, $(a,b) \neq (0,0)$, which satisfies the weakened criterion of Setup 4.54, and let L be a line bundle with $c_1(L) \in \mathbb{Q}\,t$. Then for every $m \in \mathbb{Z}$, every morphism $p\colon F \to L[m]$, every morphism $q\colon L[m] \to F$, and every $z \in H^2(\mathcal{O}_X)$,*

$$z \cdot p = 0 \quad \text{in } \operatorname{Hom}(F, L[m+2]), \qquad z \cdot q = 0 \quad \text{in } \operatorname{Hom}(L[m], F[2]),$$

where $z \cdot (\,\cdot\,)$ is the $H^*(\mathcal{O}_X)$-module structure on morphisms in $D^b(X)$. In particular, if $F = \mathcal{I}_Z \otimes L'$ for a closed subscheme Z and a line bundle L' with $c_1(L') \in \mathbb{Q}t$, then the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$ is injective.

Proof. For a constant bivector π the class $x_\pi = (\frac{d}{2}\,\pi \lrcorner t^2, 0, \pi)$ preserves $\operatorname{ch}(F)$ by Theorem 4.56(ii), so $\operatorname{ev}_F(x_\pi) = 0$ by hypothesis. The Hochschild action is by natural transformations $\mathrm{id} \to \mathrm{id}[2]$ of $D^b(X)$, so for $p\colon F \to L[m]$ we have $\operatorname{ev}_{L[m]}(x_\pi) \circ p = p \circ \operatorname{ev}_F(x_\pi) = 0$, where $\operatorname{ev}_{L[m]} = \operatorname{ev}_L[m]$. By (11), $\operatorname{ev}_L(x_\pi) = c_\pi \cdot \mathrm{id}_L$ with $c_\pi = \frac{1}{2}\,\pi \lrcorner (\lambda^2 + d\,t^2) \in H^{0,2}(X)$ and $\lambda = c_1(L) = e\,t$, that is, $c_\pi = \frac{e^2+d}{2}\,\pi \lrcorner t^2$. Since $e^2 + d > 0$ and $\pi \mapsto \pi \lrcorner t^2$ is an isomorphism from $H^0(\bigwedge^2 T_X)$ onto $H^{0,2}(X)$, as in the proof of Theorem 4.56(iii), c_π runs over all of $H^2(\mathcal{O}_X)$ as π varies; as $(c_\pi \cdot \mathrm{id}_L) \circ p = c_\pi \cdot p$, this proves the assertion for p. The assertion for q follows in the same way from the square $\operatorname{ev}_F(x_\pi) \circ q = q \circ \operatorname{ev}_{L[m]}(x_\pi)$. For the last assertion, apply the first to the inclusion $p\colon \mathcal{I}_Z \otimes L' \to L'$. After twisting by L'^{-1}, the sequence $0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$ gives an exact sequence

$$\operatorname{Ext}^2(\mathcal{O}_Z, \mathcal{O}_X) \longrightarrow H^2(\mathcal{O}_X) \longrightarrow \operatorname{Ext}^2(\mathcal{I}_Z, \mathcal{O}_X),$$

in which the second map is $z \mapsto z \cdot p$. So $z \cdot p = 0$ for all z means that the first map is surjective, and by Serre duality, ω_X being trivial, the first map is dual to the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$. ∎

Theorem 4.89 (No two-term complex of powers of the polarisation satisfies the criterion)**.** *Let (X,t) be a polarised abelian variety of dimension $n \geq 3$ and M an ample line bundle with $c_1(M) \in \mathbb{Q}t$. Let A and B be direct sums of tensor powers $M^{\otimes e}$, $e \in \mathbb{Z}$, let $\varphi\colon A \to B$ be any morphism, and let F be the cone of φ, or any shift of it. If $F \neq 0$ and $\operatorname{ch}(F) = au_t + bv_t$ with $(a,b) \neq (0,0)$, then F does not satisfy the weakened criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ by sums of powers of M satisfies it, whatever the rank.*

Proof. A pair of summands $M^{\otimes e}$ of A and $M^{\otimes e}$ of B joined by a nonzero entry of φ, which is then a nonzero scalar, can be cancelled without changing the cone up to isomorphism; do this until no such pair remains. If $A = 0$ or $B = 0$ after cancelling, F is a shifted sum of line bundles, $\operatorname{ev}_F(x_\pi)$ is diagonal with the nonzero entries $c_\pi \cdot \mathrm{id}$ of the proof of Lemma 4.88, and the criterion fails as in Corollary 4.21; so suppose both are nonzero and consider the triangle $A \xrightarrow{\varphi} B \xrightarrow{q} F \to A[1]$. Let $M^{\otimes b_j}$ be a summand of B, ι_j its inclusion into B, and $q_j = q \circ \iota_j\colon M^{\otimes b_j} \to F$. Applying $\operatorname{Hom}(M^{\otimes b_j}, -)$ to the triangle gives the exact sequence

$$\operatorname{Hom}(M^{\otimes b_j}, A[2]) \xrightarrow{\varphi_*} \operatorname{Hom}(M^{\otimes b_j}, B[2]) \xrightarrow{q_*} \operatorname{Hom}(M^{\otimes b_j}, F[2]),$$

and $z \cdot q_j = q_*(z \cdot \iota_j)$ for $z \in H^2(\mathcal{O}_X)$. Now $\operatorname{Hom}(M^{\otimes b_j}, B[2]) = \bigoplus_k H^2(M^{\otimes(b_k - b_j)})$, and $H^2(M^{\otimes m}) = 0$ for $m \neq 0$: for $m > 0$ by Kodaira vanishing, ω_X being trivial, and for $m < 0$ because a negative line bundle on an abelian variety has cohomology only in degree $n \geq 3$. So $z \cdot \iota_j$ is the element with component z in the summand $k = j$ and zero elsewhere. It lies in the image of φ_* only if there is $\psi \in \operatorname{Hom}(M^{\otimes b_j}, A[2]) = \bigoplus_i H^2(M^{\otimes(a_i - b_j)})$ with $(\varphi \circ \psi)_j = \sum_i \varphi_{ji}\psi_i = z$; but $\psi_i \neq 0$ forces $a_i = b_j$, and for such i the entry φ_{ji} is a scalar, zero after the cancellation. So $z \cdot q_j \neq 0$ for every $z \neq 0$, which contradicts Lemma 4.88. The last assertion is the case of a resolution, where $F \cong \operatorname{coker}\varphi$ is the cone of $\varphi\colon E_1 \to E_0$. ∎

The powers of the polarisation are not the only bundles on which the compensated Poisson classes act by scalars. The same holds for every bundle whose Atiyah class is a multiple of the identity, and on an abelian variety these are the semi-homogeneous bundles of Mukai [Muk78]. The argument of Theorem 4.89 goes through for them, with the cohomology of the fibre product of two isogenies in place of the cohomology of a power of Θ. We state the result in a self-contained form.

Definition 4.90 (Isogeny blocks)**.** Let (X, Θ) be a polarised abelian variety. An *isogeny block* is a vector bundle $E = \pi_* L$, where $\pi\colon Y \to X$ is an isogeny and L is a line bundle on Y with $c_1(L) = \nu\, \pi^*\Theta$ in $H^2(Y, \mathbb{Q})$, $\nu \in \mathbb{Q}$; its *slope* is ν, so that $c_1(E) = \nu \deg(\pi)\, \Theta$. The block is *simple* if $\tau_g^* L \not\cong L$ for every nonzero $g \in \ker \pi$. The powers $\mathcal{O}(e\Theta)$ are the blocks with $\pi = \mathrm{id}$.

Lemma 4.91 (Three properties of a block)**.** *Let $E = \pi_* L$ be an isogeny block of slope ν on an abelian n-fold.*

(i) $\operatorname{At}(E) = \nu\Theta \cdot \mathrm{id}_E$*; hence for a constant bivector π the compensated class x_π of Theorem 4.56(ii) acts by* $\operatorname{ev}_E(x_\pi) = \frac{\nu^2 + d}{2}\, (\pi \lrcorner \Theta^2) \cdot \mathrm{id}_E$*, and Lemma 4.88 holds with E in place of the line bundle L.*

(ii) *For a second block $E' = \pi'_* L'$ of slope ν', $\operatorname{Ext}^q(E', E) = H^q(Z, M)$ with $Z = Y' \times_X Y$, a disjoint union of abelian varieties isogenous to X, and M a line bundle whose class on each component is $(\nu - \nu')$ times the pullback of Θ. If $\nu \neq \nu'$ and $n \geq 3$ then $\operatorname{Ext}^2(E', E) = 0$. If E is simple then $\operatorname{Ext}^2(E, E) = H^2(\mathcal{O}_X) \cdot \mathrm{id}_E$.*

(iii) *A simple block is μ-stable with respect to Θ; hence a nonzero morphism between simple blocks of the same slope is an isomorphism.*

Proof. (i) Since π is étale, $\pi^*\Omega^1_X = \Omega^1_Y$ and π_* is exact, so the first jet sequence of $\pi_* L$ is the direct image of that of L, and $\operatorname{At}(\pi_* L)$ is the image of $\operatorname{At}(L)$ under $\pi_*\colon \operatorname{Ext}^1_Y(L, L \otimes \pi^*\Omega^1_X) \to \operatorname{Ext}^1_X(\pi_* L, \pi_* L \otimes \Omega^1_X)$. As $\operatorname{At}(L) = c_1(L) \cdot \mathrm{id}_L = \nu\, \pi^*\Theta \cdot \mathrm{id}_L$, the projection formula gives $\nu\Theta \cdot \mathrm{id}_{\pi_* L}$. Then (15) with a scalar Atiyah class gives the value of $\operatorname{ev}_E(x_\pi)$, and the proof of Lemma 4.88 used nothing else about L.

(ii) Adjunction for the finite étale π', flat base change along the cartesian square with vertices Z, Y', Y, X, and the projection formula give $\operatorname{Ext}^q_X(\pi'_* L', \pi_* L) = H^q(Y', L'^{-1} \otimes \pi'^*\pi_* L) = H^q(Z, p'^* L'^{-1} \otimes p^* L)$, with p, p' the two projections; the class of $M = p'^* L'^{-1} \otimes p^* L$ is $(\nu - \nu')$ times the pullback of Θ on every component, which is an abelian variety finite étale over Y'. A line bundle on an abelian n-fold whose class is a positive multiple of an ample class has cohomology only in degree 0, and one whose class is a negative multiple only in degree n, by Kodaira vanishing and Serre duality with trivial canonical bundle; so $H^2(Z, M) = 0$ when $\nu \neq \nu'$ and $n \geq 3$. When $E' = E$, the components of $Y \times_X Y$ are the graphs of the translations by $g \in \ker \pi$, on which M restricts to $L^{-1} \otimes \tau_g^* L \in \operatorname{Pic}^0(Y)$; a nontrivial element of Pic^0 has no cohomology, so $\operatorname{Ext}^q(E, E) = \bigoplus_g H^q(\mathcal{O}_Y)$ over the g with $\tau_g^* L \cong L$, which for a simple block is $g = 0$ alone, and $H^q(\mathcal{O}_Y) = H^q(\mathcal{O}_X)$ through π.

(iii) The pullback $\pi^*E = \bigoplus_{g\in\ker\pi} \tau_g^*L$ is a direct sum of line bundles of one slope, hence μ-semistable, and semistability descends along the finite étale π; so E is semistable. If $F \subset E$ is a saturated subsheaf of the same slope, then $\pi^*F \subset \pi^*E$ is a saturated subsheaf of the same slope of a direct sum of pairwise non-isomorphic line bundles, the τ_g^*L being distinct because the block is simple, hence a partial sum $\bigoplus_{g\in S} \tau_g^*L$; it is invariant under the translations by $\ker\pi$, which permute the summands transitively, so S is empty or everything and $F = 0$ or $F = E$. Finally, a nonzero morphism f between μ-stable bundles of the same slope is injective and generically surjective, so the two bundles have the same rank; for blocks of the same slope ν they then have the same c_1, since $c_1(E) = \nu\,\mathrm{rk}(E)\,\Theta$, and $\det f$ is a nonzero section of a numerically trivial line bundle, hence nowhere zero. So f is an isomorphism. ∎

Theorem 4.92 (No two-term complex of simple blocks satisfies the criterion)**.** *Let (X, Θ) be a polarised abelian variety of dimension $n \geq 3$, let A and B be direct sums of simple isogeny blocks, let $\varphi\colon A \to B$ be any morphism and let F be the cone of φ or a shift of it. If $F \neq 0$ and $\mathrm{ch}(F) = au_\Theta + bv_\Theta$ with $(a, b) \neq (0, 0)$, then F does not satisfy the weakened criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ by sums of simple isogeny blocks satisfies it, whatever the rank.*

Proof. Cancel, by Gaussian elimination in the two-term complex, every pair of isomorphic blocks of A and B joined by an entry of φ that is an isomorphism; this changes neither the cone nor the block structure of the remaining terms. By Lemma 4.91(iii), every remaining entry between blocks of the same slope is zero. If A or B is empty after cancelling, F is a shifted sum of blocks, on each of which $\mathrm{ev}(x_\pi)$ is a nonzero scalar by (i). Otherwise the rest is the proof of Theorem 4.89 with Lemma 4.91 in place of the vanishing of $H^2(M^{\otimes m})$. For a block E' of B with inclusion ι and $q' = q \circ \iota\colon E' \to F$, the class $z \cdot \iota \in \mathrm{Hom}(E', B[2])$ has component $z \cdot \mathrm{id}_{E'} \neq 0$ in $\mathrm{Ext}^2(E', E') = H^2(\mathcal{O}_X) \cdot \mathrm{id}_{E'}$ by (ii). It lies in the image of φ_* only if $\sum_i \varphi_{ji}\psi_i = z \cdot \mathrm{id}_{E'}$ for some $\psi_i \in \mathrm{Ext}^2(E', A_i)$, and $\psi_i \neq 0$ forces A_i to have the slope of E' by (ii), for which $\varphi_{ji} = 0$. So $z \cdot q' \neq 0$, contradicting Lemma 4.88 in the form given by (i). ∎

Remark 4.93 (Semi-homogeneous bundles)*.* By Mukai [Muk78], the simple isogeny blocks are exactly the simple semi-homogeneous bundles, a bundle E being semi-homogeneous if for every x there is $P \in \mathrm{Pic}^0(X)$ with $\tau_x^*E \cong E \otimes P$; and every semi-homogeneous bundle on a variety with $\mathrm{NS}(X)_\mathbb{Q} = \mathbb{Q}\Theta$ is an iterated extension of such blocks and their twists by Pic^0. We do not use this. In these terms, Theorem 4.92 says that the bundles in a Hilbert–Burch resolution of a candidate cannot all have scalar Atiyah class. The obstruction is always the same: the compensated Poisson classes act on a block by the scalar $\frac{\nu^2+d}{2}\,\pi\lrcorner\,\Theta^2$, which would vanish only for $\nu^2 = -d$, and cones of maps between blocks inherit the scalar through the summand inclusions, because Ext^2 between blocks of different slopes vanishes. A candidate must therefore be built from a bundle whose Atiyah class is not a scalar, that is, one which is not projectively flat.

Corollary 4.94 (Split resolutions and the remaining shapes)**.** *Let (X, Θ) be a principally polarised abelian fourfold.*

(i) *No secant object of rank one whose support has a resolution (27) by sums of powers of Θ satisfies the weakened criterion, for any r, any twists and any d, whether or not its matrix degenerates in codimension two.*

(ii) *If a secant object $\mathcal{I}_Z(b\Theta)$ satisfies the criterion, then Z has no point at which two smooth branches of codimension two meet transversally, no two-term resolution*

of $\mathcal{I}_Z$ splits into powers of Θ or into simple semi-homogeneous bundles, and the restriction $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_Z)$ is injective. If Z is Cohen–Macaulay, $\mathcal{I}_Z$ has projective dimension one. If $b = 3$ and Z is smooth, then $d \leq 9$, so $d \in \{1, 3, 5, 7\}$ when $N = (9 + d)/2$ is an integer as in Setup 4.51, the two terms of a resolution do not both have classes in $\mathcal{L}_\Theta$, and when X is simple the injectivity is automatic.

(iii) *At $b = 3$, smooth supports and split resolutions are disjoint, for every rank and every twist of the resolution: a smooth support forces $d \leq 9$, and a resolution by bundles with classes in $\mathcal{L}_\Theta$ forces $d \equiv 15, 23 \pmod{24}$, hence $d \geq 15$, so that, even without (i), no support could have both properties.*

Proof. Part (i) is Theorem 4.89 with $M = \mathcal{O}_X(\Theta)$, the resolution (27) presenting $\mathcal{I}_Z(b\Theta)$ as the cone of $\varphi \otimes \mathcal{O}(b\Theta)$. In (ii), the branches are excluded by Corollary 4.77(i), projective dimension one comes from Corollary 4.77(ii), the non-splitting from Theorem 4.89 and Theorem 4.92, the injectivity from the last assertion of Lemma 4.88, the bound on d from Corollary 4.68, the statement on the two terms from Corollary 4.84, and the automatic injectivity from Lemma 4.70. Part (iii) compares Corollary 4.68 with Corollary 4.84: no $d \leq 9$ is congruent to 15 or 23 modulo 24. ∎

5. Weil classes of CM fields and the polarised criterion

5.1. **Weil tori of a CM field and the pure form.** The argument of Theorem 3.5 uses the field only through the eigenspaces of its action, and it extends to every CM field. For $m \geq 2$ the Hodge ring of a very general Weil torus is larger than for an imaginary quadratic field, since products of Weil classes over a set of embeddings closed under conjugation are Hodge classes there, and a Weil torus may carry subvarieties of codimension $2n$; but it carries none of codimension n, and that is all the argument needs.

Setup 5.1 (Weil tori of a CM field)**.** Keep [I, Setup 17.1], let $n \geq 2$ and let A be of (F, n, δ)-Weil type, with $V_{\mathbb{C}} = \bigoplus_\sigma V_\sigma$ as there. Complex conjugation carries $V_{\overline{\sigma}}$ onto V_σ. A complex structure on $V_{\mathbb{R}}$ commuting with F, with $H^{1,0} \cap V_\sigma$ of dimension n for every σ, is the same as a choice, for each real place τ, of complementary n-dimensional subspaces $X_\tau, Y_\tau \subset V_{\sigma_\tau}$: then $H^{1,0} \cap V_{\sigma_\tau} = X_\tau$, $H^{0,1} \cap V_{\sigma_\tau} = Y_\tau$, and on $V_{\overline{\sigma}_\tau}$ the two parts are the conjugates of Y_τ and X_τ. Call the complex torus obtained in this way a Weil torus of F, and let S_F be the connected complex manifold of these choices, of dimension $2mn^2$. Its tangent space at A is $\mathrm{Hom}_F(H^{1,0}, H^{0,1})$, the polarised family $\mathcal{D}_F$ is the submanifold on which E is of type $(1, 1)$, and S_F contains non-algebraic tori. For $\sigma = \sigma_\tau$ the line $\bigwedge^{2n} V_\sigma = \bigwedge^n X_\tau \otimes \bigwedge^n Y_\tau$ is of type (n, n), and likewise for $\overline{\sigma}_\tau$, so W_F consists of Hodge classes on every member of S_F. For a set Σ of embeddings write $\alpha_\Sigma = \prod_{\sigma \in \Sigma} \alpha_\sigma$, with α_σ spanning $\bigwedge^{2n} V_\sigma$.

Lemma 5.2 (Hodge classes and subvarieties of a very general Weil torus)**.** *Let T be a very general member of S_F and $d = 2mn = \dim T$.*

(i) *Every Hodge class of T lies in the span of the α_Σ. In particular $\mathrm{Hdg}^k(T) = 0$ unless n divides k, and $\mathrm{Hdg}^n(T) = W_F(T)$.*

(ii) *Let κ_0 be the Kähler class of a hermitian form on the tangent space that is block diagonal for the splitting of $H^{1,0}$ into the $H^{1,0} \cap V_\sigma$. Then $\int_T \alpha_\Sigma \, \kappa_0^{d - n|\Sigma|} = 0$ unless Σ is closed under conjugation.*

(iii) *Every irreducible analytic subset of T other than a point and T itself has codimension ns with s even, hence at least $2n$.*

Proof. (i) As in the proof of Lemma 3.2: a rational class of Hodge type at a very general member is of Hodge type on all of S_F, and the Mumford–Tate group M of a very general member lies in $G = \operatorname{Res}_{F/\mathbb{Q}} \mathrm{GL}_F(V)$, since F acts by endomorphisms. Here $G(\mathbb{R}) = \prod_\tau \mathrm{GL}(V_{\sigma_\tau})$ acts transitively on S_F, and the $\operatorname{Ad} G(\mathbb{R})$-span of the complex structure of one member is an ideal of $\bigoplus_\tau \mathfrak{gl}_{2n}(\mathbb{C})$ whose projection to every factor contains a non-central element, i on X_τ and $-i$ on Y_τ. An ideal of a sum of simple real Lie algebras is a sum of some of them, and each $\mathfrak{sl}_{2n}(\mathbb{C})$ is simple as a real Lie algebra, so the span contains $\bigoplus_\tau \mathfrak{sl}_{2n}(\mathbb{C})$ and M contains $\operatorname{Res}_{F/\mathbb{Q}} \mathrm{SL}_F(V)$. Over $\mathbb{C}$ this group is $\prod_\sigma \mathrm{SL}(V_\sigma)$ acting on $\bigwedge^* V_{\mathbb{C}} = \bigotimes_\sigma \bigwedge^* V_\sigma$, and the invariants of $\mathrm{SL}(V_\sigma)$ in $\bigwedge^* V_\sigma$ are $\bigwedge^0$ and $\bigwedge^{2n}$. So the invariants are spanned by the α_Σ, of degree $2n|\Sigma|$, and a rational class in the span of the α_σ lies in W_F.

(ii) Choose the hermitian form so that $\kappa_0 = \sqrt{-1}\sum_\sigma \sum_k \xi_{\sigma,k} \wedge \overline{\xi}_{\sigma,k}$ with $\xi_{\sigma,k}$ a basis of $H^{1,0} \cap V_\sigma$; each term lies in $V_\sigma \otimes V_{\overline{\sigma}}$. A product of classes has nonzero integral only if its degree in every V_ρ is $2n$. The factor α_Σ has degree $2n$ in V_ρ for $\rho \in \Sigma$ and 0 otherwise, and every term of a power of κ_0 has the same degree in V_ρ as in $V_{\overline{\rho}}$. For $\rho \in \Sigma$ that degree must be 0, so $V_{\overline{\rho}}$ must be filled by α_Σ, that is $\overline{\rho} \in \Sigma$.

(iii) Let Z be irreducible of codimension c with $0 < c < d$. For a Kähler class κ, $\int_T [Z]\,\kappa^{d-c} = \int_Z \kappa^{d-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$, and by (i) $c = ns$ and $[Z]$ is a combination of the α_Σ with $|\Sigma| = s$. Complex conjugation permutes the embeddings without fixed points, so a set closed under it has even size; if s were odd, (ii) would give $\int_T [Z]\,\kappa_0^{d-c} = 0$. ∎

Proposition 5.3 (Coherent sheaves on a very general Weil torus of F)**.** *Let T be a very general member of S_F. Every coherent analytic sheaf $\mathcal{F}$ on T has $\mathrm{ch}_k(\mathcal{F}) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \le k \le 2n-1$.*

Proof. The Chern classes of [Gri10] are Hodge classes, additive, and compatible with restriction to open subsets, so by Lemma 5.2(i) only $k = n$ needs an argument. First let $\mathcal{F}$ be supported on a proper analytic subset W. Its restriction to $T \smallsetminus W$ is zero, so $\mathrm{ch}_n(\mathcal{F})$ comes from $H^{2n}_W(T, \mathbb{Q})$, which is isomorphic to the Borel–Moore homology $H_{2d-2n}(W, \mathbb{Q})$; by Lemma 5.2(iii) every component of W has codimension at least $2n$, so W has real dimension at most $2d - 4n < 2d - 2n$ and that group vanishes. In general, the torsion of $\mathcal{F}$ is of this kind, and so is the cokernel of the inclusion of a torsion-free sheaf in its reflexive hull. Every coherent sheaf has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$, so as in the proof of Proposition 3.4 a torsion-free sheaf has a Jordan–Hölder filtration whose factors Q_j are stable for every Kähler class, and $\mathrm{ch}_n(Q_j) = \mathrm{ch}_n(Q_j^{**})$. By Bando and Siu [BS94], $\int_T c_2(Q_j^{**})\,\kappa_0^{d-2} \ge 0$, with equality exactly when Q_j^{**} is a projectively flat bundle. Here $c_2(Q_j^{**})$ lies in $\mathrm{Hdg}^2(T)$, which is zero for $n \ge 3$ and is $W_F(T)$ for $n = 2$, where the integral vanishes by Lemma 5.2(ii) with $|\Sigma| = 1$. So Q_j^{**} is projectively flat with $c_1 = 0$, hence flat, and $\mathrm{ch}_n(Q_j) = 0$. Summing over the factors and adding the torsion gives $\mathrm{ch}_n(\mathcal{F}) = 0$. ∎

Theorem 5.4 (The pure form is never met, for any CM field)**.** *Let $n \ge 2$, let A be of (F, n, δ)-Weil type, and let E be a perfect complex on A whose Chern character stays of Hodge type on every member of S_F and has $\mathrm{ch}_n(E) \ne 0$; for instance $\mathrm{ch}(E) = c + N\omega$ with $c \in \mathbb{Q}$, $N \ne 0$ and ω a nonzero rational Weil class. Then σ_E is not injective, and $\dim \mathrm{Ext}^2(E, E) > r(\mathrm{ch}\, E)$. In particular no complex of the pure form $\mathrm{ch}(E) = N\omega_{s_0}$ in* [I, Theorem 17.7(v)]*, with or without a rank, has injective semiregularity map, whatever the CM field.*

Proof. The proof of Theorem 3.5 applies with S_F in place of S: an injective σ_E would let E deform, by the theorem of Buchweitz and Flenner in Pridham's form and Kuranishi's method, to a bounded complex of holomorphic vector bundles on every member of a neighbourhood of A in S_F, with Chern character the parallel transport of $\mathrm{ch}(E)$; at a very general member ch_n of that complex vanishes by Proposition 5.3, while it is the transport of $\mathrm{ch}_n(E) \neq 0$. A class $c + N\omega$ is of Hodge type on all of S_F by Setup 5.1. The inequality follows from the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$, since $\dim \mathrm{Ext}^2(E,E) \geq \mathrm{rk}\,\mathrm{ev}_E \geq r(\mathrm{ch}\,E)$ and equality in both forces ev_E onto with kernel the annihilator, and then σ_E injective, as in Theorem 3.9(iv). ∎

For $m = 1$ this is Theorem 3.5, and for $m = 1$, $n = 2$ the vanishing of Proposition 5.3 is Voisin's theorem [Voi02b, Theorem 10]. So the criterion of [I, Theorem 17.7(v)] needs a polynomial P in the classes η_t that is not constant, for every CM field and every $n \geq 2$. At a quartic field and $n = 2$ this pushes the least possible value of $\dim \mathrm{Ext}^2(E,E)$ for a complex meeting it from 80 to 88 (Corollary 5.15).

5.2. **Quartic CM fields at $n = 2$.** By Theorem 6.70(ii) the only form of (P2) on a minimal route to the conjecture is the one for the split families of the CM fields of degree at least four, and by Proposition 6.13(i) the Weil classes of any family of such a field in a higher dimension give those of its families at $n = 2$. For a quartic field these are families of abelian eightfolds, the smallest families of a field of degree at least four. This subsection determines how far the available mechanisms reach there.

Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, which has codimension two when δ is trivial (Proposition 5.6, Corollary 5.7). For biquadratic F, Markman's theorem reaches them on a further locus of dimension four (Proposition 5.8). For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ, Schoen's cycles on Prym varieties of cyclic covers reach them on a locus of dimension at most six, at whose very general points the Hodge group is all of G_F (Remark 5.9). The Casimir class of the quadratic form on the weight two part R_F below, the natural class of degree four attached to it, is algebraic exactly when the Weil classes are (Proposition 5.10), and correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group (Proposition 5.12). For a general shape the polarised criterion asks $\dim \mathrm{Ext}^2(E,E) = 112$ of one complex (Theorem 5.13). No Orlov product of sheaves on a principally polarised fourfold with real multiplication meets the criterion through a dimension count, whatever the quartic field (Theorem 5.22), and no Orlov product of complexes does so unless both factors have nonzero self-Ext groups in some degree at most -3 (Remark 5.24). Those products are in any case candidates only for a twisted form of the criterion, the form of Markman's reduction (Remark 5.23). Objects with pure spinor characters give no Weil class at all (Proposition 5.25). Together these results fix the loci that divisors, scalar extension and correspondences reach inside $\mathcal{D}_F$, and the numbers that one complex has to attain on the rest.

Throughout, F is a quartic CM field and F_0 its real quadratic subfield, with real places τ_1, τ_2, and (V, H) is of $(F, 2)$-Weil type with discriminant δ; so $m = n = 2$, the members are abelian eightfolds, and $\mathcal{D}_F$ has dimension $mn^2 = 8$. By [I, Proposition 17.3(i)], $H^1(A, \mathbb{C}) = \bigoplus_\sigma V_\sigma$ over the four embeddings σ of F, with $\dim V_\sigma^{1,0} = \dim V_\sigma^{0,1} = 2$. Put

$$R_F = \textstyle\bigwedge_F^2 V \;\subseteq\; \bigwedge_{\mathbb{Q}}^2 V = H^2(A, \mathbb{Q}),$$

the $\mathbb{Q}$-subspace with complexification $\bigoplus_\sigma \bigwedge^2 V_\sigma$; for $m = 1$ it is the space R of [I, Proposition 15.18]. It is an F-vector space of dimension 6, of $\mathbb{Q}$-dimension 24, with Hodge numbers

$(4, 16, 4)$. Fix a generator Ω of the F-line $W_F = \bigwedge_F^4 V$, so that $W_F = \{a\Omega : a \in F\}$, and define a symmetric F-bilinear form q on R_F by $x \wedge_F y = q(x,y)\,\Omega$. Let h be the hermitian form on R_F induced by H, $h(x_1 \wedge x_2, y_1 \wedge y_2) = \det\big(H(x_i, y_j)\big)$, and h_4 the one induced on $\bigwedge_F^4 V$; in a basis $e_1, \dots, e_4$ with $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, $h_4(\Omega, \Omega)$ is $\det H$, so its class in $F_0^\times / \mathrm{N}_{F/F_0}(F^\times)$ is δ. With dual bases (ω_k), (ω_k^*) of F for the trace form, as in [I, Theorem 17.5], put $x \star y = \sum_k (\omega_k x) \cup (\omega_k^* y)$ for $x, y \in R_F$. Then

$$x \star y = q(x,y)\,\Omega, \tag{30}$$

because the complexification of $(\omega_k x) \cup (\omega_k^* y)$ is $\sum_{\sigma,\sigma'} \sigma(\omega_k)\sigma'(\omega_k^*)\, x_\sigma \wedge y_{\sigma'}$, the sum over k of the coefficient is 1 for $\sigma = \sigma'$ and 0 otherwise, as in the proof of [I, Theorem 17.5(ii)], and $\sum_\sigma x_\sigma \wedge y_\sigma$ is the complexification of $q(x,y)\Omega$. Finally let $\theta_j \in H^{1,1}(A)$ be the component of the polarisation that pairs V_σ with $V_{\overline{\sigma}}$ for the two embeddings $\sigma, \overline{\sigma}$ over τ_j. Then $\eta_t = \sum_j \tau_j(t)\,\theta_j$ for the classes η_t of [I, Proposition 17.3(iii)], and $\mathbb{C}[\theta_1, \theta_2]$ is the complexification of the $\mathbb{Q}$-algebra generated by the η_t.

Proposition 5.5 (Hodge classes, and a quaternion algebra acting on R_F)**.** *Let A be a member of an $(F, 2, \delta)$-family.*

(i) *If $\mathrm{Hg}(A) = G_F$, the space of Hodge classes of A has dimension 2 in degree two, spanned by the η_t, and 7 in degree four, spanned by the three products of two η_t and by $W_F(A)$, which meets the span of those products in 0.*

(ii) *The map $\phi\colon R_F \to R_F$ defined by $q(x,y) = h(x, \phi y)$ satisfies $\phi(ax) = \overline{a}\,\phi(x)$ for $a \in F$ and $\phi^2 = h_4(\Omega,\Omega)^{-1}$, and it commutes with $\mathrm{SU}(V,H)$. So ϕ and the action of F are Hodge endomorphisms of R_F at every member, and the Hodge classes in R_F form a module over the algebra $\mathcal{Q}_\delta = F \oplus F\phi$.*

(iii) *$\mathcal{Q}_\delta$ is the cyclic quaternion algebra $(F/F_0, \delta)$ over F_0, which is split if and only if δ is trivial. If $\mathrm{Hg}(A) = G_F$, then $\mathcal{Q}_\delta$ is the whole algebra of Hodge endomorphisms of R_F.*

(iv) *If δ is trivial, then $R_F = W \otimes_{F_0} F$ for a rational sub-Hodge structure W, stable under F_0, of dimension 12 and with Hodge numbers $(2, 8, 2)$.*

Proof. (i) As in the proof of [I, Proposition 17.3(iii)], over $\mathbb{C}$ the group G_F is $\mathrm{SL}(U_1) \times \mathrm{SL}(U_2)$ with $\dim U_j = 4$, and $H^1(A, \mathbb{C}) = \bigoplus_j (U_j \oplus U_j^\vee)$. The invariants of $\mathrm{SL}(U)$ in $\bigwedge^a U \otimes \bigwedge^b U^\vee$ form a line if $a = b$ or $\{a, b\} = \{0, 4\}$, and are zero otherwise. In degree two this leaves the bidegree $(1,1)$ in one factor: two lines, spanned by θ_1 and θ_2. In degree four it leaves $(2,2)$ in one factor or $(1,1)$ in both, three lines spanned by θ_1^2, $\theta_1\theta_2$ and θ_2^2, which are nonzero because $\dim U_j = 4$; and $(4,0)$ or $(0,4)$ in one factor, four lines spanning $W_F \otimes \mathbb{C}$. These lie in different summands, so $W_F(A)$ meets the span of the products in 0.

(ii) Choose an H-orthogonal basis $e_1, \dots, e_4$ of V and put $a_i = H(e_i, e_i) \in F_0$. Replacing Ω by $\lambda\Omega$ replaces ϕ by $\overline{\lambda}^{-1}\phi$ and $h_4(\Omega,\Omega)$ by $\mathrm{N}(\lambda) h_4(\Omega,\Omega)$, so we may take $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, with $h_4(\Omega,\Omega) = a_1a_2a_3a_4$. For $J = \{j < j'\}$ put $e_J = e_j \wedge e_{j'}$ and $a_J = a_j a_{j'}$, and let J^c be the complement, with $e_{J^c} \wedge e_J = \epsilon_J \Omega$. The e_J form an h-orthogonal F-basis with $h(e_J, e_J) = a_J$, and $q(e_K, e_J)$ is ϵ_J for $K = J^c$ and 0 otherwise, so $\phi(e_J) = \epsilon_J a_{J^c}^{-1} e_{J^c}$. Two-forms commute, so $\epsilon_{J^c} = \epsilon_J$ and $\phi^2(e_J) = (a_J a_{J^c})^{-1} e_J = h_4(\Omega,\Omega)^{-1} e_J$. Semilinearity holds because q is linear and h antilinear in the second variable. An element of $\mathrm{SU}(V,H)$ preserves h and fixes Ω, hence preserves q and commutes with ϕ, and it commutes with F. The Hodge group of every member lies in G_F, because the circle defining the Hodge structure preserves E, commutes with F and acts on each V_σ with determinant $z^2 z^{-2} = 1$; so ϕ and F commute with it, and they are Hodge endomorphisms.

(iii) The relations $\phi a = \overline{a}\phi$ and $\phi^2 = h_4(\Omega,\Omega)^{-1}$ present $F \oplus F\phi$ as the cyclic algebra $(F/F_0, h_4(\Omega,\Omega)^{-1})$, which is $(F/F_0, \delta)$ because a square in F_0 is a norm; and a cyclic algebra $(F/F_0, c)$ is split exactly when c is a norm from F. It has $\mathbb{Q}$-dimension 8, since ϕ is not F-linear. If $\mathrm{Hg}(A) = G_F$, the Hodge endomorphisms of R_F are the commutant of G_F. Over $\mathbb{C}$, $R_F \otimes \mathbb{C} = \bigoplus_j (\bigwedge^2 U_j \oplus \bigwedge^2 U_j^\vee)$, where $\bigwedge^2 U_j \cong \bigwedge^2 U_j^\vee$ is irreducible for $\mathrm{SL}(U_j)$ and trivial for the other factor, so the commutant is $M_2(\mathbb{C}) \times M_2(\mathbb{C})$, of dimension 8.

(iv) If δ is trivial, $h_4(\Omega,\Omega) = \mathrm{N}(b)$ for some $b \in F$, and $\psi = b\phi$ satisfies $\psi^2 = b\overline{b}\,\phi^2 = 1$. Its fixed space W is an F_0-subspace and a sub-Hodge structure, ψ being a Hodge endomorphism, and an element $\beta \in F$ with $\overline{\beta} = -\beta \neq 0$ maps it isomorphically onto the (-1)-eigenspace, since $\psi\beta = -\beta\psi$. So $R_F = W \oplus \beta W = W \otimes_{F_0} F$, and W has half the dimension and half the Hodge numbers of R_F. ∎

Proposition 5.6 (What divisor classes reach)**.** *Let A be a member of an $(F, 2, \delta)$-family.*

(i) *The following are equivalent:* (a) *$W_F(A)$ lies in the subalgebra of $H^*(A,\mathbb{Q})$ generated by $\mathrm{NS}(A)_\mathbb{Q}$;* (b) *$W_F(A)$ meets that subalgebra in a nonzero class;* (c) *R_F contains a nonzero rational Hodge class. When they hold, R_F contains a rational Hodge class t with $q(t,t) \neq 0$, and $W_F(A) = \{(ft)\star t : f \in F\}$, a space of polynomials with rational coefficients in divisor classes.*

(ii) *For a nonzero $\mathcal{Q}_\delta$-submodule $M \subseteq R_F$ let $\mathcal{D}(M) \subseteq \mathcal{D}_F$ be the set of points at which M consists of Hodge classes, and let $\mathrm{NL}(R_F) \subseteq \mathcal{D}_F$ be the set of points at which* (c) *holds. Then $\mathrm{NL}(R_F) = \bigcup_M \mathcal{D}(M)$, a countable union. Each nonempty $\mathcal{D}(M)$ is a closed complex submanifold, isomorphic to a product of two bounded symmetric domains of type IV and dimension $4-d$ each, where $d = \dim_F M$, and $d = 1$ occurs only if δ is trivial. So if δ is trivial, $\mathrm{NL}(R_F)$ is the union of the nonempty $\mathcal{D}(M)$ with $d = 1$, each of dimension exactly 6; if δ is not trivial, every $\mathcal{D}(M)$ has dimension at most 4.*

(iii) *$\mathrm{NL}(R_F)$ contains the base point s_0 of* [I, Theorem 17.5]*, and for every δ it contains the locus of products $B_1 \times B_2$ of two abelian fourfolds of $(F,1)$-Weil type, which has dimension 4. The split locus, of members isogenous to $X \otimes_{\mathcal{O}_{F_0}} \mathcal{O}_F$ with X an abelian fourfold with $F_0 \subseteq \mathrm{End}^0(X)$, has dimension 6, lies in the family of trivial discriminant, and is contained in $\mathrm{NL}(R_F)$.*

Proof. (i) Since $W_F(A) \neq 0$, (a) implies (b). Let $F^\times$ act on $H^*(A,\mathbb{Q})$ by $\alpha \mapsto \iota(\alpha)^*$, which on $\bigotimes_\sigma \bigwedge^{a_\sigma} V_\sigma$ is the character $\prod_\sigma \sigma^{a_\sigma}$. The isotypic parts for the characters σ^2 in degree two and σ^4 in degree four are R_F and W_F. The projections p_R and p_W onto them are rational, the two sets of characters being Galois orbits, and they are morphisms of Hodge structures, being $\mathbb{Q}$-linear combinations of the $\iota(\alpha)^*$. The characters occurring in H^2 are the products $\rho\rho'$, and the only way to write σ^4 as a product of two of them is $\sigma^2 \cdot \sigma^2$, so $p_W(D \cup D') = q(p_R D, p_R D')\,\Omega$ for $D, D' \in H^2(A,\mathbb{Q})$. If a nonzero $w \in W_F$ equals $\sum_i D_i \cup D_i'$ with $D_i, D_i' \in \mathrm{NS}(A)_\mathbb{Q}$, then $w = p_W(w) = \sum_i q(p_R D_i, p_R D_i')\,\Omega$, so some $p_R D_i$ is a nonzero rational Hodge class in R_F, and (b) implies (c). Conversely, let $t \in R_F$ be a nonzero rational Hodge class. The part of type $(1,1)$ of $\bigwedge^2 V_\sigma$ is $V_\sigma^{1,0} \wedge V_\sigma^{0,1}$, and H is definite of opposite signs on the two factors by [I, Proposition 17.3(i)], so h is negative definite on it at each real place, and $h(t,t) \in F_0$ is totally negative. By Proposition 5.5(ii), $q(t,\phi t) = h(t,\phi^2 t) = h_4(\Omega,\Omega)^{-1} h(t,t) \neq 0$, so one of the rational Hodge classes t, ϕt, $t + \phi t$, call it t', has $q(t',t') \neq 0$. Each $\omega_k f t'$ and each $\omega_k^* t'$ is a rational Hodge class in H^2, hence a divisor class, and by (30) $(ft') \star t' = f\, q(t',t')\,\Omega$, which runs over W_F as f runs over F. So (c) implies (a), with the stated formula.

(ii) The Hodge classes in R_F form a $\mathcal{Q}_\delta$-module, so $\mathrm{NL}(R_F)$ is the union of the $\mathcal{D}(M)$, and there are countably many M. Fix a real place τ, let σ be an embedding over it, and put $R_\tau = R_F \otimes_{F_0,\tau} \mathbb{R}$, a complex vector space of dimension 6 through $F \otimes_{F_0,\tau} \mathbb{R} = \mathbb{C}$, identified with $\bigwedge^2 V_\sigma$. Since $\det H$ has sign $(-1)^n = 1$ at τ, the map ϕ induces an antilinear ϕ_τ on R_τ with $\phi_\tau^2 = \lambda^2$, $\lambda > 0$. Its eigenspace $W_\tau = \{x : \phi_\tau x = \lambda x\}$ is a real form of R_τ and a real sub-Hodge structure with $W_\tau \otimes \mathbb{C} \cong \bigwedge^2 V_\sigma$, of Hodge numbers $(1,4,1)$, on which $q = \lambda h$ is real of signature $(2,4)$, positive on the plane Π_τ of real classes of type $(2,0)+(0,2)$ and negative on those of type $(1,1)$. The group $\mathrm{SU}(V_\tau, H_\tau)$ acts on W_τ preserving q, through the isogeny $\mathrm{SU}(2,2) \to \mathrm{SO}(2,4)^0$, and the map from $\mathcal{D}_{2,2}$ to the space of oriented positive planes in W_τ that sends a point to Π_τ is the equivariant isomorphism of the symmetric domains of this group onto one component, $\mathcal{D}^{\mathrm{IV}}(W_\tau)$, of dimension 4. For a quadratic space P of signature $(2,k)$ write likewise $\mathcal{D}^{\mathrm{IV}}(P)$, of dimension k. Let M have F-dimension d. Then $M_\tau = M \otimes_{F_0,\tau} \mathbb{R}$ is stable under ϕ and F, so $M_\tau = P_\tau \otimes \mathbb{C}$ with $P_\tau = M_\tau \cap W_\tau$ of real dimension d, and M consists of Hodge classes exactly when every P_τ is of type $(1,1)$, that is, orthogonal to Π_τ. So $\mathcal{D}(M) = \prod_\tau \mathcal{D}^{\mathrm{IV}}(P_\tau^\perp)$, which is empty unless every P_τ is negative definite and then has dimension $2(4-d)$. If $d = 1$, then $M = Fx$ with $\phi x = bx$ for some $b \in F$, and $\phi^2 x = \bar{b}b\,x$ gives $h_4(\Omega,\Omega)^{-1} = \mathrm{N}(b)$, so δ is trivial. If δ is trivial, $\mathcal{Q}_\delta \cong M_2(F_0)$, every nonzero module contains one with $d = 1$, and $\mathcal{D}(M) \subseteq \mathcal{D}(M')$ for $M' \subseteq M$.

(iii) The classes $\delta_i(f)$ of [I, Theorem 17.5(ii)] have complexification in $\bigoplus_\sigma \bigwedge^2 V_\sigma$, so they lie in R_F, and they are Hodge classes at s_0. For a product, $V = V_1 \oplus V_2$ with H-orthogonal F-planes of signature $(1,1)$ at each place, and $\bigwedge^2_F V_1 \subseteq R_F$ is $W_F(B_1)$, of type $(1,1)$ by [I, Proposition 17.3(i)]. Every δ is the product of the discriminants of two such planes, for instance of $\mathrm{diag}(1,-\delta)$ and $\mathrm{diag}(1,-1)$, and each factor varies in a family of dimension $m = 2$. On the split locus $H^1(A,\mathbb{Q}) = U \otimes_{F_0} F$, with $U = H^1(X,\mathbb{Q})$ and the complex structure of X on the first factor. The polarisation class e of X lies in $\bigwedge^2_{F_0} U$, the hermitian form of A is c times the sesquilinear extension of the F_0-bilinear form e, with $c \in F$ totally imaginary, and so $\det H = c^4 \,\mathrm{Pf}(e)^2$ is a square in F_0 and δ is trivial. The class $e \otimes 1 \in (\bigwedge^2_{F_0} U) \otimes_{F_0} F = R_F$ is fixed by $\mathrm{Hg}(A) = \mathrm{Hg}(X) \subseteq \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{Sp}(U,e)$, so it is a nonzero Hodge class. The polarised abelian fourfolds with real multiplication by F_0 form families of dimension $[F_0 : \mathbb{Q}] \cdot 3 = 6$. ∎

Corollary 5.7 (A base locus of codimension two)**.** *In every $(F,2,\delta)$-family, $\Sigma_F \supseteq \mathrm{NL}(R_F)$, and at every point of $\mathrm{NL}(R_F)$ the Weil classes are the polynomials in divisor classes of Proposition 5.6(i). If δ is trivial, $\mathrm{NL}(R_F)$ is a countable union of closed complex submanifolds of $\mathcal{D}_F$ of codimension exactly two, and it contains the split locus; if δ is not trivial, they have codimension at least four, and the product loci have codimension four. In both cases $\mathrm{NL}(R_F)$ contains the base point s_0 of* [I, Theorem 17.5] *and is stable under $G_F(\mathbb{Q})$. At a point outside $\mathrm{NL}(R_F)$ no nonzero Weil class is a polynomial in divisor classes.*

Proof. Everything except the stability is Proposition 5.6. An element $g \in G_F(\mathbb{Q})$ commutes with F and with ϕ, so it carries $\mathcal{Q}_\delta$-submodules to $\mathcal{Q}_\delta$-submodules, and $g\,\mathcal{D}(M) = \mathcal{D}(gM)$. ∎

[I, Theorem 17.5] gives one CM point, whose $G_F(\mathbb{Q})$-orbit is dense but countable. When δ is trivial, Corollary 5.7 replaces it by a locus of codimension two on which the Weil classes are balanced products of divisor classes, as they are at s_0; nothing is claimed about the rest of the Hodge ring at its points. It does not decide $\mathbf{W}(F,2,\delta)$: $\mathrm{NL}(R_F)$ is a countable union of proper closed submanifolds, hence meagre, and [I, Theorem 17.7(iii)] allows Σ_F

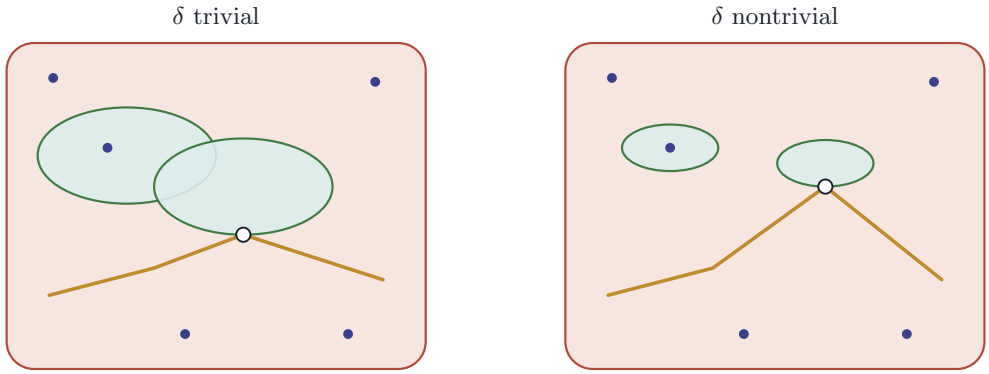


FIGURE 15. Where the Weil classes of an $(F, 2, \delta)$-family are known to be algebraic, F a quartic CM field; each panel is the period domain $\mathcal{D}_F$, of dimension 8. The grass regions are components of the Noether–Lefschetz locus $\mathrm{NL}(R_F)$, on which W_F consists of polynomials in divisor classes (Proposition 5.6, Corollary 5.7): of dimension exactly 6 when δ is trivial, where the quaternion algebra $\mathcal{Q}_\delta$ of Proposition 5.5 is split, and at most 4 otherwise. The ochre curve is the locus of scalar extensions of Weil fourfolds, of dimension 4 when F is biquadratic, on which Markman's theorem applies (Proposition 5.8); it is not contained in $\mathrm{NL}(R_F)$ but meets it (ring) where the fourfold is a product of Weil surfaces. The indigo points are CM points; the base point s_0 of [I, Theorem 17.5] lies on $\mathrm{NL}(R_F)$. For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial δ the Prym locus of Remark 5.9, of dimension at most 6, is not drawn. On the rest of the domain, the very general member included, the algebraicity of the Weil classes is the open propagation statement.

to be meagre. The last sentence of the corollary is the analogue of [I, Proposition 15.18]: no argument through divisor classes reaches a member outside $\mathrm{NL}(R_F)$. The codimension is two rather than $h^{2,0}(R_F) = 4$ (compare [I, Remark 15.21]) because Hodge classes in R_F come in $\mathcal{Q}_\delta$-modules, and a module of F-dimension one costs one condition at each real place. Figure 15 shows where the Weil classes of such a family are known to be algebraic.

Proposition 5.8 (Scalar extension from an imaginary quadratic subfield)**.** *Let $F = KF_0$ with K imaginary quadratic; these are the biquadratic quartic CM fields, and the only ones containing an imaginary quadratic field. Let C be an abelian fourfold of $(K, 2, \delta)$-Weil type and $B_F = C \otimes_{\mathcal{O}_K} \mathcal{O}_F$, which is of $(F, 2, \iota(\delta))$-Weil type by Proposition 6.13(ii); write*

(V_C, H_C) for the hermitian space of C. For $f \in F$ let $g_f \in \mathrm{Hom}(B_F, C) \otimes \mathbb{Q}$ be the element inducing $v \mapsto v \otimes f$ from $H^1(C, \mathbb{Q})$ to $H^1(B_F, \mathbb{Q}) = H^1(C, \mathbb{Q}) \otimes_K F$.

(i) *$W_F(B_F)$ lies in the $\mathbb{Q}$-span of the classes $g_f^* w$, $f \in F$, $w \in W_K(C)$, which has dimension 10. Hence $W_F(B_F)$ is algebraic, by Markman's theorem $\mathbf{W}(K, 2, \delta)$ [Mar25a] (Proposition 6.12(ii)).*

(ii) *The members B_F form a locus of dimension 4 in the $(F, 2, \iota(\delta))$-family. At those with $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ the space R_F contains no nonzero Hodge class, so this locus is not contained in $\mathrm{NL}(R_F)$; it meets $\mathrm{NL}(R_F)$, for instance where $C = C_1 \times C_2$ with C_1, C_2 abelian surfaces of Weil type, and then B_F is a product of two abelian fourfolds of $(F, 1)$-Weil type.*

Proof. (i) The map $v \mapsto v \otimes f$ commutes with the complex structures, that of B_F being the one of C tensored with F, so g_f exists. Over $\mathbb{C}$, for each embedding κ of K let V_κ be the κ-eigenspace of K on $H^1(C, \mathbb{C})$ and $U_\kappa = F \otimes_{K,\kappa} \mathbb{C} \cong \mathbb{C}^2$, the sum of the lines L_σ, $\sigma|\kappa$, on which F acts through σ. The κ-part of $H^1(B_F, \mathbb{C})$ is $V_\kappa \otimes U_\kappa$ and its σ-part is $V_\kappa \otimes L_\sigma$, so the σ-component $\bigwedge^4 V_\kappa \otimes L_\sigma^{\otimes 4}$ of $W_F(B_F) \otimes \mathbb{C}$ lies in $\bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa \subseteq \bigwedge^4 (V_\kappa \otimes U_\kappa)$. The map g_f^* sends a generator α_κ of $\bigwedge^4 V_\kappa$ to $\alpha_\kappa \otimes f_\kappa^{\otimes 4}$, with f_κ the image of f in U_κ. The fourth powers of the elements of the Zariski dense subset F of U_κ span $\mathrm{Sym}^4 U_\kappa$, so the complexified span of the $g_f^* w$ is $\bigoplus_\kappa \bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa$, of dimension $2 \cdot 5$, and it contains $W_F(B_F) \otimes \mathbb{C}$. A rational subspace whose complexification lies in that of a rational subspace lies in it, and pull-backs of algebraic classes are algebraic.

(ii) The map $C \mapsto B_F$ embeds the four-dimensional domain of the K-family into $\mathcal{D}_F$. As rational Hodge structures $H^1(B_F, \mathbb{Q}) \cong H^1(C, \mathbb{Q})^2$, so $\mathrm{Hg}(B_F) = \mathrm{Hg}(C)$, and $R_F \otimes \mathbb{C} = \bigoplus_\sigma \bigwedge^2 V_\kappa \otimes L_\sigma^{\otimes 2}$, with $\bigwedge^2 V_\kappa$ irreducible and nontrivial for $\mathrm{SL}(V_\kappa)$; so when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$, R_F has no invariants. For $C = C_1 \times C_2$ one has $B_F = (C_1 \otimes_{\mathcal{O}_K} \mathcal{O}_F) \times (C_2 \otimes_{\mathcal{O}_K} \mathcal{O}_F)$, and Proposition 5.6(iii) applies. ∎

Proposition 5.8 gives no new case of the Hodge conjecture: B_F is isogenous to $C \times C$, the classes are Hodge classes there, and when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ Proposition 6.24 already gives the conjecture for $C \times C$ from [Mar25a].

Remark 5.9 (Prym varieties of cyclic covers). For three cyclotomic fields published work gives a further locus, at whose very general points the Hodge group is the whole of G_F. Let $N \geq 3$, let $\widetilde{C} \to C$ be an étale cyclic cover of degree N of a smooth projective curve of genus $g \geq 2$, and let $P \subseteq J(\widetilde{C})$ be the abelian subvariety on whose H^1 the covering group acts through characters of exact order N. Then $F = \mathbb{Q}(\zeta_N)$ acts on P, and every embedding of F occurs in $H^{1,0}(P)$ with multiplicity $g - 1$ [Sch88, Lemma 1.6a], [PZ25, Lemma 5.1], so P is of $(F, g-1)$-Weil type ([I, Remark 17.2]). The hermitian form is split, because the part of $H_1(\widetilde{C}, \mathbb{Z})$ on which the covering group acts through characters of exact order N has a hyperbolic $\mathbb{Z}[\zeta_N]$-basis [Loo97, Proposition 2.2]; so $\delta = [(-1)^{g-1}]$, which is trivial for $g = 3$. Schoen proved that the Weil classes of P are algebraic [Sch88, Theorem 2.0 with $r = 0$, Corollary 3.1], and Patel and Zhang give a proof by unramified geometric class field theory [PZ25, Theorems 1.1, 1.2 and 5.3, Remark 3.7]. By [Loo97, Theorem 2.4], for every $g \geq 2$, the monodromy group of the family of the P over a finite cover of the moduli of such covers contains a subgroup of finite index of an arithmetic subgroup of G_F. Every factor $\mathrm{SU}(g-1, g-1)$ of $G_F(\mathbb{R})$ is noncompact, so by Borel's density theorem [Bor60] its Zariski closure contains G_F; the Hodge group of every member lies in G_F, so by André's theorem [And92, Theorem 1] a very general P has $\mathrm{Hg}(P) = G_F$,

and by [I, Proposition 17.3(iii)] the Hodge conjecture holds for it in every degree. All of this is a consequence of [Sch88, Loo97, And92], and we make no claim of priority.

For $\varphi(N) = 4$, that is for $F = \mathbb{Q}(\zeta_5) = \mathbb{Q}(\zeta_{10})$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_{12})$, and $g = 3$, the P are abelian eightfolds of $(F, 2, 1)$-Weil type. With their $G_F(\mathbb{Q})$-translates, which lie in Σ_F by [I, Theorem 17.7(ii)], they fill a countable union of analytic subvarieties of $\mathcal{D}_F$, the *Prym locus* of F, of positive dimension, since the monodromy is infinite, and of dimension at most $3g - 3 = 6$ [Sch88, pp. 25–26]. At a very general point of it $\mathrm{Hg} = G_F$, so R_F has no nonzero Hodge class there, each $\bigwedge^2 V_\sigma$ being a nontrivial irreducible representation of $\mathrm{SL}(V_\sigma)$, and the Hodge group is larger than that of a scalar extension of a Weil fourfold: such points lie outside $\mathrm{NL}(R_F)$, outside the locus of Proposition 5.8, and are not CM points. They are the only members with $\mathrm{Hg} = G_F$ at which we know the Weil classes of a quartic CM field to be algebraic. They do not decide $\mathbf{W}(F, 2, 1)$: the count $3g - 3 = 6 < 8 = \frac{1}{2}\varphi(N)(g-1)^2 = \dim \mathcal{D}_F$ of [Sch88, pp. 25–26] makes the union meagre, as $\mathrm{NL}(R_F)$ is, and [I, Theorem 17.7(iii)] allows Σ_F to be meagre. Nor does Proposition 5.12 exclude them: the cycles come from $J(\widetilde{C})$, which has P as a factor. For imaginary quadratic fields the count is $3n \geq n^2$ for $n \leq 3$, and there the Prym varieties do reach whole families: Schoen proved the algebraicity for the Weil fourfolds and for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-3})$ [Sch88, Theorem 3.2], [Sch98], and Koike for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-1})$ [Koi04, Theorem 2.1, Corollary 2.1].

Proposition 5.10 (The Casimir class)**.** *Let A be a member of an $(F, 2, \delta)$-family and $c \in F^\times$, and let*

$$C_q \in R_F \otimes_{\mathbb{Q}} R_F \subseteq H^2(A, \mathbb{Q}) \otimes H^2(A, \mathbb{Q}) \subseteq H^4(A \times A, \mathbb{Q})$$

be the Casimir element $\sum_j u_j \otimes u_j^*$ of the nondegenerate symmetric form $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$ on R_F, for a basis (u_j) and its dual basis (u_j^*). Then C_q is a Hodge class, $\Delta^* C_q = 6c\,\Omega$ for the diagonal $\Delta\colon A \to A \times A$, and C_q lies in the $\mathbb{Q}$-span of the classes f^*w, $w \in W_F(A)$, $f(x, y) = \iota(a)x + \iota(b)y$ with $a, b \in \mathcal{O}_F$. Hence $W_F(A)$ is algebraic if and only if C_q is.

Proof. For an F-basis (x_I) of R_F and its q-dual basis (x_I^q), the elements $\omega_k x_I$ and $c\,\omega_k^* x_I^q$ form dual bases for $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$. So by (30) $\Delta^* C_q = \sum_I x_I \star (c\,x_I^q) = \sum_I c\,\Omega = 6c\,\Omega$, and if C_q is algebraic so is $W_F(A)$, by [I, Proposition 17.3(ii)]. Conversely, $\mathrm{SU}(V, H)$ preserves q, so C_q is invariant under the diagonal action of G_F, which contains $\mathrm{Hg}(A \times A)$, and C_q is a Hodge class. Since q pairs $\bigwedge^2 V_\sigma$ only with itself, the complexification of C_q lies in $\bigoplus_\sigma \bigwedge^2 V_\sigma \otimes \bigwedge^2 V_\sigma$, inside the part $\bigoplus_\sigma \bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ of $H^4(A \times A, \mathbb{C})$ on which $F^\times$, acting diagonally, has the characters σ^4. The group G_F acts on V_σ through $\mathrm{SL}(V_\sigma)$, whose invariants in $\bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ are $\bigwedge^4 V_\sigma \otimes \mathrm{Sym}^4\mathbb{C}^2$ by Cauchy's formula, so the G_F-invariants in that part form the rational space $W_F(A) \otimes_F \mathrm{Sym}^4_F(F^2)$, of dimension 20. The map f^* sends w to $w \otimes (a, b)^{\otimes 4}$, and the fourth powers of the elements of $\mathcal{O}_F^2$ span $\mathrm{Sym}^4_F(F^2)$; so the classes f^*w span that space, and C_q is algebraic if $W_F(A)$ is. ∎

Remark 5.11 (What the exceptional isogeny gives). By the proof of Proposition 5.6(ii), at each real place R_F becomes $W_\tau \otimes \mathbb{C}$ with W_τ of K3 type, $\mathrm{SU}(V_\tau, H_\tau)$ acting through $\mathrm{SU}(2, 2) \to \mathrm{SO}(2, 4)$, and when δ is trivial the W_τ are the real components of the rational structure W of Proposition 5.5(iv). A route of Kuga–Satake type would start from this weight two structure and would have to return to degree four, on A or on $A \times A$; by Proposition 5.10 the natural class it meets there, the Casimir class of q, is algebraic exactly when the Weil classes are. So this mechanism restates $\mathbf{W}(F, 2, \delta)$ at each member and does not reduce it.

Proposition 5.12 (No correspondence from a small abelian variety)**.** *Let A be a member with $\mathrm{Hg}(A) = G_F$, and let X be an abelian variety such that $\mathrm{Hg}(X)$ has no normal subgroup isogenous to G_F; by the proof, this holds when every simple factor of X has dimension at most 7, in particular when $\dim X \leq 7$, and it holds when X is of CM type. Then $\mathrm{Hg}(A \times X) = \mathrm{Hg}(A) \times \mathrm{Hg}(X)$, and $\mathrm{Hdg}^*(A^a \times X^b) = \mathrm{Hdg}^*(A^a) \otimes \mathrm{Hdg}^*(X^b)$ for all $a, b \geq 0$. So the class of an algebraic cycle Z on $A^a \times X^b$ is $\sum_i \beta_i \otimes \xi_i$ with $\beta_i \in \mathrm{Hdg}^*(A^a)$ and linearly independent $\xi_i \in \mathrm{Hdg}^*(X^b)$; the correspondence Z carries $H^*(X^b, \mathbb{Q})$ into the span of the β_i; and if the Hodge conjecture holds for X^b, every β_i is algebraic.*

Proof. The group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$ is almost simple over $\mathbb{Q}$. By Goursat's lemma the kernel N of the projection $\mathrm{Hg}(A \times X) \to \mathrm{Hg}(X)$ maps onto a normal subgroup of $\mathrm{Hg}(A) = G_F$. If that subgroup is G_F, then $\mathrm{Hg}(A \times X)$ contains $G_F \times 1$ and is the product. Otherwise N is finite, $\mathrm{Hg}(X)$ has a normal subgroup H isogenous to G_F, contrary to the hypothesis, and the cocharacter μ defining the Hodge structure of X projects through $H \to G_F$ to that of A. This situation also cannot occur when every simple factor of X has dimension at most 7, or when X is of CM type, as follows. Over $\mathbb{C}$, H is $\mathrm{SL}_4 \times \mathrm{SL}_4$, and μ projects in each factor to $\mathrm{diag}(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$ modulo the centre. Since μ acts on $H^1(X, \mathbb{C})$ with two weights differing by one, the weights of its projection spread by at most one on every irreducible constituent for H; for the representation of SL_4 of highest weight $a\varpi_1 + b\varpi_2 + c\varpi_3$ the spread is $a + 2b + c$, and spreads add over the two factors. So every nontrivial constituent is the standard representation of one factor or its dual. These four are permuted transitively by $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, as the embeddings of F are; so a simple factor Y of X on whose H^1 the group H acts nontrivially has $\dim_{\mathbb{Q}} H^1(Y, \mathbb{Q}) \geq 16$, that is $\dim Y \geq 8$. A torus has no such H. Next, $\mathrm{Hg}(A^a \times X^b)$ is $\mathrm{Hg}(A) \times \mathrm{Hg}(X)$ acting factorwise on $H^*(A^a) \otimes H^*(X^b)$, and the invariants of a product of groups in a tensor product of representations of the factors are the tensor product of the invariants. Finally, the intersection pairing is perfect on $\mathrm{Hdg}^*(X^b)$, the invariants being orthogonal to the other isotypic parts, so there are $\xi_j^* \in \mathrm{Hdg}^*(X^b)$ with $\int \xi_i \xi_j^* = \delta_{ij}$; they are algebraic when the conjecture holds for X^b, and then $\beta_j = pr_{A^a *}(Z \cdot pr_{X^b}^* \xi_j^*)$ is algebraic. ∎

So a correspondence with such an X can produce a Weil class of A only from an algebraic cycle on some $A^a \times X^b$ that already yields one on A^a. At a member with full Hodge group this excludes every abelian variety of dimension at most seven, among them the fourfolds and sixfolds of Weil type of the imaginary quadratic fields on which Markman's theorems are proved, and every CM abelian variety. The same argument, with real points in place of dimensions, excludes rational Hodge structures Y of K3 type: by Zarhin's theorem [Zar83] the Hodge group of such a Y is the restriction of scalars of an orthogonal or unitary group over the endomorphism field of its transcendental part, noncompact at exactly one real place, so its real Lie algebra has at most one simple factor isomorphic to $\mathfrak{su}(2, 2)$, while that of G_F has two; hence $\mathrm{Hg}(A \times Y) = \mathrm{Hg}(A) \times \mathrm{Hg}(Y)$. This last statement rests on Zarhin's theorem, which we quote without proof.

Theorem 5.13 (The polarised criterion for a quartic field at $n = 2$)**.** *Let A be a member of an $(F, 2, \delta)$-family, let $\omega = \sum_\sigma u_\sigma \alpha_\sigma$, with α_σ spanning $\bigwedge^4 V_\sigma$ and every $u_\sigma \neq 0$, as holds for every nonzero rational Weil class, let $p \in \mathbb{C}[\theta_1, \theta_2]$, put $\gamma = \omega + p$, and let $r(\gamma)$ be the rank of the contraction $HT^2(A) \to H^*(A, \mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. Write $T_\sigma = (V_\sigma^{1,0})^\vee \subseteq T_0 A$.*

(i) $\dim HT^2(A) = 28 + 64 + 28 = 120$, *and*

$$\mathrm{Ann}_{HT^2}(\omega) = \bigoplus_\sigma \mathrm{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1}) \ \oplus \ \bigoplus_{\{\sigma, \sigma'\},\, \sigma \neq \sigma'} T_\sigma \otimes T_{\sigma'},$$

of dimension $0+16+24$*: the* F*-linear first order deformations of* A *and the bivectors mixing two embeddings. So* $r(\omega)=80$*. Unlike the case* $m=1$ *of Theorem 2.5(i), no nonzero element of* $H^{0,2}(A)$ *kills* ω*.*

(ii) $r(\gamma)\le 112$ *for every* p*. For a general* p*,* $r(\gamma)=112$*, and whenever* $r(\gamma)=112$ *the annihilator of* γ *is the tangent space* $T_{[A]}\mathcal{D}_F$ *of the family, of dimension* 8*, inside* $H^1(T_A)$*. The value* 112 *is also taken at a general element, with nonzero component in* $W_F\otimes\mathbb{C}$*, of the complexified Hodge ring of a member with* $\mathrm{Hg}(A)=G_F$*.*

(iii) $r(\gamma)\ge 80$ *for every* p*, with equality exactly when* p *is a constant.*

(iv) *The rank* $r(\gamma)$ *takes exactly the fifteen values* 80, 84, 87, 88, 91, 92, 94, 95, 96, 104, 107, 108, 110, 111, 112*; Theorem 5.14 computes it for every* p*. On the monomials* $\theta_1^i\theta_2^j$ *with a general coefficient it takes the six values* 80, 84, 88, 104, 108, 112*.*

(v) *Let* E *be a perfect complex on* A *with* $\mathrm{ch}(E)=N\omega_1+P$*, where* $N\neq 0$*,* ω_1 *is a nonzero rational Weil class and* P *is a polynomial with rational coefficients in the classes* η_t*. Then* $\dim\mathrm{Ext}^2(E,E)\ge r(\mathrm{ch}\,E)\ge 80$*. If* $\dim\mathrm{Ext}^2(E,E)=r(\mathrm{ch}\,E)$*, then* ev_E *is onto with kernel* $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)$ *and* σ_E *is injective, so that* $\mathbf{W}(F,2,\delta)$ *follows by* [I, Theorem 17.7(v)]*.*

Proof. (i) $HT^2(A)=\bigwedge^2 H^{0,1}\oplus H^{0,1}\otimes T_0A\oplus\bigwedge^2 T_0A$, with $H^{0,1}$ and T_0A of dimension 8. As in the proof of Theorem 2.5(i) the three summands map into different Hodge components, so the annihilator is the sum of the three annihilators; and the torus $(\mathbb{C}^\times)^4$ acting on each V_σ by scalars separates the images further. A block $V_\rho^{0,1}\wedge V_{\rho'}^{0,1}$ of $\bigwedge^2 H^{0,1}$ kills α_σ exactly when $\sigma\in\{\rho,\rho'\}$, and otherwise maps injectively into a weight space that no other pair of a block and an embedding reaches; there are four embeddings, so some σ lies outside $\{\rho,\rho'\}$, and no nonzero element of $\bigwedge^2 H^{0,1}$ kills ω. A block $\mathrm{Hom}(V_\rho^{1,0},V_{\rho'}^{0,1})$ of $H^1(T_A)$ kills α_σ for $\sigma\neq\rho$; it kills α_ρ when $\rho'=\rho$, because $V_\rho^{0,1}$ is already exhausted in α_ρ, and maps it injectively into its own weight space when $\rho'\neq\rho$. A bivector in $T_\rho\otimes T_{\rho'}$ with $\rho\neq\rho'$ kills every α_σ, whose two factors of type $(1,0)$ both lie in $V_\sigma^{1,0}$, and $\bigwedge^2 T_\sigma$ maps α_σ to a nonzero multiple of a generator of $\bigwedge^2 V_\sigma^{0,1}$. The dimensions are 0, $4\cdot 4$ and $6\cdot 4$.

(ii) The classes θ_j and the rational Weil classes stay of Hodge type along $\mathcal{D}_F$, so each is killed by $T_{[A]}\mathcal{D}_F\subseteq H^1(T_A)$, as in Proposition 3.11(i); hence $r(\gamma)\le 120-8$, and when equality holds the annihilator, of dimension 8, is $T_{[A]}\mathcal{D}_F$. Take a basis of $H^1(A,\mathbb{C})$ adapted to the V_σ and to the Hodge decomposition, in which the polarisation pairs the basis of $V_\sigma^{1,0}$ with that of $V_{\overline{\sigma}}^{0,1}$. The torus acting diagonally on it contains a subtorus that multiplies each θ_j by a scalar and each α_σ by an arbitrary scalar, and contraction is equivariant; so $r(\omega+p)$ depends on ω only through a rescaling of the variables of p, and each statement of the theorem holds for every ω once it holds for one. The rank is lower semicontinuous in p, and it equals 112 at $p=\theta_1^2\theta_2^2$ by Theorem 5.14(iii) below, so it equals 112 for a general p. Every element of the complexified Hodge ring of a member with $\mathrm{Hg}(A)=G_F$ is invariant under G_F, so its components stay of Hodge type along $\mathcal{D}_F$ and it is killed by $T_{[A]}\mathcal{D}_F$; its rank is therefore at most 112. The ring contains $\omega+\theta_1^2\theta_2^2$, since it contains $W_F\otimes\mathbb{C}$ and the classes θ_1, θ_2, whose span is defined over $\mathbb{Q}$, and the rank is lower semicontinuous; so it is 112 at a general element of the ring, whose component in $W_F\otimes\mathbb{C}$ is not zero.

(iii) and (iv) are Theorem 5.14(iii) and (iv) below.

(v) By the commutative square $\sigma_E\circ\mathrm{ev}_E=(\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 4.54, which holds on every abelian variety, $\dim\mathrm{Ext}^2(E,E)\ge\mathrm{rk}\,\mathrm{ev}_E\ge r(\mathrm{ch}\,E)$, and equality forces ev_E to be onto with kernel the annihilator and σ_E to be injective, as in Theorem 3.9(iv); and

$r(\operatorname{ch} E) \geq 80$ by (iii), since the complexification of $\operatorname{ch}(E)$ is a γ as above with $\omega = N\omega_1$. The last implication is [I, Theorem 17.7(v)], whose character is exactly of this form. ■

At $m = 1$ and $n = 2$ the corresponding numbers are 28, 12 and 24 (Theorem 3.9); in both cases the general value is the dimension of HT^2 minus that of the family. For the pure shape the argument of Theorem 2.5(ii) and (iv) transfers with the annihilator of Theorem 5.13(i): an injective σ_E would force $A_\theta A_{\theta'} = 0$ for $\theta \in T_\rho$, $\theta' \in T_{\rho'}$, $\rho \neq \rho'$, and a smooth component of the support of codimension at least two along which E is locally free of positive rank would be tangent to three of the four T_ρ. The pure shape is excluded, for every CM field, by Theorem 5.4.

The rank $r(\gamma)$ can be computed for every p. In the basis of the proof of Theorem 5.13(ii) write $x_{\sigma,1}, x_{\sigma,2}$ for the basis of $V_\sigma^{1,0}$, $y_{\sigma,1}, y_{\sigma,2}$ for that of $V_\sigma^{0,1}$, and $\partial_{\sigma,1}, \partial_{\sigma,2}$ for the dual basis of T_σ, so that $\theta_j = \sum_{\sigma|\tau_j} \sum_k x_{\sigma,k}\, y_{\overline{\sigma},k}$ and $\alpha_\sigma = x_{\sigma,1}x_{\sigma,2}y_{\sigma,1}y_{\sigma,2}$, and use the torus of that proof to make every u_σ equal to 1. Expand $p = \sum_{i,j=0}^4 e_{ij}\, \theta_1^i \theta_2^j/(i!\, j!)$, so that e_{ij} is a derivative of p at 0, and put:

- $\mu(p) = 0$ if $\partial_1\partial_2 p = 0$ and $\mu(p) = 1$ otherwise, the derivatives taken in $\mathbb{C}[\theta_1, \theta_2]$;
- $\rho_1(p)$ the dimension of the span of $\partial_1 p$ and $\partial_1^2 p$ modulo θ_1^3, which is the rank of the 2×15 matrix whose columns are the pairs $(e_{i,j}, e_{i+1,j})$, $1 \leq i \leq 3$, $0 \leq j \leq 4$; and $\rho_2(p)$ likewise, with the two indices exchanged;
- for $t = 1, 2$, with $\sigma, \overline{\sigma}$ the embeddings over τ_t, $W_t \subseteq HT^2(A)$ the span of the eight elements $y_{\sigma,1} \wedge y_{\sigma,2}$, $y_{\overline{\sigma},1} \wedge y_{\overline{\sigma},2}$, $\partial_{\sigma,1} \wedge \partial_{\sigma,2}$, $\partial_{\overline{\sigma},1} \wedge \partial_{\overline{\sigma},2}$ and $y_{\sigma,k} \otimes \partial_{\overline{\sigma},l}$, $y_{\overline{\sigma},k} \otimes \partial_{\sigma,l}$ with $k \neq l$, and $R_t(p)$ the rank of $\xi \mapsto \xi \lrcorner \gamma$ on W_t;
- $h_1 = e_1e_3 - e_2^2$, $h_2 = e_2e_4 - e_3^2$, $h_3 = e_1e_4 - e_2e_3$ on $\mathbb{C}^4$, and $V_\pm \subset \mathbb{C}^4$ the set where $h_3 = \pm e_3$, $2h_2 = \pm e_4$ and $2h_1 + 1 = \pm e_2$.

Theorem 5.14 (The rank of the criterion at a quartic field)**.** *In this notation:*

(i) $r(\gamma) = 64 + 16\mu(p) + 4\rho_1(p) + 4\rho_2(p) + R_1(p) + R_2(p)$.

(ii) $R_t(p) \in \{7, 8\}$. *If* $R_1(p) = 7$, *then* $(e_{10}, e_{20}, e_{30}, e_{40})$ *lies on* $V_+ \cup V_-$ *and* $\rho_1(p) = 2$; *conversely, if* $\mu(p) = 0$ *and* $(e_{10}, \ldots, e_{40})$ *lies on* $V_+ \cup V_-$, *then* $R_1(p) = 7$, *as for* $p = \theta_1^2/2$. *The same holds for* R_2 *with* $(e_{01}, \ldots, e_{04})$. *Each* $V_\pm$ *contains the surface of points* $(w\lambda \pm \frac{3}{2\lambda},\ w\lambda^2 \pm \frac{1}{2},\ w\lambda^3,\ w\lambda^4)$, $w \in \mathbb{C}$, $\lambda \in \mathbb{C}^\times$.

(iii) $r(\gamma) \geq 80$, *with equality exactly when* p *is a constant, and* $r(\gamma)$ *takes exactly the fifteen values*

$$80,\ 84,\ 87,\ 88,\ 91,\ 92,\ 94,\ 95,\ 96,\ 104,\ 107,\ 108,\ 110,\ 111,\ 112,$$

the values 87, 91, 94, 95, 107, 110 *and* 111 *only when* R_1 *or* R_2 *is* 7. *The value* 100 *is not taken.*

(iv) *For* $\omega = \sum_\sigma u_\sigma \alpha_\sigma$ *with arbitrary nonzero* u_σ, *the same holds with* e_{ij} *replaced by* $\kappa_1^i \kappa_2^j e_{ij}$, *where* $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$ *for the embeddings over* τ_t. *The four choices of* κ_t *fix* $V_+ \cup V_-$, *and* μ, ρ_1, ρ_2 *do not depend on them.*

Proof. (i) Let S be the torus of the basis that fixes θ_1, θ_2 and every α_σ. If $x_{\sigma,k}$ has weight $a_{\sigma,k}$, then $y_{\sigma,k}$ has weight $-a_{\overline{\sigma},k}$ and $\partial_{\sigma,k}$ has weight $-a_{\sigma,k}$, and the only relations are $a_{\sigma,1} + a_{\sigma,2} = a_{\overline{\sigma},1} + a_{\overline{\sigma},2}$, one for each real place; so S has dimension six. Attach to $y_{\sigma,k}$ the label $(\overline{\sigma}, k)$ and to $\partial_{\sigma,k}$ the label (σ, k). A basis element of HT^2 is a product of two of these and has weight $-(a_P + a_Q)$ for its two labels P, Q, and two such weights agree exactly when the pairs of labels coincide or are $\{(\sigma,1),(\sigma,2)\}$ and $\{(\overline{\sigma},1),(\overline{\sigma},2)\}$. So HT^2 splits into 34 weight spaces: the eight lines spanned by $y_{\sigma,k} \otimes \partial_{\overline{\sigma},k}$; sixteen spaces of dimension four, one for each pair of labels over different real places; eight of dimension four, one for

each pair $\{(\sigma,k),(\overline{\sigma},l)\}$ over one place; and W_1, W_2. The class γ is invariant under S, so contraction carries each weight space into the same weight of $H^*(A,\mathbb{C})$, and $r(\gamma)$ is the sum of the ranks on the 34 spaces.

Two rules compute these ranks. For a label $P=(\rho,k)$ with ρ over τ_t put $y_P=y_{\overline{\rho},k}$, the element of $H^{0,1}$ that carries it. Contraction by $\partial_{\rho,k}$ is a derivation that maps θ_t to $\pm y_P$ and the other $\theta_{t'}$ to 0, so it maps $f(\theta_1,\theta_2)$ to $\pm y_P\,(\partial_t f)(\theta_1,\theta_2)$. Hence a basis element with labels P,Q carries p to $\pm y_P y_Q\,(Dp)(\theta_1,\theta_2)$, where D is 1, ∂_t or $\partial_t\partial_{t'}$ according as none, one or both labels are carried by elements of T_0A, and t,t' are their places. Multiplication by y_Py_Q removes from θ_1 and θ_2 the terms that contain y_P or y_Q, and the products $\theta_1^a\theta_2^b$ of what remains are linearly independent as long as a and b do not exceed the numbers of remaining terms. Every monomial of $y_Py_Qf(\theta_1,\theta_2)$ contains, with each $x_{\sigma,i}$, its partner $y_{\overline{\sigma},i}$. The second rule is that the images of ω below contain some $x_{\sigma,i}$ without its partner, so they are independent of every image of p.

A line $y_{\sigma,k}\otimes\partial_{\overline{\sigma},k}$ carries p to $\pm y_{\sigma,k}^2(\cdots)=0$ and $\alpha_{\overline{\sigma}}$ to $\pm y_{\sigma,k}x_{\overline{\sigma},k'}y_{\overline{\sigma},1}y_{\overline{\sigma},2}\neq 0$, $k'\neq k$: rank one. On a space of the second kind, with labels $P=(\rho,k)$ over τ_1 and $Q=(\rho',l)$ over τ_2, the four basis elements are $y_P\wedge y_Q$, $y_P\otimes\partial_{\rho',l}$, $y_Q\otimes\partial_{\rho,k}$ and $\partial_{\rho,k}\wedge\partial_{\rho',l}$. They carry ω to $y_Py_Q(\alpha_\rho+\alpha_{\rho'})$, $y_P\,(\partial_{\rho',l}\lrcorner\,\alpha_{\rho'})$, $y_Q\,(\partial_{\rho,k}\lrcorner\,\alpha_\rho)$ and 0, the factors y_P and y_Q killing $\alpha_{\overline{\rho}}$ and $\alpha_{\overline{\rho}'}$; these contain $x_{\rho,k'}$ or $x_{\rho',l'}$ with $k'\neq k$, $l'\neq l$, without its partner, and the first three are independent, having two, one and one such factors x at different places. The bivector carries γ to $\pm y_Py_Q(\partial_1\partial_2p)(\theta_1,\theta_2)$, with three terms left in each θ_t, and $\partial_1\partial_2p$ has degree at most three in each variable; so this image is zero exactly when $\partial_1\partial_2p=0$: rank $3+\mu(p)$. On a space of the third kind, with labels $P=(\sigma,k)$ and $Q=(\overline{\sigma},l)$ over τ_1, the element $y_P\wedge y_Q$ carries ω to $y_Py_Q(\alpha_{\sigma'}+\alpha_{\overline{\sigma}'})$, σ' and $\overline{\sigma}'$ the embeddings over τ_2, and the other three elements kill ω: $y_P\otimes\partial_{\overline{\sigma},l}$ and $y_Q\otimes\partial_{\sigma,k}$ carry $\alpha_{\overline{\sigma}}$ and α_σ into multiples of $y_{\overline{\sigma},k}y_{\overline{\sigma},1}y_{\overline{\sigma},2}=0$ and $y_{\sigma,l}y_{\sigma,1}y_{\sigma,2}=0$, and the bivector meets no α. They carry p to $\pm y_Py_Q(\partial_1p)$, twice, and to $\pm y_Py_Q(\partial_1^2p)$, with two terms left in θ_1 and four in θ_2. So the rank is 1 plus the dimension of the span of ∂_1p and ∂_1^2p modulo θ_1^3; in the basis $\theta_1^a\theta_2^j/(a!\,j!)$, $a\le 2$, their coefficient vectors are $(e_{a+1,j})$ and $(e_{a+2,j})$, and the rank is $1+\rho_1(p)$; over τ_2 it is $1+\rho_2(p)$. The total is $8+16(3+\mu)+4(1+\rho_1)+4(1+\rho_2)+R_1+R_2$.

(ii) Over τ_1 write x_k,X_k,y_k,Y_k for $x_{\sigma,k},x_{\overline{\sigma},k},y_{\sigma,k},y_{\overline{\sigma},k}$, so that $\theta_1=\sum_k(x_kY_k+X_ky_k)$, $\alpha_\sigma=x_1x_2y_1y_2$ and $\alpha_{\overline{\sigma}}=X_1X_2Y_1Y_2$, and put $c=y_1y_2$, $C=Y_1Y_2$, $a_k=x_ky_1y_2Y_k$, $A_k=X_ky_kY_1Y_2$, $b=x_1x_2y_1y_2Y_1Y_2$ and $B=X_1X_2y_1y_2Y_1Y_2$. Let α' be the part of ω over τ_2, so that $\alpha'\theta_2=0$, and put $\langle e_i\rangle=\sum_j e_{ij}\theta_2^j/j!$. Since c kills the terms X_ky_k of θ_1, $c\,\theta_1^i/i!$ is c, a_1+a_2, $-b$ and 0 for $i=0$, 1, 2 and $i\ge 3$, so that $c\gamma=B+c(\alpha'+\langle e_0\rangle)+(a_1+a_2)\langle e_1\rangle-b\langle e_2\rangle$; in the same way, with ∂ acting as a left derivation, the other seven basis elements of W_1 carry γ to

$$\begin{aligned}
C\gamma &= b+C(\alpha'+\langle e_0\rangle)+(A_1+A_2)\langle e_1\rangle-B\langle e_2\rangle,\\
\partial_{\sigma,1}\lrcorner\,\partial_{\sigma,2}\lrcorner\,\gamma &= -c+C\langle e_2\rangle+(A_1+A_2)\langle e_3\rangle-B\langle e_4\rangle,\\
\partial_{\overline{\sigma},1}\lrcorner\,\partial_{\overline{\sigma},2}\lrcorner\,\gamma &= -C+c\langle e_2\rangle+(a_1+a_2)\langle e_3\rangle-b\langle e_4\rangle,\\
\pm y_k\,\partial_{\overline{\sigma},k'}\lrcorner\,\gamma &= A_k+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle,\\
\pm Y_k\,\partial_{\sigma,k'}\lrcorner\,\gamma &= a_k+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle,
\end{aligned}$$

where $\{k,k'\}=\{1,2\}$ and the sign is + for $k=1$. The differences of the last two pairs are A_1-A_2 and a_1-a_2. In a basis containing A_1-A_2, a_1-a_2, $c\alpha'$ and $C\alpha'$, these four coordinates form the identity matrix on the two differences and on $c\gamma$, $C\gamma$, and vanish on the other four images. So R_1 is 4 plus the rank of the two images of the bivectors and of the halves of the sums of the last two pairs, $\frac12(A_1+A_2)+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle$

and $\frac{1}{2}(a_1+a_2)+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle$. In the coordinates along c, a_1+a_2, b, C, $-(A_1+A_2)$, B times $\theta_2^j/j!$, and up to the signs of some columns, the matrix of these four vectors is $K=\left(K^0\,\middle|\,\operatorname{diag}(H,H)\right)$: the columns with $j=0$ form, with $e_k=e_{k0}$,

$$K^0=\begin{pmatrix} e_1 & e_2 & -e_3 & 0 & -\frac{1}{2} & 0\\ e_2 & e_3 & -e_4 & -1 & 0 & 0\\ 0 & \frac{1}{2} & 0 & e_1 & -e_2 & -e_3\\ -1 & 0 & 0 & e_2 & -e_3 & -e_4 \end{pmatrix},$$

and those with $j\ge 1$ form $\operatorname{diag}(H,H)$, with H the 2×12 matrix of the pairs $(e_{i,j},e_{i+1,j})$, $1\le i\le 3$, $1\le j\le 4$. So $R_1=4+\operatorname{rk}K$. Let (a,b,c,d) annihilate the columns of K^0. The first two columns give $d=ae_1+be_2$ and $c=-2(ae_2+be_3)$, and the other four then say that (a,b) is orthogonal to the rows of

$$Q=\begin{pmatrix} e_3 & e_4\\ -e_1e_2 & e_2^2-2e_1e_3-1\\ 2e_2^2-e_1e_3-\frac{1}{2} & e_2e_3\\ e_1e_4-2e_2e_3 & e_2e_4-2e_3^2 \end{pmatrix},$$

so $\operatorname{rk}K^0=2+\operatorname{rk}Q$. If Q were zero, then $e_3=e_4=0$ by its first row, $e_2^2=\frac{1}{4}$ by its third, and the second row would have the entry $-\frac{3}{4}$. So $\operatorname{rk}K\ge 3$ and $R_1\ge 7$. If $R_1=7$, then $\operatorname{rk}Q=1$, and the 2×2 minors of Q vanish exactly on $V_+\cup V_-$. Indeed, with $P=2h_1+1$ and m_{ij} the minor of rows i and j, expanding gives

$$m_{12}=e_2h_3-e_3P,\quad 2m_{13}=e_4P-2e_2h_2,\quad m_{14}=2e_3h_2-e_4h_3,$$
$$2m_{23}=e_2^2-P^2,\quad m_{24}=Ph_3-e_2e_3,\quad 2m_{34}=e_3^2-h_2-4h_1h_2.$$

On $V_\pm$, $(P,h_3,2h_2)=\pm(e_2,e_3,e_4)$, which makes the first five vanish, and the last is $e_3^2-h_2-2(\pm e_2-1)h_2=e_3^2+h_2-e_2e_4=0$. Conversely, let all six vanish. If $e_2=e_3=e_4=0$ then $P=1$ and $m_{23}\ne 0$; so $(P,h_3,2h_2)=\lambda(e_2,e_3,e_4)$ for some λ, by the first three. If $e_2\ne 0$, then $m_{23}=0$ gives $\lambda=\pm1$. If $e_2=0$, then $P=0$, so $e_1e_3=-\frac{1}{2}$, and $h_3=e_1e_4=\lambda e_3$ and $2h_2=-2e_3^2=\lambda e_4$ give $e_4=-2\lambda e_3^2$ and then $\lambda^2=1$. In both cases the point lies on V_λ. If moreover $\rho_1(p)\le 1$, then $h_1=h_2=h_3=0$, and the minors of the second and third rows and of the first and third rows become $(e_2^2-1)/2$ and, once $e_2^2=1$, $e_4/2$; so $e_4=0$, $e_3^2=e_2e_4=0$ and $e_1e_3=0\ne e_2^2$, against $h_1=0$. Hence $\rho_1(p)=2$. If $\mu(p)=0$, then $e_{ij}=0$ for $i,j\ge 1$, so $H=0$ and $R_1=6+\operatorname{rk}Q$, which is 7 on $V_\pm$; the point $(0,1,0,0)$ of $p=\theta_1^2/2$ lies on V_-. The parametrised points satisfy $h_1=\pm\frac{1}{2}w\lambda^2-\frac{1}{4}$, $h_2=\pm\frac{1}{2}w\lambda^4$ and $h_3=\pm w\lambda^3$, which are the equations of $V_\pm$. The space W_2 is the same with the indices exchanged.

(iii) A nonzero e_{ij} with $i,j\ge 1$ enters both Hankel matrices, so $\mu=1$ forces $\rho_1,\rho_2\ge 1$, and by (ii) $R_t=7$ forces $\rho_t=2$. The tuples $(\mu,\rho_1,\rho_2,R_1,R_2)$ allowed by these two conditions give exactly the fifteen values, and 80 only for $(0,0,0,8,8)$; $\rho_1=\rho_2=0$ means that $e_{ij}=0$ unless $i=j=0$, that is, p is constant. Each value is taken. For $p=0$, θ_1, θ_1^2, $\theta_1^2+\theta_2$, $\theta_1^2+\theta_2^2$, $\theta_1\theta_2$, $\theta_1\theta_2^2$ and $\theta_1^2\theta_2^2$ the points $(e_{10},\dots,e_{40})$ and $(e_{01},\dots,e_{04})$ are 0, $(1,0,0,0)$ or $(0,2,0,0)$, which lie on neither $V_\pm$, as $2h_1+1$ is 1, 1 and -7 there; so $R_1=R_2=8$, and (μ,ρ_1,ρ_2) is $(0,0,0)$, $(0,1,0)$, $(0,2,0)$, $(0,2,1)$, $(0,2,2)$, $(1,1,1)$, $(1,1,2)$ and $(1,2,2)$, giving 80, 84, 88, 92, 96, 104, 108 and 112. For $p=\theta_1^2/2+q$ with q one of 0, θ_2, θ_2^2, $\theta_2^2/2$ the point $(0,1,0,0)$ lies on V_- and $\mu=0$, so $R_1=7$, and $R_2=7$ exactly for $q=\theta_2^2/2$; this gives 87, 91, 95 and 94. For $p=\theta_1^2/2+\theta_1\theta_2+q$ with q one of 0, θ_2^2, $\theta_2^2/2$, the only nonzero e_{ij} with $i,j\ge 1$ is $e_{11}=1$, so the columns of $\operatorname{diag}(H,H)$ are multiples of $(1,0,0,0)$ and $(0,0,1,0)$; at $e=(0,1,0,0)$ the columns of K^0 span the space of $(1,0,0,0)$,

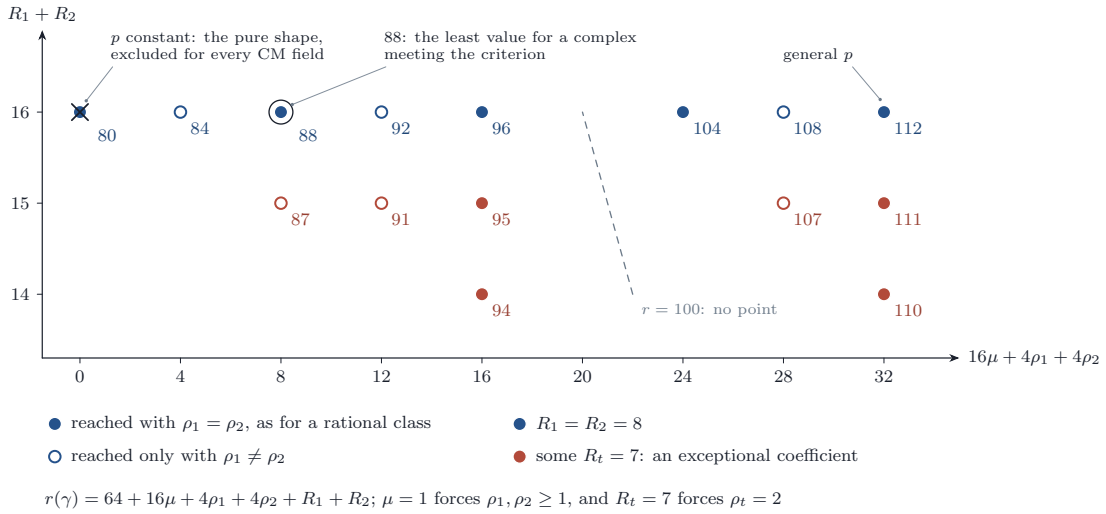


FIGURE 16. The rank $r(\gamma)$ of Theorem 5.14, over the plane of the Hankel part $16\mu + 4\rho_1 + 4\rho_2$ and of $R_1 + R_2$. The fifteen points are the tuples allowed by the two conditions of the theorem, and each carries its value. Filled points are reached with $\rho_1 = \rho_2$, as the character of a complex with rational coefficients is (Corollary 5.15(ii)); red points need an exceptional coefficient at one place at least. The value 80 is the pure shape, excluded for every CM field by Theorem 5.4; so 88 is the least rank a complex meeting the criterion can have. The level $r = 100$ meets no point.

$(0,0,1,0)$ and $(0,1,0,-1)$, so $\operatorname{rk} K = 3$ and $R_1 = 7$, and likewise $R_2 = 7$ for $q = \theta_2^2/2$. With $\mu = 1$ and $(\rho_1, \rho_2) = (2,1)$, $(2,2)$, $(2,2)$ this gives 107, 111 and 110.

(iv) The element of the torus of the proof of Theorem 5.13(ii) that multiplies $x_{\sigma,k}$ by s_σ and $y_{\sigma,k}$ by t_σ, with $s_\sigma t_{\overline{\sigma}} = \kappa_t$ for σ over τ_t, multiplies θ_t by κ_t and α_σ by $(s_\sigma t_\sigma)^2$, and these can be chosen to make every $u_\sigma (s_\sigma t_\sigma)^2$ equal to 1 exactly when $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$. It carries $\omega + p$ to the normalised class with coefficients $\kappa_1^i \kappa_2^j e_{ij}$, and contraction is equivariant. Replacing κ_t by $-\kappa_t$ changes the signs of e_1 and e_3, which fixes V_+ and V_-, and replacing it by $\sqrt{-1}\,\kappa_t$ exchanges them; scaling the columns and the second row of the Hankel matrices does not change their ranks. ∎

The count of (i), drawn in Figure 16, is the analogue of the dimension counts of Theorem 3.9 for a field of degree four: sixty-four of the rank come from the Weil class alone, the Hankel terms measure how far p is from an exponential in each variable, and the two eight-dimensional spaces W_t carry the exceptional coefficients, the analogue of the exceptional ratio of Remark 3.12. Indeed $\rho_1(p) \leq 1$ exactly when $(e_{10}, \dots, e_{40})$ lies on the cone over the twisted cubic, that is, when the part of p in θ_1 alone is, up to a constant, a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4.

Corollary 5.15 (The least object the criterion asks for)**.** *Let E be as in Theorem 5.13*(v).

(i) $\dim \operatorname{Ext}^2(E,E) \geq 81$.

(ii) *The coefficients of P satisfy $e_{ji} = \iota(e_{ij})$ for the nontrivial automorphism ι of F_0, so $\rho_1 = \rho_2$, and $r(\operatorname{ch} E)$ is one of* 80, 88, 94, 95, 96, 104, 110, 111, 112.

(iii) *If* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E)$*, then* P *is not constant and* $\dim \mathrm{Ext}^2(E,E)$ *is one of* 88, 94, 95, 96, 104, 110, 111, 112. *The value* 88 *requires* $\mu = 0$ *and* $\rho_1 = \rho_2 = 1$*, and it is the rank of* $N\omega_1 + \eta_s$ *for every* $s \in F_0^\times$.

Proof. (i) By Theorem 5.13(v), $\dim \mathrm{Ext}^2(E,E) \geq r(\mathrm{ch}\,E) \geq 80$. If $r(\mathrm{ch}\,E) = 80$, then P is a constant by Theorem 5.14(iii), so $\mathrm{ch}(E) = c + N\omega_1$ and Theorem 5.4 makes the inequality strict. (ii) Write $\eta_t = \tau_1(t)\theta_1 + \tau_2(t)\theta_2$ as before Proposition 5.5. A polynomial with rational coefficients in the η_t becomes, after exchanging θ_1 and θ_2, the polynomial whose coefficients are the images under ι of the old ones, so $e_{ji} = \iota(e_{ij})$; the Hankel matrix of ρ_2 is then the image under ι of that of ρ_1, which has the same rank. The values are those of Theorem 5.14(iii) with $\rho_1 = \rho_2$, the scaling of (iv) there changing neither μ nor the ρ_t. (iii) If P were constant, Theorem 5.4 would give $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$. For $P = \eta_s$ the pairs are $(\tau_t(s), 0)$, so $\rho_1 = \rho_2 = 1$, $\mu = 0$, and $R_t = 8$ by Theorem 5.14(ii). ∎

The Weil tori of Setup 5.1 exclude more than the pure shape. A polynomial in θ_1, θ_2 can stay of Hodge type on all of S_F, and then the argument of Theorem 5.4 applies to it.

Proposition 5.16 (The characters that the Weil tori exclude)**.** *Let* E *be as in Theorem 5.13*(v)*, and suppose that* P *lies in the span of the* $\theta_1^i\theta_2^j$ *with* $i, j \in \{0, 4\}$*, as for* $\mathrm{ch}(E) = c + N\omega_1 + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ *with* $b \in F_0$*. Then* σ_E *is not injective and* $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$*. Consequently, if* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$*, then* $P = c + f(\theta_1) + f^\iota(\theta_2)$ *with* $c \in \mathbb{Q}$*, where* $f(\theta)$ *is a nonzero multiple either of* θ *or of* $e^{\lambda\theta} - 1$ *for some* $\lambda \neq 0$*, and* f^ι *is obtained by applying* ι *to the coefficients of* f.

Proof. In the basis before Theorem 5.14, θ_1 is a sum of four products of two classes, one from V_σ and one from $V_{\overline{\sigma}}$ for the two embeddings over τ_1, in disjoint sets of basis vectors. So θ_1^4 is 4! times their product, a nonzero element of $\bigwedge^4 V_\sigma \otimes \bigwedge^4 V_{\overline{\sigma}}$, that is a nonzero multiple of $\alpha_\sigma\alpha_{\overline{\sigma}}$, and likewise for θ_2^4. Hence $\mathrm{ch}(E) = N\omega_1 + P$ lies in the span of the classes α_Σ, which are of type (p,p) on every member of S_F (Setup 5.1); being rational, it is a Hodge class on every member. Its component of degree four is $N\omega_1 \neq 0$, since P has none, and Theorem 5.4 applies. If $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$, then $\mu = 0$ and $\rho_1 = \rho_2 = 1$ by Corollary 5.15(iii), so P has no mixed monomials, and by the paragraph after Theorem 5.14 its part in θ_1 alone is a multiple of $e^{\lambda\theta_1} - 1$, of θ_1 or of θ_1^4; its part in θ_2 alone is obtained by ι, by Corollary 5.15(ii). In the third case P lies in the span of 1, θ_1^4 and θ_2^4, which the first part excludes. ∎

The shape $c + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ has rank 88, and with a multiple of $\theta_1^4\theta_2^4$ added it has rank 104, by Theorem 5.14: $\rho_1 = \rho_2 = 1$, μ is 0 or 1, and $R_1 = R_2 = 8$, since $(e_{10}, \dots, e_{40}) = (0, 0, 0, 24b)$ with $b \neq 0$ does not meet $V_+ \cup V_-$. Proposition 5.16 removes both. What remains at 88 is a Chern character whose polynomial part is exponential or linear in each θ_t separately, such as $c + \eta_s + N\omega_1$. For these the classes θ_t are not of Hodge type on the Weil tori, and the argument needs a family of complex tori along which the character stays of Hodge type and whose members are not algebraic. The next proposition, the analogue for a quartic field of Proposition 3.11(iii) and (iv), shows that there is none: the Hodge locus of such a character is the polarised family itself.

Proposition 5.17 (The Hodge locus of a character at a quartic field)**.** *Keep the notation before Theorem 5.14, with* $\gamma = \omega + p$ *and every* $u_\sigma \neq 0$*, and for the embeddings* $\sigma, \overline{\sigma}$ *over* τ_t *put* $\omega_t = u_\sigma\alpha_\sigma + u_{\overline{\sigma}}\alpha_{\overline{\sigma}}$*. Call* τ_t *a* mid place *of* p *if* p *has a monomial* $\theta_1^i\theta_2^j$ *with nonzero coefficient whose exponent of* θ_t *is* 1, 2 *or* 3*, and call a mid place* exceptional *if the only such monomial is* θ_t^2 *itself and its coefficient* c_t *in* p *satisfies* $c_t^2\theta_t^4 = \frac{3}{4}\omega_t^2$*, the ratio of Proposition 3.11(iv).*

(i) *The annihilator of γ in $H^1(T_A)$ is the space of F-linear $\xi \in \bigoplus_\sigma \mathrm{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1})$ with $\xi \lrcorner \theta_t = 0$ at every mid place, plus, at each exceptional place, a line spanned by an element of $\mathrm{Hom}(V_\sigma^{1,0}, V_{\overline{\sigma}}^{0,1}) \oplus \mathrm{Hom}(V_{\overline{\sigma}}^{1,0}, V_\sigma^{0,1})$ with both components nonzero. Its dimension is $8 + 4a + b$, where a is the number of places that are not mid places and b the number of exceptional places.*

(ii) *The connected component through A of the Hodge locus of γ, in the space of complex structures on the real torus underlying A, is S_F when neither place is a mid place, that is when p lies in the span of the $\theta_1^i\theta_2^j$ with $i, j \in \{0, 4\}$; and it is $\mathcal{D}_F$ when both places are mid places and neither is exceptional. In the second case all its members are abelian varieties polarised by $\theta_1 + \theta_2$.*

(iii) *If both places are exceptional, then near A the Hodge locus is the orbit of A under $G_1 \times G_2$, where $G_t \cong \mathrm{Spin}(4, 3)$ is the connected stabiliser of $X_t = \omega_t + c_t\theta_t^2$ in the real linear group of $V_\sigma \oplus V_{\overline{\sigma}}$, and it has dimension 10. A very general member T of it has $\mathrm{NS}(T) = 0$ and $\mathrm{Hdg}^2(T) \otimes \mathbb{C}$ contained in the span of X_1 and X_2, and on every member, for every Kähler class κ, $\int X_t\,\kappa^6$ is nonzero with the sign of c_t, in the normalisation in which θ_t is a real class with θ_t^2 positive.*

(iv) *If γ is the complexified character of Theorem 5.13(v), then both places are mid places or neither is, and both are exceptional or neither is; and an exceptional γ has $r(\gamma) = 94$, or 110 when p contains $\theta_1^4\theta_2^4$.*

Proof. (i) Grade $\bigwedge^* H^1$ by the degrees in the four V_σ, and write $\xi_{\sigma\sigma'}$ for the component of ξ in $\mathrm{Hom}(V_{\sigma'}^{1,0}, V_\sigma^{0,1})$, which raises the degree in V_σ by one and lowers that in $V_{\sigma'}$ by one. The monomial $\theta_1^i\theta_2^j$ has degree i in the two embeddings over τ_1 and j in those over τ_2, and α_σ has degree 4 in V_σ. By the proof of Theorem 5.13(i), $\xi_{\sigma\sigma}\lrcorner\omega = 0$, and for $\sigma \neq \sigma'$ the only term of $\xi_{\sigma\sigma'}\lrcorner\omega$ is $u_{\sigma'}\xi_{\sigma\sigma'}\lrcorner\alpha_{\sigma'}$, of degree 3 in $V_{\sigma'}$ and 1 in V_σ, which determines $\xi_{\sigma\sigma'}$. A term $\xi_{\rho\rho'}\lrcorner\theta_1^i\theta_2^j$ has that degree only if σ, σ' lie over the same place τ_t, (i, j) is $(2, 0)$ or $(0, 2)$ with the 2 at t, and $(\rho, \rho') = (\sigma', \sigma)$. So every component $\xi_{\sigma\sigma'}$ with σ, σ' over different places vanishes, and the equations split into the F-linear part and, at each place, the pair $\xi' = \xi_{\overline{\sigma}\sigma}$, $\xi'' = \xi_{\sigma\overline{\sigma}}$.

For F-linear ξ put $D_t = \xi\lrcorner\theta_t \in V_\sigma^{0,1} \otimes V_{\overline{\sigma}}^{0,1}$; the map $\xi \mapsto (D_1, D_2)$ is onto, with kernel $T_{[A]}\mathcal{D}_F$. Then $\xi\lrcorner\theta_1^i\theta_2^j = i\,\theta_1^{i-1}\theta_2^j D_1 + j\,\theta_1^i\theta_2^{j-1}D_2$, the monomials lie in different degrees, and the two terms have different Hodge types at the two places, so each vanishes. Now $\theta_t^3 D_t = 0$ for every D_t, as a degree count shows, while multiplication by θ_t^k is injective on $V_\sigma^{0,1} \otimes V_{\overline{\sigma}}^{0,1}$ for $k \leq 2$: $\theta_t^2 y_{\sigma,k}y_{\overline{\sigma},l}$ is 2 times the product of $y_{\sigma,k}y_{\overline{\sigma},l}$ with the two terms of θ_t that avoid $y_{\sigma,k}$ and $y_{\overline{\sigma},l}$, and different (k, l) give different monomials. Multiplication by $\theta_{t'}^j$, $j \leq 4$, acts on the other factor of $\bigwedge^* H^1 = \bigwedge^*(V_\sigma \oplus V_{\overline{\sigma}}) \otimes \bigwedge^*(V_{\sigma'} \oplus V_{\overline{\sigma}'})$ and is nonzero there. So the conditions on the F-linear part are exactly $D_t = 0$ at the mid places, a space of dimension $16 - 4(2 - a) = 8 + 4a$.

At a place τ_t let c_t be the coefficient of θ_t^2 in p. The terms involving ξ'' are $u_{\overline{\sigma}}\xi''\lrcorner\alpha_{\overline{\sigma}} + c_t\,\xi'\lrcorner\theta_t^2$, of degree 3 in $V_{\overline{\sigma}}$ and 1 in V_σ, the same with σ and $\overline{\sigma}$ and with ξ' and ξ'' exchanged, and $i\,\theta_t^{i-1}\theta_{t'}^j(\xi'\lrcorner\theta_t)$ and its analogue for ξ'' for every other monomial with exponent $i \in \{1, 2, 3\}$ of θ_t. Here $\xi'\lrcorner\theta_t$ lies in the line $\bigwedge^2 V_{\overline{\sigma}}^{0,1}$, on which multiplication by $\theta_t^{i-1}\theta_{t'}^j$ is injective for $i \leq 3$ by the same count. If such another monomial occurs, $\xi'\lrcorner\theta_t = \xi''\lrcorner\theta_t = 0$, and then $\xi' = \xi'' = 0$, the maps $\xi' \mapsto \xi'\lrcorner\alpha_\sigma$ and $\xi'' \mapsto \xi''\lrcorner\alpha_{\overline{\sigma}}$ being injective. Otherwise $\xi''\lrcorner\theta_t^2 = 2\theta_t\,(\xi''\lrcorner\theta_t)$ is a multiple of $\theta_t\,y_{\sigma,1}y_{\sigma,2}$, which is $\xi_0'\lrcorner\alpha_\sigma$ for one element ξ_0', and likewise with σ and $\overline{\sigma}$ exchanged; so ξ' and ξ'' are multiples of ξ_0' and ξ_0''. Explicitly, $\xi_0'(x_{\sigma,1}) = -y_{\overline{\sigma},2}$ and $\xi_0'(x_{\sigma,2}) = y_{\overline{\sigma},1}$ give $\xi_0'\lrcorner\alpha_\sigma = \theta_t\,y_{\sigma,1}y_{\sigma,2}$ and

$\xi_0' \lrcorner \theta_t = 2y_{\overline{\sigma},1}y_{\overline{\sigma},2}$, and symmetrically for ξ_0''. Writing $\xi' = a\xi_0'$ and $\xi'' = b\xi_0''$, the two remaining equations are $u_\sigma a + 4c_t b = 0$ and $4c_t a + u_{\overline{\sigma}} b = 0$, and their determinant vanishes exactly when $16c_t^2 = u_\sigma u_{\overline{\sigma}}$. Since $\theta_t^4 = 24\,\alpha_\sigma\alpha_{\overline{\sigma}}$ and $\omega_t^2 = 2u_\sigma u_{\overline{\sigma}}\alpha_\sigma\alpha_{\overline{\sigma}}$, that is the condition $c_t^2\theta_t^4 = \frac{3}{4}\omega_t^2$, and the solutions then form a line on which both components are nonzero.

(ii) By Proposition 3.11(i), applied to each homogeneous component, the Hodge locus of γ in the chart at A is the linear space $\mathrm{Ann}_{H^1(T_A)}(\gamma)$. If neither place is a mid place, that space is the space of all F-linear ξ, which is also the part of S_F in the chart; γ is of Hodge type on all of S_F, as in the proof of Proposition 5.16, and at every member of S_F its component of degree four is ω, whose annihilator in $H^1(T)$ is again the space of F-linear deformations (Theorem 5.13(i), whose proof uses only the decomposition into the V_σ). So S_F is open in the Hodge locus; it is closed in the space of complex structures, since commuting with F is a closed condition and the dimensions of the $H^{1,0} \cap V_\sigma$ are locally constant on it; and it is connected. If both places are mid places and neither is exceptional, the annihilator is $T_{[A]}\mathcal{D}_F$ by (i), at every member of $\mathcal{D}_F$, since the conditions of (i) are stated in terms of the flat classes θ_t and ω_t; and the argument of Proposition 3.11(iii) shows that $\mathcal{D}_F$ is open, closed and connected in the Hodge locus.

(iii) At each place X_t is, in the 8-dimensional space $V_\sigma \oplus V_{\overline{\sigma}}$ with the form θ_t, the class of Remark 3.12, and the argument there applies word for word, with the factor $\mathfrak{su}(2,2)$ at τ_t of the Lie algebra of G_F in place of the Hodge group: the stabiliser is $\mathfrak{so}(4,3)$, with an orbit of dimension 5 in the Grassmannian. The other components of γ are constants, multiples of θ_t^4, which is 24 times the volume form of $V_\sigma \oplus V_{\overline{\sigma}}$, and of $\theta_1^4\theta_2^4$, all fixed by $G_1 \times G_2$. So the orbit of A lies in the Hodge locus, and it has dimension 10, which by (i) is that of the locus near A. A rational class of Hodge type at a very general member is of Hodge type on the whole orbit, hence killed by the complex structures of all its members, which lie in $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ and generate an ideal projecting nontrivially to both simple factors, that is all of it. The invariants of $\mathfrak{so}(7,\mathbb{C})$ in the exterior algebra of its spin representation are 1, X_t and the volume form (Remark 3.12), so in degree two there are none and in degree four only X_1 and X_2. For the sign, a Kähler class is κ_h for a positive hermitian form h on the tangent space; only the terms of κ_h^6 that contain the whole volume of the other place pair with X_t, and they give $\det(h_{t'})$ times $\kappa_{h'}^2$ for the Schur complement h' of $h_{t'}$ in h, a positive hermitian form on the tangent space at τ_t. So $\int X_t\kappa_h^6$ is a positive multiple of the integral of Remark 3.12 for h', which has the sign of c_t; and $G_1 \times G_2$ fixes X_t and carries Kähler cones to Kähler cones.

(iv) An automorphism of $\overline{\mathbb{Q}}$ inducing ι on F_0 fixes the rational class γ and exchanges θ_1 with θ_2 and ω_1 with ω_2, so it carries the conditions at τ_1 to those at τ_2. An exceptional p lies in the span of 1, θ_t^2, θ_t^4 and $\theta_1^4\theta_2^4$ with $c_t \neq 0$, so $\rho_1 = \rho_2 = 2$, $\mu = 1$ exactly when $\theta_1^4\theta_2^4$ occurs, and $(e_{10}, \dots, e_{40})$ lies on $V_+ \cup V_-$, so $R_1 = R_2 = 7$ by Theorem 5.14(ii); the formula of Theorem 5.14(i) gives 94 or 110. ∎

Corollary 5.18 (The Weil tori at rank 88)**.** *Let E be as in Theorem 5.13*(v) *with* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$*, so that* $\mathrm{ch}(E) = N\omega_1 + c + f(\theta_1) + f^\iota(\theta_2)$ *as in Proposition 5.16. Then the connected component through A of the Hodge locus of* $\mathrm{ch}(E)$ *is $\mathcal{D}_F$, and every complex torus reached from A by a deformation along which* $\mathrm{ch}(E)$ *stays of Hodge type is an abelian variety of the family. So no argument that deforms E itself to a torus that is not algebraic can exclude it; the characters that such an argument excludes at* 88 *are exactly those of Proposition 5.16.*

Proof. The coefficient e_{10} of f is nonzero, both for a multiple of θ and for one of $e^{\lambda\theta}-1$, and $e_{01}=\iota(e_{10})$, so both places are mid places, with the monomials θ_1 and θ_2, and neither is exceptional. Proposition 5.17(ii) applies. ∎

As after Proposition 3.11, this concerns E itself: an autoequivalence of $D^b(A)$ changes the character, and the proposition then applies to the new one. For the exceptional characters of Proposition 5.17(iii), of rank 94 or 110, the proof of Theorem 5.4 can be started, since the locus contains tori without divisors, but it breaks down as in Remark 3.12: the classes X_t have integrals of constant sign against every Kähler class, so Bando and Siu give no vanishing: they bound the signs of the coefficients of ch_2 of a torsion-free sheaf along X_1 and X_2, which a complex, an alternating sum, need not respect. So at a quartic field the Weil tori and their exceptional analogues exclude exactly the characters of Proposition 5.16, and what the criterion still asks for, at 88 and above, is the construction of an object.

The natural candidates are Markman's secant objects [Mar25c], in the twisted form in which he states his reduction. For them a B-field twist of $\mathrm{ch}(E)$, the class κ of Theorem 5.22, is flat, and under a non-degeneracy condition its component in H^4 is a polynomial in the η_t plus a nonzero Weil class [Mar25c, Proposition 10.2.1, Theorem 1.1.2]. But $\mathrm{ch}(E)$ itself is not flat, and κ is not of the form $N\omega+P(\eta)$ in higher degrees (Remark 5.23), so they are candidates only for a twisted form of Theorem 5.13(v). The dimension count that works for imaginary quadratic fields (Theorem 3.19) can be tested on them exactly. Let X be a principally polarised abelian fourfold with real multiplication by the ring of integers $\mathcal{O}_{F_0}$, write $\theta_1,\theta_2\in H^2(X,\mathbb{R})$ for the components of the polarisation at the two real places of F_0, so that $\theta=\theta_1+\theta_2$ and $\theta_j^3=0$, and let $\mathcal{R}$ be the $\mathbb{Q}$-subalgebra of $H^{\mathrm{ev}}(X,\mathbb{Q})$ spanned by the rational combinations of the $\theta_1^a\theta_2^b$, $0\le a,b\le 2$. A class $v=\sum_{a,b}V_{ab}\,\theta_1^a\theta_2^b/(a!\,b!)$ of $\mathcal{R}$ has a 3×3 matrix V with entries in $F_0\otimes\mathbb{R}$.

Proposition 5.19 (The secant space of a quartic field)**.** *Let $F=F_0(\sqrt{-q})$ with $q\in F_0$ totally positive, let $t\in F_0$, $s_j=\sqrt{-1}\,\sqrt{\tau_j(q)}$, and for $\varepsilon\in\{\pm1\}^2$ put $e_\varepsilon=\exp\bigl(\sum_j(\tau_j(t)+\varepsilon_j s_j)\theta_j\bigr)$.*

(i) *The span $S(t,q)$ of the four e_ε is a rational subspace of $\mathcal{R}$ of dimension $4=2^{[F_0:\mathbb{Q}]}$, secant to the spinor variety. For $t=0$ it is the secant space of* [Mar25c, §11.1, (11.1.2)] *for the polarisation θ and q; replacing θ by $f\theta$, $f\in F_0^\times$, replaces q by qf^2, and a rational F_0-B-field t transports Markman's F-structure by the rational automorphism of $X\times\widehat{X}$ acting on $H^*(X)$ as e^{θ_t}, $\theta_t=\sum_j\tau_j(t)\theta_j$, and replaces $S(0,q)$ by $S(t,q)=e^{\theta_t}S(0,q)$.*

(ii) *For $v=\sum_\varepsilon w_\varepsilon e_\varepsilon\in S(t,q)$ let $\rho=\mathrm{rk}\,V$ and $N_w=\#\{\varepsilon:w_\varepsilon\neq 0\}$. The ranks of contraction from $HT^k(X)$ into v are 1, 8 and $2\rho+4N_w$ for $k=0,1,2$, with $(\rho,N_w)\in\{(1,4),(2,4),(2,2)\}$, and $N_w=2$ only for biquadratic F and v in the secant plane of one of its two imaginary quadratic subfields* [Mar25c, §11.2.1].

(iii) *For $v=e^{\theta_t}(a\,u+v_f+c\,w)$ in the basis $u=(1-\frac12 q_1\theta_1^2)(1-\frac12 q_2\theta_2^2)$, $v_f=\theta_f-\frac12\theta_{\bar f q}\theta_1\theta_2$, $w=\theta_1\theta_2$ of $e^{-\theta_t}S(t,q)$, one has $\chi(v,v)=\int_X v^\vee v=4\bigl(\mathrm{N}(q)a^2+c^2+\mathrm{Tr}(\bar q f^2)\bigr)$, which is positive definite.*

Proof. (i) The Galois group permutes the four exponentials as it permutes the CM types of F, so their span is rational; since $\theta_j^3=0$, $e^{\pm s_j\theta_j}=(1-\frac12\tau_j(q)\theta_j^2)\pm s_j\theta_j$, and the four classes are independent. The conjugates in $\mathfrak{so}\cap\mathrm{End}^0(X\times\widehat{X})$ of the standard structure have eigenvalues $t\pm s$, and the pure spinors of their eigenspaces are the e_ε.

(ii) Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspaces of F_0, and $HT^*(X)$ and $H^*(X)$ are tensor products over j, compatibly with contraction. By Lemma 4.55(i), $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi) e^{\lambda\theta_j}$, and the blocks $HT_1^a \otimes HT_2^b$ have independent images because they shift the two Hodge bidegrees by (a, b) while v is balanced. The block $(1, 1)$ contributes $4N_w$, by the invertibility of the Hadamard matrix of the signs; the blocks $(2, 0)$ and $(0, 2)$ contribute ρ each; the blocks $(1, 0)$ and $(0, 1)$ contribute 4 each in degree one. Reality gives $w_{-\varepsilon} = \overline{w_\varepsilon}$, and when F is not biquadratic the Galois group acts transitively on the CM types, so $N_w = 4$.

(iii) One has $\int_X e^{c_1\theta_1 + c_2\theta_2} = c_1^2 c_2^2$, and χ is invariant under e^{θ_t}; the form is positive definite because q is totally positive. ∎

The integral classes of $\mathcal{R}$ are described by one congruence, and the description does not depend on the field.

Lemma 5.20 (The integral classes of $\mathcal{R}$)**.** *Let F_0 be a real quadratic field with ring of integers $\mathcal{O}$ and different $\mathfrak{d}$, and let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}$. A class $v \in \mathcal{R}$ with matrix*

$$V = \begin{pmatrix} r & \bar{h} & \bar{g} \\ h & c & \bar{k} \\ g & k & m \end{pmatrix}, \qquad r, c, m \in \mathbb{Q}, \quad h, g, k \in F_0,$$

lies in $H^{\mathrm{ev}}(X, \mathbb{Z})$ if and only if $r, c, m \in \mathbb{Z}$, $h, g, k \in \mathcal{O}$ and $c \equiv g$ modulo $\mathfrak{d}$.

Proof. Integrality can be tested one prime p at a time, and after a faithfully flat extension of $\mathbb{Z}_p$. Let $M = H_1(X, \mathbb{Z})$ and let $\langle\,,\rangle$ be the polarisation form on M, that is, θ. The Rosati involution fixes $\mathcal{O}$, so $\langle\,,\rangle = \mathrm{Tr}_{F_0/\mathbb{Q}}\,\psi$ for a unique $\mathcal{O}$-bilinear alternating form ψ with values in $\mathfrak{d}^{-1}$, and ψ is perfect because θ is principal. Write $\mathcal{O} = \mathbb{Z}[\nu]$ and $\delta = \nu - \bar{\nu}$, a generator of $\mathfrak{d}$ with $\bar{\delta} = -\delta$, and put $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$, a product of at most two discrete valuation rings. The form $\delta\psi$ on $M_p = M \otimes \mathbb{Z}_p$ is perfect with values in $\mathcal{O}_p$, so M_p has a symplectic $\mathcal{O}_p$-basis e, f, e^*, f^*. Hence $M_p = \mathcal{O}_p \otimes_{\mathbb{Z}_p} W$, with W the $\mathbb{Z}_p$-span of this basis, and $\langle\,,\rangle = B \otimes \omega_0$, where ω_0 is the standard symplectic form of W and $B(\alpha, \beta) = \mathrm{Tr}(\delta^{-1}\alpha\beta)$.

Let Λ be the ring of integers of a finite extension of $\mathbb{Q}_p$ over which F_0 splits, and fix an embedding of F_0 in its field of fractions. The map $\alpha \otimes \lambda \mapsto (\alpha\lambda, \bar{\alpha}\lambda)$ identifies $\mathcal{O}_p \otimes_{\mathbb{Z}_p} \Lambda$ with the lattice of pairs $(x, y) \in \Lambda^2$ with $x \equiv y$ modulo δ, whose basis is $(1, 1), (\delta, 0)$. Let $\varepsilon_1, \varepsilon_2$ be the coordinates and $u_1, \dots, u_4$ the basis of $W^\vee$ dual to e, f, e^*, f^*. The dual lattice has basis ε_2, $(\varepsilon_1 - \varepsilon_2)/\delta$, so $H^1(X, \Lambda)$ has the basis

$$Y_i = \varepsilon_2 \otimes u_i, \qquad Z_i = \delta^{-1}(X_i - Y_i), \qquad \text{where } X_i = \varepsilon_1 \otimes u_i,$$

and $H^*(X, \Lambda)$ is the exterior algebra on the Y_i and Z_i. In these coordinates $B = \delta^{-1}(\varepsilon_1^2 - \varepsilon_2^2)$, and the components of $\theta = B \otimes \omega_0$ in the two eigenspaces of $\mathcal{O}$, the first being the one on which $\mathcal{O}$ acts through the fixed embedding, are

$$\theta_1 = \delta^{-1}(X_1X_3 + X_2X_4), \qquad \theta_2 = -\delta^{-1}(Y_1Y_3 + Y_2Y_4).$$

Substitute $X_i = Y_i + \delta Z_i$ in $v = \sum_{a,b} V_{ab}\,\theta_1^a\theta_2^b/(a!\,b!)$ and expand in the monomials in the Y_i and Z_i. Up to sign, every coefficient is one of

$$r,\ m,\ h,\ k,\ c,\ g,\ g - c,\ \delta h,\ \delta k,\ \delta g,\ \delta^2 g,\ \frac{g - c}{\delta},\ \frac{h - \bar{h}}{\delta},\ \frac{k - \bar{k}}{\delta},\ \frac{2c - g - \bar{g}}{\delta^2},$$

and each of r, m, h, k, c, g and $(g - c)/\delta$ occurs, so an integral class satisfies the conditions at p. Conversely, suppose that they hold. Every $x \in \mathcal{O}_p$ has $x - \bar{x} \in \delta\mathcal{O}_p$, which disposes of $h - \bar{h}$ and $k - \bar{k}$; and $g = c + \delta g'$ with $g' \in \mathcal{O}_p$ gives $2c - g - \bar{g} = -\delta(g' - \bar{g}') \in \delta^2\Lambda$. So

every coefficient lies in Λ. When p is unramified in F_0, δ is a unit at p and the congruence is empty. ∎

Proposition 5.21 (Integral secant classes of small Euler characteristic)**.** *Let $F_0 = \mathbb{Q}(\sqrt{D})$ with $D > 1$ squarefree, let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and let v be an integral class in a space $S(t,q)$ of Proposition 5.19, with ρ and N_w as there.*

(i) *If $\rho = 1$, then $\chi(v,v) \neq 4$.*

(ii) *If $\rho = 2$ and $N_w = 4$, then $\chi(v,v) \neq 6$.*

Proof. By Lemma 5.20 the entries of V lie in $\mathcal{O} = \mathcal{O}_{F_0}$, the diagonal ones in $\mathbb{Z}$, and $c \equiv g$ modulo $\mathfrak{d}$; conjugation is trivial on $\mathcal{O}/\mathfrak{d}$.

(i) Write $V = \lambda\alpha\bar{\alpha}^T$ with $\alpha \in \mathcal{O}^3$, and let $\mathfrak{a}$ be the ideal generated by the entries of α; replacing α by a multiple, we may take $\mathfrak{a}$ integral and prime to $\mathfrak{d}$. The entries $\lambda\alpha_i\bar{\alpha}_j$ lie in $\mathcal{O}$ and generate $\lambda\mathfrak{a}\bar{\mathfrak{a}} = \lambda\mathrm{N}(\mathfrak{a})\mathcal{O}$, so $\mu = \lambda\mathrm{N}(\mathfrak{a}) \in \mathbb{Z}$. With $D' = \alpha_1^2 - \alpha_0\alpha_2 \in \mathfrak{a}^2$ one has $\chi = 4\lambda^2\mathrm{N}(D')$, and $\mathrm{N}(D') \in \mathrm{N}(\mathfrak{a})^2\mathbb{Z}$, so $\chi = 4$ forces $\mu = \pm 1$ and $(D') = \mathfrak{a}^2$. Since conjugation is trivial modulo $\mathfrak{d}$, $\alpha_1\bar{\alpha}_1 - \alpha_2\bar{\alpha}_0 \equiv D'$ modulo $\mathfrak{a}\mathfrak{d}$, and the congruence $c \equiv g$ gives $\lambda D' \in \mathfrak{d}\,\bar{\mathfrak{a}}^{-1}$. The ideal of $\lambda D'$ is $\mathfrak{a}\bar{\mathfrak{a}}^{-1}$, so $\mathfrak{a} \subseteq \mathfrak{d}$, which is impossible for $\mathfrak{a}$ prime to $\mathfrak{d}$ and $\mathfrak{d} \neq (1)$.

(ii) The matrix $\operatorname{adj} V$ is Hermitian of rank one, so $\operatorname{adj} V = \mu\,\bar{l}\,l^T$ with $l \in \mathcal{O}^3$, $l^T V = 0$ and $\mu \in \mathbb{Q}^\times$. Put $\Delta = l_1^2 - 4l_0l_2$ and $m = \mu^2\mathrm{N}(\Delta)$. Replacing l by λl, $\lambda \in F_0^\times$, replaces μ by $\mu/\mathrm{N}(\lambda)$ and Δ by $\lambda^2\Delta$, so m depends only on V. As the entries of $\operatorname{adj} V$ are integral, μ is integral at every prime at which l is primitive, and l can be made primitive at the primes above any given p, because $\mathcal{O} \otimes \mathbb{Z}_{(p)}$ is a principal ideal domain; so $m \in \mathbb{Z}$. Let J be the matrix with rows $(0,0,1)$, $(0,-2,0)$, $(1,0,0)$. Since $\int_X \theta_1^2\theta_2^2 = 4$, one has $\chi(v,v) = \operatorname{tr}(JVJ\bar{V})$; and $\operatorname{adj} J$ has rows $(0,0,2)$, $(0,-1,0)$, $(2,0,0)$, so that $l^T(\operatorname{adj} J)\,l = -\Delta$ and

$$\sigma_2(JVJ\bar{V}) = \operatorname{tr}\operatorname{adj}(JVJ\bar{V}) = \mu^2\,\Delta\bar{\Delta} = m.$$

At a real place, $V = PWQ^T$ with $W = (w_\varepsilon)$ and P, Q the 3×2 matrices with columns $(1, \pm s_1, s_1^2)$ and $(1, \pm s_2, s_2^2)$, when $t = 0$; multiplication by e^{θ_t} replaces V by $UV\bar{U}^T$ with U unipotent and $U^TJU = J$, which conjugates $JV\bar{J}\bar{V}$, so we may take $t = 0$. As $(1,z,z^2)\,J\,(1,z',z'^2)^T = (z-z')^2$ and $w_{-\varepsilon} = \overline{w_\varepsilon}$, the nonzero eigenvalues of $JV\bar{J}\bar{V}$ are $16\mathrm{N}(q)\big(|w_{++}| \pm |w_{+-}|\big)^2$, so

$$\chi^2 - 4m = \big(64\mathrm{N}(q)\,|w_{++}|\,|w_{+-}|\big)^2 \geq 0,$$

with equality exactly when $N_w = 2$. Modulo $\mathfrak{d}$ one has $\bar{V} \equiv V$, $\operatorname{tr}(JV) = g - 2c + \bar{g} \equiv 0$ and $\sigma_2(JV) = \operatorname{tr}\big(\operatorname{adj}(V)\operatorname{adj}(J)\big) = \mu\,l^T(\operatorname{adj} J)\,\bar{l} \equiv -\mu\Delta$, with l primitive at the primes dividing $\mathfrak{d}$. Hence

$$\chi \equiv \operatorname{tr}\big((JV)^2\big) = \operatorname{tr}(JV)^2 - 2\sigma_2(JV) \equiv 2\mu\Delta \pmod{\mathfrak{d}}.$$

Now let $\chi = 6$ and $N_w = 4$. Then $1 \leq m \leq 8$, because Δ is totally negative, and m satisfies three conditions.

(a) $4m \equiv 36$ modulo $\mathfrak{d} \cap \mathbb{Z}$, which is $D\mathbb{Z}$ if $D \equiv 1$ modulo 4 and $2D\mathbb{Z}$ otherwise. As D is squarefree, this says that $m \equiv 9$ modulo the odd part D_o of D.

(b) $m \equiv 0$ or 1 modulo 4. Take l primitive at 2, so that $\mu \in \mathbb{Z}_{(2)}$. Since $\mathrm{N}(x+4y) \equiv \mathrm{N}(x)$ modulo 4 for $x, y \in \mathcal{O} \otimes \mathbb{Z}_{(2)}$, $\mathrm{N}(\Delta) \equiv \mathrm{N}(l_1^2) = \mathrm{N}(l_1)^2$, and $m \equiv \big(\mu\mathrm{N}(l_1)\big)^2$ modulo 4.

(c) If $D \equiv 2$ modulo 4, then m is odd. The prime $\mathfrak{p}$ above 2 is ramified, $v_\mathfrak{p}(2) = 2$ and $v_\mathfrak{p}(\sqrt{D}) = 1$, so $v_\mathfrak{p}(\mathfrak{d}) = 3$ while $v_\mathfrak{p}(6) = 2$; the congruence $6 \equiv 2\mu\Delta$ modulo $\mathfrak{p}^3$ forces $v_\mathfrak{p}(\mu\Delta) = 0$, so μ and $\mathrm{N}(\Delta)$ are odd.

If $D \equiv 1$ modulo 4 and $D \neq 5$, then $D \geq 13$ and (a) has no solution in $[1, 8]$. If $D \equiv 3$ modulo 4, (a) gives $m \in \{3, 6\}$ for $D = 3$, $m = 2$ for $D = 7$ and nothing for $D \geq 11$, and (b) excludes these values. If $D = 2D'$, then (b) and (c) give $m \equiv 1$ modulo 4, which with (a) says $m \equiv 9$ modulo $4D'$, and this has no solution in $[1, 8]$ unless $D' = 1$. There remain $D = 5$, with $m = 4$, and $D = 2$, with $m \in \{1, 5\}$.

For $\mathbb{Q}(\sqrt{5})$, $m = 4$ means that Δ is -2 times a unit or minus the square of a unit, since 2 is inert; and $\Delta \equiv l_1^2$ modulo 4, while neither -1 nor twice a unit is a square modulo 4 in $\mathbb{Z}[\varphi]$. For $\mathbb{Q}(\sqrt{2})$, (c) makes μ odd, so $\mu = \pm 1$ and $\mathrm{N}(\Delta) \in \{1, 5\}$; the norms of the elements of $\mathbb{Z}[\sqrt{2}]$ prime to 2 are ± 1 modulo 8, so $\mathrm{N}(\Delta) = 1$ and $\Delta = -(1 + \sqrt{2})^{2k}$, which is 3 or $1 + 2\sqrt{2}$ modulo 4, and neither is a square modulo 4 in $\mathbb{Z}[\sqrt{2}]$. ■

Theorem 5.22 (No dimension-count certificate for quartic CM fields)**.** *Let $F = F_0(\sqrt{-q})$ be a quartic CM field with real quadratic subfield F_0, let X be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and give $A = X \times \widehat{X}$ the action of F of* [Mar25c, §11.1]*, transported by the B-field e^{θ_t} for a rational $t \in F_0$. Let F_1, F_2 be perfect complexes on X with $\mathrm{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves, whose Chern characters lie in spaces $S(t', q')$ of Proposition 5.19, and let $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \mathrm{ch}(E)e^{\ell/2}$. Put $v_1 = \mathrm{ch}(F_1)$ and $v_2 = \mathrm{ch}(F_2^\vee)$.*

(i) *$r(\kappa) = r(\mathrm{ch}\, E) = r^2(v_1) + 64 + r^2(v_2)$, where $r^2(v_i) \in \{12, 18, 20\}$ is given by Proposition 5.19(ii); so $r(\kappa) \leq 104$, and $r(\kappa) \in \{100, 102, 104\}$ unless F is biquadratic and some v_i has $N_w = 2$.*

(ii) *$\dim \mathrm{Ext}^2(E, E) > r(\kappa)$. So the injectivity of σ_E cannot be obtained from a dimension count for any such E.*

(iii) *If $t = 0$ and both Chern characters lie in $S(0, q)$, then κ is invariant under G_F, and so is a Hodge class on every member of the $(F, 2, 1)$-family. This is the case $[F : \mathbb{Q}] = 4$ of* [Mar25c, Proposition 10.2.1(1)] *and of the inclusion stated before* [Mar25c, Lemma 10.1.3]*, with other conventions for Φ and for the sign of the twist.*

Proof. (i) By Lemma 4.55(i), multiplication by $e^{\ell/2}$ conjugates contraction by an automorphism of HT^2, so $r(\kappa) = r(\mathrm{ch}\, E)$; and, as in the proof of Theorem 4.56(iv), Φ identifies $r(\mathrm{ch}\, E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$. Write r^k for the rank of contraction from $HT^k(X)$. The block $HT^1 \otimes HT^1$ maps into $H^{\mathrm{odd}} \otimes H^{\mathrm{odd}}$ by the tensor product of the two contractions from HT^1, of rank $r^1(v_1)r^1(v_2) = 64$, and the other two blocks map into $H^{\mathrm{ev}} \otimes H^{\mathrm{ev}}$. If $(x_1 \lrcorner v_1) \otimes v_2 + v_1 \otimes (x_2 \lrcorner v_2) = 0$, then $x_1 \lrcorner v_1 = \lambda v_1$ for a scalar λ, and $\lambda = 0$ because contraction from HT^2 raises $q - p$ by two on classes of type (p, q) while $v_1 \neq 0$ is of type (p, p); so those two blocks contribute $r^2(v_1) + r^2(v_2)$, and $r^2(v_i) = 2\rho + 4N_w \in \{12, 18, 20\}$ by Proposition 5.19(ii), with 12 only for biquadratic F.

(ii) By the Künneth formula, as in Proposition 3.20, $\dim \mathrm{Ext}^2(E, E) = e_0e_2' + e_1e_1' + e_2e_0'$ with $e_k = \dim \mathrm{Ext}^k(F_1, F_1) \geq r^k(v_1)$, by the commutative square $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner \mathrm{ch}(F)$, and likewise for F_2. So $\dim \mathrm{Ext}^2(E, E) \geq r(\kappa)$, with equality only if each F_i is minimal, $\dim \mathrm{Ext}^k(F_i, F_i) = 1, 8, r^2(\mathrm{ch}\, F_i)$ for $k = 0, 1, 2$. For such F_i, Serre duality gives $\chi(F_i, F_i) = r^2 - 14$, that is 4, 6 or -2 for $(\rho, N_w) = (1, 4)$, $(2, 4)$, $(2, 2)$. The last is impossible because χ is positive on $S(t', q')$. The Chern character of a perfect complex on a complex torus is integral, since by the Künneth theorem the topological K-theory of a real torus is

generated by products of classes pulled back from its circle factors, whose Chern characters are integral. So the first two values are excluded by Proposition 5.21.

(iii) Over $\mathbb{R}$, $H^1(X,\mathbb{R}) = U_1 \oplus U_2$, with U_j the eigenspace of F_0 for τ_j, and the U_j are orthogonal for θ. As in the proof of Proposition 3.21, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z))e^{\ell/2}$, the Poincaré class $\ell = \ell_1 + \ell_2$, the classes η and γ and the exponentials e_ε all decompose as tensor products over j of classes of even degree, so that the image of $e_\varepsilon \boxtimes e^\vee_{\varepsilon'}$ is the tensor product over j of the images of $e^{\varepsilon_j s_j \theta_j} \boxtimes e^{-\varepsilon'_j s_j \theta_j}$ in the j-th factor. On that factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of Proposition 3.21 at $n = 2$, which are identities in the exterior algebra of a symplectic space with d a symbol, show that each image is a multiple of $e^{\pm(s_j/2\tau_j(q))\eta_j}$ or of a Weil class $(\gamma_j \mp s_j\ell_j)^2$ of the j-th factor. These are invariant under the special unitary group SU_j of the j-th factor, and $G_F(\mathbb{R}) = SU_1 \times SU_2$ acts factor by factor; so the rational class κ is invariant under $G_F(\mathbb{R})$, which is Zariski dense in G_F, and the Hodge group of every member of the family lies in G_F. ∎

Remark 5.23 (The shape of the secant characters). Let $t = 0$ in Theorem 5.22(iii), and write $\eta = \eta_1 + \eta_2$, $\gamma = \gamma_1 + \gamma_2$ and $\ell = \ell_1 + \ell_2$ for the decompositions over the real places used in its proof: $\eta_j = \tau_j(q)\theta_j + \widehat{\theta}_j$ and $\gamma_j = \tau_j(q)\theta_j - \widehat{\theta}_j$, with θ_j and $\widehat{\theta}_j$ here the pull-backs to A of the components of the polarisations of X and $\widehat{X}$. The classes η_t of A are the combinations of η_1 and η_2, and the Weil classes at the two embeddings over τ_j are the multiples of $(\gamma_j \mp s_j\ell_j)^2$. Let $\kappa_{(a,b)}$ be the component of κ of degree a at τ_1 and b at τ_2. Although κ is flat, it does not lie in $\mathbb{C}[\eta_1,\eta_2] + W_F \otimes \mathbb{C}$ once κ_4 has a nonzero Weil part, unlike the classes of Proposition 3.21. Indeed, by the proof of Theorem 5.22(iii), κ is a sum of products X_1X_2 in which each X_j is a multiple of $e^{\pm c_j\eta_j}$, $c_j = s_j/(2\tau_j(q))$, or of $(\gamma_j \mp s_j\ell_j)^2$. The components $\kappa_{(4,0)}$ and $\kappa_{(4,8)}$ receive exactly the terms whose second factor is an exponential, through its parts of degree 0 and 8, and $c_2^4 = 1/(16\tau_2(q)^2)$ for both signs; so

$$\kappa_{(4,8)} = \frac{\kappa_{(4,0)}\,\eta_2^4}{384\,\tau_2(q)^2},$$

and symmetrically. The polynomials in η_1, η_2 of this bidegree are the multiples of $\eta_1^2\eta_2^4$ and W_F has no component there, while $\kappa_{(4,0)}$ has a nonzero Weil part as soon as κ_4 has one, since a nonzero rational Weil class has nonzero components at all four embeddings. The class $\mathrm{ch}(E) = \kappa e^{-\ell/2}$ itself is not flat when $\mathrm{rk}(E) \neq 0$: for Y in the Lie algebra of G_F, $D_Y \mathrm{ch}(E) = -\frac{1}{2}(D_Y\ell)\,\mathrm{ch}(E)$, whose part of degree two is $-\frac{1}{2}\mathrm{rk}(E)\,D_Y\ell$, and ℓ is not flat. So these objects, when their rank is not zero, are candidates only for a twisted form of Theorem 5.13(v), with hypothesis a flat twisted character κ with $\kappa_4 = N\omega + P_4$, which needs the extension of Theorem 2.2 to twisted objects [Per26] and the extraction of untwisted cycles of [Mar25b, Section 2], as after Theorem 3.19. This is the form in which Markman states his reduction: he uses the normalised class $\mathrm{ch}(E)\exp(-c_1(E)/\mathrm{rk}\,E)$, which is $\kappa e^{-\kappa_1/\kappa_0}$ when $\mathrm{rk}\,E \neq 0$, asserts the shape of a polynomial plus a nonzero Weil class only for its component in H^4, and takes as propagation hypothesis that this component stays algebraic [Mar25c, Proposition 10.2.1, Corollary 10.2.3].

Remark 5.24 (Negative Ext groups). The hypothesis $\mathrm{Ext}^{<0}(F_i,F_i) = 0$ in Theorem 5.22(ii) can be replaced by a condition on the amplitude. Let F_1, F_2 be any perfect complexes with characters in spaces $S(t',q')$, let $e_k = \dim \mathrm{Ext}^k(F_1,F_1)$ and $e'_k = \dim \mathrm{Ext}^k(F_2,F_2)$, and put $\nu_i = \sum_{j\geq 3}(-1)^j \dim \mathrm{Ext}^{-j}(F_i,F_i)$. Then still $\dim \mathrm{Ext}^2(E,E) \geq r(\kappa)$, and equality forces, for each i, $\dim \mathrm{Ext}^k(F_i,F_i) = 1, 8, r^2(v_i)$ for $k = 0,1,2$, $\mathrm{Ext}^{-1}(F_i,F_i) = \mathrm{Ext}^{-2}(F_i,F_i) = 0$

and $\nu_i = (\chi(v_i, v_i) - r^2(v_i) + 14)/2$. This gives $\nu_i \geq 2$ for $(\rho, N_w) = (1, 4)$ and $(2, 2)$, and $\nu_i \geq 1$ for $(2, 4)$ unless $F_0 = \mathbb{Q}(\sqrt{3})$ and $\chi(v_i, v_i) = 4$, where $\nu_i = -1$, a case that needs integral classes of that kind. So equality needs, for both i, some $\operatorname{Ext}^{-j}(F_i, F_i) \neq 0$ with $j \geq 4$ even, or $j \geq 3$ odd in the exceptional case, and it fails as soon as one F_i has its cohomology sheaves in an interval of length at most 3, at most 2 in the exceptional case. These conditions are necessary, not sufficient. The strict inequality $\dim \operatorname{Ext}^2(E', E') > r(\operatorname{ch} E')$ passes to the images E' of such an E under derived equivalences, push-forward and pull-back along isogenies, and twists by line bundles or B-fields.

Indeed, $\dim \operatorname{Ext}^2(E, E) = \sum_{a \in \mathbb{Z}} e_a e'_{2-a}$ by the Künneth formula, and $e_a = e_{4-a}$ by Serre duality. The terms $a = 0, 1, 2$ give at least $r(\kappa)$, with equality only if e_k and e'_k are the ranks r^k for $k \leq 2$, as in the proof of Theorem 5.22(ii); the terms $a = 3, 4, -1, -2$ are $e_1 e'_{-1}$, $e_0 e'_{-2}$, $e_{-1} e'_1$ and $e_{-2} e'_0$, which must vanish. Then $\chi(F_i, F_i) = 2e_0 - 2e_1 + e_2 + 2\sum_{j \geq 1} (-1)^j e_{-j}$ gives the value of ν_i. The number $\chi(v, v)$ is a positive even integer: positive by Proposition 5.19(iii), and even because $\int_X x^2$ is even for $x \in H^4(X, \mathbb{Z}) = \bigwedge^4 H^1(X, \mathbb{Z})$. For $(1, 4)$, $\chi \in 4\mathbb{Z}$ by the proof of Proposition 5.21(i), and $\chi \neq 4$; for $(2, 2)$, $r^2 = 12$. For $(2, 4)$, $r^2 = 20$, and in the proof of Proposition 5.21(ii) $\chi^2 - 4m > 0$ with $m \geq 1$, so $\chi \geq 4$, and $\chi \neq 6$. If $\chi = 4$, then $m \in \{1, 2, 3\}$, and conditions (a) to (c) of that proof, with 36 replaced by 16, say that $m \equiv 4$ modulo the odd part of D, so that $D \in \{2, 3, 6\}$; that $m = 1$; and, for $D \equiv 2$ modulo 4, that m is even, because $v_{\mathfrak{p}}(4) = 4 \geq 3$ here gives $v_{\mathfrak{p}}(\mu\Delta) \geq 1$. Only $D = 3$ remains. A complex with cohomology sheaves in an interval of length l has $\operatorname{Ext}^{-j}(F, F) = 0$ for $j > l$. Equivalences preserve Ext algebras and, through the induced isomorphisms of HT^* and H^*, ranks of contraction ([CRV12], as in the proof of Theorem 4.56(iv)); for an isogeny g, $g^* g_* E = \bigoplus_{x \in \ker g} t_x^* E$, so $\operatorname{Ext}^2(g_* E, g_* E)$ contains $\operatorname{Ext}^2(E, E)$ while $r(\operatorname{ch} g_* E) = r(\operatorname{ch} E)$, and likewise for g^*; twists are Lemma 4.55(i).

What Theorem 5.22 and Remark 5.24 leave for $\mathbf{W}(F, 2, \delta)$ is one of three things: an Orlov product both of whose factors have nonzero Ext groups in some even degree at most -4 (or odd degree at most -3 when $F_0 = \mathbb{Q}(\sqrt{3})$); a kernel on $X \times X$ that is not an external product; or a proof of the injectivity of σ_E for some Orlov product that does not go through a dimension count, which is what Markman's question [Mar25c, Question 11.2.2] asks for. The next proposition shows that a kernel whose transform has a pure spinor as its character, for instance a line bundle on the graph of a homomorphism, gives no Weil class.

Proposition 5.25 (Pure spinors give no Weil class)**.** *Keep the notation of Remark 5.23 on $A = X \times \widehat{X}$ with $t = 0$, let $V_j \subseteq H^1(A, \mathbb{R})$ be the part at τ_j, and call a class flat if it is invariant under $G_F(\mathbb{R})$. Let $v = c e^{\beta} \Omega_N$ be a real pure spinor, with $c \in \mathbb{R}^\times$, β a real 2-form, $N \subseteq H^1(A, \mathbb{R})$ and Ω_N a generator of $\det N$. If v is flat, then N is one of 0, V_1, V_2, $H^1(A, \mathbb{R})$, and $v \in \mathbb{R}[\eta_1, \eta_2]$. This applies to the characters of line bundles on translates of abelian subvarieties, of points and of simple semi-homogeneous bundles, and of their images under derived equivalences and isogenies, twisted by any real B-field; for instance to Φ of a line bundle on the graph of a homomorphism $X \to X \times X$. The component in $W_F \otimes \mathbb{R}$ of such a class vanishes in every degree: no such object has a character, twisted as stated, of the form required by Theorem 5.13(v) or by its twisted form, whatever the quartic field.*

Proof. Over $\mathbb{R}$, $G_F(\mathbb{R}) = SU_1 \times SU_2$ with SU_j acting on V_j and trivially on the other part, and $V_j \otimes \mathbb{C}$ is the sum of the standard representation of $SU_j \otimes \mathbb{C} \cong \mathrm{SL}_4$ and its dual, which are not isomorphic (proof of Proposition 5.5(i)). So V_1 and V_2 are irreducible and not isomorphic, and 0, V_1, V_2, $H^1(A, \mathbb{R})$ are the only stable subspaces. The flat classes are

$\mathrm{Inv}_1 \otimes \mathrm{Inv}_2$, where by the same proof $\mathrm{Inv}_j \subseteq \bigwedge^* V_j$ is spanned by the powers of η_j and the two Weil classes $(\gamma_j \mp s_j \ell_j)^2$; in degree two it is $\mathbb{R}\eta_j$, and η_j^4 spans $\bigwedge^8 V_j$.

If v is real and pure over $\mathbb{C}$, then $N = \{\xi : \xi \wedge v = 0\}$ is defined over $\mathbb{R}$, and so are c and the class of β modulo $N \wedge H^1$. For $g \in G_F(\mathbb{R})$, $gv = v$ gives $gN = N$, and $g\Omega_N = \det(g|_N)\Omega_N = \Omega_N$ because the connected semisimple group $G_F(\mathbb{R})$ has no nontrivial character. Hence $(e^{g\beta-\beta} - 1)\Omega_N = 0$, whose part of lowest degree gives $g\beta - \beta \in N \wedge H^1$. So N is one of the four stable subspaces, and the class of β in $\bigwedge^2(H^1/N)$ is invariant; by complete reducibility it is the image of an invariant 2-form $a_1\eta_1 + a_2\eta_2$, and since $e^{n\wedge x}\Omega_N = \Omega_N$ for $n \in N$, $v = c\, e^{a_1\eta_1 + a_2\eta_2}\Omega_N$, with Ω_N a multiple of 1, η_1^4, η_2^4 or $\eta_1^4\eta_2^4$. The characters listed are pure spinors, and derived equivalences, isogenies and twists preserve purity; this is the classical description recalled in Setup 3.26 and after Corollary 3.17. ∎

Markman's candidate for a quartic field. Markman gives one explicit pair of sheaves for a biquadratic field [Mar25c, Example 11.2.7, Lemma 11.2.8] and asks whether the second can be chosen so that its semiregularity map is injective on the image of the evaluation map [Mar25c, Question 11.2.2]. For one object on the fourfold this is the weakened criterion of Setup 4.54, and the local method of Theorem 4.61 decides it for his pair: both sheaves fail it, at points of the curves that the construction glues in. Two local computations are needed, one for a sheaf that is locally free on a smooth support and one for the ideal of a curve in a smooth divisor. As in Lemma 4.60, for a germ of vector field θ and a perfect complex F of modules over a local ring we write $a_\theta \in \mathrm{Ext}^1(F, F)$ for the class of the jet extension along θ.

Lemma 5.26 (Free germs on smooth supports)**.** *Let $\mathcal{O}$ be the local ring of a smooth complex variety Y at a point p, or its completion, let $C \ni p$ be a smooth subvariety of codimension $c \geq 1$, put $N = T_pY/T_pC$, and let F be a perfect complex of $\mathcal{O}$-modules quasi-isomorphic to a shift of $\mathcal{O}_{C,p}^{\oplus r}$, $r \geq 1$. Then $\mathrm{Ext}^2_{\mathcal{O}}(F, F)$ is a free $\mathcal{O}_{C,p}$-module isomorphic to $M_r(\mathcal{O}_{C,p}) \otimes \bigwedge^2 N$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a sign that does not depend on θ, θ', the element $\mathrm{id} \otimes (\bar\theta \wedge \bar\theta')$, where $\bar\theta, \bar\theta'$ are the images of $\theta(p), \theta'(p)$ in N. In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p)$, $\theta'(p)$ and T_pC span a space of dimension $\dim C + 2$.*

Proof. Choose formal coordinates $x_1, \ldots, x_m$ at p with $C = \{x_1 = \cdots = x_c = 0\}$. Ext groups of finitely generated modules and jet extensions commute with the flat base change from $R = \mathbb{C}[x_1, \ldots, x_m]$ to $\mathcal{O}$, and a shift changes neither the Ext algebra nor the jet classes, so we may take $F = (R/(x_1, \ldots, x_c))^{\oplus r}$ over R. Write $R = A_1 \otimes \cdots \otimes A_c \otimes B$ with $A_i = \mathbb{C}[x_i]$ and $B = \mathbb{C}[x_{c+1}, \ldots, x_m]$; then $F = k_1 \otimes \cdots \otimes k_c \otimes B^{\oplus r}$, with k_i the residue field of A_i at the origin. On the resolution $A_i \xrightarrow{x_i} A_i$ of k_i the jet chain map of ∂_i is $-\partial_i(x_i) = -1$, as in the proof of Lemma 4.60, so its class γ_i spans $\mathrm{Ext}^1_{A_i}(k_i, k_i) = k_i$, and $\mathrm{Ext}^j_{A_i}(k_i, k_i) = 0$ for $j \geq 2$. The module $B^{\oplus r}$ is free, so its Ext algebra is $M_r(B)$ in degree zero and its jet classes vanish. By (20), applied factor by factor, $\mathrm{Ext}^*_R(F, F)$ is the graded tensor product of the exterior algebra on $\gamma_1, \ldots, \gamma_c$ with $M_r(B)$, and $a_{\partial_i} = \gamma_i \otimes \mathrm{id}$ for $i \leq c$, while $a_{\partial_i} = 0$ for $i > c$. Hence $\mathrm{Ext}^2_R(F, F) = \bigwedge^2\langle\gamma_1, \ldots, \gamma_c\rangle \otimes M_r(B)$, and $B = \mathcal{O}_C$ in these coordinates. For $\theta = \sum_i f^i\partial_i$ and $\theta' = \sum_i f'^i\partial_i$ the linearity of Lemma 4.59 gives

$$a_\theta a_{\theta'} = \sum_{i,j\leq c} f^i f'^j\, \gamma_i\gamma_j \otimes \mathrm{id} = \sum_{i<j\leq c} \left(f^i f'^j - f^j f'^i\right) \gamma_i\gamma_j \otimes \mathrm{id},$$

because the γ_i anticommute and square to zero. The functions $x_1, \ldots, x_c$ act by zero, so modulo the maximal ideal of $\mathcal{O}_C$ the coefficients are those of $\theta(p) \wedge \theta'(p)$ on $\partial_i \wedge \partial_j$,

$i < j \leq c$, which are the coordinates of $\bar{\theta} \wedge \bar{\theta}'$ in $\bigwedge^2 N$. A section of a free module with nonzero reduction is nonzero. ∎

Lemma 5.27 (A curve in a divisor)**.** *Let $\mathcal{O}$, Y and p be as in Lemma 5.26, let $D \ni p$ be a smooth divisor, let $C \subset D$ be a smooth subvariety of codimension two in D through p, and let F be quasi-isomorphic to a shift of the ideal $I_{C\subset D}$ of C in $\mathcal{O}_D$, regarded as an $\mathcal{O}$-module. Put $\nu = T_pY/T_pD$ and $N' = T_pD/T_pC$. Then $\operatorname{Ext}^2_{\mathcal{O}}(F,F) \cong \mathcal{O}_{C,p} \otimes \nu \otimes N'$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a fixed sign, the image of $\theta(p) \wedge \theta'(p)$ under*

$$\textstyle\bigwedge^2 T_pY \longrightarrow \bigwedge^2(T_pY/T_pC) \longrightarrow \bigwedge^2(T_pY/T_pC)/\bigwedge^2 N' \cong \nu \otimes N'.$$

In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ does not lie in $\bigwedge^2 T_pD + T_pC \wedge T_pY$.

Proof. With formal coordinates in which $D = \{x_1 = 0\}$ and $C = \{x_1 = x_2 = x_3 = 0\}$, and the same reduction to the polynomial ring, write $R = \mathbb{C}[x_1] \otimes \mathbb{C}[x_2, x_3] \otimes B$ with $B = \mathbb{C}[x_4, \ldots, x_m]$. Then $I_{C\subset D} = k \otimes \mathfrak{m} \otimes B$, where k is the residue field of $\mathbb{C}[x_1]$ and $\mathfrak{m} = (x_2, x_3)$. As in the proof of Lemma 4.60, $\mathfrak{m}$ has projective dimension one, $\operatorname{Ext}^1(\mathfrak{m},\mathfrak{m}) = \mathfrak{m}/\mathfrak{m}^2$ has the basis $\alpha_2 = a_{\partial_2}$, $\alpha_3 = a_{\partial_3}$ and is killed by $\mathfrak{m}$, and $\operatorname{Ext}^2(\mathfrak{m},\mathfrak{m}) = 0$; the factor k contributes γ_1 in degree one, as in Lemma 5.26, and B contributes nothing in positive degree. By (20)

$$\operatorname{Ext}^2_R(I_{C\subset D}, I_{C\subset D}) = \operatorname{Ext}^1(k,k) \otimes \operatorname{Ext}^1(\mathfrak{m},\mathfrak{m}) \otimes B = \langle \gamma_1 \rangle \otimes \langle \alpha_2, \alpha_3 \rangle \otimes B,$$

the other summands vanishing by the projective dimensions, and $a_{\partial_1} a_{\partial_j} = -a_{\partial_j} a_{\partial_1} = \pm \gamma_1 \otimes \alpha_j$ for $j = 2, 3$, while $a_{\partial_2} a_{\partial_3} = 1 \otimes \alpha_2\alpha_3 = 0$, $a^2_{\partial_1} = 0$ and $a_{\partial_i} = 0$ for $i \geq 4$. As before, $a_\theta a_{\theta'}$ reduces to the component of $\theta(p) \wedge \theta'(p)$ on $\partial_1 \wedge \langle \partial_2, \partial_3 \rangle$. In these coordinates ∂_1 spans ν, the images of ∂_2, ∂_3 span N', and the component is the image in $\bigwedge^2(T_pY/T_pC)/\bigwedge^2 N'$. Its kernel on $\bigwedge^2 T_pY$ is spanned by $\partial_2 \wedge \partial_3$ and the $\partial_i \wedge \partial_j$ with $j \geq 4$, which is $\bigwedge^2 T_pD + T_pC \wedge T_pY$. ∎

Now keep the notation of Proposition 5.19, with X an abelian fourfold with real multiplication by F_0 and $\theta = \theta_1 + \theta_2$ a polarisation in $\bigwedge^2_{F_0} H^1(X, \mathbb{Q})$. Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with U_j the eigenspace of F_0 at τ_j, and the tangent space splits accordingly as $T = T_1 \oplus T_2$, with T_j of dimension two, dual to $U_j \cap H^{1,0}$; translation identifies every T_pX with T. Let π_j span $\bigwedge^2 T_j$ and put

$$x_j = \left(\tfrac{1}{2}\tau_j(q)\, \pi_j \lrcorner\, \theta_j^2,\ 0,\ \pi_j\right) \in HT^2(X), \qquad j = 1, 2. \tag{31}$$

The six classes $(0, v, 0)$ with $v \in \operatorname{Hom}(H^{1,0}, H^{0,1})$ preserving the decomposition by places and $v \lrcorner \theta_1 = v \lrcorner \theta_2 = 0$ are the first order deformations of X with its real multiplication and polarisation; we call them the polarised F_0-linear deformations.

Theorem 5.28 (The local obstruction for a quartic secant character)**.** *In this notation, let $q \in F_0$ be totally positive and $t \in F_0$.*

(i) *$x_j \lrcorner w = 0$ for every $w \in S(0, q)$, and $(e^{-\theta_t} x_j) \lrcorner w = 0$ for every $w \in S(t, q)$, $j = 1, 2$. If $w \in S(t, q)$ has $(\rho, N_w) = (2, 4)$, then $\operatorname{Ann}_{HT^2}(w)$ is spanned by $e^{-\theta_t}x_1$, $e^{-\theta_t}x_2$ and the six polarised F_0-linear deformations, and its bivector components span exactly $\langle \pi_1, \pi_2 \rangle$.*

(ii) *Let F be a perfect complex on X with $\operatorname{ch}(F) \in S(t, q)$, and let θ, θ' be translation invariant vector fields with $\theta \wedge \theta' = \pi_j$. If $a_\theta a_{\theta'} \neq 0$ in $\operatorname{Ext}^2(F_p, F_p)$ at some point p, then $\operatorname{ev}_F(e^{-\theta_t} x_j) \neq 0$ and F does not satisfy the weakened criterion. Neither do*

$\Phi(F \boxtimes G)$ *and* $\Phi(G \boxtimes F)$ *on* $X \times \widehat{X}$*, for any nonzero perfect complex* G *on* X*, with* Φ *as in Theorem 4.56(iv).*

(iii) *The hypothesis of* (ii) *holds at* p*, for* $j = 1$ *or for* $j = 2$*, whenever* F_p *is a shift of a free module of positive rank over the local ring of a smooth curve through* p*, since* T_pC *lies in at most one of* T_1, T_2*; it holds for both* j *when* F_p *is a shift of a free module over the residue field. It holds for* j *when* F_p *is a shift of the ideal of a smooth curve* C *in a smooth divisor* D *with* $T_pC \not\subseteq T_j$ *and* $T_j \not\subseteq T_pD$*.*

Proof. (i) Let $t = 0$. By the proof of Lemma 4.55(i), for a bivector π of type $(2,0)$ in the tangent directions and a class B of type $(1,1)$, $\pi \lrcorner e^B = \frac{1}{2}(\pi \lrcorner B^2)\, e^B$. The interior products by vectors of T_j are odd derivations that vanish on $\bigwedge^* U_{j'}$ for $j' \neq j$, so $\pi_j \lrcorner (ab) = (\pi_j \lrcorner a)\, b$ for a in the subalgebra generated by U_j and b an even class generated by $U_{j'}$. Applied to $e_\varepsilon = e^{\varepsilon_1 s_1 \theta_1} e^{\varepsilon_2 s_2 \theta_2}$, with $s_j^2 = -\tau_j(q)$, this gives $\pi_j \lrcorner e_\varepsilon = -\frac{1}{2}\tau_j(q)(\pi_j \lrcorner \theta_j^2)\, e_\varepsilon$ for all four signs, and so $x_j \lrcorner e_\varepsilon = 0$. A class $(0, v, 0)$ with v preserving the places and killing θ_1 and θ_2 acts as a derivation that kills every polynomial in θ_1, θ_2, hence all of $S(0,q)$. For general t, Lemma 4.55(i) gives $y \lrcorner (we^{\theta_t}) = ((e^{\theta_t} y) \lrcorner w) e^{\theta_t}$, so $e^{-\theta_t}$ carries $\mathrm{Ann}_{HT^2}(w)$ onto $\mathrm{Ann}_{HT^2}(we^{\theta_t})$; it fixes the six classes $(0, v, 0)$, because $v \lrcorner \theta_t = 0$, and by (16) it does not change bivector components. The eight classes are independent: the six have no bivector component and are independent, and the other two have the independent components π_1, π_2. When $(\rho, N_w) = (2, 4)$ the rank of contraction from $HT^2(X)$, of dimension 28, is $2\rho + 4N_w = 20$ by Proposition 5.19(ii), so the annihilator has dimension 8 and the eight classes span it.

(ii) By (i), $y = e^{-\theta_t} x_j = (z, v, \pi_j)$ lies in $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$, and by (15) $\mathrm{ev}_F(y) = z \cdot \mathrm{id}_F + v \lrcorner \mathrm{At}(F) + \frac{1}{2}\pi_j \lrcorner \mathrm{At}(F)^2$. The last term is $A_\theta A_{\theta'}$, as in the proof of Theorem 4.61, and its germ at p is $a_\theta a_{\theta'} \neq 0$ by Lemma 4.59. The germs of the first two terms vanish: $z \cdot \mathrm{id}_F$ by Lemma 4.59, and $v \lrcorner \mathrm{At}(F)$ because it is a composite $F \to F \otimes \Omega^1_X[1] \to F \otimes \Omega^1_X \otimes T_X[2] \to F[2]$ in which the middle arrow is $\mathrm{id} \otimes v$ with $v \in \mathrm{Ext}^1(\mathcal{O}_X, T_X) = H^1(T_X)$, and v restricts to zero on every affine open set. So $\mathrm{ev}_F(y) \neq 0$ with $y \lrcorner \mathrm{ch}(F) = 0$, and F fails the weakened criterion. For $G \boxtimes F$ and $F \boxtimes G$ the class $y \otimes 1$, resp. $1 \otimes y$, preserves the Chern character, and its evaluation $\pm\, \mathrm{ev}_F(y) \otimes \mathrm{id}_G$ is nonzero in its Künneth summand, as in the proof of Theorem 4.56(iv); the transport under Φ is the one given there.

(iii) Since T_j is a plane, the image of π_j in $\bigwedge^2(T/T_pC)$ is nonzero exactly when $T_j \cap T_pC = 0$. For a curve this says $T_pC \not\subseteq T_j$, which fails for at most one j because $T_1 \cap T_2 = 0$; for a point it always holds. Lemma 5.26 then gives $a_\theta a_{\theta'} \neq 0$. In the last case the image of π_j in $\bigwedge^2(T/T_pC)$ is the nonzero decomposable vector of the plane $(T_j + T_pC)/T_pC$, and it lies in $\bigwedge^2 N'$ only if that plane is N', that is, only if $T_j + T_pC = T_pD$, which would put T_j inside T_pD. So Lemma 5.27 applies. ∎

At a point where F is locally free both sides of the local identity vanish, as at a smooth point of a surface in Proposition 4.65; the test sees only where the sheaf drops dimension or fails to be locally free on its support. Markman's construction produces exactly such points.

Proposition 5.29 (The classes of Markman's candidate)**.** *Let* X *be the Jacobian of a curve of genus four with its principal polarisation* Θ*, with real multiplication by a real quadratic field* F_0 *and an automorphism* g *with* $g^* = \hat{\eta}(f)$ *on* $H^1(X, \mathbb{Q})$*, where* $f \in F_0$ *has norm one and* $f^2 \neq 1$*; let* $q > 0$ *be rational, so that* $F = F_0(\sqrt{-q})$ *is biquadratic, and write* $f_j = \tau_j(f)$*. Let* $\alpha_0 = \Theta - \frac{q}{6}\Theta^3$*, the character of the sheaf* $E_0 = F_q$ *of* [Mar25a,

Example 8.2.4], *and* $\beta' = g^*\Theta - \frac{q}{6}(g^{-1})^*\Theta^3$, *a multiple of which is the character of the sheaf* E' *of* [Mar25c, Example 11.2.7]. *With* $A_j = 1 - \frac{q}{2}\theta_j^2$ *and* $\theta = \Theta$*:*

(i) $\alpha_0 = \theta_1 A_2 + \theta_2 A_1$ *and* $\beta' = f_1^2\theta_1 A_2 + f_2^2\theta_2 A_1$*, both in* $S(0,q)$*. With* $s = \sqrt{-q}$*, the coefficients of* β' *on the four* e_ε *are* $w_{++} = -w_{--} = (f_1^2 + f_2^2)/4s$ *and* $w_{+-} = -w_{-+} = (f_1^2 - f_2^2)/4s$*, so* β' *has* $(\rho, N_w) = (2,4)$*, while* α_0 *lies on the secant plane of* $\mathbb{Q}(\sqrt{-q})$ *and has* $(\rho, N_w) = (2,2)$*.*

(ii) *The ranks of contraction from* $HT^2(X)$ *are* 12 *for* α_0 *and* 20 *for* β'*. The annihilator of* β' *is that of Theorem 5.28(i); that of* α_0 *contains, besides it, the classes* $(\frac{q}{2}\,\pi\lrcorner\,\Theta^2, 0, \pi)$ *for every bivector* π*, by Theorem 4.56(ii) with* $t = \Theta$ *and* $d = q$*.*

(iii) $\int_X \alpha_0\beta' = -4q\,\mathrm{Tr}_{F_0/\mathbb{Q}}(f^2) \neq 0$*, which is Markman's formula for the rank of* $\Phi(E_0 \boxtimes E')$ [Mar25c, Lemma 11.2.8]*;* $\chi(\beta',\beta') = 4q\,\mathrm{Tr}_{F_0/\mathbb{Q}}(f^4)$ *and* $\chi(\alpha_0,\alpha_0) = 8q$*; and the rank of contraction from* $HT^2(X\times X)$ *into* $\alpha_0\boxtimes\beta'$ *is* $12+64+20 = 96$*, as Theorem 5.22(i) predicts.*

Proof. (i) The automorphism g acts on U_j by f_j, hence on θ_j by f_j^2, and $f_1f_2 = 1$. Since $\theta_j^3 = 0$, $\Theta^3 = 3\theta_1^2\theta_2 + 3\theta_1\theta_2^2$ and $(g^{-1})^*\Theta^3 = 3f_1^{-4}f_2^{-2}\theta_1^2\theta_2 + 3f_1^{-2}f_2^{-4}\theta_1\theta_2^2 = 3f_2^2\theta_1^2\theta_2 + 3f_1^2\theta_1\theta_2^2$, which gives the two expressions. With $t = 0$ and $e_\varepsilon = A_1A_2 - q\varepsilon_1\varepsilon_2\theta_1\theta_2 + s(\varepsilon_1\theta_1A_2 + \varepsilon_2\theta_2A_1)$, the class $\sum_\varepsilon w_\varepsilon e_\varepsilon$ equals β' exactly when $\sum w_\varepsilon = \sum \varepsilon_1\varepsilon_2 w_\varepsilon = 0$, $s\sum\varepsilon_1 w_\varepsilon = f_1^2$ and $s\sum\varepsilon_2 w_\varepsilon = f_2^2$, whose solution is the one stated. All four coefficients are nonzero because $f_1^2 = f_2^2$ with $f_1f_2 = 1$ would force $f^2 = 1$. For α_0 the same computation with $f_1 = f_2 = 1$ gives $w_{+-} = w_{-+} = 0$. The matrix V of Proposition 5.19 has rank two in both cases.

(ii) The ranks are $2\rho + 4N_w$, by Proposition 5.19(ii); $\alpha_0 = v_\Theta$ in the notation of [I, (30)], with $d = q$.

(iii) By $\int_X \theta_1^2\theta_2^2 = 4$, $\int \alpha_0\beta' = -2q(f_1^2 + f_2^2 + f_1^{-2} + f_2^{-2}) = -4q\,\mathrm{Tr}(f^2)$, which is negative; the Euler pairings are Proposition 5.19(iii) in the basis given there. The last number is Theorem 5.22(i) with $r^2(\alpha_0) = 12$ and $r^2(\beta') = 20$. ∎

Corollary 5.30 (Markman's candidate does not satisfy the weakened criterion)**.** *In Proposition 5.29, let* E' *be the sheaf of* [Mar25c, Example 11.2.7]*, obtained by gluing* N *general translates of* g^*F' *to a line bundle on a curve* C' *at the points where* C' *meets them, and assume that* C' *is reduced.*

(i) E' *does not satisfy the weakened criterion, so it does not answer* [Mar25c, Question 11.2.2].

(ii) E_0 *does not satisfy the weakened criterion.*

(iii) *For every nonzero perfect complex* G *on* X*, neither* $\Phi(E_0\boxtimes G)$ *nor* $\Phi(G\boxtimes E')$ *nor* $\Phi(G\boxtimes E'^\vee)$ *satisfies it; in particular Markman's object* $\Phi(E_0\boxtimes E')$ *does not.*

Proof. (i) Let p be a smooth point of C' that lies on none of the translates. Near p the sheaf E' is the line bundle on C', so $E'_p \cong \mathcal{O}_{C',p}$, and $\mathrm{ch}(E')$ is a multiple of $\beta' \in S(0,q)$. Theorem 5.28(ii) and (iii) apply with the j for which $T_pC' \not\subseteq T_j$.

(ii) In [Mar25a, Example 8.2.4] the sheaf $E_0 = F_q$ is the push-forward, from the theta divisor $\Theta \subset X$, of the ideal twisted by $\mathcal{O}(\Theta)$ of the union of a translate W of the surface W_2 and q translates $C_1,\dots,C_q$ of the curve W_1, each meeting W in one point, and its character is α_0. Let p be a point of C_1 outside W, the other C_i and the finite singular locus of Θ. Near p, E_0 is the ideal of the smooth curve C_1 in the smooth divisor Θ. Choose translation invariant fields with $\theta(p) \notin T_p\Theta$ and $\theta'(p) \in T_p\Theta \setminus T_pC_1$. Then $\theta(p)\wedge\theta'(p)$ has nonzero image in $\nu\otimes N'$, and Lemma 5.27 gives $a_\theta a_{\theta'} \neq 0$. By Proposition 5.29(ii) the class $x = (\frac{q}{2}\,\pi\lrcorner\,\Theta^2, 0, \pi)$ with $\pi = \theta\wedge\theta'$ preserves α_0, and the argument of Theorem 5.28(ii) gives $\mathrm{ev}_{E_0}(x) \neq 0$.

(iii) The first two cases are the last assertion of Theorem 5.28(ii), with the class of (ii) for E_0. For $E'^{\vee}$, the germ at p is the dual of $\mathcal{O}_{C'}$, a shift of a line bundle on C' because C' is a local complete intersection of codimension three near p, and $\operatorname{ch}(E'^{\vee}) = -\operatorname{ch}(E')$ lies in $S(0,q)$. ∎

Figure 17 draws the support of Markman's sheaf near a point of the glued curve, with the two eigenplanes of the real multiplication.

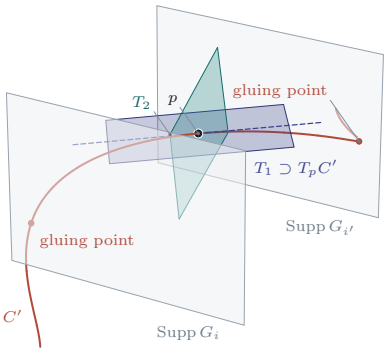
Remark 5.31 (What is left of Markman's question). The question [Mar25c, Question 11.2.2] asks for a sheaf F_2 on X whose character lies in $S(0,q)$ and forms with α_0 a pair as in [Mar25c, Lemma 11.2.1], and whose semiregularity map is injective on the image of $HT^2(X)$ in $\operatorname{Ext}^2(F_2,F_2)$. Corollary 5.30 does not settle it. It decides it for the sheaf E' of the example, and for every choice of N, of the translates and of the gluing, because the obstruction sits at any smooth point of the glued curve. By Theorem 5.28, a sheaf that answers the question, and likewise a first factor with character on the secant plane that meets the criterion, has no point at which it is locally free on a smooth curve or on a point, no point at which it is the ideal of a curve in a smooth divisor in the position of Theorem 5.28(iii), and at a point where it is locally free on a smooth surface S the tangent plane T_pS must meet both T_1 and T_2, by Lemma 5.26 with $c=2$. Corrections of the character by sheaves on curves, the device of [Mar25c, Example 11.2.7], are therefore excluded. Sheaves that are locally free on smooth divisors, and the global part of the identity $\operatorname{ev}_F(e^{-\theta_t}x_j)=0$, are not reached by this test. Even a positive answer would give an object with $\dim\operatorname{Ext}^2 \geq 97$, by Theorem 5.22(ii) and Proposition 5.29(iii), so its semiregularity could not come from a dimension count.

5.3. **Sextic CM fields.** The dimension count of Theorem 5.22 does not extend to CM fields of degree six, for a reason of parity (Remark 5.33). Let $F = F_0(\sqrt{-q})$ be a sextic CM field with totally real cubic subfield F_0, real places τ_1,τ_2,τ_3 and $q \in F_0$ totally positive, and let X be a principally polarised abelian sixfold with real multiplication by $\mathcal{O}_{F_0}$, so that $H^1(X,\mathbb{C}) = U_1 \oplus U_2 \oplus U_3$ with U_j the eigenspace of τ_j, of dimension four and Hodge type $(2,2)$. Let $\theta_j \in H^{1,1}(X)$ be the component of the polarisation on U_j, so that $\theta = \sum_j \theta_j$, $\theta_j^3 = 0$ and $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$. For $t \in F_0$ and $\varepsilon \in \{\pm 1\}^3$ put $s_j = \sqrt{-1}\sqrt{\tau_j(q)}$ and $e_\varepsilon = \exp(\sum_j(\tau_j(t)+\varepsilon_j s_j)\theta_j)$, and let $S(t,q)$ be the span of the eight e_ε. As in Proposition 5.19(i) it is a rational subspace of dimension $8 = 2^{[F_0:\mathbb{Q}]}$ of the algebra generated by the θ_j, the Galois group permuting the e_ε as it permutes the CM types of F; for $t=0$ it is the secant space of Markman's construction for F on $X \times \widehat{X}$ [Mar25c, §11.1], and $S(t,q) = e^{\sum_j \tau_j(t)\theta_j} S(0,q)$. A real class $v = \sum_\varepsilon w_\varepsilon e_\varepsilon$ has $w_{-\varepsilon} = \overline{w}_\varepsilon$. Write N_w for the number of nonzero w_ε; $A_{jj'}$ for the number of sign pairs (σ,σ') such that $w_\varepsilon \neq 0$ for some ε with $\varepsilon_j = \sigma$ and $\varepsilon_{j'} = \sigma'$; ρ_j for the rank of the 2×4 matrix (w_ε) whose rows are indexed by ε_j and whose columns by the two other signs; and, for $\{j,j',j''\} = \{1,2,3\}$ and $\sigma' = \pm 1$, $M_{\sigma'}^{(j;j')}$ for the 2×2 matrix $(w_\varepsilon)_{\varepsilon_j,\varepsilon_{j''}}$ with $\varepsilon_{j'} = \sigma'$ fixed. Let $r^k(v)$ be the rank of the contraction $HT^k(X) \to H^*(X,\mathbb{C})$, $\xi \mapsto \xi \lrcorner v$.

Proposition 5.32 (The count in degree six). *Let $v \in S(t,q)$ be a nonzero real class.*

(i) $r^1(v) = 12$,

$$r^2(v) = 4\,(A_{12}+A_{13}+A_{23}) + \rho_1+\rho_2+\rho_3, \qquad r^3(v) = 8N_w + 2\sum_{j\neq j'} (\operatorname{rk} M_+^{(j;j')} + \operatorname{rk} M_-^{(j;j')}).$$



$T_0X = T_1 \oplus T_2$: the eigenplanes of F_0.
T_pC' lies in at most one of them; here in T_1.
$T_2 \cap T_pC' = 0$, so π_2 has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, and the germ of $\mathrm{ev}_{E'}(x_2)$ at p is that image:
$x_2 \lrcorner \mathrm{ch}(E') = 0$ and $\mathrm{ev}_{E'}(x_2) \neq 0$.

		pairs $\{k,l\}$ with $a_{\partial_k} a_{\partial_l} \neq 0$						
		12	13	14	23	24	34	pd
(i)	$\mathcal{O}_C$, a line bundle on a smooth curve	■	■		■			3
(ii)	$k(p)$, a skyscraper	■	■	■	■	■	■	4
(iii)	a curve ideal in a smooth divisor	■	■					2
(iv)	a point ideal in a smooth divisor	■	■	■	■	■	■	3
(v)	$\mathcal{O}_D$, a smooth divisor							1
(vi)	two planes meeting at a point		■	■	■	■		2
(vii)	maximal Cohen–Macaulay on a node							1
(viii)	the glued germ of Example 11.2.7				■	■	■	3
(ix)	reflexive, non-Cohen–Macaulay, node	■		■	■		■	2
(x)	$\mathcal{O}_C^2$, rank two on a smooth curve	■	■		■			3

coordinates adapted to the germ: a filled cell is a nonzero Yoneda square in the local Ext^2; at a point of C' the pairs of normal directions are filled, and one of π_1, π_2 projects onto them.

FIGURE 17. Theorem 5.28 and Corollary 5.30. Left: the support of Markman's sheaf E' near a point p of the glued curve C', between two of the translated supports that the curve crosses at its gluing points. The tangent space at p is the sum of the eigenplanes T_1 and T_2 of the real multiplication; the tangent line of C' lies in at most one of them, here in T_1, so π_2 has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, which is the germ of $\mathrm{ev}_{E'}(x_2)$ at p. Right: for ten germs at the origin of $\mathbb{C}^4$, the pairs $\{k,l\}$ of coordinate directions whose jet classes have nonzero product in the local Ext^2, and the projective dimension; rows (i), (ii) and (x) are Lemma 5.26, row (iii) is Lemma 5.27, row (vi) is Lemma 4.60.

So $r^2(v) \leq 54$ and $r^3(v) \leq 112$, with equality when every w_ε is nonzero and every $M_{\sigma'}^{(j;j')}$ is invertible, which holds outside a proper Zariski closed subset of $S(t,q)$. When F contains an imaginary quadratic field K and v lies in the secant plane of K, so that $N_w = 2$, one has $(r^2, r^3) = (30, 40)$.

(ii) *$\chi(v,v) = \int_X v^\vee v = -64\,\mathrm{N}_{F_0/\mathbb{Q}}(q) \sum_\varepsilon |w_\varepsilon|^2 < 0$. For a CM field of degree $2m$, with X of dimension $2m$ and 2^m exponentials, the same computation gives $(-4)^m \mathrm{N}(q) \sum_\varepsilon |w_\varepsilon|^2$: the Euler form on the secant space is definite, of sign $(-1)^m$.*

(iii) *Let F_1, F_2 be perfect complexes on X with $\mathrm{Ext}^{<0}(F_i, F_i) = 0$ and characters v_i in spaces $S(t_i, q_i)$, let $A = X \times \widehat{X}$ carry the action of F of [Mar25c, §11.1], transported by the exponential of a rational F_0-B-field, and put $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \mathrm{ch}(E) e^{\ell/2}$. Then $r(\kappa) = r(\mathrm{ch}\, E) = r^2(v_1) + 144 + r^2(v_2) \leq 252$, while every class that stays of Hodge type over $\mathcal{D}_F$ has rank at most $\dim HT^2(A) - \dim \mathcal{D}_F = 276 - 12 = 264$. Moreover $\dim \mathrm{Ext}^2(E,E) \geq r(\kappa)$, with equality if and only if each F_i has*

$$\dim \mathrm{Ext}^k(F_i, F_i) = 1,\ 12,\ r^2(v_i),\ e_3,\ r^2(v_i),\ 12,\ 1 \quad (k = 0, \ldots, 6),$$

where $e_3 = 2r^2(v_i) - 22 - \chi(v_i, v_i)$, and this profile is compatible with the bound $e_3 \geq r^3(v_i)$ exactly when $\chi(F_i, F_i) \leq -(r^3(v_i) - 2r^2(v_i) + 22)$, that is $\chi(F_i, F_i) \leq -26$ for v_i general in its secant space, and $\chi(F_i, F_i) \leq -2$ when $N_w = 2$.

(iv) *If $t_1 = t_2 = 0$ and $q_1 = q_2 = q$, then κ is invariant under G_F, and so is a Hodge class on every member of the $(F, 2, 1)$-family.*

Proof. (i) Choose a basis x_{j1}, x_{j2} of $U_j^{1,0}$ and y_{j1}, y_{j2} of $U_j^{0,1}$ with $\theta_j = x_{j1}y_{j1} + x_{j2}y_{j2}$, and let $\omega_j = y_{j1}y_{j2}$ span $\bigwedge^2 U_j^{0,1}$. Then $H^*(X, \mathbb{C})$ and $HT^*(X) = \bigwedge^*(H^{0,1} \oplus T)$ are the tensor products over j of the corresponding algebras of the three factors, $HT^k(X) = \bigoplus_{a_1+a_2+a_3=k} HT_1^{a_1} \otimes HT_2^{a_2} \otimes HT_3^{a_3}$, and contraction is compatible with the factorisation; $v = \sum_\varepsilon w_\varepsilon \bigotimes_j e^{\lambda_j(\varepsilon_j)\theta_j}$ with $\lambda_j(\pm 1) = \tau_j(t) \pm s_j$, two distinct numbers. The blocks of $HT^k(X)$ shift the three Hodge bidegrees of $H^*(X)$ by different amounts, so their images are independent and $r^k(v)$ is the sum of their ranks. Inside a block the bookkeeping rests on two facts about one factor: the two classes $e^{\lambda_j(\pm 1)\theta_j}$ are linearly independent, and so are the subspaces $U_j^{0,1}e^{\lambda_j(\pm 1)\theta_j}$ of $H^*(U_j)$, since $u\, e^{\lambda\theta_j} = u + \lambda\, u\theta_j$ for $u \in U_j^{0,1}$; whereas $\omega_j e^{\lambda\theta_j} = \omega_j$ for every λ, so that the summands of a block through HT_j^2 share the factor ω_j and are combined, not juxtaposed. On one factor, contraction into $e^{\lambda\theta_j}$ is as follows. For $\xi \in HT_j^1$, $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi)\, e^{\lambda\theta_j}$ with $y^\lambda(\xi) = \xi^{0,1} + \lambda\, \xi_T \lrcorner \theta_j \in U_j^{0,1}$, as in Lemma 4.55(i); y^λ is onto, its kernel K_j^λ has dimension two, and $HT_j^1 = K_j^{\lambda_j(+1)} \oplus K_j^{\lambda_j(-1)}$ because $\lambda_j(+1) \neq \lambda_j(-1)$ and $\xi_T \lrcorner \theta_j$ determines ξ_T. For $\xi \in HT_j^2$, since $\omega_j\theta_j = 0$, one finds $\omega_j \lrcorner e^{\lambda\theta_j} = \omega_j$, $(y_{ja} \otimes \partial_{x_{jb}}) \lrcorner e^{\lambda\theta_j} = \pm\lambda\, \omega_j$ for $a \neq b$ and 0 for $a = b$, and $(\partial_{x_{j1}} \wedge \partial_{x_{j2}}) \lrcorner e^{\lambda\theta_j} = \pm\lambda^2\omega_j$: the image of HT_j^2 is the line $\mathbb{C}\omega_j$, $\xi \lrcorner e^{\lambda\theta_j} = c(\xi, \lambda)\, \omega_j$, and the pairs $\big(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1))\big)$ fill $\mathbb{C}^2$. For $k \geq 3$, HT_j^k kills $e^{\lambda\theta_j}$, because every element of it contains ω_j or $\partial_{x_{j1}} \wedge \partial_{x_{j2}}$ together with a further factor, and the further factor kills ω_j.

Now count. The block HT_j^1 acts on v by $\xi \mapsto \sum_\varepsilon w_\varepsilon\, y^{\lambda_j(\varepsilon_j)}(\xi) \bigotimes_{j'} e^{\lambda_{j'}(\varepsilon_{j'})\theta_{j'}}$; on the summand $K_j^{\lambda_j(-\sigma)}$ only the ε with $\varepsilon_j = \sigma$ contribute, and they map it isomorphically onto $U_j^{0,1}$ tensored with a nonzero class as soon as one such w_ε is nonzero. Both signs occur in the support of a nonzero real w, since $w_{-\varepsilon} = \overline{w}_\varepsilon$, so the block has rank 4 and $r^1(v) = 12$. The block $HT_j^1 \otimes HT_{j'}^1$ is the sum of the four summands $K_j^{\lambda_j(-\sigma)} \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$, each of dimension four; the summand indexed by (σ, σ') maps isomorphically onto $U_j^{0,1} \otimes U_{j'}^{0,1}$ tensored with the class $\sum_{\sigma''} w_\varepsilon e^{\lambda_{j''}(\sigma'')\theta_{j''}}$, which is nonzero exactly when the pair (σ, σ') occurs in the support, and to zero otherwise; so the block has rank $4A_{jj'}$. The block HT_j^2 maps ξ to $\sum_\sigma c(\xi, \lambda_j(\sigma))\, z_\sigma$ with $z_\sigma = \omega_j \otimes \sum_\mu w_{\sigma\mu} e_\mu$, where μ runs over the four sign patterns of the other two places and the e_μ are independent; as $(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1)))$ fills $\mathbb{C}^2$, the rank is the dimension of the span of z_+, z_-, which is ρ_j. This gives $r^2(v)$. In degree three the block $HT_1^1 \otimes HT_2^1 \otimes HT_3^1$ is the sum of eight summands $\bigotimes_j K_j^{\lambda_j(-\varepsilon_j)}$, of dimension eight, on each of which exactly one term of v acts, isomorphically when $w_\varepsilon \neq 0$ and by zero otherwise: rank $8N_w$. The block $HT_j^2 \otimes HT_{j'}^1$ factors through $\mathbb{C}^2 \otimes HT_{j'}^1$ by $c \otimes \mathrm{id}$, which is onto, and on $\mathbb{C}^2 \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$ the map is $M_{\sigma'}^{(j;j')} \otimes y^{\lambda_{j'}(\sigma')}$ followed by tensoring with independent classes, of rank $2 \operatorname{rk} M_{\sigma'}^{(j;j')}$; the two values of σ' give independent images. The blocks HT_j^3 contribute nothing. This gives $r^3(v)$. The maximal values are $48 + 6 = 54$ and $64 + 6 \cdot 8 = 112$; the conditions for them are the nonvanishing of the w_ε and of the determinants of the $M_{\sigma'}^{(j;j')}$, which are polynomials in the coordinates of v, nonzero on $S(t, q)$. If $F \supseteq K$ and v lies in the secant plane of K, the support of w is a pair $\{\varepsilon_0, -\varepsilon_0\}$,

the two CM types of F induced from K; then $A_{jj'} = 2$, $\rho_j = 2$, $N_w = 2$ and every $M_{\sigma'}^{(j;j')}$ has one nonzero entry, so $(r^2, r^3) = (24 + 6, 16 + 24)$.

(ii) Since $\theta_j^3 = 0$, $\theta^{2m} = \frac{(2m)!}{2^m}\theta_1^2 \cdots \theta_m^2$, and $\int_X \theta^{2m}/(2m)! = 1$ gives $\int_X \theta_1^2 \cdots \theta_m^2 = 2^m$ and $\int_X \exp(\sum_j c_j\theta_j) = \prod_j c_j^2$. The dual of $e^{\lambda\theta_j}$ is $e^{-\lambda\theta_j}$, so $\int_X e_\varepsilon^\vee e_{\varepsilon'} = \prod_j(\lambda_j(\varepsilon_j') - \lambda_j(\varepsilon_j))^2$, which is 0 unless $\varepsilon' = -\varepsilon$, where it is $\prod_j(2s_j)^2 = (-4)^m\mathrm{N}(q)$. Hence $\chi(v, v) = (-4)^m\mathrm{N}(q)\sum_\varepsilon w_\varepsilon w_{-\varepsilon} = (-4)^m\mathrm{N}(q)\sum_\varepsilon |w_\varepsilon|^2$.

(iii) As in the proof of Theorem 5.22(i), $r(\kappa) = r(\mathrm{ch}\, E)$ by Lemma 4.55(i), and Φ identifies $r(\mathrm{ch}\, E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$; the block $HT^1 \otimes HT^1$ contributes $r^1(v_1)r^1(v_2) = 144$ and the other two contribute $r^2(v_1) + r^2(v_2)$, by the type argument given there. The tangent space $T_{[A]}\mathcal{D}_F \subseteq H^1(T_A)$, of dimension $mn^2 = 12$, kills every class that stays of Hodge type along $\mathcal{D}_F$, as in Proposition 3.11(i), and $\dim HT^2(A) = 66 + 144 + 66$. By the Künneth formula, with $\mathrm{Ext}^{<0} = 0$, $\dim \mathrm{Ext}^2(E, E) = e_0e_2' + e_1e_1' + e_2e_0'$ where $e_k = \dim \mathrm{Ext}^k(F_1, F_1)$ and $e_k' = \dim \mathrm{Ext}^k(F_2, F_2)$; the commutative square $\sigma \circ \mathrm{ev} = (\,\cdot\,)\lrcorner\,\mathrm{ch}$ of Setup 4.54, valid in every degree, gives $e_k \geq r^k(v_1)$ and $e_k' \geq r^k(v_2)$, and $e_0, e_0' \geq 1$. So $\dim \mathrm{Ext}^2(E, E) \geq r^2(v_2) + 144 + r^2(v_1)$, with equality if and only if $e_0 = e_0' = 1$, $e_1 = e_1' = 12$, $e_2 = r^2(v_1)$ and $e_2' = r^2(v_2)$. Serre duality on the sixfold gives $e_k = e_{6-k}$, so $\chi(F_i, F_i) = 2(e_0 - e_1 + e_2) - e_3 = 2r^2(v_i) - 22 - e_3$, and by Riemann–Roch on an abelian variety $\chi(F_i, F_i) = \chi(v_i, v_i)$, which is (ii); this determines e_3, and $e_3 \geq r^3(v_i)$ is the stated inequality. With the maximal ranks it reads $\chi \leq -(112 - 108 + 22) = -26$, and with $N_w = 2$ it reads $\chi \leq -(40 - 60 + 22) = -2$.

(iv) The proof of Theorem 5.22(iii) applies with three factors in place of two: for $t = 0$ the classes $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee$, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z))e^{\ell/2}$ and the Poincaré class decompose as tensor products over j of classes of even degree, on each factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with d replaced by $\tau_j(q)$, and the identities of Proposition 3.21 at $n = 2$ show that each image is a multiple of an exponential of η_j or of a Weil class of the j-th factor, invariant under the special unitary group of that factor; $G_F(\mathbb{R})$ is the product of the three, and is Zariski dense in G_F. Statement (iv) is the case $[F : \mathbb{Q}] = 6$ of [Mar25c, Proposition 10.2.1(1)], with the inclusion $\widetilde{\phi}(B \otimes B) \subset A$ of [Mar25c, p. 33], which Markman proves for every CM field in his conventions for Φ and for the twist; we make no claim of priority. ∎

Remark 5.33 (The special role of degree four). In Theorem 5.22 the equality $\dim \mathrm{Ext}^2(E, E) = r(\kappa)$ determines the whole Ext profile of each factor, because on a fourfold Ext^2 is the middle degree and Serre duality supplies the rest; the Euler characteristic of a minimal factor is then the small positive integer $r^2 - 14$, the Euler form on the secant space being positive definite, and the arithmetic of Proposition 5.21 excludes it. On a sixfold the same equality fixes Ext^0, Ext^1 and Ext^2 and leaves the middle group Ext^3 free, and the Euler form has the opposite sign: $\chi(F_i, F_i) = -64\mathrm{N}(q)\sum |w_\varepsilon|^2$ is a negative integer that e_3 absorbs. So no dimension count excludes a minimal Orlov product over a sextic field, and none does over any CM field of degree $2m \geq 6$, where $\mathrm{Ext}^3, \ldots, \mathrm{Ext}^m$ are free and the Euler form has sign $(-1)^m$. What Proposition 5.32(iii) leaves for such an object is the profile displayed there, with $\chi(F_i, F_i) \leq -26$ for a general secant character, and the flat twisted character of (iv), which stays of Hodge type over the whole twelve-dimensional period domain. Whether a sheaf with that profile exists on a sixfold with real multiplication by a cubic field is not known; nothing in Theorem 5.22 bears on it. A certificate of the kind of Theorem 3.19 for a sextic field would be an Orlov

product of such sheaves, with $\dim \mathrm{Ext}^2(E,E) = r(\kappa) \le 252$ on the twelvefold, whose twisted character has a nonzero component in the Weil classes $W_F(A)$; flatness does not give that component, and Lemma 5.35 computes it. In degree six as in degree four, the Orlov products never reach the rank of a class of general shape: $r(\kappa) \le 252$ against 264, and $r(\kappa) \le 104$ against 112.

The count of Proposition 5.32(iii) is a condition on the Euler characteristic of the character alone, so it can be tested on the lattice of integral characters, as Proposition 5.21 tests the quartic one. In degree four the test excludes every minimal character. In degree six it can exclude an integral character only if $-26 < \chi(v,v) < 0$ (Proposition 5.34(i),(ii)), and it excludes neither of the integral classes $\mathrm{Re}\, e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\, e^{\sqrt{-1}\,\theta}$ (Proposition 5.34(iv)). For $t = 0$, the case in which Proposition 5.32(iv) makes κ flat, put $B_j = 1 - \frac{1}{2}\tau_j(q)\,\theta_j^2$, so that $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, and for $c_0, c_3 \in \mathbb{Q}$ and $f, g \in F_0$ let

$$v(c_0, f, g, c_3) = c_0 B_1B_2B_3 + \sum_j \tau_j(f)\,\theta_j B_k B_l + \sum_j \tau_j(g)\, B_j\theta_k\theta_l + c_3\,\theta_1\theta_2\theta_3,$$

where $\{j,k,l\} = \{1,2,3\}$ in each sum.

Proposition 5.34 (Integral flat characters in degree six)**.** *Keep the notation of Proposition 5.32.*

(i) *For every perfect complex F on a complex torus of dimension six, $\chi(F,F)$ is even. So the condition of Proposition 5.32(iii) holds for every F whose character has $N_w = 2$.*

(ii) *For every nonzero real $v \in S(t,q)$, $r^3(v) - 2r^2(v) \le 4$, with equality only if $(r^2(v), r^3(v)) = (54, 112)$. So for every shape the condition of Proposition 5.32(iii) is implied by $\chi(F_i, F_i) \le -26$.*

(iii) *The rational points of $S(0,q)$ are the classes $v(c_0, f, g, c_3)$, and*

$$\chi(v,v) = -8\big(\mathrm{N}(q)\,c_0^2 + \mathrm{Tr}\big(\mathrm{N}(q)\,q^{-1}f^2\big) + \mathrm{Tr}(q\,g^2) + c_3^2\big).$$

(iv) *For $q = 1$, the classes $\mathrm{Re}\, e^{\sqrt{-1}\,\theta} = v(1,0,-1,0)$ and $\mathrm{Im}\, e^{\sqrt{-1}\,\theta} = v(0,1,0,-1)$ are integral points of $S(0,1)$ with $\chi = -32$ and $N_w = 2$, so they satisfy the condition of Proposition 5.32(iii).*

Proof. (i) The Chern character $v = \mathrm{ch}(F)$ is integral, as in the proof of Theorem 5.22(ii), and $\chi(F,F) = \int_X v^\vee v$ by Riemann–Roch. Write $v = \sum_k v_{2k}$, so that $\int_X v^\vee v = \sum_k (-1)^k \int_X v_{2k} v_{12-2k}$. The terms k and $6-k$ are equal, since $(-1)^k = (-1)^{6-k}$ and classes of even degree commute, so they contribute an even number when $k \ne 3$. For $k = 3$ write $v_6 = \sum_I a_I e_I$ in the basis of $\bigwedge^6 H^1(X,\mathbb{Z})$ given by the products e_I of six elements of a basis of $H^1(X,\mathbb{Z})$; since $e_I e_I = 0$ and $e_I e_J = e_J e_I$, $v_6^2 = 2\sum_{I<J} a_I a_J e_I e_J$ is even. For $N_w = 2$ the condition of Proposition 5.32(iii) is $\chi(F,F) \le -2$, and $\chi(F,F)$ is a negative even integer by Proposition 5.32(ii).

(ii) For j and $\sigma = \pm 1$ let $W_{j,\sigma}$ be the 2×2 slice $(w_\varepsilon)_{\varepsilon_j = \sigma}$, with rows and columns indexed by the signs at the two other places. The matrices $M_\sigma^{(j';j)}$ are $W_{j,\sigma}$ and its transpose, so $\sum_{j\ne j'} (\mathrm{rk}\, M_+^{(j;j')} + \mathrm{rk}\, M_-^{(j;j')}) = 2\sum_{j,\sigma} \mathrm{rk}\, W_{j,\sigma}$. Since $w_{-\varepsilon} = \overline{w}_\varepsilon$, $W_{j,-}$ is the complex conjugate of $W_{j,+}$ with its rows and its columns reversed, so the two slices have the same rank k_j. Let S be the support of w and $N = N_w = |S|$. By Proposition 5.32(i),

$$r^3(v) - 2r^2(v) = 8N - 8\,(A_{12} + A_{13} + A_{23}) + \sum_j (8k_j - 2\rho_j). \tag{32}$$

The set S is stable under $\varepsilon \mapsto -\varepsilon$, which exchanges the two slices along each place; so N is even, each slice has $N/2$ nonzero entries, $k_j \geq 1$, and $A_{jj'} \in \{2,4\}$, the projection of S to the signs at j and j' being stable under negation. Also $1 \leq \rho_j \leq 2$ and $k_j \leq 2$, and a slice whose nonzero entries lie in one row or one column has rank one. If $\rho_j = 1$, then $w_\varepsilon = a_{\varepsilon_j} W(\varepsilon')$, with ε' the signs at the two other places, so every slice along a place $j' \neq j$ is $a \otimes$ (a row of W) and $k_{j'} \leq 1$. Now take the four values of N in turn.

If $N = 8$, every $A_{jj'}$ is 4 and (32) is $-32 + \sum_j (8k_j - 2\rho_j)$. When every ρ_j is 2 this is at most $-32 + 3 \cdot 12 = 4$, with equality exactly when every k_j is 2, and then $r^2 = 48 + 6 = 54$ and $r^3 = 64 + 48 = 112$. When some ρ_j is 1, the two other k are at most 1 and (32) is at most $-32 + 14 + 6 + 6 = -6$.

If $N = 6$, S misses one pair $\{\pm\varepsilon_0\}$. As ε_0 and $-\varepsilon_0$ differ at every place, they never have the same signs at two places, so every sign pair at two places occurs in S, every $A_{jj'}$ is 4, and (32) is at most $48 - 96 + 3 \cdot 14 = -6$.

If $N = 4$, $S = \{\pm\varepsilon, \pm\varepsilon'\}$, and after replacing ε' by $-\varepsilon'$ if necessary, ε and ε' differ at exactly one place j. For the two other places k and l one has $A_{kl} = 2$ and $A_{jk} = A_{jl} = 4$. The two nonzero entries of a slice along k are $w_{\pm\varepsilon}$ and $w_{\pm\varepsilon'}$ with the same sign, whose indices agree at l; they lie in one row or one column, so $k_k = 1$, and likewise $k_l = 1$. Hence (32) is at most $32 - 80 + 8\,(2 + 1 + 1) - 6 = -22$.

If $N = 2$, $S = \{\pm\varepsilon_0\}$, every $A_{jj'}$ is 2, every k_j is 1, and every ρ_j is 2, the two slices along j having their nonzero entries at different positions; so (32) is $16 - 48 + 24 - 12 = -20$, the value $40 - 60$ of Proposition 5.32(i).

So $r^3 - 2r^2 \leq 4$, with equality only at $(54, 112)$, and the threshold $r^3 - 2r^2 + 22$ of Proposition 5.32(iii) is at most 26.

(iii) Since $\theta_j^3 = 0$ and $s_j^2 = -\tau_j(q)$, $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, so the eight exponentials and the eight products $\prod_j \beta_j$ with $\beta_j \in \{B_j, \theta_j\}$ span the same complex space, the change of basis being the tensor cube of an invertible 2×2 matrix. The Galois group permutes the τ_j, the θ_j and the B_j compatibly, so $\sum_a C_a \prod_j \beta_j$, indexed by the subsets a of $\{1,2,3\}$ with $\beta_j = \theta_j$ for $j \in a$, is rational exactly when $C_{\pi(a)} = \sigma(C_a)$ for every σ inducing π; the stabiliser of $\{j\}$, and of its complement, is the stabiliser of τ_j, which gives the form $v(c_0, f, g, c_3)$. The involution $v \mapsto v^\vee$ and the integral are multiplicative over the three factors, with $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$; on one factor $B_j^\vee B_j = 1 - \tau_j(q)\theta_j^2$, $\theta_j^\vee\theta_j = -\theta_j^2$ and the mixed products are odd in θ_j. So $\chi(v,v) = -8\sum_a C_a^2 \prod_{j\notin a} \tau_j(q)$, which is the formula.

(iv) For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = \prod_j e^{\sqrt{-1}\,\theta_j} = e^{\sqrt{-1}\,\theta}$. The polarisation is principal, so each $\theta^k/k!$ is integral, and so are the real and imaginary parts $\sum_k (-1)^k\theta^{2k}/(2k)!$ and $\sum_k (-1)^k\theta^{2k+1}/(2k+1)!$. They are $\frac{1}{2}(e_{(1,1,1)} + e_{(-1,-1,-1)})$ and $\frac{1}{2\sqrt{-1}}(e_{(1,1,1)} - e_{(-1,-1,-1)})$, so $N_w = 2$ and $\chi = -64\,(\frac{1}{4} + \frac{1}{4}) = -32$ by Proposition 5.32(ii); expanding $\prod_j (B_j + \sqrt{-1}\,\theta_j)$ gives the coordinates. ∎

The flatness of Proposition 5.32(iv) makes κ a Hodge class on every member of the family, but in degree four the Hodge classes of a very general member are the products of divisor classes and the Weil classes $W_F(A)$ ([I, Proposition 17.3(iii)]), and only a nonzero component of κ in $W_F(A)$ bears on $\mathbf{W}(F, 2, 1)$. We compute that component. Keep the notation of Proposition 5.32, with $t_1 = t_2 = 0$, $q_1 = q_2 = q$ and trivial B-field, put $d_j = \tau_j(q)$, and let $\Psi(Z) = \mathrm{ch}(\Phi(Z))e^{\ell/2}$ as in Proposition 3.21, so that $\kappa(v_1, v_2) = \Psi(v_1 \boxtimes v_2^\vee)$ for the characters v_i of F_i. As in the proof of Proposition 5.32(iv), $H^1(A, \mathbb{C})$ is the sum of three factors $U_j \oplus U_j^\vee$, on each of which $\sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = d_j$. On the j-th factor write $\widehat{\theta}_j$ and ℓ_j for the components of the

dual polarisation and of the Poincaré class, and put $\gamma_j = d_j\theta_j - \widehat{\theta}_j$, as in [I, (33)], and $\eta_j = d_j\theta_j + \widehat{\theta}_j$, a positive multiple of the polarisation of the factor for which $\sqrt{-q}$ is anti-self-adjoint ([I, Proposition 15.19(i)]). For $\sigma = \pm 1$ let $\tau_{j,\sigma}$ be the embedding of F over τ_j with $\tau_{j,\sigma}(\sqrt{-q}) = \sigma\sqrt{-1}\,\sqrt{\tau_j(q)} = \sigma s_j$, and $V_{j,\sigma} \subset H^1(A,\mathbb{C})$ its eigenspace, of dimension four. By [I, Theorem 15.2(ii)] with $n = 2$ and $d = d_j$, the line $\bigwedge^4 V_{j,\sigma}$ is spanned by

$$\omega_{j,\sigma} = -\frac{1}{2d_j^2}\big(\gamma_j - \sigma s_j \ell_j\big)^2, \tag{33}$$

so the six classes $\omega_{j,\sigma}$ span $W_F(A)\otimes\mathbb{C}$. Since η_j pairs $V_{j,+}$ with $V_{j,-}$, the products $\eta_i\eta_k$ lie in summands of $H^4(A,\mathbb{C}) = \bigwedge^4\big(\bigoplus_{j,\sigma} V_{j,\sigma}\big)$ other than the six lines $\bigwedge^4 V_{j,\sigma}$. Let $\kappa_{j,\sigma}(v_1,v_2)$ be the coefficient of $\omega_{j,\sigma}$ in the component of $\kappa(v_1,v_2)$ in $\bigwedge^4 V_{j,\sigma}$; the *F-Weil part* of κ is $\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, its component in $W_F(A)\otimes\mathbb{C}$. For $\varepsilon \in \{\pm1\}^3$ let $\varepsilon^{(j)}$ be the sign vector that agrees with ε at j and only there.

Lemma 5.35 (The F-Weil part of an Orlov product)**.** *Let $v_i = \sum_\varepsilon w_\varepsilon^i e_\varepsilon \in S(0,q)$ for $i = 1,2$.*

(i) *The degree-four part of $\kappa(v_1,v_2)$ is a quadratic form in the η_j plus $\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, with*

$$\kappa_{j,\sigma}(v_1,v_2) = -16\mathrm{N}(q)\,\tau_j(q)\sum_{\varepsilon_j=\sigma} w_\varepsilon^1\, w_{\varepsilon^{(j)}}^2.$$

(ii) *Let $v = v(c_0,f,g,c_3)$ be rational, and for $\{j,k,l\} = \{1,2,3\}$ write $f_j = \tau_j(f)$, $g_j = \tau_j(g)$ and $q_j = \tau_j(q)$. The numbers*

$$R_j = c_0^2 + \frac{f_k^2}{q_k} + \frac{f_l^2}{q_l} + \frac{g_j^2}{q_kq_l} - \frac{1}{q_j}\Big(f_j^2 + \frac{g_l^2}{q_k} + \frac{g_k^2}{q_l} + \frac{c_3^2}{q_kq_l}\Big),$$
$$I_j = c_0f_j + \frac{f_kg_l}{q_k} + \frac{f_lg_k}{q_l} + \frac{c_3g_j}{q_kq_l}$$

are $\tau_j(R)$ and $\tau_j(I)$ for elements $R, I \in F_0$. Put $\Omega(v) = R + 2I/\sqrt{-q} \in F$. Then

$$\kappa_{j,\sigma}(v,v) = -\mathrm{N}(q)\,\tau_j(q)\,\tau_{j,\sigma}\big(\Omega(v)\big).$$

So the F-Weil part of $\kappa(v,v)$ is zero if $R = I = 0$, and otherwise all six $\kappa_{j,\sigma}(v,v)$ are nonzero.

(iii) *If v_1 and v_2 are supported on one pair $\{\varepsilon_0,-\varepsilon_0\}$, the F-Weil part of $\kappa(v_1,v_2)$ is zero. For $q = 1$, F contains $K = \mathbb{Q}(\sqrt{-1})$, the classes $\mathrm{Re}\,e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ span the plane P_θ of* [I, Definition 14.4] *with $d = 1$, and for $v_i = a_i\,\mathrm{Re}\,e^{\sqrt{-1}\,\theta} + b_i\,\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ with $a_i, b_i \in \mathbb{Q}$ the class $\kappa(v_1,v_2)$ lies in $\mathbb{Q}[\eta]\oplus W_K(A)$, where $\eta = \theta + \widehat{\theta}$ and $W_K(A) \subseteq H^{12}(A,\mathbb{Q})$ are the Weil classes of K on the twelvefold (*[I, Proposition 17.10]*); its part of degree at most four is*

$$-32(a_1a_2 + b_1b_2) + 16(a_1b_2 - a_2b_1)\,\eta + 4(a_1a_2 + b_1b_2)\,\eta^2.$$

Proof. (i) The classes $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee = \bigotimes_j (e^{\varepsilon_j s_j\theta_j} \boxtimes e^{-\varepsilon'_j s_j\theta_j})$ and Ψ decompose over the three factors, as in the proof of Proposition 5.32(iv), and on the j-th factor the two identities of Proposition 3.21 at $n = 2$ read, with their constants,

$$\Psi_j\big(e^{\sigma s_j\theta_j}\boxtimes e^{\sigma s_j\theta_j}\big) = -4d_j\,e^{\sigma(s_j/2d_j)\eta_j}, \qquad \Psi_j\big(e^{\sigma s_j\theta_j}\boxtimes e^{-\sigma s_j\theta_j}\big) = \tfrac{1}{2}\big(\gamma_j - \sigma s_j\ell_j\big)^2 = -d_j^2\,\omega_{j,\sigma},$$

since $s_j^2 = -d_j$. So $\Psi(e_\varepsilon \boxtimes e_{\varepsilon'}^\vee)$ is the product over j of the first class where $\varepsilon'_j = -\varepsilon_j$ and of the second where $\varepsilon'_j = \varepsilon_j$. In degree four a product with two or three factors of the

second kind vanishes, one with none is a quadratic form in the η_j, and one with exactly one, at j, has $\varepsilon' = \varepsilon^{(j)}$ and equals $-d_j^2 \prod_{i\neq j}(-4d_i)\,\omega_{j,\varepsilon_j} = -16\mathrm{N}(q)\,d_j\,\omega_{j,\varepsilon_j}$. Summing $w^1_\varepsilon w^2_{\varepsilon'} \Psi(e_\varepsilon \boxtimes e^\vee_{\varepsilon'})$ over ε and ε' gives (i).

(ii) An element h of the Galois group that fixes the place j permutes the two others, so it fixes R_j and I_j, which are symmetric in k and l; hence $R_j, I_j \in \tau_j(F_0)$, and moving j by the Galois group shows that they are the conjugates of two elements of F_0. Write $\varepsilon = (\sigma, \mu)$ with μ the signs at k and l, so that $\varepsilon^{(j)} = (\sigma, -\mu)$. Since $e^{\pm s_i\theta_i} = B_i \pm s_i\theta_i$, the class $v = \sum_a C_a \prod_{i\in a}\theta_i \prod_{i\notin a} B_i$, with $C_\emptyset = c_0$, $C_{\{i\}} = f_i$, $C_{\{1,2,3\}\setminus\{i\}} = g_i$ and $C_{\{1,2,3\}} = c_3$, has $w_\varepsilon = \frac{1}{8}\sum_a C_a \prod_{i\in a}\varepsilon_i/s_i$. Grouping $a = b$ and $a = b\cup\{j\}$ for $b \subseteq \{k,l\}$ gives $w_{(\sigma,\mu)} = \frac{1}{8}\sum_b D_b \prod_{i\in b}\mu_i/s_i$ with $D_b = C_b + u\,C_{b\cup\{j\}}$ and $u = \sigma/s_j = \tau_{j,\sigma}(1/\sqrt{-q})$. Since $\sum_\mu \prod_{i\in b}\mu_i \prod_{i\in b'}(-\mu_i) = 4(-1)^{|b|}\delta_{bb'}$ and $s_i^2 = -q_i$,

$$\sum_{\varepsilon_j=\sigma} w_\varepsilon w_{\varepsilon^{(j)}} = \frac{1}{16}\Big((c_0+uf_j)^2 + \frac{(f_k+ug_l)^2}{q_k} + \frac{(f_l+ug_k)^2}{q_l} + \frac{(g_j+uc_3)^2}{q_kq_l}\Big),$$

and expanding with $u^2 = -1/q_j$ gives $\frac{1}{16}(R_j + 2uI_j) = \frac{1}{16}\tau_{j,\sigma}(\Omega(v))$. With (i) this is the formula. The embedding $\tau_{j,\sigma}$ is injective and 1, $1/\sqrt{-q}$ are independent over F_0, so one $\kappa_{j,\sigma}(v,v)$ vanishes if and only if $R = I = 0$, and then all do.

(iii) If w^1 and w^2 vanish off $\{\varepsilon_0, -\varepsilon_0\}$, no term of (i) survives, since $\varepsilon^{(j)}$ is neither ε nor $-\varepsilon$. For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = e^{\sqrt{-1}\,\theta}$, whose real and imaginary parts are u_θ and v_θ of [I, (30)] with $d = 1$, supported on $\{\pm(1,1,1)\}$. On each factor $\sqrt{-1} = \sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = 1$, so K acts on A by the action [I, (29)] determined by θ with $d = 1$, and Proposition 3.21 with $n = 6$ and $d = 1$ gives $\kappa \in \mathbb{Q}[\eta] \oplus W_K(A)$. For the constants write $v_1 = w_1 e^{\sqrt{-1}\,\theta} + \overline{w}_1 e^{-\sqrt{-1}\,\theta}$ and $v_2^\vee = a_2 \operatorname{Re} e^{\sqrt{-1}\,\theta} - b_2 \operatorname{Im} e^{\sqrt{-1}\,\theta} = w_2 e^{\sqrt{-1}\,\theta} + \overline{w}_2 e^{-\sqrt{-1}\,\theta}$, with $w_1 = \frac{1}{2}(a_1 - \sqrt{-1}\,b_1)$ and $w_2 = \frac{1}{2}(a_2 + \sqrt{-1}\,b_2)$. By the first identity in the proof of (i), $\Psi(e^{\pm\sqrt{-1}\,\theta} \boxtimes e^{\pm\sqrt{-1}\,\theta}) = -64\,e^{\pm(\sqrt{-1}/2)\eta}$, while the two mixed terms have degree twelve; so in degrees below twelve κ is $-128\operatorname{Re}\big(w_1 w_2\, e^{(\sqrt{-1}/2)\eta}\big)$, whose terms of degree 0, 2 and 4 are the stated ones. ∎

Markman states his reduction for the normalised class $\mathrm{ch}(E)\exp(-c_1(E)/\operatorname{rk} E)$, and it is the component of degree four of that class, a polynomial in the invariant divisor classes plus a Weil class, whose algebraicity on the deformations he needs [Mar25c, Proposition 10.2.1, Corollary 10.2.3]. When $\operatorname{rk} E \neq 0$ that class is $\kappa\exp(-\kappa_1/\operatorname{rk} E)$, with κ_1 the degree-two part of κ, which is a linear form in the η_j by the proof of Lemma 5.35(i); so it has the same F-Weil part as κ.

For $q = 1$ the Orlov products of complexes with characters in $\mathbb{Q}\operatorname{Re} e^{\sqrt{-1}\,\theta} \oplus \mathbb{Q}\operatorname{Im} e^{\sqrt{-1}\,\theta}$ are therefore the template of Proposition 3.22 at $(n,d) = (6,1)$, seen on the twelvefold through the restriction of the action of F to K. Their twisted characters are flat for the family of K and have a nonzero K-Weil part in H^{12} when $F_1 = F_2 \neq 0$ (Proposition 3.21), and they are not an input to $\mathrm{P2}_{\mathrm{split}}$. By [I, Proposition 17.10], $\mathbf{W}(F,2,1)$ gives the K-Weil classes on the members of the F-family, and the converse is not available; a consequence for K through Proposition 6.13 would still need a twisted form of the propagation of Theorem 2.3.

Remark 5.36 (Natural objects). Let X be very general with real multiplication by $\mathcal{O}_{F_0}$, so that $\mathrm{NS}(X)_\mathbb{Q} = \{\theta_t = \sum_j \tau_j(t)\theta_j : t \in F_0\}$. Call *natural* the line bundles, the simple semi-homogeneous bundles of [Muk78], the skyscraper sheaves of points, and the objects obtained from these by shifts, duals, twists by line bundles, pull-back by automorphisms

commuting with $\mathcal{O}_{F_0}$, and $\Phi_{\mathcal{P}}$ followed by $\widehat{X} \cong X$. On the j-th factor write classes in $\langle 1, \theta_j, \theta_j^2/2\rangle$ as (r, c, s), and put $\nu(t) = e^{t\theta_j} = (1, t, t^2)$ and $\nu(\infty) = (0, 0, 1)$. The character of a natural object is a multiple of $\bigotimes_j \nu(\tau_j(t))$ for one $t \in F_0 \cup \{\infty\}$, its slope: a simple semi-homogeneous bundle has character $r\, e^{\theta_t}$ [Muk78], a point has slope ∞, and the operations act on slopes by $t \mapsto -t$, $t + t'$, tu^2 with $u \in \mathcal{O}_{F_0}^{\times}$, and $-1/t$, the last by [I, Theorem 6.5] applied factor by factor. Then

(a) no natural character lies in $S(0, q)$, for any totally positive q;
(b) if a real combination of natural characters of k pairwise distinct slopes is a nonzero element of $S(0, q)$, then $k \geq 6$, and if $k = 6$ it has $N_w = 2$.

So no cone of a morphism between two natural objects has a nonzero character in $S(0, q)$, and by Lemma 5.35(iii) a class whose Orlov square has a nonzero F-Weil part needs at least seven slopes. Six are attained: $\operatorname{Re} e^{\sqrt{-1}\,\theta} = \frac{5}{2} - \frac{5}{6}(e^{\theta} + e^{-\theta}) + \frac{1}{12}(e^{2\theta} + e^{-2\theta}) - 10[\mathrm{pt}]$.

For (a): $S(0, q)$ is the tensor product of the planes $P_j = \langle B_j, \theta_j\rangle = \{s = -d_j r\}$, the form $c^2 - rs$, for which $\nu(t)$ and $\nu(\infty)$ are isotropic, is positive definite on P_j, and a decomposable tensor lies in $\bigotimes_j P_j$ only if each factor does. For (b), let $T = \sum_i a_i \bigotimes_j \nu(\tau_j(t_i))$ and $\lambda_1 = s + d_1 r$, which is positive on $\nu(t)$ and $\nu(\infty)$. Applying λ_1 to the first factor gives a relation with nonzero coefficients among the images, under the embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ by forms of bidegree $(2, 2)$, of the k points $(\tau_2(t_i), \tau_3(t_i))$, no two of which share a coordinate since each τ_j is injective. Any five such points are independent, since the product of the two vertical and the two horizontal lines through four of them misses the fifth; so $k \geq 6$. If $k = 6$, a form of bidegree $(2, 2)$ vanishing at five of the points vanishes at the sixth; applied to CD, with C the curve of bidegree $(1, 1)$ through three of the points and D running over the pencil of such curves through two others, whose base locus is those two points, this puts the sixth point on C. So the six points lie on the graph of a real Möbius transformation, and likewise for the other pairs of factors. Moving the second and third factors by these transformations, which act through GL_2 on $\langle 1, \theta_j, \theta_j^2/2\rangle$, turns T into a binary sextic, embedded in the tensor cube by $l^6 \mapsto (l^2)^{\otimes 3}$, that is annihilated by three definite binary quadrics, the transforms of the λ_j. Two non-proportional definite binary quadrics generate every binary form of degree at least three, and so would annihilate the sextic; as $T \neq 0$, the three quadrics are proportional to one quadric Λ, and the sextic lies in the kernel of contraction with Λ, spanned by l_+^6 and l_-^6 with $l_\pm$ the isotropic lines of Λ. Moving back, T is a combination of two conjugate decomposable tensors in $S(0, q)$, which are e_ε and $e_{-\varepsilon}$ for one ε; so $N_w = 2$.

6. The reach of the obstructions

An obstruction theorem is useful only if its boundary is drawn exactly. This section draws the boundary of each of the main results, then states the statements through which the conjecture is reached and proves the implications of the literature that join them.

6.1. **Limits of the divisor obstruction.**

Remark 6.1 (Markman's construction). The objects through which Markman proves the Weil classes of the sixfolds of trivial discriminant algebraic [Mar25a, Mar25b] are not obtained by restriction from an ambient projective space, and they are not K-equivariant. [I, Section 13] analyses exactly which hypotheses they escape and why, and [I, Proposition 13.5] lists the three. No obstruction of this paper bears on those objects, and none needs to: their conclusion is a theorem. The candidates proposed for cases that remain open, in

[Mar25b, Section 12] and [Mar25c, Example 11.2.7], are another matter, and Theorem 4.61 and Corollary 5.30 decide them.

Remark 6.2 (Twisted objects and gerbes). The hypothesis of [I, Theorem 7.3] is that $\mathcal{F}$ is an object of $D^b(\mathbb{P}^{N-1})$, so a twisted sheaf on a gerbe over $\mathbb{P}^{N-1}$, or an object of a derived category of twisted sheaves, is not covered. What is covered, and what the proof uses, is that the Chern character of the restricted object lies in the subring of $H^*(A^\vee, \mathbb{Q})$ generated by divisor classes. Any construction whose input has that property is subject to the conclusion, whatever category it is phrased in.

Remark 6.3 (Other ambient varieties). Replacing $\mathbb{P}^{N-1}$ by another ambient variety P changes the conclusion exactly when $H^*(P, \mathbb{Q})$ is not generated by divisor classes, since the proof only uses $\mathrm{ch}(\mathcal{F}) \in D^*(P)$ and the fact that i^* carries $D^*(P)$ into $D^*(A^\vee)$. For P a Grassmannian, a flag variety, or any variety whose cohomology is generated in degree two, nothing changes. For P with exceptional Hodge classes of its own the argument does not apply, but then the problem has been moved rather than solved.

6.2. **Limits of the character obstruction.**

Remark 6.4 (Constructions that are not equivariant). [I, Theorem 5.6] constrains eigenclasses. A construction whose output is not a $K^\times$-eigenclass is not constrained, and by [I, Remark 5.8] such a construction must break the naturality of the K-action in an essential way: it must depend on a choice that is not K-invariant, and its output must have components in both extreme summands of the grading at once. These obstructions therefore point to constructions of that kind.

Remark 6.5 (Abelian varieties that are not very general). At special points of the moduli of Weil type the endomorphism algebra grows, the Néron–Severi group grows with it, and the Weil line can meet the divisor subring. This is the split case, where Weil classes are known to be algebraic by elementary means, and it is consistent with everything here: our statements about $W(A) \cap D^{2n}(A)$ assume $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, which fails there. [I, Theorem 5.6] itself makes no such assumption and holds at every point, but at special points a class can be a sum of eigenclasses for several characters and the conclusion applies to each summand separately.

6.3. **Limits of the semiregularity statement.**

Remark 6.6 (Objects of positive rank). [I, Theorem 9.7] and [I, Corollary 9.18] both require the Chern character to lie on the Weil line, so that $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$ has no degree-two term. An object of nonzero rank has $\mathrm{rk}(\mathcal{E})\, z$ in that degree, the map is injective, and nothing here applies. Markman's sheaves are of this kind, and [I, Remark 9.9] records why the distinction is not incidental to his argument.

Remark 6.7 (The weaker semiregularity hypothesis). What [I, Corollary 9.18] obstructs is the injectivity of $\sigma_{\mathcal{E}}$ on the whole of $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$. The variant proposed in [Mar25b, Question 11.4], injectivity on the image of an evaluation map, asks less. What we prove about it is confined to three groups of results. In [I, Theorem 9.19], Proposition 4.18, [I, Theorem 9.20] and Corollary 4.21 it fails. In Theorem 4.56, for a secant object of positive rank, it is shown to be equivalent to two conditions on the factor. In Theorem 4.61 and Corollary 4.63 it fails for Markman's candidate in dimension eight and for every secant object whose support has two transversal branches of codimension two through a point. We do not decide the variant for any other object of positive rank, nor whether an affirmative answer to the question would follow from it.

Remark 6.8 (Deformation of the class itself). A kernel is not an obstruction class. [I, Corollary 9.18] says that a sufficient criterion for the Weil class to stay algebraic is unavailable at the objects in question; it does not say the class fails to stay algebraic, and no statement in this paper does. [I, Question 15.13] is asked in both directions.

6.4. **Limits of the hyperplane obstruction.**

Remark 6.9 (Families that are not linear systems on X). [I, Theorem 10.3] is about smooth members of an ample linear system on X, and about the maps j^* and j_* they carry. It does not obstruct an argument that deforms X itself, that passes to a different variety by a correspondence, or that uses a family in which X appears as a fibre and the other fibres have the same dimension. In the last case the inductive parameter does not decrease, which is a different difficulty and the one recorded in [I, Remark 10.10].

Remark 6.10 (Coniveau). A Hodge class of positive coniveau is supported on a proper closed subvariety, and the reduction to that subvariety is legitimate and lowers the dimension. Nothing here obstructs it. The obstruction concerns classes of coniveau zero, which is where the primitive middle cohomology of a general variety sits, and the generalised Hodge conjecture predicts that these are precisely the ones a coniveau argument cannot reach.

Remark 6.11 (The Kuga–Satake route). For a weight-two Hodge structure of K3 type the Kuga–Satake construction produces an abelian variety $\mathrm{KS}(T)$ and an injection of T, twisted once, into the endomorphisms of its first cohomology, carrying the Hodge class in question to a class of type $(0,0)$, that is to an endomorphism, whose graph is a cycle. The conclusion is transported back by a correspondence, not by a restriction, so [I, Theorem 10.3] does not apply; this is [I, Proposition 11.9]. The route is limited by the requirement $h^{2,0}=1$, and [I, Proposition 11.11] locates that limit exactly for abelian varieties of Weil type: no rational sub-Hodge structure of $H^2(A,\mathbb{Q})$ is of K3 type unless $n=1$. [I, Section 11] treats the construction in full, including the algebraicity of the correspondence itself.

6.5. **The Weil statements and the closure map.** The results above close several routes and narrow one. The statements through which the conjecture is reached form a list, and we record each entry with its exact statement and with what this paper proves about it. Proposition 6.13 records the one way in which the entries on Weil classes are linked.

Fix an imaginary quadratic field $K=\mathbb{Q}(\sqrt{-d})$, an integer $n\geq 2$, and a class $\delta\in \mathbb{Q}^{\times}_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^{\times})$, the discriminant of the hermitian form. Write

> $\mathbf{W}(K,n,\delta)$: every abelian $2n$-fold of $(K,-1,n)$-Weil type with discriminant δ has algebraic Weil classes.

Proposition 6.12 (The Weil statements, and how they are linked)**.** *For each triple (K,n,δ) the following hold.*

(i) $\mathbf{W}(K,n,\delta)$ *is equivalent to the algebraicity of the single class ω_s at every point s of the period domain $\mathcal{D}_{n,n}$, and that domain is connected; this is* [I, Corollary 15.35].

(ii) $\mathbf{W}(K,3,\delta_0)$ *holds for the trivial discriminant δ_0* [Mar25a, Mar25b]*, and with it, by Proposition 6.13,* $\mathbf{W}(K,2,\delta)$ *for every K and every δ, which is how Markman obtains the fourfolds.*

(iii) *The period domains attached to distinct triples are disjoint, and* [I, Theorem 15.14] *and* [I, Corollary 15.35] *are statements about one triple at a time. The triples are nevertheless not independent: by Proposition 6.13,* $\mathbf{W}(K,N,\delta'')$ *implies* $\mathbf{W}(K,n,\delta)$ *for every $n<N$ and every δ, so the families of trivial discriminant in infinitely many dimensions give every imaginary quadratic family, and the families of a CM field of degree at least four give those of the imaginary quadratic fields it contains.*

Proof. (i) is [I, Corollary 15.35] together with [I, Proposition B.8]. (ii) is the cited work together with Proposition 6.13(i); [I, Remark 15.20] records that the construction of [I, Section 15.1] reaches only δ_0. For (iii), the discriminant is a discrete invariant of the hermitian form and is constant on $\mathcal{D}_{n,n}$ by [I, Proposition 15.19(ii)], while n and K are fixed by the endomorphism algebra, so the domains are disjoint; the links are Proposition 6.13, which passes between families through a product with a Weil surface or through a scalar extension, and not through a deformation. The product argument of Schoen that Markman uses to pass from sixfolds to fourfolds is an instance of such a link. ∎

Proposition 6.13 (Descent and scalar extension)**.** *Let F be a CM field of degree $2m$ and write $\mathbf{W}(F,n,\delta)$ for the statement that every abelian variety of (F,n,δ)-Weil type has algebraic Weil classes, so that $\mathbf{W}(K,n,\delta)$ is the statement above when $m=1$.*

(i) Descent. *Let B be of (F,n,δ)-Weil type and Y of $(F,1,\delta')$-Weil type. Then $X = B\times Y$, with the diagonal action of F and the product polarisation, is of $(F,n+1,\delta\delta')$-Weil type, and the correspondence*

$$P(x) = pr_{B*}(x\cdot pr_Y^*(\eta_Y^{2m-2}\, y')),$$

with η_Y the polarisation of Y and y' a nonzero Weil class of Y, maps the Weil classes of X isomorphically onto those of B. Since y' lies in $H^2(Y)$ it is algebraic, so P is algebraic, and every discriminant, a totally positive class in $F_0^\times/\mathrm{N}_{F/F_0}(F^\times)$, is that of some Y. Hence $\mathbf{W}(F,N,\delta'')$ for one N and one δ'' implies $\mathbf{W}(F,n,\delta)$ for every $n<N$ and every δ.

(ii) Scalar extension. *Let $K\subset F$ be CM fields and B of (K,n,δ)-Weil type. Then $B_F = B\otimes_{\mathcal{O}_K}\mathcal{O}_F$ is of $(F,n,\iota(\delta))$-Weil type, and for $j\colon B\to B_F$, $b\mapsto b\otimes 1$, one has $j^*\,\mathrm{Tr}_{F/\mathbb{Q}}(a\det_F) = \mathrm{Tr}_{K/\mathbb{Q}}(\mathrm{Tr}_{F/K}(a)\det_K)$, so j^* maps the Weil classes of B_F onto those of B. Hence $\mathbf{W}(F,n,\iota(\delta))$ implies $\mathbf{W}(K,n,\delta)$; every imaginary quadratic K lies in the quartic CM field $K(\sqrt{2})$.*

Proof. (i) $H^1(X) = H^1(B)\oplus H^1(Y)$ as F-spaces and the hermitian form of X is the orthogonal sum, of signature $(n+1,n+1)$ at every real place; its normalised discriminant is the product, and $H_Y = \mathrm{diag}(1,-t)$ with t totally positive realises every class. The Weil classes of X are $\bigoplus_\sigma \bigwedge^{2n} V_{B,\sigma}\otimes\bigwedge^2 V_{Y,\sigma}$, because the top exterior power of a direct sum is the tensor product of the top powers. On the σ-component, P is multiplication by $\int_Y w_Y^\sigma\,\eta_Y^{2m-2}\,y'^{\bar\sigma}$, which is nonzero since $w_Y^\sigma\wedge y'^{\bar\sigma}$ fills the $(\sigma,\bar\sigma)$ block and η_Y is nondegenerate on the others; the remaining pairings vanish for weight reasons under the norm-one torus of F. So P is rational, diagonal over $\mathbb{C}$ with nonzero entries, and algebraic, and $\omega(B)$ is algebraic whenever $\omega(X)$ is; [I, Lemma 16.3] passes from one Weil class to all of them.

(ii) $H^1(B_F) = H^1(B)\otimes_K F$, each eigenspace of F is an eigenspace of K of signature (n,n), the hermitian form is $H_B\otimes F$ with determinant $\det H_B$, and a norm from K is a norm from F. The identity for j^* is the multiplicativity of the determinant under restriction of scalars, and $\mathrm{Tr}_{F/K}$ is onto. ∎

The case K imaginary quadratic, $n=2$ and $N=3$ of (i) is Proposition 10 of Schoen's addendum [Sch98], the argument through which Markman obtains every Weil fourfold, at every discriminant, from the sixfolds of trivial discriminant [Mar25a]. Schoen multiplies a Weil fourfold A by a Weil surface A' of the complementary discriminant, identifies $A\times A'$ up to isogeny with a member of a family of Weil sixfolds of fixed discriminant, and pushes forward the cycles $z\cdot(A\times D)$, for D a divisor on A' whose class spans its Weil classes; this is the correspondence P of (i) with $m=1$, and it involves no degeneration. What (i) adds is that the same argument works in every dimension and for every CM field. As

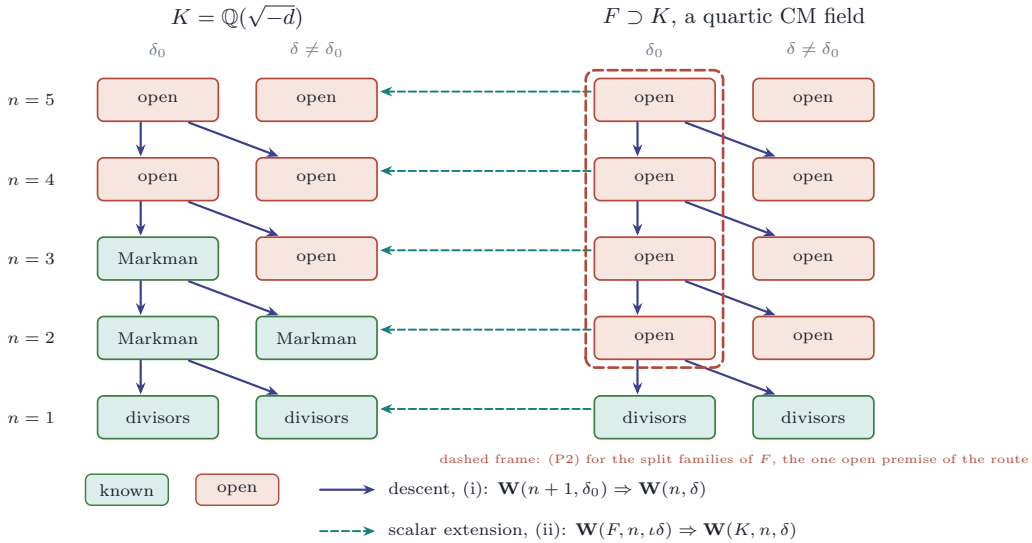


FIGURE 18. The Weil families of an imaginary quadratic field K and of a quartic CM field $F \supset K$, one cell for each n and each kind of discriminant, grass where the Weil classes are proved algebraic (divisor classes at $n = 1$; Markman [Mar25a, Mar25b] at $n = 2$ and at $n = 3$, $\delta = \delta_0$) and clay where they rest on the open propagation statement. The indigo arrows are the descent of Proposition 6.13(i), from the trivial discriminant in dimension $n + 1$ to every discriminant in dimension n; the dashed teal arrows are the scalar extension of Proposition 6.13(ii). The dashed frame is the propagation statement for the split families of the field of higher degree, which by Theorem 6.70(ii) is the only form of (P2) that any minimal route to the conjecture uses: every clay cell of the left panel lies below it.

a consequence the triples (K, n, δ) are not separate problems. The trivial discriminant families in infinitely many dimensions give every imaginary quadratic family, and by (ii) the split families of the CM fields of degree at least four give the imaginary quadratic fields as well. In particular the base points of nontrivial discriminant supplied by [I, Theorem 15.27] are not needed on any route to the conjecture, since the split families already carry base points [I, (34)]; what [I, Theorem 15.27] shows is that propagation can be started in every family, not that it has to be. Figure 18 draws the web of families that Proposition 6.13 links.

The list is as follows.

(C1) *One secant object on one fourfold.* There should exist a principally polarised abelian fourfold (X, Θ) and a perfect complex $\mathcal{F}$ on it with $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$ satisfying (19). By Theorem 4.56 and Corollary 4.57 the transform $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ would then satisfy the weakened criterion, and by Proposition 4.52 the three conditions of Markman's strategy that concern only the Chern character hold at the point $(1, 3)$ for any such $\mathcal{F}$, since they depend on $\mathrm{ch}(\mathcal{F})$ alone. Turning this into algebraicity needs one further input. The weakened criterion enters in place of the hypothesis Markman verifies, and the survival of the semiregularity theorem under the substitution is exactly [Mar25b, Question 11.4]. So (C1) is a demand on a construction, and an affirmative

answer to that question is a separate prerequisite standing beside it; Theorem 6.70 places both. What this paper contributes is the exact form of the demand. The local part of the criterion is a condition on depth and nothing else: if the support is Cohen–Macaulay there is no local obstruction at any point (Theorem 4.75); if it is not, the local Ext^2 is nonzero at the bad points, and where two smooth branches of codimension two meet transversally the criterion fails outright (Theorem 4.61, Corollary 4.63, Corollary 4.77). The candidate of [Mar25b, Section 12] is of the second kind, and it cannot be moved to the first: separating the branches is what would restore the depth, and Remark 4.78 shows that doing so moves the Chern character off the secant plane. What is needed is therefore a support without such points. For a Cohen–Macaulay support the identity (19) is the single vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$, and the sheaf must in addition extend to first order along the polarised deformations (Theorem 4.56(iii)). Such a support has one of two shapes (Corollary 4.94): a smooth one, where the invariants are forced outright and $d \leq b^2$ for the twist $b\Theta$, so $d \leq 9$ for the twist 3Θ of Setup 4.51 (Theorem 4.67, Corollary 4.68), and a singular one. In neither may the Hilbert–Burch resolution split into powers of Θ or into simple semi-homogeneous bundles (Theorem 4.89, Theorem 4.92), and no complete intersection occurs (Theorem 4.81).

(C2) *The same for every n and every discriminant.* A positive answer to (C1) reaches one triple, and the discriminant of that triple is the trivial one, because the object lives on $X \times \widehat{X}$ and every split member has trivial discriminant by [I, Proposition 15.19(ii)]. The other triples are reached through Proposition 6.13: a positive answer in infinitely many dimensions gives $\mathbf{W}(K, n, \delta)$ for every n and every δ, so only uniformity in n is asked for, and (C1) at $n = 4$, with Question 11.4, already gives every sixfold. No answer of this kind reaches the CM fields of degree at least four. What this paper contributes: three of the four ingredients of [I, Question 13.7] are supplied for every n and every K ([I, Theorem 14.15]), including the existence of the secant sheaves themselves ([I, Theorem 14.13]); the fourth is isolated exactly ([I, Remark 14.16]).

(C3) *CM fields of degree larger than two.* For a CM field F of degree $2m > 2$ acting on an abelian variety, the Weil classes span a space of dimension $2m$, and the secant framework of [I, Section 14], which is built on a plane meeting the spinor variety in a conjugate pair, does not apply to them as it stands. Markman has extended it to these fields [Mar25c]: for an abelian variety X with real multiplication by the totally real subfield F_0 of F, a polarisation $\Theta \in \bigwedge^2_{F_0} H^1(X, \mathbb{Q})$ and $q \in F_0$ with $F = F_0(\sqrt{-q})$, he makes F act on $X \times \widehat{X}$ with split Weil type, replaces the plane by a rational subspace B of dimension $2^{[F_0:\mathbb{Q}]}$ of the half-spin representation that is secant to the spinor variety, and shows that for sheaves F_1, F_2 on X with characters in B the class $\kappa(E) = \mathrm{ch}(E) \exp(-c_1(E)/\operatorname{rk} E)$ of $E = \Phi(F_1 \boxtimes F_2^\vee)$ stays of Hodge type on every member of the family, and that where it stays algebraic all Weil classes are algebraic [Mar25c, Theorem 1.1.2, Proposition 10.2.1]. He exhibits such a pair meeting his non-degeneracy criterion when F is biquadratic and X is the Jacobian of a curve of genus four with real multiplication by F_0 and an automorphism acting through a unit of norm one [Mar25c, Example 11.2.7, Lemma 11.2.8]. That $\kappa(E)$ stays algebraic would follow from a semiregularity property of E, which is not addressed there [Mar25c, Question 11.2.2]; for the pair of that example the weaker property asked in the question fails at the points of the glued curves (Corollary 5.30). No result of this paper depends on a theorem of

[Mar25c]; Section 5.2 and Section 5.3 use its construction as their setting. What this paper contributes: everything except the propagation. By [I, Section 17] each such family has a connected period domain, a flat space of Weil classes, a dense set of rational orbits, a base point at a CM member where the classes are written as polynomials in divisor classes ([I, Theorem 17.5]), and the two propagation criteria ([I, Theorem 17.7]); so the higher CM fields are the same propagation problem as the quadratic ones, and not a separate entry. For a quartic field at $n = 2$, the smallest case of higher degree, Section 5.2 determines how far divisor classes, scalar extension and correspondences reach, and computes the number that the polarised criterion asks of one complex. For a sextic field, Section 5.3 shows that the dimension count, which excludes a minimal Orlov product over a quartic field, does not exclude one over a sextic field, and gives the profile such an object would have. [I, Appendix C] places the imaginary quadratic case inside Albert's classification.

(C4) *The non-split sixfolds.* $\mathbf{W}(K, 3, \delta)$ for $\delta \neq \delta_0$; the case $\delta = \delta_0$ is Markman's [Mar25a]. By Proposition 6.13(i) it would follow from $\mathbf{W}(K, 4, \delta_0)$, and by Proposition 6.13(ii) from $\mathbf{W}(K(\sqrt{2}), 3, \iota(\delta))$. What this paper contributes: a base point in every family, at a member with quaternionic multiplication ([I, Theorem 15.27]); the locus of such members is a Lagrangian Grassmannian of codimension $n(n-1)/2$ ([I, Proposition 15.32]), and the rest of the family can be reached from it only by propagation.

(C5) *Exceptional classes that are not of Weil type.* On an abelian variety the Hodge ring can contain exceptional classes that the Weil construction does not produce. Through dimension five it does not [MZ99], and in dimension eight it does: the self-product of an abelian fourfold of Mumford type carries two Hodge classes that are neither products of divisor classes nor Weil classes of any quotient (Proposition 6.74). So the Hodge conjecture for abelian varieties does not reduce to Weil classes, and the statement needed is the algebraicity of these classes themselves. What this paper contributes: the example, made exact, and the forms of their algebraicity studied in Section 7.

(C6) *Varieties that are not abelian.* A reduction of the Hodge conjecture for arbitrary smooth projective varieties to abelian varieties cannot be of Hodge-theoretic kind: the H^3 of a very general quintic threefold lies outside the tensor category generated by the cohomology of abelian varieties (Proposition 6.75), so its Hodge structure cannot be realised inside that of an abelian variety by any correspondence. [I, Theorem 10.3] and [I, Theorem 10.8] close the two routes through hyperplane sections that would be the natural candidates. What this paper contributes: those three obstructions and, on the positive side, special cases only (Proposition 6.26, Remark 6.68). Nor is this entry the complement of the others. The varieties that are not abelian include $A \times \mathbb{P}^1$ for every abelian variety A, and the conjecture for $A \times \mathbb{P}^1$ gives it for A, so the conjecture for the varieties that are not abelian is the whole conjecture (Proposition 6.33).

6.6. **Implications from the literature.** The theorems of this paper are not the only implications that end in the conjecture. The literature supplies others, and a frontier drawn without them overstates what is needed. This subsection proves those implications, or quotes them where they are theorems of the literature, so that Section 6.9 runs over both. Nothing in it is new except the assembly, but the assembly changes the answer.

Definition 6.14 (Three statements about all varieties)**.** Here a variety is smooth, projective and defined over $\mathbb{C}$, except the base S in (V).

(L) $B(X)$ holds for every variety X ([I, Definition 10.6]).

(M) Every Hodge class on every variety X is *motivated* in the sense of André [And96b, §2.1]: it equals $\mathrm{pr}_*(\alpha \cup *\beta)$ for some variety Y, algebraic classes α, β on $X \times Y$, the projection $\mathrm{pr}\colon X \times Y \to X$, and the Lefschetz involution $*$ of $H^*(X \times Y, \mathbb{Q})$ with respect to some ample class.

(V) For every smooth connected complex variety S, every smooth projective morphism $f\colon \mathcal{X} \to S$ and every global section ξ of $R^{2p} f_* \mathbb{Q}$ over S^{an}: if ξ_s is algebraic at one point $s \in S$, then ξ_t is algebraic at every point $t \in S$.

The Lefschetz involution of a variety Z of dimension e with ample class L sends $L^r x$, for x primitive of degree j and $0 \le r \le e - j$, to $\pm L^{e-j-r} x$, with a sign that depends only on j [And96b]; the sign plays no role below.

The statement (V) is the variational Hodge conjecture for algebraic classes; it asks nothing about Hodge classes, only that algebraicity, once present, survives deformation.

Lemma 6.15 (Kleiman)**.** *Let Z be a variety of dimension e with ample class L, and suppose that $B(Z)$ holds. Then the Künneth projectors π_k of $H^*(Z, \mathbb{Q})$, the projectors onto the summands $L^r P^j$ of the Lefschetz decomposition, and the Lefschetz involution $*$ are all induced by algebraic cycles on $Z \times Z$.*

Proof. Operators induced by algebraic cycles on $Z \times Z$ are closed under sums and rational multiples, and under composition, because the composite of two algebraic correspondences is again algebraic [Kle68]. The operator L is induced by the image of a divisor of class L under the diagonal, and Λ by $B(Z)$. With $[L, \Lambda] = H$ and H acting on H^k by $k - e$ (the proof of [I, Lemma 10.7]), $H = L\Lambda - \Lambda L$ is induced by an algebraic cycle, and since the numbers $k - e$ for $0 \le k \le 2e$ are distinct,

$$\pi_k = \prod_{i \neq k} \frac{H - (i - e)}{k - i}$$

is as well. Put $\Omega = H^2 + 2H + 4\Lambda L$. For x primitive of degree j put $m = e - j \ge 0$. Then $\Lambda x = 0$ and $Hx = -mx$, and the relation

$$\Lambda L^s x = s(m - s + 1)\, L^{s-1} x \qquad (s \ge 1) \tag{34}$$

follows by induction from $\Lambda L = L\Lambda - H$: the case $s = 1$ is [I, Lemma 10.7], and $\Lambda L^s x = L\Lambda L^{s-1} x - HL^{s-1} x = \big((s-1)(m-s+2) - (2s-2-m)\big) L^{s-1} x = s(m-s+1) L^{s-1} x$. So $\Omega x = (m^2 - 2m + 4m)x = m(m+2)x$, and Ω commutes with L, since $[L, H] = -2L$ and $[L, \Lambda L] = HL$ give $[L, \Omega] = -2(LH + HL) - 4L + 4HL = 0$; hence Ω acts on $L^r P^j$ by $m(m+2)$. Inside H^k the summands are $L^r P^{k-2r}$, with $m = e - k + 2r$, and $m \mapsto m(m+2)$ is injective on $m \ge 0$, so Lagrange interpolation in Ω composed with π_k gives the projector $p_{k,r}$ onto $L^r P^{k-2r}$ as an algebraic operator. Finally let $x \in P^j$ and $0 \le r \le m$. If $2r \le m$, then $L^{m-2r}(L^r x) = L^{m-r} x$. If $2r > m$, put $t = 2r - m$, so $1 \le t \le r$; by (34),

$$\Lambda^t L^r x = \prod_{i=0}^{t-1} (r - i)(m - r + i + 1)\; L^{m-r} x,$$

and every factor is at least 1, because $r - i \ge r - t + 1 \ge 1$ and $m - r + i + 1 \ge m - r + 1 \ge 1$. So on $L^r P^j$ the involution is $\pm L^{m-2r}$ or a nonzero rational multiple of Λ^{2r-m}, and

$* = \sum_{k,r} c_{k,r} T_{k,r}\, p_{k,r}$ with rational $c_{k,r}$ and $T_{k,r}$ a power of L or of Λ. Every term is algebraic. ∎

Proposition 6.16 (The conjecture is (L) and (M))**.** *The Hodge conjecture for every variety is equivalent to* (L) *and* (M) *together.*

Proof. Assume the conjecture. Then (L) holds by [I, Theorem 12.2]. An algebraic class ξ on X of dimension d is motivated: take Y a point, $\alpha = \xi$, and $\beta = \varepsilon h^d$ with h the ample class and $\varepsilon = \pm 1$ the sign for which $*(h^d) = \varepsilon$, which exists because the unit $1 \in H^0$ is primitive of degree 0; then $\alpha \cup *\beta = \xi$. Every Hodge class is algebraic, so (M) holds.

Conversely assume (L) and (M), and let ξ be a Hodge class on X. By (M), $\xi = \mathrm{pr}_*(\alpha \cup *\beta)$ with α, β algebraic on $X \times Y$. By (L), $B(X \times Y)$ holds, so $*$ is induced by an algebraic cycle (Lemma 6.15); hence $*\beta$, the product $\alpha \cup *\beta$ and its push-forward ξ are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [And96b, §2.1]. ∎

Proposition 6.17 (Two sources of (V))**.** *The Hodge conjecture implies* (V)*, and so does* (L)*.*

Proof. Let $f\colon \mathcal{X} \to S$, ξ and s be as in (V), with ξ_s algebraic. As S is smooth and connected it is irreducible, so two affine open subsets meet, and it suffices to treat S affine, joining s to an arbitrary t through a point of the intersection of an affine neighbourhood of each. Then $\mathcal{X}$ is quasi-projective and, by resolution of singularities, has a smooth projective compactification $\overline{\mathcal{X}}$.

Assume the conjecture. Write $n = 2p$ and $\rho\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^0(S^{\mathrm{an}}, R^n f_* \mathbb{Q})$ for restriction; it is surjective by the theorem of the fixed part [Del71, 4.1.1(ii)]. For $t \in S$ the restriction $r_t\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^n(\mathcal{X}_t, \mathbb{Q})$ is ρ followed by evaluation at t, and evaluation is injective because S is connected; so $K = \ker r_t = \ker \rho$ does not depend on t. It is a sub-Hodge structure, and the Hodge structure $H^n(\overline{\mathcal{X}}, \mathbb{Q})$ is polarisable, hence semisimple, so $H^n(\overline{\mathcal{X}}, \mathbb{Q}) = K \oplus K'$ with K' a sub-Hodge structure. Choose $\eta \in K'$ with $\rho(\eta) = \xi$. Then $r_s|_{K'}$ is an injective morphism of Hodge structures and $r_s(\eta) = \xi_s$ is of type (p,p), so every component of η of type other than (p,p) maps to zero and is zero: η is a Hodge class. So $\xi_t = r_t(\eta)$ is a Hodge class on $\mathcal{X}_t$ for every t, and algebraic by the conjecture.

Assume (L). The algebraic class ξ_s is motivated (proof of Proposition 6.16), so ξ_t is motivated for every t by André's deformation theorem [And96b, Théorème 0.5], and a motivated class is algebraic under (L) (proof of Proposition 6.16). ∎

Theorem 6.18 (Deligne, André; Abdulali, André)**.** (V) *implies the Hodge conjecture for every abelian variety, and so does* (L)*.*

Proof. Milne calls a class *accessible* if it lies in the least family of rational classes that contains the classes of algebraic cycles, is stable under pull-back by morphisms of varieties, and contains t_t for every t whenever $(t_s)_{s \in S}$ is a global section of $R^r f_* \mathbb{Q}$ for a smooth projective family over a smooth connected base and contains t_s for one s; and he proves, following Deligne and André, that every Hodge class on an abelian variety is accessible [Mil20, §5.1, Theorem 3]. The algebraic classes contain the classes of algebraic cycles and are stable under pull-back, and under (V) they have the third property. So they contain every accessible class, hence every Hodge class on an abelian variety; this is [Mil20, Remark 2]. The statement for (L) follows by Proposition 6.17, and is the theorem of Abdulali and André in the form of [Mil20, Theorem 4]. ∎

Remark 6.19 (The three in the literature). $B(X)$ is known for curves, where it is immediate, and for abelian varieties [Kle68, §2], and Corollary 6.22 below adds the total space of every Mumford family. (M) holds for the Hodge classes on abelian varieties [And96b, Théorème 0.6.2]. (V) holds in degree 2, where by the argument of Proposition 6.17 the class ξ_t is a Hodge class and the Lefschetz theorem on $(1,1)$ classes applies; the one general mechanism for it in higher degree is the semiregularity theorem [Blo72, BF03, Pri24], which is the mechanism of Theorem 2.3. Theorem 6.18 says what is at stake for this paper: the single statement (V), which the conjecture itself implies, gives the Weil classes of every CM field and the classes beyond the Weil lines at once, by a route that uses none of the theorems proved here.

6.7. **The Lefschetz standard conjecture for abelian schemes over curves.** The statement $B(X)$ is known for abelian varieties X. This subsection proves it for the total space of an abelian scheme over a curve whose invariant cycles are algebraic, and so for the total space of every Mumford family. The proof assembles the operator Λ from a relative part, which is a Pontryagin product and is algebraic on every abelian scheme, and a part along the base, which is algebraic exactly when the invariant cycles are.

Proposition 6.20 (Λ as a Pontryagin product)**.** *Let A be an abelian variety of dimension g, $\ell \in H^2(A,\mathbb{Q})$ the class of an ample line bundle, $D = \int_A \ell^g/g!$ and $\gamma = \ell^{g-1}/(g-1)!$. Let $\mu\colon A\times A \to A$ be the addition and $x \star y = \mu_*(p_1^*x \cup p_2^*y)$ the Pontryagin product. Then the operator Λ of* [I, Definition 10.6] *for the class ℓ is*

$$\Lambda(x) = D^{-1}\, x \star \gamma.$$

In particular Λ is induced by the algebraic cycle $D^{-1}(p_1,\mu)_*(p_2^*Z)$ on $A\times A$, for any cycle Z of class γ.

Proof. Put $V = H_1(A,\mathbb{Q})$, so that $H^*(A,\mathbb{Q}) = \bigwedge^* V^\vee$ under cup product and $H_*(A,\mathbb{Q}) = \bigwedge^* V$ under the Pontryagin product of homology, and let $\mathrm{PD}(x) = x \cap [A]$. Cap product makes H_* a module over H^*, and for a torus it is contraction; so $\mathrm{PD}(\ell \cup x) = \ell \cap \mathrm{PD}(x)$ is the contraction ι_ℓ applied to $\mathrm{PD}(x)$. Poincaré duality commutes with μ_* and carries $p_1^*x \cup p_2^*y$ to the cross product of $\mathrm{PD}(x)$ and $\mathrm{PD}(y)$, without sign when y has even degree, so $\mathrm{PD}(x\star\gamma) = \mathrm{PD}(x)\cdot\beta$ with $\beta = \mathrm{PD}(\gamma) \in \bigwedge^2 V$.

Choose a basis $x_1, y_1, \dots, x_g, y_g$ of $V^\vee$ with $\ell = \sum_i d_i\, x_i \wedge y_i$, $d_i > 0$, oriented so that $\int_A x_1\wedge y_1 \wedge \cdots \wedge x_g \wedge y_g = 1$; then $\int_A \ell^g = g!\, d_1 \cdots d_g$, so $D = d_1 \cdots d_g$. Let e_i, f_i be the dual basis of V. Then $\gamma = \sum_j \big(\prod_{i\neq j} d_i\big) \prod_{i\neq j} x_i \wedge y_i$ and $\beta = \varepsilon D \sum_j d_j^{-1} e_j \wedge f_j$ with $\varepsilon = \pm 1$ independent of j, since the permutations of the pairs preserve the orientation and exchange the terms. The algebra $\bigwedge^* V$ is the graded tensor product of the algebras $\bigwedge^*\langle e_i, f_i\rangle$, and ι_ℓ and multiplication by $D^{-1}\beta$ are sums over i of even operators on the i-th factor, namely $d_i\, \iota_{x_i\wedge y_i}$ and $\varepsilon d_i^{-1} e_i \wedge f_i$. Operators on different factors commute, and on the i-th factor, with basis $1, e_i, f_i, e_i\wedge f_i$ and $\iota_{x_i\wedge y_i}(e_i\wedge f_i) = \sigma$ for a sign σ fixed by the convention for contraction, the commutator $[\iota_\ell, D^{-1}\beta]$ acts by $\varepsilon\sigma$ on 1, by 0 on e_i and f_i, and by $-\varepsilon\sigma$ on $e_i\wedge f_i$, that is by $\varepsilon\sigma(1-k_i)$ in degree k_i. Summing over i, $[\iota_\ell, D^{-1}\beta] = \varepsilon\sigma(g-k)$ on $\bigwedge^k V$. As PD carries H^q to $\bigwedge^{2g-q} V$, the operator $\Lambda'(x) = D^{-1}x\star\gamma$ satisfies $[L,\Lambda'] = \varepsilon\sigma(q-g) = \varepsilon\sigma H$ on $H^q(A)$.

The sign is fixed by one value. For x and y of complementary degrees, $x\star y \in H^0(A)$ is $\int_{A\times A} p_1^*x \cup p_2^*y \cup \mu^*[\mathrm{pt}]$, and $\mu^*[\mathrm{pt}]$ is the class of the antidiagonal $\{(a,-a)\}$, so $x\star y = \int_A x \cup [-1]^* y$; for $y = \gamma$, of even degree, this is $\int_A x\cup\gamma$. Hence $\Lambda'(\ell) = D^{-1}\int_A \ell\cup\gamma = g$, while $\Lambda(\ell) = \Lambda(L\cdot 1) = g$ by [I, Lemma 10.7]. If $\varepsilon\sigma = -1$, then $-\Lambda'$ would be the operator of [I, Definition 10.6] and $\Lambda'(\ell) = -g$. So $\varepsilon\sigma = 1$, and $\Lambda' = \Lambda$ by the uniqueness

in [I, Definition 10.6]. Finally, the correspondence $(p_1,\mu)_*(p_2^*Z)$ acts by $x \mapsto p_{2*}\big(p_1^*x \cup (p_1,\mu)_*p_2^*[Z]\big) = \mu_*(p_1^*x \cup p_2^*\gamma)$, by the projection formula. ∎

Theorem 6.21 (The Lefschetz operator of an abelian scheme over a curve; cf. Tankeev [Tan03])**.** *Let C be a smooth projective connected curve and $\pi\colon W \to C$ an abelian scheme of relative dimension g with W projective. Suppose that every class in $H^*(W_c,\mathbb{Q})$ invariant under the monodromy of $R^*\pi_*\mathbb{Q}$ is the restriction of an algebraic class on W. Then there is an ample class L_W on W whose operator Λ is induced by an algebraic cycle on $W \times W$: $B(W)$ holds.*

Proof. Step 1: the decomposition. Let $[n]\colon W \to W$ be multiplication by n over C. It acts on $R^q\pi_*\mathbb{Q} = \bigwedge^q R^1\pi_*\mathbb{Q}$ by n^q. The Leray spectral sequence degenerates [Del71, 4.1.1(i)], and $[2]^*$ preserves the Leray filtration and acts on its graded pieces $H^p(C,R^q\pi_*\mathbb{Q})$ by 2^q. So $H^k(W,\mathbb{Q}) = \bigoplus_q H^k_{(q)}$, where $H^k_{(q)}$ is the 2^q-eigenspace of $[2]^*$ and maps isomorphically onto $H^{k-q}(C,R^q\pi_*\mathbb{Q})$; only $k-q \in \{0,1,2\}$ occurs. The projectors $P_q = \prod_{q'\neq q}([2]^* - 2^{q'})/(2^q - 2^{q'})$ are induced by algebraic cycles, polynomials in the graph of $[2]$. As $[2]^*$ is a ring homomorphism, $H_{(q)} \cup H_{(q')} \subset H_{(q+q')}$; as $\int_W [2]^*z = 2^{2g}\int_W z$, the integral vanishes on $H_{(q)}$ for $q \neq 2g$, so Poincaré duality pairs $H_{(q)}$ with $H_{(2g-q)}$ only; and $[2]_* = 2^{2g}([2]^*)^{-1}$ acts on $H_{(q)}$ by 2^{2g-q}.

Step 2: the ample class. Let $\mathcal{L}$ be a relatively ample line bundle, replaced by $\mathcal{L}\otimes[-1]^*\mathcal{L}$ so that it is symmetric. Then $[-1]^*$ fixes $c_1(\mathcal{L})$ and acts on $H_{(q)}$ by $(-1)^q$, so $c_1(\mathcal{L})$ has no component in $H^2_{(1)}$, while $H^2_{(0)} = H^2(C,\mathbb{Q})$ is spanned by the class f of a fibre. So $c_1(\mathcal{L}) = \ell + af$ with $\ell \in H^2_{(2)}$ algebraic and $a \in \mathbb{Q}$. Put $L_W = \ell + mf = c_1(\mathcal{L}) + (m-a)f$ with $m \in \mathbb{Q}$ so large that L_W is a positive rational multiple of the class of an ample line bundle, which is possible because $\mathcal{L}$ is relatively ample. Write L_{rel} and L_C for cup product with ℓ and with mf, so that $L_W = L_{\mathrm{rel}} + L_C$; on $H^p(C,R^q\pi_*\mathbb{Q})$ the first raises q by two and the second raises p by two.

Step 3: the relative operator. Put $\gamma = \ell^{g-1}/(g-1)!$, which is algebraic and lies in $H_{(2g-2)}$, and $D = \int_{W_c}\ell^g/g!$. Let $\mu,\mathrm{pr}_1,\mathrm{pr}_2\colon W \times_C W \to W$ be the addition and the projections, and $T(x) = \mu_*(\mathrm{pr}_1^*x \cup \mathrm{pr}_2^*\gamma)$. As in Proposition 6.20, T is induced by the algebraic cycle $(\mathrm{pr}_1,\mu)_*(\mathrm{pr}_2^*Z)$ on $W \times W$. Since $\pi\circ\mathrm{pr}_1 = \pi\circ\mu$, the projection formula gives $T(x\cup\pi^*b) = T(x)\cup\pi^*b$, so T commutes with L_C. Since $[2]\circ\mu = \mu\circ([2]\times_C[2])$, $[2]_*T(x) = \mu_*(\mathrm{pr}_1^*[2]_*x \cup \mathrm{pr}_2^*[2]_*\gamma)$, and for $x \in H_{(q)}$ this is $2^{2g-q}\,2^2\,T(x)$; so T maps $H_{(q)}$ into $H_{(q-2)}$. A relative correspondence acts on the Leray spectral sequence through the maps it induces on the local systems, so on $H^p(C,R^q\pi_*\mathbb{Q})$ the operator T is $H^p(C,T_\bullet)$, with T_c the Pontryagin product with $\gamma|_{W_c}$ on the fibre. By Proposition 6.20, $D^{-1}T_c$ is the operator Λ of the fibre, so $\Lambda_{\mathrm{rel}} = D^{-1}T$ satisfies $[L_{\mathrm{rel}},\Lambda_{\mathrm{rel}}] = H_{\mathrm{rel}}$, where H_{rel} acts on $H_{(q)}$ by $q-g$.

Step 4: the invariant classes. Restriction maps $H^q(W,\mathbb{Q})$ onto the monodromy invariants $I_q \subset H^q(W_c,\mathbb{Q})$ [Del71, 4.1.1(ii)], and it kills $H^q_{(q')}$ for $q' < q$, which lies in positive Leray degree; so it maps $H^q_{(q)}$ isomorphically onto I_q. Every element of $H^q_{(q)}$ is algebraic: if $v \in I_q$ is the restriction of an algebraic class z, then P_qz is algebraic, lies in $H^q_{(q)}$ and restricts to v. The subspace I_q is a sub-Hodge structure, as the image of a morphism of Hodge structures, and it is stable under cup product with ℓ_c and under the operator Λ of the fibre, which are the restrictions of L_{rel} and Λ_{rel}. The polarisation of $H^q(W_c,\mathbb{Q})$ attached to ℓ_c has the form $(v,w) \mapsto \int_{W_c} v\cup S(w)$, with S a polynomial in cup product with ℓ_c, the operator Λ and the projectors of the fibre, and its restriction to the sub-Hodge structure I_q is again a polarisation [Voi02]. Since $S(I_q) \subset I_{2g-q}$, the pairing $I_q \times I_{2g-q} \to \mathbb{Q}$ is nondegenerate.

Step 5: the operator along the base. For $y \in H^q_{(q)}$ and $w \in H^{2g-q}_{(2g-q)}$ one has $\int_W (y \cup f) \cup w = \int_{W_c} y|_{W_c} \cup w|_{W_c}$. By Step 4 and Step 1 this shows that $L_C \colon H^q_{(q)} \to H^{q+2}_{(q)}$ is injective, and it is onto because $\dim H^{q+2}_{(q)} = \dim H^{2g-q}_{(2g-q)} = \dim I_{2g-q} = \dim I_q$, by Poincaré duality on W and on the invariants. Choose bases (a_j) of $H^q_{(q)}$ and (b_k) of $H^{2g-q}_{(2g-q)}$, all algebraic by Step 4, let $M_{jk} = \int_W a_j \cup mf \cup b_k$, which is invertible, and put

$$\Lambda_C(x) = \sum_{j,k} (M^{-1})_{kj} \Big(\int_W x \cup b_k \Big) a_j \qquad (x \in H^{q+2}(W, \mathbb{Q})),$$

for every q. It is induced by the algebraic cycle $\sum_{j,k} (M^{-1})_{kj}\, b_k \times a_j$. By Step 1, $\int_W x \cup b_k$ vanishes for $x \in H^{q+2}_{(q')}$ with $q' \neq q$, so Λ_C is zero on $H^p(C, R^q\pi_*\mathbb{Q})$ for $p \neq 2$, and $\Lambda_C(L_C a_i) = a_i$. Hence $[L_C, \Lambda_C]$ is -1 on $H^q_{(q)}$, 0 on $H^{q+1}_{(q)}$ and $+1$ on $H^{q+2}_{(q)}$: it is the operator H_C that acts on $H^p(C, R^q\pi_*\mathbb{Q})$ by $p - 1$.

Step 6: the cross relations and the conclusion. $[L_C, \Lambda_{\mathrm{rel}}] = 0$ by Step 3. Both $L_{\mathrm{rel}}\Lambda_C$ and $\Lambda_C L_{\mathrm{rel}}$ vanish off the summands with $p = 2$, and on $L_C y$ with $y \in H^q_{(q)}$ both give $\ell \cup y$, because $\ell \cup y \in H^{q+2}_{(q+2)}$ and L_C commutes with L_{rel}; so $[L_{\mathrm{rel}}, \Lambda_C] = 0$. Put $\Lambda = \Lambda_{\mathrm{rel}} + \Lambda_C$, of degree -2. Then

$$[L_W, \Lambda] = [L_{\mathrm{rel}}, \Lambda_{\mathrm{rel}}] + [L_C, \Lambda_C] = H_{\mathrm{rel}} + H_C,$$

which acts on $H^p(C, R^q\pi_*\mathbb{Q}) \subset H^{p+q}(W)$ by $(q - g) + (p - 1) = (p + q) - \dim W$. So Λ is the operator of [I, Definition 10.6] for L_W, by its uniqueness, and it is induced by an algebraic cycle by Steps 3 and 5. ∎

Figure 19 draws the summands of the proof in relative dimension four.

Corollary 6.22 (Mumford families)**.** *Let $\pi \colon W \to C$ be a non-isotrivial abelian scheme of relative dimension four over a smooth projective connected curve, W projective, whose very general fibre X is a fourfold as in Proposition 6.74, with Hodge group the group G of Mumford's construction* [Mum69]*, the restriction of scalars from a totally real cubic field F of the norm one group of a quaternion algebra over F. Then $B(W)$ holds. Such families exist, over the compact Shimura curves of* [Mum69].

Proof. By André's theorem [And92, Theorem 1] the connected monodromy group is a normal subgroup of the derived Mumford–Tate group of X, which is G. It is not trivial, because the family is not isotrivial, and G is $\mathbb{Q}$-simple, since over $\overline{\mathbb{Q}}$ its three factors are indexed by the embeddings of F and permuted transitively by the Galois group; so it is G. The monodromy invariants in $H^*(X, \mathbb{Q})$ are therefore contained in the invariants of G, which are the Hodge classes of X, and these are the powers of the polarisation by Proposition 6.74(i). The powers of ℓ restrict to the powers of the polarisation, so every invariant class is the restriction of an algebraic class, and Theorem 6.21 applies. ∎

Remark 6.23 (Scope of the argument). The total space W is a fivefold fibred in abelian fourfolds whose Hodge groups are not defined by endomorphisms; it is not a product and not an abelian variety. For abelian schemes over curves the mechanism of Theorem 6.21 goes back to Tankeev [Tan03], who proves B for the total space when certain canonically defined Hodge cycles are algebraic; the argument here does not use his hypotheses, and we make no claim of priority for Corollary 6.22. The difficulty lies in the hypothesis of Theorem 6.21. Applied to the fibre square $W \times_C W \to C$ the theorem gives $B(W \times_C W)$ as soon as every invariant class of $X \times X$ is the restriction of an algebraic class on $W \times_C W$;

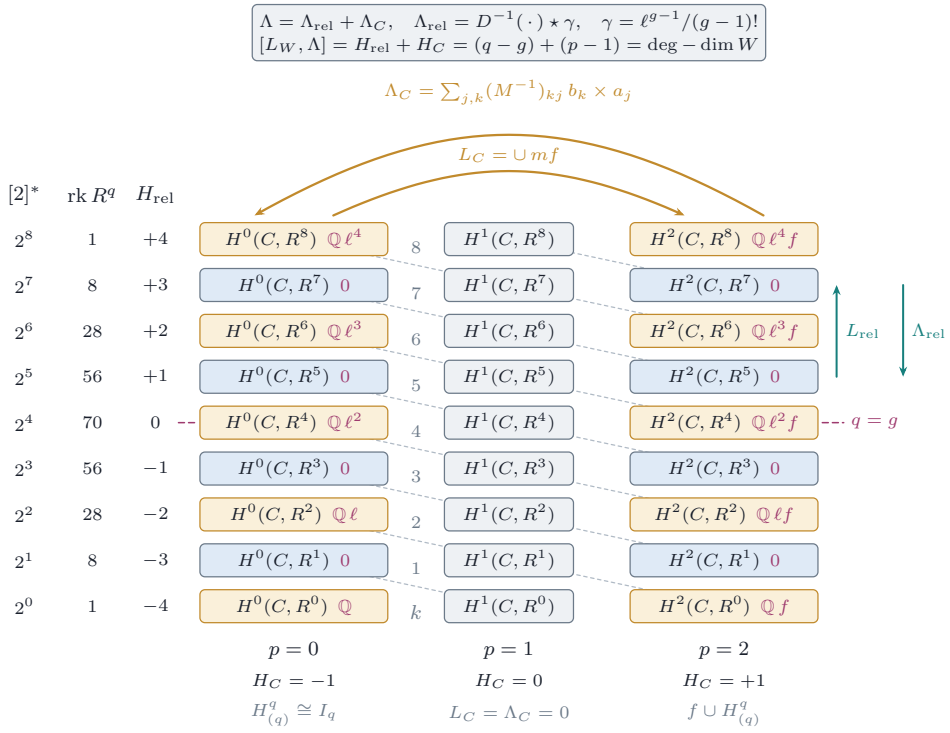


FIGURE 19. The proof of Theorem 6.21 for relative dimension four. The boxes are the summands $H^p(C, R^q\pi_*\mathbb{Q})$ of $H^*(W, \mathbb{Q})$, split off by the eigenvalue 2^q of $[2]^*$ shown on the left; the dotted diagonals join the summands of one $H^k(W, \mathbb{Q})$, $k = p + q$, and the dashed magenta line is the axis $q = g$ of hard Lefschetz on the fibres. The relative operators L_{rel} and Λ_{rel}, in teal, move within a column; Λ_{rel} is a Pontryagin product with an algebraic class on every abelian scheme (Proposition 6.20). The base operators L_C and Λ_C, in ochre, move along a row between the outer columns and kill the middle one; Λ_C is the displayed cycle built from bases (a_j), (b_k) of the invariant summands, and is algebraic exactly when the invariant cycles are. Below each column is the weight of $H = [L_W, \Lambda]$ on it, and the panel at the top is Step 6 of the proof. The ochre boxes carry the invariant cycles and their images under L_C; for a Mumford family they are the lines listed in magenta on the right, spanned by $\ell^i f$ (and by ℓ^i in the left column), and the other boxes of the outer columns are zero (Corollary 6.22).

the invariant classes include the two exceptional classes of Proposition 6.74, so this is the smallest case of (F2) in a global form. Theorem 6.21 does not bear on (P2), on the Kuga–Satake class of Theorem 7.4, or on (L) for varieties that are not fibred in abelian varieties over a curve.

Proposition 6.24 (Powers of a Weil variety; cf. [Mil22])**.** *Let X be of $(K,-1,n)$-Weil type with $n \geq 2$ and $\mathrm{Hg}(X) = \mathrm{SU}(V,H)$. For every $k \geq 1$ the $\mathbb{Q}$-algebra $\mathrm{Hdg}^*(X^k)$ is generated by the divisor classes of X^k and the pull-backs $f_a^*\omega$ of the Weil classes ω of X along the homomorphisms $f_a = \sum_j a_j\, pr_j \colon X^k \to X$, $a \in \mathbb{Z}^k$. Consequently, if the Weil classes of X are algebraic, the Hodge conjecture holds for every power of X. By [Mar25a] this is so for every such X with $n = 2$, and with $n = 3$ and trivial discriminant.*

Proof. Write $H^1(X,\mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of K and $W = V_+$. The polarisation makes V_+ and V_- isotropic and pairs them perfectly, so $V_- = W^\vee$, and restriction to V_+ identifies $\mathrm{SU}(V,H)_{\mathbb{C}}$ with $\mathrm{SL}(W)$. As $\mathrm{Hg}(X^k)$ is $\mathrm{Hg}(X)$ acting diagonally on $H^1(X^k,\mathbb{C}) = (W\otimes U)\oplus(W^\vee\otimes U)$, $U = \mathbb{C}^k$, the Hodge classes of X^k, tensored with $\mathbb{C}$, are the $\mathrm{SL}(W)$-invariants in $\bigwedge^*\big((W\otimes U)\oplus(W^\vee\otimes U)\big)$. That algebra is an $\mathrm{SL}(W)$-equivariant quotient of a tensor algebra, with an equivariant section given by antisymmetrisation, so Weyl's first fundamental theorem for $\mathrm{SL}(W)$, in its tensor form and applied before antisymmetrisation, shows that its invariants are generated by the images of the contractions and the determinant tensors: the classes $D(u,u') = \sum_i (e_i\otimes u)\wedge(e_i^\vee\otimes u')$ and the polarised determinants, which by polarisation are combinations of $\det_u = \bigwedge_i (e_i\otimes u)$ and $\det_u^\vee = \bigwedge_i (e_i^\vee\otimes u)$, $u,u' \in U$. For $2n \geq 3$ the $D(u,u')$ span the invariants of degree two, which are the divisor classes tensored with $\mathbb{C}$. With $\alpha_\pm$ spanning $\bigwedge^{2n} V_\pm$ ([I, Proposition 3.7]) one has $f_a^*\alpha_+ = \det_a$ and $f_a^*\alpha_- = \det_a^\vee$, and since $u \mapsto \det_u$ is polynomial and $\mathbb{Z}^k$ is Zariski dense in U, these span the same space as all $\det_u$. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}(X^k)_{\mathbb{Q}}$ and the $f_a^*\omega$ is contained in $\mathrm{Hdg}^*(X^k)$ and has the same complexification. The generators are algebraic when the Weil classes of X are, and they are for $n = 2$ and for $n = 3$ at trivial discriminant by [Mar25a]. ∎

The statement for powers is known (compare [Mil22], and [Li26] for the powers of CM fourfolds); we include the short proof because the next corollary uses it.

Corollary 6.25 (Abelian schemes of small relative dimension)**.** *Let $\pi \colon W \to C$ be an abelian scheme of relative dimension g over a smooth projective connected curve, with W projective, and suppose that every monodromy invariant class in $H^*(W_c,\mathbb{Q})$ is a Hodge class, as happens when the identity component of the Zariski closure of the monodromy group is the Hodge group of a very general fibre.*

(i) *If $g \leq 4$, then $B(W)$ holds.*

(ii) *If the fibres are of $(K,-1,n)$-Weil type with connected monodromy group $\mathrm{SU}(V,H)$, and $n = 2$, or $n = 3$ with trivial discriminant, then B holds for every fibre power $W \times_C \cdots \times_C W$. In particular $B(W_C)$ of Corollary 6.31 holds in these cases.*

Proof. By the theorem of the fixed part the invariant classes form a sub-Hodge structure of every fibre, with a Hodge structure independent of the point, so they are Hodge classes on every fibre.

(i) The Hodge conjecture holds for abelian varieties of dimension at most four, by the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz up to dimension three and by [Mar25a] in dimension four, so each invariant class is algebraic on every fibre; Lemma 6.27 makes it the restriction of an algebraic class on W, and Theorem 6.21 applies.

(ii) The fibre power is an abelian scheme with projective total space and the monodromy acting diagonally. Its invariant classes are $\mathrm{SU}(V,H)$-invariants, hence Hodge classes at every member with $\mathrm{Hg} = \mathrm{SU}(V,H)$, an uncountable set of points, and there the Hodge conjecture holds for the fibre by Proposition 6.24; Lemma 6.27 and Theorem 6.21 conclude as before. For W_C the connected monodromy group is $\mathrm{SU}(V,H)$ when it is Zariski dense, which is the hypothesis of Corollary 6.31(ii). ∎

The hypothesis on the monodromy in Corollary 6.25 can be checked with André's theorem [And92, Theorem 1]: the identity component of the Zariski closure of the monodromy group is always a normal subgroup of the derived Mumford–Tate group of a very general fibre, so the hypothesis holds whenever the family is not isotrivial and the Hodge group of a very general fibre is semisimple and $\mathbb{Q}$-simple, as for the Mumford families of Corollary 6.22.

Proposition 6.26 (The total space of a Mumford family)**.** *Let $W \to C$ be as in Corollary 6.22, with C the compact Shimura curve of* [Mum69] *of torsion-free level, or a finite étale cover of it. Then the Hodge conjecture holds for the fivefold W: its Hodge classes of degree $2p$ are spanned by ℓ^p and $f\ell^{p-1}$, with f the class of a fibre.*

Proof. By Step 1 of the proof of Theorem 6.21, $H^k(W,\mathbb{Q}) = \bigoplus_q H^k_{(q)}$ with $H^k_{(q)} \cong H^{k-q}(C, R^q\pi_*\mathbb{Q})$, the summands being sub-Hodge structures cut out by algebraic projectors. The summand with $k = q$ is the space of invariants, which by the proof of Corollary 6.22 is spanned by the powers of the polarisation, restrictions of the powers of ℓ. The summand with $k = q+2$ is $f \cup H^q_{(q)}$ by Step 5, spanned by the $f\ell^{q/2}$. It remains to see that $H^1(C, R^q\pi_*\mathbb{Q})$ carries no Hodge class. For q even its weight $q+1$ is odd. For q odd, over $\mathbb{C}$ the local system splits as a sum of $S^{k_1}V_1 \otimes U_\lambda \otimes M_\lambda$, where V_1 is the uniformising variation of the factor of G split at the real place, U_λ is a unitary variation of type $(0,0)$, M_λ is constant of type (a,a) with $2a + k_1 = q$, and $k_1 \equiv q \pmod 2$ because the central element $(-1,1,1)$ acts on V by -1. When C is étale over the Shimura curve the Kodaira–Spencer map of V_1 is an isomorphism, and then $H^1(C, S^{k_1}V_1 \otimes U_\lambda)$ has only the types $(k_1+1,0)$ and $(0,k_1+1)$ [Zuc79]; after the twist by M_λ these are never balanced, since $k_1 \geq 1$. ∎

Statements of the kind of Proposition 6.26 are studied for Kuga fibre varieties, and we make no claim of priority. The étale hypothesis is used only at the end of the proof, and it cannot be dropped from the argument: over a ramified cover the Kodaira–Spencer map of V_1 vanishes at the branch points, the Higgs field of $S^{k_1}V_1$ is no longer an isomorphism, and the torsion of its cokernel contributes to $H^1(C, R^q\pi_*\mathbb{Q})$ in balanced types.

6.8. **The two targets as cases of (L).** The argument of Section 6.7 does more than add a case of (L). For an abelian scheme over a curve it shows that (L) for the total space is exactly the statement that algebraicity propagates along the curve. The base points of this paper then turn each of the two targets of Remark 6.73, one Weil family and the first classes beyond the Weil lines, into (L) for one explicit variety. The reductions are equivalences, so they locate the difficulty rather than remove it. Along the way the Hodge conjecture is proved for the self-product of a Mumford fourfold at every CM point (Theorem 6.29), and the two branches of Remark 2.10 are shown not to be realisable by two objects (Theorem 6.32), an implication whose hypothesis Theorem 3.5 shows is never met.

Lemma 6.27 (Spreading along a curve)**.** *Let $\pi\colon W \to C$ be a smooth projective morphism onto a smooth projective connected curve, W projective, and ξ a global section of $R^{2p}\pi_*\mathbb{Q}$.*

If ξ_t is algebraic for uncountably many $t \in C$, there are an algebraic class $\tilde{\xi}$ on W and an integer $d \geq 1$ with $\tilde{\xi}|_{W_t} = d\,\xi_t$ for every $t \in C$.

Proof. The proof of [I, Theorem 16.4] applies verbatim with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by (W, C, ξ): the set of t at which ξ_t is algebraic is the union, over countably many tuples of Hilbert polynomials and rational coefficients, of the images of unions T of connected components of fibre products of relative Hilbert schemes, projective over C, on which $\sum_i q_i[Z_i] = \xi$. Uncountably many t force one image to be infinite, hence equal to C. Choose an irreducible component of that T surjecting onto C, two of its points over distinct points of C, and an irreducible curve through them; its normalisation C' maps finitely onto C, of some degree d. Pulling the universal subschemes back to $W' = W \times_C C'$ gives a cycle $z = \sum_i q_i[\mathcal{Z}_i]$ whose restriction to $W'_{c'} = W_{g(c')}$ is $\xi_{g(c')}$ for every $c' \in C'$. The projection $h\colon W' \to W$ is finite and flat of degree d, and by base change $h_*[z]$ restricts on W_t to the sum of the restrictions over the points of C' above t, counted with multiplicity, which is $d\,\xi_t$. ∎

Theorem 6.28 ((L) for an abelian scheme over a curve is propagation)**.** *Let $\pi\colon W \to C$ be as in Theorem 6.21, but with no hypothesis on its invariant cycles, and keep the notation of its proof. Let Λ_C be the operator that is zero on $H^p(C, R^q\pi_*\mathbb{Q})$ for $p \neq 2$ and inverts $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$.*

(i) *The operator Λ of* [I, Definition 10.6] *for L_W is $\Lambda_{\mathrm{rel}} + \Lambda_C$. So $B(W)$ holds for L_W if and only if Λ_C is induced by an algebraic cycle.*

(ii) *If $B(W)$ holds for L_W, then for every $c \in C$ and every algebraic class y on W_c the class $\Lambda_C(i_{c*}y)$ is algebraic; up to the factor m it lies in $H^0(C, R^q\pi_*\mathbb{Q})$ and restricts on W_c to the unique monodromy invariant class v with $\int_{W_c} v \cup w = \int_{W_c} y \cup w$ for every invariant w of complementary degree. In particular every global section of $R^{2p}\pi_*\mathbb{Q}$ that is algebraic at one point is the restriction of an algebraic class on W, and is algebraic at every point.*

(iii) *If at one point c_0 every monodromy invariant class of $H^*(W_{c_0}, \mathbb{Q})$ is algebraic, then $B(W)$ holds for L_W if and only if every monodromy invariant class is the restriction of an algebraic class on W.*

Proof. (i) Steps 4 and 5 of the proof of Theorem 6.21 show that $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$ is an isomorphism without using the hypothesis on invariant cycles: the nondegeneracy of $I_q \times I_{2g-q} \to \mathbb{Q}$ comes from the Hodge–Riemann relations alone. So Λ_C is defined, and Steps 5 and 6, which use only that Λ_C vanishes off $p = 2$ and inverts L_C, give $[L_W, \Lambda_{\mathrm{rel}} + \Lambda_C] = H$. By the uniqueness in [I, Definition 10.6] this is Λ. As Λ_{rel} is induced by an algebraic cycle (Step 3), Λ is if and only if Λ_C is.

(ii) The Gysin map i_{c*} commutes with $[2]_*$, which acts on $H^q(W_c)$ by 2^{2g-q}, so $i_{c*}y \in H^{q+2}_{(q)}$, the summand with $p = 2$. If y is algebraic so is $i_{c*}y$, and by (i) so is $\Lambda_C(i_{c*}y)$. Put $u = m\,\Lambda_C(i_{c*}y) \in H^q_{(q)}$, so $f \cup u = i_{c*}y$. For $w \in H^{2g-q}_{(2g-q)}$ the projection formula gives $\int_{W_c} u|_{W_c} \cup w|_{W_c} = \int_W (f \cup u) \cup w = \int_W i_{c*}y \cup w = \int_{W_c} y \cup w|_{W_c}$, and $w|_{W_c}$ runs over the invariants of complementary degree (Step 4); by the nondegeneracy of Step 4 this determines $u|_{W_c}$. If $y = \xi_c$ for a global section ξ, then $v = \xi_c$ has the property, so $u|_{W_c} = \xi_c$, and u, being a global section of $R^q\pi_*\mathbb{Q}$ equal to ξ at c, restricts to ξ_t at every t.

(iii) If every invariant class is a restriction of an algebraic class, $B(W)$ holds by Theorem 6.21. Conversely, under $B(W)$, (ii) applied at c_0 to each invariant class makes it the restriction of an algebraic class. ∎

Figure 20 draws part (ii) along a compact Shimura curve.

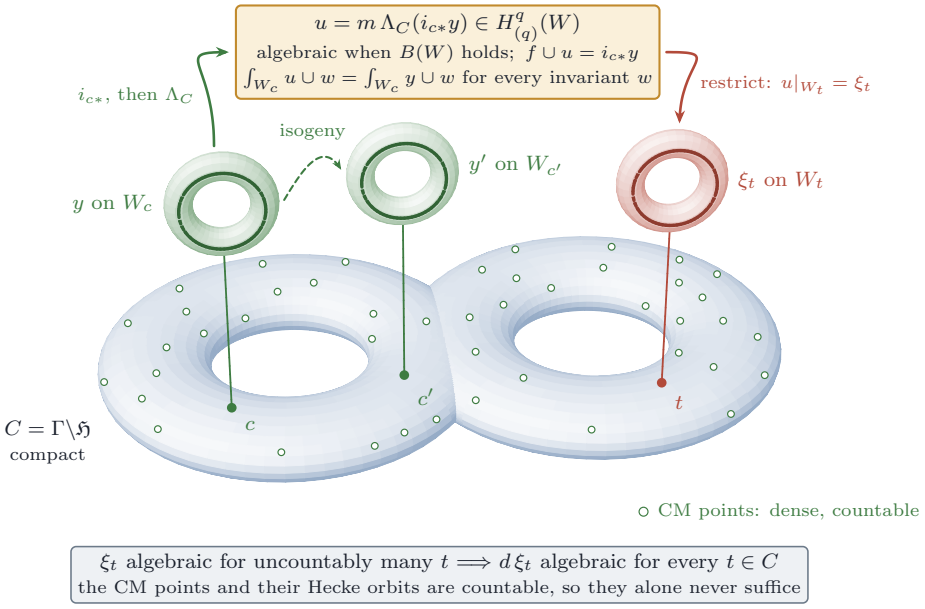


FIGURE 20. Theorem 6.28(ii) along a compact Shimura curve C. An algebraic class y on one fibre, at a base point of a Weil family or at a CM point of a Mumford family (hollow dots: the CM points, countable and dense), is pushed into W by the Gysin map and brought back to the invariant summand by the base operator Λ_C. When $B(W)$ holds the result u is an algebraic class on W, pinned down by the displayed pairing against the invariant classes, and its restriction to a very general fibre is the invariant part of y. The dashed arcs are isogenies between CM members along Hecke orbits: they carry algebraic classes to algebraic classes but reach only countably many points, and Lemma 6.27, in the box, needs uncountably many. This is how Corollary 6.31 and Corollary 6.30 turn each target into one case of (L).

Theorem 6.29 (The Mumford square at a CM point)**.** *Let $W \to C$ be a family as in Corollary 6.22 and $c \in C$ a CM point. Then every Hodge class on $X_c \times X_c$ is algebraic. In particular the two exceptional classes of Proposition 6.74 are algebraic on $X_c \times X_c$.*

Proof. The Hodge group T of X_c is a $\mathbb{Q}$-torus contained in G. It lies in a maximal $\mathbb{Q}$-torus of $G = \mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{SL}_1(D)$, which is the restriction of scalars of the norm one torus of a quadratic extension of F inside D, of rank three, one factor for each embedding of F. The cocharacter of the Hodge structure lies in the factor of the embedding at which D splits, and the Galois group permutes the embeddings of F transitively, so the smallest $\mathbb{Q}$-subtorus containing it has rank three: $T_{\mathbb{C}}$ is a maximal torus of $G_{\mathbb{C}} \cong \mathrm{SL}_2^3$. Its eight weights on $V = H^1(X_c, \mathbb{C}) = V_1 \otimes V_2 \otimes V_3$ are the vectors $(\pm 1, \pm 1, \pm 1)$, pairwise distinct. Hence $\mathrm{End}^0(X_c) \otimes \mathbb{C} = \mathrm{End}_T(V)$ is the algebra of diagonal matrices in a weight basis (x_w). For $n \geq 1$, $\mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ is the space of n-tuples $(\varphi_1, \ldots, \varphi_n)$ of such matrices, acting by $x_w \mapsto x_w \otimes u_w$ with $u_w = (\varphi_{1,w}, \ldots, \varphi_{n,w}) \in \mathbb{C}^n$, so the vectors u_w can be prescribed independently. The Hodge classes of X_c^n, tensored with $\mathbb{C}$, are spanned by the products of vectors $x_w \otimes u$, $u \in \mathbb{C}^n$, in $\bigwedge^*(V \otimes \mathbb{C}^n)$ in which the weights w, counted with multiplicity, sum to zero.

We show that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of X_c along homomorphisms $X_c^n \to X_c$. Put $w_1 = (1,1,1)$, $w_2 = (1,-1,-1)$, $w_3 = (-1,1,-1)$ and $w_4 = (-1,-1,1)$; the weights are the $\pm w_i$, $w_1 + w_2 + w_3 + w_4 = 0$, and any three of the w_i are linearly independent, so the integer relations among them are the multiples of $(1,1,1,1)$. Let α_i and β_i be the multiplicities of w_i and $-w_i$ in a nonempty multiset of weights with sum zero. Then $\alpha_i - \beta_i$ is independent of i; so either some α_i and β_i are both positive, or all α_i are positive, or all β_i are. In each case the multiset contains one of the sets $\{w_i, -w_i\}$, $\{w_1, w_2, w_3, w_4\}$, $\{-w_1, -w_2, -w_3, -w_4\}$, each of sum zero, and by induction every product above is, up to sign, a product of monomials $\bigwedge_{w \in P}(x_w \otimes u_w)$ over such sets P, whose weights are pairwise distinct. Choosing $f \in \mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ with $f^* x_w = x_w \otimes u_w$ for $w \in P$ gives $\bigwedge_{w \in P}(x_w \otimes u_w) = f^*(\bigwedge_{w \in P} x_w)$, with $\bigwedge_{w \in P} x_w$ a Hodge class of X_c tensored with $\mathbb{C}$. The map $f \mapsto f^*(t)$ is polynomial and the rational homomorphisms are Zariski dense in their complexification, so the same span is reached with f defined over $\mathbb{Q}$ and t rational. The Hodge classes of the abelian fourfold X_c are algebraic because the Hodge conjecture holds for abelian fourfolds [Mar25a], and pull-backs and products of algebraic classes are algebraic. The case $n = 2$ is the theorem. ■

Corollary 6.30 (The first classes beyond the Weil lines are one case of (L))**.** *Let $W \to C$ be as in Corollary 6.22 and $W^{(2)} = W \times_C W \to C$, an abelian scheme of relative dimension eight with $W^{(2)}$ projective of dimension nine. The following are equivalent.*

(a) *$B(W^{(2)})$ holds for the class $L_{W^{(2)}}$.*
(b) *The two exceptional classes of $X_t \times X_t$ are algebraic for uncountably many $t \in C$.*
(c) *The Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*
(d) *The Hodge conjecture holds for $W^{(2)}$; for this item C is taken to be the compact Shimura curve of torsion-free level, or a finite étale cover of it.*

Proof. As in the proof of Corollary 6.22, the connected monodromy group is G, so the monodromy invariants in $H^*(X_t \times X_t, \mathbb{Q})$ are contained in the G-invariants, which are the Hodge classes of $X_t \times X_t$ for t very general; conversely a G-invariant class is flat, being fixed by the monodromy. The Hodge group of every fibre lies in G, so the invariant classes are Hodge classes on every fibre. At a point t that is not a CM point the Hodge group H_t of X_t is G. Indeed, if H_t were smaller, the Hodge classes of the powers of X_t that are not G-invariant would stay Hodge only on a discrete subset of the period domain, the upper half plane; every point of the orbit of the Hodge structure at t under $H_t(\mathbb{R})^+$ carries them, so that orbit is a point, the cocharacter of the Hodge structure is central in H_t, and H_t, the smallest $\mathbb{Q}$-group containing it, is a torus: t would be a CM point. So at such t the Hodge classes of $X_t \times X_t$ are the invariant ones, and $\mathrm{End}^0(X_t) = \mathrm{End}_G(V) = \mathbb{Q}$, because $V \otimes \bar{\mathbb{Q}} = V_1 \otimes V_2 \otimes V_3$ is an irreducible representation of G; thus X_t is as in Setup 7.1.

(c) implies (b) trivially. (b) implies (c): the exceptional classes are invariant, hence flat, so by Lemma 6.27 a multiple of each is the restriction of an algebraic class on $W^{(2)}$, and they are algebraic on every fibre. At a point that is not a CM point, Theorem 7.7 with $n = 2$ makes every Hodge class of $X_t \times X_t$ algebraic; at a CM point this is Theorem 6.29. (c) implies (a): by (c) every invariant class is algebraic on every fibre, so by Lemma 6.27 a multiple of it is the restriction of an algebraic class on $W^{(2)}$, and Theorem 6.21 gives $B(W^{(2)})$. (a) implies (c): at a CM point c_0 every Hodge class of $X_{c_0} \times X_{c_0}$ is algebraic by Theorem 6.29, so by Theorem 6.28(iii) every invariant class is the restriction of an algebraic class, and is algebraic on every fibre. This is (c) at the points that are not CM points, where the Hodge classes are the invariant ones; at a CM point (c) is Theorem 6.29.

For (d), split $H^*(W^{(2)},\mathbb{Q})$ into the summands $H^p(C,R^q)$, $p\in\{0,1,2\}$, as in the proof of Proposition 6.26. The summands with $p=1$ carry no Hodge class, by the argument given there, applied to the fibre square. Those with $p=2$ are $f\cup H^q_{(q)}$, and $f\cup u=i_{c*}(u|_{W_c^{(2)}})$ for any c; taking c a CM point, where $u|_{W_c^{(2)}}$ is a Hodge class and so algebraic by Theorem 6.29, they are algebraic. The summands with $p=0$ are the invariants, all of Hodge type. So (d) holds exactly when every invariant class is the restriction of an algebraic class, which by Theorem 6.28(iii), applied at a CM point, is (a). ∎

Corollary 6.31 (One Weil family is one case of (L))**.** *Keep* [I, Setup 16.1]*, let $s_0\in\mathcal{S}$ be the image of a base point of* [I, Theorem 15.27]*, and let $t_1\in\mathcal{S}$ lie outside every proper Zariski closed set $\Sigma_{\underline{P},\underline{q}}$ of* [I, Theorem 16.4]*. Let $\kappa\colon C\to\mathcal{S}$ be a morphism from a smooth projective connected curve whose image contains s_0 and t_1, and $W_C=\mathcal{A}\times_{\mathcal{S}}C$.*

(i) *If $B(W_C)$ holds for L_{W_C}, then $\Sigma=\mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with $\mathrm{Hg}(A)=\mathrm{SU}(V,H)$.*

(ii) *Conversely, if $\Sigma=\mathcal{D}_{n,n}$ and the monodromy of W_C is Zariski dense in $\mathrm{SU}(V,H)$, then $B(W_C)$ holds for L_{W_C}.*

Proof. (i) The class $\kappa^*\omega$ is a global section of $R^{2n}\pi_*\mathbb{Q}$ on C, algebraic at a point over s_0 by [I, Theorem 15.27]. By Theorem 6.28(ii) it is algebraic at a point over t_1, so t_1 lies in some closed set $\Sigma_{\underline{P},\underline{q}}$, which is therefore all of $\mathcal{S}$. [I, Theorem 16.4] gives $\Sigma=\mathcal{D}_{n,n}$, and [I, Theorem 15.14(a)] the rest. (ii) With dense monodromy the invariants are the $\mathrm{SU}(V,H)$-invariants, which are the Hodge classes of a member with $\mathrm{Hg}(A)=\mathrm{SU}(V,H)$: the powers of the polarisation and the Weil classes, by [I, Theorem 4.14]. All are algebraic on every fibre when $\Sigma=\mathcal{D}_{n,n}$, so they are restrictions of algebraic classes by Lemma 6.27, and Theorem 6.21 applies. ∎

Curves κ as in Corollary 6.31 exist: the Baily–Borel compactification of $\mathcal{S}$ is projective [BB66], its boundary has codimension $2n-1\geq 3$, the rational boundary components of the domain of type $\mathrm{I}_{n,n}$ being of type $\mathrm{I}_{n-r,n-r}$ with $r\geq 1$, and a general complete intersection curve through s_0 and t_1 avoids it.

Theorem 6.32 (Two branches cannot be two objects)**.** *Let A be of $(K,-1,n)$-Weil type with $n\geq 2$, $\omega\neq 0$ a rational class on the Weil line, and E a perfect complex with $\mathrm{ch}(E)=N\omega$, $N\neq 0$, and σ_E injective. Let $E\cong\bigoplus_i F_i[k_i]$ be any decomposition. Then every summand F_i with $\mathrm{ch}(F_i)\neq 0$ has a Yoneda product $\bigwedge^2\mathrm{Ext}^1(F_i,F_i)\to\mathrm{Ext}^2(F_i,F_i)$ that is not injective. In particular E is not a direct sum of shifts of line bundles, of skyscraper sheaves of points, or of their images under autoequivalences of $D^b(A)$, since the Ext algebra of each of these is the exterior algebra on its Ext^1.*

Proof. By Theorem 2.11(ii) the kernel of ev_E in degree two is $P\wedge Q$. The characteristic morphism is compatible with the decomposition, so $\mathrm{ev}_{F_i}(p\wedge q)=\mathrm{ev}_{F_i}(p)\,\mathrm{ev}_{F_i}(q)=0$ for all $p\in P$ and $q\in Q$. Suppose the Yoneda product of $F=F_i$ is injective on $\bigwedge^2\mathrm{Ext}^1$. Then the images $\bar P$ and $\bar Q$ of P and Q in $\mathrm{Ext}^1(F,F)$ satisfy $\bar p\wedge\bar q=0$ for all $\bar p,\bar q$, so either $\bar P=0$, or $\bar Q=0$, or $\bar P=\bar Q$ is a line. In the first case ev_F kills $P\wedge HH^1$, in the second $Q\wedge HH^1$, in the third all of $HH^2=HH^1\wedge HH^1$; by the compatibility $\sigma_F\circ\mathrm{ev}_F=(\,\cdot\,)\lrcorner\,\mathrm{ch}(F)$ of Setup 2.7 the same subspace annihilates $x=\mathrm{ch}(F)$.

These annihilators are computed in the free module of rank one that $H^*(A,\mathbb{C})$ is over $HH^*(A)=\bigwedge^*(P\oplus Q)$, with generator the holomorphic volume form Ω: an element $y\cdot\Omega$ is killed by $P\wedge HH^1$ exactly when $p\wedge y$ has top degree for every $p\in P$, that is when $y\in\det P\wedge\bigwedge^*Q+\bigwedge^{\geq 4n-1}(P\oplus Q)$. Since $\det P\cdot\Omega=\pm\alpha_+$, the first summand gives the

classes $\bigwedge^j Q \cdot \alpha_+$, of type $(n-b, n+a)$ with a wedges by $V_-^{0,1}$ and b contractions by T_+, and the second gives classes of type $(1, 2n)$, $(0, 2n-1)$ and $(0, 2n)$. The components of x are of type (k, k), so $x \in \mathbb{C}\alpha_+$. Symmetrically the second case gives $x \in \mathbb{C}\alpha_-$ and the third $x = 0$. Finally x is rational, complex conjugation carries V_+ to V_- and $H^{1,0}$ to $H^{0,1}$, so $\bar{\alpha}_+ \in \mathbb{C}\alpha_-$, and a rational multiple of α_+ or of α_- is zero. So $\operatorname{ch}(F) = 0$. As $\sum_i (-1)^{k_i} \operatorname{ch}(F_i) = N\omega \neq 0$, some summand has nonzero Chern character, and the first statement follows; the second is the special case in which every summand has an exterior Ext algebra. ■

6.9. **Which hypotheses suffice.** The implications proved above, together with those of Section 6.6, determine exactly which combinations of statements yield the conjecture. The logical skeleton of the paper and of the literature it quotes is finite: a set of statements, each of them proved here, quoted from the literature, or taken as a hypothesis, and a set of inference rules, each of them one theorem. Consequences can then be taken.

The answer is only as strong as the rule set. A set of statements the rules do not carry to the conjecture is *insufficient* in the sense of the rules, and that means no more than that no rule derives the conjecture from it. An implication that is true and left out makes a set look necessary when it is not. One such implication is elementary, and it decides the shape of the answer, so we prove it first.

Proposition 6.33 (The conjecture for the varieties that are not abelian is the conjecture)**.** *Let A be a complex abelian variety of dimension $g \geq 1$ and let $X = A \times \mathbb{P}^1$. Then X is a smooth complex projective variety that is not an abelian variety, and if every Hodge class of degree $2p$ on X is algebraic, then so is every Hodge class of degree $2p$ on A. Consequently the statement*

> *the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety*

implies the Hodge conjecture for every abelian variety, and it is equivalent to **HC**.

Proof. X is smooth and projective of dimension $g+1$. By the Künneth formula $h^{1,0}(X) = h^{1,0}(A) + h^{1,0}(\mathbb{P}^1) = g$, while an abelian variety of dimension $g+1$ has $h^{1,0} = g+1$; so X is not an abelian variety. (Equally, X contains the rational curves $\{a\} \times \mathbb{P}^1$, and every morphism from $\mathbb{P}^1$ to a complex torus is constant, since it lifts to the universal cover $\mathbb{C}^{g+1}$ [BL04, Chapter 1].)

Let $\mathrm{pr}\colon X \to A$ be the projection and $D = A \times \{0\}$, a smooth divisor that pr maps isomorphically onto A, so that $\mathrm{pr}_*[D] = 1$ in $H^0(A, \mathbb{Q})$. Let α be a Hodge class of degree $2p$ on A. Pull-back along a morphism is a morphism of Hodge structures, so $\mathrm{pr}^*\alpha$ is a Hodge class of degree $2p$ on X, and by hypothesis $\mathrm{pr}^*\alpha = [Z]$ for an algebraic cycle Z on X with rational coefficients. The projection formula gives

$$\mathrm{pr}_*\big([Z] \cup [D]\big) = \mathrm{pr}_*\big(\mathrm{pr}^*\alpha \cup [D]\big) = \alpha \cup \mathrm{pr}_*[D] = \alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [Ful98, Chapter 19], so the left side is the class of the cycle $\mathrm{pr}_*(Z \cdot D)$ and α is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. ■

Since the conjecture for the varieties that are not abelian is the whole conjecture, the statement that belongs on a frontier is weaker: it asks for algebraicity only modulo what abelian varieties supply.

Definition 6.34 (The conjecture modulo abelian varieties)**.** For a smooth complex projective variety X and $p \geq 0$ let $\mathrm{Ab}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ be the span of the classes $\Gamma_*\gamma = \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^*\gamma)$ of degree $2p$, where B runs over the complex abelian varieties, the point included, γ over the Hodge classes on B, and Γ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3′) is: for every X and every p,

$$\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X).$$

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward preserve Hodge classes, so $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$; and taking B a point shows that $\mathrm{Ab}^p(X)$ contains every algebraic class. So (F3′) says that every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences, which is the form of (F3) that Remark 6.72 asks for.

Proposition 6.35 (What is known of (F3′))**.**

(i) **HC** *implies* (F3′)*, and* (F3′) *together with the conjecture for abelian varieties implies* **HC**.

(ii) (F3′) *holds on every abelian variety and on every product* $A \times \mathbb{P}^1$ *with* A *abelian; the argument of Proposition 6.33 gives nothing for it.*

(iii) *Let* $\mathcal{A}$ *be the class of smooth complex projective varieties* X *for which* $H^*(X, \mathbb{Q})$ *is spanned by classes* $\Gamma_* y$ *with* B *abelian,* Γ *an algebraic cycle on* $B \times X$ *and* $y \in H^*(B, \mathbb{Q})$*. Every* $X \in \mathcal{A}$ *satisfies* (F3′)*. The class* $\mathcal{A}$ *contains every abelian variety and every smooth projective curve, and it is closed under products and under images of surjective morphisms. So it contains every product of curves, every symmetric product of a curve, and every smooth projective variety onto which a product of curves or an abelian variety maps surjectively.*

(iv) *Every* X *of dimension* n *satisfies* (F3′) *in the degrees* $2p$ *with* $p \in \{0, 1, n-1, n\}$*, so every* X *of dimension at most three satisfies it in every degree.*

Proof. (i) If **HC** holds, then $\mathrm{Hdg}^p(X)$ consists of algebraic classes, which lie in $\mathrm{Ab}^p(X)$. Conversely, if the conjecture holds for abelian varieties, every γ in Definition 6.34 is algebraic, so every $\Gamma_*\gamma$ is, since composition with a correspondence carries algebraic classes to algebraic classes [Ful98, Chapter 16]; then (F3′) says that every Hodge class is algebraic.

(ii) On an abelian variety A take $B = A$ and Γ the diagonal. On $X = A \times \mathbb{P}^1$ the Künneth formula gives $H^*(X, \mathbb{Q}) = \mathrm{pr}^*H^*(A, \mathbb{Q}) \oplus \mathrm{pr}^*H^*(A, \mathbb{Q}) \cup \eta$, with η the class of a point of $\mathbb{P}^1$ pulled back to X, which is algebraic. The first summand is $\Gamma_*H^*(A, \mathbb{Q})$ for Γ the transpose of the graph of pr, and the second is $\Gamma'_*H^*(A, \mathbb{Q})$ for $\Gamma' = \Gamma \cdot \mathrm{pr}_X^*\eta$. So $X \in \mathcal{A}$, and (iii) applies.

(iii) Let $X \in \mathcal{A}$ and fix p. Since $H^{2p}(X, \mathbb{Q})$ is finite dimensional, finitely many classes $\Gamma_{j*}y_j$ span it, and we may take each Γ_j of pure codimension c_j on $B_j \times X$ and each y_j of pure degree k_j, with $k_j + 2c_j - 2\dim B_j = 2p$. Then $\Gamma_{j*}\colon H^{k_j}(B_j, \mathbb{Q}) \to H^{2p}(X, \mathbb{Q})$ is a morphism of Hodge structures up to a Tate twist, and the sum of these maps is a surjective morphism of polarisable Hodge structures onto $H^{2p}(X, \mathbb{Q})$. Polarisable Hodge structures are semisimple, so its kernel has a complement that is a sub-Hodge structure and maps isomorphically onto $H^{2p}(X, \mathbb{Q})$; hence every Hodge class of degree $2p$ is the image of a Hodge class, that is $\sum_j \Gamma_{j*}\gamma_j$ with each γ_j a Hodge class on B_j. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$.

An abelian variety lies in $\mathcal{A}$ through its diagonal. A curve C of genus $g \geq 1$ lies in $\mathcal{A}$ through an Abel–Jacobi embedding $\iota\colon C \to J(C)$: the transpose of its graph induces ι^*, which is an isomorphism on H^0 and H^1 and sends the class of the theta divisor to g times the class of a point, so it is surjective; the projective line is the image of an elliptic curve under a double cover, and is covered by the last closure property. For products, if $H^*(X_i, \mathbb{Q})$ is spanned by the $\Gamma_{i*}y$, then by the Künneth formula $H^*(X_1 \times X_2, \mathbb{Q})$ is spanned by the $(\Gamma_1 \times \Gamma_2)_*(y_1 \otimes y_2)$, with $\Gamma_1 \times \Gamma_2$ an algebraic cycle on $(B_1 \times B_2) \times (X_1 \times X_2)$. For images, let $f\colon Y \to X$ be a surjective morphism with $Y \in \mathcal{A}$, let $r = \dim Y - \dim X$ and let h be an ample class on Y. The correspondence $\Psi = [\Gamma_f] \cdot \mathrm{pr}_Y^* h^r$ on $Y \times X$ induces $\Psi_* y = f_*(y \cup h^r)$, and $\Psi_* f^* x = x \cup f_* h^r = c\,x$ with $c > 0$ the degree of h^r on a general fibre. So Ψ_* is surjective, and $H^*(X, \mathbb{Q})$ is spanned by the classes $(\Psi \circ \Gamma)_* y$, with $\Psi \circ \Gamma$ algebraic [Ful98, Chapter 16]. A symmetric product of a curve is the image of a power of it.

(iv) Every class of degree 0 or $2n$ is algebraic, and every Hodge class of degree 2 is algebraic by the Lefschetz theorem on $(1,1)$ classes. For $n \geq 2$, the hard Lefschetz theorem makes cup product with h^{n-2}, for h an ample class, an isomorphism $H^2(X, \mathbb{Q}) \to H^{2n-2}(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so every Hodge class of degree $2n-2$ is h^{n-2} times a Hodge class of degree 2, and is algebraic. Algebraic classes lie in $\mathrm{Ab}^p(X)$. When $n \leq 3$ every p between 0 and n is one of $0, 1, n-1, n$. ■

Remark 6.36 (The reach of algebraic correspondences). (F3′) is the statement that the Hodge conjecture reduces to abelian varieties along algebraic correspondences, and Proposition 6.35 records the cases of such a reduction that are theorems: the varieties of $\mathcal{A}$, whose whole cohomology is reached from abelian varieties by algebraic correspondences, and the degrees in which the conjecture itself is a theorem. The next degree is degree four on fourfolds outside $\mathcal{A}$ and outside the scope of Proposition 6.43 and Proposition 6.44. The square $S \times S$ of a K3 surface is the standard example when $\mathrm{End}_{\mathrm{Hdg}}(T(S))$ is a totally real field of degree at least two, that is, when S has real multiplication. When that field is $\mathbb{Q}$ or a CM field, the Hodge conjecture holds on $S \times S$: by Proposition 6.38(i) in the first case, and in the second by [Bus19] and the fact that the elements of norm one span a CM field, as in [RM08] (Proposition 6.39). Its Hodge classes in $H^2(S) \otimes H^2(S)$ include the classes that induce the Hodge endomorphisms of the transcendental lattice $T(S)$, and $S \times S$ lies in $\mathcal{A}$ as soon as the Kuga–Satake embedding of $T(S)$ is induced by an algebraic cycle, the Kuga–Satake Hodge conjecture for S: the transpose of such a cycle maps the cohomology of the Kuga–Satake variety onto $T(S)$, and the rest of $H^*(S, \mathbb{Q})$ is algebraic. The rational Hodge isometries are algebraic [Bus19], but a real multiplication is not an isometry, and Remark 7.5 meets exactly this obstacle for the surfaces S_λ; the dihedral family of correspondences of [BB26] crosses it for two cyclic cubic fields and reaches no other field.

For varieties whose cohomology is not of abelian type the position is starker. By Proposition 6.75 no correspondence moves the H^3 of a very general quintic threefold into the cohomology of an abelian variety, so such a variety does not lie in $\mathcal{A}$. That is no obstruction to (F3′) itself, since a Hodge class spans a Hodge structure of Tate type, which is of abelian type, and nothing in Hodge theory forbids it from lying in $\mathrm{Ab}^p(X)$. Nor is it a route: to put a Hodge class into $\mathrm{Ab}^p(X)$ one must construct an algebraic cycle on $B \times X$ whose class carries it, which is the kind of construction the Hodge conjecture asks for, now on $B \times X$ instead of on X. The cases of (F3′) proved below outside $\mathcal{A}$ come from the Hodge conjecture itself, from hypotheses on zero-cycles, and, in degree four, from

correspondences with members of $\mathcal{A}$ (Proposition 6.39 to Proposition 6.46); none of them reaches a variety in general position, and (F3′) enters Theorem 6.70 as a hypothesis.

Two further facts fix the position of (F3′). The first says how strong it is.

Proposition 6.37 (The strength of (F3′))**.** (F3′) *implies* (M)*: every Hodge class on every smooth complex projective variety is motivated in the sense of André. In particular it implies that every such class is absolutely Hodge. Adding the implication* (F3′) ⇒ (M) *to the rule set of Setup 6.69 leaves the five minimal sufficient sets of Theorem 6.70(ii) unchanged.*

Proof. Let $\gamma \in \mathrm{Hdg}^p(X)$. By (F3′), $\gamma = \sum_j \Gamma_{j*}\beta_j$ with β_j a Hodge class on an abelian variety B_j and Γ_j an algebraic cycle on $B_j \times X$. Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], and the motivated classes form a graded $\mathbb{Q}$-algebra that contains the algebraic classes and is stable under pull-back and push-forward along morphisms [And96b, §2]. So $\mathrm{pr}_{B_j}^*\beta_j \cup [\Gamma_j]$ is motivated, and so is its push-forward $\Gamma_{j*}\beta_j$; hence γ is motivated. Motivated classes are absolutely Hodge [And96b, §2]. For the last assertion, the new rule has the premise (F3′) and the conclusion (M). The sets C_1 and C_2 of the proof of Theorem 6.70 omit (F3′), and C_3 and C_4 contain both statements, so the four sets stay closed and that proof applies without change; the five sets remain sufficient, since adding a rule only enlarges closures. ∎

So on the route $\{\mathrm{L}, \mathrm{F3}'\}$ the second member already contains (M), and that route is never easier than $\{\mathrm{L}, \mathrm{M}\}$. The second fact locates, on the powers of a variety of orthogonal type, the first class that (F3′) has to reach.

Proposition 6.38 (Powers of a variety of orthogonal type)**.** *Let Y be a smooth complex projective variety of even dimension m. Suppose that $H^k(Y,\mathbb{Q})$ is spanned by algebraic classes for $k \neq m$, that $H^m(Y,\mathbb{Q}) = N \oplus V$ with N spanned by algebraic classes, the intersection form ψ nondegenerate on N and $V = N^\perp$, and that the Hodge group of V is $\mathrm{SO}(V,\psi)$, with $t = \dim V \geq 3$. Let $\det(V) \in H^{mt}(Y^t,\mathbb{Q})$ be the image of a generator of $\bigwedge^t V \subseteq V^{\otimes t}$ under the Künneth embedding $V^{\otimes t} \subseteq H^{mt}(Y^t,\mathbb{Q})$.*

(i) *The Hodge conjecture holds for Y^n for every $n < t$.*

(ii) $\det(V)$ *is a Hodge class, and* $\mathrm{Hdg}^*(Y^t)$ *is spanned by algebraic classes and* $\det(V)$.

(iii) $\det(V)$ *does not lie in the $\mathbb{Q}$-algebra generated by the pull-backs along the projections of the Hodge classes of the Y^j, $j < t$.*

(iv) *The Hodge conjecture holds for every power of Y if and only if* $\det(V)$ *is algebraic, and* (F3′) *holds for every power of Y if and only if* $\det(V) \in \mathrm{Ab}(Y^t)$.

Proof. Algebraic classes have even degree, so $H^k(Y,\mathbb{Q}) = 0$ for odd k. Let $[\Delta] = \sum_k \pi_k$ be the Künneth decomposition of the class of the diagonal, $\pi_k \in H^{2m-k}(Y,\mathbb{Q}) \otimes H^k(Y,\mathbb{Q})$. For $k \neq m$, $\pi_k = \sum_i e_i^\vee \otimes e_i$ for a basis (e_i) of $H^k(Y,\mathbb{Q})$ and the Poincaré dual basis of $H^{2m-k}(Y,\mathbb{Q})$, both spanned by algebraic classes, so π_k is algebraic; likewise the part π_N of π_m in $N \otimes N$, since ψ is nondegenerate on N. So $\pi_V = [\Delta] - \sum_{k\neq m} \pi_k - \pi_N$, the class of ψ^{-1} in $V \otimes V \subseteq H^{2m}(Y \times Y,\mathbb{Q})$, is algebraic.

The Hodge group of Y^n is $\mathrm{SO}(V,\psi)$, acting on $H^*(Y^n,\mathbb{Q}) = \big(H^*(Y,\mathbb{Q})\big)^{\otimes n}$ through the factors V and trivially on the rest, which is spanned by algebraic classes; so the Hodge classes are the $\mathrm{SO}(V)$-invariants. By the first fundamental theorem for the special orthogonal group [GW09, Chapter 5], the invariants of $\mathrm{SO}(V)$ in $V^{\otimes j}$ are spanned by the complete contractions with ψ^{-1} when $j < t$, and, when $j = t$, by these and the determinant

tensor; in general they are spanned by products of contractions and determinant tensors on disjoint sets of t factors. A contraction of the factors a and b of Y^n is the pull-back of π_V along the projection to those two factors, a determinant on a set I of t factors is the pull-back of $\det(V)$ along the projection to Y^I, and products of these with pull-backs of algebraic classes span the Hodge classes. This gives (i), (ii) and the first half of (iv). For (iii), the orthogonal group $\mathrm{O}(V,\psi)$ acts on $H^*(Y^j,\mathbb{Q})$ in the same way, and for $j<t$ every $\mathrm{SO}(V)$-invariant is $\mathrm{O}(V)$-invariant; pull-backs along projections and cup products commute with the diagonal action of $\mathrm{O}(V)$, so every element of the algebra in (iii) is $\mathrm{O}(V)$-invariant, while a reflection acts on $\det(V)$ by -1. For the second half of (iv), $\mathrm{Ab}^*(Y^n)$ contains the algebraic classes and is stable under pull-back along projections and under cup product: $\Gamma_*\gamma\cup\Gamma'_*\gamma'$ is the image of $\gamma\otimes\gamma'$ on $B\times B'$ under the correspondence $(\Gamma\times\Gamma')$ followed by the diagonal of Y^n. ■

For a K3 surface S with $\mathrm{End}_{\mathrm{Hdg}}(T(S))=\mathbb{Q}$ the Hodge group of $T(S)$ is $\mathrm{SO}(T(S))$ [Zar83]; for a very general one, $t=21$. So the Hodge conjecture holds for S^n with $n\le 20$, and on S^{21} it is equivalent to the algebraicity of the single class $\det(T(S))$. The argument is the classical one from invariant theory, and we make no claim of priority for it. What it adds here is the location of the first instance of (F3′) on the powers of such a variety that invariant theory does not reach: one class, in one degree, on one power.

The same invariant theory, with the motivic decompositions of the literature and the theorems of Buskin and Markman on Hodge isometries, gives the Hodge conjecture itself, and so (F3′), on the other varieties built from a K3 surface and on varieties of $\mathrm{K3}^{[n]}$ type, up to a real multiplication and one determinant class.

Proposition 6.39 (Powers, Hilbert schemes and moduli spaces of a K3 surface)**.** *Let S be a projective K3 surface, $T=T(S)_{\mathbb{Q}}$, $t=\dim T$ and $E=\mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field* [Zar83].

(i) *If E is a CM field, the Hodge conjecture holds for every power S^k* [RM08, Theorem 5.4]*,* [Bus19]*, hence for every Hilbert scheme $S^{[n]}$* [dCM02]*, for every smooth projective moduli space of Gieseker stable (twisted) sheaves on S, or of σ-stable objects in $D^b(S,\alpha)$ for a generic stability condition σ and a Brauer class α* [Bül20, Theorem 0.1, Remark 2.1]*, and for the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of σ-semistable objects in $D^b(S,\alpha)$, with v_0 primitive, $v_0^2=2$ and σ v_0-generic* [FFZ21, Theorem 1.3].

(ii) *If $E=\mathbb{Q}$, the Hodge conjecture holds for $S^{[n]}$ for every $n<t(t+1)/2$, and for every moduli space M as in* (i) *with $\dim M<t$. For $N=t(t+1)/2$ the following are equivalent: the Hodge conjecture for $S^{[N]}$; the algebraicity of the class $\det(T)$ on S^t of Proposition 6.38; the Hodge conjecture for every power of S; and the Hodge conjecture for every $S^{[n]}$. For a very general S, the Hodge conjecture holds for $S^{[n]}$ with $n\le 230$, and for $S^{[231]}$ if and only if $\det(T(S))$ on S^{21} is algebraic.*

(iii) *If E is totally real of degree at least two, the Hodge conjecture for $S\times S$, the Hodge conjecture for $S^{[2]}$, and the statement that every element of E acts on T through an algebraic cycle on $S\times S$ are equivalent.*

Wherever the Hodge conjecture holds, so does (F3′)*. This is a consequence of* [RM08, Bus19, dCM02, Bül20, FFZ21] *and Proposition 6.38; we make no claim of priority.*

Proof. We use three facts. First, $H^*(S,\mathbb{Q})=A\oplus T$ with $A=H^0\oplus\mathrm{NS}(S)_{\mathbb{Q}}\oplus H^4$ spanned by algebraic classes; the Künneth components of the diagonal of S are algebraic [Mur90], and so is its part π_T in $T\otimes T$, as in the proof of Proposition 6.38. The Hodge classes of S^k

are the invariants in $(A\oplus T)^{\otimes k}$ of the Hodge group of T, which acts trivially on A. Second, if $H^*(X,\mathbb{Q})$ is spanned by the images of the $H^*(Y_i,\mathbb{Q})$ under algebraic correspondences and the Hodge conjecture holds for every Y_i, it holds for X: by the semisimplicity argument of the proof of Proposition 6.35(iii) every Hodge class of X is a sum of images of Hodge classes of the Y_i. Third, the correspondences Γ_ν of the proof of Proposition 6.42(ii) give an isomorphism of Chow motives between $h(S^{[n]})$ and $\bigoplus_\nu h(S^{(\nu)})(l(\nu)-n)$ [dCM02], so each $h(S^{(\nu)})(l(\nu)-n)$ is a direct summand of $h(S^{[n]})$, and the lift $\widetilde{\Gamma}_{\nu *}$ takes the same value on a class of $S^{l(\nu)}$ and on its average over G_ν. Hence the Hodge conjecture holds for $S^{[n]}$ if and only if, for every partition ν of n, every G_ν-invariant Hodge class of $S^{l(\nu)}$ is algebraic. By [Bül20, Theorem 0.1, Remark 2.1] the Chow motive of a moduli space M as in (i) is a direct summand of a sum of Tate twists of motives $h(S^{k_i})$ with $1\le k_i\le \dim M$, and by [FFZ21, Corollary 4.5] the Chow motive of a crepant resolution as in (i) lies in the tensor subcategory generated by $h(S)$, whose objects are direct summands of sums of Tate twists of the $h(S^k)$, the dual of $h(S)$ being $h(S)(2)$.

(i) Over $\mathbb{C}$ the Hodge group of T is a product of groups $\mathrm{GL}(W_\sigma)$ acting on $T_\mathbb{C}=\bigoplus_\sigma(W_\sigma\oplus W_\sigma^\vee)$ [Zar83], and by the first fundamental theorem for the general linear group [GW09, Chapter 5] its invariants in $T_\mathbb{C}^{\otimes b}$ are spanned by products of contractions of one factor with another. Under the isomorphism $\mathrm{End}(T)\cong T\otimes T$, $\varphi\mapsto\tau_\varphi=\sum_i t_i^\vee\otimes\varphi(t_i)$, with $(t_i^\vee)$ the dual basis of a basis (t_i) for the intersection form, the Hodge classes of $T\otimes T$ are the τ_φ with $\varphi\in E$, and these contractions lie in their complex span. The rational span of the products of the τ_φ on pairs of factors has the same complexification as the space of invariants, so the Hodge classes of S^k are spanned by such products and pull-backs of algebraic classes. For $u\in E$ with $u\bar u=1$ the map $\mathrm{id}_{\mathrm{NS}}\oplus u$ is a rational Hodge isometry of $H^2(S,\mathbb{Q})$, since the adjoint of u is $\bar u$ [Zar83]; it is induced by an algebraic cycle [Bus19], and composing that cycle with π_T on both sides gives τ_u, which is therefore algebraic. The elements of norm one span E. Indeed, write $E=E_0(y_0)$ with $y_0^2=-\delta$, $\delta\in E_0$, and let $y\in E$ with $\bar y=-y$; for $s\in\mathbb{Q}$ the element $u_s=(1+sy)(1-sy)^{-1}$ has norm one. A linear form on E vanishing on all u_s gives a rational function of s vanishing on $\mathbb{Q}$, so it vanishes on the coefficients 1, $2y$, $2y^2$ of the expansion of u_s in powers of s. As y runs over E_0y_0, the y span E_0y_0 and the $y^2=-a^2\delta$, $a\in E_0$, span E_0, since $2a=(a+1)^2-a^2-1$. So every τ_u, $u\in E$, is algebraic, and the Hodge conjecture holds for every S^k; this is the argument of [RM08, Lemma 5.3, Theorem 5.4], which takes the algebraicity of the isometries from Mukai's announcement; [Bus19] proves it in general. We use [RM08] only when E is a CM field: for a totally real E the invariants of the Hodge group are not generated by those of degree two, since they include determinants, as Proposition 6.38 shows for $E=\mathbb{Q}$, and the powers do not reduce to the square. The rest of (i) follows from the second and third facts.

(ii) Here the Hodge group of T is $\mathrm{SO}(T)$ [Zar83], and $t\ge 3$, since a Hodge structure of K3 type of dimension two has complex multiplication. The Hodge conjecture holds for S^k, $k<t$, by Proposition 6.38(i), which gives the assertion on M by the second fact. Let ν be a partition of n with a_j parts equal to j, so that $G_\nu=\prod_j\mathfrak{S}_{a_j}$ permutes the factors of $S^{l(\nu)}$ within blocks of sizes a_j, one block for each distinct part size. By the proof of Proposition 6.38, the Hodge classes of $S^{l(\nu)}$ are spanned by algebraic classes and by products $\beta=\det_I\cup\gamma$, where $\det_I$ is the pull-back of $\det(T)$ along the projection to a set I of t factors and γ is pulled back from the other factors. If two factors of I lie in one block, the transposition in G_ν exchanging them fixes γ and changes the sign of the alternating class $\det_I$, all classes involved being of even degree; so it takes β to $-\beta$, and the average of β over G_ν is zero. If ν has fewer than t distinct part sizes, every set of t

factors meets some block twice, so a G_ν-invariant Hodge class, being its own average, is the average of an algebraic class, and is algebraic. A partition with t distinct part sizes has $n \geq 1+2+\cdots+t = t(t+1)/2$, so by the third fact the Hodge conjecture holds for $S^{[n]}$ when $n < t(t+1)/2$ (Figure 21). For $n = N$ the partition $(1, 2, \ldots, t)$ has $l(\nu) = t$ and $G_\nu = 1$, so the Hodge class $\det(T)$ of S^t is algebraic if the Hodge conjecture holds for $S^{[N]}$. If $\det(T)$ is algebraic, the Hodge conjecture holds for every power of S by Proposition 6.38(iv), and for every $S^{[n]}$ by the second fact; and the conjecture for S^t contains the algebraicity of $\det(T)$. A very general S has $E = \mathbb{Q}$ and $t = 21$, and $N = 231$.

(iii) The Hodge structure T has no nonzero Hodge class, so the Hodge classes of $A \otimes T$ and $T \otimes A$ vanish, and those of $S \times S$ are the algebraic classes and the τ_φ, $\varphi \in E$. A cycle inducing φ on T, composed with π_T on both sides, gives τ_φ, and τ_φ induces φ on T; this proves the equivalence for $S \times S$. By [dCM02], $h(S^{[2]}) = h(S^{(2)}) \oplus h(S)(-1)$, and the Hodge conjecture holds for S. The exchange of the two factors takes τ_φ to τ_{φ^*}, and the adjoint involution of a totally real E is trivial [Zar83], so every τ_φ is invariant under $\mathfrak{S}_2$ and the Hodge conjecture for $S^{[2]}$ asks for the same classes as that for $S \times S$. ■

Proposition 6.40 (Varieties of $\mathrm{K3}^{[n]}$ type)**.** *Let X be a smooth projective hyperkähler variety deformation equivalent to the Hilbert scheme of $n \geq 2$ points on a K3 surface, q the Beauville–Bogomolov form on $H^2 = H^2(X, \mathbb{Q})$, $\mathrm{NS} = \mathrm{NS}(X)_\mathbb{Q}$, $T = \mathrm{NS}^\perp$ and $E = \mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field whose adjoint involution for q is complex conjugation* [Zar83]*. Let $\mathrm{SH}(X) \subseteq H^*(X, \mathbb{Q})$ be the subalgebra generated by H^2, and for $\varphi \in E$ with $\varphi^* = \varphi$ let $c_\varphi = \sum_i t_i^\vee \cup \varphi(t_i) \in H^4(X, \mathbb{Q})$, with (t_i) a basis of T and $(t_i^\vee)$ its dual basis for q.*

(i) *The Hodge classes of $\mathrm{SH}(X)$ are spanned by products of divisor classes and classes c_φ.*

(ii) *If E is $\mathbb{Q}$ or a CM field, every Hodge class of $\mathrm{SH}(X)$ is algebraic.*

(iii) *Let $n = 2$. Then $\mathrm{SH}(X) = H^*(X, \mathbb{Q})$, with $b_4(X) = 276 = \dim \mathrm{Sym}^2 H^2$, and $c_2(X) = \frac{6}{5} q^\vee$, where $q^\vee \in \mathrm{Sym}^2 H^2$ is dual to q. So the Hodge conjecture holds for X when E is $\mathbb{Q}$ or a CM field. When E is totally real of degree at least two, it holds for X if and only if every element of E, acting on T and by zero on NS, is induced by an algebraic cycle on $X \times X$.*

This is a consequence of [Mar24, CM13]*; we make no claim of priority.*

Proof. (i) Multiplication $\mathrm{Sym}^k H^2 \to \mathrm{SH}^{2k}(X)$ is a surjective morphism of polarisable Hodge structures, so by semisimplicity every Hodge class of $\mathrm{SH}^{2k}(X)$ is the image of one of $\mathrm{Sym}^k H^2 = \bigoplus_{a+b=k} \mathrm{Sym}^a \mathrm{NS} \otimes \mathrm{Sym}^b T$, whose Hodge classes are $\mathrm{Sym}^a \mathrm{NS} \otimes (\mathrm{Sym}^b T)^{\mathrm{Hdg}}$. The isomorphism $\mathrm{End}(T) \cong T \otimes T$ of the proof of Proposition 6.39, now for q, carries the adjoint to the exchange of the factors, so $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}}$ consists of the tensors of the self-adjoint $\varphi \in E$, whose images in H^4 are the c_φ. Over $\mathbb{C}$ the Hodge group of T is a product of groups $\mathrm{SO}(T_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma T_\sigma$, with $\dim T_\sigma = \dim_E T \geq 3$, when E is totally real, and of groups $\mathrm{GL}(W_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ when E is a CM field [Zar83]. By the first fundamental theorems [GW09, Chapter 5], the invariant polynomials of one vector of T_σ, or of one vector of W_σ and one of $W_\sigma^\vee$, are generated by the quadratic form, or by the pairing: a determinant needs $\dim T_\sigma$ distinct vectors. So $(\mathrm{Sym}^b T)^{\mathrm{Hdg}} \otimes \mathbb{C}$ is spanned by products of elements of $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}} \otimes \mathbb{C}$, and the rational span of the products of classes c_φ, which has the same complexification, is $(\mathrm{Sym}^b T)^{\mathrm{Hdg}}$.

(ii) Divisor classes are algebraic, so by (i) it suffices that every c_φ is. By [CM13, Theorem 1.1] the Lefschetz standard conjecture holds for X, so the Künneth projectors

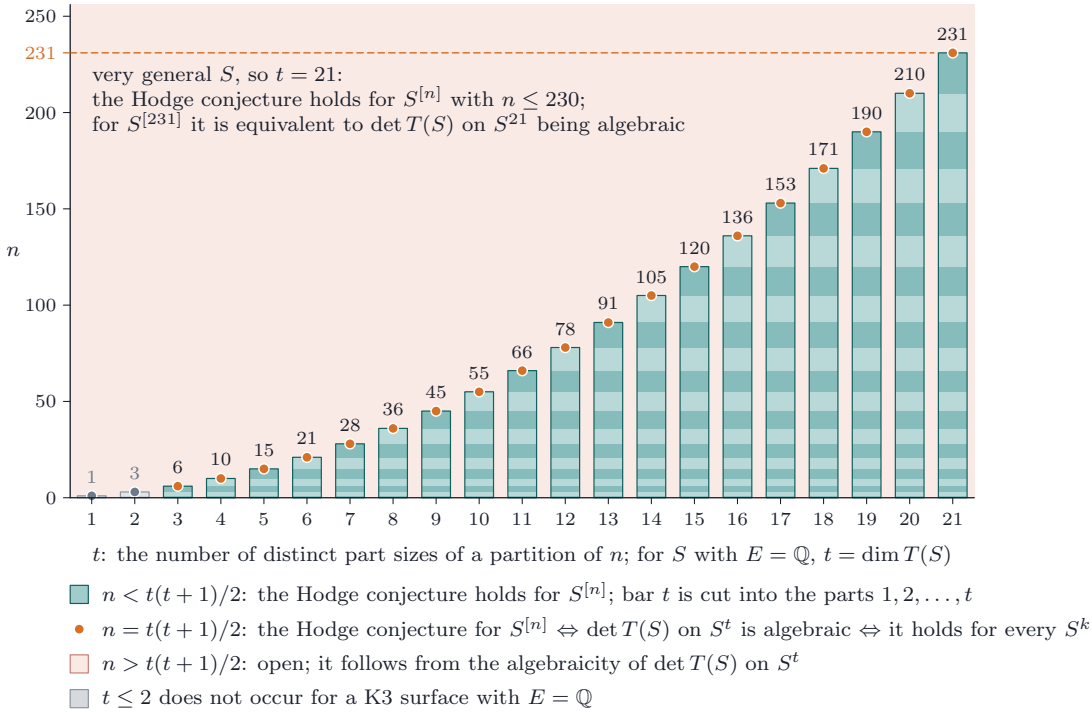


FIGURE 21. The threshold of Proposition 6.39(ii). For $t = 1, \dots, 21$ the bar rises to $t(t+1)/2$, the least n that has a partition with t distinct part sizes, and is cut into the parts $1, 2, \dots, t$ of the partition that attains it. For a K3 surface S with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S)) = \mathbb{Q}$ and $t = \dim T(S)$, which is at least 3, the Hodge conjecture holds for $S^{[n]}$ with n below the bar; at its top it is equivalent to the algebraicity of the class $\det T(S)$ on S^t of Proposition 6.38, and to the Hodge conjecture for every power of S; above it, it follows from that algebraicity. A very general S has $t = 21$: the conjecture holds for $S^{[n]}$ with $n \le 230$, and for $S^{[231]}$ it is equivalent to the algebraicity of $\det T(S)$ on S^{21}, one of the frontier cases of (F3′) (Remark 6.68).

and the inverse of the isomorphism $L^{2n-2}\colon H^2 \to H^{4n-2}$, L the cup product with an ample class h, are induced by algebraic cycles [Kle68, Kle94]. Let $u \in E$ with $u\bar{u} = 1$, and $u = 1$ if $E = \mathbb{Q}$. The map $f_u = \mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of H^2, so by [Mar24, Theorem 1.1] there is an algebraic class Z_u on $X \times X$ (the diagonal if $u = 1$) inducing on H^2 a nonzero rational multiple of f_u: the class of the correspondence $\tilde{f}$ of that theorem if it preserves degrees, and the class of $\tilde{f} \circ (c_2(X)^{n-1} \cup \cdot\,)$ if it reverses them. The class Z_u may act in other degrees too; its Künneth component in $H^{4n-2}(X) \otimes H^2(X)$ is algebraic and induces a nonzero multiple of f_u on H^2 and zero elsewhere. Applying $(L^{2n-2})^{-1} \otimes \mathrm{id}$ gives, up to a nonzero rational factor, the algebraic class $\sum_i b_i \otimes f_u(a_i)$, for a basis (a_i) of H^2 and the dual basis (b_i) for $B(x,y) = \int_X xyh^{2n-2}$. Polarising the relation $\int_X \alpha^{2n} = c_X q(\alpha)^n$ of Fujiki and Beauville [Bea83, Fuj87] gives

$$B(x,y) = \frac{c_X}{2n-1}\, q(h)^{n-2}\big(q(h)q(x,y) + 2(n-1)q(x,h)q(y,h)\big),$$

so NS and T are orthogonal for B, and $B = \kappa q$ on T with $\kappa = c_X q(h)^{n-1}/(2n-1) \neq 0$. With (a_i) adapted to $\mathrm{NS} \oplus T$, the pull-back along the diagonal is $\sum_i b_i \cup f_u(a_i) = \nu + \kappa^{-1} c_{(u+\bar{u})/2}$ with $\nu \in \mathrm{Sym}^2\mathrm{NS}$ algebraic, because cup product on H^2 is commutative and the exchange of factors turns u into $\bar{u}$. So $c_{(u+\bar{u})/2}$ is algebraic. When E is a CM field its elements of norm one span it (proof of Proposition 6.39(i)), so the $(u+\bar{u})/2$ span E_0, the self-adjoint part of E; when $E = \mathbb{Q}$, $u = 1$ suffices. So every c_φ is algebraic.

(iii) The variety X has the Betti numbers of $S^{[2]}$, which by [dCM02] are those of $\mathrm{Sym}^2 H^*(S) \oplus H^*(S)(-1)$: $H^{\mathrm{odd}}(X) = 0$ and $b_4(X) = 253 + 1 + 22 = 276 = \dim \mathrm{Sym}^2 H^2$. Here $c_X = 3$, and the relation of Fujiki and Beauville gives $\int_X abcd = q(a,b)q(c,d) + q(a,c)q(b,d) + q(a,d)q(b,c)$ for $a, b, c, d \in H^2$. This form on $\mathrm{Sym}^2 H^2$ is nondegenerate: in a q-orthogonal basis (e_i) with $q(e_i, e_i) = d_i$, the $e_i e_j$ with $i < j$ are orthogonal to each other and to the squares, with $\int_X (e_i e_j)^2 = d_i d_j$, and on the squares the form has the matrix $D(J + 2I)D$, with $D = \mathrm{diag}(d_i)$ and J the matrix of ones, of determinant $2^{22} \cdot 25 \prod_i d_i^2$. So $\mathrm{Sym}^2 H^2 \to H^4$ is injective, hence bijective, and with $H^6 = h^2 \cup H^2$ by the hard Lefschetz theorem, $\mathrm{SH}(X) = H^*(X, \mathbb{Q})$. By [Fuj87], $\int_X c_2(X) xy = C\, q(x,y)$ for a constant C, and the Riemann–Roch polynomial $\chi(L) = \binom{q(L)/2+3}{2}$ [EGL01] gives $C = 30$; the Fujiki relation gives $\int_X q^\vee xy = 25\, q(x,y)$, so $c_2(X) = \frac{6}{5} q^\vee$ by nondegeneracy. For $E = \mathbb{Q}$ this makes $c_1 = q^\vee - q^\vee_{\mathrm{NS}}$ algebraic without [CM13], with $q^\vee_{\mathrm{NS}} \in \mathrm{Sym}^2\mathrm{NS}$ the class dual to $q|_{\mathrm{NS}}$. With (i) and (ii) this proves the Hodge conjecture for X when E is $\mathbb{Q}$ or a CM field. Let E be totally real and $\varphi \in E$. If $\varphi \oplus 0$ is induced by an algebraic cycle, the computation of (ii) with $\varphi \oplus 0$ in place of f_u makes c_φ algebraic. Conversely, if c_φ is algebraic, so is the correspondence $x \mapsto (L^2)^{-1}(c_\varphi \cup x)$. The Fujiki relation gives $\int_X c_\varphi xy = \mathrm{Tr}(\varphi) q(x,y) + 2q(\varphi x, y)$ for $x \in T$ and $y \in H^2$, and $B(z,y) = q(h)q(z,y)$ for $z \in T$, so that correspondence acts on T as $(\mathrm{Tr}(\varphi) + 2\varphi)/q(h)$. The projector onto T, the Künneth projector π_2 minus $\sum_j n_j^\vee \otimes n_j$ over a basis (n_j) of NS and the Hodge classes $n_j^\vee \in H^6$ dual to it, is algebraic by Proposition 6.35(iv); composing with it, $\varphi \oplus 0$ is induced by an algebraic cycle. By (i) the Hodge conjecture for X is the algebraicity of the c_φ, which proves the equivalence. ∎

For varieties of generalised Kummer type every Hodge class in the subalgebra generated by H^2 is algebraic, whatever the field, and the Hodge conjecture holds in dimension four [FV25]. The preprint [O'G26] announces, with [Mar24], that rational Hodge isometries between varieties of the types $\mathrm{K3}^{[a_1^2+1]}$ and $\mathrm{K3}^{[a_2^2+1]}$ are induced by algebraic cycles; it is not used here.

Corollary 6.41 (Cubic fourfolds and their Fano varieties of lines)**.** *Let $Y \subset \mathbb{P}^5$ be a smooth cubic fourfold, $T(Y) \subset H^4(Y, \mathbb{Q})$ the orthogonal complement of the Hodge classes, $E(Y) = \mathrm{End}_{\mathrm{Hdg}}(T(Y))$, and $F = F(Y)$ the Fano variety of lines on Y. The Hodge conjecture for F, the Hodge conjecture for $Y \times Y$, and the statement that every element of $E(Y)$ acts on $T(Y)$ through an algebraic cycle on $Y \times Y$ are equivalent. They hold when $E(Y)$ is $\mathbb{Q}$ or a CM field, for instance when Y is very general.*

Proof. The Hodge conjecture holds for Y [Zuc77], and also by Proposition 6.43, since Y is covered by lines. So $H^*(Y, \mathbb{Q})$ is $T(Y)$ plus a space spanned by algebraic classes, on which the intersection form is nondegenerate by the Hodge–Riemann relations, and as in the proof of Proposition 6.38 the part π_T of the diagonal in $T(Y) \otimes T(Y)$ is algebraic. The Hodge structure $T(Y)$ is of K3 type up to a Tate twist and has no nonzero Hodge class, so $E(Y)$ is a field [Zar83], and as in Proposition 6.39(iii) the Hodge conjecture for $Y \times Y$ is the third statement. By [BD85], F is a hyperkähler fourfold of $\mathrm{K3}^{[2]}$ type, and the universal line

$P \subset F \times Y$ induces an isomorphism of Hodge structures $\alpha = P_*\colon H^4(Y,\mathbb{Q}) \to H^2(F,\mathbb{Q})$ up to a Tate twist. Both $T(Y)$ and $T(F)$ are the sums of the isotypic components that are not of Tate type, so $\alpha(T(Y)) = T(F)$ and $E(F) = \alpha E(Y)\alpha^{-1}$. Let g be the Plücker class on F and $e = P_*^t \circ (g^2 \cup \cdot\,) \circ \alpha$, an algebraic correspondence that maps $T(Y)$ to itself, so that $e|_{T(Y)} \in E(Y)$. For $0 \neq \sigma \in H^{3,1}(Y)$, $\int_Y e(\sigma)\bar\sigma = \int_F g^2\alpha(\sigma)\alpha(\bar\sigma) = q(g)\,q(\alpha(\sigma),\alpha(\bar\sigma)) \neq 0$ by the Fujiki relation, since $q(g) > 0$ and $\alpha(\sigma)$ spans $H^{2,0}(F)$. So $e|_{T(Y)}$ is invertible, its inverse is a polynomial in it and is algebraic, and the inverse of α on $T(F)$, which is $e^{-1} \circ P_*^t \circ (g^2 \cup \cdot\,)$, is induced by an algebraic cycle. Hence $\varphi \in E(Y)$ acts through an algebraic cycle on $Y \times Y$ if and only if $\alpha\varphi\alpha^{-1}$ acts through one on $F \times F$, and Proposition 6.40(iii) proves the equivalences and the case where $E(Y) = E(F)$ is $\mathbb{Q}$ or a CM field. A very general Y has $E(Y) = \mathbb{Q}$. ∎

The class $\mathcal{A}$ of Proposition 6.35(iii) is larger than the closure properties recorded there show. Four further operations preserve it, the last of them domination by correspondences, which brings in the varieties whose motives are known to be built from those of abelian varieties, and two classical families lie in it.

Proposition 6.42 (More varieties of $\mathcal{A}$)**.** *Let $\mathcal{A}$ be as in Proposition 6.35(iii).*

(i) *If $Y \in \mathcal{A}$ and $Z \subset Y$ is a smooth closed subvariety with $Z \in \mathcal{A}$, then the blow-up $\mathrm{Bl}_Z Y$ lies in $\mathcal{A}$. If $Y \in \mathcal{A}$ and V is a vector bundle on Y, then $\mathbb{P}(V) \in \mathcal{A}$.*

(ii) *If $S \in \mathcal{A}$ is a surface, then the Hilbert scheme $S^{[n]}$ lies in $\mathcal{A}$ for every $n \geq 1$.*

(iii) *Every Fermat hypersurface $X_m^r = \{x_0^m + \cdots + x_{r+1}^m = 0\} \subset \mathbb{P}^{r+1}$, $r \geq 0$, $m \geq 1$, lies in $\mathcal{A}$.*

(iv) *A K3 surface S lies in $\mathcal{A}$ as soon as its Kuga–Satake correspondence is induced by an algebraic cycle, that is, as soon as some algebraic cycle Γ on $S \times B$, with B an abelian variety, induces a nonzero map $T(S)_{\mathbb{Q}} \to H^2(B,\mathbb{Q})$.*

(v) *If $H^*(X,\mathbb{Q})$ is the sum of the images of the $H^*(Y_i,\mathbb{Q})$, with $Y_i \in \mathcal{A}$, under algebraic correspondences, then $X \in \mathcal{A}$; in particular $X \in \mathcal{A}$ when its Chow motive is a direct summand of a sum of Tate twists of motives of members of $\mathcal{A}$, for instance when it is of abelian type. So $\mathcal{A}$ contains the $2n$-dimensional generalised Kummer varieties $K_n(A)$ of abelian surfaces A* [Xu15, Theorem 2.7, Corollary 2.8], [FTV19, Theorem 1.5]*; the smooth projective moduli spaces of Gieseker stable (twisted) sheaves, or of σ-stable objects for a generic stability condition σ, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [Bül20, Theorem 0.1, Remark 2.1]*; O'Grady's six-dimensional varieties $\widetilde{K}_v(A,H)$ obtained from moduli spaces of sheaves on an abelian surface A, with $v = 2v_0$, v_0 primitive, effective and of square 2, and H a v-generic polarisation* [Flo23, Theorem 1.1]*; the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of σ-semistable objects, with v_0 primitive, $v_0^2 = 2$ and σ v_0-generic, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [FFZ21, Theorem 1.3, Corollary 4.5]*; and the cubic fourfolds whose lattice of algebraic 2-cycles has rank greater than 19, with their Fano varieties of lines, the eightfolds of Lehn, Lehn, Sorger and van Straten of those containing no plane, and the tenfolds of Laza, Saccà and Voisin of infinitely many Pfaffian cubic fourfolds* [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

Consequently (F3′) *holds on every smooth projective variety obtained from abelian varieties, curves, Fermat hypersurfaces and K3 surfaces as in* (iv) *by products, surjective images, blow-ups along smooth centres in $\mathcal{A}$, projective bundles and Hilbert schemes of points on surfaces, and on every variety dominated by these as in* (v).

Proof. (i) Let c be the codimension of Z, $E=\mathbb{P}(N_{Z/Y})$ the exceptional divisor, $j\colon E\to\widetilde{Y}=\mathrm{Bl}_Z Y$ its inclusion, $\rho\colon E\to Z$ and $\pi\colon\widetilde{Y}\to Y$ the projections and $\xi=c_1(\mathcal{O}_E(1))$. By the blow-up formula [Voi02, §7.3.3],

$$H^*(\widetilde{Y},\mathbb{Q})=\pi^*H^*(Y,\mathbb{Q})\oplus\bigoplus_{i=0}^{c-2}j_*\big(\xi^i\cup\rho^*H^*(Z,\mathbb{Q})\big).$$

The first summand is the image of the transpose of the graph of π, and the i-th of the others is the image of the algebraic cycle $(\rho\times j)_*\xi^i$ on $Z\times\widetilde{Y}$. Composing with the correspondences that exhibit Y and Z as members of $\mathcal{A}$, which is allowed [Ful98, Chapter 16], shows that $H^*(\widetilde{Y},\mathbb{Q})$ is spanned by classes Γ_*y with B abelian. For $\mathbb{P}(V)$ the projective bundle formula $H^*(\mathbb{P}(V),\mathbb{Q})=\bigoplus_i\xi^i\cup\pi^*H^*(Y,\mathbb{Q})$ [Voi02, §7.3.3] does the same with the algebraic cycles $\Gamma^t_\pi\cdot\mathrm{pr}^*\xi^i$.

(ii) By de Cataldo and Migliorini [dCM02], for every partition ν of n with $l(\nu)$ parts the incidence correspondence $\Gamma_\nu\subset S^{(\nu)}\times S^{[n]}$ between the product $S^{(\nu)}$ of symmetric products attached to ν and the Hilbert scheme induces, after a Tate twist by $n-l(\nu)$, an isomorphism $\bigoplus_\nu H^*(S^{(\nu)},\mathbb{Q})\to H^*(S^{[n]},\mathbb{Q})$. Let $q_\nu\colon S^{l(\nu)}\to S^{(\nu)}$ be the quotient by the finite group G_ν. With rational coefficients the cycles and the cohomology of $S^{(\nu)}$ are the G_ν-invariants of those of $S^{l(\nu)}$ [Ful98, Example 1.7.6], so Γ_ν pulls back to a cycle $\widetilde{\Gamma}_\nu$ with rational coefficients on the smooth variety $S^{l(\nu)}\times S^{[n]}$, whose action on $H^*(S^{l(\nu)},\mathbb{Q})$ is $\Gamma_{\nu*}\circ q_{\nu*}$ up to a nonzero factor. As $H^*(S^{(\nu)},\mathbb{Q})=q_{\nu*}H^*(S^{l(\nu)},\mathbb{Q})$, the ν-th summand is the image of $H^*(S^{l(\nu)},\mathbb{Q})$ under $\widetilde{\Gamma}_\nu$. Products of members of $\mathcal{A}$ lie in $\mathcal{A}$, so $S^{[n]}\in\mathcal{A}$.

(iii) By induction on r, following the inductive structure of Shioda and Katsura [SK79]. For $r=0$, X^0_m is a finite set of points, and X^1_m is a curve. Let $r\geq 2$ and $r=a+b$ with $a,b\geq 1$; fix ϵ with $\epsilon^m=-1$ and put

$$\varphi\big((x),(y)\big)=\big(x_0y_{b+1}:\cdots:x_ay_{b+1}:\epsilon x_{a+1}y_0:\cdots:\epsilon x_{a+1}y_b\big),$$

a rational map $X^a_m\times X^b_m\dashrightarrow\mathbb{P}^{r+1}$; since $\sum_{i\leq a}x_i^m=-x_{a+1}^m$ and $\sum_{k\leq b}y_k^m=-y_{b+1}^m$, its image lies in X^r_m. It is defined by sections of $\mathcal{O}(1,1)$ whose common zero scheme is $\{x_{a+1}=0\}\times\{y_{b+1}=0\}$, the product $Z\cong X^{a-1}_m\times X^{b-1}_m$ of two smooth hyperplane sections, transversal because the Fermat hypersurfaces are smooth. Blowing up that reduced scheme resolves the map, so φ extends to a morphism $\widetilde{\varphi}\colon\mathrm{Bl}_Z(X^a_m\times X^b_m)\to X^r_m$. It is dominant: at a point z of X^r_m with $\sum_{i\leq a}z_i^m\neq 0$ the preimages with $x_{a+1}=y_{b+1}=1$ are the points $x_i=\lambda z_i$, $y_k=\lambda z_{a+1+k}/\epsilon$ with $\lambda^m\sum_{i\leq a}z_i^m=-1$, and both equations reduce to this one because $\sum_{\mathrm{all}}z_i^m=0$. So $\widetilde{\varphi}$ is surjective, being proper. By induction X^a_m, X^b_m, X^{a-1}_m and X^{b-1}_m lie in $\mathcal{A}$, hence so do their products, the blow-up by (i) and X^r_m as a surjective image; one may take $a=r-1$, $b=1$.

(iv) The transcendental part $T=T(S)_\mathbb{Q}$ is an irreducible Hodge structure, and $H^*(S,\mathbb{Q})=T\oplus T^\perp$ with $T^\perp$ spanned by algebraic classes. Let h be an ample class on B, $d=\dim B$, and let $\phi=\Gamma^t_*\circ(\,\cdot\cup h^{d-2})\circ\Gamma_*$, an algebraic correspondence from S to itself and a morphism of Hodge structures. It preserves T, because a morphism from the irreducible structure T to the algebraic part, which is of Tate type, vanishes; so $\phi|_T$ lies in the field $\mathrm{End}_{\mathrm{Hdg}}(T)$. For $0\neq\sigma\in H^{2,0}(S)$, $\Gamma_*\sigma\neq 0$ because T is irreducible and $\Gamma_*|_T\neq 0$; it is a class of type $(2,0)$ on B, hence primitive, and by the Hodge–Riemann relations $\int_S\phi(\sigma)\,\bar\sigma=\int_B\Gamma_*\sigma\,\overline{\Gamma_*\sigma}\,h^{d-2}\neq 0$. So $\phi|_T\neq 0$, it is invertible, and $T\subseteq\Gamma^t_*H^*(B,\mathbb{Q})$. With $T^\perp$ algebraic, $S\in\mathcal{A}$.

(v) If $H^*(X,\mathbb{Q})=\sum_i\Psi_{i*}H^*(Y_i,\mathbb{Q})$ and each $H^*(Y_i,\mathbb{Q})$ is spanned by classes $\Gamma_{ij*}y$ with B_{ij} abelian, then $H^*(X,\mathbb{Q})$ is spanned by the classes $(\Psi_i\circ\Gamma_{ij})_*y$, and composites of

algebraic correspondences are algebraic [Ful98, Chapter 16]. If $h(X)$ is a direct summand of $\bigoplus_i h(Y_i)(n_i)$, through α and β with $\beta\circ\alpha=\mathrm{id}$, then β_* is surjective on cohomology, and its components are the Ψ_i; a motive of abelian type is such a summand with abelian varieties as the Y_i. The objects of the tensor subcategory generated by $h(Y)$ are direct summands of sums of Tate twists of the motives $h(Y^k)$, since the dual of $h(Y)$ is a Tate twist of $h(Y)$, and $Y^k\in\mathcal{A}$ when $Y\in\mathcal{A}$. The Chow motive of $K_n(A)$ lies in the tensor subcategory generated by $h(A)$ [Xu15, Theorem 2.7, Corollary 2.8], see also [FTV19, Theorem 1.5]; that of a moduli space as in (v) is a direct summand of a sum of Tate twists of motives of powers of the surface [Bül20, Theorem 0.1]; those of the varieties $\widetilde{K}_v(A,H)$ and of the ten-dimensional crepant resolutions lie in the tensor subcategory generated by the motive of the surface [Flo23, Theorem 1.1], [FFZ21, Corollary 4.5]; and the cubic fourfolds and hyperkähler varieties of [ABLP26] listed in (v) have Chow motives of abelian type [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

The last assertion is Proposition 6.35(iii). ■

The known cases of (iv) are recalled in Remark 7.5; for the K3 surfaces among them whose powers are covered by [Var22] the Hodge conjecture itself holds on every power, which is stronger. What (iii) adds is a family on which (F3′) holds in every degree and dimension, while the Hodge conjecture for Fermat hypersurfaces is proved in a range of degrees and dimensions [Shi79]. Three recent preprints, if correct, extend that range to every Fermat variety of degree less than 65 other than 44, 51 and 52 [MMMV26], to the Fermat fourfolds of every odd degree at most 199, by a computer-assisted proof [Jum26], and to the Fermat fourfold of degree 33 [FK26]; (iii) is new for the degrees and dimensions outside these ranges. By (iii) and Proposition 6.35(i) the conjecture holds for X^r_m as soon as it holds for the powers of the Jacobian of the Fermat curve of degree m. The inductive structure is Shioda and Katsura's, and the consequence for (F3′) is immediate; we claim no priority for it.

Item (v) collects consequences of published motivic results, and we claim no priority for them either. For a generalised Kummer variety $K_n(A)$ it is needed in general. The restriction from $A^{[n+1]}$, which lies in $\mathcal{A}$ by (ii), reaches exactly the $A[n+1]$-invariant part of $H^*(K_n(A),\mathbb{Q})$, since $A\times K_n(A)\to A^{[n+1]}$ is an étale Galois cover with group $A[n+1]$. When $n+1$ is prime the rest is spanned by the classes of the punctual Hilbert schemes at the points of $A[n+1]$, which are algebraic, and (ii) suffices; when $n+1$ is composite the rest contains transcendental classes, and [Xu15, FTV19] are needed. These results concern the varieties $K_n(A)$ themselves, not every variety of generalised Kummer type; for products of varieties $K_n(A)$ the Hodge conjecture itself is known [Xu15, Theorem 3.3], and for the varieties $\widetilde{K}_v(A,H)$ it holds on all self-products [Flo23, Corollary 1.2]. If the preprint [Ped26] is correct, (v) also puts in $\mathcal{A}$ the tenfolds of Laza, Saccà and Voisin of every cubic fourfold in $\mathcal{A}$, whose motives it makes direct summands of Tate twists of the motive of the fifth power of the cubic fourfold. In low dimension the algebraicity results of Bloch–Srinivas and Conte–Murre give more.

Proposition 6.43 (Varieties whose zero-cycles are small)**.** *Let X be a smooth complex projective variety of dimension n whose group $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subvariety of dimension at most three. If $n\leq 5$, the Hodge conjecture holds for X, and so does* (F3′)*. This applies to every fourfold whose group of zero-cycles is supported on a proper closed subset, in particular every uniruled fourfold, and to every fivefold whose maximal rationally connected quotient has dimension at most three, in particular every rationally connected fivefold.*

Proof. By Bloch and Srinivas [BS83] every Hodge class of degree 4 on X is algebraic. By Proposition 6.35(iv) the Hodge classes of degree 0, 2, $2n-2$ and $2n$ are algebraic, which settles $n \leq 4$. For $n = 5$ the remaining degree is $6 = 2n-4$, and cup product with h, for h ample, is an isomorphism $H^4(X,\mathbb{Q}) \to H^6(X,\mathbb{Q})$ of Hodge structures up to a Tate twist by hard Lefschetz, so every Hodge class of degree 6 is h times one of degree 4. A uniruled fourfold has CH_0 supported on any ample divisor D, because a rational curve through a point meets D and any two points of $\mathbb{P}^1$ are rationally equivalent; this case is the theorem of Conte and Murre [CM78]. A proper closed subset of a fourfold has dimension at most three.

Let $\phi\colon X \dashrightarrow R$ be the maximal rationally connected fibration of Campana and of Kollár, Miyaoka and Mori [Cam92, KMM92]: R, the maximal rationally connected quotient, is smooth and projective and unique up to birational equivalence, and ϕ is dominant with connected fibres, the general one rationally connected. Resolving its indeterminacy [Hir64] gives a smooth projective variety $\widetilde{X}$, a birational morphism $\pi\colon \widetilde{X} \to X$, an isomorphism over the domain of definition of ϕ, and a surjective morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ whose general fibre is smooth, by generic smoothness, and birational to a general fibre of ϕ, hence rationally connected. Suppose $\dim R \leq 3$, and let $W \subset \widetilde{X}$ be a subvariety that $\widetilde{\phi}$ maps generically finitely onto R, of degree e. Any two points of a general fibre F are joined by a rational curve, so every $x \in F$ satisfies $e[x] = [W \cap F]$ in $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$. The general fibres cover a dense open subset of $\widetilde{X}$, and every zero-cycle is rationally equivalent to one supported on a given dense open subset, so $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ is supported on W and $\mathrm{CH}_0(X)_{\mathbb{Q}} = \pi_*\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ on $\pi(W)$, of dimension $\dim R \leq 3$. A rationally connected fivefold is the case where R is a point. ∎

In degree four the hypothesis on zero-cycles can be relaxed to one on the fibres of a map.

Proposition 6.44 (Degree four along rationally connected fibrations)**.** *Let $f\colon X \to Y$ be a surjective morphism of smooth complex projective varieties whose general fibre is rationally connected. Then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$ if and only if $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, and every Hodge class of degree four on X is algebraic if and only if every one on Y is. Both properties are therefore birational invariants of smooth projective varieties, and they hold for X exactly when they hold for a smooth projective model of its maximal rationally connected quotient. Consequently* (F3′) *holds for every fourfold and every fivefold whose maximal rationally connected quotient is birational to a variety R with $\mathrm{Hdg}^2(R) = \mathrm{Ab}^2(R)$, for instance a member of $\mathcal{A}$. With the Hodge conjecture for abelian fourfolds* [Mar25a, Mar25b]*, the Hodge conjecture holds for every fourfold birational to an abelian fourfold, and for every fivefold admitting a dominant rational map with rationally connected general fibre onto an abelian fourfold.*

Proof. Let $n = \dim X$ and $m = \dim Y$. Let W be a component, dominating Y, of the intersection of X with a general linear space of codimension $n-m$, so that $W \to Y$ is generically finite, of degree $e \geq 1$; let $i_W\colon \widetilde{W} \to X$ be a resolution of W [Hir64] and $g = f \circ i_W$. The correspondence $\Phi = \Gamma_f^t \circ \Gamma_g \circ \Gamma_{i_W}^t$ on $X \times X$ acts by $\Phi_* = f^* g_* i_W^*$ [Ful98, Chapter 16]. Let η be the generic point of X and K its field of functions. The restriction of Φ^t to $\{\eta\} \times X$ is the zero-cycle $W_{f(\eta)}$, of degree e, that W cuts on the fibre of X_K over the point $f(\eta)$, and that of $[\Delta_X]$ is the point η of the same fibre. Any two points of a general fibre of f are rationally equivalent, and the argument of Bloch and Srinivas [BS83] (a countable algebraically closed field of definition, an embedding of an algebraic closure of K into $\mathbb{C}$, specialisation and a norm) gives $e[\eta] = [W_{f(\eta)}]$ in $\mathrm{CH}_0(X_K)_{\mathbb{Q}}$. As $\mathrm{CH}_0(X_K)$ is

the direct limit of the groups $\mathrm{CH}^n(U \times X)$ over the dense open subsets $U \subseteq X$, the cycle $e[\Delta_X] - \Phi^t$ vanishes on some $U \times X$, and by the localisation sequence [Ful98, Chapter 1] it is supported on $D \times X$ for a divisor $D \supseteq X \smallsetminus U$. The diagonal is symmetric, so $e[\Delta_X] - \Phi$ is supported on $X \times D$. Let $i\colon D' \to X$ be a resolution of D. Each component of a cycle supported on $X \times D$ is, up to a nonzero factor, the image of a subvariety of $X \times D'$ mapping generically finitely onto it, so $e[\Delta_X] - \Phi = (\mathrm{id} \times i)_* \Psi$ for a cycle Ψ with rational coefficients on $X \times D'$, and the projection formula gives, for every $a \in H^k(X, \mathbb{Q})$,

$$e\, a = f^* g_* i_W^* a + i_* \Psi_* a, \qquad \Psi_* a \in H^{k-2}(D', \mathbb{Q}).$$

Let $a \in \mathrm{Hdg}^2(X)$. Then $g_* i_W^* a \in \mathrm{Hdg}^2(Y)$, and $\Psi_* a$ is a Hodge class of degree two on D', algebraic by the Lefschetz theorem on $(1,1)$ classes. If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $f^* g_* i_W^* a \in \mathrm{Ab}^2(X)$, because composites of algebraic correspondences are algebraic, and so $a \in \mathrm{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, a is algebraic. Conversely, as in the proof of Proposition 6.35(iii), for h ample on X the correspondence $[\Gamma_f] \cdot \mathrm{pr}_X^* h^{n-m}$ sends $f^* b$ to $c\, b$, with $c > 0$ the degree of h^{n-m} on a general fibre; so $b \in \mathrm{Hdg}^2(Y)$ lies in $\mathrm{Ab}^2(Y)$, or is algebraic, as soon as $f^* b$ does, or is.

A birational morphism is surjective with general fibre a point, and two birational smooth projective varieties are dominated through birational morphisms by a resolution of the closure of the graph [Hir64]; so both properties are birational invariants. The morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ of the proof of Proposition 6.43, from a variety birational to X onto its maximal rationally connected quotient, has rationally connected general fibre, which gives the statement on R. For $n \leq 5$ the degrees 0, 2, $2n-2$ and $2n$ are settled by Proposition 6.35(iv), and for $n = 5$ cup product with h is an isomorphism $H^4(X, \mathbb{Q}) \to H^6(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so that $\mathrm{Hdg}^3(X) = h \cup \mathrm{Hdg}^2(X)$; members of $\mathcal{A}$ satisfy (F3′) by Proposition 6.35(iii). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b]. A dominant rational map onto B with rationally connected general fibre becomes, after its indeterminacy is resolved, a morphism whose general fibre is smooth and rationally connected, as in the proof of Proposition 6.43; so the fourfolds and fivefolds of the last assertion have every Hodge class of degree four algebraic, and the other degrees are settled as before. ∎

Proposition 6.44 reaches varieties that Proposition 6.42 does not: conic bundles with singular fibres over members of $\mathcal{A}$ of dimension three or four, and fourfolds and fivefolds birational to members of $\mathcal{A}$ without being known to lie in $\mathcal{A}$. A projective bundle over a member of $\mathcal{A}$ already lies in $\mathcal{A}$ by Proposition 6.42(i), and so, by Proposition 6.42(v) and the Leray–Hirsch theorem, does the source of a smooth morphism onto a member of $\mathcal{A}$ on which algebraic classes restrict to a basis of the cohomology of a fibre, such as a Severi–Brauer bundle. Proposition 6.44 is the relative form of the argument of Bloch and Srinivas, and we claim no priority for it.

Remark 6.45 (Zero-cycles alone go no further). Hypotheses on zero-cycles cannot carry (F3′) further without the fourfolds. If $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subset Z, with resolution $j\colon Z' \to X$, the proof of Proposition 6.44, with the closure in $X \times Z$ of a zero-cycle on Z_K rationally equivalent to the generic point in place of Φ^t, gives a divisor with resolution $i\colon D' \to X$ and cycles Φ on $Z' \times X$ and Ψ on $X \times D'$ with $a = \Phi_* j^* a + i_* \Psi_* a$ for every $a \in H^k(X, \mathbb{Q})$, where $\Psi_* a$ has degree $k-2$; this is the transpose of the decomposition of Bloch and Srinivas [BS83]. When $Z \neq X$ and a has type $(n,0)$, $n = \dim X$, the first term vanishes, as $\dim Z' < n$, and the second has no component of type $(n,0)$; so $H^{n,0}(X) = 0$. A uniruled fivefold has its zero-cycles supported on an ample divisor, and for a fourfold Y

and $b \in \mathrm{Hdg}^2(Y)$ the class pr^*b is a Hodge class on $Y \times \mathbb{P}^1$ with $b = \mathrm{pr}_*(\mathrm{pr}^*b \cup \eta)$, for η the class of $Y \times \{0\}$. So (F3′), and likewise the Hodge conjecture, holds for every uniruled fivefold if and only if it holds in degree four for every fourfold.

It also holds for every rational sixfold if and only if it holds for every fourfold. A rational sixfold has its zero-cycles supported on a point, so the proof of Proposition 6.43 settles every degree but six; and it is the image under a birational morphism of a variety obtained from $\mathbb{P}^6$ by blow-ups along smooth centres [Hir64], on which, by the blow-up formula [Voi02, §7.3.3], the Hodge classes of degree six are images under algebraic correspondences of Hodge classes of degree at most four on the centres, of dimension at most four, and lie in Ab^3, or are algebraic, when the fourfolds satisfy (F3′), or the conjecture. Conversely, a fourfold Y is birational to a hypersurface in $\mathbb{P}^5$; place it in a hyperplane of $\mathbb{P}^6$, and let $\widetilde{Y}$ be its smooth strict transform under an embedded resolution $M \to \mathbb{P}^6$ [Hir64]. On the rational sixfold $X = \mathrm{Bl}_{\widetilde{Y}} M$, with exceptional divisor $j\colon E \to X$ and projection $\rho\colon E \to \widetilde{Y}$, the normal bundle of E is $\mathcal{O}_E(-1)$, so $a = -\rho_* j^* j_* \rho^* a$ for $a \in H^4(\widetilde{Y}, \mathbb{Q})$; this recovers $\mathrm{Hdg}^2(\widetilde{Y})$ from $\mathrm{Hdg}^3(X)$ through algebraic correspondences, and Proposition 6.44 passes from $\widetilde{Y}$ to Y.

Among smooth hypersurfaces, ordered by dimension and degree, the first family that these results do not cover is that of the sextic fourfolds: those of dimension at most three are covered by Proposition 6.35(iv), the fourfolds of degree at most five are Fano, hence rationally connected [Cam92, KMM92], and covered by Proposition 6.43, while a sextic fourfold has $h^{4,0} = 1$, so that its zero-cycles are not supported on a proper closed subset and it is its own maximal rationally connected quotient; the Fermat sextic lies in $\mathcal{A}$ by Proposition 6.42(iii). For a smooth sextic fourfold Y in a hyperplane of $\mathbb{P}^6$ no resolution is needed, and the rational sixfold $\mathrm{Bl}_Y \mathbb{P}^6$, with $h^{6,0} = 0$, $h^{5,1} = 1$, $h^{4,2} = 426$ and $h^{3,3} = 1753$ by the blow-up formula [Voi02, §7.3.3] and the Hodge numbers 1, 426, 1752 of Y, satisfies (F3′) if and only if Y does. For the Hodge conjecture these equivalences are folklore, and we claim no priority for them.

Surjective morphisms carry $\mathcal{A}$ to itself (Proposition 6.35(iii)). In degree four the morphism can be replaced by any dominant rational map, which reaches some of these sextic fourfolds.

Proposition 6.46 (Degree four along dominant rational maps)**.** *Let $\varphi\colon Y \dashrightarrow X$ be a dominant rational map of smooth complex projective varieties.*

(i) *If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$; if every Hodge class of degree four on Y is algebraic, so is every one on X.*

(ii) (F3′) *holds on every fourfold and every fivefold that admits a dominant rational map from a variety Y with $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, for instance from a member of $\mathcal{A}$. The Hodge conjecture holds on every fourfold that admits a dominant rational map from an abelian fourfold, such as a smooth projective model of the quotient of an abelian fourfold by a finite group, and on every fivefold that admits one from the product of an abelian fourfold and a projective space.*

Proof. (i) Let $\Gamma \subset Y \times X$ be the closure of the graph of φ and $\widetilde{Y} \to \Gamma$ a resolution of singularities [Hir64]. The projections give a birational morphism $\widetilde{Y} \to Y$ and a morphism $g\colon \widetilde{Y} \to X$, which is surjective, being proper and dominant. Both properties are birational invariants (Proposition 6.44), so they hold for $\widetilde{Y}$. Let $x \in \mathrm{Hdg}^2(X)$, let h be ample on $\widetilde{Y}$ and $r = \dim \widetilde{Y} - \dim X$, and let $\Psi = [\Gamma_g] \cdot \mathrm{pr}_{\widetilde{Y}}^* h^r$, so that $\Psi_* g^* x = c\,x$ with $c > 0$, as in the proof of Proposition 6.35(iii). The class g^*x is a Hodge class of degree four on $\widetilde{Y}$, hence

a sum of classes $\Gamma_{j*}\gamma_j$ as in Definition 6.34, or algebraic. So $x = c^{-1}\sum_j(\Psi\circ\Gamma_j)_*\gamma_j$ lies in $\mathrm{Ab}^2(X)$, or x is algebraic, since composites of algebraic correspondences are algebraic [Ful98, Chapter 16].

(ii) On a fourfold or a fivefold the degrees other than four and six are settled by Proposition 6.35(iv), and the same argument, the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz, makes the Hodge classes of those degrees algebraic; on a fivefold cup product with an ample class is an isomorphism from H^4 onto H^6 of Hodge structures up to a Tate twist, so degree six follows from degree four. Members of $\mathcal{A}$ satisfy (F3′) (Proposition 6.35(iii)). An abelian fourfold B satisfies the Hodge conjecture [Mar25a, Mar25b], and by the Künneth formula a Hodge class of degree four on $B\times\mathbb{P}^k$ is a sum of pull-backs of Hodge classes of degrees four, two and zero on B multiplied by powers of the hyperplane class, so it is algebraic. A quotient B/G by a finite group is dominated by B, and so is every smooth projective model of it. ■

Lefschetz theory carries the two statements from a variety to its ample complete intersections, outside the middle degree always and in the middle degree for a very general member.

Proposition 6.47 (Ample complete intersections)**.** *Let X be a smooth complex projective variety of dimension N, L an ample line bundle on X, and $Y = Y_c \subset Y_{c-1}\subset\cdots\subset Y_0 = X$ with each Y_j a smooth divisor of Y_{j-1} in the linear system of $L^{m_j}|_{Y_{j-1}}$, $m_j\geq 1$; put $r = \dim Y = N-c\geq 1$ and let $i\colon Y\to X$ be the inclusion.*

(i) *If $2p\neq r$, every Hodge class of degree $2p$ on Y is $i^*\alpha$ for a Hodge class α of degree $2p$ on X. So if* (F3′)*, or the Hodge conjecture, holds for X in degree $2p$, it holds for Y in degree $2p$; if r is odd and it holds for X in every degree, it holds for Y in every degree.*

(ii) *Let r be even, $c=1$ and L^{m_1} very ample, and let $V\subset H^r(Y,\mathbb{Q})$ be the vanishing cohomology, the kernel of the Gysin map i_*. If V is not of type $(r/2,r/2)$, then for a very general Y in the linear system every Hodge class of degree r on Y is $i^*\alpha$ for a Hodge class α on X, and the conclusion of (i) holds in degree r as well.*

So the Hodge conjecture holds for every smooth ample divisor of odd dimension in $B\times\mathbb{P}^k$, for B an abelian variety of dimension at most five, and for the very general member of every very ample linear system on an abelian fivefold; and (F3′) *holds for every smooth ample divisor of odd dimension in a member of $\mathcal{A}$.*

Proof. (i) By the Lefschetz hyperplane theorem [Voi03, Chapter 1], applied to each $Y_j\subset Y_{j-1}$, the restriction $i^*\colon H^k(X,\mathbb{Q})\to H^k(Y,\mathbb{Q})$ is an isomorphism for $k<r$ and injective for $k=r$. Let $2p<r$. Then i^* is an isomorphism of Hodge structures in degree $2p$, so every Hodge class on Y is i^* of a Hodge class on X. Let $2p>r$. The Gysin map $i_*\colon H^{2p}(Y,\mathbb{Q})\to H^{2p+2c}(X,\mathbb{Q})$ is Poincaré dual to i^* in degree $2r-2p<r$, hence an isomorphism, and i_*i^* is cup product with $[Y] = m_1\cdots m_c\,c_1(L)^c$. By the hard Lefschetz theorem [Voi02, Chapter 6], cup product with $c_1(L)^c$ maps $H^{2p}(X,\mathbb{Q})$ onto $H^{2p+2c}(X,\mathbb{Q})$ when $2p\geq N-c$, so i^* is onto in degree $2p$. A surjective morphism of polarisable Hodge structures has a section that is a morphism of Hodge structures, so again every Hodge class on Y is i^* of a Hodge class on X. The restriction i^* is induced by the transpose of the graph of i, so it carries $\mathrm{Ab}^p(X)$ into $\mathrm{Ab}^p(Y)$ and algebraic classes to algebraic classes, which gives the rest of (i).

(ii) The projection formula shows that V is the orthogonal complement of $i^*H^r(X,\mathbb{Q})$ for the intersection form, and $H^r(Y,\mathbb{Q}) = i^*H^r(X,\mathbb{Q})\oplus V$ as Hodge structures [Voi03,

Chapter 2]; the Hodge classes in the first summand are the $i^*\alpha$ with α a Hodge class on X. Over the open set U of the linear system parametrising smooth members, V is a local system, and its monodromy group Γ acts irreducibly [Voi03, Chapter 3]. For each class v of the lattice of V at a point of the universal cover of U, the set where the flat continuation of v is of Hodge type is a closed analytic subset, either everything or nowhere dense; so at a very general point the Hodge classes of V are those whose continuation is of Hodge type everywhere. They form a subspace W stable under Γ, hence $W = 0$ or $W = V$, and $W = V$ would make V of type $(r/2, r/2)$. So $W = 0$.

For the consequences, $B \times \mathbb{P}^k$ satisfies the Hodge conjecture by the Künneth formula, since B does [Mar25a, Mar25b, MZ99]. For a very ample divisor Y of an abelian fivefold B, $r = 4$, and V is not of type $(2,2)$: $h^{4,0}(Y) = h^0(K_Y) = h^0(Y, \mathcal{O}_Y(Y)) = h^0(B, \mathcal{O}_B(Y)) + 4$, by the sequence $0 \to \mathcal{O}_B \to \mathcal{O}_B(Y) \to \mathcal{O}_Y(Y) \to 0$ and $h^1(\mathcal{O}_B(Y)) = 0$, exceeds $h^{4,0}(B) = 5$ because $h^0(\mathcal{O}_B(Y)) \geq 2$, so V has a nonzero part of type $(4,0)$. Members of $\mathcal{A}$ satisfy (F3′) by Proposition 6.35(iii). ∎

For the Hodge conjecture these transfers are classical, and we claim no priority. What they add to (F3′) is a large class outside $\mathcal{A}$: the vanishing cohomology of an ample divisor of odd dimension is not reached from abelian varieties in general, yet it carries no Hodge class in even degree, since its only degree is the odd middle one.

A hypersurface of Delsarte type is defined by as many monomials as variables. Shioda observed that such a surface is dominated by a Fermat surface [Shi86]; the same monomial substitution works in every dimension, and with Proposition 6.46 it gives the following.

Proposition 6.48 (Delsarte fourfolds)**.** *Let $A = (a_{ij})_{0 \leq i,j \leq r+1}$ be an invertible matrix of non-negative integers whose rows all sum to $m \geq 2$, let $F_A = \sum_i \prod_j x_j^{a_{ij}}$, suppose that $X_A = \{F_A = 0\} \subset \mathbb{P}^{r+1}$ is smooth, and let d be a positive integer with dA^{-1} integral, such as $|\det A|$.*

(i) *There is a dominant rational map from the Fermat hypersurface X_d^r onto X_A, given by monomials.*

(ii) $\mathrm{Hdg}^2(X_A) = \mathrm{Ab}^2(X_A)$. *For $r = 4$, X_A satisfies* (F3′)*, and it satisfies the Hodge conjecture if X_d^4 does.*

(iii) *If $m \geq 3$ and F_A is a sum, in disjoint sets of variables, of terms x^m, chains $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^m$ and loops $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^{m-1}x_1$ with $k \geq 2$, then A is invertible and X_A is smooth. For $r = 4$ and $m = 6$ these polynomials have 29 shapes up to a renumbering of the variables, the Fermat sextic being one; each defines a smooth sextic fourfold, with $h^{4,0} = 1$, $h^{3,1} = 426$ and primitive $h^{2,2}$ equal to 1751. The least d is 6 for the Fermat sextic, 24 or 30 for six others, those whose terms other than x^m are all loops of length two or all chains of length two, at least 120 for the remaining twenty-two, and $5^6 - 1 = 15624$ for the loop $x_0^5x_1 + x_1^5x_2 + \cdots + x_5^5x_0$.*

Proof. (i) The matrix dA^{-1} is integral for $d = |\det A|$, by the formula for the inverse through the adjugate. Let $B = dA^{-1}$. From $A\mathbf{1} = m\mathbf{1}$, with $\mathbf{1}$ the vector of ones, $B\mathbf{1} = (d/m)\mathbf{1}$, so the rows of the integral matrix B all sum to the integer d/m, and $AB = d \cdot I$. Choose $c \geq 0$ with every $b_{jk} + c \geq 0$ and let $\psi(y)_j = \prod_k y_k^{b_{jk}+c}$; the coordinates of ψ are monomials of the same degree, so ψ is a rational map $\mathbb{P}^{r+1} \dashrightarrow \mathbb{P}^{r+1}$, and

$$F_A(\psi(y)) = \sum_i \prod_k y_k^{\sum_j a_{ij}(b_{jk}+c)} = (y_0 \cdots y_{r+1})^{cm} \sum_i y_i^d.$$

On the torus T of points with all coordinates nonzero, ψ is induced by the homomorphism $y \mapsto (\prod_k y_k^{b_{jk}})_j$ of $\mathbb{G}_m^{r+2}$, which is surjective because $\det B \neq 0$. So every $z \in X_A \cap T$ is $\psi(y)$ for some $y \in T$, and the identity puts y on X_d^r. A smooth hypersurface of degree at least two and positive dimension is irreducible, so it is not contained in a coordinate hyperplane, and $X_A \cap T$ and $X_d^r \cap T$ are dense. Hence ψ restricts to a dominant rational map $X_d^r \dashrightarrow X_A$.

(ii) X_d^r lies in $\mathcal{A}$ by Proposition 6.42(iii), so (i) and Proposition 6.46 give both assertions, the degrees other than four being settled as in the proof of Proposition 6.46(ii).

(iii) The exponent matrix is block diagonal, with blocks m, $(m-1)I+N+E$ for a chain, N the nilpotent shift and E the matrix whose only nonzero entry is a 1 in the last diagonal place, and $(m-1)I+P$ for a loop, P the cyclic shift; their determinants are m, $m(m-1)^{k-1}$ and $(m-1)^k-(-1)^k$, which are nonzero. The partial derivatives of a sum in disjoint sets of variables are those of the summands, so it suffices that the partials of each summand vanish together only at 0. For x^m this is clear. For a chain, $\partial F/\partial x_1 = (m-1)x_1^{m-2}x_2$ vanishes only if $x_1 = 0$ or $x_2 = 0$, and if $x_2 = 0$ then $\partial F/\partial x_2 = x_1^{m-1}$; so $x_1 = 0$, the same argument applies to $x_2, \dots, x_{k-1}$, and then $\partial F/\partial x_k = m x_k^{m-1}$ gives $x_k = 0$. For a loop, $\partial F/\partial x_j = (m-1)x_j^{m-2}x_{j+1} + x_{j-1}^{m-1}$, indices modulo k. If one coordinate vanishes, so does the one before it, hence all of them; if none vanishes, the product of the relations $x_{j-1}^{m-1} = -(m-1)x_j^{m-2}x_{j+1}$ gives $\prod_j x_j^{m-1} = (1-m)^k \prod_j x_j^{m-1}$, which is impossible because $|1-m| \geq 2$. The number of shapes, drawn in Figure 22, is the coefficient of t^6 in $(1-t)^{-1}\prod_{k\geq 2}(1-t^k)^{-2}$, which is 29, and the least d is the least common multiple of the orders of the cyclic groups attached to the terms in the proof of Corollary 6.53(iii): 6 for x^6, $6\cdot 5^{k-1}$ for a chain and $5^k-(-1)^k$ for a loop of length k; so 24 or 30 when the terms other than x^6 are all loops or all chains of length two, and at least $\operatorname{lcm}(24,30) = 120$, 5^3+1 or $6\cdot 5^2$ when both kinds of length two, or a loop or chain of length at least three, occur. The Hodge numbers are those of every smooth sextic fourfold, read off from the Jacobian ring [Voi03, Chapter 6]. ∎

A sextic fourfold has $h^{4,0} = 1$, so no hypothesis on zero-cycles or on rationally connected fibrations reaches it (Remark 6.45), and among the smooth sextic fourfolds only the Fermat one is covered by Proposition 6.42. The Delsarte sextics are finitely many up to isomorphism, since the torus rescales the coefficients of a Delsarte polynomial to 1, so they are isolated points in the 426-dimensional moduli of smooth sextic fourfolds: Proposition 6.48 adds points, not a family. By Proposition 6.48(ii) the Hodge conjecture for a Delsarte fourfold follows from that for one Fermat fourfold. For six of the sextic shapes that Fermat fourfold has degree 24 or 30, inside the range claimed in the preprint [MMMV26], so if it is correct the Hodge conjecture itself holds on these six; for the others the degree ranges from 120 to 18750, and from 120 to 3750 with the lattice degree of Corollary 6.53, beyond every known or claimed range [Shi79, MMMV26, Jum26]. The monomial cover is Shioda's, and we claim no priority for it.

Proposition 6.48 uses the covering map only in degree four, where algebraicity modulo abelian varieties descends along every dominant rational map. In higher degrees a birational modification can add classes, so the boundary of a compactification has to be controlled. For a hypersurface of a torus defined by affinely independent monomials the boundary is again of the same kind, and the whole cohomology can be traced back to Fermat varieties of a single degree. We use toric varieties as in [CLS11]: a torus T with character lattice M and cocharacter lattice N, a fan Σ in $N_{\mathbb{R}}$ and its toric variety Y_Σ, and for a cone $\tau \in \Sigma$ the orbit $O(\tau)$, whose character lattice is $\tau^\perp \cap M$, and its closure $V(\tau)$, the toric variety of

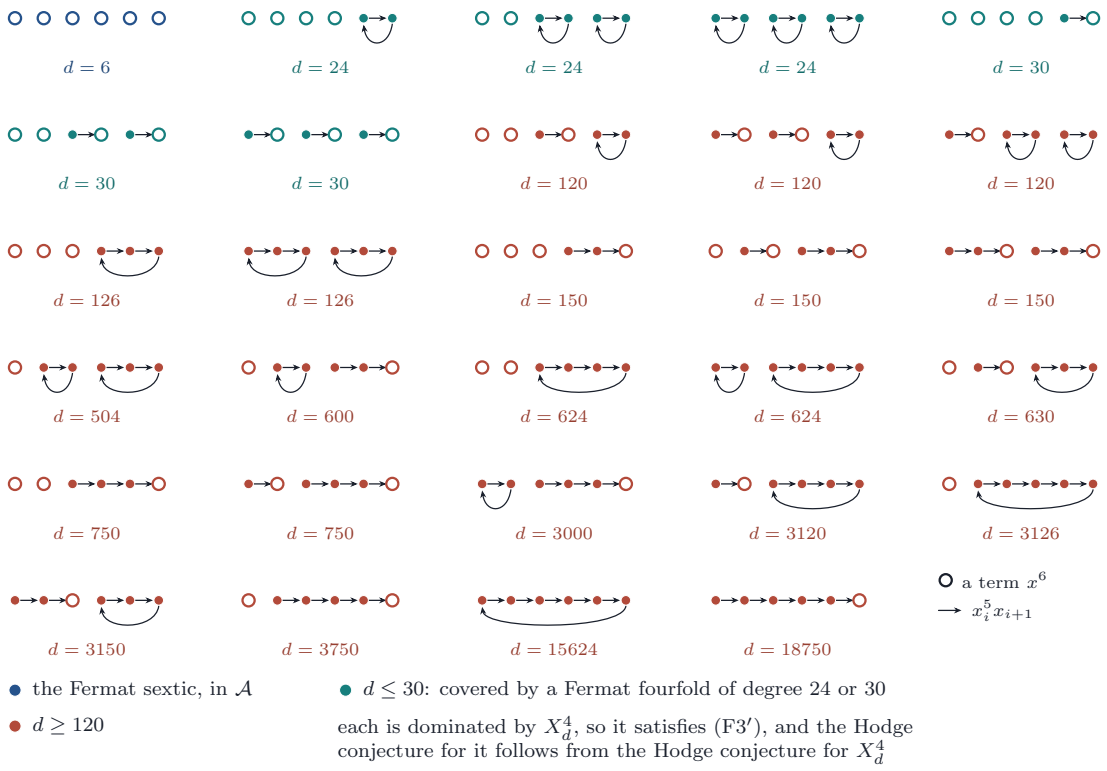


FIGURE 22. The 29 smooth sextic fourfolds of Proposition 6.48(iii), up to a renumbering of the variables. Each is drawn on its six variables: an arrow from x_i to x_j is a term $x_i^5 x_j$, and a ring is a term x^6, so a chain is a path of arrows ending at a ring and a loop is a cycle of arrows. Under each is the least d with dA^{-1} integral, the degree of the Fermat fourfold that covers it. All of them satisfy (F3′).

the fan formed by the images of the cones containing τ in $N_{\mathbb{R}}/\mathbb{R}\tau$ [CLS11, Chapter 3]; for a ray ρ we write $D_\rho = V(\rho)$. In what follows varieties need not be connected: a smooth projective variety lies in $\mathcal{A}$ when each of its connected components does, and the closure properties of Proposition 6.35(iii) and Proposition 6.42 hold component by component.

Definition 6.49 (Simplex type)**.** Let T be a torus of dimension n with character lattice M. A Laurent polynomial $f = \sum_{i=0}^{k} c_i \chi^{u_i}$ on T is of *simplex type* if every c_i is nonzero and $u_0, \dots, u_k \in M$ are affinely independent. Let $L' \subseteq M$ be the lattice spanned by the $u_i - u_0$, L its saturation, the set of $x \in M$ with a nonzero multiple in L', and $e(f)$ the exponent of the finite group L/L'; put $V_f = \{f = 0\} \subseteq T$. A fan Σ in $N_{\mathbb{R}}$ is *adapted* to f if for every cone $\sigma \in \Sigma$ there is an index i_σ with $\langle u_{i_\sigma}, v\rangle \leq \langle u_j, v\rangle$ for all j and all $v \in \sigma$, that is, if every cone of Σ lies in a cone of the normal fan of the simplex spanned by the u_i. For a cone τ of an adapted fan let I_τ be the set of indices i with $\langle u_i, v\rangle = \min_j \langle u_j, v\rangle$ for every $v \in \tau$, fix $i_\tau \in I_\tau$, and let $f_\tau = \sum_{i \in I_\tau} c_i \chi^{u_i - u_{i_\tau}}$, a Laurent polynomial of simplex type on the torus $O(\tau)$, since its exponents lie in $\tau^\perp \cap M$.

The compactification is Khovanskii's [Kho77]; for a simplex the proof is short, and we give it.

Lemma 6.50 (The compactification)**.** *Let f be of simplex type on T, let Σ be a smooth fan adapted to f, and let Z be the closure of V_f in Y_Σ.*

(i) *V_f is empty if $k = 0$. If $k \geq 1$, it is a finite étale cover, of degree $[L : L']$, of the product of a torus of dimension $n - k$ and the complement of $k + 1$ hyperplanes in general position in $\mathbb{P}^{k-1}$; in particular it is smooth.*

(ii) *Z is smooth, and for every $\tau \in \Sigma$, $Z \cap O(\tau) = V_{f_\tau}$ and Z meets $V(\tau)$ transversally: $Z \cap V(\tau)$ is empty or smooth of codimension $\dim \tau$ in Z, and it is the closure of V_{f_τ} in $V(\tau)$. So the divisors $Z \cap D_\rho$ of Z are smooth and cross normally, and $Z \cap D_\rho$ is empty when I_ρ has one element.*

(iii) *The fan of $V(\tau)$ is smooth and adapted to f_τ.*

Proof. (i) If $k = 0$, f is a monomial and has no zero on T. Let $k \geq 1$, choose a complement C of the saturated lattice L in M and a basis $b_{k+1}, \dots, b_n$ of C. The characters $u_1 - u_0, \dots, u_k - u_0, b_{k+1}, \dots, b_n$ span a sublattice of index $[L : L']$ in M, so the map $T \to (\mathbb{C}^\times)^n$ that they define is an isogeny of that degree. The function $\chi^{-u_0} f$ is the pull-back of $c_0 + c_1 z_1 + \cdots + c_k z_k$, so V_f is the full preimage of the product of $(\mathbb{C}^\times)^{n-k}$ and the set H of points of $(\mathbb{C}^\times)^k$ with $c_0 + \sum_i c_i z_i = 0$. The projective closure of this affine hyperplane is a hyperplane $\mathbb{P}^{k-1}$ of $\mathbb{P}^k$, with homogeneous coordinates $w_0, \dots, w_k$, on which the $k + 1$ linear forms w_i satisfy the single relation $\sum_i c_i w_i = 0$, with every $c_i \neq 0$; so any k of them are independent, the hyperplanes $\{w_i = 0\}$ of $\mathbb{P}^{k-1}$ are in general position, and H is their complement.

(ii) Let $\sigma \in \Sigma$, spanned by part $v_1, \dots, v_s$ of a basis $v_1, \dots, v_n$ of N, let $w_1, \dots, w_n$ be the dual basis of M, and let $t_j = \chi^{w_j}$, the coordinates of $U_\sigma \cong \mathbb{C}^s \times (\mathbb{C}^\times)^{n-s}$. Put $i = i_\sigma$. Every $u_j - u_i$ lies in the dual cone of σ, so $g = \chi^{-u_i} f = \sum_j c_j \chi^{u_j - u_i}$ is a regular function on U_σ, and $Z \cap U_\sigma \subseteq \{g = 0\}$. A face τ of σ is spanned by the v_j with j in a subset J, and $O(\tau)$ is the locus where exactly the t_j with $j \in J$ vanish. A character χ^x regular on U_σ restricts to $O(\tau)$ as χ^x if $x \in \tau^\perp$ and as 0 otherwise [CLS11, Chapter 3]; since $\langle u_j - u_i, v \rangle \geq 0$ on σ, $u_j - u_i \in \tau^\perp$ exactly when $j \in I_\tau$. So $h = g|_{O(\tau)}$ is $\chi^{u_{i_\tau} - u_i} f_\tau$, a unit times f_τ. Let $p \in O(\tau)$ with $h(p) = 0$ and put $a_j = c_j \chi^{u_j - u_i}(p)$ for $j \in I_\tau$. The derivative of h at p along the one-parameter subgroup of $O(\tau)$ given by y in its cocharacter lattice is $\sum_j a_j \langle u_j - u_i, y \rangle$. If all of these vanished, then, with $\sum_j a_j = h(p) = 0$, the vector $\sum_j a_j (1, u_j - u_i)$ of $\mathbb{C} \oplus (\tau^\perp \cap M) \otimes \mathbb{C}$ would be zero, as the cocharacter lattice of $O(\tau)$ is dual to $\tau^\perp \cap M$; affine independence would give every $a_j = 0$, which is impossible at a point of the torus $O(\tau)$. So $dh(p) \neq 0$. Hence $\{g = 0\}$ is smooth at p and transversal to $O(\tau)$ and to every $\{t_j = 0,\ j \in J'\}$ with $J' \subseteq J$, whose tangent space at p contains that of $O(\tau)$. No component of $\{g = 0\}$ lies in a divisor $\{t_j = 0\}$, since g restricted to the orbit of the ray of v_j is a unit times a nonzero Laurent polynomial; so every component meets T, where $\{g = 0\}$ is V_f, and $\{g = 0\} = Z \cap U_\sigma$. This shows that Z is smooth, that $Z \cap O(\tau) = V_{f_\tau}$ and that $Z \cap V(\tau)$ is smooth of codimension $\dim \tau$. As $Z \cap O(\tau')$ has codimension at least $\dim \tau'$ in Z for every cone $\tau' \supsetneq \tau$, no component of $Z \cap V(\tau)$ lies in the smaller orbits, and $Z \cap V(\tau)$ is the closure of V_{f_τ}. If I_ρ has one element, f_ρ is a nonzero constant.

(iii) The fan of $V(\tau)$ is smooth because Σ is. Let $\sigma \supseteq \tau$. The index i_σ attains the minimum on σ, hence on τ, so $i_\sigma \in I_\tau$, and for $j \in I_\tau$ and $v \in \sigma$, $\langle u_{i_\sigma} - u_{i_\tau}, v \rangle \leq \langle u_j - u_{i_\tau}, v \rangle$. Both sides depend only on the image of v in $N_\mathbb{R}/\mathbb{R}\tau$, because the exponents of f_τ lie in $\tau^\perp$; so the image of σ satisfies the condition of adaptedness for f_τ. ∎

Lemma 6.51 (Cohomology across a toric modification)**.** *Let f be of simplex type on T, let Σ_1 and Σ_0 be smooth projective fans adapted to f, with Σ_1 refining Σ_0, let $\varphi \colon Y_1 =$*

$Y_{\Sigma_1} \to Y_0 = Y_{\Sigma_0}$ be the toric morphism, and let Z_1 and Z_0 be the closures of V_f. For each ray ρ of Σ_1 that is not a ray of Σ_0 let $E_\rho = Z_1 \cap D_\rho$, with inclusion i_ρ. Then the kernel of $\varphi_*\colon H^k(Z_1,\mathbb{Q}) \to H^k(Z_0,\mathbb{Q})$ is contained in $\sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q})$, and

$$H^k(Z_1,\mathbb{Q}) = \varphi^*H^k(Z_0,\mathbb{Q}) + \sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q}).$$

Proof. Let $W \subseteq Y_0$ be the union of the open sets U_τ with $\tau \in \Sigma_0 \cap \Sigma_1$. A cone of Σ_1 contained in such a τ is a face of it, since two cones of a fan meet in a common face; so $\varphi^{-1}(U_\tau) = U_\tau$, and φ is an isomorphism over W. A cone of Σ_1 whose rays are all rays of Σ_0 lies in a cone τ_0 of Σ_0, its rays are rays of τ_0, and τ_0 is simplicial, so it is a face of τ_0 and lies in Σ_0. Hence $Y_1 \smallsetminus \varphi^{-1}(W)$ is the union of the orbits of the cones with a ray not in Σ_0, that is, of the divisors D_ρ of those rays. So $U_1 = Z_1 \smallsetminus \bigcup_\rho E_\rho$ is the closure of V_f in $\varphi^{-1}(W)$, and φ maps it isomorphically onto $U_0 = Z_0 \cap W$, with $U_1 = Z_1 \cap \varphi^{-1}(U_0)$. The morphism $Z_1 \to Z_0$ is birational on each component, so $\varphi_*\varphi^* = \mathrm{id}$ by the projection formula, and every class is $\varphi^*\varphi_*a$ plus a class in the kernel of φ_*. Let $\varphi_*b = 0$. Push-forward along a proper map commutes with restriction to an open set and its preimage, so $b|_{U_1} = (\varphi|_{U_1})^*((\varphi_*b)|_{U_0}) = 0$. By Lemma 6.50(ii), $\bigcup_\rho E_\rho$ is a divisor with normal crossings and smooth components in the smooth projective variety Z_1, and by Deligne [Del71, §3.2] the weight spectral sequence of its complement U_1 degenerates at E_2; its term $E_2^{0,k}$, the cokernel of the Gysin map $\bigoplus_\rho H^{k-2}(E_\rho,\mathbb{Q}) \to H^k(Z_1,\mathbb{Q})$, is the image of $H^k(Z_1,\mathbb{Q})$ in $H^k(U_1,\mathbb{Q})$. So b lies in the image of the Gysin map. ∎

Theorem 6.52 (Hypersurfaces of simplex type)**.** *Let f be a Laurent polynomial of simplex type on a torus T of dimension n, with $e = e(f)$, let Y be a projective toric variety with torus T, not necessarily smooth, and let Z be the closure of V_f in Y. Suppose that Z is smooth.*

(i) *$Z \in \mathcal{A}$, so (F3′) holds on Z in every degree.*

(ii) *If the Hodge conjecture holds for the Fermat varieties X_e^j with $j \le \dim Z$, then it holds for Z.*

Proof. By induction on n. If $k = 0$, Z is empty; let $k \ge 1$.

The Fermat form. With C and b_j as in the proof of Lemma 6.50(i), let $\widetilde{M}$ be the lattice with basis $w_i = (u_i - u_0)/e$ for $1 \le i \le k$ and $w_j = b_j$ for $k < j \le n$. It contains M, since $eL \subseteq L'$ gives $L \subseteq \frac{1}{e}L'$. Let $\widetilde{T}$ be the torus with character lattice $\widetilde{M}$, $\widetilde{N} \subseteq N$ its cocharacter lattice and $\pi\colon \widetilde{T} \to T$ the isogeny dual to $M \subseteq \widetilde{M}$. In the coordinates $z_i = \chi^{w_i}$ of $\widetilde{T}$, $\pi^*(\chi^{-u_0}f) = c_0 + \sum_{i\le k} c_i z_i^e$; call it $\widetilde{f}$. It is of simplex type, $V_{\widetilde{f}} = \pi^{-1}(V_f)$, and after the z_i are rescaled it is $c_0(1 + \sum_{i\le k} z_i^e)$.

The model. Let Σ_0 be the fan in $\widetilde{N}_\mathbb{R}$ of $Y_0 = \mathbb{P}^k \times \mathbb{P}^{n-k}$, on whose torus $z_1,\dots,z_k$ are the affine coordinates of the first factor and $z_{k+1},\dots,z_n$ those of the second. The closure of $V_{\widetilde{f}}$ in Y_0 is, after the rescaling, $Z_0 = X_e^{k-1} \times \mathbb{P}^{n-k}$, since the part of the Fermat variety in the torus is dense in it; $Z_0 \in \mathcal{A}$ by Proposition 6.42(iii) and Proposition 6.35(iii), $\mathbb{P}^{n-k}$ being the Fermat variety X_1^{n-k}. The fan Σ_0 is adapted to $\widetilde{f}$: $\min(0, e\langle w_1, v\rangle, \dots, e\langle w_k, v\rangle)$ depends only on the image of v in the fan of $\mathbb{P}^k$, the normal fan of the standard simplex, on whose cones it is linear.

The common refinement. Let Σ_Y be the fan of Y, the normal fan of a polytope P, as Y is projective [CLS11, Chapter 2], and Σ_0 that of a polytope Q. The face of $P + Q$ on which a vector v attains its minimum is the sum of those of P and of Q, so the normal fan of $P + Q$ consists of the intersections of cones of Σ_Y and Σ_0; it has a smooth projective

refinement Σ_1 with respect to $\widetilde{N}$ [CLS11, Chapter 11]. It is adapted to $\widetilde{f}$, because Σ_0 is. Let $Y_1 = Y_{\Sigma_1,\widetilde{N}}$ and Z_1 the closure of $V_{\widetilde{f}}$ in it, smooth by Lemma 6.50(ii). The inclusion $\widetilde{N} \subseteq N$ maps each cone of Σ_1 into a cone of Σ_Y and defines a proper toric morphism $\psi\colon Y_1 \to Y$ extending π [CLS11, Chapter 3]; $\psi(Z_1)$ is closed, contains $\pi(V_{\widetilde{f}}) = V_f$ and lies in Z, so ψ maps Z_1 onto Z. The identity of $\widetilde{N}$ defines the toric morphism $\varphi\colon Y_1 \to Y_0$, which maps Z_1 onto Z_0 and is an isomorphism over $V_{\widetilde{f}}$.

The boundary. By Lemma 6.50, each $E_\rho = Z_1 \cap D_\rho$ is the smooth closure of $V_{\widetilde{f}_\rho}$ in the smooth projective toric variety D_ρ of dimension $n-1$. The exponents of $\widetilde{f}_\rho$ are the $e(w_i - w_{i_\rho})$ with $i \in I_\rho$, where $w_0 = 0$. The vectors $w_i - w_{i_\rho}$ with $i \le k$, $i \ne i_\rho$, together with $w_{k+1}, \dots, w_n$, form a basis of $\widetilde{M}$, so those with $i \in I_\rho$ span a saturated sublattice of $\widetilde{M}$, which lies in $\rho^\perp \cap \widetilde{M}$ and is saturated there. Hence $e(\widetilde{f}_\rho) = e$ if I_ρ has at least two elements, and E_ρ is empty otherwise. By induction every E_ρ lies in $\mathcal{A}$, and it satisfies the Hodge conjecture if the X_e^j with $j \le n-2$ do.

(i) By Lemma 6.51, $H^*(Z_1, \mathbb{Q})$ is spanned by the images of $H^*(Z_0, \mathbb{Q})$ and of the $H^*(E_\rho, \mathbb{Q})$ under φ^* and the $i_{\rho *}$, which are induced by algebraic cycles, the transposed graph of φ and the graphs of the i_ρ. So $Z_1 \in \mathcal{A}$ by Proposition 6.42(v), and $Z \in \mathcal{A}$ by Proposition 6.35(iii), as the image of Z_1 under ψ.

(ii) By the Künneth formula every Hodge class on $X_e^{k-1} \times \mathbb{P}^{n-k}$ is a sum of pull-backs of Hodge classes on X_e^{k-1} multiplied by powers of the hyperplane class, so the conjecture holds on Z_0, as $k-1 \le n-1 = \dim Z$; it holds on the E_ρ by induction. Let a be a Hodge class on Z_1. Then $\varphi_* a$ is a Hodge class, hence algebraic, and so is $\varphi^*\varphi_* a$. The class $b = a - \varphi^*\varphi_* a$ is a Hodge class in the kernel of φ_*, hence in the image of the Gysin map of Lemma 6.51, a morphism of polarisable Hodge structures after a Tate twist; as in the proof of Proposition 6.35(iii) it is the image of a Hodge class, which is algebraic, so b is algebraic. So the conjecture holds on Z_1, and it passes to Z along ψ: for a Hodge class x on Z, $\psi^* x$ is algebraic, and $\Psi_* \psi^* x = c\, x$ with $c > 0$ for the algebraic correspondence Ψ of the proof of Proposition 6.35(iii). ∎

Corollary 6.53 (Delsarte hypersurfaces of every dimension)**.** *Let $X_A \subset \mathbb{P}^{r+1}$, $r \ge 1$, be a smooth Delsarte hypersurface of degree m as in Proposition 6.48, let $M = \{x \in \mathbb{Z}^{r+2} : \sum_j x_j = 0\}$, and let e_A be the exponent of M/M', where M' is spanned by the differences $a_i - a_0$ of the rows of A.*

(i) *$X_A \in \mathcal{A}$, so (F3′) holds on X_A in every degree, and the Hodge conjecture holds on X_A if it holds on the Fermat varieties $X_{e_A}^j$ with $j \le r$. The number e_A divides every d of Proposition 6.48, and $[M : M'] = |\det A|/m$.*

(ii) *The same holds, with $j \le r+1$, for the cyclic cover $\{y^k = F_A(x)\}$ of $\mathbb{P}^{r+1}$ in the weighted projective space $\mathbb{P}(1, \dots, 1, m/k)$, for every k dividing m, with $M = \{x \in \mathbb{Z}^{r+3} : \sum_{j \le r+1} x_j + (m/k) x_{r+2} = 0\}$ and M' spanned by the differences of the exponents of $F_A - y^k$; it is smooth because X_A is.*

(iii) *For the 29 sextic shapes of Proposition 6.48(iii), e_A is the least d there, except for the chain $x_0^5 x_1 + x_1^5 x_2 + \cdots + x_4^5 x_5 + x_5^6$ and the loop, for which it is 3125 and 2604 instead of 18750 and 15624. For the Klein quartic $x^3 y + y^3 z + z^3 x$ it is 7.*

Proof. (i) On the torus of $\mathbb{P}^{r+1}$, with character lattice M, X_A is the closure of V_f for $f = \chi^{-a_0} F_A$, as in the proof of Proposition 6.48(i). The rows of A are linearly independent and lie on the affine hyperplane $\sum_j x_j = m$, which does not contain 0, so they are affinely independent, f is of simplex type, $L' = M'$ has rank $r+1$ and $L = M$; so $e(f) = e_A$,

and Theorem 6.52 gives the first two assertions. If dA^{-1} is integral and $x \in M$, then $dx = \lambda A$ with $\lambda = dxA^{-1}$ integral and $\sum_i \lambda_i = d\sum_j x_j/m = 0$, as $A^{-1}\mathbf{1} = \mathbf{1}/m$; so $dx = \sum_i \lambda_i(a_i - a_0) \in M'$, and e_A divides d. The lattice M' is the intersection of M with the lattice R spanned by the rows, whose image under $x \mapsto \sum_j x_j$ is $m\mathbb{Z}$; the exact sequence $0 \to M/M' \to \mathbb{Z}^{r+2}/R \to \mathbb{Z}/m\mathbb{Z} \to 0$ gives $[M : M'] = |\det A|/m$.

(ii) The exponents of $F_A - y^k$ are linearly independent and lie on the hyperplane of weighted degree m, so the polynomial is of simplex type on the torus of $\mathbb{P}(1, \dots, 1, m/k)$, a projective toric variety whose character lattice is the given M. On the affine cone, the partial derivatives vanish together with $y^k - F_A$ only where $y = 0$ and x is a singular point of the cone over X_A, that is $x = 0$; so the cover is quasi-smooth, and it misses the only singular point $(0 : \cdots : 0 : 1)$ of the weighted projective space, where $y^k = F_A(0)$ would force $y = 0$. So it is smooth. It is irreducible, F_A being irreducible, and not contained in a coordinate hyperplane, so it is the closure of its part in the torus, and Theorem 6.52 applies.

(iii) The least d with dA^{-1} integral is the exponent of $\mathbb{Z}^{r+2}/R$, and the map $x \mapsto \sum_j x_j$ modulo m is well defined on it because the rows sum to m. For a polynomial as in Proposition 6.48(iii) the group $\mathbb{Z}^{r+2}/R$ is the direct sum of one group for each term. The relations of a chain are $x_{j+1} = (1-m)x_j$ and $mx_k = 0$, those of a loop are $x_{j+1} = (1-m)x_j$, indices modulo k, and that of x^m is $mx = 0$. So each summand is cyclic, generated by the class of its first variable, of order $N = m(m-1)^{k-1}$, $(m-1)^k - (-1)^k$ and m respectively, and the sum map sends that generator to 1, since $1 - m \equiv 1$ modulo m; each N is divisible by m. The least d is the least common multiple of these orders, which gives the values of Proposition 6.48(iii). The group M/M' is the kernel of the sum map. With one term only, the loop or the chain in six variables, the kernel is $m\mathbb{Z}/N\mathbb{Z}$, of exponent N/m, which is $15624/6 = 2604$ and $18750/6 = 3125$; for the Klein quartic, a loop with $m = 4$ and $k = 3$, it is $28/4 = 7$. With at least two terms, of orders $N_1, N_2, \dots$, the kernel contains the differences of the generators of the first and of the b-th summand, of order $\operatorname{lcm}(N_1, N_b)$; so its exponent is the least common multiple of all the N_b, which is the least d. ∎

For the Klein quartic the construction is classical: the isogeny has kernel of order $e_A^2/[M : M'] = 7$, and the Klein quartic is the quotient of the Fermat septic curve by a group of order seven; Figure 23 draws its Newton triangle and the smooth adapted fan of the proof. For $r = 4$, Corollary 6.53(i) recovers Proposition 6.48(ii), with the Fermat degree lowered to e_A. Its content lies in dimension five and more, where otherwise only the Fermat hypersurfaces are covered (Proposition 6.42(iii)), and in the cyclic covers, among them the Calabi–Yau varieties $\{y^2 = F_A(x)\}$ with F_A of degree $2r + 4$ in $r + 2$ variables. Khovanskii's compactification and Shioda's cover are classical, and we claim no priority for the two lemmas. What Theorem 6.52 adds is that the boundary of the compactification is again of simplex type with the same exponent e, so that the whole cohomology, not only its part of top Hodge level, comes from Fermat varieties of one degree.

Remark 6.54 (Quotients by diagonal automorphisms). Let $G \subset T$ be a finite subgroup of the torus of $\mathbb{P}^{r+1}$ on which every character $\chi^{a_i - a_0}$ is trivial, so that G acts on X_A; these G are the subgroups of the group G_A dual to M/M'. Then f descends to a Laurent polynomial of simplex type on T/G, whose character lattice M_G, the characters trivial on G, contains M'; so $L' = M'$, $L = M_G$, and $e(f)$ becomes the exponent e_G of M_G/M', a divisor of e_A. As $\mathbb{P}^{r+1}/G$ is a projective toric variety with torus T/G in which X_A/G is the closure of V_f/G, Theorem 6.52 applies to every projective model Z of X_A/G that is the closure of V_f/G in a projective toric variety with torus T/G and is smooth, for instance in one whose fan is

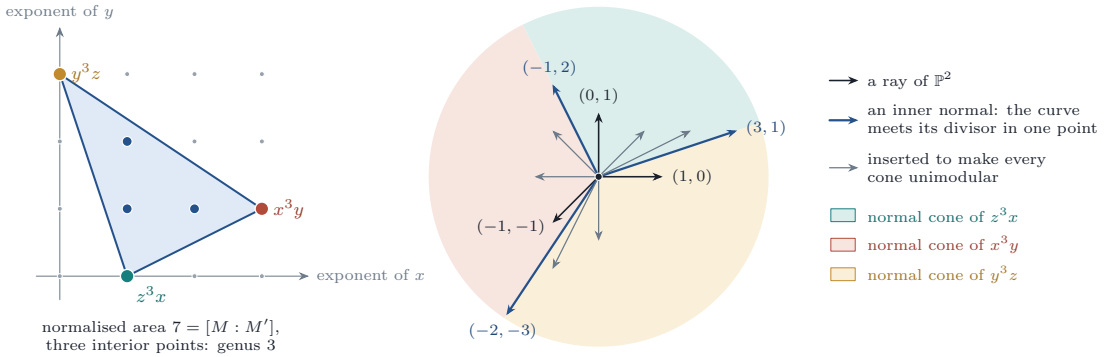


FIGURE 23. The Klein quartic $x^3y+y^3z+z^3x$ in the proof of Theorem 6.52. Left: its Newton triangle in the lattice M of the torus of $\mathbb{P}^2$, in the chart $z = 1$, with normalised area $7 = [M : M']$ and three interior points, the genus. Right: the rays of a smooth fan adapted to it, the three rays of $\mathbb{P}^2$, the three inner normals of the triangle and the six rays inserted to make every cone unimodular. The curve meets only the three divisors of the inner normals, in one point each, the lattice length of the edge, so its Euler number is $-7+3=-4$. With $e_A = 7$ the isogeny of the proof makes it the quotient of the Fermat septic by a group of order seven.

smooth and adapted to f (Lemma 6.50): such a Z lies in $\mathcal{A}$, and the Hodge conjecture holds on it if it holds on the $X^j_{e_G}$ with $j \le r$. For $G = G_A$, $M_G = M'$ and $e_G = 1$, so the conjecture holds on Z. For the Fermat hypersurface of degree $m = r+2$ and the group of diagonal automorphisms of determinant one modulo scalars, the starting point of the mirror construction for X_A, $M' = mM$ and M_G is the set of $x \in M$ whose coordinates are congruent modulo m, so $M_G/M' \cong \mathbb{Z}/m$ and $e_G = m = e_A$: the quotient needs the same Fermat varieties as X_A, and nothing is gained. For any finite group G of automorphisms of X_A, every smooth projective model X' of X_A/G has $\mathrm{Hdg}^2(X') = \mathrm{Ab}^2(X')$, so (F3′) holds on it when it is a fourfold or a fivefold, by Proposition 6.46 and Corollary 6.53(i).

The Fermat hypersurfaces are the complete intersections of one diagonal hypersurface. Complete intersections of several diagonal hypersurfaces of the same degree lie in $\mathcal{A}$ as well when their coefficients are of Vandermonde type, because they are quotients of powers of one curve. Fix $d \ge 2$, $N \ge 2$ and $1 \le c \le N-1$, put $r = N-c$, and call a $c \times (N+1)$ matrix (a_{ki}) *of Vandermonde type* if there are distinct $\lambda_0, \dots, \lambda_N \in \mathbb{C}$, nonzero $\mu_0, \dots, \mu_N$ and $g \in \mathrm{GL}_c(\mathbb{C})$ with $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl} \lambda_i^{l-1}$; for $c \ge 2$ this says that the columns are distinct points of a rational normal curve of degree $c-1$ in $\mathbb{P}^{c-1}$. Put $w_i = \prod_{l \ne i} (\lambda_i - \lambda_l)^{-1}$ and

$$X_{d,r}(\lambda) = \Big\{ x \in \mathbb{P}^N : \sum_{i=0}^{N} w_i \lambda_i^k x_i^d = 0, \quad 0 \le k \le c-1 \Big\}.$$

For $r = 1$ this is the generalised Fermat curve of type (d, N) of González-Diez, Hidalgo and Leyton [GDHL09], the Galois cover of $\mathbb{P}^1$ with group $H = \mu_d^{N+1}/\mu_d$, acting by $u_i \mapsto h_i u_i$, branched over the λ_i. The characters of H are the $\chi_a(h) = \prod_i h_i^{a_i}$ with $a \in (\mathbb{Z}/d)^{N+1}$ and $\sum_i a_i = 0$; write $V_a \subset H^1(C, \mathbb{C})$ for the eigenspace on which h^* acts by $\chi_a(h)$, $[a]$ for the orbit of a under $(\mathbb{Z}/d)^\times$, and $B_{[a]}$ for the image of a multiple of the idempotent of $\mathbb{Q}[H]$

attached to $[a]$, an abelian subvariety of the Jacobian $J(C)$ whose first cohomology is the part $\bigoplus_{a'\in[a]} V_{a'}$ of $H^1(C,\mathbb{C})$.

Theorem 6.55 (Diagonal complete intersections of Vandermonde type)**.** *Let $X \subset \mathbb{P}^N$ be the complete intersection of c hypersurfaces $\sum_i a_{ki}x_i^d = 0$, $1 \le k \le c$, with (a_{ki}) of Vandermonde type, and let $C = X_{d,1}(\lambda)$ for the λ_i of (a_{ki}).*

(i) *X is smooth of dimension r and isomorphic to $X_{d,r}(\lambda)$, and C is a smooth curve of genus $1 + \frac{1}{2}d^{N-1}\big((N-1)(d-1)-2\big)$.*

(ii) *The map $\Phi(u^{(1)},\dots,u^{(r)}) = \big(\prod_j u_i^{(j)}\big)_i$ is a surjective morphism $C^r \to X_{d,r}(\lambda)$, and it induces an isomorphism $C^r/G \cong X_{d,r}(\lambda)$, where G is the group of order $r!\,d^{N(r-1)}$ generated by the permutations of the factors and the $(h_1,\dots,h_r) \in H^r$ with $h_1\cdots h_r = 1$.*

(iii) *$X \in \mathcal{A}$, so (F3′) holds on X in every degree.*

(iv) *$\dim V_a = \#\{i : a_i \ne 0\} - 2$ for $a \ne 0$, and the Hodge conjecture holds on X as soon as it holds on the abelian varieties $B_{[a]}$ with $\dim V_a \ge r$, which have dimension $\frac{1}{2}|[a]|\dim V_a$.*

Proof. Everything rests on the Lagrange identity: interpolating a polynomial f of degree at most N at the λ_i and comparing the coefficients of t^N gives $\sum_i w_i f(\lambda_i) = 0$ when $\deg f \le N-1$.[27]

(i) Replacing the equations by their combinations under g^{-1} and x_i by $\nu_i x_i$ with $\nu_i^d\mu_i = w_i$ turns them into those of $X_{d,r}(\lambda)$. Every $c\times c$ minor of (a_{ki}) is $\det g\prod\mu_i$ times a Vandermonde determinant in distinct λ_i, so any c columns are independent. At a point x of X the Jacobian matrix is $(d\,a_{ki}x_i^{d-1})$; if $S = \{i : x_i \ne 0\}$ had at most c elements, $\sum_{i\in S} x_i^d a_i = 0$ would be a relation between at most c columns, so $|S| > c$ and the Jacobian has rank c.[28] Hence X is a smooth complete intersection of dimension r, connected because $r \ge 1$, and the same holds for C. The genus follows from the Riemann–Hurwitz formula for the cover $C \to \mathbb{P}^1$ of degree d^N described below, which has $N+1$ branch points of ramification index d.

(ii) For $1 \le k \le N-1$ let Λ_k be the space of $y \in \mathbb{C}^{N+1}$ with $\sum_i w_i\lambda_i^l y_i = 0$ for $0 \le l \le N-k-1$. The vectors $(F(\lambda_i))_i$ with $\deg F \le k$ lie in Λ_k by the identity, since $\deg t^l F \le N-1$; they form a space of dimension $k+1$, since $k < N+1$; and Λ_k has dimension $k+1$, the $N-k$ equations having independent Vandermonde rows. So $\Lambda_k = \{(F(\lambda_i))_i : \deg F \le k\}$, and C and $X_{d,r}(\lambda)$ are the sets of points whose d-th powers lie in Λ_1 and in Λ_r. The point $u \in C$ determines $(s:t) \in \mathbb{P}^1$ with $u_i^d = \kappa(t-\lambda_i s)$ for all i; this is the quotient $C \to C/H = \mathbb{P}^1$, whose fibres are the H-orbits. If $(u_i^{(j)})^d = t_j - \lambda_i s_j$, then $\Phi(u)_i^d = F(\lambda_i)$ for $F(t) = \prod_j(t_j - ts_j)$, a nonzero polynomial of degree at most $r < N+1$; so $\Phi(u) \in X_{d,r}(\lambda)$ and $\Phi(u) \ne 0$, and Φ, given by forms of multidegree $(1,\dots,1)$ with no common zero, is a morphism. It is G-invariant. Conversely, if $x \in X_{d,r}(\lambda)$, then

[27] Lagrange interpolation writes $f(t) = \sum_i f(\lambda_i)\prod_{l\ne i}(t-\lambda_l)/(\lambda_i-\lambda_l)$ for $\deg f \le N$, and the coefficient of t^N on the right is $\sum_i w_i f(\lambda_i)$. For $N=2$ and $f=1$ it says that $\sum_i \prod_{l\ne i}(\lambda_i-\lambda_l)^{-1} = 0$ for any three distinct numbers. The identity says that the vector $(w_i\lambda_i^k)_i$ is orthogonal to the values of every polynomial of degree at most $N-1-k$, which is how the equations of $X_{d,r}(\lambda)$ become the condition that the d-th powers of the coordinates are the values of one polynomial of degree at most r.

[28] The columns of the Jacobian matrix with $i \in S$ are the columns of (a_{ki}) multiplied by the nonzero numbers $d\,x_i^{d-1}$, and any c of the columns of (a_{ki}) are independent; so c of them already give rank c. Without the Vandermonde condition this fails: for two diagonal quadrics in $\mathbb{P}^3$ with proportional columns $i = 0, 1$, the points with $x_2 = x_3 = 0$ and $a_0x_0^2 + a_1x_1^2 = 0$ are singular.

$(x_i^d)_i = (F(\lambda_i))_i$ for a unique nonzero F of degree at most r, which factors as $\prod_j (t_j - t s_j)$, uniquely up to order and to rescaling the factors. Choosing $u^{(j)}$ over $(s_j : t_j)$ gives $\Phi(u)_i = \epsilon_i x_i$ with $\epsilon_i \in \mu_d$, and multiplying $u^{(1)}$ by (ϵ_i^{-1}) gives a preimage. Two preimages have the same roots, so after a permutation $u'^{(j)} = h_j u^{(j)}$ with $h_j \in H$, and $\prod_j h_j$ acts trivially on the coordinates where $x_i \neq 0$; where $x_i = 0$ some $u_i^{(j)}$ vanishes and $h_{j,i}$ may be changed freely, so we may take $\prod_j h_j = 1$. Hence the fibres of Φ are the G-orbits, Φ induces a finite bijective morphism from the normal variety C^r/G to the smooth variety $X_{d,r}(\lambda)$, and by Zariski's main theorem this is an isomorphism.[29] The order of G is $r!\,|H|^{r-1}$.

(iii) C is a curve, so $C^r \in \mathcal{A}$, and X is the image of C^r under a surjective morphism (Proposition 6.35(iii)).

(iv) For $a \neq 0$, the forms $t^j \prod_i (t - \lambda_i)^{-b_i/d}\, dt$ with $b_i \in \{0, \dots, d-1\}$, $b_i \equiv -a_i$, and $0 \leq j \leq \sum_i b_i/d - 2$, are holomorphic on C and span the χ_a-eigenspace of $H^{1,0}(C)$;[30] its dimension $\sum_i \langle -a_i/d \rangle - 1$ and that of the χ_{-a}-eigenspace add up to $\#\{i : a_i \neq 0\} - 2$, and $H^1(C)^H = H^1(\mathbb{P}^1) = 0$. The group H acts trivially on $H^0(C)$ and $H^2(C)$, so a tensor in $H^r(C^r) = \bigoplus \bigotimes_j H^{k_j}(C)$ invariant under the (h_j) with $\prod_j h_j = 1$ has all its factors in H^1, with one character, or none: taking h in one factor and h^{-1} in another shows that two factors carry the same character, and a factor in H^0 or H^2 carries the trivial one. Taking the invariants of the permutations,

$$H^r(C^r, \mathbb{C})^G = \mathbb{C}\,\eta \;\oplus \bigoplus_{a \neq 0} \bigwedge^r V_a,$$

with η the symmetrised product of $r/2$ classes of points when r is even and $\eta = 0$ otherwise, and $W_{[a]} = \bigoplus_{a' \in [a]} \bigwedge^r V_{a'}$ is a sub-Hodge structure defined over $\mathbb{Q}$, zero unless $\dim V_a \geq r$. Let $\psi \colon C \to B_{[a]}$ be an Abel–Jacobi map followed by the projection, so that ψ^* maps $H^1(B_{[a]}, \mathbb{Q})$ isomorphically onto the $[a]$-part of $H^1(C, \mathbb{Q})$, and let $\sigma \colon C^r \to B_{[a]}$ be $\sum_j \psi \circ \mathrm{pr}_j$. Then $\sigma^*(\alpha_1 \wedge \dots \wedge \alpha_r) = \prod_k \sum_j \mathrm{pr}_j^* \psi^* \alpha_k$, whose component in $H^1(C)^{\otimes r}$ is the alternating sum of the $\psi^*\alpha_1 \otimes \dots \otimes \psi^*\alpha_r$. The Künneth projectors of a curve are algebraic,[31] and so is the average e_G over G, so $e_G \circ \pi_1^{\otimes r} \circ \sigma^*$ is induced by an algebraic cycle on $B_{[a]} \times C^r$, and it maps $H^r(B_{[a]}, \mathbb{Q})$ onto $W_{[a]}$: the alternating tensors in the $[a]$-part are the image of $\pi_1^{\otimes r} \circ \sigma^*$, and e_G keeps those of one character. A surjective morphism

[29] In characteristic zero a finite morphism that is bijective on closed points is generically étale of degree one, hence birational, and a finite birational morphism onto a normal variety is an isomorphism [Har77, Chapter III, Corollary 11.4]. The group G acts faithfully on C^r, since H is the group of deck transformations of $C \to \mathbb{P}^1$, so Φ has degree $|G|$.

[30] On the affine chart $s = 1$ put $u_i^d = t - \lambda_i$, so that the form is $t^j \prod_i u_i^{-b_i}\, dt$ and $h \in H$ multiplies it by $\prod_i h_i^{-b_i} = \chi_a(h)$; it is well defined on $H = \mu_d^{N+1}/\mu_d$ because $\sum_i b_i \equiv 0$. Over λ_i the cover is ramified of index d, with a local parameter z satisfying $z^d = t - \lambda_i$, and the form is a unit times $z^{-b_i}\, d(z^d) = d\, z^{d-1-b_i}\, dz$, holomorphic as $b_i \leq d-1$. Over $t = \infty$, with $w = 1/t$ and $v^d = w$, it is a unit times $v^{d(\sum_i b_i/d - j - 2) + d - 1}\, dv$, holomorphic exactly when $j \leq \sum_i b_i/d - 2$, the exponent of v being an integer. Forms with different characters, or with different j, are independent, and their number is $\frac{1}{2}\sum_{a \neq 0}\big(\#\{i : a_i \neq 0\} - 2\big) = \frac{1}{2}\big((N+1)(d^N - d^{N-1}) - 2(d^N - 1)\big)$, which is the genus of (i); so they form a basis of $H^{1,0}(C)$, and each eigenspace is spanned by the forms of its character. For $d = 2$ and $N = 4$ they are the five forms $dt/\sqrt{(t - \lambda_{i_1}) \cdots (t - \lambda_{i_4})}$, one for each choice of four of the five indices, on a curve of genus five.

[31] For a curve C with a point o they are $\pi_0 = [o \times C]$, $\pi_2 = [C \times o]$ and $\pi_1 = [\Delta_C] - \pi_0 - \pi_2$, and $\pi_1^{\otimes r}$ is their exterior product on $C^r \times C^r$. The average $e_G = |G|^{-1} \sum_{g \in G} [\Gamma_g]$ of the graphs of the elements of G acts on $H^*(C^r)$ as the projector onto the invariants, and Φ^* identifies $H^*(X_{d,r}(\lambda), \mathbb{Q})$ with $H^*(C^r, \mathbb{Q})^G$.

of polarisable Hodge structures maps Hodge classes onto Hodge classes,[32] so the Hodge classes of $W_{[a]}$ are algebraic when those of $B_{[a]}$ are. Now let x be a Hodge class of degree r on $X = X_{d,r}(\lambda)$, the other degrees being multiples of powers of the hyperplane class by the Lefschetz hyperplane theorem. Then $\Phi^* x$ is a G-invariant Hodge class, its components in the $W_{[a]}$ and along η are Hodge classes, being cut out by the algebraic projectors attached to the action of H^r, and they are algebraic; so $\Phi^* x$ is algebraic, and so is $x = \Phi_* \Phi^* x / |G|$. Finally $\dim B_{[a]}$ is half of $\sum_{a' \in [a]} \dim V_{a'} = |[a]| \dim V_a$. ∎

Corollary 6.56 (Two diagonal hypersurfaces)**.**

(i) *Every smooth complete intersection of two diagonal hypersurfaces* $\sum_i a_i x_i^d = \sum_i b_i x_i^d = 0$ *of the same degree in* $\mathbb{P}^N$ *is of Vandermonde type, so it lies in* $\mathcal{A}$ *and satisfies* (F3′) *in every degree.*

(ii) *Let* $d \in \{3, 4, 6\}$ *and* $N \leq 6$*, or* $d = 2$ *and* $N \leq 12$*. Then the Hodge conjecture holds on every complete intersection of Vandermonde type in* $\mathbb{P}^N$ *of* $N - r$ *diagonal hypersurfaces of degree* d*. In particular it holds on every smooth complete intersection of two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$*.*

(iii) *For* $d \in \{3, 4, 6\}$ *and* $N = 6$*, each orbit* $[a]$ *of order at least three with six nonzero coordinates and* $\sum_i \langle a_i/d \rangle = 3$ *makes* $B_{[a]}$ *an abelian fourfold of Weil type for* $K = \mathbb{Q}(\zeta_d)$*, and* $W_{[a]}$ *is its space of Weil classes. On each of these fourfolds* X *the Weil classes of the* 70*,* 490 *and* 6125 *such orbits, for* $d = 3, 4, 6$*, give Hodge classes of degree four outside* $\mathbb{Q}h^2$*, so the Hodge classes of* X *form a space of dimension at least* 141*,* 981 *and* 12251*, whereas the divisor classes of* X *are the multiples of* h*.*

Proof. (i) If two columns (a_i, b_i) and (a_j, b_j) were proportional, or one of them zero, X would contain a point supported on $\{i, j\}$, or the coordinate point e_i, at which the Jacobian has rank at most one; so the $N+1$ columns are distinct points of $\mathbb{P}^1$. Sending a point of $\mathbb{P}^1$ outside them to infinity writes them as $\mu_i(1, \lambda_i)$ with distinct λ_i, which is the Vandermonde form for $c = 2$, and Theorem 6.55(iii) applies. (ii) For $d \in \{2, 3, 4, 6\}$ every a has order 1, 2, 3, 4 or 6, so $|[a]| \leq 2$, with $|[a]| = 1$ for the a of order two, whose V_a is real, so that $\dim B_{[a]} \leq \dim V_a \leq N - 1$. When $d = 2$ every $a \neq 0$ has order two and an even number of nonzero coordinates, at most 12 when $N \leq 12$, so $\dim B_{[a]} = \frac{1}{2} \dim V_a \leq 5$. So every $B_{[a]}$ has dimension at most five,[33] and the Hodge conjecture holds on abelian varieties of dimension at most five [Mar25a, Mar25b, MZ99]; Theorem 6.55(iv) gives the conclusion, and (i) the last statement. (iii) For such a the eigenspaces V_a and V_{-a} have dimension four and contain two-dimensional parts of type $(1, 0)$, so K acts on $H^{1,0}(B_{[a]})$ with multiplicities $(2, 2)$ and $B_{[a]}$, of dimension four, is of Weil type; $\bigwedge^4 V_a$ and $\bigwedge^4 V_{-a}$ are its Weil lines, of type $(2, 2)$ for every λ, and they lie in $H^4(C^4)^G = \Phi^* H^4(X)$. The zero coordinate of a is one of seven, and the other six $a_i \in \{1, \dots, d-1\}$ add up to $3d$, which they do in 20, 141 and 1751 ways for $d = 3, 4, 6$, the coefficients of x^{3d-6} in $(1 + x + \cdots + x^{d-2})^6$; removing the character of order two with every $a_i = d/2$ leaves 20, 140 and 1750, and $(\mathbb{Z}/d)^\times = \{\pm 1\}$ pairs them off, since $-a$ again has coordinate sum $3d$; this gives $7 \cdot 20/2 = 70$, $7 \cdot 140/2 = 490$ and $7 \cdot 1750/2 = 6125$ orbits. Distinct orbits give independent classes, and the hyperplane

[32] The category of polarisable rational Hodge structures is semisimple, so the kernel of a surjective morphism $f \colon V \to W$ has a complement V', a sub-Hodge structure that f maps isomorphically onto W; a Hodge class of W is then the image of a Hodge class of V'.

[33] Both bounds are sharp: for $d = 3$ and $N = 7$ a character with eight nonzero coordinates has $\dim V_a = 6$ and $|[a]| = 2$, so $\dim B_{[a]} = 6$; for $d = 2$ and $N = 13$ a character with fourteen nonzero coordinates has $\dim B_{[a]} = 6$.

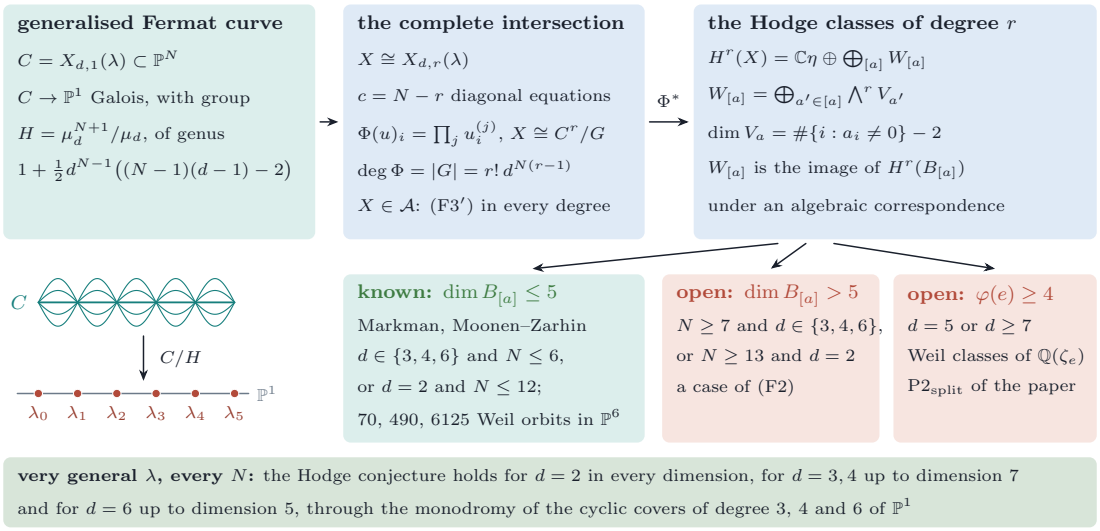


FIGURE 24. Theorem 6.55 and Corollary 6.56. Left: the generalised Fermat curve C over $\mathbb{P}^1$, its sheets meeting over the branch points $\lambda_0, \ldots, \lambda_N$ (five sheets stand for the d^N of the cover). Top: C, the complete intersection $X \cong C^r/G$ with the map Φ of degree $|G|$, and the decomposition of $H^r(X)$ through Φ^*. Bottom: where the Hodge conjecture for X stands at an arbitrary configuration, according to the abelian varieties $B_{[a]}$ that carry its Hodge classes: known when they have dimension at most five, which happens for $d \in \{3,4,6\}$ and $N \leq 6$ and for $d = 2$ and $N \leq 12$, and open when they are larger, a case of (F2), or carry Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, the case $\mathrm{P2}_{\mathrm{split}}$ of the paper. Foot: the very general configuration, Theorem 6.66.

class lies in the part of the trivial character, which gives the bounds; the Picard group of a complete intersection of dimension at least three is generated by h. ■

Figure 24 draws the construction. For $c = 1$ the theorem is Proposition 6.42(iii) again, by a different route, and for $r = 1$ it restates the definition of C.

At a very general configuration the abelian varieties $B_{[a]}$ carry only the Hodge classes that the monodromy of the curves allows, and the conjecture can then be decided in every dimension for $d = 2$, up to dimension seven for $d = 3$ and $d = 4$, and up to dimension five for $d = 6$. Write $U \subset \mathbb{C}^{N+1}$ for the space of $(N+1)$-tuples of distinct numbers. The varieties $X_{d,r}(\lambda)$ with $\lambda \in U$ form a smooth projective family over U, and so do the curves C and the maps Φ, with H acting on the family of curves.

Lemma 6.57 (Hodge classes of a very general member)**.** *Let $\mathbb{V}$ be a polarised variation of $\mathbb{Z}$-Hodge structure of weight $2p$ on a smooth connected quasi-projective variety S, and let $\Pi = \pi_1(S, s)$ act on $\mathbb{V}_s$. If s lies outside a countable union of proper Zariski closed subsets of S, the space $\mathrm{Hdg}(\mathbb{V}_s)$ of rational classes of type (p,p) is stable under Π, and a subgroup of finite index of Π fixes it pointwise.*

Proof. The locus of Hodge classes has countably many components ([I, Definition B.3], [I, Lemma B.4]), and the image in S of each is Zariski closed ([I, Theorem B.6]); let s avoid the images that are not all of S. For $\alpha \in \mathrm{Hdg}(\mathbb{V}_s)$ the component through (s, α) maps

onto S, locally biholomorphically onto its image, so the flat section $\tilde{\alpha}$ through α on the universal cover $\tilde{S}$ is of type (p,p) on a nonempty open set. The set where it is of type (p,p) is a closed analytic subset of the connected manifold $\tilde{S}$, so it is all of $\tilde{S}$, and every transport of α along a loop at s is a Hodge class at s. The group Π therefore preserves $\mathrm{Hdg}(\mathbb{V}_s)$, the lattice $\mathbb{V}_{s,\mathbb{Z}} \cap \mathrm{Hdg}(\mathbb{V}_s)$ and the flat form $\tilde{Q}$ of the proof of [I, Lemma B.5], which is positive definite on the real classes of type (p,p); so it acts on $\mathrm{Hdg}(\mathbb{V}_s)$ through a finite group, whose kernel is the subgroup of the statement. ∎

For the characters of order three, four and six the monodromy of the curves D_a is determined by induction on the number of branch points. The step lets two branch points collide, and a lemma on two copies of SL of a hyperplane closes it. The base has four branch points, two of them with opposite exponents, whose full twist is a transvection (Figure 25).

Setup 6.58 (Cyclic covers)**.** Let $m \in \{3,4,6\}$, $\zeta = e^{2\pi i/m}$ and $K = \mathbb{Q}(\zeta)$, and let $a = (a_1,\dots,a_k)$ with $a_i \in \mathbb{Z}/m$ nonzero, $\sum_i a_i = 0$ and $\gcd(m, a_1,\dots,a_k) = 1$, so that a has order m. For λ in the space Conf_k of k-tuples of distinct complex numbers let D_a be the cyclic cover $y^m = \prod_i (t-\lambda_i)^{a_i}$ of $\mathbb{P}^1$, with representatives $0 < a_i < m$; it is unramified over ∞. Let $V_a \subset H^1(D_a, \mathbb{C})$ be the eigenspace of the deck group on which $H^{1,0}$ is spanned by the forms $t^j \prod_i (t-\lambda_i)^{-b_i/m}\, dt$, where $0 < b_i < m$, $b_i \equiv -a_i$ and $0 \le j \le \sum_i b_i/m - 2$ [DM86, §2], and put $n = \dim V_a$, $p = \dim V_a^{1,0} = \sum_i b_i/m - 1$ and $q = n - p$. Let $\Gamma_a \subset \mathrm{GL}(V_a)$ be the image of the pure braid group $P_k = \pi_1(\mathrm{Conf}_k)$ and G_a^0 the identity component of its Zariski closure.

Lemma 6.59 (A model of the monodromy)**.** *Let F_k be the free group on loops $x_1,\dots,x_k$ around $\lambda_1,\dots,\lambda_k$, put $t_i = \zeta^{a_i}$, let $\partial\colon K^k \to K$ map e_i to $t_i - 1$, and let $E_a = \ker\partial / K\ell$, where $\ell = \sum_i t_1 \cdots t_{i-1} e_i$. A pure braid β acts on F_k by Artin's automorphism and on K^k by $v \mapsto vJ_\beta$, where J_β is the matrix of Fox derivatives $\partial\beta(x_i)/\partial x_j$ evaluated at $x_j \mapsto t_j$. This action preserves $\ker\partial$ and fixes ℓ, and the representation of P_k on E_a, of dimension $k-2$, is isomorphic to V_a, to its dual or to its complex conjugate. So $n = k-2$, and $G_a^0 \supseteq \mathrm{SL}(V_a)$ if and only if the identity component of the Zariski closure of the image of P_k in $\mathrm{GL}(E_a)$ contains $\mathrm{SL}(E_a)$.*

Proof. Fox's formula $\sum_j (\partial w/\partial x_j)(x_j - 1) = w - 1$ [Fox53], evaluated at $x_j \mapsto t_j$, shows that the vector $\sum_j (\partial w/\partial x_j)(t)\, e_j$ of a word w with $t(w) = 1$ lies in $\ker\partial$. It is the class of the lift of the loop w in the summand of $H_1(D_a^\circ, K)$ on which the deck group acts through $x_i \mapsto t_i$, where D_a° is the unramified part of the cover, and that summand is $H_1(F_k, K_t) = \ker\partial$, computed from the free resolution of length one of the augmentation ideal of a free group.[34] Filling in the points over λ_i adds no relation: the loop around such a point lifts x_i^e, e the order of t_i, whose vector $(1 + t_i + \cdots + t_i^{e-1})e_i$ is zero. Filling in the m points over ∞, where the cover is unramified, kills the lift of $x_1 \cdots x_k$, whose vector is $\ell \neq 0$. So E_a is the summand of $H_1(D_a, K)$ for the character $x_i \mapsto t_i$, of dimension $k-2$. A pure braid fixes $x_1 \cdots x_k$ and maps each x_i to a conjugate of itself, so it preserves t, and the chain rule for Fox derivatives makes $\beta \mapsto J_\beta$ the action of P_k on this summand.[35]

[34] By Shapiro's lemma $H_1(D_a^\circ, K) = H_1(F_k, K[\mu_m])$, and its summand for the character $x_i \mapsto t_i$ is $H_1(F_k, K_t)$. The augmentation ideal of $\mathbb{Z}[F_k]$ is free on the $x_i - 1$, so $H_1(F_k, K_t)$ is the kernel of $\bigoplus_i K e_i \to K$, $e_i \mapsto t_i - 1$, the chain e_i standing for the loop x_i and a word w for $\sum_j (\partial w/\partial x_j)(t) e_j$.

[35] Artin's automorphism of the half twist σ_i of the i-th and $(i+1)$-st points is $x_i \mapsto x_i x_{i+1} x_i^{-1}$, $x_{i+1} \mapsto x_i$, the other generators being fixed; P_k is generated by the full twists $A_{st} = \sigma_{t-1} \cdots \sigma_{s+1} \sigma_s^2 \sigma_{s+1}^{-1} \cdots \sigma_{t-1}^{-1}$. For

Poincaré duality and the choice of the generator of the deck group identify it with V_a, its dual or its conjugate, and the automorphisms $g \mapsto {}^t g^{-1}$ and $g \mapsto \bar{g}$ of GL preserve SL. ■

Lemma 6.60 (Collision of two branch points)**.** *In Lemma 6.59 suppose that $a_1 + a_2 \neq 0$, and let $a' = (a_1 + a_2, a_3, \dots, a_k)$. Let $I = (1-t_2)e_1 - (1-t_1)e_2$, the cycle of a disc around λ_1 and λ_2, and let $Y \subset E_a$ be the image of the cycles outside that disc, the vectors of* $\ker \partial$ *with $v_2 = t_1 v_1$. Then $E_a = Y \oplus KI$, and $\iota(v_1, v_3, \dots, v_k) = (v_1, t_1 v_1, v_3, \dots, v_k)$ induces an isomorphism $E_{a'} \to Y$. If $\beta \in P_k$ is obtained from $\beta' \in P_{k-1}$ by doubling its first strand, then β preserves Y and KI, acts on Y as β' acts on $E_{a'}$, and multiplies I by a power of ζ.*

Proof. The vectors of Y have (v_1, v_2) proportional to $(1, t_1)$, and $(1-t_2, t_1-1)$ is proportional to $(1, t_1)$ only if $t_1 t_2 = 1$; so $I \notin Y$. The map ι carries the kernel of ∂', where $\partial'(e_1) = t_1 t_2 - 1$, onto the cycles outside the disc, because $v_1(t_1 t_2 - 1) = v_1(t_1 - 1) + t_1 v_1 (t_2 - 1)$, and it carries the vector ℓ' of a' to ℓ; as $\dim E_{a'} + 1 = \dim E_a$, both assertions about Y follow. The doubled braid moves a disc containing λ_1, λ_2 rigidly. Its automorphism therefore maps $x_1 \mapsto w x_1 w^{-1}$, $x_2 \mapsto w x_2 w^{-1}$ and $x_j \mapsto u_j$ for $j \geq 3$, where $y \mapsto w' y w'^{-1}$, $x_j \mapsto u'_j$ is the automorphism of β' on the free group on $y, x_3, \dots, x_k$, and w, u_j are w', u'_j with y replaced by $x_1 x_2$. For such a word u the chain rule gives $\partial u / \partial x_1 = \partial u / \partial y$ and $\partial u / \partial x_2 = (\partial u / \partial y)\, x_1$, which at t is the relation $v_2 = t_1 v_1$; together with $\partial (w x_i w^{-1}) / \partial x_j = (1 - t_i)\, \partial w / \partial x_j + t(w) \delta_{ij}$ at t, this shows that the rows of J_β for $j \geq 3$, and the first row plus t_1 times the second, are the images under ι of the rows of $J_{\beta'}$; this is the equivariance of ι. The same formula gives $I J_\beta = (1-t_2)\big((1-t_1)\partial w + t(w) e_1\big) - (1-t_1)\big((1-t_2)\partial w + t(w) e_2\big) = t(w) I$, and $t(w)$ is a power of ζ. ■

Lemma 6.61 (Two hyperplanes)**.** *Let $E = Y \oplus L$ be a complex vector space with* $\dim Y \geq 2$ *and* $\dim L = 1$*, let $S \subset \mathrm{SL}(E)$ be the subgroup acting on Y through* $\mathrm{SL}(Y)$ *and trivially on L, and let $\gamma \in \mathrm{GL}(E)$ satisfy $\gamma L \neq L$ and $\gamma Y \neq Y$. Then the group T generated by S and $\gamma S \gamma^{-1}$ acts irreducibly on E. If* $\dim Y \geq 3$*, then $T = \mathrm{SL}(E)$. If* $\dim Y = 2$*, every connected semisimple subgroup of* $\mathrm{SL}(E)$ *that contains T is* $\mathrm{SL}(E)$*.*

Proof. The group T is connected. As a representation of S, E is the sum of the irreducible nontrivial Y and the trivial L, so its S-stable subspaces are 0, L, Y and E, and those of $\gamma S \gamma^{-1}$ are 0, γL, γY and E; by the hypotheses and the dimensions only 0 and E are in both lists, and T is irreducible.

Let $\dim Y \geq 3$. As a representation of $\mathfrak{sl}(Y)$,

$$\mathfrak{sl}(E) = \mathfrak{sl}(Y) \oplus \mathrm{Hom}(L, Y) \oplus \mathrm{Hom}(Y, L) \oplus \mathbb{C} z,$$

z acting by 1 on Y and by $-\dim Y$ on L, and the four summands, the adjoint representation, Y, Y^* and the trivial one, are pairwise non-isomorphic since $\dim Y \geq 3$. The Lie algebra $\mathfrak{t}$ of T contains $\mathfrak{sl}(Y)$ and is stable under it, so it is $\mathfrak{sl}(Y)$ plus some of the other three summands. Without $\mathrm{Hom}(L, Y)$ it preserves L, and without $\mathrm{Hom}(Y, L)$ it preserves Y; both are excluded by irreducibility. The bracket of a nonzero map $L \to Y$ with a map $Y \to L$ not vanishing on its image has a nonzero component along z, so $\mathfrak{t} = \mathfrak{sl}(E)$ and $T = \mathrm{SL}(E)$.

Let $\dim Y = 2$ and let $G \supseteq T$ be connected and semisimple. Then E is a faithful irreducible representation of dimension 3 of the Lie algebra $\mathfrak{g}$ of G. An irreducible representation of a product of simple Lie algebras is a tensor product of irreducible representations of the factors, and 3 is prime, so $\mathfrak{g}$ is simple. It lies in $\mathfrak{sl}(E)$, of dimension

a pure braid the chain rule reads $J_{\beta\beta'} = J_{\beta'} J_\beta$ or $J_\beta J_{\beta'}$, according to the convention for composing braids, with no twist of the coefficients, since $t \circ \beta = t$.

8 and rank 2, and the simple Lie algebras of rank at most 2 are $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, $\mathfrak{so}_5$ and $\mathfrak{g}_2$, of dimensions 3, 8, 10 and 14. If $\mathfrak{g} \cong \mathfrak{sl}_2$, then $\mathfrak{g}$ equals the Lie algebra of S, both being of dimension 3, so $G = S$, which preserves L, against the irreducibility of T. Hence $\mathfrak{g} = \mathfrak{sl}(E)$ and $G = \mathrm{SL}(E)$. ■

Lemma 6.62 (Merging two branch points)**.** *Let m and a be as in Setup 6.58, with $k \geq 6$ and $p, q \geq 1$. Then there are $i \neq j$ with $a_i + a_j \neq 0$ such that the vector a' obtained by replacing a_i and a_j by $a_i + a_j$ has order m, and its numbers p' and q' are at least 1.*

Proof. Put $f_i = b_i/m \in [1/m, 1 - 1/m]$, so that $p + 1 = \sum_i f_i$ and $q + 1 = \sum_i (1 - f_i)$. Merging a_i and a_j keeps p and lowers q by one if $f_i + f_j < 1$, and lowers p by one and keeps q if $f_i + f_j > 1$. Since $a_i + a_j = -\sum_{l \neq i,j} a_l$, the order of a' is that of the subgroup generated by the a_l with $l \neq i, j$, the rest of a. Replacing a by $-a$ exchanges p and q, so we may assume $q \geq p$, hence $q \geq 2$ as $p + q = k - 2 \geq 4$; if $p \geq 2$, every merge keeps $p', q' \geq 1$ and only the order matters.

The order. For $m = 3$ two equal coordinates exist, their sum is not 0, and any nonzero rest generates $\mathbb{Z}/3$. For $m = 4$ the number of odd coordinates is even and positive; if it is at least 4, merge two equal odd ones, and if it is 2, merge an odd one with one of the at least four coordinates equal to 2. For $m = 6$ the rest generates $\mathbb{Z}/6$ exactly when it contains an odd coordinate and one prime to 3. The odd coordinates are even in number and the coordinates prime to 3 have sum divisible by 3, so each kind occurs at least twice. If exactly two coordinates are odd, the at least four others lie in $\{2, 4\}$; merge two equal ones among them. If exactly two are prime to 3, the others equal 3; merge a 3 with one of the two. Otherwise each kind occurs at least three times, and any two coordinates with nonzero sum can be merged; they exist, since pairwise sums 0 would force every coordinate to be 3.

The case $p = 1$. Then $\sum_i b_i = 2m$. Some b_i equals 1, since otherwise $\sum_i b_i \geq 2k \geq 12 \geq 2m$, with equality only if $m = 6$ and every coordinate is 4, when a has order three. Keep such a coordinate, $a_i = -1$, which generates $\mathbb{Z}/m$. The other $k - 1 \geq 5$ coordinates have b-sum $2m - 1$, so the two smallest values of b among them add up to at most $2(2m - 1)/5 < m$; merge these two coordinates. Their sum is not 0, $p' = p = 1$, $q' = q - 1 = k - 4 \geq 2$, and the rest contains a_i, so a' has order m. ■

Lemma 6.63 (Five branch points)**.** *Let m and a be as in Setup 6.58, with $k = 5$ and $p, q \geq 1$. Then $a_i + a_j = 0$ for some $i \neq j$, and the other three coordinates can be labelled x, y, z so that $x + y \neq 0$ and $a' = (x + y, z, a_i, a_j)$ has order m and $p' = q' = 1$.*

Proof. Replacing a by $-a$ we may assume $p = 1$, so that $\sum_l b_l = 2m$ with five terms in $[1, m - 1]$; and $a_i + a_j = 0$ means $b_i + b_j = m$. For $m = 3$ the terms lie in $\{1, 2\}$ and exactly one is 2; take it and a 1. For $m = 4$, take a 3 and a 1 if both occur. If no term is 3, the terms lie in $\{1, 2\}$ and at least three are 2, which gives $2 + 2$; if no term is 1, they lie in $\{2, 3\}$ and add up to at least 10, which is impossible. For $m = 6$ suppose that no two terms add up to 6. Then at most one term is 3, and neither 1 and 5 nor 2 and 4 both occur. Terms in $\{1, 2, 3\}$ with at most one 3 add up to at most 11, and terms in $\{3, 4, 5\}$ to at least 19. Terms in $\{1, 3, 4\}$, respectively $\{2, 3, 5\}$, cannot add up to 12, since the number y of terms 4, respectively 5, would satisfy $3y \in \{7, 5\}$, respectively $3y \in \{2, 1\}$. This proves the first assertion.

The other three coordinates add up to 0 and have b-sum m, so any two of them, x and y, have $x + y = -z \neq 0$ and b-sum less than m; merging them keeps $p' = 1$, so $q' = 1$, and a' contains $c = a_i$ and $a_j = -c$. The order of a' is that of the subgroup generated by z and c. If c generates $\mathbb{Z}/m$, any choice of z will do. Otherwise c is even, or $m = 6$ and $c = 3$.

As a has order m, one of x, y, z is then odd, respectively prime to 3, and we take it as z; then z and c generate $\mathbb{Z}/m$. ∎

Proposition 6.64 (Monodromy of cyclic covers)**.** *In Setup 6.58, if $p = 0$ or $q = 0$ then Γ_a is finite, and if $n \geq 3$ and $p, q \geq 1$ then $G_a^0 \supseteq \mathrm{SL}(V_a)$.*

Proof. If $pq = 0$, the polarisation is definite on V_a up to sign, and Γ_a preserves it and the $\mathbb{Z}[\zeta]$-lattice spanned by $H^1(D_a, \mathbb{Z})$; a discrete subgroup of a compact group is finite. Assume $p, q \geq 1$ and use the model of Lemma 6.59. Three facts are used throughout. If $u \in \Gamma_a$ is unipotent, the Zariski closure of its powers is the group $\exp(\tau \log u)$, so $\log u$ lies in the Lie algebra of G_a^0. The group Γ_a normalises G_a^0. And G_a^0 is semisimple: by André's theorem [And92, Theorem 1] the identity component of the Zariski closure of the monodromy group of $H^1(D_a, \mathbb{Q})$ is a normal subgroup of the derived group of the Mumford–Tate group of a very general fibre, hence semisimple, and G_a^0 is its image in $\mathrm{GL}(V_a)$, the monodromy commuting with the deck group.

Four points. Let $k = 4$, so $n = 2$, and suppose that two coordinates of a are opposite; relabel them a_1 and a_2, so that $t_1 t_2 = 1$. The full twist A_{12} maps $x_i \mapsto w x_i w^{-1}$ for $i = 1, 2$, with $w = x_1 x_2$, and fixes x_3, x_4. As $t(w) = 1$, the formula for $\partial(w x_i w^{-1})/\partial x_j$ in the proof of Lemma 6.60 gives $v J_{A_{12}} = v + \varphi(v)\,\delta$, with $\delta = e_1 + t_1 e_2$ and $\varphi(v) = (1 - t_1)v_1 + (1 - t_2)v_2$, which equals $\sum_{j \geq 3}(t_j - 1)v_j$ on $\ker \partial$. Here $\delta \in \ker \partial$, $\varphi(\delta) = 0$, $\varphi(\ell) = t_1 \cdots t_k - t_1 t_2 = 0$, the vector δ is not a multiple of ℓ, whose third coordinate is $t_1 t_2 = 1$, and $\varphi((t_3 - 1)e_1 - (t_1 - 1)e_3) = -(t_1 - 1)(t_3 - 1) \neq 0$. So A_{12} acts on E_a as a unipotent $u \neq 1$, and G_a^0 is a connected semisimple subgroup of $\mathrm{SL}(E_a) \cong \mathrm{SL}_2$ containing $\exp(\tau \log u)$. The connected subgroups of SL_2 of dimension one or two are solvable, so $G_a^0 = \mathrm{SL}(E_a)$.

The induction. Let $n \geq 3$. If $n = 3$, Lemma 6.63 gives, after a relabelling, a vector $a' = (a_1 + a_2, a_3, a_4, a_5)$ of order m with $a_1 + a_2 \neq 0$, $p' = q' = 1$ and two opposite coordinates, so the identity component of the Zariski closure of the monodromy of $E_{a'}$ is $\mathrm{SL}(E_{a'})$ by the case of four points. If $n \geq 4$, Lemma 6.62 and a relabelling give $a_1 + a_2 \neq 0$ and $a' = (a_1 + a_2, a_3, \ldots, a_k)$ of order m with $p', q' \geq 1$, and by induction on n that identity component contains $\mathrm{SL}(E_{a'})$. Let H be the identity component of the Zariski closure of the doubled braids of Lemma 6.60. Then $H \subseteq G_a^0$, H preserves Y and KI and acts trivially on KI, where the doubled braids act through a finite group, and its image in $\mathrm{GL}(Y)$ is the identity component of the Zariski closure of the monodromy of $E_{a'}$, which contains $\mathrm{SL}(Y)$. So G_a^0 contains the group S of Lemma 6.61 for $E_a = Y \oplus KI$. Let $\gamma = A_{23}$, the full twist of the second and third points; its automorphism fixes x_1 and maps $x_2 \mapsto x_2 x_3 x_2 x_3^{-1} x_2^{-1}$ and $x_3 \mapsto x_2 x_3 x_2^{-1}$. Then $\gamma(I) = I + (1 - t_1)t_2 I_{23}$ with $I_{23} = (1 - t_3)e_2 - (1 - t_2)e_3$, and $\gamma(KI) \neq KI$, since I, I_{23} and ℓ are independent, ℓ having a nonzero fourth coordinate. The functional $v \mapsto v_2 - t_1 v_1$ vanishes on ℓ and cuts out Y, and at $\gamma(v)$ for $v \in Y$ it equals $(t_3 - 1)(t_1 t_2 v_1 - v_3)$, which is nonzero for the cycle $(1 - t_4)e_3 - (1 - t_3)e_4$ of Y; so $\gamma Y \neq Y$. As γ normalises G_a^0, this group contains S and $\gamma S \gamma^{-1}$, and Lemma 6.61, with $\dim Y = n - 1$, gives $G_a^0 = \mathrm{SL}(E_a)$: for $n \geq 4$ directly, and for $n = 3$ because G_a^0 is connected and semisimple. ∎

For $m = 3$ the proposition also follows from the ℓ-adic theorem of Achter and Pries [AP07, Corollary 3.10], which identifies the monodromy of trielliptic curves with a special unitary group.

Lemma 6.65 (The discriminant of the new part)**.** *In Setup 6.58 let n be even with $p = q = n/2$, let $L = H^1(D_a, \mathbb{Z})$ with the cup product Q, and let $L_a = L \cap (V_a \oplus \overline{V_a})$. Then*

L_a is a free $\mathbb{Z}[\zeta]$-module of rank n, the K-hermitian form H with $Q = \frac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}H)$ on it, $\delta = \sqrt{-1}$ for $m = 4$ and $\delta = \sqrt{-3}$ otherwise ([I, Lemma 4.5]), has signature $(n/2, n/2)$, and the class of $(-1)^{n/2}\det H$ in $\mathbb{Q}^\times/\mathrm{N}(K^\times)$ is trivial for $m = 3$ and $m = 4$; for $m = 6$ it is trivial if and only if the number of i with $a_i = 3$ is even.

Proof. The deck transformation g is an isometry of the unimodular lattice L, so its eigenspaces for eigenvalues μ, ν are orthogonal unless $\mu\nu = 1$. Hence L_a, the part for the primitive m-th roots of unity, and the part L^{old} for the other eigenvalues are orthogonal primitive sublattices, each the orthogonal complement of the other, and $|\det(L_a)| = |\det(L^{\mathrm{old}})|$, which is 1 for $m = 3$, when $L^{\mathrm{old}} = 0$.[36] For a quotient map $\pi\colon D_a \to D'$ of prime degree e that is ramified, $\pi^*H^1(D', \mathbb{Z})$ is primitive in L and the cup product on it is e times that of D'.[37] For $m = 4$, $L^{\mathrm{old}} = \pi^*H^1(D', \mathbb{Z})$ for the quotient D' by g^2, ramified over the λ_i with a_i odd, so $|\det L^{\mathrm{old}}| = 2^{2g'}$, g' the genus of D'. For $m = 6$, the quotients D_2 by g^2 and D_3 by g^3 are ramified over the λ_i with $3 \nmid a_i$ and with a_i odd, which exist since a has order six, and L^{old} contains $A \oplus B$ with $A = \pi_2^*H^1(D_2, \mathbb{Z})$ and $B = \pi_3^*H^1(D_3, \mathbb{Z})$, of determinants 3^{2g_2} and 2^{2g_3}; the quotient $L^{\mathrm{old}}/(A \oplus B)$ embeds in $A^\vee/A$ and in $B^\vee/B$, of coprime orders, so $L^{\mathrm{old}} = A \oplus B$ and $|\det L^{\mathrm{old}}| = 3^{2g_2}2^{2g_3}$.

On L_a the deck transformation satisfies the cyclotomic equation of ζ, so L_a is a torsion-free module over the principal ideal domain $\mathbb{Z}[\zeta]$, free of rank n; the signature of H is (p, q) ([I, Remark 4.6]). The determinant of the trace form of a hermitian lattice gives $|\det(L_a)| = |d_K|^n\mathrm{N}(\frac{1}{2\delta})^n(\det H)^2$, that is $(\det H)^2$ for $K = \mathbb{Q}(i)$ and $(\det H)^2/4^n$ for $K = \mathbb{Q}(\sqrt{-3})$.[38] Hence $(-1)^{n/2}\det H$ equals 2^n for $m = 3$, $2^{g'}$ for $m = 4$ and $2^{n+g_3}3^{g_2}$ for $m = 6$, the sign being that of a form of signature $(n/2, n/2)$. For $m = 3$ this is $\mathrm{N}(2^{n/2})$, and for $m = 4$ a norm as $2 = \mathrm{N}(1 + i)$. For $m = 6$, $3 = \mathrm{N}(\sqrt{-3})$ and $4 = \mathrm{N}(2)$ are norms and 2 is not, since $x^2 + xy + y^2$ is even only when x and y are, so that every norm has even 2-adic valuation; so the class is trivial exactly when g_3 is even. By the Riemann–Hurwitz formula D_3, the triple cover $y^3 = \prod_i (t - \lambda_i)^{a_i}$ branched over the i with $3 \nmid a_i$, has genus $k - c_3 - 2 = n - c_3$, where c_3 is the number of i with $a_i = 3$; as n is even, this is the assertion. ∎

Theorem 6.66 (Very general diagonal complete intersections)**.** *Keep the notation of Theorem 6.55, let $r \geq 2$ be even, and let $\lambda \in U$ lie outside a countable union of proper Zariski closed subsets of U. For a Hodge class x of degree r on $X = X_{d,r}(\lambda)$, write $x_{[a]}$ for the component of Φ^*x in $W_{[a]}$.*

(i) *If a has order two, then $x_{[a]}$ is a rational multiple of the image of $\theta_a^{r/2}$, where θ_a is the class of a polarisation of $B_{[a]}$, under the correspondence of the proof of Theorem 6.55(iv); in particular $x_{[a]}$ is algebraic.*

[36] If $M \subset L$ is primitive and L is unimodular, restriction $L \to M^\vee$ is onto with kernel $M^\perp$, so $L/(M \oplus M^\perp) \cong M^\vee/M$, and likewise $\cong (M^\perp)^\vee/M^\perp$; the order of $M^\vee/M$ is $|\det M|$.

[37] A loop in D' avoiding the branch points, composed with a loop around a branch point, whose monodromy generates the group of order e, lifts to a closed loop with the same image in $H_1(D', \mathbb{Z})$; so $\pi_*\colon H_1(D_a, \mathbb{Z}) \to H_1(D', \mathbb{Z})$ is onto, and its transpose π^* is a split injection. The factor e is $\pi_*\pi^* = e$.

[38] For a free $\mathcal{O}_K$-module with hermitian Gram matrix M and $c \in K^\times$, the $\mathbb{Z}$-bilinear form $\operatorname{Tr}_{K/\mathbb{Q}}(c\,v^TM\bar{w})$ has determinant of absolute value $|d_K|^n\mathrm{N}(c)^n(\det M)^2$, where d_K is the discriminant of K: for $n = 1$ this is the discriminant of the trace form of $\mathcal{O}_K$ scaled by $\mathrm{N}(cM)$. Changing the basis by $P \in \mathrm{GL}_n(K)$ multiplies both sides by $\mathrm{N}_{K/\mathbb{Q}}(\det P)^2$, and a basis in which M is diagonal makes the form an orthogonal sum of forms of rank one. Here $d_K = -4$ and $\mathrm{N}(\frac{1}{2i}) = \frac{1}{4}$, or $d_K = -3$ and $\mathrm{N}(\frac{1}{2\sqrt{-3}}) = \frac{1}{12}$.

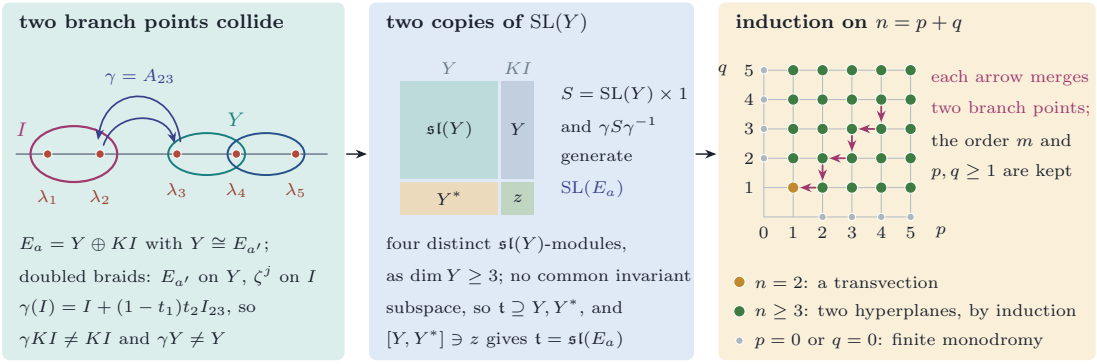


FIGURE 25. The proof of Proposition 6.64. Left: two branch points λ_1, λ_2 collide inside a disc (Lemma 6.60); the cycles outside the disc form the hyperplane $Y \cong E_{a'}$, the cycle I of the disc spans the complementary line, and the twist γ of the second and third points moves both. Middle: the two copies of $\mathrm{SL}(Y)$ of Lemma 6.61, which generate $\mathrm{SL}(E_a)$ when $\dim Y \geq 3$. Right: the induction on $n = k - 2$, each merge lowering p or q by one (Lemma 6.62, Lemma 6.63), down to $n = 2$, where the full twist of two points with opposite exponents is a transvection.

(ii) *If a has order $m \in \{3, 4, 6\}$ and $x_{[a]} \neq 0$, then $\dim V_a = r$ and $\dim V_a^{1,0} = r/2$. In that case $B_{[a]}$ is an abelian variety of dimension r of Weil type for $\mathbb{Q}(\zeta_m)$, and $W_{[a]}$ is the image of its space of Weil classes. Its discriminant is trivial for $m = 3, 4$, and for $m = 6$ exactly when an even number of the coordinates of a equal $d/2$.*

(iii) *The Hodge conjecture holds on $X_{d,r}(\lambda)$ for every N when $d = 2$, when $d \in \{3, 4\}$ and $r \leq 6$, and when $d = 6$ and $r \leq 4$. For $d = 6$ and $r = 6$ the Hodge classes are algebraic modulo the spaces $W_{[a]}$ of the characters of order six of (ii) with an odd number of coordinates 3, which are spaces of Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ and occur for every N. For $d \in \{2, 3, 4, 6\}$ the Hodge classes of degree r form a space of dimension*

$$1 + \binom{N+1}{r+2} T_d(r+2) + \epsilon_d \sum_{j \geq r/2+2} \binom{N+1}{2j},$$

where $T_d(k)$ is the number of $(s_1, \ldots, s_k) \in \{1, \ldots, d-1\}^k$ with $\sum_i s_i = dk/2$, and ϵ_d is 1 for d even and 0 for d odd; thus $T_2(k) = 1$, $T_3(k) = \binom{k}{k/2}$, $T_4(6) = 141$ and $T_6(6) = 1751$.

For odd r there is nothing to prove, since every Hodge class of X is then a multiple of a power of the hyperplane class. So the theorem gives the Hodge conjecture for the very general complete intersection of Vandermonde type of diagonal quadrics in every dimension, of diagonal cubics or quartics up to dimension seven, and of diagonal sextics up to dimension five, in every $\mathbb{P}^N$.

Proof. Apply Lemma 6.57 to $H^r(X_{d,r}(\lambda), \mathbb{Z})$ over U, polarised by the hyperplane class, and let $\Pi' \subset \pi_1(U, \lambda)$ be the subgroup of finite index that fixes the Hodge classes. The monodromy commutes with H and with Φ^*, so Π' fixes $x_{[a]}$ and each of its parts in the spaces $\bigwedge^r V_{a'}$. Let m be the order of a, take representatives $0 \leq a_i < d$, and put $a_i' = ma_i/d$.

The function $y = \prod_i u_i^{a_i}$ is invariant under the kernel of χ_a and satisfies $y^m = \prod_i (t-\lambda_i)^{a_i'}$, so the quotient of C by that kernel is the cyclic cover D_a of $\mathbb{P}^1$ of degree m given by this equation, and pull-back identifies $H^1(D_a, \mathbb{Q})$ with the part $\bigoplus_{j\neq 0} V_{ja}$ of $H^1(C, \mathbb{Q})$, compatibly with the monodromy. The image $B_{[a]}$ is isogenous to the abelian subvariety of $J(D_a)$ whose first cohomology is $\bigoplus_{j\in(\mathbb{Z}/m)^\times} V_{ja}$, all of $J(D_a)$ for $m \leq 3$, its polarisation being a rational multiple of the restriction of the principal one, since pull-back along a map of degree e multiplies the cup product by e.

(i) For $m = 2$, D_a is the hyperelliptic curve $y^2 = \prod_{i\in S}(t-\lambda_i)$ over the support S of a, of genus $g = |S|/2 - 1$, and $H^1(D_a, \mathbb{Q}) = V_a$. Move the base point so that the λ_i with $i \in S$ are real and consecutive, $\lambda_{s_1} < \cdots < \lambda_{s_{2g+2}}$, the others lying away from the segment they span. The loop in which λ_{s_k} and $\lambda_{s_{k+1}}$ turn once around each other, inside a disc that contains no other λ_i, makes D_a acquire a node with vanishing cycle c_k, the lift of the segment $[\lambda_{s_k}, \lambda_{s_{k+1}}]$, and acts by the square of the Picard–Lefschetz transvection, $T_k^2(z) = z + 2\langle z, c_k\rangle c_k$ up to orientation [Voi03, Chapter 3].[39] The classes $c_1, \ldots, c_{2g}$ satisfy $\langle c_k, c_{k+1}\rangle = \pm 1$ and $\langle c_k, c_l\rangle = 0$ for $|k-l| \geq 2$, so their intersection matrix has determinant 1 and they form a basis of $H_1(D_a, \mathbb{Q})$; they are the distinguished basis of type A of [ACa79]. Some power T_k^{2e} with $e \geq 1$ lies in the image of Π'; it is unipotent, so the Zariski closure of that image contains the one-parameter group $\exp(\tau N_k)$, $N_k(z) = \langle z, c_k\rangle c_k$, and its Lie algebra contains $N_1, \ldots, N_{2g}$. These generate the Lie algebra $\mathfrak{sp}(V_a)$.[40] So the Zariski closure contains $\mathrm{Sp}(V_a)$, whose invariants in $\bigwedge^r V_a$ are the multiples of $\omega_a^{r/2}$, ω_a the polarisation form [FH91, Theorem 17.5].[41] The correspondence of the proof of Theorem 6.55(iv) maps $H^r(B_{[a]}, \mathbb{Q}) = \bigwedge^r H^1(B_{[a]}, \mathbb{Q})$ isomorphically and equivariantly onto $\bigwedge^r V_a$; the class θ_a goes to an invariant nondegenerate form, a nonzero multiple of ω_a, and $\theta_a^{r/2}$ to a nonzero multiple of $\omega_a^{r/2}$, which is algebraic.

(ii) For $m \in \{3, 4, 6\}$, D_a is the cyclic cover of Setup 6.58 for the exponents $a_i' \in \{1, \ldots, m-1\}$ on the support S of a, and V_a is its eigenspace, of dimension $n = |S| - 2$ (Lemma 6.59). The map from U to $\mathrm{Conf}_{|S|}$ that forgets the λ_i outside S is a fibration with connected fibres, so it is onto on fundamental groups, and Π' maps onto a subgroup of finite index of $P_{|S|}$; the Zariski closure of its image in $\mathrm{GL}(V_a)$ therefore contains G_a^0. If $n \leq 2$, then $\bigwedge^r V_a$ is zero unless $r = n = 2$. If $V_a^{1,0}$ or $V_a^{0,1}$ is zero, then $\bigwedge^r V_a$ has type $(r, 0)$ or $(0, r)$ and contains no class of type $(r/2, r/2)$. Otherwise $n \geq 3$, and Proposition 6.64 gives $G_a^0 \supseteq \mathrm{SL}(V_a)$, whose invariants in $\bigwedge^r V_a$ vanish for $0 < r < n$. In every case $x_{[a]} \neq 0$ forces $r = n$, and then $\bigwedge^r V_a$ is a line of type $(p_a, r - p_a)$ with $p_a = \dim V_a^{1,0}$, which contains a class of type $(r/2, r/2)$ only if $p_a = r/2$. The field $\mathbb{Q}(\zeta_m)$ then acts on $B_{[a]}$, of dimension $|[a]| \dim V_a/2 = r$, with multiplicities $(r/2, r/2)$ on the tangent space and compatibly with the polarisation, the automorphism of D_a preserving the cup product; so $B_{[a]}$ is of Weil type ([I, Definition 3.1]), and $W_{[a]} = \bigwedge^r V_a \oplus \bigwedge^r V_{-a}$ is the image of $\bigwedge^r_{\mathbb{Q}(\zeta_m)} H^1(B_{[a]}, \mathbb{Q})$,

[39] Near the collision the curve is $y^2 = (t-\lambda_{s_k})(t-\lambda_{s_{k+1}})\, v$ with v a unit, analytically $y^2 = t^2 - \epsilon$ with ϵ a multiple of $(\lambda_{s_{k+1}} - \lambda_{s_k})^2$; one turn of the two points around each other is two turns of ϵ around 0, and one turn of ϵ acts by the Dehn twist along the vanishing cycle, whose action on H_1 is the transvection T_k.

[40] Identify $\mathfrak{sp}(V)$ with $\mathrm{Sym}^2 V$, the product uv acting by $z \mapsto \langle z, u\rangle v + \langle z, v\rangle u$, so that $N_k = \frac{1}{2}c_k^2$ and $[uv, wz] = \langle u, w\rangle vz + \langle u, z\rangle vw + \langle v, w\rangle uz + \langle v, z\rangle uw$ up to an overall sign. Then $[c_k^2, c_{k+1}^2] = \pm 4 c_k c_{k+1}$, and if all products $c_i c_j$ with $i, j \leq k$ lie in the generated algebra, so do the $c_i c_{k+1} = \pm\frac{1}{2}[c_i c_k, c_{k+1}^2]$ for $i < k$, because c_{k+1} meets only c_k among $c_1, \ldots, c_k$. By induction every $c_i c_j$ lies in it, and these span $\mathrm{Sym}^2 V$.

[41] The exterior algebra of V is a representation of $\mathrm{Sp}(V)$ and of the $\mathfrak{sl}_2$ spanned by $\omega\wedge$ and its adjoint, and $\bigwedge^r V = \bigoplus_j \omega^j \wedge P^{r-2j}$, where P^k, the kernel of the contraction with ω in degree $k \leq g$, is irreducible and nontrivial for $k \geq 1$, so the invariants are the multiples of $\omega^{r/2}$, for r even.

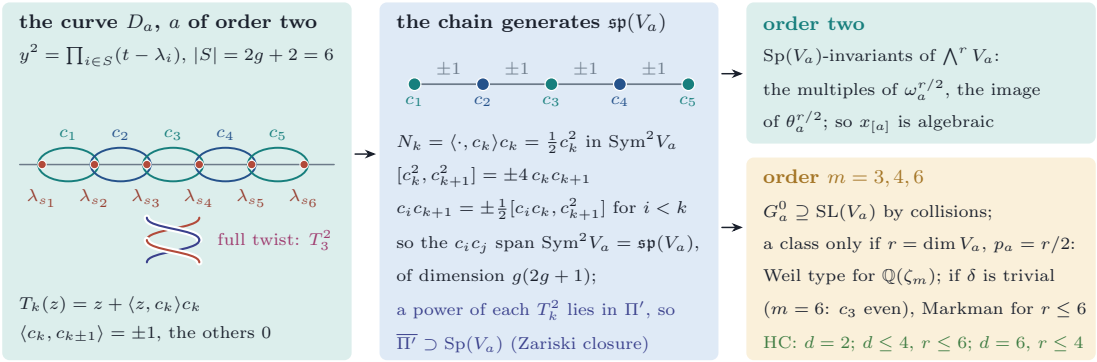


FIGURE 26. The proof of Theorem 6.66. Left: the hyperelliptic curve D_a of a character of order two, drawn over the six points of its support for $g = 2$, with the chain of vanishing cycles $c_1, \dots, c_5$; a full twist of two neighbouring branch points, drawn as a braid, acts by the square of the transvection T_k. Middle: the chain as a graph, and the induction in $\mathrm{Sym}^2 V_a$ by which the logarithms N_k generate $\mathfrak{sp}(V_a)$. Right: what the monodromy leaves of the Hodge classes, the powers of the polarisation for characters of order two, and for characters of order three, four and six the Weil classes of an abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$.

its space of Weil classes. Its discriminant ([I, Definition 15.22]) is that of the lattice L_a of Lemma 6.65: the isogeny with the abelian subvariety of $J(D_a)$ whose first cohomology is $L_a \otimes \mathbb{Q}$ scales the polarisation by a rational number and $\det H$ by its r-th power, a norm. The coordinates of a equal to $d/2$ are those with $a_i' = 3$ when $m = 6$, and Lemma 6.65 gives the assertion.

(iii) Let x be a Hodge class of degree r. The component of $\Phi^* x$ along η is algebraic. For $d \in \{2, 3, 4, 6\}$ every $a \neq 0$ has order 2, 3, 4 or 6, and by (i) and (ii) the components $x_{[a]}$ are images of powers of polarisations or of Weil classes of abelian r-folds of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$. These are algebraic for $r = 2$ by the theorem of Lefschetz, for $r = 4$ by Markman's theorem on abelian fourfolds of Weil type, and for $r = 6$ by Markman's theorem on abelian sixfolds of split Weil type [Mar25a, Mar25b] when the discriminant is trivial: always for $d = 3, 4$, and for $d = 6$ except along the characters of order six with an odd number of coordinates 3. So $\Phi^* x$ is algebraic in the cases stated, and so is $x = \Phi_* \Phi^* x / |G|$. For the dimension, the classes found are Hodge at every λ, and by (i) and (ii) there are no others. Each character of order two with $|S| \geq r + 2$ gives one class, the image of $\theta_a^{r/2}$, nonzero since $r \leq \dim V_a$. A character of order at least three gives a class only if $|S| = r + 2$ and $\dim V_a^{1,0} = \sum_{i\in S}(d - a_i)/d - 1 = r/2$, that is $\sum_i a_i = d(r + 2)/2$ with representatives $0 \leq a_i < d$, and then exactly one, in $\bigwedge^r V_a$. The characters with $|S| = r + 2$ whose coordinates on S average $d/2$ are counted by $T_d(r + 2)$; for d even they include the one of order two with all coordinates $d/2$, and the characters of order two with $|S| \geq r + 4$ give the last sum. ∎

Figure 26 draws the steps of the proof. The dimensions of (iii) never exceed $h^{r/2,r/2}(X)$, and they equal it for a quadric, for the Fermat hypersurfaces (for cubics 7, 21 and 71 in dimensions 2, 4 and 6, for quartics 142 and 1108 in dimensions 4 and 6, for sextics 1752 in dimension 4) and for two quadrics in $\mathbb{P}^N$ with N even $(N + 2)$, where every class of

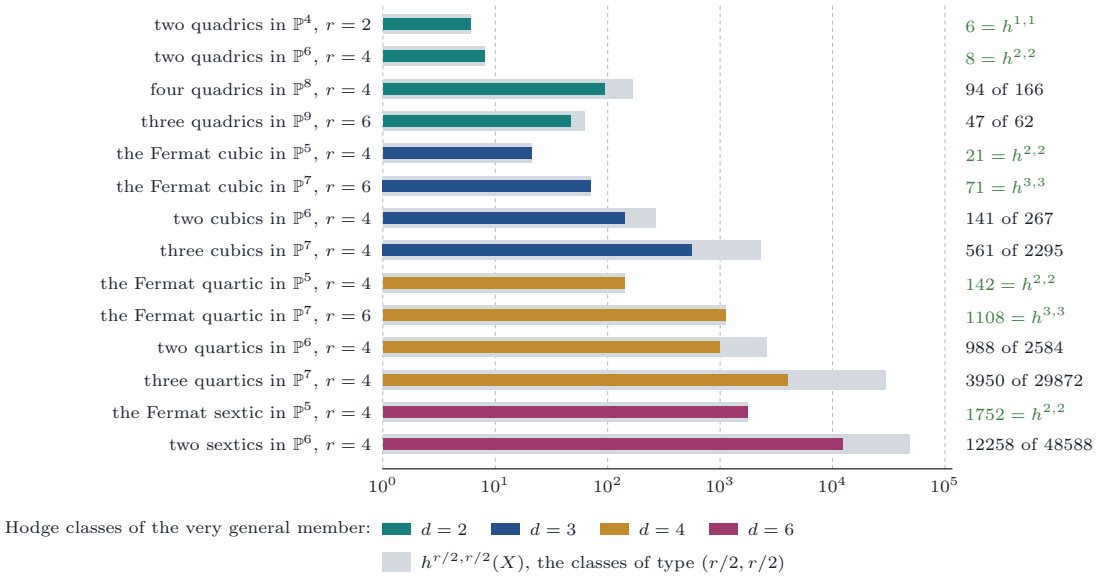


FIGURE 27. The dimension of the space of Hodge classes of degree r on the very general member, from Theorem 6.66(iii), against the Hodge number $h^{r/2,r/2}$, on a logarithmic scale. Where the two agree, the space $H^{r/2,r/2}$ is spanned by Hodge classes at every configuration; elsewhere the very general member has fewer Hodge classes, and the members with more lie over a countable union of proper closed subsets of the space of configurations.

type $(r/2, r/2)$ is algebraic. For two cubics in $\mathbb{P}^6$ the very general member has 141 Hodge classes of degree four against $h^{2,2} = 267$, and for three quartics in $\mathbb{P}^7$ it has 3950 against $h^{2,2} = 29872$ (Figure 27). Figure 28 shows, for each degree d and dimension r, what is known for the very general member.

Remark 6.67 (What the diagonal complete intersections give). Theorem 6.55 adds to the domain of (F3′) a family of complete intersections of every dimension and every degree, parametrised by configurations of $N+1$ points on the line, among them all smooth complete intersections of two diagonal hypersurfaces; for $c \geq 3$ and $r \geq 2$ the condition on the columns is a real restriction. Corollary 6.56 shows on them how (F3′) and (F2) together give the conjecture: once the Hodge classes of X are carried to the abelian varieties $B_{[a]}$, the conjecture for X is the algebraicity of their Hodge classes, a question of type (F2), and for the fourfolds in $\mathbb{P}^6$ of degrees 3, 4 and 6 it is answered by Markman's theorem: their exceptional classes are Weil classes of abelian fourfolds of Weil type, 490 families of them for $d = 4$. For $d = 3$ these fourfolds are Fano and the conjecture also follows from the theorem of Conte and Murre on uniruled fourfolds [CM78]; for $d = 4$ and $d = 6$ they are of general type, with $h^{2,2} = 2584$ and 48588. Complete intersections of diagonal hypersurfaces, and of quadrics, are also the subject of Terasoma's paper [Ter88]. At a very general configuration Theorem 6.66 goes further, in every $\mathbb{P}^N$: for $d = 2$ in every dimension, for $d = 3$ and $d = 4$ up to dimension seven, and for $d = 6$ up to dimension five, through the monodromy of cyclic covers of degree three, four and six (Proposition 6.64). For $d = 3$ and $r \geq 8$ the conjecture for the very general member is the algebraicity of the Weil classes of the abelian r-folds $B_{[a]}$ of split Weil type for $\mathbb{Q}(\sqrt{-3})$, which is open from dimension

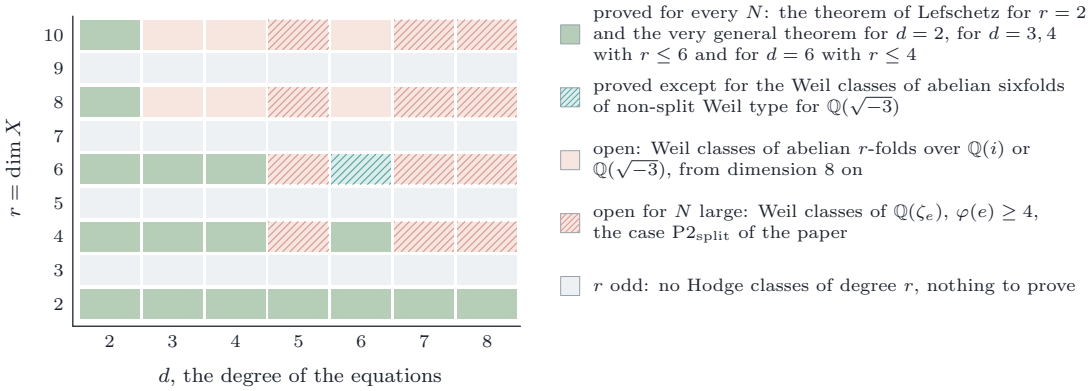


FIGURE 28. The Hodge conjecture for the very general complete intersection $X_{d,r}(\lambda)$ of Vandermonde type, for every N. Grass: proved, by the theorem of Lefschetz for $r = 2$ and by Theorem 6.66 for $d = 2$, for $d = 3, 4$ with $r \leq 6$ and for $d = 6$ with $r \leq 4$. Teal hatching: proved except for the Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$. Clay: open, the Weil classes of abelian varieties of Weil type over $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$ in dimension at least eight, or, hatched, of a CM field of degree at least four (Remark 6.67). For odd r there is nothing to prove.

eight on, and the same holds for $d = 4$ with $\mathbb{Q}(i)$. For $d = 6$ and $r = 6$ the open classes are those of the abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ of Theorem 6.66(iii), such as the one of the character $(1, 1, 2, 2, 3, 5, 5, 5)$ on eight points. Beyond these cases the same reduction meets (F2) itself: for $d = 5$ or $d \geq 7$ and N large enough, some $B_{[a]}$ with $\dim V_a = r$ carries an action of $\mathbb{Q}(\zeta_e)$, $e \mid d$, with $\varphi(e) \geq 4$, and $W_{[a]}$ is the top exterior power of its first cohomology over that CM field, which consists of Hodge classes when the types are balanced: Weil classes for a CM field of degree at least four, of the kind treated in [I, Theorem 1.1]; and for larger N the $B_{[a]}$ have dimension above five.

Remark 6.68 (Where (F3′) stops). With Proposition 6.35, Proposition 6.42 and Proposition 6.43, (F3′) holds on every variety of dimension at most three, on every fourfold whose zero-cycles are supported in dimension three (in particular every uniruled fourfold), on every fivefold whose zero-cycles are supported in dimension at most three, on every fourfold and fivefold whose maximal rationally connected quotient is birational to a variety satisfying (F3′) in degree four (Proposition 6.44), on every fourfold and fivefold dominated by a rational map from such a variety (Proposition 6.46), and on every member of $\mathcal{A}$, a class containing the Fermat hypersurfaces, the smooth Delsarte hypersurfaces with the cyclic covers branched along them (Corollary 6.53), and the complete intersections of diagonal hypersurfaces of Vandermonde type, among them every smooth complete intersection of two diagonal hypersurfaces (Theorem 6.55, Corollary 6.56), of every degree and dimension, the Hilbert schemes of points on abelian surfaces, on products of curves and on the K3 surfaces of Proposition 6.42(iv), the generalised Kummer varieties $K_n(A)$ and the other varieties of Proposition 6.42(v), and every blow-up and projective bundle built from these. Through the Hodge conjecture itself it holds on the complete intersections of Corollary 6.56(ii), among them the fourfolds of general type cut out by two diagonal quartics or sextics in $\mathbb{P}^6$, on the very general complete intersections of diagonal quadrics, of diagonal cubics and quartics

up to dimension seven and of diagonal sextics up to dimension five, of Theorem 6.66, and on the varieties of Proposition 6.39, Proposition 6.40 and Corollary 6.41: for a K3 surface S with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S))$ a CM field, on every power, Hilbert scheme and moduli space of S; for $E = \mathbb{Q}$ and $t = \dim T(S)$, on S^k for $k < t$, on $S^{[n]}$ for $n < t(t+1)/2$ and on the moduli spaces of dimension less than t; on every fourfold of $\mathrm{K3}^{[2]}$ type whose field E is $\mathbb{Q}$ or a CM field; on $F(Y)$ and $Y \times Y$ for every cubic fourfold Y with such a field; and on every fourfold of generalised Kummer type [FV25]. Its frontier cases are these (Figure 29), the first two in degree four on fourfolds whose zero-cycles are not supported on a proper closed subset and which are not birational to a member of $\mathcal{A}$ by any construction above: $S \times S$ and $S^{[2]}$ for a K3 surface whose real multiplication is not induced by a cycle constructed so far, such as S_λ (the known cases are recalled in Remark 7.5); the fourfolds of $\mathrm{K3}^{[2]}$ type with real multiplication, among them $F(Y)$ for a cubic fourfold Y with real multiplication, and with it $Y \times Y$; and, for a very general K3 surface, the single class $\det T(S)$ on S^{21} of Proposition 6.38, equivalently the Hodge conjecture for $S^{[231]}$. In the first two the rational Hodge isometries [Bus19, Mar24] act on T by ± 1, and a composite of Hodge similarities that returns to T is an element $g \in E$ with $g^2 \in \mathbb{Q}$, the adjoint involution of E being trivial; such elements span only the compositum of the quadratic subfields of E, which is $\mathbb{Q}$ for the cubic field of S_λ. Apart from proving the conjecture itself where zero-cycles are small, none of these results touches the step that Remark 6.36 identifies: putting a Hodge class into $\mathrm{Ab}^p(X)$ for a variety that no correspondence connects to an abelian variety. Theorem 6.70 is unchanged, since every statement proved here is a special case of one of the hypotheses of its rule set, not a new rule.

Setup 6.69 (The rule set)**.** The rule set has forty-two statements and twenty-two rules. Ten statements are proved in this paper: the base point with an explicit cycle in every Weil family of an imaginary quadratic field ([I, Theorem 15.27]) and of a CM field of degree at least four ([I, Theorem 17.5]); for each of the two kinds of field, the reduction of the Weil classes of a family to one class at every point of one connected domain, and the stability of the algebraic locus under the rational points, with dense orbits ([I, Section 12], [I, Proposition 16.7], [I, Theorem 17.7]); and four statements on secant objects: the form of the weakened criterion for the transform (Theorem 4.56), the conditions of Markman's strategy that bear on the Chern character, the description of the local obstruction as a failure of Cohen–Macaulayness (Corollary 4.77), and three of the four ingredients of the construction in every dimension. Four are quoted: $\mathbf{W}(K, 2, \delta)$ for every K and δ, $\mathbf{W}(K, 3, \delta_0)$ [Mar25a, Mar25b], the deformation theorem for motivated classes [And96b, Théorème 0.5], and the accessibility of Hodge classes on abelian varieties [Mil20, §5.1, Theorem 3]. Sixteen are hypotheses: (F3), (F2) of Corollary 6.71 and (F3′) of Definition 6.34; (L), (M), (V) of Definition 6.14; (P2), in the polarised form of Remark 3.6, for four groups of families, the split and the other families of the imaginary quadratic fields and of the CM fields of degree at least four, the third group giving $\mathrm{P2}_{\mathrm{split}}$; secant objects meeting the weakened criterion in every dimension, and an affirmative answer to [Mar25b, Question 11.4]; and, for a smooth and for a singular Cohen–Macaulay support on an abelian fourfold, its existence with the required extension property and the vanishing of the image of the product of translation classes. The other twelve are derived: **HC**, the conjecture for every abelian variety, written **AB**, and the Weil classes of every CM field, of the imaginary quadratic fields, of their families of trivial and of nontrivial discriminant, of the CM fields of degree at least four and of their split and their other families, together with one secant object on an abelian fourfold, $\mathbf{W}(K, 4, \delta_0)$, and the range settled in the literature. Each

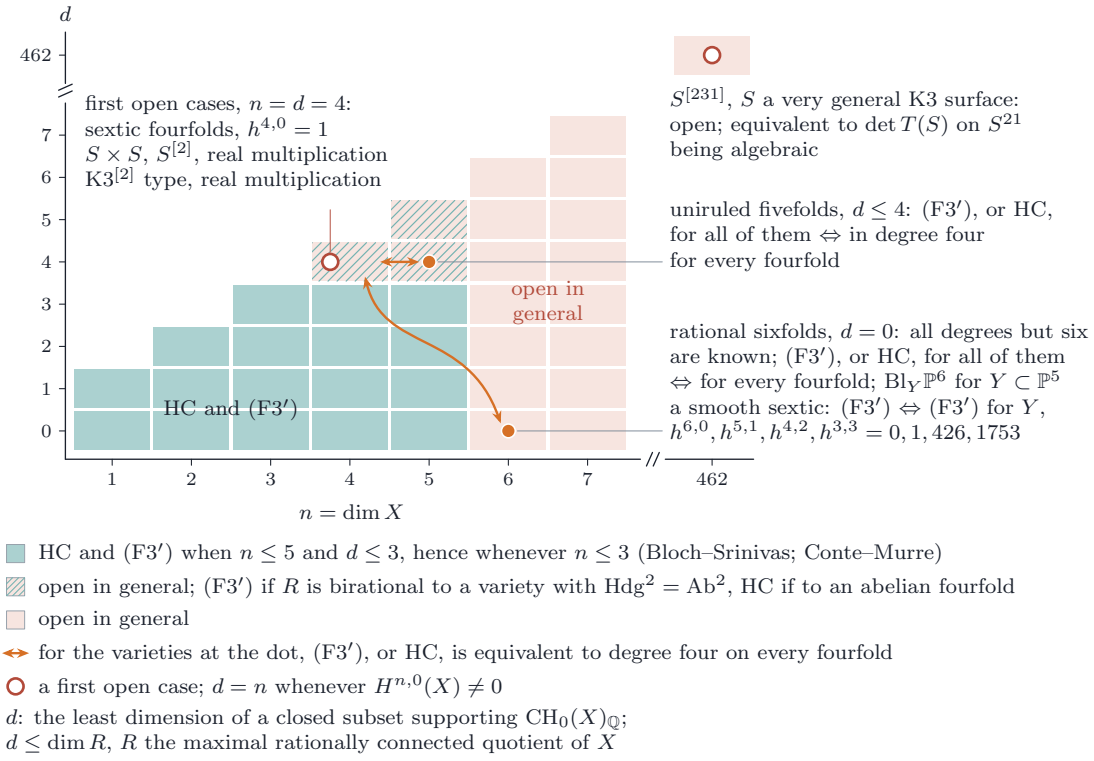


FIGURE 29. What arguments on zero-cycles and on rationally connected fibrations give for (F3′) and for the Hodge conjecture (HC in the figure) on a variety X, over the plane of $n = \dim X$ and the least dimension d of a closed subset supporting $\mathrm{CH}_0(X)_{\mathbb{Q}}$; d is at most the dimension of the maximal rationally connected quotient R of X, and $d = n$ whenever $H^{n,0}(X) \neq 0$ (Remark 6.45). Both hold when $n \le 5$ and $d \le 3$ (Proposition 6.43), hence on every variety of dimension at most three. In the hatched cells (F3′) holds when R is birational to a variety with $\mathrm{Hdg}^2 = \mathrm{Ab}^2$, for instance a member of $\mathcal{A}$, and the Hodge conjecture when R is birational to an abelian fourfold (Proposition 6.44). The double arrows are the equivalences of Remark 6.45: (F3′), and the Hodge conjecture, hold for every uniruled fivefold, and for every rational sixfold, if and only if they hold in degree four for every fourfold; the sixfold $\mathrm{Bl}_Y\mathbb{P}^6$ of a smooth sextic fourfold Y in a hyperplane carries the Hodge numbers listed. The rings are the frontier cases (Remark 6.68): the sextic fourfolds, the first family of smooth hypersurfaces beyond these results (the 29 Delsarte sextics lie in $\mathcal{A}$ by Corollary 6.53); $S \times S$ and $S^{[2]}$ for a K3 surface S with real multiplication, such as S_λ; the fourfolds of $K3^{[2]}$ type with real multiplication, among them $F(Y)$; and, beyond the breaks of the axes, $S^{[231]}$ for a very general K3 surface, which is equivalent to the algebraicity of $\det T(S)$ on S^{21} (Proposition 6.39, Figure 21).

rule is a finite set of premises with one conclusion, and each is a theorem of this paper: Table 2 lists twenty-one of them, and the twenty-second derives the range settled in the literature from the two quoted Weil statements. For a set T of statements write $\mathrm{Cn}(T)$ for the least set that contains T and is closed under the rules, which exists because each rule has a single conclusion and the consequence operator is therefore monotone. Write $\mathbf{T}$ for the set of statements proved or quoted. Call a set S of hypotheses *sufficient* if $\mathbf{HC} \in \mathrm{Cn}(\mathbf{T} \cup S)$, and *minimal* if no proper subset of S is sufficient.

Theorem 6.70 (The closure of what is proved here)**.** *With the notation of Setup 6.69:*

(i) $\mathbf{HC} \notin \mathrm{Cn}(\mathbf{T})$. *Nothing in this paper, combined with the theorems it quotes, implies the Hodge conjecture.*

(ii) *There are exactly five minimal sufficient sets:*

$$\{\mathrm{F3}\}, \quad \{\mathrm{L}, \mathrm{M}\}, \quad \{\mathrm{L}, \mathrm{F3}'\}, \quad \{\mathrm{V}, \mathrm{F3}'\}, \quad \{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\},$$

with (L), (M), (V) *as in Definition 6.14,* (F2), (F3) *as in Corollary 6.71, (F3′) as in Definition 6.34, and* $\mathrm{P2}_{\mathrm{split}}$ *the propagation statement* (P2)*, read in the polarised form of Remark 3.6, for the split families of the CM fields of degree at least four. No minimal set contains* (P2) *for an imaginary quadratic field or for a family of nontrivial discriminant: Proposition 6.13 derives those. The single hypothesis (F3) is sufficient, and it is equivalent to* **HC** *(Proposition 6.33); no other single hypothesis is. The sets* $\{\mathrm{L}, \mathrm{F3}\}$, $\{\mathrm{V}, \mathrm{F3}\}$ *and* $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}\}$ *are sufficient and not minimal. The last of the five is the only one with three elements and the only one whose derivation of* **HC** *uses a statement proved in this paper.*

(iii) *In each of the five, every element is necessary: removing any one leaves* **HC** *outside the closure.*

(iv) *The secant route lies in no minimal sufficient set. Granting both of its demands, a secant object satisfying the weakened criterion in every dimension and an affirmative answer to* [Mar25b, Question 11.4]*, puts* $\mathbf{W}(K, n, \delta_0)$ *in the closure, and with it, by Proposition 6.13,* $\mathbf{W}(K, n, \delta)$ *for every* δ*; it leaves the Weil classes of the CM fields of degree at least four outside.*

(v) *No statement proved in this paper rests, at any depth, on a hypothesis, and the rule set is acyclic.*

(vi) *Every element of a minimal sufficient set other than* $\mathrm{P2}_{\mathrm{split}}$ *is a consequence of* **HC***, and* $\{\mathrm{F3}\}$ *and* $\{\mathrm{L}, \mathrm{M}\}$ *are each equivalent to it; no implication from* **HC** *to* (P2) *is known.*

Proof. (i) and (v). No rule has a proved or quoted statement as its conclusion, so each of them lies in $\mathrm{Cn}(\mathbf{T})$ by itself, on the strength of its proof in this paper or in the literature; this is the first assertion of (v). The twenty-second rule is the only one whose premises all lie in $\mathbf{T}$: every other rule has among its premises a hypothesis or a derived statement other than the range settled in the literature. Hence $\mathrm{Cn}(\mathbf{T})$ is $\mathbf{T}$ together with that range, and it does not contain **HC**, which is (i). For acyclicity, list first the thirty statements that are proved, quoted or hypotheses other than (V); then one secant object, $\mathbf{W}(K, 4, \delta_0)$ and the range settled in the literature; the split and then the other families of the CM fields of degree at least four; the families of trivial and then of nontrivial discriminant of the imaginary quadratic fields; the imaginary quadratic fields, the CM fields of degree at least four and every CM field; and last (V), **AB** and **HC**. The premises of every rule come before its conclusion in this list.

premises	conclusion	by
the conjecture for abelian varieties, (F3)	**HC**	Section 6.5
(F3)	abelian varieties	Proposition 6.33
the conjecture for abelian varieties, (F3′)	**HC**	Proposition 6.35
the Weil classes of every CM field, (F2)	abelian varieties	Section 6.5
K imaginary quadratic, the fields of degree at least four	every CM field	[I, Appendix C]
$\delta = \delta_0$, $\delta \neq \delta_0$	every discriminant	Proposition 6.12
a base point, the reduction, the orbit, (P2) for the family	that family	Theorem 2.3
the same three for a CM field of degree at least four, (P2) for the family	that family	[I, Theorem 17.7]
its split families, its other families	the fields of degree at least four	[I, Proposition 17.3]
$\delta = \delta_0$	$\delta \neq \delta_0$	Proposition 6.13
the split families of the fields of degree at least four	their other families; $\delta = \delta_0$	Proposition 6.13
a secant object in every dimension, Question 11.4	$\delta = \delta_0$	[I, Proposition 15.19]
a Cohen–Macaulay support, smooth or singular	one secant object	Corollary 4.77
one secant object, Question 11.4	$\mathbf{W}(K, 4, \delta_0)$	Corollary 4.57
(L), (M)	**HC**	Proposition 6.16
(L), the deformation theorem for motivated classes	(V)	Proposition 6.17
(V), the accessibility of Hodge classes on abelian varieties	abelian varieties	Theorem 6.18

TABLE 2. The rules of Setup 6.69 whose conclusion is derived, with the theorem that proves each; the rows for (P2) and for the Cohen–Macaulay support each stand for two rules, the row for descent over the fields of degree at least four for two, descent within a field and scalar extension down to the imaginary quadratic fields, the second row is Proposition 6.33, the third is Proposition 6.35(i) for the weaker statement (F3′) of Definition 6.34, the last three rows are the rules of Section 6.6, one of whose conclusions, (V), is itself a hypothesis of the rule set, and the one remaining rule records what the literature supplies on Weil classes. The bounded criterion (P1) is absent: by [I, Theorem 16.17] it is equivalent to its own conclusion, so it is a restatement and not a premise.

(ii) and (iii). The five sets are sufficient: F3 $\Rightarrow$ **AB** (Proposition 6.33) and **AB** $\wedge$ F3 $\Rightarrow$ **HC**; L $\wedge$ M $\Rightarrow$ **HC** (Proposition 6.16); (L) and the deformation theorem give (V) (Proposition 6.17), (V) and accessibility give **AB** (Theorem 6.18), and **AB** $\wedge$ F3′ $\Rightarrow$ **HC** (Proposition 6.35(i)), which settles $\{\mathrm{L}, \mathrm{F3}'\}$ and $\{\mathrm{V}, \mathrm{F3}'\}$; and $\mathrm{P2}_{\mathrm{split}}$ with the CM base point, the reduction and the orbit gives the split families of the CM fields of degree at least four ([I, Theorem 17.7]), descent gives their other families and the families of trivial discriminant of the imaginary quadratic fields (Proposition 6.13), hence the fields of degree at least four ([I, Proposition 17.3]), the nontrivial discriminant (Proposition 6.13), the imaginary quadratic fields (Proposition 6.12) and every CM field ([I, Appendix C]), and with (F2) this is **AB** and with (F3′) it is **HC**.

Let C_1, respectively C_2, be the set of all statements other than **HC**, (F3), (F3′) and (L), respectively (M). Let C_3 be the set of all statements other than **HC**, **AB**, the Weil classes of every CM field, of the fields of degree at least four and of their split families, (F3), (L), (V) and $\mathrm{P2}_{\mathrm{split}}$; and let C_4 be the set of all statements other than **HC**, **AB**, (F3), (L), (V) and (F2). Each C_j contains **T** and not **HC**, and each is closed under the rules: every rule whose conclusion it omits has a premise it omits. For C_1 and C_2 the only omitted conclusion is **HC**, and the three rules with that conclusion have the premises (F3), (F3′), and (L) and (M). For C_3 and C_4 the rules with conclusion **HC** have the premises **AB** or (L); those with conclusion **AB** have the premises (F3), the Weil classes of every CM field together with (F2), and (V); the only rule with conclusion (V) has the premise (L); and for C_3 the rule for every CM field needs the fields of degree at least four, the rule for these needs their split families, and the rule for the split families needs $\mathrm{P2}_{\mathrm{split}}$. Since Cn is monotone, a set of hypotheses contained in some C_j has its closure, together with **T**, inside C_j, and is not sufficient. So a sufficient set meets each of

$$H_1 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{L}\}, \qquad H_2 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{M}\},$$
$$H_3 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{P2}_{\mathrm{split}}\}, \qquad H_4 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{F2}\},$$

the hypotheses omitted from $C_1, \ldots, C_4$. Let S meet all four. If F3 $\in S$, then $S \supseteq \{\mathrm{F3}\}$. If F3 $\notin S$ and L $\in S$, then S meets H_2 in (F3′) or (M). If neither is in S, then S contains (F3′), from H_1, and (V) or both $\mathrm{P2}_{\mathrm{split}}$ and (F2), from H_3 and H_4. So a set of hypotheses is sufficient exactly when it contains one of the five sets, and as none of the five contains another, they are the minimal sufficient sets. The other assertions of (ii) can be read off the list, and the derivations above show that only the last uses a statement proved here; (iii) is the minimality.

(iv) Granting the two demands of the secant route, the rule of [I, Proposition 15.19] gives the families of trivial discriminant, descent those of nontrivial discriminant, and Proposition 6.12 the imaginary quadratic fields. The set formed by **T**, the range settled in the literature, the two demands and these three statements is closed, since every other rule has a premise outside it: a hypothesis not granted, one secant object, or a Weil statement about the fields of degree at least four or about every CM field. Every base point the secant route produces is a member $X \times \widehat{X}$, whose discriminant is trivial whatever β is chosen ([I, Proposition 15.19(ii)]); the deformation argument stays inside the family of trivial discriminant, and it is Proposition 6.13(i), not the deformation, that carries the result to the other discriminants. The route has no base point over a field of degree at least four, so it reaches none of them, and none of its hypotheses lies in one of the five sets.

(vi) (L) follows from **HC** by [I, Theorem 12.2], (M) by Proposition 6.16, (V) by Proposition 6.17, (F3′) by Proposition 6.35(i), and (F2) and (F3) are special cases of **HC**.

The equivalence of {F3} with **HC** is Proposition 6.33, and {L, M} is equivalent to it by Proposition 6.16 and the first two implications. ■

Corollary 6.71 (The three statements, one of which is the conjecture)**.** *The Hodge conjecture follows from the results of this paper together with the following three statements. No rule of Setup 6.69 derives it from (F1) and (F2) together, and it follows from (F3) alone, which is equivalent to it (Proposition 6.33); so the list is not a reduction of the conjecture to three smaller problems. With (F3) replaced by the weaker statement* (F3′) *of Definition 6.34, the conjecture modulo abelian varieties, the three statements (F1), (F2),* (F3′) *still suffice, and no rule of Setup 6.69 derives the conjecture from a proper subset of them. With the rule of Proposition 7.49, which the rule set does not contain, (F2) and* (F3′) *already suffice (Remark 7.50).*

(F1) One propagation statement, for every CM field. (P2)*, in the polarised form of Remark 3.6: at one base point of each Weil family of each CM field some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation has injective semiregularity map. Its first form, with Chern character exactly a multiple of the Weil class, is false for every imaginary quadratic field and every $n \geq 2$ (Theorem 3.5): such a complex would have to deform to non-algebraic Weil tori, whose coherent sheaves have no Chern character in degrees* 1 *to* $2n-1$*. The polarised form closes* $\mathbf{W}(K, n, \delta)$ *for every imaginary quadratic* K*, every* n *and every* δ *at once, by Theorem 2.3, and* $\mathbf{W}(F, n, \delta)$ *for every CM field* F *of higher degree, by* [I, Theorem 17.7]*, because those theorems put* $\Sigma = \mathcal{D}$*, so the Weil class is algebraic at every member of the family and not only at the general one, and because* [I, Theorem 15.27] *and* [I, Theorem 17.5] *supply the base point in every one of those families. By Proposition 6.13 it is enough to have it for the split families of the CM fields of degree at least four, in infinitely many dimensions. Theorem 2.5, Theorem 2.13, Theorem 6.32, Theorem 2.11 and Theorem 2.14 describe the object of the first form and are vacuous by Theorem 3.5; for the polarised form Theorem 3.9 computes the number that* $\dim \mathrm{Ext}^2(E, E)$ *has to reach,* $7n^2 - 2n$ *for a general shape. For one family its conclusion is equivalent to the Lefschetz standard conjecture for one variety (Corollary 6.31). The bounded criterion* (P1) *is not a second form of the same demand: by* [I, Theorem 16.17] *it is equivalent to the conclusion it was to imply, and* [I, Theorem 16.15] *shows that the one cycle it could have been tested on, the transport of the base cycle, does not satisfy it.*

(F2) The classes beyond the Weil lines. *Every Hodge class on an abelian variety that lies outside the subring generated by divisor classes and by Weil classes of quotients is algebraic. Through dimension five there is no such class* [MZ99]*; in dimension eight there are (Proposition 6.74), so this is not a reduction of the conjecture for abelian varieties to Weil classes, which is false as a statement about Hodge rings: it is the conjecture for those classes, and by Proposition 7.49 it is equivalent to the conjecture for every abelian variety. Its smallest case is brought down to one Kuga–Satake class in Theorem 7.4; it holds at every CM point (Theorem 6.29), and on all members of a Mumford family it is equivalent to the Lefschetz standard conjecture for one ninefold (Corollary 6.30). Its classes are rigid along the base curve (Theorem 7.21), and one complex at one CM point with* Ext^2 *of dimension* 119 *would close it (Theorem 7.27), as would a bound on the cycles that represent them at the closed points of the reductions of the curve modulo primes (Theorem 7.43).*

(F3) The varieties that are not abelian. *The Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety. This is the Hodge conjecture itself. The varieties it covers include $A \times \mathbb{P}^1$ for every abelian variety A, and the conjecture for $A \times \mathbb{P}^1$ gives it for A (Proposition 6.33); so (F3) implies the conjecture for abelian varieties, hence (F1) and (F2), and hence* **HC**. *Nor is it a reduction to abelian varieties in the other direction: no realisation of the cohomology of an arbitrary variety inside that of abelian varieties exists (Proposition 6.75), and* [I, Theorem 10.3] *and* [I, Theorem 10.8] *close the two routes through hyperplane sections that would be the natural candidates for one.*

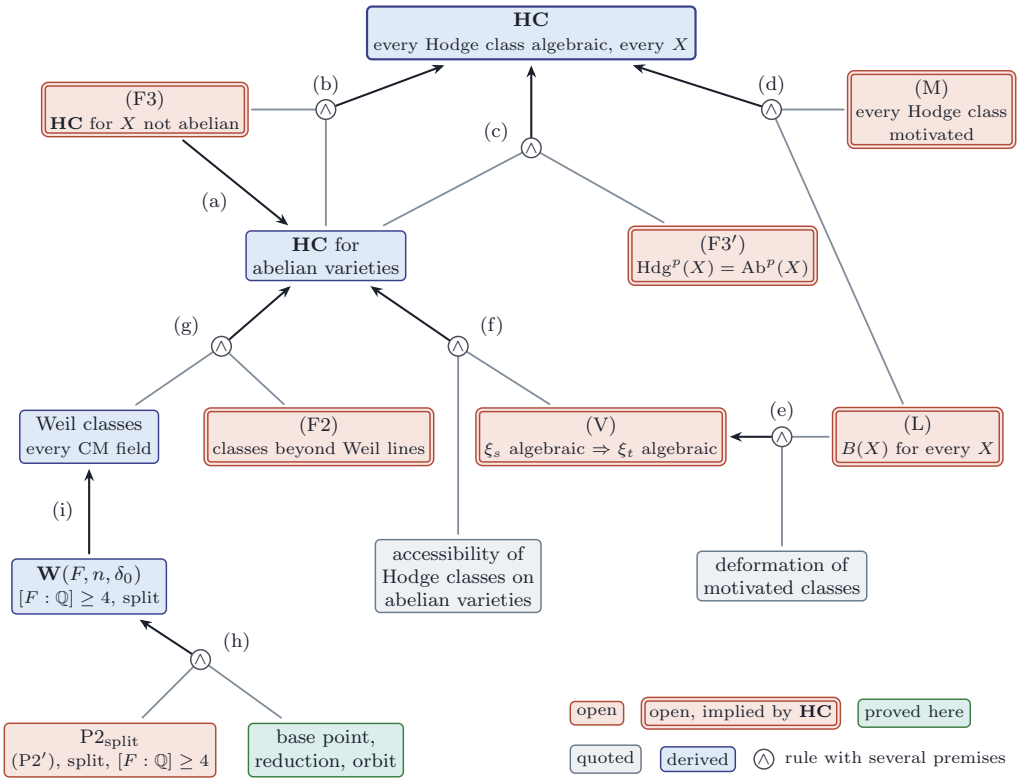
Proof. The three together are sufficient by the derivation of Theorem 6.70, and (F3) alone by Proposition 6.33. That (F1) and (F2) together are not sufficient in the rule set is Theorem 6.70(ii), since every minimal sufficient set contains (F3), (F3′) or (M); the statement about (F1), (F2), (F3′) is Theorem 6.70(ii) and (iii) for the last of the five minimal sets. The Weil classes of the CM fields of degree at least four are not a fourth statement: they are derived from (F1) by [I, Theorem 17.7]. Nor is the bounded criterion a premise, since by [I, Theorem 16.17] it is a restatement of its own conclusion. ■

Remark 6.72 (Necessity relative to a rule set, and the wording of (F2)). A rule set can show that a set of statements suffices; it can show that a set is needed only relative to the implications written into it. A rule set that used (F3) only through (abelian varieties) ∧ (F3) ⇒ **HC**, without the elementary rule (F3) ⇒ (abelian varieties) of Proposition 6.33, would make the three statements of Corollary 6.71 look jointly necessary, although (F3) alone suffices.

The wording of (F2) has the same feature, one degree at a time. Let $R(A) \subseteq H^*(A, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients. Suppose some Hodge class β of degree $2p$ on A lies outside $R(A)$, and let α be any Hodge class of degree $2p$. If $\alpha \notin R(A)$, (F2) makes it algebraic. If $\alpha \in R(A)$, then $\alpha + \beta \notin R(A)$, because $R(A)$ is closed under subtraction, so (F2) makes $\alpha + \beta$ and β algebraic, and with them α. So in every degree in which (F2) says anything it is the whole conjecture for that degree: on the square of a Mumford fourfold it gives every Hodge class of H^4, the six products of divisor classes included. Multiplying by an exceptional class of that square moves every Hodge class of every abelian variety outside the subring, and Proposition 7.49 deduces from this that (F2) is the whole conjecture for abelian varieties. The rule set does not contain that implication; Remark 7.50 says how Theorem 6.70(ii) changes when it is adjoined.

The form of (F3) that is not the conjecture itself asks only for algebraicity modulo classes coming from abelian varieties. It is (F3′) of Definition 6.34; Proposition 6.35 says what is known of it, and Remark 6.36 why it is not closed.

Remark 6.73 (The target, and what it would take). With the rules of Section 6.6, Proposition 6.33 and Proposition 6.35 in place, Theorem 6.70 changes the reading of Corollary 6.71 in three ways. First, the route through this paper is a route only with (F3) replaced by (F3′). As stated, its last statement (F3) is the conjecture (Proposition 6.33), so (F1) and (F2), and everything this paper proves about Weil classes, drop out of every minimal derivation that uses (F3). With (F3′) in its place the route $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$ is minimal, and it is not the shortest: (F1) and (F2), which between them give the conjecture for abelian varieties along this route, are replaced on the routes of the literature by the single statement (V) (Theorem 6.18), or by (L), which implies it (Proposition 6.17). Second, every element of every minimal sufficient set except $\mathrm{P2}_{\mathrm{split}}$ is a consequence of the conjecture (Theorem 6.70(vi)), and two of the sets, $\{\mathrm{F3}\}$ and $\{\mathrm{L}, \mathrm{M}\}$, are each equivalent to it. (P2) is



minimal set	(F3)	(L)	(M)	(V)	(F3′)	(F2)	$\mathrm{P2}_{\mathrm{split}}$	rules used
S_1	●	·	·	·	·	·	·	(a), (b)
S_2	·	●	●	·	·	·	·	(d)
S_3	·	●	·	·	●	·	·	(c), (e), (f)
S_4	·	·	·	●	●	·	·	(c), (f)
S_5	·	·	·	·	●	●	●	(c), (g), (h), (i)
implied by **HC**	yes	yes	yes	yes	yes	yes	not known	

FIGURE 30. The rule set of Setup 6.69 with the rules of Section 6.6, drawn as a hypergraph. A rule with several premises is a junction marked $\wedge$, joined to its premises and with one arrow to its conclusion. The letters name the results that prove the rules: (a) is Proposition 6.33; (b) and (g) are the closure map of Section 6.5; (c) is Proposition 6.35(i); (d) is Proposition 6.16; (e) is Proposition 6.17; (f) is Theorem 6.18; (h) is [I, Theorem 17.7]; and (i) is the chain of Proposition 6.13, Proposition 6.12, [I, Proposition 17.3] and [I, Appendix C]. Clay marks a hypothesis, a double frame a hypothesis that the conjecture implies, grass what this paper proves, slate what it quotes, and indigo a derived statement. The table lists the five minimal sufficient sets of Theorem 6.70(ii), the rules each of them uses, and which of the hypotheses the conjecture implies; only $\mathrm{P2}_{\mathrm{split}}$, which asks for a particular complex, is drawn without that frame. The secant route, which reaches the imaginary quadratic families only, is drawn in Figure 31.

a statement about semiregularity maps, and a proof of the conjecture need not prove it; in its first form, with a Chern character that is exactly a Weil class, it is false (Theorem 3.5), and the route runs through the polarised form of Remark 3.6. Third, every minimal sufficient set contains (F3), (F3′) or (M): the varieties that are not abelian are reached only through the conjecture for them, which is the whole conjecture, through the conjecture modulo abelian varieties (Remark 6.36), or through motivated classes.

So the target at the centre is (L). It lies in two of the five minimal sets, it gives (V) and with it the whole abelian half, the conjecture implies it, and with (M) it is the conjecture. For each X it asks for one explicit Hodge class on $X \times X$ to be algebraic, the class inducing Λ. This paper proves (L), (M), (V) and (F3′) on the classes of varieties named here: Corollary 6.22 adds the total spaces of Mumford families to the known cases of (L), Proposition 6.35 records the known cases of (F3′), and the routes of the literature use no other theorem of this paper.

Two of the three are not reductions. One could hope that (F2) and (F3) are reductions in the ordinary sense, statements to the effect that every Hodge class is carried, by some correspondence, to a class on a Weil type variety or on an abelian variety, so that the whole conjecture would follow from the Weil classes alone. Both hopes are refutable, and we refute them, because a frontier that contains a false statement is not a frontier.

Proposition 6.74 (The Hodge ring is not generated by divisor and Weil classes)**.** *Let X be a complex abelian fourfold whose Hodge group is a $\mathbb{Q}$-form of an almost direct product of three copies of SL_2, acting on $V = H^1(X, \mathbb{Q})$ through the tensor product $V_1 \otimes V_2 \otimes V_3$ of the three standard representations, with $\mathrm{End}^0(X) = \mathbb{Q}$; such fourfolds exist and form families of dimension one, by Mumford* [Mum69]*, and by* [MZ99, 2.5(i)] *they are exactly the fourfolds with $\mathrm{End}^0(X) = \mathbb{Q}$ and Hodge group smaller than Sp_8. Then:*

(i) *the Hodge ring of X is generated by the polarisation:* $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^2(X) = 1$*;*

(ii) $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X \times X) = 3$ *and* $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X \times X) = 8$*, while the products of divisor classes span a subspace of dimension* 6 *in $H^4(X \times X, \mathbb{Q})$;*

(iii) *the two remaining classes lie in $H^4(X \times X, \mathbb{Q})$ and are Weil classes of no CM field acting on $X \times X$, on a quotient of it, or on an abelian subvariety of it.*

In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five [MZ99].

Proof. Hodge classes are the invariants of the Hodge group, so $\mathrm{Hdg}^k(Y) \otimes \mathbb{C}$ is the space of $\mathrm{Hg}(Y)_{\mathbb{C}}$-invariants in $\bigwedge^{2k} H^1(Y, \mathbb{C})$, and $\mathrm{Hg}(X \times X) = \mathrm{Hg}(X)$ acting diagonally on $V \oplus V$. Over $\mathbb{C}$ the group is SL_2^3 and the weights of V under its maximal torus are the eight sign vectors $(\pm 1, \pm 1, \pm 1)$. Write $V = V_1 \otimes W$ with $W = V_2 \otimes V_3$, on which $\mathrm{SL}_2^2 = \mathrm{Spin}(4)$ acts orthogonally. The Cauchy formula $\bigwedge^k (V_1 \otimes W) = \bigoplus_{|\lambda|=k} S_\lambda V_1 \otimes S_{\lambda'} W$ [FH91, §6.1], with $S^2 V_1 = S_{(3,1)} V_1 = V(2)$, $\bigwedge^2 V_1 = S_{(2,2)} V_1 = \mathbf{1}$ and $S^4 V_1 = V(4)$, gives $\bigwedge^2 V = V(2) \otimes \bigwedge^2 W \oplus S^2 W$ and $\bigwedge^4 V = V(4) \otimes \bigwedge^4 W \oplus V(2) \otimes S_{(2,1,1)} W \oplus S_{(2,2)} W$. Under $\mathrm{Spin}(4)$, $\bigwedge^2 W = V(2,0) \oplus V(0,2)$, $S^2 W = V(2,2) \oplus \mathbf{1}$, $\bigwedge^4 W = \mathbf{1}$, $S_{(2,1,1)} W \cong \mathfrak{sl}(W) = \bigwedge^2 W \oplus S^2_0 W$, and $S_{(2,2)} W = V(4,0) \oplus V(0,4) \oplus V(2,2) \oplus \mathbf{1}$, the decomposition of the algebraic curvature tensors in dimension four into the self-dual and anti-self-dual Weyl parts, the traceless Ricci part and the scalar part. Hence

$$\bigwedge^2 V = V(2,2,0) \oplus V(2,0,2) \oplus V(0,2,2) \oplus \mathbf{1},$$

$$\textstyle\bigwedge^4 V = V(2,2,2) \oplus \bigoplus_{\mathrm{cyc}} V(4,0,0) \oplus \bigoplus_{\mathrm{cyc}} V(2,2,0) \oplus \mathbf{1},$$

each with one trivial summand, which is (i). For (ii), $\bigwedge^4(V \oplus V) = \bigoplus_a \bigwedge^a V \otimes \bigwedge^{4-a} V$ and the invariants in $\bigwedge^a V \otimes \bigwedge^{4-a} V$ are $\mathrm{Hom}_{\mathrm{SL}_2^3}(\bigwedge^a V, \bigwedge^{4-a} V)$, since every summand is self-dual; the counts are $1, 1, 4, 1, 1$, total 8, the 4 coming from the four distinct irreducible summands of $\bigwedge^2 V$. Likewise the invariants in $\bigwedge^2(V \oplus V)$ number $1 + 1 + 1 = 3$. The divisor classes of $X \times X$ are the three symplectic forms ψ_{11}, ψ_{12}, ψ_{22} on $V \oplus V$ built from the polarisation ψ of V. Their six pairwise products are linearly independent: they lie in the summands $\bigwedge^a V \otimes \bigwedge^{4-a} V$ with $a = 4, 3, 2, 2, 1, 0$, each is nonzero, and in the summand $a = 2$ the classes $\psi_{11}\psi_{22}$ and ψ_{12}^2 correspond, as maps $\bigwedge^2 V \to \bigwedge^2 V$, to a multiple of the projection onto $\mathbb{C}\psi$ and to a multiple of the identity. The Sp_8-invariants of $\bigwedge^4(V \oplus V)$ number $1 + 1 + 2 + 1 + 1 = 6$ by the same count, since $\bigwedge^1 V = V$, $\bigwedge^2 V = \bigwedge^2_0 V \oplus \mathbf{1}$ and $\bigwedge^3 V = \bigwedge^3_0 V \oplus V$ under Sp_8; so the six products are all the classes a Hodge group equal to Sp_8 would allow. For (iii), the Weil classes of an abelian variety Y with respect to a CM field $F \subseteq \mathrm{End}^0(Y)$ are, by definition, the classes in $\bigwedge^{2k}_F H^1(Y,\mathbb{Q}) \subseteq H^{2k}(Y,\mathbb{Q})$ with $2k = \dim_F H^1(Y,\mathbb{Q})$ ([I, Setup 17.1]). A CM field acting on $X \times X$ lies in $\mathrm{End}^0(X \times X) = M_2(\mathbb{Q})$, so it is imaginary quadratic, $2k = 8$, and its Weil classes lie in H^8, while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of $X \times X$ is isogenous to X, because X is simple, and $\mathrm{End}^0(X) = \mathbb{Q}$ contains no CM field, so X carries no Weil classes at all. This is the statement of [MZ99, 0.1(iii)] that the exceptional classes of $X \times X$ are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [MZ99] for dimension at most five. ∎

Proposition 6.75 (No Hodge-theoretic reduction to abelian varieties)**.** *Let Y be a very general smooth quintic threefold in $\mathbb{P}^4$, and let $V = H^3(Y,\mathbb{Q})$ with its polarised Hodge structure of weight three. Then V is not isomorphic to a subquotient of any Hodge structure of the form $\bigoplus_i T_i$, where each T_i is obtained from the H^1 of finitely many abelian varieties by tensor products, duals and Tate twists. In particular the Hodge structure $H^*(Y,\mathbb{Q})$ is not of abelian type, and no correspondence between Y and an abelian variety, or a product of abelian varieties, induces an injection of V into the cohomology of the latter.*

Proof. Write $\mathrm{MT}(W)$ for the Mumford–Tate group of a rational Hodge structure W and $\mu_W \colon \mathbb{G}_m \to \mathrm{MT}(W)_{\mathbb{C}}$ for its Hodge cocharacter, acting on $W^{p,q}$ by z^p. The adjoint action of μ_W on $\mathrm{Lie}\,\mathrm{MT}(W)_{\mathbb{C}} \subseteq \mathrm{End}(W_{\mathbb{C}})$ has weights among the differences $p - p'$ of two Hodge indices of W.

If W has weight one, these differences lie in $\{-1, 0, 1\}$. If W' is a subquotient of a tensor construction T on W, then $\mathrm{MT}(W')$ is the image of $\mathrm{MT}(W)$ acting on W', and $\mu_{W'}$ is the image of μ_W; so $\mathrm{Lie}\,\mathrm{MT}(W')$ is a quotient of $\mathrm{Lie}\,\mathrm{MT}(W)$ as a representation of $\mathbb{G}_m$ through μ, and its weights are a subset of $\{-1, 0, 1\}$. The same holds with $W = \bigoplus H^1(A_i)$. Hence for every Hodge structure of abelian type the adjoint weights of the Hodge cocharacter lie in $\{-1, 0, 1\}$.

For $V = H^3(Y,\mathbb{Q})$ we show that the adjoint weight 3 occurs. The canonical bundle of a quintic threefold is trivial by adjunction, so $h^{3,0}(Y) = 1$; pick $0 \neq e \in H^{3,0}$. The intersection form ψ on V is symplectic and pairs $H^{3,0}$ with $H^{0,3}$ perfectly, so the rank one endomorphism $x \mapsto \psi(x,e)\,e$ of $V_{\mathbb{C}}$ is nonzero, sends $H^{0,3}$ to $H^{3,0}$ and kills the rest, and lies in $\mathfrak{sp}(V_{\mathbb{C}},\psi)$ because $\psi(\psi(x,e)e,y) + \psi(x,\psi(y,e)e) = \psi(x,e)(\psi(e,y) + \psi(y,e)) = 0$; its adjoint weight is $3 - 0 = 3$. It remains to see that $\mathfrak{sp}(V_{\mathbb{C}},\psi) \subseteq \mathrm{Lie}\,\mathrm{MT}(V)_{\mathbb{C}}$. Let S be the parameter space of smooth quintic threefolds, connected and nonsingular, and $\mathbb{V}$ the polarisable variation of Hodge structure R^3 of the universal family. The connected

algebraic monodromy group G^0 of $\mathbb{V}$ contains the connected monodromy group of any Lefschetz pencil of quintics, that is of hyperplane sections of the fifth Veronese image of $\mathbb{P}^4$; for a Lefschetz pencil of odd fibre dimension the image of the monodromy representation on the ℓ-adic vanishing cohomology $E\otimes\mathbb{Q}_\ell$ is open in $\mathrm{Sp}(E\otimes\mathbb{Q}_\ell,\psi)$ [Del74, Theorem 5.10], so the Zariski closure of the monodromy group over $\mathbb{Q}$ is $\mathrm{Sp}(E,\psi)$; here $E=H^3(Y_t,\mathbb{Q})$, the vanishing cohomology being the orthogonal complement of the image of $H^3(\mathbb{P}^4,\mathbb{Q})=0$. Hence $G^0\supseteq\mathrm{Sp}(V,\psi)$, and $G^0\subseteq\mathrm{Sp}(V,\psi)$ since the monodromy preserves ψ. At a very general point Y the group G^0 is a normal subgroup of the derived group of $\mathrm{MT}(V)$ [And92, Theorem 1], so $\mathrm{Sp}(V,\psi)\subseteq\mathrm{MT}(V)^{\mathrm{der}}\subseteq\mathrm{Sp}(V,\psi)$, and the adjoint weight 3 occurs in $\mathrm{Lie}\,\mathrm{MT}(V)$. So V is not of abelian type, and neither is any Hodge structure containing it as a subquotient, in particular $H^*(Y,\mathbb{Q})$. A correspondence inducing an injection of V into the cohomology of an abelian variety would exhibit V as a sub-Hodge structure of a Tate twist of that cohomology, which is of abelian type. ■

6.10. **Routes from outside algebraic geometry.** By Proposition 7.49 and Proposition 6.35(i), the two statements (F2) and (F3′) are the two halves of the conjecture: the conjecture for abelian varieties, and the conjecture modulo abelian varieties. A route from another domain must therefore either produce algebraic cycles or arrive at one of the hypotheses of Setup 6.69. This subsection records what the routes most often proposed do. One of them, the analytic route through positive currents, can be made exact: the conjecture is the vanishing of an integrality gap in a mass minimisation problem whose real version is solved, and positivity, which is what analysis sees first, cannot distinguish an algebraic class from any other.

Throughout, X is a smooth complex projective variety of dimension n, $h\in H^2(X,\mathbb{Z})$ is the class of an ample line bundle, ω_h is a Kähler form with class h, and for $c\in H^{2p}(X,\mathbb{Q})$ we put

$$L_h(c)=\int_X c\cup\frac{h^{n-p}}{(n-p)!}\in\mathbb{Q}.$$

A real (p,p)-form is *strongly positive* if at every point it is a nonnegative combination of forms $\mathrm{i}\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\mathrm{i}\beta_p\wedge\overline{\beta}_p$ with β_j of type $(1,0)$; the closed positive currents of bidegree (p,p) are the currents nonnegative on such forms of the complementary degree, and every smooth closed strongly positive form is one [Dem12, Chapter III].

Proposition 6.76 (Positivity is no test)**.** *For every class $\alpha\in H^{p,p}(X,\mathbb{R})$, $0\le p\le n$, there is an integer N such that $\alpha+Nh^p$ is the class of a smooth closed strongly positive (p,p)-form. Consequently every real (p,p)-class becomes the class of a closed positive current after the addition of a multiple of h^p, and, since h^p is algebraic, a Hodge class is algebraic if and only if the positive class $\alpha+Nh^p$ is: no property of a class implied by positivity, or by its being a limit of classes of effective cycles, separates the algebraic Hodge classes from the others.*

Proof. Let $V=T_xX$ with the hermitian form $\omega_h(x)$, and let $\mathrm{SP}_p\subseteq\Lambda^{p,p}_{\mathbb{R}}V^\vee$ be the closed convex cone generated by the forms $\mathrm{i}^p\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\beta_p\wedge\overline{\beta}_p$. These forms span $\Lambda^{p,p}_{\mathbb{R}}V^\vee$, so SP_p has nonempty interior, and $\omega_h(x)^p$ lies in that interior: in a unitary basis $dz_1,\dots,dz_n$ of $V^\vee$,

$$\omega_h^p=p!\sum_{|J|=p}\Big(\frac{\mathrm{i}}{2}\Big)^p dz_J\wedge d\overline{z}_J,$$

and a generator $D=\mathrm{i}^p\beta_1\wedge\overline{\beta}_1\wedge\cdots\wedge\beta_p\wedge\overline{\beta}_p$ equals $|\det B|^2\,2^p(\mathrm{i}/2)^p dz_1\wedge d\overline{z}_1\wedge\cdots\wedge dz_p\wedge d\overline{z}_p$ after a unitary change of coordinates taking the span of the β_j to that of $dz_1,\dots,dz_p$,

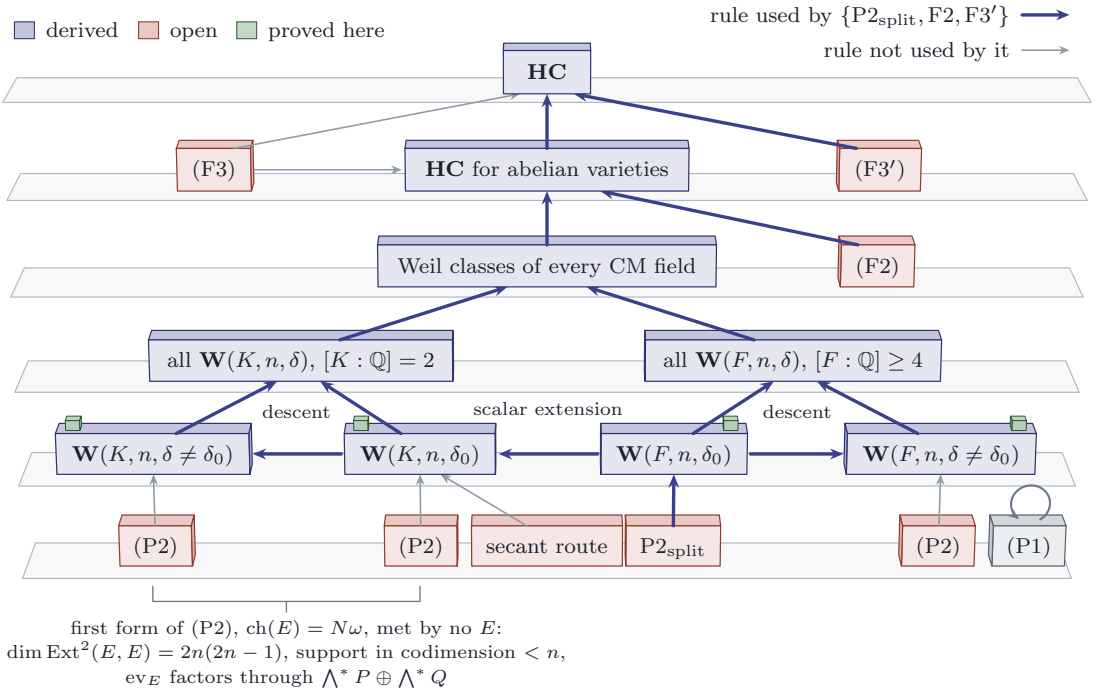


FIGURE 31. The route of this paper through the rule set of Theorem 6.70, drawn as a stack of plates, one per level of the derivation; the other routes are drawn in Figure 30. Heavy arrows are the rules used by the minimal set $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$, light arrows the rules it does not use. The bottom plate carries the propagation statement, read in the polarised form of Remark 3.6, once for each of the four kinds of Weil family, with the secant route beside it. The small grass blocks on the next plate are the base points of [I, Theorem 15.27] and [I, Theorem 17.5], one in every family. The three arrows inside that plate are Proposition 6.13: descent from the trivial discriminant to the others, and scalar extension from the fields of degree at least four to the quadratic fields. With them (P2) is needed only for the split families of the fields of higher degree, and the other three (P2) blocks are not used. (F2) is the conjecture for the Hodge classes that no Weil line reaches, and (F3) is the whole conjecture (Proposition 6.33), so the route is minimal only with (F3) replaced by (F3′) of Definition 6.34. The bracket under the two quadratic (P2) blocks records the first form of the criterion, in which the Chern character is exactly a Weil class, as Theorem 2.11, Theorem 2.13 and Corollary 2.9 describe it; by Theorem 3.5 that form is met by no complex. The secant route reaches the trivial discriminant, since the member it is built on is split, and through descent the rest of the quadratic case; it has no edge to the fields of higher degree. The bounded criterion (P1) stands apart, closed on itself: it is equivalent to its own conclusion.

with B the matrix of the β_j in that basis; so $\omega_h^p - \delta D \in \mathrm{SP}_p$ for $0 < \delta \leq p!/(2^p|\det B|^2)$. As every real (p,p)-form is a finite real combination of generators, $\omega_h^p + \varepsilon\gamma \in \mathrm{SP}_p$ for every γ and all small ε, which is interiority. The cone, the form $\omega_h(x)^p$ and the norm defined by $\omega_h(x)$ are carried into each other by unitary frames, so the radius $r > 0$ of a ball around $\omega_h(x)^p$ contained in SP_p does not depend on x. Represent α by a smooth closed real (p,p)-form a, for instance its harmonic representative, and let $N \geq \max_X |a|/r$. Then $a + N\omega_h^p = N(\omega_h^p + a/N)$ is strongly positive at every point, closed, and represents $\alpha + Nh^p$. The remaining assertions follow, the last because a class of the form $\alpha + Nh^p$ is positive for every α. ∎

The proposition says what positivity cannot do. Demailly studied the cone of closed positive currents against the conjecture and constructed extremal closed strongly positive currents that are not integration currents on analytic sets [Dem82], so extremality does not single out the algebraic cycles either. What positivity can do is turn the conjecture into a variational statement, and the statement is exact. It is the Kähler case of a formulation that Harvey and Lawson give for every calibrated manifold [HL09, Section 6, Proposition 6.13 and Remarks 6.9 and 6.11]; we state it for the classes of this paper, with the effective representative that Lemma 6.77 supplies, and then apply it to the Mumford square. For a closed integral current T of dimension $2(n-p)$ on X, Wirtinger's inequality [Fed69, 1.8.2], [HK74], [HL82, II.3] gives

$$\mathbf{M}(T) \;\geq\; T\Big(\frac{\omega_h^{n-p}}{(n-p)!}\Big) \;=\; L_h(c), \tag{35}$$

where $\mathbf{M}$ is the mass and $c \in H^{2p}(X,\mathbb{Z})$ is the Poincaré dual of the homology class of T, with equality if and only if the approximate tangent plane of T is a complex subspace, positively oriented, at $\|T\|$-almost every point. The right side depends on c alone.

Lemma 6.77 (Complete intersection classes are big)**.** *For every algebraic class $\beta \in H^{2p}(X,\mathbb{Q})$ there is an integer M such that $\beta + Mh^p$ is a positive rational combination of classes of irreducible subvarieties of codimension p; so some positive integer multiple of $\beta + Mh^p$ is the class of an effective algebraic cycle.*

Proof. Replacing h by a multiple, which changes M, we may assume h very ample. Let $Z \subseteq X$ be irreducible of codimension p, and let d be such that the ideal sheaf $I_Z(d)$ is globally generated. The base locus of the linear system of divisors of degree d containing Z is exactly Z, so by Bertini p general members $D_1, \ldots, D_p$ of it meet properly away from Z. Every component of $D_1 \cap \cdots \cap D_p$ has codimension at most p; a component not contained in Z has codimension exactly p, because the D_i meet properly off Z, and a component contained in Z has codimension at least p and so equals Z. The intersection is therefore the union of Z and a closed set W' of pure codimension p. Hence $d^p h^p = [D_1]\cdots[D_p] = m[Z] + [W']$ with $m \geq 1$ and W' effective [Ful98, Chapter 7], and $h^p - \varepsilon[Z]$ is an effective rational class for $0 < \varepsilon \leq m/d^p$. Write $\beta = \sum_i n_i[Z_i]$ with $n_i \in \mathbb{Q}$ and choose ε_i for each Z_i with $n_i < 0$; then for $M \geq \sum_{n_i<0} |n_i|/\varepsilon_i$ the class $\beta + Mh^p$ is $\sum_{n_i>0} n_i[Z_i] + \sum_{n_i<0}(|n_i|/\varepsilon_i)(h^p - \varepsilon_i[Z_i]) + (M - \sum_{n_i<0}|n_i|/\varepsilon_i)h^p$, a positive rational combination of effective classes, h^p itself being the class of a complete intersection. ∎

Theorem 6.78 (The conjecture as a mass gap)**.** *Let $\alpha \in \mathrm{Hdg}^p(X)$ be a rational Hodge class.*

(i) *Let $N \geq 1$ be an integer with $N\alpha$ integral, let M be an integer for which $\alpha + Mh^p$ is the class of a smooth closed strongly positive form, as every large M is by Proposition 6.76,*

and put $c = N(\alpha + Mh^p)$. The minimum of the mass over closed real normal currents of dimension $2(n-p)$ in the class dual to c is $L_h(c)$, and it is attained by a smooth strongly positive form. The minimum over closed integral currents in the same class exists and is at least $L_h(c)$.

(ii) *The class α is algebraic if and only if there are integers $N \geq 1$ and $M \geq 0$, with $N\alpha$ integral, for which the minimum of the mass over closed integral currents in the class dual to $N(\alpha + Mh^p)$ equals $L_h(N(\alpha + Mh^p))$; and then every mass minimising integral current in that class is the integration current of an effective algebraic cycle.*

(iii) *Write $\|c\|_{\mathbb{Z}}$ for the minimum in* (i) *over integral currents. By a theorem of Federer, recorded in this form by Harvey and Lawson* [HL09, Section 6], [Fed69, Fed74], *$\lim_{k\to\infty} \|kc\|_{\mathbb{Z}}/k = L_h(c)$: the integrality gap closes asymptotically for every class. The class α is algebraic if and only if, for every sufficiently large M, the gap is closed at some finite k, that is, $\|kc\|_{\mathbb{Z}} = kL_h(c)$ for some $k \geq 1$.*

In particular the conjecture for X is the statement that for every rational Hodge class the integrality gap of this problem vanishes for some N and M; (F2) *is that statement for abelian varieties. Teh and Yang give a different sufficient condition in the same setting, through Lipschitz chains that are closed for d and d^c with positive (k,k)-part* [TY21, Theorem 4.2].

Proof. (i) The form $\omega_h^{n-p}/(n-p)!$ has comass one and calibrates exactly the complex $2(n-p)$-planes, which is Wirtinger's inequality [Fed69, 1.8.2]; so every closed real normal current S in the class has $\mathbf{M}(S) \geq S(\omega_h^{n-p}/(n-p)!) = L_h(c)$. By Proposition 6.76 the class is represented by a smooth closed strongly positive form a, and for a strongly positive (p,p)-form the mass of the current it defines is $\int_X a \wedge \omega_h^{n-p}/(n-p)!$, because a strongly positive $2(n-p)$-vector is a nonnegative combination of complex unit planes, each of mass one and calibrated. So the minimum over real currents is $L_h(c)$ and a attains it. Closed normal currents represent every real homology class and closed integral currents every integral one [FF60, 5.11], and by the compactness theorem of Federer and Fleming the mass attains its minimum on the closed integral currents in a given class [FF60, 9.6]; the lower bound is (35).

(ii) If α is algebraic, Lemma 6.77 gives M and N with $N(\alpha + Mh^p) = \sum_j n_j[V_j]$, $n_j \in \mathbb{Z}_{>0}$, V_j irreducible of codimension p. The integration current $T = \sum_j n_j[V_j]$ is a closed locally integral current [Kin71, Theorem 3.1.1], its tangent planes are complex and positively oriented, so equality holds in (35) and $\mathbf{M}(T) = L_h(c)$ with $c = N(\alpha + Mh^p)$. Conversely let T be a closed integral current in the class dual to c with $\mathbf{M}(T) = L_h(c)$, for instance a mass minimiser when the minimum is $L_h(c)$. Equality in (35) makes T a closed locally rectifiable current of bidimension $(n-p, n-p)$ which is positive, and by King's theorem [Kin71, Theorem 5.2.1] such a current is a holomorphic cycle $\sum_j n_j[V_j]$ with n_j nonzero integers and V_j irreducible analytic subvarieties of dimension $n-p$; positivity makes every n_j positive. (Harvey and Shiffman characterised the holomorphic chains among closed rectifiable currents of bidimension $(n-p, n-p)$ without positivity, under a hypothesis on the support that Alexander removed [HS74, Ale97].) By Chow's theorem [GH78, p. 167] the V_j are algebraic, so c is the class of an effective algebraic cycle and α is algebraic. This is the argument of [HL09, Proposition 6.13].

(iii) The limit is Federer's theorem on the real and integral masses of a class, as stated in the final note of [HL09, Section 6]; with (i) the limit is $L_h(c)$, since the real minimum is attained by a smooth form. If the gap is closed at k, then (ii), applied with kN in place of N, makes α algebraic. Conversely, if α is algebraic, Lemma 6.77 gives M_0 such that

$\alpha + M_0 h^p$ is a positive rational combination of classes of subvarieties; for $M \geq M_0$ so is $\alpha + M h^p$, since h^p is the class of a complete intersection, and some multiple kc is the class of an effective cycle, whose integration current attains the bound. For a fixed small M the equivalence is not asserted, since it would need an effective multiple of every algebraic class represented by a strongly positive form, in codimension at least two. The last sentence of the theorem is the definition of the conjecture for X, and Proposition 7.49 for (F2). ■

Example 6.79 (The Mumford square)**.** Let $Y = X \times X$ be the square of a Mumford fourfold at a very general point of its curve, with the product principal polarisation θ_Y, and let ω be one of the two exceptional classes of Proposition 6.74, normalised in $H^2(X) \otimes H^2(X)$ with $a_0 = 0$ in the notation of Setup 7.18, so that it corresponds to an endomorphism of $H^2(X)$ killing θ with image in the primitive part. Then $L_{\theta_Y}(\omega) = 0$: the pairing of ω with $\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$ sees only the term $20\,\theta^3 \otimes \theta^3$, and $\int_X x \cup \theta^3 = 0$ for every primitive $x \in H^2(X)$. Also $L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$, and $4\theta_Y^2$ is the class of the complete intersection $D \cap D'$ of two general members of $|2\theta_Y|$, a base point free system, whose integration current has mass 224. By Theorem 6.78 and Corollary 7.45, the Hodge conjecture for every member of the Mumford family, and with it $B(W \times_C W)$, is equivalent to the following: for some integers $N \geq 1$ and $M \geq 1$ with $N\omega$ integral, the class dual to $N(M\theta_Y^2 + \omega)$ contains an integral current of mass exactly $56NM$; and any such current is an effective algebraic cycle representing the class. For M divisible by four the class $NM\theta_Y^2$ contains one, $NM/4$ copies of $D \cap D'$, of mass $56NM$; so the question is whether the perturbation by $N\omega$, which changes the Wirtinger bound by nothing, can be absorbed without mass. By Theorem 6.29 the answer is yes at every CM point of the curve, with integers N and M depending on the point; and if the same N and M served at infinitely many CM points the answer would be yes everywhere, by Corollary 7.28, because the components of an effective cycle of degree $56NM$ have degree at most $56NM$ and the Hilbert polynomials of subvarieties of bounded degree form a finite set.

Figure 32 draws the theorem and the example.

6.11. **The routes at a glance.**

Table 3: The routes considered, and the boundary of each result.

Route	**Result**	**Beyond the result**
Ambient object, restricted to $A^\vee$, transported by $\Phi_{\mathcal{P}}$	closed by [I, Theorem 7.3]	ambient varieties with exceptional classes
Any K-equivariant construction with the norm character	closed by [I, Theorem 5.6]	constructions that are not eigenclasses
Secant dimension count	no obstruction, [I, Corollary 8.4]	none
Semiregularity dimension count	no obstruction, [I, Corollary 9.3]	none

continued on the next page

Table 3, continued.

Route	Result	Beyond the result
Semiregularity of an object whose Chern character sits on the Weil line	closed by [I, Theorem 9.7] and [I, Corollary 9.18] for direct sums, and for every object by Theorem 3.5, for every CM field by Theorem 5.4	objects of positive rank; the weaker hypothesis of [Mar25b, Question 11.4]
The polarised criterion at a quartic CM field, $n = 2$	computed for every shape, Theorem 5.14: a complex meeting it has $\dim \mathrm{Ext}^2(E, E) \geq 88$, Corollary 5.15	existence of such a complex
Induction through hyperplane sections	closed by [I, Theorem 10.3], [I, Theorem 10.8]	correspondences, degenerations, coniveau
Kuga–Satake, from a weight two structure of K3 type	not obstructed, [I, Proposition 11.9]	available only at $n = 1$, [I, Corollary 11.12]
Divisors, on a member with quaternionic multiplication	succeeds, [I, Theorem 15.25]	fails at full Hodge group, [I, Proposition 15.18]
Propagation off the base locus	reduced to one bound or one rank, [I, Theorem 16.9], Theorem 2.3	the bound or the rank, Remark 3.7
Objects with effective Chern character at a base point	closed by Theorem 4.5	objects indecomposable in D^b, Remark 4.8
The weakened criterion for a secant object of positive rank	reduced to two conditions on the factor, Theorem 4.56, the first automatic for Markman's construction, Corollary 4.57; the identity fails for Markman's candidate, Theorem 4.61, and at every transversal double point, Corollary 4.63	objects whose support has no such point, Remark 4.64

continued on the next page

Table 3, continued.

Route	Result	Beyond the result
A secant object $\mathcal{I}_S(b\Theta)$ with smooth support	invariants forced and $d \le b^2$, so $d \le 9$ at $b = 3$, Theorem 4.67, Corollary 4.68	existence for $d \in \{1, 3, 5, 7\}$ at $b = 3$, one vanishing in $H^1(S, N_{S/X})$ and first-order extension, Remark 4.69
The local part of the weakened criterion	it is a condition on depth: no obstruction at a Cohen–Macaulay point, Theorem 4.75, Corollary 4.77	Cohen–Macaulay supports, where the identity becomes one global vanishing beside first-order extension
A support that is a complete intersection	closed for every b and every d, Theorem 4.81	supports of higher Hilbert–Burch rank, Proposition 4.86
A support with a Hilbert–Burch resolution by line bundles or by simple semi-homogeneous bundles	closed for every rank, twist and d, Theorem 4.89, Theorem 4.92	resolutions over bundles that are not projectively flat
The secant route, granted every demand it makes	reaches $\delta = \delta_0$, and every δ by Proposition 6.13, Theorem 6.70(iv)	the CM fields of degree at least four, which give the quadratic ones by Proposition 6.13(ii)
Propagation from a base point	closes every imaginary quadratic family at once, [I, Theorem 16.9], Theorem 2.3	(P2) for the family, Corollary 6.71
The bounded form (P1), read on the transported cycle	closed: the transport spreads each component by c^{2n}/m_i and the data is unbounded whenever a component has finite stabiliser, [I, Lemma 16.14], [I, Theorem 16.15]	nothing; the other reading is closed too
The bounded form (P1), read on the minimal representative	closed: equivalent to the conjecture for the family, [I, Theorem 16.17]	nothing; it is a restatement, so it is not a route

continued on the next page

Table 3, continued.

Route	Result	Beyond the result
The pure form of (P2), for a given representative	forces the evaluation map to vanish on the $4n^2$-dimensional annihilator, and excludes every representative that is a sheaf on a smooth germ of codimension two of its support, Theorem 2.5, Corollary 2.6	nothing: no representative exists, Theorem 3.5
The same form, tested by a number	$\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$ at one base point would suffice, Theorem 2.11, and no representative attains it, Theorem 3.5	nothing
The same form, for a representative in codimension n	closed at every member of every family: the support must have a component of codimension less than n, and at a general member every component of codimension at least two has multiplicity zero, Lemma 2.12, Theorem 2.13	nothing: no representative exists, Theorem 3.5
The Hochschild action on such a representative	the annihilator is the mixed part of a splitting of HH^1 and the action factors through a ring of dimension $2^{2n+1}-1$ in one of dimension 2^{4n}, Theorem 2.8, Corollary 2.9	nothing: no such complex exists, Theorem 3.5
The shape of such a representative	an indecomposable summand carries the class; at $n=2$, and at $n=3$ granting the compatibility of Corollary 2.9(iii), its endomorphism algebra has dimension at least three, so no simple object serves, Theorem 2.14	nothing: no object of any shape meets the pure form, Theorem 3.5

continued on the next page

Table 3, continued.

Route	Result	Beyond the result
The pure form of (P2), Chern character exactly a Weil class	false for every imaginary quadratic field and every $n \geq 2$: a semiregular such complex would deform to very general non-algebraic Weil tori, whose coherent sheaves have no Chern character below the top degree, Proposition 3.4, Theorem 3.5	the polarised form (P2′), whose Chern character may carry powers of the polarisation, Remark 3.6
The reduction to Weil classes, and to abelian varieties	false as a statement about Hodge rings in dimension eight, Proposition 6.74; impossible Hodge-theoretically, Proposition 6.75	the algebraicity of the classes beyond the Weil lines, which is (F2), and (F3), which is the whole conjecture, Proposition 6.33
The two classes on the self-product of a Mumford fourfold	a real multiplication by a cubic field on the primitive part of H^2, algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is, Proposition 7.2, Theorem 7.4	that Kuga–Satake class, Remark 7.5
Other routes to the same two classes	degeneration and larger families closed off by rigidity; semiregularity reduced to $\dim \mathrm{Ext}^2 = 119$ at one CM point; specialisation reduced to a degree bound, Theorem 7.21, Theorem 7.27, Corollary 7.28	an object with $\dim \mathrm{Ext}^2 = 119$, or the degree bound, Remark 7.32

continued on the next page

Table 3, continued.

Route	Result	Beyond the result
The cycles at a CM point, for the same two classes	the fourfold is of Weil type for $k_c \subset \mathrm{End}^0(X_c)$ with the two exotic classes as its Weil line; pull-backs of that line along homomorphisms with components in k_c span, with divisor products, 110 of the 132 Hodge classes of the square and no exceptional class, and an exceptional class is reached exactly by components generating the endomorphism algebra over k_c, Proposition 7.29	a degree bound for cycles using the real multiplication, at infinitely many CM points, Remark 7.31
Objects built from line bundles, for the same two classes	closed at every point of the curve: the Chern characters stay invariant under the Lefschetz group, Theorem 7.36	steps that break that symmetry, Remark 7.37
Twistor deformation of the square, for the same two classes	closed: the component of the Hodge locus among all complex tori is the Mumford curve, which contains no twistor line, Proposition 7.24	the Kuga–Satake class, which moves in a seven-dimensional family
Five further routes, for the same two classes	one statement: three of them are equivalent to the target and two imply it, so any one of them suffices, Corollary 7.45	the target itself, which is only the first case of (F2), Remark 7.46
Pull-back from a hyperkähler manifold, for the same two classes	the square is holomorphic symplectic and the classes are the twisted dual forms of the part of H^2 its symplectic class generates; a symplectic rational map to some $S_\mu^{[n]}$, $n \geq 5$, suffices, and no K3 surface or hyperkähler eightfold serves, Proposition 7.25	such a map, Remark 7.26

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Table 3, continued.

Route	**Result**	**Beyond the result**
Reduction modulo p, for the same two classes	represented by cycles at every closed point of every reduction; the target is equivalent to a bound on their Hilbert data on a Zariski dense set, Proposition 7.42, Theorem 7.43	the bound, Remark 7.44
The Weil classes of every CM field	the same propagation problem, with a CM base point in every family, [I, Theorem 17.5], [I, Theorem 17.7]	(P2) for the split families
Routes from outside algebraic geometry	positivity of currents sees nothing; the conjecture is the vanishing of the integrality gap of a mass minimisation whose real version is solved, Proposition 6.76, Theorem 6.78	a calibrated integral current in one class, Example 6.79
The closure, as a whole	five minimal sets of hypotheses: $\{\mathrm{F3}\}$ and $\{\mathrm{L}, \mathrm{M}\}$, each equivalent to the conjecture, and three through the conjecture modulo abelian varieties, $\{\mathrm{L}, \mathrm{F3}'\}$, $\{\mathrm{V}, \mathrm{F3}'\}$ and this route, $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$, Theorem 6.70, Proposition 6.33, Corollary 6.71, Remark 6.73	the Lefschetz standard conjecture, (M), (V), (F2), (F3′)

continued on the next page

Table 3, continued.

Route	**Result**	**Beyond the result**
Convolutions of the Weil pieces on E_0^{2n}	at $n = 3$ every convolution has $\dim \mathrm{Ext}^2 \geq 69 > 57$, Theorem 4.35; at $n = 4$, with the pieces at two shifts, $\dim \mathrm{Ext}^2 \geq 40960 > 104$, Theorem 4.37, at three consecutive shifts $\dim \mathrm{Ext}^2 \geq 22528$, Theorem 4.38, with the shifts within six consecutive values $\dim \mathrm{Ext}^2 \geq 640$, Corollary 4.48, and with each parity at one shift, at any odd distance, $\dim \mathrm{Ext}^2 \geq 1792$, Proposition 4.46	the other arrangements of the shifts at $n = 4$, Remark 4.50
Diagonal complete intersections of Vandermonde type	quotients of powers of a generalised Fermat curve, so in $\mathcal{A}$; the conjecture holds when it holds on the abelian varieties $B_{[a]}$, and holds for two diagonal hypersurfaces of degree 3, 4, 6 in $\mathbb{P}^6$, Theorem 6.55, Corollary 6.56; it holds on the very general member for $d = 2$ in every dimension, for $d = 3, 4$ up to dimension seven and for $d = 6$ up to dimension five, Theorem 6.66	$B_{[a]}$ of dimension above five, the Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, and the very general member for $d = 3, 4, 6$ from dimension eight on and for $d = 6$ in dimension six, Remark 6.67

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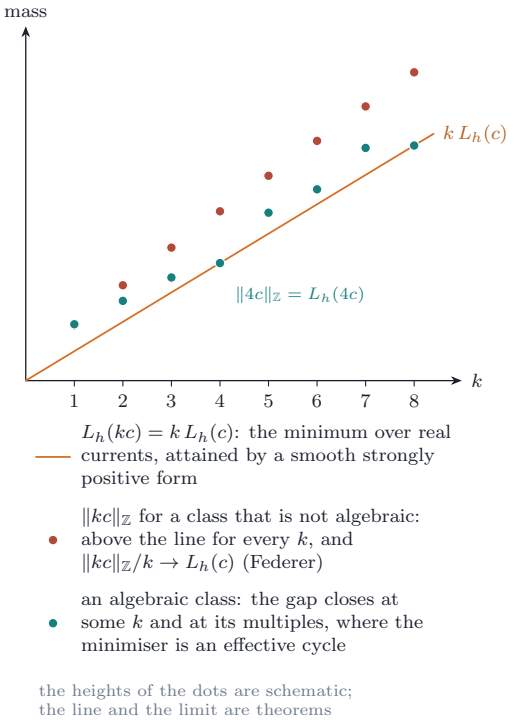
Table 3, continued.

Route	Result	Beyond the result
The conjecture modulo abelian varieties, (F3′)	holds on the class $\mathcal{A}$ of varieties whose cohomology is reached from abelian varieties by algebraic correspondences (curves, abelian varieties, products, surjective images), in degrees 0, 2, $2n-2$, $2n$, and in dimension at most three, Proposition 6.35; on the Delsarte hypersurfaces and the diagonal complete intersections of Vandermonde type, Corollary 6.53, Theorem 6.55; on the fourfolds and fivefolds of Proposition 6.43, Proposition 6.44 and Proposition 6.46; through the Hodge conjecture on the varieties of Proposition 6.39 and Proposition 6.40	degree four on fourfolds outside $\mathcal{A}$ not covered by these results, and beyond, for instance squares of K3 surfaces with real multiplication, such as S_λ, Remark 6.36, Remark 6.68

7. Hodge classes beyond the Weil lines

7.1. **The first classes beyond the Weil lines.** Statement (F2) is the Hodge conjecture for the Hodge classes on abelian varieties that no Weil line reaches, and Proposition 6.74 exhibits the first of them: two classes in $H^4(X \times X, \mathbb{Q})$ for an abelian fourfold X of Mumford type. None of the constructions of the preceding sections acts on them. In this subsection we identify them as a real multiplication on the primitive part of $H^2(X)$, and we prove that they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to X is algebraic. We do not prove that the Kuga–Satake class is algebraic, and we know of no proof. The reduction changes the kind of question: two classes on an eightfold are traded for one correspondence of a well-studied type, between a surface and its Kuga–Satake variety. The cubic field of the real multiplication enters through the Clifford relations, which force the blocks of the Kuga–Satake embedding to span it.

Setup 7.1 (Mumford type)**.** Let X be as in Proposition 6.74, with $V = H^1(X, \mathbb{Q})$, polarisation ψ and polarisation class $\theta \in H^2(X, \mathbb{Q})$, and let $G = \mathrm{Hg}(X)$. Over $\bar{\mathbb{Q}}$ the Hodge cocharacter is nontrivial on exactly one of the three factors of G: on V_i it has weights $\pm a_i/2$ with $a_i \geq 0$, and V has only the two weights $\pm\frac{1}{2}$, so exactly one a_i is nonzero. Since $\mathrm{End}^0(X) = \mathbb{Q}$, it follows that G is $\mathbb{Q}$-simple: the Lie algebra of a $\mathbb{Q}$-factor not containing



The Mumford square $Y = X \times X$, $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$

$\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$

k	0	1	2	3	4	5	6
$\binom{6}{k}$	1	6	15	20	15	6	1
$\int_Y \pi_0\, \theta^k \otimes \theta^{6-k}$	0	0	0	144	0	0	0
$\int_Y \pi_{ij}\, \theta^k \otimes \theta^{6-k}$	0	0	0	0	0	0	0

$\int_Y \pi_0\, \theta_Y^6 = 20 \cdot 144 = 2880$, so
$L_{\theta_Y}(\omega_a) = 2880\, a_0/6! = 4a_0$

$L_{\theta_Y}(\omega) = 0$ for an exceptional class with $a_0 = 0$

$L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$; $4\theta_Y^2 = [D \cap D']$ for $D, D' \in |2\theta_Y|$, of mass 224

open: an integral current of mass $56NM$ in the class dual to $N(M\theta_Y^2 + \omega)$; known at every CM point with N, M depending on the point; the same N, M at infinitely many CM points would give every member

FIGURE 32. Theorem 6.78 and Example 6.79. Left: the mass of a current in the class dual to kc against k. The line $k\,L_h(c)$ is the minimum over closed real normal currents, attained by a smooth strongly positive form; the minima over closed integral currents lie on or above it, with $\|kc\|_{\mathbb{Z}}/k \to L_h(c)$ by Federer's theorem, and for M large the class is algebraic exactly when one of them lies on the line, where the minimiser is an effective algebraic cycle. The heights of the dots are schematic. Right: the numbers of the Mumford square, in the split model. Of the seven terms $\binom{6}{k}\theta^k \otimes \theta^{6-k}$ of θ_Y^6 only $k = 3$ pairs with $\pi_0 = \frac{1}{4}\theta \otimes \theta$, giving $20 \cdot 144 = 2880$, and none pairs with the primitive summands; so $L_{\theta_Y}(\omega_a) = 4a_0$, the Wirtinger bound of an exceptional class with $a_0 = 0$ is zero, and the perturbation of $NM\theta_Y^2$ by $N\omega$ changes the bound $56NM$ by nothing.

that geometric factor would consist of Hodge classes in $\mathrm{End}(V)$, that is, of endomorphisms of X. So G is the image in $\mathrm{GL}(V)$ of $\mathrm{Res}_{F/\mathbb{Q}} D^1$, where F is a totally real cubic field, D is a quaternion algebra over F split at one real place σ_3 and definite at the other two, σ_1 and σ_2, with $\mathrm{Cor}_{F/\mathbb{Q}} D \cong M_8(\mathbb{Q})$, and D^1 is the group of elements of reduced norm one [Mum69, Gal00]; the kernel is the central subgroup of triples of signs with product one. Over $\bar{\mathbb{Q}}$, $V = V_1 \otimes V_2 \otimes V_3$, where V_i is the standard representation of $D \otimes_{F,\sigma_i} \bar{\mathbb{Q}} = M_2(\bar{\mathbb{Q}})$; the Galois group permutes the three factors as it permutes the three embeddings, and $H^{1,0}(X) = V_1 \otimes V_2 \otimes V_3^{1,0}$. Put

$$T = \mathrm{Lie}\, G = D^0, \qquad q_\lambda(x, y) = \mathrm{Tr}_{F/\mathbb{Q}}(\lambda\, \mathrm{trd}(xy)) \quad (\lambda \in F^\times),$$

where D^0 is the space of quaternions of reduced trace zero, regarded as a $\mathbb{Q}$-vector space of dimension nine, and trd is the reduced trace. Over $\bar{\mathbb{Q}}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i = \mathfrak{sl}(V_i)$,

and $a \in F$ acts on T_i by $\sigma_i(a)$. Write m_a for multiplication by a. The space $T \subset \mathrm{End}(V)$ is a sub-Hodge structure of weight zero.

Proposition 7.2 (The two classes are a real multiplication)**.** *In Setup 7.1:*

(i) $H^2(X,\mathbb{Q}) = \mathbb{Q}\theta \oplus U$, *where* U *is the primitive part, of dimension* 27, *and over* $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \qquad U_{ij} = S^2V_i \otimes S^2V_j \otimes \textstyle\bigwedge^2 V_k, \quad \{i,j,k\} = \{1,2,3\},$$

each of dimension nine. The summand U_{12} *is of type* $(1,1)$, *and* U_{13} *and* U_{23} *have Hodge numbers* $(3,3,3)$.

(ii) *The algebra of Hodge endomorphisms of* $H^2(X,\mathbb{Q})$ *is* $\mathbb{Q} \times F$, *acting by scalars on* $\mathbb{Q}\theta$, *with* $a \in F$ *acting on* U_{ij} *by* $\sigma_k(a)$. *Through Poincaré duality and the Lefschetz isomorphism* $\theta^2\colon H^2 \to H^6$ *this algebra is the space of Hodge classes in the Künneth component* $H^2(X) \otimes H^2(X)$ *of* $H^4(X \times X, \mathbb{Q})$; *the subalgebra* $\mathbb{Q} \times \mathbb{Q}$ *is spanned by the products of divisor classes that lie there, and the two classes of Proposition 6.74(ii) may be taken to be the elements of* F *of trace zero.*

(iii) $T(-1)$ *is a Hodge structure of K3 type with Hodge numbers* $(1,7,1)$, *and its Hodge endomorphisms are the* m_a, $a \in F$. *The* G*-invariant symmetric forms on* T *are the* q_λ, *and* q_λ *polarises* $T(-1)$ *if and only if* λ *is totally positive.*

Proof. (i) is the decomposition of $\bigwedge^2(V_1 \otimes V_2 \otimes V_3)$ in the proof of Proposition 6.74, where U_{ij} appears as the summand of highest weight 2 on the i-th and j-th factors. The Hodge cocharacter acts trivially on V_1 and V_2 and splits $V_3 = V_3^{1,0} \oplus V_3^{0,1}$, so $\bigwedge^2 V_3$ has type $(1,1)$ and S^2V_3 has one dimension of each of the types $(2,0)$, $(1,1)$, $(0,2)$.

(ii) A Hodge endomorphism commutes with the Mumford–Tate group $G \cdot \mathbb{G}_m$, and $\mathbb{G}_m$ acts on H^2 by scalars, so the Hodge endomorphisms are the G-endomorphisms. Over $\bar{\mathbb{Q}}$ the four summands $\mathbb{Q}\theta$, U_{12}, U_{13}, U_{23} are irreducible and pairwise non-isomorphic, so the G-endomorphisms are the combinations of the four projections. The Galois group fixes the projection onto $\mathbb{Q}\theta$ and permutes the other three as it permutes the indices k, that is, as it permutes the embeddings of F. For $a \in F$ an endomorphism acting on U_{ij} by $\sigma_k(a)$ is therefore Galois invariant, hence rational, and with the scalars on $\mathbb{Q}\theta$ these exhaust the rational points, which form $\mathbb{Q} \times F$. Poincaré duality identifies $\mathrm{End}\, H^2(X,\mathbb{Q})$ with $H^6(X) \otimes H^2(X)$, and $\theta^2 \otimes 1$ carries $H^2 \otimes H^2$ onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [Kle68, §2]. The products of divisor classes in $H^2 \otimes H^2$ are $\psi_{11}\psi_{22}$ and ψ_{12}^2 in the notation of the proof of Proposition 6.74. They are invariant under the whole of $\mathrm{Sp}(V,\psi)$, whose endomorphisms of $H^2 = \mathbb{Q}\theta \oplus U$ are spanned by the identity and the projection onto $\mathbb{Q}\theta$, and they are linearly independent, being a multiple of the projection and a multiple of the identity (proof of Proposition 6.74); so they span $\mathbb{Q} \times \mathbb{Q}$, and any complement of $\mathbb{Q} \times \mathbb{Q}$ in $\mathbb{Q} \times F$ may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on T_1 and T_2 and has weights -1, 0, 1 on T_3, so T has Hodge numbers $(1,7,1)$ in the types $(-1,1)$, $(0,0)$, $(1,-1)$, and $T(-1)$ is of K3 type. Its Hodge endomorphisms are again the G-endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands T_i, gives F acting by multiplication. On each T_i the invariant symmetric forms are the multiples of $\mathrm{tr}(xy)$, and $\mathrm{tr}(x_iy_i) = \sigma_i(\mathrm{trd}(xy))$ for $x, y \in D^0$, so the Galois-invariant ones are the q_λ. At σ_1 and σ_2 the algebra $D \otimes \mathbb{R}$ is Hamilton's quaternions, T_i is of type $(1,1)$, and $\mathrm{trd}(x^2) = -2\,\mathrm{nrd}(x)$ is negative on nonzero pure quaternions. At σ_3 the form $\mathrm{tr}(xy)$ on $\mathfrak{sl}_2(\mathbb{R})$ is negative on the line of type $(1,1)$, the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types

$(2,0)$ and $(0,2)$. The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by $\sigma_i(\lambda)$, so q_λ polarises $T(-1)$ precisely when every $\sigma_i(\lambda)$ is positive. ∎

The two classes are thus the endomorphisms of U given by the elements of F of trace zero. A Hodge structure of K3 type with the same field sits beside them, and the next lemma connects the two.

Lemma 7.3 (The product map)**.** *For $x, y \in T$ let $\mu(x \otimes y) \in H^2(X, \mathbb{Q})$ be the class that corresponds under ψ to the alternating form $(v, w) \mapsto \psi(xv, yw) + \psi(yv, xw)$ on V. Then μ is a G-equivariant map $T \otimes T \to H^2(X, \mathbb{Q})$, and:*

(i) *for $i \neq j$ it restricts to an isomorphism $T_i \otimes T_j \to U_{ij}$, and it carries $T_i \otimes T_i$ onto $\mathbb{Q}\theta$;*

(ii) *for $e, \lambda \in F^\times$ put $\mu_e = \mu \circ (m_e \otimes m_e)$ and let $\mu_e^\dagger$ be its adjoint for the form $q_\lambda \otimes q_\lambda$ on $T \otimes T$ and the Lefschetz pairing $B(\alpha, \beta) = \int_X \alpha\beta\,\theta^2$ on $H^2(X, \mathbb{Q})$. Then $\mu_e\mu_e^\dagger$ preserves $\mathbb{Q}\theta$ and acts on U as the element*

$$\nu\, \mathrm{N}_{F/\mathbb{Q}}(g)\, g^{-1} \in F, \qquad g = e^2/\lambda,$$

where ν is a nonzero rational number independent of e and λ.

Proof. Equivariance is clear, since μ is built from ψ and the action of $\mathrm{End}(V)$ on V. An element $x \in T_i$ acts on V through the i-th factor and lies in the symplectic Lie algebra of $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, so $\psi(xv, yw) = -\psi(v, xyw)$ and $\mu(x \otimes y)$ is the form $-\psi(v, (xy + yx)w)$. If $x \in T_i$ and $y \in T_j$ with $i \neq j$, then $xy = yx$ acts as $x \otimes y$ on $V_i \otimes V_j$, and the form is -2 times the product of $\epsilon_i(\cdot, x\,\cdot)$, $\epsilon_j(\cdot, y\,\cdot)$ and ϵ_k; the first two are symmetric because x and y lie in $\mathfrak{sp}(\epsilon) = \mathfrak{sl}_2$, and $x \mapsto \epsilon(\cdot, x\,\cdot)$ is an isomorphism $\mathfrak{sl}_2 \to S^2V^\vee$, so μ maps $T_i \otimes T_j$ isomorphically onto U_{ij}. If $i = j$, then $xy + yx = \mathrm{tr}(xy)$ on V_i, because x and y are trace-free 2×2 matrices, and $\mu(x \otimes y) = -\mathrm{tr}(xy)\,\theta$.

For (ii), m_e is self-adjoint for q_λ, and the adjoint of a map out of $(T \otimes T, q_\lambda \otimes q_\lambda)$ is $(m_\lambda \otimes m_\lambda)^{-1}$ times its adjoint for $q_1 \otimes q_1$, so $\mu_e\mu_e^\dagger = \mu\,(m_g \otimes m_g)\,\mu^{\dagger_1}$ with $\mu^{\dagger_1}$ the adjoint for $q_1 \otimes q_1$. This map is G-equivariant, so it preserves $\mathbb{Q}\theta$ and each U_{ij}, and $\mu^{\dagger_1}$ carries U_{ij} into $(T_i \otimes T_j) \oplus (T_j \otimes T_i)$, on which $m_g \otimes m_g$ is multiplication by $\sigma_i(g)\sigma_j(g) = \mathrm{N}(g)/\sigma_k(g)$. Hence $\mu_e\mu_e^\dagger$ is $\mathrm{N}(g)\sigma_k(g)^{-1}$ times $\mu\mu^{\dagger_1}$ on U_{ij}. The summands of $H^2(X, \bar{\mathbb{Q}})$ are pairwise non-isomorphic and self-dual, so B is nondegenerate on U_{ij}, and by Schur's lemma the pull-back of B along the isomorphism $\mu_0 = \mu|_{T_i \otimes T_j}$ is $c_k\, q_1 \otimes q_1$ with $c_k \neq 0$; then $\mu_0\mu_0^{\dagger_1} = c_k$ on U_{ij}. As μ is symmetric in its two arguments and the exchange of the factors is an isometry of $q_1 \otimes q_1$, $\mu^{\dagger_1}u = \mu_0^{\dagger_1}u + s\,\mu_0^{\dagger_1}u$ with s that exchange, and $\mu\mu^{\dagger_1} = 2c_k$ on U_{ij}. On the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, $\psi = \epsilon \otimes \epsilon \otimes \epsilon$, the permutations of the three factors preserve ψ, hence θ, B and μ, and also $q_1 = \sum_i \mathrm{tr}(x_i y_i)$, while they permute the U_{ij}; so $c_1 = c_2 = c_3$, and $\mu\mu^{\dagger_1}$ is one and the same nonzero number ν on the three summands U_{ij}, rational because $\mu\mu^{\dagger_1}$ is defined over $\mathbb{Q}$. By Proposition 7.2(ii) an endomorphism of U acting on U_{ij} by $\nu\mathrm{N}(g)/\sigma_k(g)$ is the element $\nu\mathrm{N}(g)g^{-1}$ of F. ∎

Figure 33 draws Proposition 7.2 and Lemma 7.3 over the three factors of the Hodge group.

By the Kuga–Satake class of a projective K3 surface S we mean the Hodge class on $S \times \mathrm{KS}(S) \times \mathrm{KS}(S)$ that induces the embedding of [I, Proposition 11.7] for $T(S)_\mathbb{Q}$; its algebraicity is the Kuga–Satake Hodge conjecture for S ([I, Remark 11.8]).

Theorem 7.4 (One K3 surface for the two classes)**.** *In Setup 7.1, let $\lambda \in F$ be totally positive.*

(i) *There is a projective K3 surface S_λ of Picard number thirteen and an isomorphism of Hodge structures $T(S_\lambda)_\mathbb{Q} \cong T(-1)$ carrying the intersection form to q_λ. Its Kuga–Satake variety is isogenous to X^{32}.*

(ii) *If the Kuga–Satake class of S_λ is algebraic, then the real multiplication by F on $T(S_\lambda)$ is induced by algebraic cycles on $S_\lambda \times S_\lambda$, and every Hodge class in $H^4(X \times X, \mathbb{Q})$ is algebraic; in particular the two classes of Proposition 6.74(ii) are algebraic.*

Proof. (i) The quadratic space (T, q_λ) has dimension nine and signature $(2, 7)$ by Proposition 7.2(iii), and H^2 of a K3 surface with its intersection form has dimension 22 and signature $(3, 19)$. Over $\mathbb{Q}_p$ a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over $\mathbb{R}$ the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [OMe00, §§63, 66] there is an isometric embedding of (T, q_λ) into $\Lambda_\mathbb{Q}$, where Λ is the K3 lattice. The line $\ell = T(-1)^{2,0} \subset T_\mathbb{C} \subset \Lambda_\mathbb{C}$ satisfies $(\ell, \ell) = 0$ and $(\sigma, \bar\sigma) > 0$ for $0 \neq \sigma \in \ell$, so by the surjectivity of the period map there is a K3 surface S_λ with a marking $H^2(S_\lambda, \mathbb{Z}) \cong \Lambda$ that carries $H^{2,0}(S_\lambda)$ to ℓ [Huy16, Chs. 6–7]. A rational class is of type $(1, 1)$ exactly when it is orthogonal to ℓ, and its component in T is then a rational class of type $(1, 1)$ in $T(-1)$, that is a G-invariant vector of $T = \operatorname{Lie} G$; there is none, since G is semisimple. So $\mathrm{NS}(S_\lambda)_\mathbb{Q} = T^\perp$, of dimension 13 and signature $(1, 12)$, and $T(S_\lambda)_\mathbb{Q} = T$ with the form q_λ. A class of positive square in $\mathrm{NS}(S_\lambda)$ makes S_λ projective [Huy16, Ch. 8].

For the Kuga–Satake variety, note first that over $\bar\mathbb{Q}$ the space T is the orthogonal sum of the three ternary spaces T_i, so $C(T)$ is the graded tensor product of the $C(T_i)$. Scaling a ternary form does not change its even Clifford algebra, so D^1 is the spin group of $(D^0, \lambda \operatorname{tr} d)$ for every λ, and $\operatorname{Res}_{F/\mathbb{Q}} D^1$ maps to $\mathrm{Spin}(T, q_\lambda)$ by multiplication in the Clifford algebra, with the same kernel as its map to $\mathrm{GL}(V)$, so that G is a subgroup of $\mathrm{Spin}(T, q_\lambda)$; as a representation of the i-th factor, $C(T_i)$ is four copies of V_i. Hence $C^+(T)$ is $V^{\oplus 32}$ as a representation of G over $\bar\mathbb{Q}$, and over $\mathbb{Q}$ as well, because V is absolutely irreducible. The Hodge structures on $C^+(T) = H^1(\mathrm{KS}(S_\lambda), \mathbb{Q})$ and on $V^{\oplus 32} = H^1(X^{32}, \mathbb{Q})$ are both given by lifts of weight one of the same homomorphism $\mathbb{S} \to \mathrm{MT}(T)_\mathbb{R}$ through $G \cdot \mathbb{G}_m$, and two such lifts coincide.[42] So $H^1(\mathrm{KS}(S_\lambda), \mathbb{Q}) \cong H^1(X^{32}, \mathbb{Q})$ as Hodge structures, and $\mathrm{KS}(S_\lambda)$ is isogenous to X^{32}. For the form used by Galluzzi this is her theorem [Gal00]; van Geemen proves the corresponding splitting for every K3 surface with real multiplication [vG08].

(ii) Write $S = S_\lambda$ and $A = \mathrm{KS}(S)$, and let Z_S be an algebraic cycle on $S \times A \times A$ inducing the Kuga–Satake embedding $T(S) \to \operatorname{End} H^1(A)$. Composing Z_S with the graph of an isogeny $A \to X^{32}$ and with the graphs of the inclusions and projections of the factors of X^{32} gives algebraic cycles on $S \times X \times X$ inducing the 32^2 block components of the embedding, which are morphisms of Hodge structures $T(S) \to \mathrm{End}(V)$, up to a Tate twist; both lie in the tensor category generated by $H^1(X)$, so the blocks are equivariant for $\mathrm{MT}(X) = G \cdot \mathbb{G}_m$. Over $\bar\mathbb{Q}$, $\mathrm{End}(V) = S^2V \oplus \bigwedge^2 V$, and $T_i \cong S^2V_i$ occurs once in S^2V, as $S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and not at all in $\bigwedge^2 V$. So each block takes values in T, and through the identification of (i) it is a G-endomorphism of T, that is $m_{e_{ab}}$ for some $e_{ab} \in F$ by Proposition 7.2(iii). Over $\bar\mathbb{Q}$ the embedding is therefore $v \mapsto v \otimes N_i$ on T_i, acting on $H^1(A) = V \otimes \mathbb{Q}^{32}$, with $N_i = (\sigma_i(e_{ab}))_{a,b}$. The embedding Ψ of [I, Proposition 11.7] satisfies the Clifford relation $\Psi(v)\Psi(w) + \Psi(w)\Psi(v) = 2q_\lambda(v_0, v_0)\, q_\lambda(v, w)$, because $\Psi(v)\Psi(w)x = vwx\, v_0^2$. Take nonzero $v \in T_i$ and $w \in T_j$ with $i \neq j$. As endomorphisms of V they commute and $vw \neq 0$, so the left side is $vw \otimes (N_iN_j + N_jN_i)$, while the right side is zero because T_i and T_j are

[42] They differ by a homomorphism from the anisotropic torus $\mathbb{S}/\mathbb{G}_m$ to the centre of $G \cdot \mathbb{G}_m$, whose identity component is $\mathbb{G}_m$, and such a homomorphism is trivial.

orthogonal; hence N_i and N_j anticommute. For $v \in T_i$ with $q_\lambda(v,v) \neq 0$ the relation reads $v^2 \otimes N_i^2 = q_\lambda(v_0,v_0)\, q_\lambda(v,v)$, and v^2 is a nonzero scalar on V_i, so N_i^2 is a nonzero scalar. Three pairwise anticommuting invertible matrices are linearly independent: multiplying a relation $\sum_i c_i N_i = 0$ by N_j on either side and adding gives $2c_j N_j^2 = 0$. If the e_{ab} spanned a subspace of F of dimension at most two, the three N_i would lie in the span of at most two rational matrices, which is impossible; so the e_{ab} span F, and two of them, e and e', are not rational multiples of each other. Then $(e/e')^2$ is irrational, since a cubic field contains no irrational element with rational square, so e^2/λ and e'^2/λ are not both rational. Fix a block $\kappa = m_e$ with $e^2/\lambda \notin \mathbb{Q}$, induced by a cycle Γ.

Consider first the real multiplication. The trace form $(a,b) \mapsto \operatorname{tr}(ab)$ on $\operatorname{End}(V)$ is induced by an algebraic class: for correspondences a, b on X the Lefschetz trace formula gives $\operatorname{tr}(a_* b_* \mid H^1) = -\deg((a \circ b \circ \pi_1) \cdot \Delta_X)$, where π_1 is the Künneth projector onto H^1, algebraic on an abelian variety [Kle68, §2]. So the adjoint $\kappa^\dagger$ of κ, for this form and the intersection form on S, is induced by a cycle built from the transpose of Γ, π_1 and Δ_X, and $\kappa^\dagger\kappa$ is an algebraic endomorphism of $T(S)$. An element x of T_i acts on the eight-dimensional space V through a two-dimensional factor, with multiplicity four, so $\operatorname{tr}_V(xy) = 4 \operatorname{Tr}_{F/\mathbb{Q}} \operatorname{trd}(xy)$, and

$$q_\lambda(\kappa^\dagger\kappa x, y) = \operatorname{tr}_V(ex \cdot ey) = 4 \operatorname{Tr}_{F/\mathbb{Q}}(e^2 \operatorname{trd}(xy)) = q_\lambda(4e^2\lambda^{-1}x, y),$$

that is $\kappa^\dagger\kappa = m_{4e^2/\lambda}$. By the choice of e this element is irrational, so it generates the cubic field F and every m_a is a polynomial in $\kappa^\dagger\kappa$. Composing with the projector of $H^2(S)$ onto $T(S)$, which is algebraic because it is the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_\mathbb{Q}$, gives algebraic cycles on $S \times S$ inducing the real multiplication.

Consider next the classes on $X \times X$. The map $\mu_\kappa = \mu \circ (\kappa \otimes \kappa)$ from $T(S) \otimes T(S)$ to $H^2(X, \mathbb{Q})$ is induced by an algebraic cycle on $S \times S \times X$, obtained from $\Gamma \times \Gamma$, the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$, whose class is ψ in $H^1 \otimes H^1$, and the diagonal of X by pull-back, push-forward and intersection, because μ uses nothing but ψ and the composition of endomorphisms of V. The form $q_\lambda \otimes q_\lambda$ is the intersection form of $S \times S$ on $T(S) \otimes T(S)$ and B is the intersection form of X twisted by θ^2, so the adjoint $\mu_\kappa^\dagger$ is induced by cup product with θ^2 followed by the transposed cycle, and $E = \mu_\kappa \mu_\kappa^\dagger$ is an algebraic endomorphism of $H^2(X, \mathbb{Q})$. Since $\kappa = m_e$, Lemma 7.3(ii) says that E preserves $\mathbb{Q}\theta$ and acts on U as $\nu \mathrm{N}(g) g^{-1}$ with $g = e^2/\lambda$, and this element of F is not rational, since g is not. The projection π_θ onto $\mathbb{Q}\theta$ is algebraic, being given by the class $\theta \times \theta$ suitably normalised, so the algebraic Hodge endomorphisms of $H^2(X, \mathbb{Q})$ contain 1, π_θ and $E(1 - \pi_\theta)$, which generate $\mathbb{Q} \times F$, that is all of them by Proposition 7.2(ii). By the same part every Hodge class in $H^2(X) \otimes H^2(X)$ is then algebraic. The Hodge classes of $H^4(X \times X, \mathbb{Q})$ in the other Künneth components number one in each of $H^a \otimes H^{4-a}$, $a \neq 2$, by the count in the proof of Proposition 6.74, and are products of divisor classes. ∎

Remark 7.5 (The Kuga–Satake class in the literature). Theorem 7.4 takes the algebraicity of the Kuga–Satake class of S_λ as its input. The cases of the Kuga–Satake Hodge conjecture proved in the literature are these. They include the projective K3 surfaces with a Shioda–Inose structure, which relates the surface to an abelian surface; by Morrison these are the surfaces of Picard number 19 or 20, those of Picard number 18 with $T(S) \cong U \oplus T'$ and those of Picard number 17 with $T(S) \cong U^2 \oplus T'$ [Mor84, Theorem 6.3 and Corollary 6.4], and for these the proof uses in addition the Hodge conjecture for the powers of an abelian surface, whose Hodge classes are generated by divisor classes (see [MZ99]), and the fact that the Kuga–Satake variety of an abelian surface is isogenous to a power of it [Mor85].

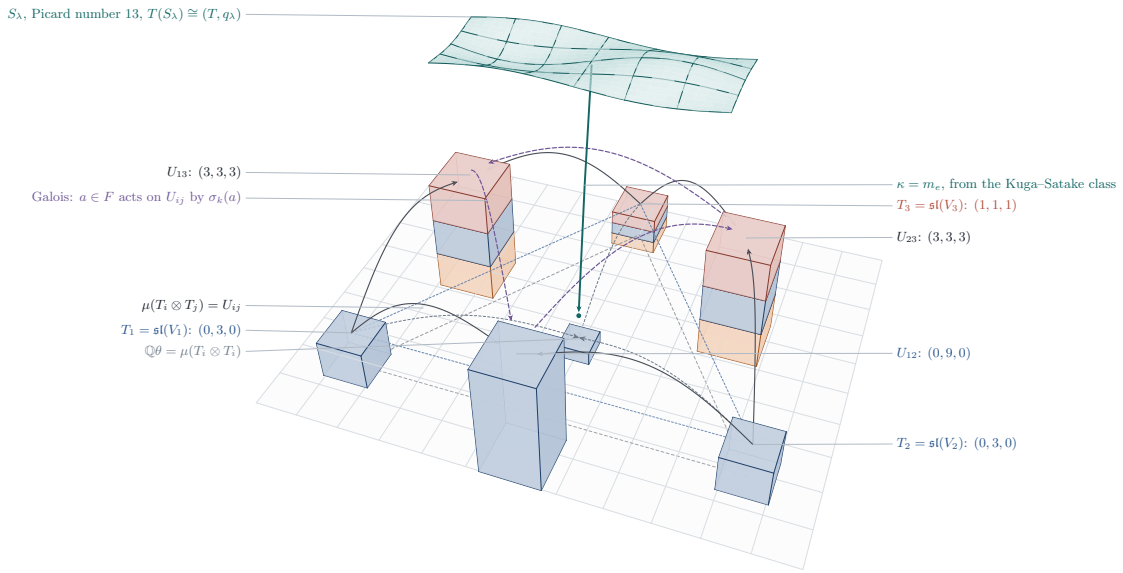


FIGURE 33. Proposition 7.2 and Lemma 7.3, drawn over the triangle of the three factors of the Hodge group; every height is a dimension. At the corners stand the summands $T_i = \mathfrak{sl}(V_i)$ of $T = \operatorname{Lie} G$, banded by Hodge type after the Tate twist $T(-1)$: blue for $(1,1)$, clay for $(2,0)$ and amber for $(0,2)$, so that only T_3 carries the K3 type. On the edges stand the three summands U_{ij} of the primitive part of $H^2(X)$, of dimension nine, U_{12} of pure type $(1,1)$ and the other two in three bands of three; in the middle is the line of θ. The solid arcs are the product map μ, which carries $T_i \otimes T_j$ onto U_{ij}, and the dashed ones carry $T_i \otimes T_i$ to $\mathbb{Q}\theta$. The violet arcs are the Galois group permuting the U_{ij} as it permutes the embeddings of F, which is why the two exceptional classes may be taken to be the trace-zero part of a cubic field. Above, the K3 surface S_λ of Theorem 7.4, whose Kuga–Satake class, if algebraic, comes down to T as the map $\kappa = m_e$.

With Buskin's theorem that rational Hodge isometries between K3 surfaces are algebraic [Bus19], this reaches the surfaces of Picard number at least 19, those of Picard number 18 with $T(S)_{\mathbb{Q}}$ isotropic, and those of Picard number 17 with $T(S)_{\mathbb{Q}}$ of Witt index two (Remark 7.17). It does not reach every surface of Picard number seventeen, as Morrison notes [Mor84, Remark 7.2]: for $T(S) = U \oplus \langle 6\rangle \oplus \langle -2\rangle \oplus \langle -2\rangle$, the transcendental lattice of the very general K3 surface of a family of dimension three [Mor84, Corollary 2.10], $T(S)_{\mathbb{Q}}$ has Witt index one, since $\langle 6, -2, -2\rangle$ is anisotropic over $\mathbb{Q}_3$: a primitive solution of $3x^2 = y^2 + z^2$ in $\mathbb{Z}_3$ would have $3 \mid y$ and $3 \mid z$, as -1 is not a square modulo 3, and then $3 \mid x$. The double planes branched along six lines, of Picard number at least sixteen, are settled by Paranjape's cycle; see [vG00]. Floccari proves the conjecture for the K3 surfaces attached to projective sixfolds of generalised Kummer type [Flo24], and then, for the K3 surfaces attached to all projective varieties of generalised Kummer type and for every projective K3 surface S with $T(S)_{\mathbb{Q}}$ embedding isometrically in $(U^3 \oplus \langle -m\rangle)_{\mathbb{Q}}$ for some $m > 0$, both the conjecture and the Hodge conjecture for all powers [Flo25, Theorems 1.2 and 5.11]; their transcendental lattices embed, after scaling in the first case, in a rational quadratic space of rank seven and signature $(3,4)$, so their transcendental rank is at most

six. Ingalls, Logan and Patashnick prove the conjecture, with the Hodge conjecture for the square, for a general K3 surface in $\mathbb{P}^4$ with fifteen ordinary double points and for the K3 surfaces joined to it by chains of rational maps of finite degree [ILP22, Theorems 1.5, 4.8 and 6.4]. Bolognesi and Laterveer prove it for the triple covers of a quadric branched along a curve of genus four, which form two families of dimension nine; these carry a non-symplectic automorphism of order three, so their fields of Hodge endomorphisms are CM fields [BL24]. The surfaces S_λ, by contrast, have transcendental rank nine and real multiplication by a cubic field, and none of these cases contains them: a Shioda–Inose structure needs $T(S) \cong T(A)$ for an abelian surface A, hence a transcendental rank of at most five; the surfaces of Floccari, of Paranjape and of Ingalls, Logan and Patashnick have transcendental rank at most six; and the field of Hodge endomorphisms of S_λ is totally real, not CM.

Schlickewei made the real multiplication algebraic on those double planes that have it, necessarily by a field of degree at most two, from Paranjape's cycle and van Geemen's description of their Kuga–Satake varieties [Sch10, vG08], and Varesco showed that when the Kuga–Satake Hodge conjecture holds for two hyperkähler varieties, every Hodge similarity between their transcendental lattices is algebraic up to the Lefschetz isomorphism [Var23], which reaches a quadratic field through its square root. More generally, if the Hodge endomorphisms of $T(S)$ form a totally real field E, then E acts by self-adjoint maps [Zar83], so $g \in E$ is a similarity, with multiplier μ, exactly when $g^2 = \mu \in \mathbb{Q}$; the Hodge similarities of $T(S)$ to itself, and with them all composites of Hodge similarities between transcendental lattices of K3 surfaces that start and end at $T(S)$, therefore span exactly the maximal multiquadratic subfield E_{mq} of E, the compositum of its quadratic subfields. For a cubic field $E_{\mathrm{mq}} = \mathbb{Q}$: a cubic field contains no irrational element with rational square, so no irrational element of it acts as a similarity. In the proof of Theorem 7.4 the element $\kappa^\dagger\kappa = m_{4e^2/\lambda}$ plays the role that the similarities play for quadratic fields, and the Clifford relations are what make it irrational. Buskin's theorem [Bus19] does not give the real multiplication either. A Hodge isometry of $T(S_\lambda)$ is a Hodge endomorphism, hence some m_a by Proposition 7.2(iii), and it is an isometry only when $a^2 = 1$. For $b \in F^\times$ Buskin's theorem does give algebraic cycles between S_λ and $S_{\lambda b^{-2}}$ that realise m_b, but every composite of them that returns to S_λ is again a Hodge isometry of $T(S_\lambda)$, so it multiplies by an element of square one.

The surfaces S_λ include those of Zhu, who computes the integral structure that the fourfold induces on T and studies the K3 surfaces of Picard number thirteen that carry it [Zhu18]; the form of that structure is the corestriction of the reduced norm of D^0, which up to sign is $\frac{1}{2}q_1$. Theorem 7.4 applies to them as it stands, because the irrationality of e^2/λ comes from the anticommuting blocks and not from λ. The explicit families closest to these are those of van Geemen and Schütt, with real multiplication by the cyclic cubic fields $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ and Picard numbers four and ten, on which the real multiplication is induced by explicit algebraic cycles [vGS25, Theorem 1.2 and Section 4.8]; their Picard numbers are not thirteen, and the Kuga–Satake class, which Theorem 7.4 needs and algebraic real multiplication does not supply, is not treated there.

By Theorem 7.4, then, the first case of (F2) is implied by one instance of the Kuga–Satake Hodge conjecture, for one K3 surface of Picard number thirteen; by Proposition 7.12 below that instance is equivalent to the first case of (F2) at X together with a link, and it is not a reduction of it. The sixfold link of [BB26] replaces the Kuga–Satake variety of dimension 2^7 by an abelian sixfold and shows what the replacement does and does not give.

Theorem 7.4 trades the two exceptional classes for one Kuga–Satake class. The next results make this exchange precise. If the exceptional classes are algebraic, so is every Hodge class on every power of X (Theorem 7.7); if the Kuga–Satake class is algebraic, so is every Hodge class on every product $S_\lambda^a \times X^b$ (Corollary 7.11). The Kuga–Satake class is algebraic exactly when the exceptional classes are and, in addition, some algebraic cycle links S_λ to a power of X (Proposition 7.12), and no such link can pass through an abelian variety of dimension at most five (Proposition 7.13). The theorem and the corollary are implications, and Remark 7.17 collects the forms they give to the first case of (F2).

Proposition 7.6 (Where T occurs)**.** *In Setup 7.1, write* $\mathrm{Hom}_{\mathrm{Hdg}}$ *for morphisms of rational Hodge structures.*

(i) *T is isomorphic to no subquotient of $H^k(X,\mathbb{Q})(j)$, for any k and j.*

(ii) *The F-vector space $\mathrm{Hom}_{\mathrm{Hdg}}(T, H^2(X\times X,\mathbb{Q})(1))$ is one-dimensional, and the copy of T in $H^2(X\times X,\mathbb{Q})(1)$ lies in $H^1(X)\otimes H^1(X)$. For n odd and for $n\in\{0,16\}$ the Hodge structure T is isomorphic to no subquotient of any $H^n(X\times X,\mathbb{Q})(j)$.*

(iii) *$\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{End}\, V)$ is a one-dimensional F-vector space, spanned by the inclusion. For $\lambda\in F$ totally positive, after an isogeny $\mathrm{KS}(S_\lambda)\to X^{32}$ the Kuga–Satake embedding of $T(S_\lambda)$ is a 32×32 matrix of maps $m_{e_{ab}}$ with $e_{ab}\in F$, and $\mathrm{KS}(S_\lambda)$ is isogenous to B^4, where B is isogenous to X^8 and $\mathrm{End}^0(B)=M_8(\mathbb{Q})$.*

Proof. After the twists all the Hodge structures involved have weight zero and lie in the tensor category generated by V, whose Mumford–Tate group is $G\cdot\mathbb{G}_m$ with $\mathbb{G}_m$ acting trivially in weight zero; so their morphisms and subquotients are those of representations of G. Over $\bar{\mathbb{Q}}$, $T=T_1\oplus T_2\oplus T_3$ with $T_i\cong S^2V_i$, and the Galois group permutes the three summands transitively. Hence T is irreducible, a representation of G defined over $\mathbb{Q}$ contains T_1, T_2 and T_3 with one and the same multiplicity m, and its space of G-maps from T, of $\mathbb{Q}$-dimension $3m$, is a vector space over $\mathrm{End}_G(T)=F$ (Proposition 7.2(iii)) of dimension m.

For (i), write $V=V_1\otimes W$ with $W=V_2\otimes V_3$, on which $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$ acts through $\mathrm{SO}(W,q)$, q the product of the two symplectic forms. By the skew Cauchy formula $\bigwedge^k V=\bigoplus_\lambda S_\lambda V_1\otimes S_{\lambda'}W$, over the partitions λ of k with at most two rows and at most four columns, and $S_\lambda V_1\cong S^{\lambda_1-\lambda_2}V_1$ as a representation of $\mathrm{SL}(V_1)$. So T_1 occurs in $\bigwedge^k V$ only through $\lambda=(2,0)$, $(3,1)$ or $(4,2)$, and only if $S_{\lambda'}W$ has a nonzero invariant under $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$. As representations of $\mathrm{SL}(W)$, $S_{(1,1)}W=\bigwedge^2 W$, $S_{(2,2,1,1)}W\cong\bigwedge^2 W$, and $S_{(2,1,1)}W\cong\mathfrak{sl}(W)$, the kernel of the contraction $\bigwedge^3 W\otimes W\to\bigwedge^4 W$ with $\bigwedge^3 W\cong W^*$. Restricted to $\mathrm{SO}(W,q)$, $\bigwedge^2 W=\mathfrak{so}(W)=\mathfrak{sl}(V_2)\oplus\mathfrak{sl}(V_3)$ and $\mathfrak{sl}(W)=\mathfrak{so}(W)\oplus S^2V_2\otimes S^2V_3$, the second summand being the trace-free part of S^2W; none of these has a nonzero invariant. So T_1 occurs in no $\bigwedge^k V$, and by the same argument with the factors exchanged neither do T_2 and T_3. As G is reductive, the subquotients of $\bigwedge^k V$ are its summands; that is (i).

For (ii), $H^n(X\times X)=\bigoplus_{a+b=n}\bigwedge^a V\otimes\bigwedge^b V$. The element of G that is -1 on the first factor $\mathrm{SL}(V_1)$ and 1 on the other two acts on V by -1, hence on $H^n(X\times X)$ by $(-1)^n$, and trivially on T; and H^0, H^{16} are trivial representations. In $H^2(X\times X)=\bigwedge^2 V\oplus(V\otimes V)\oplus\bigwedge^2 V$ the summands $\bigwedge^2 V$ contain no T_i by (i), and $V\otimes V=S^2V\oplus\bigwedge^2 V$ with

$$S^2(V_1\otimes V_2\otimes V_3)=S^2V_1\otimes S^2V_2\otimes S^2V_3\oplus\bigoplus_{i=1}^{3}S^2V_i\otimes\bigwedge^2 V_j\otimes\bigwedge^2 V_k,$$

$\{i,j,k\}=\{1,2,3\}$, where $\bigwedge^2 V_j$ is trivial. So each T_i occurs exactly once in $H^2(X\times X)$, inside $V\otimes V$, and $m=1$. For (iii), $\mathrm{End}\, V=S^2V\oplus\bigwedge^2 V$ contains each T_i once, and the

proof of Theorem 7.4(ii) shows that the blocks of the Kuga–Satake embedding are maps $m_{e_{ab}}$. Finally $\mathrm{KS}(S_\lambda)$ is isogenous to $X^{32} = (X^8)^4$ by Theorem 7.4(i), and $\mathrm{End}^0(X^8) = M_8(\mathrm{End}^0 X) = M_8(\mathbb{Q})$. This is the splitting $\mathrm{KS} \sim B^{2^{d-1}}$, $d = [F : \mathbb{Q}]$, which van Geemen proves for K3 surfaces with real multiplication [vG08]; there $H^1(B, \mathbb{Q})$ is the corestriction to $\mathbb{Q}$ of the even Clifford algebra of T over F, here D, and over $\bar{\mathbb{Q}}$ one has $\mathrm{Cor}_{F/\mathbb{Q}}(D) = \bigotimes_k D \otimes_{F,\sigma_k} \bar{\mathbb{Q}} = \bigotimes_k \mathrm{End}(V_k) = \mathrm{End}(V)$, which is $V^{\oplus 8}$ under left multiplication by G. ∎

Theorem 7.7 (Every power of a Mumford fourfold)**.** *In Setup 7.1, suppose that the two exceptional classes of Proposition 6.74(ii) are algebraic on $X \times X$. Then every Hodge class on X^n is algebraic, for every $n \geq 1$.*

Proof. Over $\bar{\mathbb{Q}}$ keep $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, as in the proof of Lemma 7.3. For $m \geq 1$ regard $V^{\otimes m} = H^1(X)^{\otimes m}$ as the Künneth component of $H^m(X^m, \mathbb{Q})$ of multidegree $(1, \dots, 1)$, whose projector is algebraic [Kle68, §2]. Call an endomorphism of $V^{\otimes m}$ algebraic if it is the action on $V^{\otimes m}$ of an algebraic cycle on $X^m \times X^m$ that preserves $V^{\otimes m}$. Composing with the Künneth projector, the algebraic endomorphisms form a $\mathbb{Q}$-algebra, and they carry algebraic classes in $V^{\otimes m}$ to algebraic classes.

Step 1: reduction to multilinear invariants. Let $n \geq 1$ and $p \geq 0$. The map $V^{\otimes 2p} \otimes (\mathbb{Q}^n)^{\otimes 2p} \to \bigwedge^{2p}(V \otimes \mathbb{Q}^n) = H^{2p}(X^n, \mathbb{Q})$ sending $(v_1 \otimes \cdots \otimes v_{2p}) \otimes (a_1 \otimes \cdots \otimes a_{2p})$ to $(v_1 \otimes a_1) \wedge \cdots \wedge (v_{2p} \otimes a_{2p})$ is surjective and G-equivariant for the trivial action on $\mathbb{Q}^n$, so it is surjective on G-invariants, G being reductive. For $a_1, \dots, a_{2p} \in \mathbb{Z}^n$ the image of $t \otimes (a_1 \otimes \cdots \otimes a_{2p})$ is f^*t, where $f\colon X^n \to X^{2p}$ is the homomorphism with rows $a_1, \dots, a_{2p}$, and such tensors span $(\mathbb{Q}^n)^{\otimes 2p}$. So $\mathrm{Hdg}^p(X^n)$ is spanned by the classes f^*t with f a homomorphism and $t \in (V^{\otimes 2p})^G$, and it suffices to show that every element of $(V^{\otimes 2p})^G$ is algebraic.

Step 2: degree four. The space $(V^{\otimes 4})^G$ has dimension eight, the product of the dimensions of the three spaces of SL_2-invariants in $V_k^{\otimes 4}$. Let $\mathcal{P}_0 \subset (V^{\otimes 4})^G$ be the span of the multilinear components of the classes $f^*\xi$, for $\xi \in \mathrm{Hdg}^2(X \times X)$ and $f\colon X^4 \to X^2$ a homomorphism, and of the products of two divisor classes of X^4. By Proposition 6.74(ii) $\mathrm{Hdg}^2(X \times X)$ is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of $\mathcal{P}_0$ is algebraic. For $t \in V^{\otimes 4}$, read as a correspondence from the first two factors of X^4 to the last two, let $\hat{t}$ be the endomorphism $v \mapsto t_*((\theta^3 \times \theta^3) \cup v)$ of $V \otimes V$. The pairing $(v, w) \mapsto \int_X v \cup w \cup \theta^3$ on V is nondegenerate by hard Lefschetz and G-invariant, so $t \mapsto \hat{t}$ is an isomorphism $(V^{\otimes 4})^G \to \mathrm{End}_G(V \otimes V)$, and $\hat{t}$ is algebraic when t is. We show that the $\hat{t}$ and the composites $\hat{u}\hat{t}$, for $t, u \in \mathcal{P}_0$, span $\mathrm{End}_G(V \otimes V)$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V \otimes V) = \bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k \otimes V_k)$ is the algebra of functions on the eight sign patterns $\varepsilon \in \{\pm 1\}^3$, where ε_k is the eigenvalue of the exchange s_k of the two copies of V_k; the patterns with $\varepsilon_1\varepsilon_2\varepsilon_3 = -1$, the odd ones, make up $\bigwedge^2 V$.

(a) Let $f = (x_1 + x_2, x_3 + x_4)\colon X^4 \to X^2$. The multilinear component of $f^*\xi$ depends only on the Künneth component of ξ in $H^2(X) \otimes H^2(X)$, and it is that component read as a tensor $t \in \bigwedge^2 V \otimes \bigwedge^2 V$, so $\hat{t}$ vanishes on S^2V and takes values in $\bigwedge^2 V$. These components fill $(H^2 \otimes H^2)^G = \mathrm{End}_G(H^2)$, of dimension 4 by Proposition 7.2(ii); so the $\hat{t}$ span the functions supported on the four odd patterns.

(b) The map $g = (x_1 + x_3, x_2 + x_4)$ is f composed with the exchange of the factors 2 and 3 of X^4, which permutes the factors of $V^{\otimes 4}$ up to sign. On $V_k^{\otimes 4}$ the SL_2-invariants are spanned by $\epsilon_{13}\epsilon_{24}$, $\epsilon_{14}\epsilon_{23}$ and $\epsilon_{12}\epsilon_{34} = \epsilon_{13}\epsilon_{24} - \epsilon_{14}\epsilon_{23}$, where ϵ_{ab} is ϵ_k on the factors a and b; up to a common sign their hats are 1, s_k and $1 - s_k$. The exchange of the factors 2 and 3 swaps the first and the third and changes the sign of the second, so on functions of ε_k it

carries $\frac{1}{2}(1-s_k)$ to $\frac{1}{2}$ and $\frac{1}{2}(1+s_k)$ to $\frac{1}{2}-s_k$. Applied to the indicator of the odd pattern $(1,1,-1)$ it gives $\frac{1}{2}(\frac{1}{2}-s_1)(\frac{1}{2}-s_2)$. The identity is $\hat{t}$ for the multilinear component of a product of two divisor classes, the pull-backs of the class ψ on $X\times X$ along the projections of X^4 to the factors $(1,3)$ and $(2,4)$, so $h=(1-2s_1)(1-2s_2)-1$ lies in the span of the $\hat{t}$, $t\in\mathcal{P}_0\otimes\bar{\mathbb{Q}}$. As $(\varepsilon_1,\varepsilon_2)$ runs through $(1,1)$, $(1,-1)$, $(-1,1)$, $(-1,-1)$, h takes the values $0,-4,-4,8$, so $h(h+4)/96$ is the indicator of $\varepsilon_1=\varepsilon_2=-1$; subtracting the indicator of $(-1,-1,-1)$, from (a), leaves that of $(-1,-1,1)$. The same argument with the factors V_k permuted gives the other two even patterns with two minus signs, and the indicator of $(1,1,1)$ is 1 minus the other seven.

So the $\hat{t}$ and $\hat{u}\hat{t}$ span $\mathrm{End}_G(V\otimes V)\otimes\bar{\mathbb{Q}}$. Since composition is bilinear and $\mathcal{P}_0\otimes\bar{\mathbb{Q}}$ is spanned by the same pull-backs and products of a basis of Hodge classes, the $\mathbb{Q}$-algebra of algebraic G-endomorphisms of $V\otimes V$ has dimension 8: every G-endomorphism of $V\otimes V$ is algebraic.

Step 3: the partial swaps. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V\otimes V)=\bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$, and $\mathrm{End}_{\mathrm{SL}_2}(V_k\otimes V_k)$ is spanned by 1 and the exchange s_k of the two copies of V_k. In a basis of V_k the matrix units E_{ab} and E_{ba} are dual for the trace form of $\mathfrak{gl}(V_k)$, and $\sum_{a,b}E_{ab}\otimes E_{ba}=s_k$; removing the centre, spanned by the identity 1 with $\mathrm{tr}(1\cdot 1)=2$, gives $s_k=\mathrm{Cas}_k+\frac{1}{2}$, with $\mathrm{Cas}_k=\sum_c x_c\otimes x^c$ for dual bases (x_c), (x^c) of $\mathfrak{sl}(V_k)$ under the trace form. So the s_k come from the three summands T_k of T. The Galois group permutes the s_k as it permutes the embeddings σ_k, so for $a\in F$ the endomorphism $r_a=\sum_k\sigma_k(a)s_k$ is rational, hence algebraic by Step 2. For $m\geq 2$ and $1\leq i<m$ let $r_a^{(i)}$ and $s_k^{(i)}$ act as r_a and s_k on the factors i and $i+1$ of $V^{\otimes m}$ and as the identity on the others; $r_a^{(i)}$ is algebraic, being induced by the product of a cycle inducing r_a with the diagonals of the other factors. By Schur–Weyl duality $\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$ is the image of $\bar{\mathbb{Q}}[S_m]$, generated by the adjacent transpositions, and the commutant of SL_2^3 in $\bigotimes_k V_k^{\otimes m}$ is the tensor product of the three commutants; so $\mathrm{End}_G(V^{\otimes m})\otimes\bar{\mathbb{Q}}$ is generated by the $s_k^{(i)}$. For a basis a_1,a_2,a_3 of F the matrix $(\sigma_k(a_j))$ is invertible, the square of its determinant being the discriminant of F, so each $s_k^{(i)}$ is a $\bar{\mathbb{Q}}$-linear combination of the $r_{a_j}^{(i)}$. The $\mathbb{Q}$-algebra generated by the $r_a^{(i)}$ therefore lies in $\mathrm{End}_G(V^{\otimes m})$ and spans it over $\bar{\mathbb{Q}}$, so it is $\mathrm{End}_G(V^{\otimes m})$: every G-endomorphism of $V^{\otimes m}$ is algebraic. Its dimension is the cube of $\dim\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$, the sum of the squares of the multiplicities in $V_k^{\otimes m}$: $8=2^3$ for $m=2$ and $125=5^3$ for $m=3$, where $V_k^{\otimes 3}=S^3V_k\oplus V_k^{\oplus 2}$.

Step 4: all multilinear invariants. By the first fundamental theorem for SL_2 the invariants in $V_k^{\otimes 2m}$ are spanned by the images of $\epsilon_k^{\otimes m}$ under the permutations of the $2m$ factors. So $(V^{\otimes 2m})^G\otimes\bar{\mathbb{Q}}$ is spanned by the images of $\psi^{\otimes m}=\bigotimes_k\epsilon_k^{\otimes m}$ under the operators that permute the factors of the three $V_k^{\otimes 2m}$ separately, which lie in $\mathrm{End}_G(V^{\otimes 2m})\otimes\bar{\mathbb{Q}}$, and the map $\mathrm{End}_G(V^{\otimes 2m})\to(V^{\otimes 2m})^G$, $e\mapsto e(\psi^{\otimes m})$, is surjective. The class ψ in $V\otimes V$ is that of the divisor $m^*\theta-p_1^*\theta-p_2^*\theta$ on $X\times X$ (proof of Theorem 7.4(ii)), up to a nonzero factor, and $\psi^{\otimes m}$ is, up to sign, the product of its pull-backs along the projections of X^{2m} to the pairs of factors $(2j-1,2j)$; so it is algebraic, and by Step 3 so is every element of $(V^{\otimes 2m})^G$. Step 1 concludes. ∎

Remark 7.8 (Why a composite is needed). Step 2 cannot be carried out with pull-backs and products alone. The skew Cauchy decomposition $\bigwedge^4(V\otimes\mathbb{Q}^4)=\bigoplus_\lambda S_\lambda V\otimes S_{\lambda'}\mathbb{Q}^4$, over the partitions λ of 4 with conjugates λ', is functorial in $\mathbb{Q}^4$, and its summand $S^4V\otimes\bigwedge^4\mathbb{Q}^4$ contains exactly one line of Hodge classes of X^4: S_4 acts on the two-dimensional space of SL_2-invariants in $V_k^{\otimes 4}$ through its two-dimensional irreducible representation σ, and the trivial representation occurs once in $\sigma^{\otimes 3}$. That line is spanned by a class $\mathrm{Det}\in(V^{\otimes 4})^G$

on which the permutations of the factors of X^4 act by the sign, and whose restriction to the diagonal, on the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, is an invariant quartic form of SL_2^3, hence a nonzero multiple of Cayley's hyperdeterminant. Pull-backs along homomorphisms $X^4 \to X^2$ have no component in that summand, because $\bigwedge^4 \mathbb{Q}^2 = 0$, and products of divisor classes have none, because they are invariant under $\mathrm{Sp}(V,\psi)$ and S^4V has no $\mathrm{Sp}(V,\psi)$-invariant. In the notation of Step 2 the hats of the elements of $\mathcal{P}_0$ therefore satisfy one linear condition, which is $v_{(1,-1,-1)} + v_{(-1,1,-1)} + v_{(-1,-1,1)} = 3v_{(1,1,1)}$ for the values v on the even patterns, since the functions found in (a) and (b), those supported on the odd patterns, the constant, h and its two images under the permutations of the factors V_k, satisfy it and span a space of dimension 7. So $\mathcal{P}_0$ has dimension 7 and is the space of Hodge classes of X^4 in the multilinear parts of the other summands, and Det is reached only through a composite of correspondences, which involves a push-forward. The composite h^2 of Step 2 is one: the indicator of $(-1,-1,1)$ violates the condition. This is where Theorem 7.7 differs from Proposition 6.24, in which the first fundamental theorem for $\mathrm{SL}(W)$ lets pull-backs and products suffice. We have not found Theorem 7.7 in the literature; it is an implication between two cases of the Hodge conjecture.

At the CM points of a Mumford family the conclusion holds without any hypothesis.

Proposition 7.9 (Powers at a CM point)**.** *Let $W \to C$ be a family as in Corollary 6.22 and $c \in C$ a CM point. Then the Hodge conjecture holds for X_c^n for every $n \geq 1$.*

Proof. The proof of Theorem 6.29 shows that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of the fourfold X_c along homomorphisms $X_c^n \to X_c$, and these are algebraic [Mar25a]. ∎

The statement is also contained in the theorem of [Li26] on the powers of CM fourfolds.

Corollary 7.10 (Powers and fibre powers of a Mumford family)**.** *Let $W \to C$ be as in Corollary 6.22 and, for $n \geq 1$, let $W^{(n)} = W \times_C \cdots \times_C W$, with n factors. The conditions of Corollary 6.30 are equivalent to each of the following.*

(e) *The Hodge conjecture holds for X_t^n for every $t \in C$ and every $n \geq 1$.*

(f) *$B(W^{(n)})$ holds for $L_{W^{(n)}}$ for every $n \geq 1$.*

Proof. (b) implies (e): by Corollary 6.30, (b) gives (c), so the exceptional classes of $X_t \times X_t$ are algebraic for every t. At a point t that is not a CM point the Hodge group of X_t is G, as shown in the proof of Corollary 6.30, and Theorem 7.7 applies; at a CM point Proposition 7.9 does. (e) implies (f): $W^{(n)}$ is an abelian scheme over C with projective total space, and, as in the proof of Corollary 6.25, its monodromy invariant classes are Hodge classes on every fibre; by (e) they are algebraic on every fibre, hence restrictions of algebraic classes on $W^{(n)}$ by Lemma 6.27, and Theorem 6.21 gives $B(W^{(n)})$. (f) implies (a): take $n = 2$. ∎

Corollary 7.11 (Products with the K3 surface)**.** *In Setup 7.1, let $\lambda \in F$ be totally positive and $S = S_\lambda$ as in Theorem 7.4. If the Kuga–Satake class of S is algebraic, then the Hodge conjecture holds for $S^a \times X^b$ for all $a, b \geq 0$.*

Proof. By Theorem 7.4(ii) the exceptional classes are algebraic, so by Theorem 7.7 the Hodge conjecture holds for every power of X. In the proof of Theorem 7.4(ii) a block $\kappa = m_e$, $e \neq 0$, of the Kuga–Satake embedding is induced by a cycle on $S \times X \times X$; it is a map from $T(S)$ to $\mathrm{End}\, V$, which up to a twist is a Künneth component of $H^*(X \times X, \mathbb{Q})$. Its adjoint $\kappa^\dagger$ is algebraic, $\kappa^\dagger\kappa = m_{4e^2/\lambda}$, and every m_a, $a \in F$, is induced by a cycle on

$S \times S$. So $L = m_{\lambda/(4e^2)}\kappa^\dagger$ is algebraic and $L\kappa$ is the identity of $T(S)$. The decomposition $H^*(S,\mathbb{Q}) = \mathbb{Q} \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus T(S) \oplus H^4(S,\mathbb{Q})$ has algebraic projectors, as in that proof, so $H^*(S^a \times X^b,\mathbb{Q})$ is the sum of the summands $N \otimes T(S)^{\otimes r} \otimes H^*(X^b,\mathbb{Q})$, N a tensor product of $a-r$ of the three other pieces, each cut out by an algebraic projector. Each such N is spanned by products of algebraic classes and is of Tate type, so a Hodge class in the summand is a sum of products $p^*\eta \cup q^*\xi$ with η algebraic on S^{a-r} and ξ a Hodge class in $T(S)^{\otimes r} \otimes H^*(X^b,\mathbb{Q}) \subset H^*(S^r \times X^b,\mathbb{Q})$. For such ξ the class $(\kappa^{\otimes r} \otimes 1)\xi$ is a Hodge class on X^{2r+b}, hence algebraic, and $\xi = (L^{\otimes r} \otimes 1)(\kappa^{\otimes r} \otimes 1)\xi$ is algebraic. ∎

Corollary 7.11 strengthens Theorem 7.4(ii), which gives $X \times X$ only. Varesco proves an implication of the same kind, from the algebraicity of the Kuga–Satake correspondence to the Hodge conjecture for all powers, for families of K3 surfaces of generic Picard number sixteen [Var22]. The converse question, what the Kuga–Satake class asks for beyond the target, has a precise answer.

Proposition 7.12 (What the Kuga–Satake route asks)**.** *In the situation of Corollary 7.11, the Kuga–Satake class of S is algebraic if and only if*

(i) *the two exceptional classes of $X \times X$ are algebraic, and*
(ii) (*a link*) *for some $n \geq 1$ there is an algebraic cycle Z on $S \times X^n$ whose action Z_* is nonzero on $T(S)$.*

Proof. If the Kuga–Satake class is algebraic, (i) is Theorem 7.4(ii), and a nonzero block of the embedding, induced by a cycle on $S \times X \times X$, is a link with $n = 2$. Conversely assume (i) and (ii). By Theorem 7.7 the Hodge conjecture holds for every power of X. Since $T(S) \cong T(-1)$ is irreducible (proof of Proposition 7.6), Z_* maps $T(S)$ isomorphically onto a sub-Hodge structure of some $H^{2k}(X^n,\mathbb{Q})(k-1)$, which is a direct summand, polarisable Hodge structures being semisimple. Projecting onto it and composing with an isomorphism onto the copy of $T(-1)$ in $H^2(X \times X,\mathbb{Q})$ (Proposition 7.6(ii)) gives a morphism of Hodge structures $H^{2k}(X^n,\mathbb{Q})(k-1) \to H^2(X \times X,\mathbb{Q})$, that is a Hodge class on X^{n+2}, which is algebraic. Its composite with Z_* is an algebraic nonzero morphism $T(S) \to T(-1) \subset H^2(X \times X,\mathbb{Q})$, which is some m_e, $e \neq 0$, by Proposition 7.2(iii). The endomorphisms of $H^2(X \times X,\mathbb{Q})$ that act by m_a on this copy of $T(-1)$ and by zero on a Hodge complement are Hodge classes on X^4, hence algebraic, so every $m_a\colon T(S) \to H^1(X) \otimes H^1(X)$, $a \in F$, is algebraic, and so is its composite with cup product with θ^3 on one factor, which carries $H^1(X) \otimes H^1(X)$ onto the Künneth component of $H^8(X \times X,\mathbb{Q})$ that represents $\operatorname{End} V$. By Proposition 7.6(iii), after an isogeny $\mathrm{KS}(S) \to X^{32}$ the Kuga–Satake embedding is a matrix of such maps, and the Kuga–Satake class is recovered from its blocks through the graphs of the isogeny and of the inclusions and projections of the factors of X^{32}; so it is algebraic. ∎

So the Kuga–Satake route is at least as strong as the first case of (F2) and is not a reduction of it: at the point X it asks for (i), which is the first case of (F2) there, and in addition for a link. The Hodge conjecture implies both (i) and (ii). The route helps only if the Kuga–Satake class of S_λ can be made algebraic by arguments proper to K3 surfaces. A link has to be a correspondence of a new kind: T occurs in no $H^k(X)$ (Proposition 7.6(i)), and the next proposition shows that it occurs in the cohomology of no abelian variety of dimension below six.

Proposition 7.13 (No small abelian carrier of T)**.** *In Setup 7.1, let Y be an abelian variety such that $T(-1)$ is isomorphic to a subquotient of $H^k(Y,\mathbb{Q})(j)$ for some k and j. Then* $\dim Y \geq 6$. *Consequently, if* $\dim Y \leq 5$ *and $S = S_\lambda$ is as in Theorem 7.4, every Hodge*

class on $S\times Y$, algebraic or not, acts by zero on $T(S)$, and no link as in Proposition 7.12(ii) is a composite $Z_2\circ Z_1$ of algebraic cycles on $S\times Y$ and on $Y\times X^n$.

Proof. Write $W=T(-1)$. As W is a subquotient of a tensor construction on $H^1(Y)$, $\mathrm{MT}(Y)$ maps onto $\mathrm{MT}(W)$, whose derived group is the adjoint group G^{ad} of G, acting on W through the adjoint representation. So the derived group of $\mathrm{Hg}(Y)$, an almost direct product of $\mathbb{Q}$-simple groups, maps onto G^{ad}. The image of each factor is a normal subgroup of the $\mathbb{Q}$-simple group G^{ad}, the images commute, and G^{ad} is not commutative, so exactly one factor M maps onto G^{ad}, with finite kernel. Over $\bar{\mathbb{Q}}$, M is isogenous to SL_2^3, with factors indexed by the embeddings $\sigma_1,\sigma_2,\sigma_3$ and permuted transitively by the Galois group, and the irreducible constituents of $H^1(Y,\bar{\mathbb{Q}})$ under M are products $R_1\otimes R_2\otimes R_3$ of irreducible representations of the three factors.

Let μ_Y be the Hodge cocharacter of Y, with the two weights 0 and 1 on $H^1(Y)$, and write $\mu_Y=\mu'\mu''$ with fractional cocharacters μ' of M and μ'' commuting with M. The image of μ'' in $\mathrm{MT}(W)$ commutes with G^{ad}, so it is central, and the image of μ' in G^{ad} is the component there of the Hodge cocharacter of W, which by Proposition 7.2(iii) is trivial on the factors at σ_1 and σ_2 and has weights $-1,0,1$ on T_3. So μ' is trivial on the first two factors of M and has weights $\pm\frac{1}{2}$ on the standard representation of the third. On $R_1\otimes R_2\otimes R_3$, tensored with an eigenvector of μ'' in its multiplicity space, the weights of μ_Y therefore fill an interval of length $\dim R_3-1$, so $\dim R_3\le 2$. The constituents form a Galois-stable multiset and the Galois group carries the third factor to each of the others, so every constituent is $V_I=\bigotimes_{i\in I}V_i$ for a subset $I\subseteq\{1,2,3\}$, with a multiplicity m_I that depends only on $|I|$, and $m_I\neq 0$ for some nonempty I, since M acts nontrivially.

If $m_I\neq 0$ for some I with $|I|=2$, then $\dim H^1(Y)\ge 3\cdot 4=12$. If $m_{\{1\}}\neq 0$, let $U=V_1\otimes N$, $\dim N=m_{\{1\}}$, be the V_1-isotypic summand of $H^1(Y,\mathbb{C})$. The embedding σ_1 is real, so complex conjugation preserves the first factor of M and U is defined over $\mathbb{R}$. The connected group $\mathrm{MT}(Y)$ acts on M preserving each factor, and SL_2 has a single irreducible representation of dimension two, so U is stable under $\mathrm{MT}(Y)(\mathbb{R})$ and is a real sub-Hodge structure of $H^1(Y,\mathbb{R})$, and $U^{1,0}$ and $U^{0,1}$ are complex conjugate and both nonzero. Since μ' acts trivially on U, μ_Y acts on $V_1\otimes N$ through N, so N carries both weights, $m_{\{1\}}\ge 2$, and $\dim H^1(Y)\ge 3\cdot 2\cdot 2=12$. Otherwise $H^1(Y,\bar{\mathbb{Q}})$ is $V^{\oplus m}$, $m\ge 1$, plus a trivial representation of M. If $m=1$, every $H^k(Y)$ is, as a representation of M, a sum of copies of the $\bigwedge^a V$, none of which contains an S^2V_i (Proposition 7.6(i)), while W contains all three; so $m\ge 2$ and $\dim H^1(Y)\ge 16$. In every case $\dim Y=\frac{1}{2}\dim H^1(Y)\ge 6$.

For the rest, a Hodge class on $S\times Y$ acting nontrivially on the irreducible $T(S)\cong W$ carries it isomorphically onto a sub-Hodge structure of some $H^k(Y,\mathbb{Q})(j)$, and if $Z_2\circ Z_1$ acts nontrivially on $T(S)$, so does Z_1. ∎

Among the powers of X the smallest carrier of T is $X\times X$, of dimension eight, where $T(-1)$ sits in H^2 (Proposition 7.6(ii)). Whatever is known about abelian varieties of dimension at most five, in particular Markman's theorem in dimension four [Mar25a], no link passes through one.

Remark 7.14 (The bound six as a statement about Hodge structures). The bound of Proposition 7.13 is, as far as we know, the best that Hodge theory gives. Let $K\subset D$ be a CM quadratic extension of F, χ_i the character of the i-th factor of the torus $\mathrm{Res}_{F/\mathbb{Q}}K^1$ over $\bar{\mathbb{Q}}$, and $H=\bigoplus_i(V_i\chi_i\oplus V_i\chi_i^{-1})$, of dimension twelve, with types $(1,0)$ on $V_i\chi_i$ and $(0,1)$ on $V_i\chi_i^{-1}$ for $i=1,2$, and with the Hodge cocharacter of V_3 on the two summands with $i=3$: the signature $(2,0)$, $(2,0)$, $(1,1)$. The characters $\chi_i^{\pm1}\chi_j^{\pm1}$ with $i\neq j$ and $\chi_i^{\pm2}$ are nontrivial,

so the part of $\bigwedge^2 H$ on which the torus acts trivially is $\bigoplus_i V_i\chi_i \otimes V_i\chi_i^{-1} = \bigoplus_i (S^2V_i \oplus \bigwedge^2 V_i)$, that is, $T_1 \oplus T_2 \oplus T_3$ plus three trivial summands. For $i = 1, 2$ the summand T_i is of type $(1, 1)$, and S^2V_3 has one dimension in each of the types $(2, 0)$, $(1, 1)$, $(0, 2)$, so this copy of T has Hodge numbers $(1, 7, 1)$, those of $T(S_\lambda)$. We recall without proof that these are the Hodge structures on $H^1(Z, \mathbb{Q}) \cong D$ of the abelian sixfolds Z with multiplication by K over the Shimura curves of PEL type attached to K and D, which go back to Shimura, and that at a suitable member the copy of T is isomorphic to $T(S_\lambda)$, the Hodge structure on it depending only on the image of the Hodge cocharacter in G^{ad}. Nothing in the paper uses this, and it would not give a link: that would still need an algebraic cycle on $S_\lambda \times Z$ acting nontrivially on $T(S_\lambda)$.

Remark 7.15 (The K3 surfaces at the CM points). Apply the construction of Theorem 7.4(i) at a CM point c of a family as in Corollary 6.22, with the Hodge structure that X_c induces on T, and call the surface S_c. By the proof of Theorem 6.29 the Hodge group of X_c is $\mathrm{Res}_{F/\mathbb{Q}} L^1$ for a quadratic extension L of F contained in D; L is a CM field, since the torus is compact at σ_1 and σ_2, where D is definite, and at σ_3, where it contains the nontrivial image of the circle of the Hodge structure. The torus acts on $T = D^0$ by conjugation, with invariants $D^0 \cap L$, the three-dimensional space L^0 of elements of L of reduced trace zero. So $\mathrm{NS}(S_c)_\mathbb{Q} = T^\perp \oplus L^0$ has rank sixteen. Write $D = L \oplus Lj$ with $jx = \bar{x}j$ for $x \in L$. Then Lj is the q_λ-orthogonal complement of L^0 in D^0, because $\mathrm{trd}(yj) = 0$ for $y \in L$, so $T(S_c)_\mathbb{Q} = Lj$ has rank six; $t \in L^1$ acts on it by multiplication by t^2, so left multiplication by L consists of Hodge endomorphisms and $T(S_c)$ has complex multiplication by the sextic CM field L. At the CM points the Kuga–Satake question therefore lies among K3 surfaces of Picard number sixteen, among which the known cases recalled in Remark 7.5 are particular families. Membership of some S_c in one of them would not help: at c the exceptional classes are already algebraic (Theorem 6.29), and what is missing is the propagation along C.

The last reduction replaces the eightfold $X \times X$ by a fourfold.

Proposition 7.16 (A surface in X)**.** *In Setup 7.1, let $i\colon \Sigma \to X$ be a smooth surface that is the intersection of two ample divisors, and L the cup product with θ. Then $i_*i^* = cL^2$ on $H^2(X, \mathbb{Q})$ for a rational number $c > 0$. For $a \in F$ let ρ_a be the Hodge endomorphism of $H^2(X, \mathbb{Q})$ that is zero on $\mathbb{Q}\theta$ and acts on U as a (Proposition 7.2(ii)), and put $\rho'_a = c^{-1}i^*\rho_a L^{-2}i_*$, a Hodge endomorphism of $H^2(\Sigma, \mathbb{Q})$, where L^{-2} inverts $L^2\colon H^2(X) \to H^6(X)$. Then ρ_a is induced by an algebraic cycle on $X \times X$ if and only if ρ'_a is induced by an algebraic cycle on $\Sigma \times \Sigma$. So the two exceptional classes of $X \times X$ are algebraic if and only if the Hodge classes in $H^2(\Sigma) \otimes H^2(\Sigma) \subset H^4(\Sigma \times \Sigma, \mathbb{Q})$ that correspond to the ρ'_a, $a \in F$, are algebraic on the fourfold $\Sigma \times \Sigma$.*

Proof. By Proposition 6.74(i) the two divisors have classes $d_1\theta$ and $d_2\theta$ with $d_1, d_2 > 0$, so $i_*i^*\alpha = \alpha \cup [\Sigma] = d_1d_2\theta^2\alpha$ and $c = d_1d_2$. By hard Lefschetz $L^2\colon H^2(X) \to H^6(X)$ is an isomorphism, so i^* is injective on $H^2(X)$, and L^{-2} is induced by an algebraic cycle, X being an abelian variety [Kle68, §2]. For $\alpha \in H^2(X)$, $\rho'_a(i^*\alpha) = c^{-1}i^*\rho_a L^{-2}(cL^2\alpha) = i^*\rho_a\alpha$. If $\beta \in H^2(\Sigma)$ is orthogonal to $i^*H^2(X)$, then $\int_X i_*\beta \cup \gamma = \int_\Sigma \beta \cup i^*\gamma = 0$ for every $\gamma \in H^2(X)$, so $i_*\beta = 0$ and $\rho'_a\beta = 0$. Hence ρ'_a is $i^*\rho_a(i^*)^{-1}$ on $i^*H^2(X)$ and zero on its orthogonal complement, and $\rho_a = c^{-1}L^{-2}i_*\rho'_a i^*$. Both formulas are composites of correspondences with the graph of i, its transpose and L^{-2}, so ρ_a is algebraic if and only if ρ'_a is. The exceptional classes are algebraic if and only if every ρ_a is, by Proposition 7.2(ii), since 1 and the projection onto $\mathbb{Q}\theta$ are algebraic. The last statement uses Poincaré duality on Σ and the algebraicity of the Künneth projectors of a surface [Mur90]. ∎

This is the usual reduction to a complete intersection surface, and it does not make the problem easier. By adjunction K_Σ is the restriction of the sum of the two ample divisors, so Σ is a surface of general type, and the ρ'_a are Hodge classes of degree four on the self-product of a surface of general type.

Remark 7.17 (The forms of the first case of (F2)). Theorem 7.7, Corollary 7.11 and one direction of Proposition 7.12 are implications between cases of the Hodge conjecture; Proposition 7.9 follows from Markman's theorem and is also contained in [Li26]; the rest of the results from Proposition 7.6 to Proposition 7.16 are statements about Hodge structures and correspondences. Together they give the first case of (F2) many equivalent forms. By Corollary 6.30, Corollary 7.10 and Proposition 7.16 it is equivalent to each of the following: the algebraicity of the exceptional classes of $X_t \times X_t$ for uncountably many $t \in C$; the Hodge conjecture for every power of every fibre X_t; $B(W \times_C W)$; B for every fibre power of W; the Hodge conjecture for $W \times_C W$, for the curves of Corollary 6.30(d); and, for uncountably many t, the algebraicity of the classes ρ'_a on $\Sigma_t \times \Sigma_t$ for one complete intersection surface $\Sigma_t \subset X_t$. For a family defined over a number field it is also equivalent to the algebraicity, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, of one cycle on one power of X_t whose class is not fixed by the cyclic permutation of the three tensor factors of V (Proposition 7.47). It is implied by the Kuga–Satake Hodge conjecture for the surfaces S_λ attached to uncountably many fibres (Corollary 7.45(ii)), which by Proposition 7.12 is equivalent to it together with a link; the cases of the Kuga–Satake Hodge conjecture recalled in Remark 7.5 are of other shapes, and none has transcendental rank nine with real multiplication by a cubic field.

Two limits govern the constructions at hand. First, the Kuga–Satake class has a nonzero component in $T(S) \otimes H^*(A)$, A a power of X, whereas a product $p^*\alpha \cup q^*\beta$ of pull-backs of algebraic classes of S and of A has none, the algebraic classes of S lying in $H^0 \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus H^4$; so no product of pull-backs gives it, whatever is known on A, and a new correspondence between S and an abelian variety is needed. Second, $\mathrm{Sp}(V,\psi)$ acts on $H^*(X^m,\mathbb{Q}) = \bigwedge^*(V \otimes \mathbb{Q}^m)$ through V, compatibly with products, with pull-back and push-forward along homomorphisms and with integration, hence with the composition of correspondences. The divisor classes of X^m, the Hodge classes of X and the Weil classes of every abelian variety isogenous to a power of X are invariant under it: for a CM field $E \subset M_m(\mathbb{Q}) = \mathrm{End}^0(X^m)$, each component over $\mathbb{C}$ of the top exterior power over E of $V \otimes \mathbb{Q}^m$ is the top exterior power of $V \otimes U$ for a complex vector space U on which V plays no part, that is $(\det V)^{\otimes \dim U} \otimes (\det U)^{\otimes 8}$, and $\mathrm{SL}(V)$ acts trivially on it. So every class obtained from these by the operations above is $\mathrm{Sp}(V,\psi)$-invariant, while the exceptional classes are not, the $\mathrm{Sp}(V,\psi)$-invariants of $H^4(X \times X,\mathbb{Q})$ being the six products of divisor classes (proof of Proposition 6.74). Proposition 7.48 extends this to every abelian variety, with the Hodge classes of the abelian varieties without isogeny factor X and the Hodge classes of the powers of X fixed by the cyclic permutation of the three tensor factors among the generators, and with a subgroup $G \cdot A_3$ of $\mathrm{Sp}(V,\psi)$ in place of $\mathrm{Sp}(V,\psi)$. What escapes it are classes on products $X^m \times Z$ that are not invariant under $G \cdot A_3$ acting through V, for instance, with $T(-1)$ in $H^2(Z)$ as in Remark 7.14, the Hodge classes of $X \times X \times Z$ that pair the two copies of $T(-1)$.

The statement needed can be taken in either of two forms. One is (V) along C for the exceptional classes, from a CM point, where they are algebraic (Theorem 6.29), and Section 7.2 studies it. The other is (V) for the Kuga–Satake class along the seven-dimensional family of K3 surfaces whose Néron–Severi group contains $\mathrm{NS}(S_\lambda)$, on which it stays a Hodge class, from a member at which it is algebraic; it implies the Kuga–Satake Hodge conjecture for S_λ and with it everything above.

The Kuga–Satake class is algebraic at some members of this family. Call the family $\mathcal{K}$, keep the embedding $T \subset \Lambda_{\mathbb{Q}}$ of the proof of Theorem 7.4(i), and for $S \in \mathcal{K}$ let κ_S be the Kuga–Satake class of T with the Hodge structure that S induces on it, a Hodge class on $S \times A_S \times A_S$ with $H^1(A_S, \mathbb{Q}) = C^+(T)$; at S_λ it is the class of Theorem 7.4. If $P = T(S)_{\mathbb{Q}}$ is smaller than T, then $N = P^\perp \cap T$ lies in $\mathrm{NS}(S)_{\mathbb{Q}}$. The complex structure on $C^+(T)$ is left multiplication by an element of $C^+(P \otimes \mathbb{R})$, which commutes with right multiplications; so $C^+(T) = C^+(P) \otimes C^+(N) \oplus C^-(P) \otimes C^-(N)$ is, as a Hodge structure, a sum of copies of $C^+(P)$, right multiplication by an anisotropic vector of P carrying $C^-(P)$ onto $C^+(P)$, and the map Ψ of [I, Proposition 11.7], with $v_0 \in P$, carries N to Hodge endomorphisms of A_S, left multiplication by N commuting with that of $C^+(P \otimes \mathbb{R})$. So κ_S is algebraic if and only if the Kuga–Satake class of P is. By the argument recalled in Remark 7.5, κ_S is therefore algebraic at the members of Picard number at least 19, which are dense in $\mathcal{K}$, at those of Picard number 18 with $T(S)_{\mathbb{Q}}$ isotropic and at those of Picard number 17 with $T(S)_{\mathbb{Q}}$ of Witt index two. In the last two cases $T(S)_{\mathbb{Q}} \cong L_{\mathbb{Q}}$ for an even lattice $L = U \oplus \langle 2a \rangle \oplus \langle -2b \rangle$ or $U^2 \oplus \langle -2b \rangle$, which embeds primitively in $U^3 \subset \Lambda$. The surjectivity of the period map gives a K3 surface S' with $T(S') = L$, which has a Shioda–Inose structure [Mor84, Corollary 6.4], and a Hodge isometry $T(S)_{\mathbb{Q}} \cong T(S')_{\mathbb{Q}}$; by Witt's theorem it extends to a rational Hodge isometry $H^2(S, \mathbb{Q}) \cong H^2(S', \mathbb{Q})$, which is algebraic by [Bus19]. The case of Picard number 17 fills subfamilies of dimension three. The space T has Witt index two: its signature $(2, 7)$ allows no totally isotropic subspace of dimension three, and an anisotropic kernel of dimension at least seven would be indefinite, hence isotropic by the Hasse–Minkowski theorem [OMe00]. So $T = H \oplus H \oplus K$ with H a hyperbolic plane and K negative definite, and each of the countably many rational subspaces $W \subset T$ of signature $(2, 3)$ and Witt index two, for instance $H \oplus H \oplus \mathbb{Q}k$ with $k \in K$, defines a subfamily $\mathcal{K}_W = \{S : T(S)_{\mathbb{Q}} \subseteq W\}$ of dimension three. At a very general point of a component of $\mathcal{K}_W$ one has $T(S)_{\mathbb{Q}} = W$, so κ is algebraic there. After a finite cover with level structure the component carries a family of K3 surfaces and of Kuga–Satake abelian varieties, on which κ is a flat section of Hodge classes, and, as in the proof of Lemma 6.27, the points at which it is algebraic form a countable union of images of components of relative Hilbert schemes, closed by properness; by the Baire category theorem one of them is the whole component, so κ is algebraic at every point of $\mathcal{K}_W$. The Mumford curve meets none of these loci: its surfaces have Picard number thirteen, and sixteen at the CM points (Theorem 7.4(i), Remark 7.15). What is missing is (V) for κ along $\mathcal{K}$ from them, or a bound on the Hilbert data of cycles representing κ over a Zariski dense union of the $\mathcal{K}_W$, which would give κ on the whole of $\mathcal{K}$ by the same argument; neither is known.

7.2. **Rigidity and a numerical criterion for the Mumford classes.** The first case of (F2) is a propagation problem along one compact Shimura curve: the exceptional classes are algebraic at every CM point (Theorem 6.29), and they are algebraic on every member exactly when $B(W \times_C W)$ holds (Corollary 6.30). We ask here whether the propagation can be obtained without the Lefschetz standard conjecture, by one of the mechanisms that have proved other cases of the Hodge conjecture: degeneration, deformation into a larger family on which the class stays of Hodge type, specialisation to special points, or semiregularity. The first two cannot work; the third works exactly when the representing cycles have bounded degree; and the fourth reduces to one dimension count at one CM point. We do not prove that the classes are algebraic at a very general point.

Setup 7.18 (The classes ω_a)**.** Keep Setup 7.1 and let $W \to C$ be a family as in Corollary 6.22, with fibres X_t. Over $\bar{\mathbb{Q}}$ let $\pi_0, \pi_{12}, \pi_{13}, \pi_{23} \in H^2(X) \otimes H^2(X) \subset H^4(X \times X)$ be the classes that correspond, under the isomorphism $H^2 \otimes H^2 \cong \operatorname{End} H^2$ given by the form $\bigwedge^2 \psi$ on $H^2(X) = \bigwedge^2 V^\vee$, to the projections onto $\mathbb{Q}\theta$, U_{12}, U_{13} and U_{23} of Proposition 7.2(i), and for $a = (a_0, a_1, a_2, a_3)$ put

$$\omega_a = a_0\pi_0 + a_1\pi_{12} + a_2\pi_{13} + a_3\pi_{23}, \qquad L_0(a) = a_0 + a_1 + a_2 + a_3.$$

The four tensors are invariant under G and under the exchange τ of the two factors of $X \times X$, because $\bigwedge^2 \psi$ is symmetric; so each ω_a is a flat section over C of Hodge type at every point. The argument of Proposition 7.2(ii) shows that the rational ones are those with $a_0 \in \mathbb{Q}$ and $(a_1, a_2, a_3) = (\sigma_3(\alpha), \sigma_2(\alpha), \sigma_1(\alpha))$ for some $\alpha \in F$; we call such a class *exceptional* when $\alpha \notin \mathbb{Q}$. The products of divisor classes in $H^2 \otimes H^2$ are the rational ω_a with $\alpha \in \mathbb{Q}$, and π_0 is a multiple of $\mathrm{pr}_1^*\theta \cup \mathrm{pr}_2^*\theta$; so adding a rational multiple of π_0 to an exceptional class changes L_0 and not the question of algebraicity. For $c \in C$ write $Y = X_c \times X_c$, let $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$, of dimension $28 + 64 + 28 = 120$, act on $H^*(Y, \mathbb{C})$ by contraction as in Setup 4.54, and let $\kappa \in H^1(T_Y)$ be the Kodaira–Spencer class at c of the family $W \times_C W \to C$. Put

$$e(\omega) = \dim HT^2(Y) - \dim \operatorname{Ann}_{HT^2(Y)}(\omega) = 120 - \dim \operatorname{Ann}_{HT^2(Y)}(\omega).$$

Lemma 7.19 (Linear relations among conjugates)**.** *Let F be a cubic number field with embeddings $\sigma_1, \sigma_2, \sigma_3$, let $\alpha \in F \setminus \mathbb{Q}$, and let $c_0, c_1, c_2, c_3, a_0 \in \mathbb{Q}$. The three conjugates $\sigma_i(\alpha)$ are pairwise distinct and nonzero, and if $c_0a_0 + \sum_i c_i\sigma_i(\alpha) = 0$ then $c_1 = c_2 = c_3$.*

Proof. The minimal polynomial of α has degree three, since $[F : \mathbb{Q}]$ is prime, and is separable, so the conjugates are distinct; they are nonzero because $\alpha \neq 0$. Let Γ be the Galois group of a normal closure; it permutes the σ_i transitively by $\gamma \circ \sigma_i = \sigma_{\gamma(i)}$. Put $\beta = \alpha - \operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3$, so that $\beta \neq 0$ has trace zero, and $r = -c_0a_0 - (c_1 + c_2 + c_3)\operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3 \in \mathbb{Q}$; the hypothesis reads $\sum_i c_i\sigma_i(\beta) = r$. Applying every $\gamma \in \Gamma$ and averaging, each σ_j occurs $|\Gamma|/3$ times in the place of each σ_i, so $r = \frac{1}{3}(\sum_i c_i)\sum_j \sigma_j(\beta) = 0$. The relations $c \in \mathbb{Q}^3$ with $\sum_i c_i\sigma_i(\beta) = 0$ form a Γ-stable subspace N containing $(1, 1, 1)$. The plane $\{\sum c_i = 0\}$ is an irreducible representation of Γ over $\mathbb{Q}$: being transitive on three letters, Γ is the alternating or the full symmetric group, so it contains the 3-cycles, which have no rational eigenvector in that plane. If N were larger than the line of $(1, 1, 1)$, it would therefore contain the plane and be all of $\mathbb{Q}^3$, and $\sigma_1(\beta) = 0$. Hence $c \in \mathbb{Q}(1, 1, 1)$. ∎

Lemma 7.20 (One exceptional class gives the others)**.** *Let $t \in C$. If one rational exceptional class ω is algebraic on $X_t \times X_t$, then every Hodge class in $H^2(X_t) \otimes H^2(X_t)$ is algebraic; in particular so are the two exceptional classes of Proposition 6.74.*

Proof. At a CM point every Hodge class of $X_t \times X_t$ is algebraic (Theorem 6.29). Let t be any other point, so that the Hodge group of X_t is G (proof of Corollary 6.30), and write $X = X_t$. For $x \in H^2(X) \otimes H^2(X)$ let E_x be the endomorphism $y \mapsto x_*(\theta^2 \cup y)$ of $H^2(X, \mathbb{Q})$, where x_* is the action of x as a correspondence. By Poincaré duality and the hard Lefschetz isomorphism $\theta^2 \colon H^2 \to H^6$, the map $x \mapsto E_x$ is a G-equivariant isomorphism onto $\operatorname{End} H^2(X, \mathbb{Q})$, so it carries the Hodge classes onto the Hodge endomorphisms, which are $\mathbb{Q} \times F$ by Proposition 7.2(ii). It carries the products of divisor classes, which are invariant under $\mathrm{Sp}(V, \psi)$, onto the $\mathrm{Sp}(V, \psi)$-equivariant endomorphisms, which are $\mathbb{Q} \times \mathbb{Q}$; hence it carries ω to an element (r, β) with $\beta \notin \mathbb{Q}$. The algebraic Hodge classes form a subspace A containing the products of divisor classes, and its image under $x \mapsto E_x$ is closed under composition: if x_1, x_2 are algebraic, then the composite correspondence

$x_1 \circ \Delta_*(\theta^2) \circ x_2$ is algebraic, and so is its Künneth component in $H^2 \otimes H^2$, since the Künneth projectors of $X \times X$ are polynomials in the graphs of $[2] \times 1$ and $1 \times [2]$, which act on $H^i(X) \otimes H^j(X)$ by 2^i and 2^j (as in Step 1 of the proof of Theorem 6.21); that component x has $E_x = E_{x_1} E_{x_2}$. So the image of A is a subalgebra of $\mathbb{Q} \times F$ containing $\mathbb{Q} \times \mathbb{Q}$ and (r, β), hence $\mathbb{Q} \times \mathbb{Q}[\beta] = \mathbb{Q} \times F$, because β generates the cubic field F. Thus A is the space of all Hodge classes in $H^2(X) \otimes H^2(X)$. ∎

Theorem 7.21 (Rigidity of the exceptional classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, and let $\omega = \omega_a$ be a rational exceptional class.*

(i) $\operatorname{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$ *and* $\operatorname{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

(ii) $\operatorname{Ann}_{H^0(\bigwedge^2 T_Y)}(\omega) = 0$ *if and only if* $L_0(a) = a_0 + \operatorname{Tr}_{F/\mathbb{Q}}(\alpha) \neq 0$. *In particular* $\operatorname{Ann}_{HT^2(Y)}(\omega) = \mathbb{C}\kappa$ *and* $e(\omega) = 119$ *whenever* $L_0(a) \neq 0$, *which can always be arranged by adding a rational multiple of* π_0.

(iii) *Let $\mathfrak{S}$ be the Siegel space of weight one Hodge structures on $H^1(Y, \mathbb{Q})$ polarised by $\psi \oplus \psi$, of dimension* 36. *The germ at Y of the locus in $\mathfrak{S}$ where ω stays of Hodge type is the germ of the curve $\{X_t \times X_t\}$ through Y.*

Proof. Notation. Over $\mathbb{C}$ write $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, as in the proof of Lemma 7.3, and identify V with $H^1(X_c, \mathbb{C})$ and with its dual through ψ. As at every point of C, the Hodge cocharacter lies in the third factor: $V_3^{1,0} = \mathbb{C}e$, $V_3^{0,1} = \mathbb{C}f$, $V^{1,0} = V_1 \otimes V_2 \otimes e$ and $V^{0,1} = V_1 \otimes V_2 \otimes f$. Let $N \in \mathfrak{sl}(V_3)$ be the nilpotent element with $Ne = f$, $Nf = 0$. Then $H^1(Y, \mathbb{C}) = V_{(1)} \oplus V_{(2)}$, two copies of V, with the form $\psi \oplus \psi$, and $\omega \in \bigwedge^2 V_{(1)} \otimes \bigwedge^2 V_{(2)}$.

Step 1: operators. For $J \subseteq \{1, 2, 3\}$ let $E_J \subset \operatorname{End} V$ be the tensor product of $\mathfrak{sl}(V_i)$ for $i \in J$ and of $\mathbb{C} \cdot 1$ for $i \notin J$. Then $\operatorname{End} V = \bigoplus_J E_J$, and the E_J are irreducible and pairwise non-isomorphic under conjugation by G. The ψ-adjoint of an element of $\mathfrak{sl}(V_i)$ is its negative, so E_J consists of ψ-skew operators when $|J|$ is odd and of ψ-symmetric ones when $|J|$ is even. The map $v \wedge w \mapsto \psi(v, \cdot)w - \psi(w, \cdot)v$ identifies $\bigwedge^2 V$ with the ψ-symmetric operators $E_\emptyset \oplus E_{12} \oplus E_{13} \oplus E_{23}$, which are $\mathbb{C}\theta$, U_{12}, U_{13}, U_{23}; a direct expansion shows that $\bigwedge^2 \psi$ becomes the form $\langle M, M' \rangle = \frac{1}{2} \operatorname{tr}(MM')$, and that the derivation action of $X \in \operatorname{End} V$ becomes $M \mapsto XM + MX^\psi$, that is $D_X M = [X, M]$ if X is ψ-skew and $D_X M = XM + MX$ if X is ψ-symmetric. In particular $D_X^\dagger = \mp D_X$ in the two cases, $\dagger$ denoting the adjoint for $\langle\,,\rangle$. By the Cayley–Hamilton theorem $XY = \frac{1}{2}[X, Y] + \frac{1}{2} \operatorname{tr}(XY)$ for $X, Y \in \mathfrak{sl}(V_i)$. Finally, by Setup 7.18, $\omega = \sum_j M_j \otimes P_a M^j$ for a basis (M_j) of $\bigwedge^2 V$ with dual basis (M^j), where P_a acts on $E_\emptyset, E_{12}, E_{13}, E_{23}$ by a_0, a_1, a_2, a_3. As P_a commutes with G, it preserves the Hodge types.

The space $\bigwedge^2 V^{0,1}$ consists of the ψ-symmetric operators that kill $V^{0,1}$ and take values in it, namely $\mathfrak{sl}(V_1) \otimes 1 \otimes N \oplus 1 \otimes \mathfrak{sl}(V_2) \otimes N \subset E_{13} \oplus E_{23}$, on which P_a is a_2 and a_3. The map $v \otimes w \mapsto \psi(v, \cdot)w$ identifies $V^{0,1} \otimes V^{0,1}$ with $\operatorname{Hom}(V^{1,0}, V^{0,1}) = \operatorname{End} V_1 \otimes \operatorname{End} V_2 \otimes \mathbb{C}N = \bigoplus_{J \ni 3} E_J^-$, where $E_J^- = E_J \cap \operatorname{Hom}(V^{1,0}, V^{0,1})$ has dimension $1, 3, 3, 9$ for $J = \{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$.

Step 2: the pieces. The space $HT^2(Y)$ acts as follows: $H^{0,2} = \bigwedge^2 H^{0,1}$ by the wedge product, $H^1(T_Y) = \operatorname{Hom}(H^{1,0}, H^{0,1})$ by derivations, and $H^0(\bigwedge^2 T_Y) = \bigwedge^2 (H^{1,0})^\vee$, which $\psi \oplus \psi$ identifies with $\bigwedge^2 H^{0,1}$, by contraction. Write $\eta \in \bigwedge^2 H^{0,1}$ as $\eta_{11} + \eta_{12} + \eta_{22}$ with $\eta_{11} \in \bigwedge^2 V_{(1)}^{0,1}$, $\eta_{12} \in V_{(1)}^{0,1} \otimes V_{(2)}^{0,1}$, $\eta_{22} \in \bigwedge^2 V_{(2)}^{0,1}$, and $\xi \in H^1(T_Y)$ as a matrix (ξ_{st}) with $\xi_{st} \in \operatorname{Hom}(V_{(t)}^{1,0}, V_{(s)}^{0,1})$. The products $\eta_{11} \wedge \omega$, $\eta_{12} \wedge \omega$, $\eta_{22} \wedge \omega$ lie in different summands $\bigwedge^p V_{(1)} \otimes \bigwedge^q V_{(2)}$, and so do the three contractions, and $\xi_{11}\omega + \xi_{22}\omega$, $\xi_{12}\omega$, $\xi_{21}\omega$. So each annihilator is computed piece by piece, and τ exchanges the pieces 11 and 22, 12 and 21.

For a rational exceptional class, Lemma 7.19 gives: a_1, a_2, a_3 are pairwise distinct and nonzero, and no linear form in $a_0, \dots, a_3$ with rational coefficients that are not all equal on a_1, a_2, a_3 vanishes. We use this for $a_i \pm a_j$ $(i \neq j)$, $a_2 + 3a_3$, $3a_2 + a_3$ and the three forms in Step 4.

Step 3: $H^{0,2}$. Let (x_j) be a basis of $\bigwedge^2 V^{1,0}$ and (x^j) the dual basis of $\bigwedge^2 V^{0,1}$. The component of ω in $\bigwedge^2 V^{1,0}_{(1)} \otimes \bigwedge^2 V^{0,1}_{(2)}$ is $\sum_j x_j \otimes K x^j$, where $K = P_a$ on $\bigwedge^2 V^{0,1}$ is invertible because $a_2 a_3 \neq 0$. For $\eta_{11} \neq 0$ the part of $\eta_{11} \wedge \omega$ in $H^{2,2}(X_c) \otimes H^{0,2}(X_c)$ is $\sum_j (\eta_{11} \wedge x_j) \otimes K x^j$, which is nonzero because $\bigwedge^2 V^{0,1} \otimes \bigwedge^2 V^{1,0} \to \bigwedge^4 V$ is injective; by τ the same holds for η_{22}. For $\eta_{12} = \sum_r v_r \otimes w_r$ the part of $\eta_{12} \wedge \omega$ in $H^{2,1} \otimes H^{0,3}$ is $\sum_{r,j} (v_r \wedge x_j) \otimes (w_r \wedge K x^j)$. If it vanishes, then, as $V^{0,1} \otimes \bigwedge^2 V^{1,0} \to \bigwedge^3 V$ is injective and the $K x^j$ span $\bigwedge^2 V^{0,1}$, $\sum_r \phi(v_r)\, w_r \wedge y = 0$ for every linear form ϕ on $V^{0,1}$ and every $y \in \bigwedge^2 V^{0,1}$; as $\dim V^{0,1} = 4$ this forces $\sum_r \phi(v_r) w_r = 0$ for every ϕ, so $\eta_{12} = 0$. Hence $\mathrm{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

Step 4: $H^0(\bigwedge^2 T_Y)$. Up to signs and one nonzero factor fixed by the normalisation of the contraction, $\iota_{\eta_{11}}\omega = \sum_j \langle \eta_{11}, M_j \rangle P_a M^j = P_a \eta_{11}$, which is nonzero for $\eta_{11} \neq 0$ because $a_2 a_3 \neq 0$; likewise for η_{22}. For $\eta_{12} = v \otimes w$ the contraction is $\sum_j M_j v \otimes (P_a M^j) w$, and through $v \otimes w \mapsto \psi(v, \cdot) w$ it becomes the operator $Q_a(R) = \sum_j (P_a M^j)\, R\, M_j$ with $R = \psi(v, \cdot) w$, because the M_j are ψ-symmetric. For a basis (X_k) of $\mathfrak{sl}(V_i)$ and the dual basis (X^k) for the trace form, $\sum_k X^k R X_k$ is $\frac{3}{2} R$ for $R = 1$ and $-\frac{1}{2} R$ for $R \in \mathfrak{sl}(V_i)$ (take $X_k = h, e, f$ and $X^k = h/2, f, e$). The dual basis of E_{ij} for $\langle\, , \rangle$ is the tensor product of these dual bases, and that of $E_\emptyset$ is $\frac{1}{4}$. Hence Q_a acts on E_J by the scalar

$$q_J(a) = \tfrac{1}{4} a_0 + \sum_{\{i,j\}} a_{\{i,j\}}\, \lambda_i \lambda_j, \qquad \lambda_i = -\tfrac{1}{2}\ (i \in J), \quad \lambda_i = \tfrac{3}{2}\ (i \notin J),$$

where $a_{\{1,2\}} = a_1$, $a_{\{1,3\}} = a_2$, $a_{\{2,3\}} = a_3$; on the four pieces E_J^- this is

$$4q_3 = a_0 + 9a_1 - 3a_2 - 3a_3, \qquad 4q_{13} = a_0 - 3a_1 + a_2 - 3a_3,$$
$$4q_{23} = a_0 - 3a_1 - 3a_2 + a_3, \qquad 4q_{123} = L_0(a).$$

The first three do not vanish. So the annihilator in $H^0(\bigwedge^2 T_Y)$ is zero when $L_0(a) \neq 0$, and is the piece E^-_{123} of η_{12}, of dimension nine, when $L_0(a) = 0$.

Step 5: $H^1(T_Y)$, the diagonal pieces. Put $\Phi(x \otimes y)(z) = \langle z, x \rangle y$, so that $\Phi(\omega) = P_a$, $\Phi((D \otimes 1)\omega) = P_a D^\dagger$ and $\Phi((1 \otimes D)\omega) = D P_a$. Thus $\xi_{11}\omega + \xi_{22}\omega = 0$ if and only if $P_a D^\dagger_{\xi_{11}} + D_{\xi_{22}} P_a = 0$. This condition is G-equivariant in $(\xi_{11}, \xi_{22}) \in \mathrm{End}\, V \oplus \mathrm{End}\, V$, so its solutions are the pairs $(b_{11} X, b_{22} X)$ with $X \in E_J$ and (b_{11}, b_{22}) in a subspace $M_J \subseteq \mathbb{C}^2$; we need $J \ni 3$. For $X = \bigotimes X_i \in E_J$ and $M = \bigotimes M_i \in E_I$, the product rule above gives $XM \pm MX$ as $2^{1-|I \cap J|}$ times the sum, over $T \subseteq I \cap J$ with $|T| \equiv |J|$ modulo 2, of the tensor with factor $[X_i, M_i]$ for $i \in T$, $\mathrm{tr}(X_i M_i)$ for $i \in (I \cap J) \setminus T$, and the remaining factor of X or M elsewhere; this term lies in $E_{(I \triangle J) \cup T}$ and is nonzero for suitable X and M. On the block from E_I to $E_{I'}$ the condition reads $b_{22} a_I = b_{11} a_{I'}$ for $|J|$ odd and $b_{22} a_I = -b_{11} a_{I'}$ for $|J|$ even, with $a_\emptyset = a_0$, $a_{12} = a_1$, $a_{13} = a_2$, $a_{23} = a_3$.

- $J = \{3\}$: the blocks from E_{13} to E_{13} and from E_{23} to E_{23} give $(b_{22} - b_{11}) a_2 = 0$, so $M_J = \mathbb{C}(1, 1)$.
- $J = \{1, 2, 3\}$: there is a block between any two distinct E_{12}, E_{13}, E_{23}, so $b_{22} a_1 = b_{11} a_2$ and $b_{22} a_2 = b_{11} a_1$, whence $(b_{11} + b_{22})(a_1 - a_2) = 0$ and $b_{11} = -b_{22}$; then $b_{22}(a_1 + a_2) = b_{22}(a_1 + a_3) = 0$, and $b_{22} \neq 0$ would give $a_2 = -a_1 = a_3$. So $M_J = 0$.
- $J = \{1, 3\}$: the blocks between E_{12} and E_{23} give $b_{22} a_1 + b_{11} a_3 = b_{22} a_3 + b_{11} a_1 = 0$, of determinant $a_1^2 - a_3^2 \neq 0$, so $M_J = 0$; and $J = \{2, 3\}$ likewise, with $a_1^2 - a_2^2 \neq 0$.

So on the diagonal pieces the annihilator is the line of (N,N), with N acting on both copies of V.

Step 6: $H^1(T_Y)$, the mixed pieces. The map ξ_{12} kills $V^{0,1}_{(2)}$, so the part of $\xi_{12}\omega$ in $H^{0,3}(X_c)\otimes H^{1,0}(X_c)$ comes from the component $\sum_j Kx^j\otimes x_j$ of ω in $\bigwedge^2 V^{0,1}_{(1)}\otimes\bigwedge^2 V^{1,0}_{(2)}$ only, and is $\sum_j Kx^j\wedge\xi_{12}(x'_j)\otimes x''_j$, where $x\mapsto x'\otimes x''$ is the coproduct $\bigwedge^2\to V\otimes V$. Identify $V^{1,0}$ and $V^{0,1}$ with $W=V_1\otimes V_2$ through e and f, with the symmetric form $q=\epsilon_1\otimes\epsilon_2$ and a volume form; then $\bigwedge^2 W$ is the space of q-skew operators $\mathfrak{sl}(V_1)\otimes 1\oplus 1\otimes\mathfrak{sl}(V_2)$, the Hodge star is $+1$ on one summand and -1 on the other, and K is a_2 on the first and a_3 on the second. Pair the part above with $b\in W$ in the first factor, through the volume form, and with $\beta\in W$ in the second, through q. Writing $\mathrm{vol}(y\wedge y')=q(*y,y')$ on $\bigwedge^2 W$ turns the pairing into the trace of $x\mapsto K*\big(\xi_{12}(\iota_\beta x)\wedge b\big)$ on $\bigwedge^2 W$, and summing over orthogonal bases of the two summands gives, up to a nonzero factor and a sign, $q\big(c_a(\xi_{12})\beta,b\big)$ with

$$c_a(\xi)=a_2\sum_k (X^k\otimes 1)\,\xi\,(X_k\otimes 1)-a_3\sum_k(1\otimes X^k)\,\xi\,(1\otimes X_k)$$

for $\xi\in\operatorname{End}W=\operatorname{End}V_1\otimes\operatorname{End}V_2$. By the scalars of Step 4, c_a acts on $1\otimes 1$, $\mathfrak{sl}(V_1)\otimes 1$, $1\otimes\mathfrak{sl}(V_2)$, $\mathfrak{sl}(V_1)\otimes\mathfrak{sl}(V_2)$, that is on E_3^-, E_{13}^-, E_{23}^-, E_{123}^-, by

$$\tfrac{3}{2}(a_2-a_3),\qquad -\tfrac{1}{2}(a_2+3a_3),\qquad \tfrac{1}{2}(3a_2+a_3),\qquad -\tfrac{1}{2}(a_2-a_3),$$

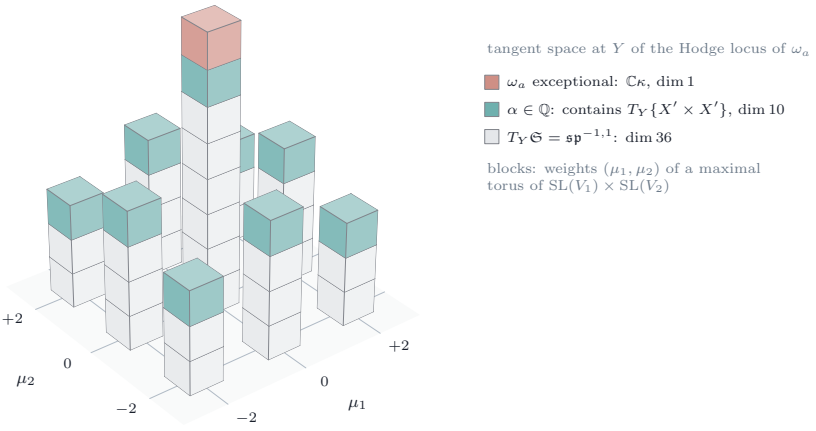
none of which vanishes. So $\xi_{12}\omega\neq 0$ for $\xi_{12}\neq 0$, and by τ the same holds for ξ_{21}.

Step 7: κ. The family of Hodge structures along C is the orbit of the Hodge structure at c under the real points of the third factor of G, so the Kodaira–Spencer class κ is the component of type $(-1,1)$ of an element of $\mathfrak{sl}(V_3)$ acting on both copies of V, a nonzero multiple of (N,N); it is nonzero because the Hodge structures along C vary. By Steps 5 and 6, $\mathrm{Ann}_{H^1(T_Y)}(\omega)=\mathbb{C}\kappa$. With Steps 3 and 4 this proves (i) and (ii).

For (iii), a point of $\mathfrak{S}$ is a Hodge filtration F^1 on $H^1(Y,\mathbb{C})$, and ω is of Hodge type at it when $\omega\in F^2H^4$, a holomorphic condition. For weight one every tangent direction of $\mathfrak{S}$ at Y is an element ξ of $\mathfrak{sp}(\psi\oplus\psi)^{-1,1}\subset H^1(T_Y)$, and it moves F^2H^4 by the derivation ξ of $\bigwedge^* H^1$, which maps $H^{2,2}$ into $H^{1,3}$; so the Zariski tangent space of the Hodge locus is $\{\xi:\xi\lrcorner\,\omega=0\}$, which is $\mathbb{C}\kappa$ by (i). This is the usual description of the tangent space of the Hodge locus of a class of type (p,p) as the kernel of the contraction into $H^{p-1,p+1}$ [Voi03, Chapter 5], in its simplest case: a variation of weight one is subject to no horizontality condition, so every direction of $\mathfrak{S}$ enters. The Hodge locus is an analytic germ whose Zariski tangent space is a line, so it lies in a smooth curve germ; it contains the smooth curve germ $\{X_t\times X_t\}$, whose tangent line is $\mathbb{C}\kappa$; so the two germs are equal. ∎

Remark 7.22 (Comparison with the divisor classes). For $\alpha\in\mathbb{Q}$ the class ω_a is a combination of products of divisor classes and is invariant under the whole of $\mathrm{Sp}(V,\psi)$ acting diagonally on $V\oplus V$, so its Hodge locus in $\mathfrak{S}$ contains the diagonal Siegel space $\{X'\times X'\}$, of dimension 10; in the proof above this shows in the scalars a_2-a_3 of Step 6, which vanish when $a_1=a_2=a_3$. The products of divisor classes therefore move with $X'\times X'$ for every X', whereas the exceptional classes move with $X_t\times X_t$ for $t\in C$ only and cease to be of Hodge type under every deformation of Y off the curve, however small.

Corollary 7.23 (No degeneration and no larger family)**.** *Let ω be a rational exceptional class on $X_c\times X_c$. Every connected complex analytic family of polarised abelian eightfolds through $X_c\times X_c$ on which ω stays of Hodge type lies, near each point of the curve, inside the curve $\{X_t\times X_t : t\in C\}$, which is compact. In particular no such family degenerates,*



	E_3^-, dim 1	E_{13}^-, dim 3	E_{23}^-, dim 3	E_{123}^-, dim 9
$H^1(T_Y)$ diagonal	$b_{11} = b_{22}$: κ	$a_1^2 - a_3^2$	$a_1^2 - a_2^2$	a_1, a_2, a_3 distinct
$H^1(T_Y)$ mixed	$\frac{3}{2}(a_2 - a_3)$	$-\frac{1}{2}(a_2 + 3a_3)$	$\frac{1}{2}(3a_2 + a_3)$	$-\frac{1}{2}(a_2 - a_3)$
$H^0(\bigwedge^2 T_Y)$ mixed	$\frac{1}{4}(a_0 + 9a_1 - 3a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 + a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 - 3a_2 + a_3)$	$\frac{1}{4}L_0(a)$

$H^{0,2}$ and the unmixed pieces of $\bigwedge^2 T_Y$: the block $K = (a_2, a_3)$ of P_a. At a rational exceptional class no entry vanishes, by the lemma on conjugates, except $b_{11} - b_{22}$, which leaves κ, and L_0, which can be made nonzero by adding a multiple of π_0.

FIGURE 34. Theorem 7.21. Top: the tangent space $\mathfrak{sp}^{-1,1}$ of the Siegel space $\mathfrak{S}$ at $Y = X_c \times X_c$, of dimension 36, split into nine blocks by the weights of a maximal torus of $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$, one cube for each dimension. The tangent space of the Hodge locus of an exceptional class is the line $\mathbb{C}\kappa$ of the Mumford curve, in the block of weight $(0, 0)$; for $\alpha \in \mathbb{Q}$ it contains the tangent space of the diagonal Siegel space, of dimension 10 (Remark 7.22). Bottom: the scalars of Steps 4 to 6 of the proof on the pieces E_J^- of $HT^2(Y)$; the pieces of $H^{0,2}$ and the unmixed pieces of $H^0(\bigwedge^2 T_Y)$ are controlled by the block K of P_a (Steps 3 and 4).

and on no member of it other than the CM points of the curve is ω a combination of products of divisor classes or of Weil classes of CM fields acting on the member or on its quotients.

Proof. By Theorem 7.21(iii) the Hodge locus of ω coincides with the curve in a neighbourhood of every point of the curve, so a connected family through Y on which ω stays of Hodge type maps into the curve near each of its points over the curve; by connectedness and the identity theorem its image lies in the curve. The curve is compact because D is a division algebra, so that the arithmetic group of the Shimura curve is cocompact [Mum69]. Indeed, by Setup 7.1 the algebra D is definite at σ_1 and σ_2, so $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and D is not $M_2(F)$: it is a division algebra, split at the one real place σ_3. The norm one group of an order in D, embedded in $\mathrm{SL}_2(\mathbb{R})$ through σ_3, is a Fuchsian group of finite covolume with no parabolic element, since a parabolic $\gamma \neq \pm 1$ would make $\gamma \mp 1$ a nonzero nilpotent element of D; and a Fuchsian group of finite covolume without parabolic elements has compact quotient.

For the last assertion: a member of the curve other than a CM point has Hodge group G (proof of Corollary 6.30), and on it ω lies outside the span of the products of divisor classes and of the Weil classes of quotients, by Proposition 6.74(ii) and (iii). At a CM point the class is already algebraic (Theorem 6.29), and the CM points, being countably many, do not carry algebraicity to the other members (Lemma 6.27 needs uncountably many points). ■

The same annihilator controls deformations that are not algebraic at all, and so decides whether the twistor lines along which Markman moves sheaves in his proof for Weil classes can be used here.

Proposition 7.24 (The Hodge locus among all complex tori, and twistor lines)**.** *Let ω be a rational exceptional class, $Y = X_c \times X_c$ with $c \in C$, and $\Lambda = H_1(Y, \mathbb{Z})$. Let $\mathcal{T}$ be the space of complex structures on the real torus $\Lambda_{\mathbb{R}}/\Lambda$, a complex manifold of dimension 64 with tangent space $H^1(T_Y)$ at Y, and $\mathcal{T}_\omega \subset \mathcal{T}$ the closed analytic subset on which ω is of type $(2,2)$.*

(i) *The connected component of $\mathcal{T}_\omega$ that contains Y is the universal cover $\mathfrak{H}$ of the Mumford curve, the set of squares of members of the family, biholomorphic to the upper half plane. Every connected complex analytic family of complex tori, algebraic or not, through a member of the Mumford family on which ω stays of Hodge type lies in the family.*

(ii) *No twistor line through a member of the Mumford family, and no compact connected complex curve in $\mathcal{T}$ through it, keeps ω of Hodge type.*

(iii) *Nevertheless $\mathcal{T}_\omega$ contains twistor lines. The algebra $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and acts on $V_{\mathbb{R}}$ through the first factor of $G(\mathbb{R})$; for a unit quaternion j with $j^2 = -1$, the complex structure $j \oplus j$ on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ makes ω of type $(2,2)$, and these structures form a twistor line. On each factor $\psi(x, jy)$ has signature $(4,4)$, and a member whose Mumford–Tate group is G is not an abelian variety. These twistor lines lie in components of $\mathcal{T}_\omega$ other than $\mathfrak{H}$.*

Proof. (i) The complex structures on $\Lambda_{\mathbb{R}}$ for which all the tensors invariant under G are of Hodge type are those J with $J^2 = -1$ in the real points of $G \cdot \mathbb{G}_m$, since a reductive group containing the homotheties is the stabiliser of its invariant tensors. Let $\mathfrak{H}$ be the connected component of this closed set that contains Y; it is closed in $\mathcal{T}$ and consists of the $J = 1 \otimes 1 \otimes j_3$ with $j_3 \in \mathrm{SL}_2(\mathbb{R})$, $j_3^2 = -1$, in the half plane that contains the structure of Y, that is, of the squares of the members of the Mumford family. At each of its points the first order condition for ω to stay of type $(2,2)$ along $\xi \in H^1(T_Y)$ is $\xi \lrcorner \omega = 0$, as in the proof of Theorem 7.21(iii), where no polarisation was used, and by Theorem 7.21(i) the solutions form the line $\mathbb{C}\kappa$ tangent to $\mathfrak{H}$. So near each of its points $\mathcal{T}_\omega$ is the smooth curve germ of $\mathfrak{H}$, and $\mathfrak{H}$ is open in $\mathcal{T}_\omega$. Being also closed and connected, it is a connected component. A connected analytic family on which ω stays of Hodge type maps into $\mathcal{T}_\omega$, hence into $\mathfrak{H}$ once one of its points does.

(ii) A twistor line is a nonconstant holomorphic image of $\mathbb{P}^1$ in $\mathcal{T}$. If ω stayed of Hodge type along one through a point of $\mathfrak{H}$, or along any compact connected curve through it, the curve would lie in $\mathfrak{H}$ by (i), and a holomorphic map from a compact connected curve to the upper half plane is constant.

(iii) At σ_1 the quaternion algebra D is definite (Setup 7.1), so the first factor of $G(\mathbb{R})$ is the group of unit quaternions, and $V_{\mathbb{R}}$ restricted to it is a module over $D \otimes_{F,\sigma_1} \mathbb{R} \cong \mathbb{H}$. The unit quaternions j with $j^2 = -1$ form a two-sphere, and the complex structures given

by left multiplication by them on a quaternionic vector space form a twistor line. Each j lies in $G(\mathbb{R})$, and so does the circle $\cos\vartheta + j\sin\vartheta$, which fixes ω; so ω is of type $(2,2)$. Inside $V_1 \otimes V_2 \otimes V_3$ the space $V_{\mathbb{R}}$ is the fixed space of a real structure that is quaternionic on the first two factors and real on the third, so $V_{\mathbb{R}} = U \otimes V_{3,\mathbb{R}}$ with U the real form of $V_1 \otimes V_2$. On U the form $g = \epsilon_1 \otimes \epsilon_2$ is symmetric and invariant under the compact group of pairs of unit quaternions, which acts irreducibly on $U \otimes \mathbb{C} = V_1 \otimes V_2$; so g is definite. The complex structures of the Mumford family are $1 \otimes h$, with h a complex structure on $V_{3,\mathbb{R}}$ for which $\epsilon_3(\cdot, h\cdot)$ is definite, and $\psi(x,(1\otimes h)y) = g \otimes \epsilon_3(\cdot, h\cdot)$ is definite. For $J = j \otimes 1 \otimes 1$ the form $g(\cdot,(j\otimes 1)\cdot)$ is alternating, $j \otimes 1$ being an isometry of g with square -1, and $\psi(x,Jy) = g(\cdot,(j\otimes 1)\cdot) \otimes \epsilon_3$ is the product of nondegenerate alternating forms on spaces of dimensions 4 and 2; as the product of two alternating forms on $\mathbb{R}^2$ is the determinant on $\mathbb{R}^2 \otimes \mathbb{R}^2 = M_2(\mathbb{R})$, of signature $(2,2)$, it has signature $(4,4)$. At a member whose Mumford–Tate group is G, the Hodge classes of degree two on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ are the forms $\psi \otimes s$ with s a symmetric 2×2 matrix, by Proposition 6.74(ii), and the signature of $\psi(\cdot, J\cdot) \otimes s$ is $(4(p+q), 4(p+q))$ when s has signature (p,q), so none is definite, none is a polarisation, and the member is not an abelian variety. The members of the Mumford family are not on these lines, since $\psi(x,Jy)$ is definite at the former and of signature $(4,4)$ at the latter, so by (i) the lines lie in other components. ∎

Markman's proof for Weil classes on abelian fourfolds moves a sheaf along twistor lines on which its Chern character stays of Hodge type. By Proposition 7.24 no such line passes through the square of a member of the Mumford family: twistor lines on which ω is of Hodge type exist, but they carry non-algebraic tori whose Hodge cocharacter lies in a definite factor of D, and they lie in components of the Hodge locus that the Mumford family never meets. The Kuga–Satake class of Theorem 7.4 behaves differently. The Kuga–Satake construction applies to every Hodge structure of K3 type on (T, q_λ), so that class stays of Hodge type over the whole period domain of (T, q_λ), of dimension seven, and not only along the curve; among the classes attached to the target, it is the only one for which deformation methods of this kind are not excluded.

The same space T also makes the square a holomorphic symplectic variety, and this determines what the mechanism of van Geemen and Rapagnetta would have to supply here. They prove the Hodge conjecture for the general abelian fourfold of Weil type with trivial discriminant by pulling back the second Chern class of a hyperkähler sixfold along a map from the fourfold, which gives an algebraic class of codimension two that is not a product of divisor classes [vGR26]. For a projective hyperkähler manifold M write σ_M for its symplectic form, q_M for its Beauville–Bogomolov form and $T(M)$ for the orthogonal complement of $\mathrm{NS}(M)_{\mathbb{Q}}$ in $H^2(M,\mathbb{Q})$ for q_M.

Proposition 7.25 (The square as a holomorphic symplectic variety)**.** *Let $c \in C$ be a point at which the Mumford–Tate group of $X = X_c$ is $G \cdot \mathbb{G}_m$, let $Y = X \times X$, and let $\Psi \in \bigwedge^2 V$ be the G-invariant bivector dual to ψ. For $x \in T$ put $\iota(x) = (x\otimes 1)\Psi$ in $V \otimes V = H^1(X,\mathbb{Q}) \otimes H^1(X,\mathbb{Q})$, a Künneth component of $H^2(Y,\mathbb{Q})$, and let $\iota_2 \colon S^2T \to H^4(Y,\mathbb{Q})$ be the map $xy \mapsto \iota(x) \cup \iota(y)$. Let $C_i \in S^2T_i$ be the Casimir element of the trace form of $\mathfrak{sl}(V_i)$, and for $\nu \in F$ let $c_\nu = \sum_i \sigma_i(\nu) C_i$, a rational element of S^2T; for $\nu \neq 0$ it is the element dual to the form $q_{1/\nu}$.*

(i) *ι is an embedding of Hodge structures $T(-1) \to H^2(Y,\mathbb{Q})$, its image is the only irreducible sub-Hodge structure of $H^2(Y,\mathbb{Q})$ with $h^{2,0} = 1$, and the holomorphic 2-form σ_0 spanning its part of type $(2,0)$ is symplectic.*

(ii) $\iota_2(c_\nu) = \rho\,\omega_a$ *with* $a_0 = 3\,\mathrm{Tr}_{F/\mathbb{Q}}(\nu)$ *and* $\alpha = 4\nu - \mathrm{Tr}_{F/\mathbb{Q}}(\nu)$*, in the notation of Setup 7.18, where* ρ *is a nonzero rational number independent of* ν*. So* $\iota_2(c_\nu)$ *is exceptional if and only if* $\nu \notin \mathbb{Q}$*, and every rational exceptional class is some* $\iota_2(c_\nu)$ *plus a rational multiple of* π_0*.*

(iii) *Let* M *be a projective hyperkähler manifold and* $f\colon Y \dashrightarrow M$ *a rational map with* $f^*\sigma_M \neq 0$*. Then* f^* *maps* $T(M)$ *isomorphically onto* $\iota(T)$ *and is compatible with products of classes in* $T(M)$*;* f *is generically finite onto a subvariety of dimension eight on which* σ_M *is generically nondegenerate;* $b_2(M) \geq 10$*; and* $\dim M \geq 10$*.*

(iv) *Suppose in (iii) that* $M = S^{[n]}$ *is the Hilbert scheme of* n *points on a projective K3 surface* S*, and let* q_λ *be the form on* T *that corresponds to the intersection form of* S *on* $T(S) = T(M)$ *under* $\iota^{-1}f^*$ *(Proposition 7.2(iii)). If* $\lambda \notin \mathbb{Q}$*, or if the real multiplication of* $T(S)$ *is induced by algebraic cycles on* $S \times S$*, then the Hodge conjecture holds for* Y*.*

Proof. (i) The map ι is G-equivariant because Ψ is invariant, and injective. Its image lies in S^2V: for x in the symplectic Lie algebra of ψ one has $(x \otimes 1)\Psi = -(1 \otimes x)\Psi$, and the exchange of the two factors carries $(x \otimes 1)\Psi$ to $-(1 \otimes x)\Psi = (x \otimes 1)\Psi$, because Ψ is alternating. An element g of $G \cdot \mathbb{G}_m$ multiplies Ψ by the multiplier $\chi(g)$ of ψ, so $(g \otimes g)\iota(x) = \chi(g)\,\iota(gxg^{-1})$: the map ι is equivariant for the adjoint action on T twisted by χ, which is the action defining $T(-1)$. Both Hodge structures come from the Hodge cocharacter through $G \cdot \mathbb{G}_m$, so ι is a morphism of Hodge structures from $T(-1)$. The sub-Hodge structures of $H^2(Y, \mathbb{Q})$ are its G-stable subspaces, since $\mathbb{G}_m$ acts by scalars. Over $\bar{\mathbb{Q}}$, $H^2(Y) = \bigwedge^2(V \otimes \bar{\mathbb{Q}}^2) = S^2V \oplus (\bigwedge^2 V)^{\oplus 3}$, and expanding $V \otimes V = \bigotimes_k (S^2V_k \oplus \bigwedge^2 V_k)$ gives $\bigwedge^2 V$ as the sum of the pieces of highest weights $(0,0,0)$, $(2,2,0)$, $(2,0,2)$, $(0,2,2)$ (an odd number of factors $\bigwedge^2 V_k$) and S^2V as the sum of those of highest weights $(2,2,2)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$. So the isotypic pieces of $H^2(Y)$ have these eight highest weights and dimensions $3, 27, 27, 27, 27, 3, 3, 3$. The Hodge cocharacter acts through V_3, on which it has one weight of each type, so S^2V_3 has one dimension of type $(2,0)$ and $\bigwedge^2 V_3$ none; the parts of type $(2,0)$ of the eight pieces have dimensions $0, 0, 9, 9, 9, 0, 0, 1$. The last three pieces occur once, in S^2V, and are $\iota(T_1)$, $\iota(T_2)$, $\iota(T_3)$. So an irreducible summand of highest weight $(2,0,2)$ or $(0,2,2)$ contributes 3 to $h^{2,0}$, one of weight $(2,2,2)$ contributes 9, one of weight $(0,0,2)$ contributes 1, and the others contribute nothing. A rational sub-Hodge structure W is stable under the Galois group, which permutes the highest weights as it permutes the factors, transitively. So if $W \otimes \bar{\mathbb{Q}}$ has a summand of weight $(2,2,0)$, $(2,0,2)$ or $(0,2,2)$, it has summands of all three and $h^{2,0}(W) \geq 6$; if it has one of weight $(2,2,2)$, then $h^{2,0}(W) \geq 9$. If $h^{2,0}(W) = 1$, therefore, $W \otimes \bar{\mathbb{Q}}$ contains the piece $\iota(T_3)$ of weight $(0,0,2)$, which has multiplicity one, and with it $\iota(T_1)$ and $\iota(T_2)$; so $W \supseteq \iota(T)$, and $W = \iota(T)$ if W is irreducible. The same argument shows that $\iota(T)$ is irreducible: a nonzero sub-Hodge structure of it contains one of the $\iota(T_i)$ over $\bar{\mathbb{Q}}$, hence all three. The part of type $(2,0)$ of $\iota(T)$ is spanned by $\sigma_0 = \iota(E_3)$, where $E_3 \in \mathfrak{sl}(V_3) \otimes \mathbb{C}$ maps $V_3^{0,1}$ onto $V_3^{1,0}$. Write $\Psi = \Psi_1 \otimes \Psi_2 \otimes \Psi_3$ with $\Psi_k \in \bigwedge^2 V_k$ dual to ϵ_k, and $\Psi_3 = e_+ \otimes e_- - e_- \otimes e_+$ with $e_\pm$ spanning $V_3^{1,0}$ and $V_3^{0,1}$. Then $(E_3 \otimes 1)\Psi_3$ is a nonzero multiple of $e_+ \otimes e_+$, so σ_0 is a nonzero multiple of $\Psi_1 \otimes \Psi_2 \otimes e_+ \otimes e_+$: it lies in $H^{1,0}(X) \otimes H^{1,0}(X)$ and pairs $H^{1,0}$ of the first factor with $H^{1,0}$ of the second through the nondegenerate form dual to $\epsilon_1 \otimes \epsilon_2$ on $V_1 \otimes V_2$. So σ_0 has rank eight: it is a holomorphic symplectic form on Y.

(ii) For $x, y \in T$ the class $\iota(x) \cup \iota(y)$ lies in $H^2(X) \otimes H^2(X)$ and comes from the tensor $(x \otimes 1)\Psi \otimes (y \otimes 1)\Psi$ in $V^{\otimes 4}$ by antisymmetrising its two slots in the first factor of Y and its two slots in the second. Read it, as in Setup 7.18, as a map from the first-factor slots to the

second-factor slots. Contraction with ϵ_k turns $\Psi_k \otimes \Psi_k$ into γ times the identity of $V_k \otimes V_k$ and, for $x, y \in T_k$, the tensor $(x \otimes 1)\Psi_k \otimes (y \otimes 1)\Psi_k$ into $\gamma\,(x \otimes y)$, with one and the same constant $\gamma \neq 0$, since x and y are antisymmetric for ϵ_k and the two signs cancel. So $\iota_2(C_1)$ corresponds, up to a nonzero constant γ' independent of the factor, to the restriction to $\bigwedge^2 V$ of $\mathrm{Cas}_1 \otimes 1 \otimes 1$, where $\mathrm{Cas}_1 = \sum_c x_c \otimes x^c = s_1 - \frac{1}{2}$ is the operator of Step 3 of the proof of Theorem 7.7, s_1 the exchange of the two copies of V_1. On $\bigwedge^2 V$ the exchange s_1 is -1 on $\mathbb{Q}\theta$ and U_{23} and $+1$ on U_{12} and U_{13}, so $\iota_2(C_1) = \rho\,(3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23})$ with $\rho = -2\gamma'$, and permuting the factors gives the two cyclic permutations of this identity. Summing with the coefficients $\sigma_i(\nu)$ gives $\iota_2(c_\nu) = \rho(3\,\mathrm{Tr}(\nu)\pi_0 + \sum \sigma_k(4\nu - \mathrm{Tr}(\nu))\pi_{ij})$, the sum over $\{i,j,k\} = \{1,2,3\}$ with $i < j$, which is $\rho\,\omega_a$ with the stated a_0 and α. For rational $\nu \neq 0$ both $\iota_2(c_\nu)$ and ω_a are rational and nonzero, so $\rho \in \mathbb{Q}$. Since $\mathrm{Tr}(\nu) \in \mathbb{Q}$, the element α is rational exactly when ν is; and $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\nu)$, so $\nu = (\alpha + \mathrm{Tr}(\alpha))/4$ recovers ν from any $\alpha \in F$. Two rational classes with the same α differ by a multiple of π_0, and $\iota_2(c_{\nu/\rho}) = \omega_a$ because c_ν is linear in ν; so every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of π_0. The Casimir element C_i is dual to the trace form on T_i, so c_ν is dual to $\sum_i \sigma_i(\nu)^{-1}\,\mathrm{tr}(x_i y_i) = q_{1/\nu}(x, y)$.

(iii) Resolve the indeterminacy of f by a sequence of blow-ups along smooth centres, $\pi\colon \tilde{Y} \to Y$ and $\tilde{f}\colon \tilde{Y} \to M$, and put $f^* = \pi_* \tilde{f}^*$, a morphism of Hodge structures. The kernel of π_* on $H^2(\tilde{Y}, \mathbb{Q})$ is spanned by the classes of the exceptional divisors, so $\alpha \mapsto \tilde{f}^*\alpha - \pi^* f^* \alpha$ is a morphism of Hodge structures from $H^2(M, \mathbb{Q})$ to a space of Hodge classes. The Hodge structure $T(M)$ is irreducible: a sub-Hodge structure $W \subseteq T(M)$ with $W^{2,0} = 0$ consists of rational classes of type $(1,1)$, which lie in $\mathrm{NS}(M)_\mathbb{Q} \cap T(M) = 0$; if $W^{2,0} \neq 0$, then $\sigma_M \in W_\mathbb{C}$, the orthogonal complement of W in $H^2(M, \mathbb{Q})$ is a sub-Hodge structure orthogonal to σ_M and $\bar{\sigma}_M$, hence of type $(1,1)$ and contained in $\mathrm{NS}(M)_\mathbb{Q}$, and $W \supseteq T(M)$. Since $T(M)$ is not spanned by Hodge classes, that morphism vanishes on it, and $\tilde{f}^*\alpha = \pi^* f^* \alpha$ for $\alpha \in T(M)$. Because $\tilde{f}^*$ is a ring homomorphism and $\pi_*\pi^*$ is the identity, $f^*(\alpha\beta) = f^*\alpha \cup f^*\beta$ for $\alpha, \beta \in T(M)$. The kernel of f^* on $T(M)$ is a sub-Hodge structure not containing σ_M, hence zero, and the image is irreducible with $h^{2,0} = 1$, hence $\iota(T)$ by (i). So $f^*\sigma_M$ is a nonzero multiple of σ_0 and has rank eight at every point of Y where f is defined; the rank of a pulled-back form is at most the dimension of the image, so f is generically finite onto its image Z, of dimension eight, and σ_M is generically nondegenerate on Z. As M is projective, $b_2(M) = \dim T(M) + \dim \mathrm{NS}(M)_\mathbb{Q} \geq 9 + 1$; this excludes the generalised Kummer varieties, whose second Betti number is seven [Bea83]. Finally $\dim M \geq \dim Z = 8$; suppose that $\dim M = 8$. Then f is dominant, and for $\alpha \in T(M)_\mathbb{C}$ the relation of Fujiki and Beauville [Bea83, Fuj87] gives

$$\int_Y (f^*\alpha)^8 = \int_{\tilde{Y}} \tilde{f}^*(\alpha^8) = \deg(f) \int_M \alpha^8 = \deg(f)\, c_M\, q_M(\alpha)^4, \qquad c_M > 0.$$

Write $f^*\alpha = \iota(x)$ with $x = x_1 + x_2 + x_3 \in T_\mathbb{C}$ and $N_i = -\det x_i$. The form $x \mapsto q_M((f^*)^{-1}\iota(x))$ is invariant under G, so it is $q_\lambda(x, x) = \sum_i \sigma_i(\lambda)\,\mathrm{tr}(x_i^2) = 2\sum_i \sigma_i(\lambda) N_i$ for some $\lambda \in F^\times$ by Proposition 7.2(iii). On the other side, in a basis (e_i) of V one has $\iota(x) = \sum_{i,j} M_{ij}\, e_i^{(1)} \wedge e_j^{(2)}$, with M the product of the matrix of x and the constant invertible matrix of Ψ, and the 2-forms $e_i^{(1)} \wedge e_j^{(2)}$ commute, so $\iota(x)^8 = 8!\det(M) \prod_i e_i^{(1)} \wedge e_i^{(2)}$ is $8!\det(x)$ times a fixed volume form, and $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$ with $\Delta = N_1^2 + N_2^2 + N_3^2 - 2N_1N_2 - 2N_1N_3 - 2N_2N_3$; this is also the product of the eight numbers $\pm r_1 \pm r_2 \pm r_3$, where $\pm r_i$ are the eigenvalues of x_i and $r_i^2 = N_i$. The N_i are algebraically independent functions on $T_\mathbb{C}$, so the two expressions for $\int_Y (f^*\alpha)^8$ would make Δ^2 a nonzero constant multiple of the fourth power of the linear form $\sum_i \sigma_i(\lambda) N_i$, and

then, by unique factorisation, Δ a multiple of its square. But Δ is a nondegenerate ternary quadratic form, hence irreducible. So $\dim M \neq 8$, and since the dimension of a hyperkähler manifold is even, $\dim M \geq 10$.

(iv) By [Bea83] the map $H^2(S,\mathbb{Q}) \to H^2(S^{[n]},\mathbb{Q})$ induced by the universal subscheme $Z_n \subset S \times S^{[n]}$ is injective, carries the intersection form to the Beauville–Bogomolov form, and has $H^2(S^{[n]},\mathbb{Q}) = H^2(S,\mathbb{Q}) \oplus \mathbb{Q}\delta$ with δ algebraic, so $T(M) = T(S)$. Let Δ_T be the Künneth component in $T(S) \otimes T(S)$ of the class of the diagonal of S; it is algebraic, being the class of the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_\mathbb{Q}$. Its image under the correspondence $Z_n \times Z_n$, pulled back along the diagonal of M, is the class $\sum_a e_a \cup e_a^\vee$ over dual bases of $T(S)$ for the intersection form, which is therefore algebraic on M, and by (iii) its image under f^* is ι_2 of the element of S^2T dual to q_λ, that is $\iota_2(c_{1/\lambda})$, an algebraic class on Y. If $\lambda \notin \mathbb{Q}$ it is exceptional by (ii). If instead the real multiplication is algebraic, apply it to the first factor of Δ_T: for $b \in F$ the class $\sum_a (m_b e_a) \cup e_a^\vee$ is algebraic, and its image is $\iota_2(c_{b/\lambda})$, which is exceptional for $b \notin \mathbb{Q}\lambda$. In both cases Lemma 7.20 makes every Hodge class in $H^2(X) \otimes H^2(X)$ algebraic; the Hodge classes in the other Künneth components of $H^4(Y,\mathbb{Q})$ are products of divisor classes, as at the end of the proof of Theorem 7.4; and the Hodge classes of Y form a ring generated in degrees two and four (proof of Corollary 6.30). So every Hodge class of Y is algebraic. ■

Remark 7.26 (The mechanism of van Geemen and Rapagnetta). By Proposition 7.25(ii) the exceptional classes are, up to rational multiples of π_0, the classes $\iota_2(c_\nu)$ with $\nu \notin \mathbb{Q}$, the images of the elements dual to the forms q_λ, $\lambda = 1/\nu$, on the part $\iota(T)$ of $H^2(Y)$ generated by the symplectic class; the form with λ rational gives a product of divisor classes. They are the analogues for (Y,σ_0) of the dual Beauville–Bogomolov class, whose pull-back is what [vGR26] uses. By part (iii), a transfer of that argument needs a projective hyperkähler manifold of dimension at least ten with transcendental part $T(-1)$, such as $S_\mu^{[n]}$ with $n \geq 5$ for a surface S_μ of Theorem 7.4, and a rational map from Y onto a symplectic subvariety of dimension eight. For $M = S_\mu^{[n]}$ the form of (iv) is q_{μ/e^2} for the $e \in F^\times$ with $\iota^{-1} f^* = m_e$ on $T(S_\mu) = T$, so $\lambda \notin \mathbb{Q}$ for every such map as soon as $\mu \notin \mathbb{Q}^\times F^{\times 2}$. No such map is known. It would give an algebraic cycle on $S_\mu \times X \times X$ inducing an isomorphism of $T(S_\mu)$ onto $\iota(T) \subset H^1(X) \otimes H^1(X)$, a correspondence of the kind that the Kuga–Satake class supplies in the proof of Theorem 7.4; so this route is the Kuga–Satake route in geometric form rather than an independent one. By (iii), maps to K3 surfaces and to hyperkähler eightfolds cannot serve.

Theorem 7.27 (A numerical criterion for the Mumford classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) $\dim \mathrm{Ext}^2(E,E) \geq 119$.

(ii) *If $\dim \mathrm{Ext}^2(E,E) = 119$, then $\mathrm{ev}_E \colon HT^2(Y) \to \mathrm{Ext}^2(E,E)$ is surjective with kernel $\mathbb{C}\kappa$, and σ_E is injective. Conversely, if σ_E is injective, then ev_E has rank 119 and kernel $\mathbb{C}\kappa$, and E extends to first order along the curve.*

(iii) *If some such E at one point c has $\dim \mathrm{Ext}^2(E,E) = 119$, then ω_t is algebraic for every $t \in C$, the Hodge conjecture holds for $X_t \times X_t$ for every t, and $B(W \times_C W)$ holds.*

(iv) *At every CM point c some perfect complex E on Y has $\mathrm{ch}(E) = N\omega$ for some $N \neq 0$ and no other component.*

Proof. (i) and (ii). By the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 4.54, $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$ by Theorem 7.21(ii), so ev_E has rank at least 119, which is (i). If $\dim \mathrm{Ext}^2(E,E) = 119$, then ev_E is onto, its kernel has dimension one and lies in $\mathbb{C}\kappa$, so equals it; and $\sigma_E(\mathrm{ev}_E(x)) = N\,x\lrcorner\omega$ vanishes only for $x \in \mathbb{C}\kappa = \ker \mathrm{ev}_E$, so σ_E is injective on $\mathrm{im}\,\mathrm{ev}_E = \mathrm{Ext}^2(E,E)$. Conversely, if σ_E is injective, then $\sigma_E(\mathrm{ev}_E(\kappa)) = N\kappa\lrcorner\omega = 0$ forces $\mathrm{ev}_E(\kappa) = 0$, so the kernel is $\mathbb{C}\kappa$ and the rank is 119; and ev_E restricted to $H^1(T_Y)$ is the obstruction map [Tod09], which therefore vanishes at κ, so that E extends to first order along the curve.

(iii) By (ii) σ_E is injective. The Chern character of E is $N\omega$, a flat section of Hodge type over the whole of C, so Theorem 2.2 applies to every infinitesimal deformation of Y inside $W \times_C W \to C$. The proof of Theorem 2.3 then applies word for word, with the stack of perfect complexes on the fibres of $W \times_C W \to C$ [Lie06] in place of that of $\mathcal{A} \to \mathcal{S}$, or the analytic Kuranishi germ if $\mathrm{Ext}^{<0}(E,E) \neq 0$: it is smooth over C at $[E]$, its image contains a nonempty open subset $U \subseteq C$, and for $t \in U$ a perfect complex E_t on $X_t \times X_t$ in the same connected component has $\mathrm{ch}(E_t) = N\omega_t$, so ω_t is algebraic, and by Lemma 7.20 so are both exceptional classes. Since U is uncountable, Corollary 6.30 gives the rest.

(iv) By Theorem 6.29 there is $\xi \in \mathrm{CH}^2(Y)_{\mathbb{Q}}$ with class ω, and the argument of Proposition 2.1 gives E: the Chern character is an isomorphism $K_0(Y)_{\mathbb{Q}} \to \mathrm{CH}^*(Y)_{\mathbb{Q}}$ [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so $\xi = \mathrm{ch}(u)$ for a unique u, and Nu is the class of a perfect complex for a suitable N. ∎

Part (iii) is the Mumford analogue of Theorem 2.11(iii): one object, at one point, and one number, which is 119 here and $2n(2n-1)$ for a Weil family. For a Weil family that number is attained by no object (Theorem 3.5), because an object attaining it would have to deform to non-algebraic tori; the classes here are rigid, so that argument does not apply (Remark 3.6). The objects of (iv) exist, and the criterion asks that one of them, or another representative of the same class in K_0, attain the bound of (i). The rigidity of Theorem 7.21 is no obstacle: semiregularity moves an object along the Hodge locus of its Chern character, and the curve is exactly that locus, so a semiregular object would propagate along the curve and nowhere else, which is all that is needed.

Corollary 7.28 (Bounded data at CM points)**.** *Fix a relatively ample line bundle on $W \times_C W \to C$, and let ω be a rational exceptional class. The following are equivalent.*

(a) *ω_t is algebraic for every $t \in C$; equivalently, by Lemma 7.20 and Corollary 6.30, the Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*

(b) *There are a finite set Φ of pairs $(\underline{P}, \underline{q})$, each a tuple of Hilbert polynomials with a tuple of rational numbers of the same length, and infinitely many points $c \in C$ at each of which $\omega_c = \sum_i q_i[Z_i]$ for subschemes $Z_i \subset X_c \times X_c$ with Hilbert polynomials P_i, for some $(\underline{P}, \underline{q}) \in \Phi$.*

In particular the classes are algebraic on every member as soon as they are represented at infinitely many CM points by cycles of bounded degree.

Proof. For a pair $(\underline{P}, \underline{q})$ let $\Sigma_{\underline{P},\underline{q}} \subseteq C$ be the set of points at which ω is represented with these data. The proof of [I, Theorem 16.4] applies verbatim to $(W \times_C W, C, \omega)$, as in Lemma 6.27, and shows that each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed. If (b) holds, then, Φ being finite, one pair of Φ serves at infinitely many points, so its Σ is an infinite closed subset of the irreducible curve C, hence C, and (a) holds. If (a) holds, C is the union of the countably many $\Sigma_{\underline{P},\underline{q}}$, and C is uncountable, so one of them is infinite, hence equal to C; then (b) holds with Φ that one pair and every point of C. ∎

The proof of Theorem 6.29 represents the classes at every CM point, through Markman's theorem for abelian fourfolds [Mar25a] and the ring generated by divisor classes and pull-backs of Hodge classes of X_c along homomorphisms $X_c \times X_c \to X_c$, but with no control of degree; isogenies between CM members, which move CM points along their Hecke orbits, do not bound the degree either. By Corollary 7.28 such a bound, at infinitely many CM points, is equivalent to the target; without it the CM points, being countably many, say nothing about a very general member (Lemma 6.27 needs uncountably many points).

The next proposition says which endomorphisms those pull-backs have to use. At a CM point the fourfold is of Weil type, its two Hodge classes of degree four that are not products of divisor classes form its Weil line, and the exceptional classes of the square are out of reach of everything built from that line with the Weil field alone: they need the real multiplication by the totally real quartic algebra of Rosati-fixed endomorphisms, and all of it.

Proposition 7.29 (The Weil structure at a CM point, and what the cycles must use)**.** *Let $c \in C$ be a CM point, $X = X_c$, $Y = X \times X$, and let e_w, $w \in \{\pm 1\}^3$, be a weight basis of $V = H^1(X, \mathbb{C})$ for the Hodge torus T of X, as in the proof of Theorem 6.29, normalised so that $\psi(e_w, e_{-w}) = \pm 1$, with $e_w^{(1)}$, $e_w^{(2)}$ its two copies in $H^1(Y, \mathbb{C})$. Let T_+ be the set of the four weights with an even number of entries -1, let $T_- = -T_+$ be the set of the other four, let $t_\pm = \prod_{w \in T_\pm} e_w \in H^4(X, \mathbb{C})$, the factors in a fixed order, and let $\dagger$ be the Rosati involution of θ on $\mathrm{End}^0(X)$. For $\varphi \in \mathrm{End}^0(X)$ write φ_w for its eigenvalue on e_w.*

(i) *T_+ and T_- are the only sets of four weights with sum zero that contain no pair of opposite weights, and $\mathrm{Hdg}^2(X) \otimes \mathbb{C}$ is the direct sum of the span of the products of divisor classes and of $\mathbb{C}t_+ \oplus \mathbb{C}t_-$. The elements of $\mathrm{End}^0(X)$ with φ_w constant on T_+ and constant on T_- form an imaginary quadratic field k_c; (X, k_c, θ) is of $(k_c, -1, 2)$-Weil type, and its Weil line is $W(X) = (\mathbb{C}t_+ \oplus \mathbb{C}t_-) \cap H^4(X, \mathbb{Q})$. The fixed algebra K_c^+ of $\dagger$ consists of the elements with $\varphi_w = \varphi_{-w}$ for every w; it is a product of totally real fields of total degree four, and $\mathrm{End}^0(X) = K_c^+ \otimes_{\mathbb{Q}} k_c$.*

(ii) *For $\varphi, \psi \in \mathrm{End}^0(X)$ let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X, \mathbb{Q}) \to H^*(Y, \mathbb{Q})$ that sends $e \in H^1(X, \mathbb{Q})$ to $\mathrm{pr}_1^*\varphi^* e + \mathrm{pr}_2^*\psi^* e$; for $\varphi, \psi \in \mathrm{End}(X)$ it is the pull-back along the homomorphism $\varphi \circ \mathrm{pr}_1 + \psi \circ \mathrm{pr}_2 \colon Y \to X$, and every homomorphism $Y \to X$ is of this form. For a $\mathbb{Q}$-subalgebra $R \subseteq \mathrm{End}^0(X)$ let $\Sigma_R \subseteq \mathrm{Hdg}^2(Y)$ be the $\mathbb{Q}$-span of the products of two divisor classes of Y and of the classes $f^*_{\varphi,\psi} W(X)$ with $\varphi, \psi \in R$. Then $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$, the span obtained from the maps $(x, y) \mapsto ax + by$ alone, has dimension 110 and meets the span of $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ in the plane of the classes with $a_1 = a_2 = a_3$; it contains no exceptional class.*

(iii) *For a $\mathbb{Q}$-subalgebra $R \subseteq \mathrm{End}^0(X)$ the following are equivalent:* (a) *Σ_R contains an exceptional class;* (b) *$\Sigma_R = \mathrm{Hdg}^2(Y)$, which has dimension 132;* (c) *R separates the weights of T_+, that is, for any two distinct $w, w' \in T_+$ some $\varphi \in R$ has $\varphi_w \neq \varphi_{w'}$; equivalently, R and k_c generate $\mathrm{End}^0(X)$. In particular $\Sigma_{K_c^+} = \mathrm{Hdg}^2(Y)$, while Σ_R contains no exceptional class if R is a proper subalgebra of K_c^+ or a subalgebra of k_c.*

Proof. (i) The four weights $d_1 = (1, 1, 1)$, $d_2 = (1, -1, -1)$, $d_3 = (-1, 1, -1)$, $d_4 = (-1, -1, 1)$ of T_+ have sum zero, and any three of them are linearly independent, so every linear relation among them is a multiple of $d_1 + d_2 + d_3 + d_4 = 0$. A set of four weights containing no opposite pair contains exactly one of d_i, $-d_i$ for each i, say $\varepsilon_i d_i$, and has sum zero only if $\sum_i \varepsilon_i d_i = 0$, that is, if all ε_i are equal: the set is T_+ or T_-. The Hodge classes of X of degree four, tensored with $\mathbb{C}$, are spanned by the monomials $\prod_{w \in S} e_w$ over

the sets S of four weights with sum zero (proof of Theorem 6.29); the divisor classes are spanned by the four monomials e_we_{-w} of weight zero and degree two, their products are the six monomials of the sets that are unions of two opposite pairs, and the two remaining sets are T_+ and T_-.

The torus T is defined over $\mathbb{Q}$, so the Galois group of $\bar{\mathbb{Q}}$ acts linearly on its character lattice and permutes the set of the eight weights of V. A linear map that permutes the weights carries opposite pairs to opposite pairs and sets with sum zero to sets with sum zero, hence carries $\{T_+, T_-\}$ to itself. Over $\bar{\mathbb{Q}}$ the algebra $\mathrm{End}^0(X)\otimes\bar{\mathbb{Q}} = \mathrm{End}_T(V_{\bar{\mathbb{Q}}})$ consists of the diagonal matrices $\sum_w x_wp_w$ in the weight basis, p_w the projection to the weight space of w, and $\gamma\in\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ acts by $\gamma(\sum_w x_wp_w) = \sum_w \gamma(x_w)p_{\gamma w}$, because γ carries the weight space of w to that of γw. As ψ is T-invariant, $\psi(e_w, e_{w'}) = 0$ unless $w' = -w$, so the adjoint of $\sum_w x_wp_w$ for ψ, which is the Rosati involution, is $\sum_w x_{-w}p_w$. The subalgebra A_W of the diagonal matrices that are constant on T_+ and on T_-, and the subalgebra A^+ of those with $x_w = x_{-w}$ for all w, are therefore stable under the Galois group and under †; so $k_c = A_W\cap\mathrm{End}^0(X)$ and $K_c^+ = A^+\cap\mathrm{End}^0(X)$ are $\mathbb{Q}$-forms of A_W and A^+, of dimensions 2 and 4, and K_c^+ is the fixed algebra of †. On A_W the involution † exchanges the two scalars. The Rosati involution is positive: $\mathrm{Tr}(xx^\dagger) > 0$ for $x\neq 0$, with Tr the trace on V [BL04, Section 5.1]. If k_c were not a field it would contain the idempotent $p_+ = \sum_{w\in T_+} p_w$, and $\mathrm{Tr}(p_+p_+^\dagger) = \mathrm{Tr}(p_+p_-) = 0$; so k_c is a quadratic field on which † is the nontrivial automorphism. If k_c were real, $x = \sqrt{d}\in k_c$ would satisfy $x^\dagger = -x$ and $\mathrm{Tr}(xx^\dagger) = -\mathrm{Tr}(x^2) = -8d < 0$. So k_c is imaginary quadratic and † is complex conjugation on it, which is condition (iii) of [I, Definition 3.1]. The eigenspaces of $k_c\otimes\mathbb{C}$ on V are $V_\pm = \bigoplus_{w\in T_\pm}\mathbb{C}e_w$. The Hodge decomposition of V is the eigenspace decomposition of the Hodge cocharacter, which takes values in the diagonal torus $T_{\mathbb{C}}\cdot\mathbb{G}_m$, so every e_w is of pure type $(1,0)$ or $(0,1)$, and t_+ is of type $(p, 4-p)$ with $p = \dim(V_+\cap H^{1,0}(X))$. Since t_+ is a Hodge class, $p = 2$: each eigenvalue of $\sqrt{-d}$ has multiplicity two on $H^{1,0}(X)$, which is condition (ii) of [I, Definition 3.1], and condition (i) is $\dim X = 4$. The Weil line of [I, Definition 3.6] is $(\bigwedge^4 V_+\oplus\bigwedge^4 V_-)\cap H^4(X,\mathbb{Q}) = (\mathbb{C}t_+\oplus\mathbb{C}t_-)\cap H^4(X,\mathbb{Q})$; it has dimension two by [I, Proposition 3.7(i)], so it is a rational complement of the products of divisor classes in $\mathrm{Hdg}^2(X)$. The algebra K_c^+ is commutative and semisimple, being a subalgebra of $\mathrm{End}^0(X)$, whose complexification is commutative, and its trace form $\mathrm{Tr}(xy) = \mathrm{Tr}(xy^\dagger)$ is positive definite; so every factor field E of K_c^+ has $\mathrm{Tr}_{E/\mathbb{Q}}(x^2) > 0$ for $x\neq 0$ and is totally real. Finally the multiplication map $K_c^+\otimes_{\mathbb{Q}} k_c\to\mathrm{End}^0(X)$ is an isomorphism, because over $\bar{\mathbb{Q}}$ it is the isomorphism $\bar{\mathbb{Q}}^{\{\text{opposite pairs}\}}\otimes\bar{\mathbb{Q}}^{\{T_+,T_-\}}\to\bar{\mathbb{Q}}^{\{\text{weights}\}}$ induced by the bijection that sends a weight to the pair and the tetrahedron containing it, a bijection because the two weights of a pair lie in different tetrahedra, T_- being $-T_+$.

(ii) Extend the notation: for diagonal $\varphi,\psi\in\mathrm{End}^0(X)\otimes\mathbb{C}$ with entries φ_w, ψ_w let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X,\mathbb{C})\to H^*(Y,\mathbb{C})$ with $e_w\mapsto\varphi_we_w^{(1)} + \psi_we_w^{(2)}$; for $\varphi,\psi\in\mathrm{End}^0(X)$ this is the map of the statement, since $\varphi^*e_w = \varphi_we_w$, and every homomorphism $Y\to X$ is $\varphi\circ\mathrm{pr}_1 + \psi\circ\mathrm{pr}_2$ for its restrictions φ,ψ to the two factors. Then

$$f^*_{\varphi,\psi}t_\pm = \prod_{w\in T_\pm}(\varphi_we_w^{(1)} + \psi_we_w^{(2)}) = \sum_{S\subseteq T_\pm}\pm\Big(\prod_{w\in S}\varphi_w\Big)\Big(\prod_{w\in T_\pm\setminus S}\psi_w\Big)\,e_S^{(1)}\,e_{T_\pm\setminus S}^{(2)}, \qquad (36)$$

where $e_S^{(1)}e_{T\setminus S}^{(2)}$ is the product of the $e_w^{(1)}$, $w\in S$, and the $e_w^{(2)}$, $w\in T\setminus S$, taken in the order of T, and the sign is that of the shuffle. We call these 32 products the tetrahedron monomials. The Hodge classes of Y of degree four, tensored with $\mathbb{C}$, are spanned by the monomials of weight zero in the sixteen classes $e_w^{(i)}$ (proof of Theorem 6.29). Such a

monomial whose four weights contain an opposite pair is the product of two monomials of weight zero and degree two, that is, of two divisor classes: there are four of the form $e_w^{(1)}e_w^{(2)}e_{-w}^{(1)}e_{-w}^{(2)}$ and 96 with two distinct opposite pairs and a choice of copy for each of the four weights. A weight repeated twice, $2w + w' + w'' = 0$, forces $w' = w'' = -w$, so the remaining monomials have four distinct weights with no opposite pair, and by (i) they are the tetrahedron monomials. Hence $\dim \mathrm{Hdg}^2(Y) = 132$, the products of divisor classes span the 100 monomials of the first two kinds, and the tetrahedron monomials form a basis of the quotient Ξ of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$ by the span of the products of divisor classes. Two distinct weights of a tetrahedron T differ in exactly two coordinates, and for $S = \{a, b\} \subseteq T$ the complement $T \setminus S$ has the same set of differing coordinates, because $a + b$ and the sum of the complement are opposite. Writing $D(a, b) \subseteq \{1, 2, 3\}$ for the set of coordinates in which two weights differ, the six tetrahedron monomials of T with two factors from each copy fall into three pairs $\{e_S^{(1)}e_{T\setminus S}^{(2)}, e_{T\setminus S}^{(1)}e_S^{(2)}\}$, indexed by the two-element set $D(S) = D(T \setminus S)$.

For $\varphi, \psi \in k_c \otimes \mathbb{C} = A_W$, with values $\varphi_\pm$, $\psi_\pm$ on $T_\pm$, (36) reads $f^*_{\varphi,\psi}t_\pm = \sum_{k=0}^4 \varphi_\pm^k \psi_\pm^{4-k} s_{\pm,k}$, where $s_{\pm,k}$ is the sum, with the signs of (36), of the tetrahedron monomials of $T_\pm$ with k factors from the first copy. Conversely the classes $(a\,\mathrm{pr}_1 + b\,\mathrm{pr}_2)^* t_\pm$ for five values of a/b span the $s_{\pm,k}$, a Vandermonde matrix being invertible. Since $W(X) \otimes \mathbb{C} = \mathbb{C}t_+ \oplus \mathbb{C}t_-$, the complexifications of Σ_{k_c} and of $\Sigma_{\mathbb{Q}}$ are both spanned by the products of divisor classes and the ten classes $s_{\pm,k}$, which are supported on disjoint nonempty sets of tetrahedron monomials; so $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$ has dimension $100 + 10 = 110$.

For the four π we use the description of Setup 7.18. Fix a total order on the weights, and for an endomorphism A of $H^2(X, \mathbb{C}) = \bigwedge^2 V$ let

$$c(A) = \sum_{a<b} A(e_a e_b) \otimes (e_a e_b)^\vee \in H^2(X) \otimes H^2(X)$$

be its class, where $(e_a e_b)^\vee$ is the dual basis for the form $\bigwedge^2\psi$. As $\bigwedge^2\psi(e_ae_b, e_ce_d) = \psi(e_a, e_c)\psi(e_b, e_d) - \psi(e_a, e_d)\psi(e_b, e_c)$ vanishes unless $\{c, d\} = \{-a, -b\}$, we have $(e_ae_b)^\vee = \chi(a)\chi(b)\, e_{-a}e_{-b}$ with $\chi(w) = \psi(e_w, e_{-w}) = \pm 1$, the two factors in the order matching (a, b). Let s_t, $t = 1, 2, 3$, be the involution of $V \otimes V = \bigotimes_{u=1}^3 (V_u \otimes V_u)$ that exchanges the two factors V_t. The three s_t commute, and $s_1s_2s_3$ is the exchange of the two slots of $V \otimes V$, which is -1 on $\bigwedge^2 V$; so each s_t preserves $\bigwedge^2 V$, on which $s_1s_2s_3 = -1$, and $\bigwedge^2 V$ is the sum of the four joint eigenspaces with eigenvalues $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ of product -1: these are $\mathbb{C}\theta = \bigwedge^2 V_1 \otimes \bigwedge^2 V_2 \otimes \bigwedge^2 V_3$ for $(-1, -1, -1)$ and $U_{ij} = S^2V_i \otimes S^2V_j \otimes \bigwedge^2 V_k$ for $\varepsilon_i = \varepsilon_j = 1$, $\varepsilon_k = -1$. The projection onto the eigenspace of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is $\prod_t (1 + \varepsilon_t s_t)/2$, which by the relation $s_1s_2s_3 = -1$ equals $(1 + \varepsilon_1 s_1 + \varepsilon_2 s_2 + \varepsilon_3 s_3)/4$. So

$$\pi_0 = c\Big(\frac{1 - s_1 - s_2 - s_3}{4}\Big), \qquad \pi_{12} = c\Big(\frac{1 + s_1 + s_2 - s_3}{4}\Big),$$

and π_{13}, π_{23} likewise, with the plus signs at s_1, s_3 and at s_2, s_3. Let a^t be the weight a with its t-th coordinate changed; then $s_t(e_ae_b) = e_ae_b$ if $t \notin D(a, b)$ and $s_t(e_ae_b) = e_{a^t}e_{b^t}$ if $t \in D(a, b)$. Under the Künneth map to $H^4(Y, \mathbb{C})$ the term of the pair $\{a, b\}$ in $c(s_t)$ is, up to sign, the monomial $e_{a'}^{(1)}e_{b'}^{(1)}e_{-a}^{(2)}e_{-b}^{(2)}$ with $\{a', b'\} = \{a, b\}$ if $t \notin D(a, b)$ and $\{a', b'\} = \{a^t, b^t\}$ otherwise. If $t \notin D(a, b)$, or $D(a, b) = \{t\}$, in which case $a^t = b$, or $D(a, b) = \{1, 2, 3\}$, in which case $b = -a$ and the monomial is $e_{a^t}^{(1)}e_{-a^t}^{(1)}e_{-a}^{(2)}e_a^{(2)}$, the monomial contains an opposite pair and is a product of two divisor classes. In the remaining case $D(a, b) = \{t, t'\}$ with $t' \neq t$; then $b = a^{tt'}$, and the four weights a^t, $b^t = a^{t'}$, $-a$, $-b = a^{t''}$, where $\{t, t', t''\} = \{1, 2, 3\}$, are pairwise non-opposite, since any two of them differ in exactly

two coordinates, and they have sum zero, because $a^t + b^t$ and $a + b$ agree outside the t-th coordinate, where both vanish. So

$$m_{ab} = \chi(a)\chi(b)\, e^{(1)}_{a^t} e^{(1)}_{b^t} e^{(2)}_{-a} e^{(2)}_{-b}, \qquad t = \min D(a,b),$$

is a tetrahedron monomial with $D(a^t, b^t) = D(-a,-b) = D(a,b)$, and since $s_{t'}(e_a e_b) = e_{a^{t'}} e_{b^{t'}} = e_{b^t} e_{a^t} = -e_{a^t} e_{b^t}$, the term of the pair in $c(s_{t'})$ is $-m_{ab}$, while its term in $c(s_{t''})$ is a product of divisor classes. For each two-element set D put $u_D = \sum_{D(a,b)=D} m_{ab}$, a sum over the four pairs with difference set D: two of these pairs lie in each tetrahedron and a^t lies in the other one, so u_D has two monomials in each tetrahedron, all with coefficients ± 1, and the three u_D are supported on disjoint sets of monomials; the two monomials of u_D in T are those of the pair of T indexed by D. In Ξ this gives $c(1) \equiv 0$, $c(s_1) \equiv u_{12} + u_{13}$, $c(s_2) \equiv -u_{12} + u_{23}$, $c(s_3) \equiv -u_{13} - u_{23}$, hence

$$\pi_0 \equiv 0, \quad \pi_{12} \equiv \tfrac{1}{2}(u_{13} + u_{23}), \quad \pi_{13} \equiv \tfrac{1}{2}(u_{12} - u_{23}), \quad \pi_{23} \equiv -\tfrac{1}{2}(u_{12} + u_{13}), \qquad (37)$$

and $\omega_a \equiv \frac{1}{2}((a_2 - a_3)u_{12} + (a_1 - a_3)u_{13} + (a_1 - a_2)u_{23})$ in Ξ: on the monomials of the pair of $T_\pm$ indexed by $\{1,2\}$, $\{1,3\}$, $\{2,3\}$ the class ω_a has coefficients of absolute values $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$.

Now suppose that $\omega_a \in \Sigma_{\mathbb{Q}} \otimes \mathbb{C}$. Its image in Ξ is a combination of the $s_{\pm,k}$, and since every monomial of ω_a has two factors from each copy, only $s_{+,2}$ and $s_{-,2}$ occur: $\omega_a \equiv x\, s_{+,2} + y\, s_{-,2}$. On the six monomials of T_+ the right side has coefficients of absolute value $|x|$. So the three differences $a_2 - a_3$, $a_1 - a_3$, $a_1 - a_2$ have a common absolute value r, and as $a_1 - a_3 = (a_1 - a_2) + (a_2 - a_3)$, $r \in \{0, 2r\}$; hence $r = 0$ and $a_1 = a_2 = a_3$. Conversely the classes with $a_1 = a_2 = a_3$ are the combinations of π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$, which are invariant under $\mathrm{Sp}(V, \psi)$ and so are combinations of the products of divisor classes $\psi_{11}\psi_{22}$ and ψ_{12}^2 (proof of Proposition 7.2(ii)); they lie in $\Sigma_{\mathbb{Q}}$. An exceptional class has for (a_1, a_2, a_3) the three distinct conjugates of an element of $F \setminus \mathbb{Q}$ (Lemma 7.19), so it does not lie in $\Sigma_{\mathbb{Q}}$.

(iii) We first show that (c) is equivalent to R and k_c generating $\mathrm{End}^0(X)$, and that R separates the weights of T_+ if and only if it separates those of T_-. For $\varphi \in \mathrm{End}^0(X)$ the eigenvalue on e_{-w} is $\varphi_{-w} = (\varphi^\dagger)_w$; since $\mathrm{End}^0(X)$ is a product of CM fields on which $\dagger$ is complex conjugation, and every embedding of a CM field into $\mathbb{C}$ commutes with complex conjugation, $\varphi_{-w} = \overline{\varphi_w}$. Hence $\varphi_w = \varphi_{w'}$ if and only if $\varphi_{-w} = \varphi_{-w'}$, and R separates T_+ if and only if it separates T_-. The elements of k_c are constant on each tetrahedron and separate the two, and a subalgebra of $\bar{\mathbb{Q}}^{\{\text{weights}\}}$ containing the constants is the algebra of functions constant on the blocks of the partition it induces; so R separates T_+ exactly when $R \cdot k_c \otimes \bar{\mathbb{Q}}$ separates all eight weights, that is, when $R \cdot k_c = \mathrm{End}^0(X)$.

(c) implies (b). If R separates T_+, then $R \otimes \mathbb{C}$ maps onto $\mathbb{C}^{T_+}$ under $\varphi \mapsto (\varphi_w)_{w \in T_+}$, so the entries φ_w, ψ_w, $w \in T_+$, are eight independent coordinates on the image of $(R \otimes \mathbb{C})^2$. The sixteen coefficients in (36) are distinct monomials in these coordinates, hence linearly independent functions of $(\varphi, \psi) \in (R \otimes \mathbb{C})^2$; so no nonzero linear form on the span of the sixteen tetrahedron monomials of T_+ vanishes on all $f^*_{\varphi,\psi} t_+$, and these classes span that space. The same holds for T_-, and with the 100 products of divisor classes this is all of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$. The map $(\varphi, \psi) \mapsto f^*_{\varphi,\psi} t_\pm$ is polynomial and $R \times R$ is Zariski dense in $(R \otimes \mathbb{C})^2$, so a linear form vanishing on its values at rational pairs vanishes on all values, and finitely many pairs $\varphi, \psi \in R$ give the same span. The $\mathbb{Q}$-span of the rational classes $f^*_{\varphi,\psi} W(X)$, $\varphi, \psi \in R$, and of the products of divisor classes is then a $\mathbb{Q}$-subspace of $\mathrm{Hdg}^2(Y)$ whose complexification is the whole space, hence is $\mathrm{Hdg}^2(Y)$.

(b) implies (a), as $\mathrm{Hdg}^2(Y)$ contains the exceptional classes.

(a) implies (c). Suppose that two distinct weights $u, u' \in T_+$ have $\varphi_u = \varphi_{u'}$ for every $\varphi \in R$, and let v, v' be the other two weights of T_+. In (36) the sets $S = \{u, v\}$ and $S' = \{u', v\}$ then have the same coefficient for every $\varphi, \psi \in R$, up to the signs of the shuffles, which do not depend on φ, ψ. Hence every class in $\Sigma_R \otimes \mathbb{C}$ has coefficients of equal absolute value on the tetrahedron monomials $e^{(1)}_S e^{(2)}_{T_+\setminus S}$ and $e^{(1)}_{S'} e^{(2)}_{T_+\setminus S'}$, the products of divisor classes involving no tetrahedron monomial. The pairs $\{S, T_+ \setminus S\}$ and $\{S', T_+ \setminus S'\}$ are different partitions of T_+, so $D(S) \neq D(S')$, and for an exceptional class ω_a in Σ_R two of the three numbers $|a_2 - a_3|$, $|a_1 - a_3|$, $|a_1 - a_2|$ would be equal. Each of the six resulting equations, $a_2 - a_3 = \pm(a_1 - a_3)$ and the four others, is a linear relation $c_1a_1 + c_2a_2 + c_3a_3 = 0$ with rational coefficients not all equal, which Lemma 7.19 forbids for the conjugates of an element of $F \setminus \mathbb{Q}$. So (a) fails, and this proves the equivalence.

For the last assertions: $K_c^+ \otimes \bar{\mathbb{Q}}$ consists of all functions on the opposite pairs, and T_+ meets every pair once, so K_c^+ separates T_+; a proper subalgebra R of K_c^+ has $R \otimes \bar{\mathbb{Q}}$ a proper subalgebra of the functions on the four pairs, which identifies two pairs and hence two weights of T_+; and a subalgebra of k_c is constant on T_+. ∎

Figure 35 draws the weights of part (i), and Figure 36 the Hodge classes of parts (ii) and (iii).

The CM points fall into two kinds, according to whether the fourfold is simple; at the points of the second kind the real multiplication comes from the fixed cubic field F.

Lemma 7.30 (Split CM points)**.** *Let $c \in C$ be a CM point, and let $L \subset D$ be the quadratic extension of F whose norm one torus $\mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{L/F})$ is the Hodge group of X_c (proof of Theorem 6.29). Then L is a CM field, and the following are equivalent:* (a) *X_c is not simple;* (b) *K_c^+ is not a field;* (c) *$L = Fk$ for an imaginary quadratic field $k \subseteq L$. In that case $k_c \cong k$, $K_c^+ \cong \mathbb{Q} \times F$, and X_c is isogenous to $E \times B$, where E is an elliptic curve with complex multiplication by k and B is an abelian threefold of CM type on which the factor F of K_c^+ acts, a real multiplication by F; the Weil line $W(X_c)$ lies in $H^1(E) \otimes H^3(B)$. At every other CM point K_c^+ is a totally real quartic field. Points of the second kind exist for infinitely many k: for every imaginary quadratic field k such that $k \otimes_{\mathbb{Q}} F_v$ is a field at each finite place v of F at which D is ramified.*

Proof. Write T for the Hodge group. By Proposition 7.29(i), $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a product of CM fields on which † is complex conjugation, and T lies in the centraliser of $\mathrm{End}^0(X_c)$, which over $\bar{\mathbb{Q}}$ is the diagonal torus, and in $\mathrm{Sp}(V, \psi)$; so $T(\mathbb{R})$ lies in the compact group of the $x \in (\mathrm{End}^0(X_c) \otimes \mathbb{R})^\times$ with $xx^\dagger = 1$. If $L \otimes_{F,\sigma} \mathbb{R}$ were $\mathbb{R} \times \mathbb{R}$ at a real place σ, the factor of $T(\mathbb{R})$ at σ would be the norm one group of $\mathbb{R} \times \mathbb{R}$ over $\mathbb{R}$, which is not compact. Hence $L \otimes_{F,\sigma} \mathbb{R} = \mathbb{C}$ at each real place, and L is a CM field.

Over $\bar{\mathbb{Q}}$, $T = \prod_\sigma T_\sigma$ with $T_\sigma = \ker(\mathrm{N}_{L/F}) \otimes_{F,\sigma} \bar{\mathbb{Q}}$ the diagonal torus of $\mathrm{SL}_2(\bar{\mathbb{Q}}) = (D \otimes_{F,\sigma} \bar{\mathbb{Q}})^1$ for the decomposition $L \otimes_{F,\sigma} \bar{\mathbb{Q}} = \bar{\mathbb{Q}} \times \bar{\mathbb{Q}}$ by the two embeddings of L over σ, one of them chosen and called τ_σ, the other $\bar{\tau}_\sigma$. The factor V_σ of $V = \bigotimes_\sigma V_\sigma$ (Setup 7.1) has the weights τ_σ and $\bar{\tau}_\sigma = \tau_\sigma^{-1}$ on T_σ, with eigenvectors $v_{\sigma,\pm}$, and $e_w = \bigotimes_\sigma v_{\sigma,w_\sigma}$. An element γ of the Galois group carries V_σ to $V_{\gamma\sigma}$ and $v_{\sigma,+}$ to $v_{\gamma\sigma,\varepsilon_\gamma(\sigma)}$, where $\varepsilon_\gamma(\sigma) = 1$ if $\gamma\tau_\sigma = \tau_{\gamma\sigma}$ and -1 if $\gamma\tau_\sigma = \bar{\tau}_{\gamma\sigma}$; so it acts on the weights by $(\gamma w)_{\gamma\sigma} = \varepsilon_\gamma(\sigma) w_\sigma$.

If $L = Fk$, then $\mathrm{Hom}(L, \bar{\mathbb{Q}}) = \mathrm{Hom}(F, \bar{\mathbb{Q}}) \times \mathrm{Hom}(k, \bar{\mathbb{Q}})$ with the Galois group acting on both factors, and with $\tau_\sigma = \sigma \otimes \iota_0$ for a fixed embedding ι_0 of k the sign $\varepsilon_\gamma(\sigma) = \varepsilon_\gamma$ does not depend on σ: it is 1 if γ fixes k and -1 otherwise. Then γ permutes the coordinates of a weight and changes all its signs when it moves k. It preserves each tetrahedron if it fixes k and exchanges the two otherwise, so $k_c \otimes \bar{\mathbb{Q}} = \bar{\mathbb{Q}}^{\{T_+, T_-\}}$ is the algebra of the Galois

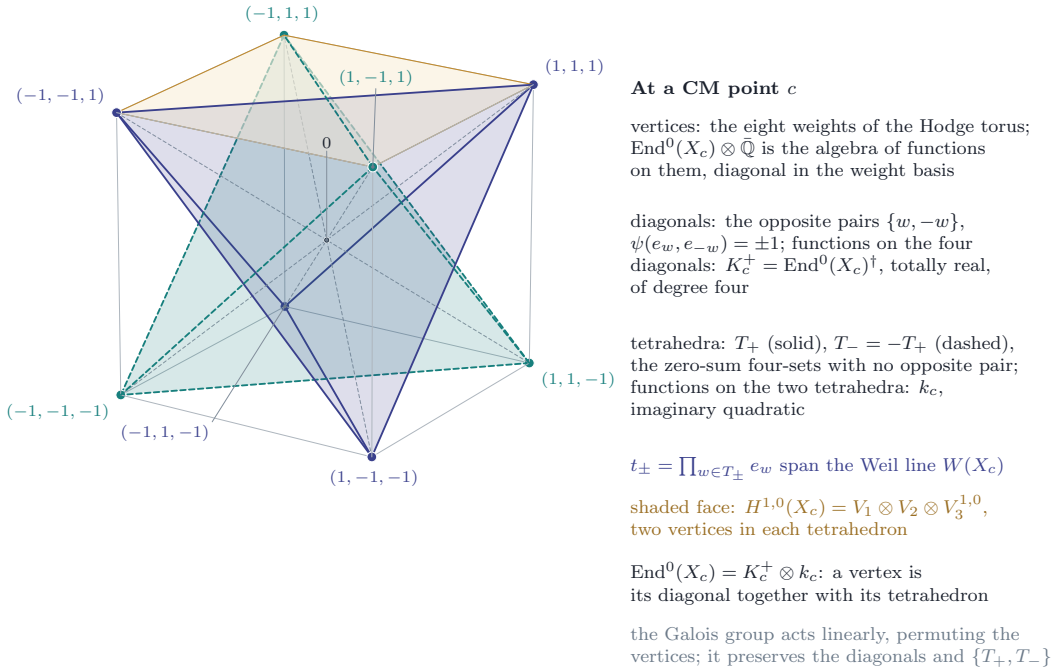
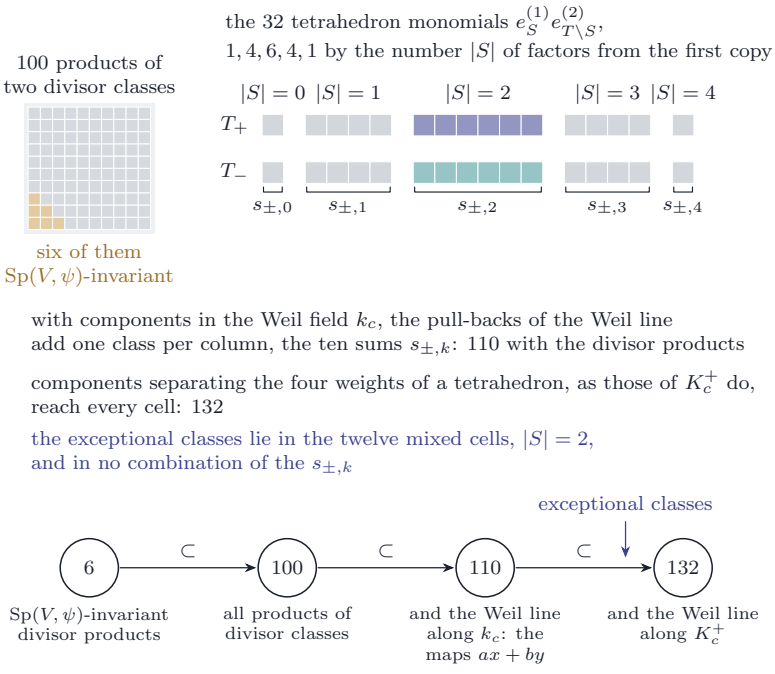


FIGURE 35. Proposition 7.29(i). The eight weights $(\pm1, \pm1, \pm1)$ of the Hodge torus at a CM point are the vertices of a cube. The four diagonals are the pairs of opposite weights, paired by the polarisation; the two inscribed tetrahedra T_+ (an even number of entries -1) and $T_- = -T_+$ are the only zero-sum four-sets without an opposite pair, and $t_\pm = \prod_{w\in T_\pm} e_w$ span the Weil line. The Galois group acts by linear maps permuting the vertices, so it preserves the diagonals and the pair of tetrahedra: the functions constant on the tetrahedra form the imaginary quadratic Weil field k_c, those constant on the diagonals the totally real quartic algebra K_c^+ of Rosati-fixed endomorphisms, and $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ because a vertex is determined by its diagonal and its tetrahedron.

set $\mathrm{Hom}(k, \bar{\mathbb{Q}})$, and $k_c \cong k$. It carries the pair $\{\pm(1,1,1)\}$ to itself, so the projection p_E onto $\mathbb{C}e_{(1,1,1)} \oplus \mathbb{C}e_{(-1,-1,-1)}$ lies in $\mathrm{End}^0(X_c)$, and by Poincaré reducibility X_c is isogenous to $E \times B$ with $H^1(E, \mathbb{C})$ the image of p_E, of dimension two, so E is an elliptic curve, on whose H^1 the field k_c acts by two conjugate scalars: E has complex multiplication by k. The threefold B has $\mathrm{End}^0(B) \otimes \bar{\mathbb{Q}}$ containing the diagonal algebra of the six remaining weights, of dimension $6 = 2\dim B$, so B is of CM type. Each of the six remaining weights has exactly one coordinate different from the other two, and γ carries the opposite pair of the weight with distinguished coordinate t to the pair with distinguished coordinate γt; so the Galois set of the four opposite pairs is the fixed point $\{\pm(1,1,1)\}$ together with a copy of $\mathrm{Hom}(F, \bar{\mathbb{Q}})$, and $K_c^+ \cong \mathbb{Q} \times F$, the factor $\mathbb{Q}$ acting on $H^1(E)$ and the factor F on $H^1(B)$. Finally T_+ consists of $(1,1,1)$ and three weights of B, and T_- of $(-1,-1,-1)$ and three weights of B, so $t_\pm \in H^1(E) \otimes H^3(B)$ and $W(X_c)$ lies there. This proves that (c) implies (a) and (b), with the asserted structure.

X_c is simple exactly when $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a field, and since every factor field E' of K_c^+ is totally real, $E' \otimes k_c$ is a field; so (a) and (b) are equivalent. Suppose (b). Then



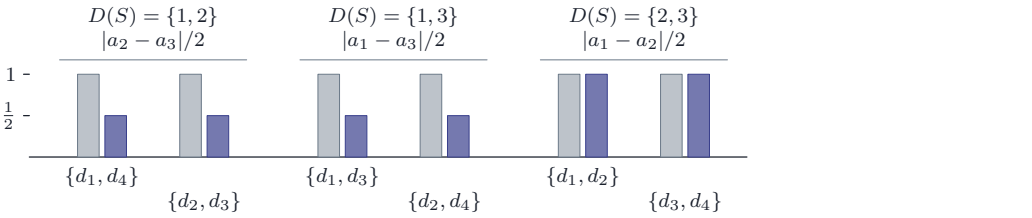


FIGURE 36. Proposition 7.29(ii) and (iii). Top: the 132 Hodge classes of degree four on the square at a CM point, one cell per monomial of weight zero. The 100 products of divisor classes are the cells with an opposite pair; the 32 tetrahedron monomials $e_S^{(1)} e_{T\setminus S}^{(2)}$ are arranged by tetrahedron and by the number $|S|$ of factors from the first copy. Pull-backs of the Weil line along homomorphisms with components in k_c add only the ten symmetric sums $s_{\pm,k}$, one per column, giving 110; components separating the four weights of a tetrahedron, as those of K_c^+ do, separate the monomials and give all 132. Bottom: on the six mixed monomials of one tetrahedron, grouped in the three pairs by the coordinates in which the two weights of S differ, the symmetric sum $s_{+,2}$ has coefficients ± 1 throughout, while an exceptional class ω_a has coefficients of absolute value $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$, drawn for $\pi_{12} - \pi_{13}$; two equal absolute values would be a linear relation among the conjugates of an irrational element of F.

the Galois group does not act transitively on the four opposite pairs, whose algebra is K_c^+. It acts transitively on the three coordinates, as on the embeddings of F, so some γ permutes them cyclically. For each coordinate t the four pairs fall into two sets of two, according to the value of the coordinate t on the representatives in T_+, and this assigns the three coordinates bijectively to the three partitions of the four pairs into two sets of two, compatibly with the Galois action, which preserves or exchanges T_+ and T_-. So the permutation of the pairs induced by γ has order divisible by three, hence equal to three: it is a cycle of length three and fixes exactly one pair. The orbits of the Galois group on the pairs are therefore either one orbit or two, of sizes one and three, and by hypothesis some pair $\{w, -w\}$ is fixed: $\gamma w = \delta_\gamma w$ with $\delta_\gamma = \pm 1$ for every γ. Replace the choice of τ_σ by $\tau'_\sigma = \tau_\sigma$ if $w_\sigma = 1$ and $\tau'_\sigma = \bar\tau_\sigma$ if $w_\sigma = -1$; for the new choice w becomes $(1, 1, 1)$, and the new signs are $\varepsilon'_\gamma(\sigma) = \varepsilon_\gamma(\sigma) w_\sigma w_{\gamma\sigma} = \delta_\gamma$ for all σ. So γ carries the set $\Phi = \{\tau'_\sigma\}$ of three embeddings of L to Φ if $\delta_\gamma = 1$ and to $\bar\Phi$ if $\delta_\gamma = -1$. The map $\gamma \mapsto \delta_\gamma$ is a character of the Galois group; it is nontrivial, for otherwise Φ would be a Galois stable set of three of the six embeddings and L would not be a field; and complex conjugation carries every embedding τ of the CM field L to $\bar\tau$, hence Φ to $\bar\Phi$, so the quadratic field k cut out by the character is imaginary. Let ι_0, $\bar\iota_0$ be the embeddings of k, with ι_0 fixed by the kernel of the character. Then $\tau \mapsto (\tau|_F, \iota_0$ if $\tau \in \Phi$ and $\bar\iota_0$ otherwise) is a Galois equivariant bijection $\mathrm{Hom}(L, \bar{\mathbb{Q}}) \to \mathrm{Hom}(F, \bar{\mathbb{Q}}) \times \mathrm{Hom}(k, \bar{\mathbb{Q}})$, so $L \cong F \otimes_{\mathbb{Q}} k$, that is, $L = Fk$ for a copy of k in L, which is (c). At a CM point where (c) fails, K_c^+ is therefore a field, quartic and totally real by Proposition 7.29(i).

Existence. Let k be imaginary quadratic. By the theorem of Albert, Brauer, Hasse and Noether [PR94, §1.5], the CM field Fk splits D, and so embeds in D over F, exactly when $Fk \otimes_F F_v$ is a field at every place v of F at which D is ramified, a quadratic extension of a local field splitting the local quaternion division algebra. At the two real places where D is definite this holds because k is imaginary, and at the finite ramified places it is the hypothesis. Given the finitely many residue characteristics p of these places, choose for each a quadratic extension k_p of $\mathbb{Q}_p$ not contained in F_v for any v above p: this is possible since the completions F_v above p have degrees over $\mathbb{Q}_p$ adding up to three, so at most one of them contains a quadratic extension of $\mathbb{Q}_p$, and then exactly one, while $\mathbb{Q}_p$ has at least three. Infinitely many imaginary quadratic fields k have the completions k_p at these p, and for them $F_v \otimes_{\mathbb{Q}_p} k_p$ is a field. For such k let $T_k = \mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{Fk/F}) \subset G$, a maximal torus with $T_k(\mathbb{R})$ compact. A compact subgroup of $G(\mathbb{R})$ fixes a point x of the symmetric space of $G(\mathbb{R})$, and the stabiliser of x is the centraliser of $h_x(\mathbb{S})$; so $h_x(\mathbb{S})$ centralises $T_k(\mathbb{R})$, which is Zariski dense in T_k, and lies in $T_k(\mathbb{R})$. The Hodge group of the fibre X_c over the image c of x is the smallest $\mathbb{Q}$-subgroup of G containing $h_x(U(1))$, so it lies in T_k; it has rank three (proof of Theorem 6.29), so it equals T_k, and c is a CM point with $L = Fk$. ■

Remark 7.31 (What the degree bound has to control).

(1) Markman's theorem [Mar25a], which the proof of Theorem 6.29 invokes, produces the cycles on X_c through its Weil structure for k_c. Proposition 7.29 says that this structure is blind to the exceptional classes of the square: everything obtained from the Weil line and from divisors by pull-back along homomorphisms with components in k_c, in particular along the maps $(x, y) \mapsto ax + by$, and by products, spans 110 of the 132 Hodge classes and misses every exceptional class; and a representation of an exceptional class by pull-backs of the Weil line and products of divisors must use homomorphisms whose components, with k_c, generate the whole endomorphism algebra. At a CM point the exceptional classes are thus a statement about the real

multiplication K_c^+, as they are at a general point about the real multiplication by F (Proposition 7.2).

(2) For $\varphi, \psi \in \mathrm{End}(X_c)$ and a cycle Z of codimension two on X_c, the degree $\int_Y f_{\varphi,\psi}^*[Z] \cup \theta_Y^6$ is, by (36) applied to the monomials of $[Z]$, a polynomial of degree four in the eigenvalues φ_w, ψ_w, which are algebraic integers. For cycles of this shape the bound of Corollary 7.28 therefore asks, at infinitely many CM points, for endomorphisms with bounded eigenvalues whose components generate $\mathrm{End}^0(X_c)$ over k_c. Only finitely many algebraic integers of degree at most eight have bounded conjugates, so the same finitely many minimal polynomials would have to occur at infinitely many points. At the split points of Lemma 7.30 the algebra $K_c^+ = \mathbb{Q} \times F$ does not depend on the point and the real multiplication is by F on the threefold factor; the representation of the proof of Theorem 6.29 then uses, besides elements of F, the isogeny $X_c \to E \times B$, a cycle representing the Weil line of $E \times B$ in $H^1(E) \otimes H^3(B)$, and divisor classes, and these three are what varies with the point.

(3) At a split point the Weil line is a Hodge class in $H^1(E) \otimes H^3(B)$, a correspondence between an elliptic curve with complex multiplication by k and a CM threefold with real multiplication by F, and its algebraicity is again Markman's theorem for the fourfold $E \times B$. A construction of that correspondence with a degree bounded independently of k, together with a bound on the isogeny and on the divisor classes, would give the target through Corollary 7.28. We do not have one.

Remark 7.32 (Routes to the Mumford target). We list the routes to the Mumford target that avoid the Lefschetz standard conjecture, with what is known about each; Figure 37 summarises them.

(1) *Degeneration.* Closed off: the base is compact and the Hodge locus is the base (Corollary 7.23), so no family on which ω stays of Hodge type approaches the boundary, and there is no limit mixed Hodge structure to specialise to.

(2) *Deformation into a larger Hodge locus*, on which ω would become a product of divisor classes or a Weil class, hence algebraic by a known theorem, and could then be carried back. Closed off by Theorem 7.21(iii), already to first order.

(3) *Hochschild deformations*: twisting $D^b(Y)$ by a B-field or a Poisson bivector, as in the gerby and noncommutative deformations used for Weil classes. Closed off: an object with $\mathrm{ch}(E) = N\omega$ extends along no direction of $HH^2(Y)$ except that of the curve (Corollary 7.41(i)).

(4) *Constructions from line bundles*: cones, tensor products, duals, pull-backs and push-forwards along homomorphisms, and Fourier–Mukai functors whose kernels are built in the same way. Closed off at every point of the curve: everything so built has a Chern character invariant under the Lefschetz group, and the exceptional classes are not (Theorem 7.36).

(5) *Twistor and hyperholomorphic deformation*, the mechanism of Markman's proof for Weil classes: move a sheaf along twistor lines on which its Chern character stays of Hodge type. Closed off on the square: no twistor line through a member of the Mumford family keeps ω of Hodge type, and the twistor lines that do keep it of Hodge type carry non-algebraic tori and lie in other components of the Hodge locus (Proposition 7.24).

(6) *Specialisation.* The class is algebraic at every CM point (Theorem 6.29), on every power of X_c [Li26, Theorem 1.3]; this suffices exactly when the degrees are bounded

(Corollary 7.28), and no bound is known. The cycles that represent the class at a CM point must use the real multiplication by the Rosati-fixed algebra K_c^+, all of it; the Weil structure of the fourfold, with which Markman's theorem produces its cycles, reaches 110 of the 132 Hodge classes and none of the exceptional ones (Proposition 7.29, Remark 7.31).

(7) *Reduction modulo p.* At every closed point of every reduction of the curve the class is a $\mathbb{Q}_\ell$-combination of algebraic cycles (Proposition 7.42), by Li's theorem [Li26, Theorem 1.1]; this suffices, with no lifting step, exactly when the Hilbert data of the cycles are bounded on a Zariski dense set of closed points of the arithmetic model (Theorem 7.43), and no bound is known.

(8) *Semiregularity.* Reduced to one dimension count at one CM point (Theorem 7.27); an object attaining it has large odd self-extensions (Theorem 7.34), is not built from line bundles (Proposition 7.35) and splits only in a constrained way (Corollary 7.41); no such object is known.

(9) *Kuga–Satake.* Reduced to the Kuga–Satake class of one K3 surface of Picard number thirteen (Theorem 7.4), which the known cases of the Kuga–Satake Hodge conjecture do not reach (Remark 7.5). Its geometric form is the mechanism of van Geemen and Rapagnetta [vGR26], pull-back of the dual Beauville–Bogomolov class of a hyperkähler manifold: up to multiples of π_0, the exceptional classes are exactly such twisted dual forms of the part of H^2 of the square generated by its symplectic class, and a rational map onto a symplectic subvariety of some $S_\mu^{[n]}$, $n \geq 5$, would suffice; K3 surfaces and hyperkähler eightfolds cannot serve (Proposition 7.25, Remark 7.26).

(10) *Motivated classes.* Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], so ω_t is motivated for every t; a motivated class is algebraic once the Lefschetz involutions of the auxiliary varieties are (Proposition 6.16). Along the family the auxiliary variety that suffices is $W \times_C W$ (Theorem 6.28), and its Lefschetz standard conjecture is equivalent to the target (Corollary 6.30): the route moves the problem and does not shorten it.

Five of the ten routes are closed off by theorems. Of the other five, two work at special points only, the CM points and the closed points of the reductions modulo p, and each suffices exactly when a degree bound holds; two are reduced to a single object, one Kuga–Satake class or one perfect complex with $\dim \mathrm{Ext}^2 = 119$ at one CM point; and one is equivalent to the target. They are not five separate obligations: three of them are the target itself in other forms and the other two are stronger inputs that imply it, so any one of them would suffice (Corollary 7.45, Remark 7.46).

7.3. **Objects, reduction modulo p and the motivic group.** Theorem 7.27 reduces the semiregularity route to one object at one CM point. We first derive what any object with the right Chern character must satisfy, whether or not it is semiregular: no sequence of standard operations produces it from line bundles; it extends along no first order deformation of $D^b(Y)$ other than the direction of the curve; if it is semiregular, it splits only into summands with no Ext^2 between them; and its self-extensions are large in every degree, with an odd part that the Euler characteristic forces beyond what Hochschild cohomology supplies. We then turn to reduction modulo p. By Li's proof of the Tate conjecture for powers of abelian fourfolds over finite fields, the classes are represented by cycles at every closed point of every reduction of the curve, and the target is equivalent to a bound on those cycles on a Zariski dense set of closed points. Finally, the motivic

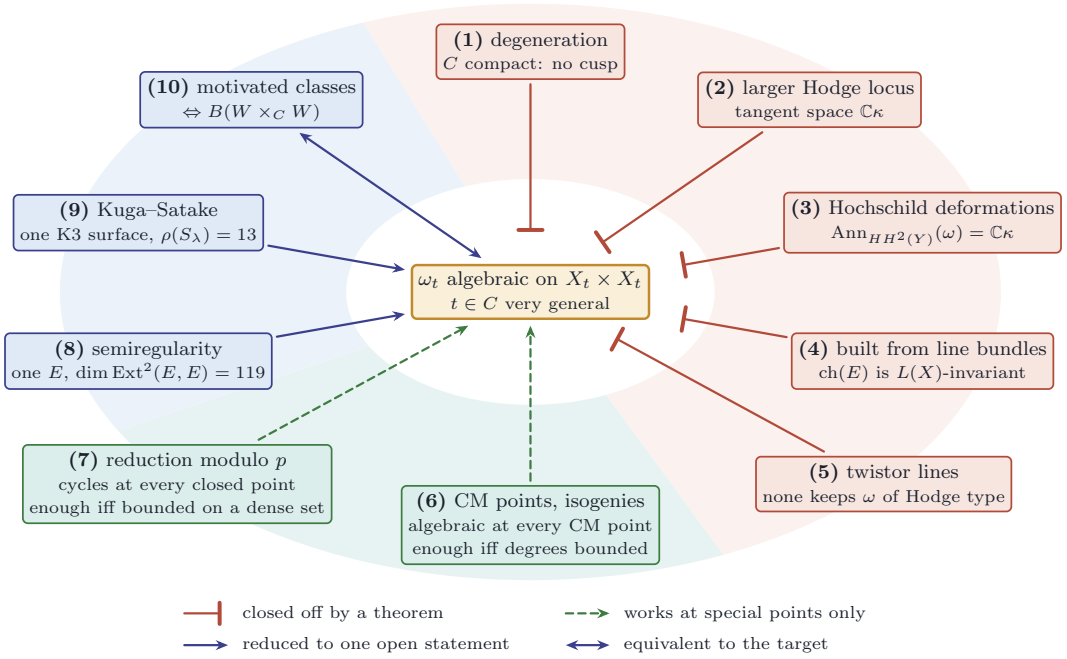


FIGURE 37. Remark 7.32: the ten routes to the Mumford target that avoid the Lefschetz standard conjecture. Clay, with a crossed dashed spoke: a route closed off by a theorem of Section 7.2 or Section 7.3. Grass: a route that works at special points only, at the CM points or at the closed points of the reductions modulo p, and suffices exactly when a degree bound holds (Corollary 7.28, Theorem 7.43). Indigo: a route reduced to one statement, of the same kind as the target or equivalent to it.

group of X is one of four groups, and the kinds of algebraic classes known on products of X with other abelian varieties generate no exceptional class. None of this constructs such an object or proves the target.

Lemma 7.33 (The contraction ranks are symmetric)**.** *Let A be an abelian variety of dimension g and $c \in H^*(A,\mathbb{C})$ a class each of whose components has Hodge type (p,p) for some p. For $k \geq 0$ put $r_k(c) = \dim(HT^k(A) \lrcorner c)$, with the action of Setup 2.7. Then $r_k(c) = r_{g-k}(c)$ for $0 \leq k \leq g$, and $r_k(c) = 0$ for $k > g$.*

Proof. Write $H^{1,0} = H^{1,0}(A)$ and $A' = H^{0,1}(A)$, both of dimension g, so that $H^*(A,\mathbb{C}) = \bigwedge^* H^{1,0} \otimes \bigwedge^* A'$ and $HT^*(A) = \bigwedge^*(A' \oplus T)$ with $T = (H^{1,0})^\vee$. Fix a generator of $\bigwedge^g A'$ and let $\star\colon \bigwedge^q A' \to \bigwedge^{g-q} A'^\vee$ send η to $\zeta \mapsto \eta \wedge \zeta$. Under $\Phi = \mathrm{id} \otimes \star$, wedging with $z \in A'$ becomes, up to a sign fixed on each bigraded piece, contraction with z on the second factor, while T still contracts the first. So Φ identifies the action of $HT^*(A)$ with the action of $\bigwedge^* U^\vee$ by interior products on $\bigwedge^* U$, where $U = H^{1,0} \oplus A'^\vee$ has dimension $2g$, up to signs that do not change ranks. A class of type (p,p) goes to $\bigwedge^p H^{1,0} \otimes \bigwedge^{g-p} A'^\vee \subset \bigwedge^g U$, so $\eta = \Phi(c)$ lies in the middle degree whatever the p of its components. Hence $r_k(c)$ is the rank of $\bigwedge^k U^\vee \to \bigwedge^{g-k} U$, $x \mapsto \iota_x \eta$, which is zero for $k > g$. Its transpose for the natural pairings is $\bigwedge^{g-k} U^\vee \to \bigwedge^k U$, $y \mapsto \pm \iota_y \eta$, because $\langle \iota_x \eta, y\rangle = \langle \eta, x \wedge y\rangle = \pm\langle \iota_y \eta, x\rangle$. A map and its transpose have the same rank. ∎

Theorem 7.34 (The self-extensions of an object with an exceptional Chern character)**.** *Let $c \in C$ and $Y = X_c \times X_c$, and write $e_k = \dim \operatorname{Ext}^k(E,E)$ for a perfect complex E on Y.*

(i) *For every E and every k, $e_k \geq r_k(\operatorname{ch} E)$ and $e_k = e_{8-k}$.*

(ii) *For a rational exceptional class $\omega = \omega_a$ with $L_0(a) \neq 0$: $r_0 = r_8 = 1$, $r_1 = r_7 = 16$ and $r_2 = r_6 = 119$.*

(iii) *Let ω be a rational exceptional class and $\operatorname{ch}(E) = N\omega$ with $N \neq 0$. Then*

$$e_1 + e_3 = e_0 + e_2 + \tfrac{1}{2}e_4 + \sum_{j\geq 1}(-1)^j e_{-j}.$$

If $\operatorname{Ext}^{<0}(E,E) = 0$, for instance if E is a sheaf, then $\sum_{k \text{ odd}} e_k = \sum_{k \text{ even}} e_k$, and $e_1 + e_3 \geq 120 + \frac{1}{2}e_4$ when $L_0(a) \neq 0$.

Proof. (i) The identity $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \operatorname{ch}(E)$ of Setup 2.7 holds in every degree, so the kernel of ev_E on $HT^k(Y)$ lies in the annihilator of $\operatorname{ch}(E)$, and the image of $HT^k(Y)$ in $\operatorname{Ext}^k(E,E)$ has dimension at least $r_k(\operatorname{ch} E)$. Serre duality with the trivial canonical bundle gives $e_k = e_{8-k}$.

(ii) $r_0 = 1$ because $\omega \neq 0$, and $r_2 = 119$ is Theorem 7.21(ii). If $w \in HT^1(Y)$ annihilates ω, so does $w' \wedge w$ for every $w' \in HT^1(Y)$, since the annihilator is a left ideal; for $w \neq 0$ these products span a space of dimension 15, which cannot lie in $\operatorname{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$. So $r_1 = 16$, and Lemma 7.33 gives r_8, r_7, r_6.

(iii) Since $\operatorname{ch}(E)$ has only a component in H^4, $\operatorname{ch}(E^\vee)\operatorname{ch}(E)$ lies in $H^8(Y)$ and $\chi(E,E) = \int_Y \operatorname{ch}(E^\vee)\operatorname{ch}(E) = 0$, the Todd class being trivial. By (i) and Serre duality, which also pairs Ext^{-j} with Ext^{8+j},

$$0 = \chi(E,E) = 2(e_0 - e_1 + e_2 - e_3) + e_4 + 2\sum_{j\geq 1}(-1)^j e_{-j},$$

which is the displayed identity. With no negative extensions the odd and even parts are equal, and $e_0 \geq 1$, $e_2 \geq 119$ by (i) and (ii) when $L_0(a) \neq 0$. ∎

Proposition 7.35 (Divisor classes do not reach the exceptional classes)**.** *Let $t \in C$ and let ω be a rational exceptional class. Then ω_t is not in the $\mathbb{Q}$-span of products of divisor classes of $X_t \times X_t$: that span meets the rational Hodge classes of $H^2 \otimes H^2$ exactly in the classes with $\alpha \in \mathbb{Q}$. Consequently no perfect complex whose Chern character lies in the subring generated by divisor classes, in particular no bounded complex of direct sums of line bundles, has $\operatorname{ch}(E) = N\omega$ with $N \neq 0$. At a CM point c the fourfold X_c has eight Hodge classes in H^4, of which the products of divisor classes span six.*

Proof. At a point with Hodge group G, the divisor classes of $X_t \times X_t$ are the three classes of Proposition 6.74(ii), all invariant under $\operatorname{Sp}(V,\psi)$, and so are their products; the $\operatorname{Sp}(V,\psi)$-invariant part of $H^2 \otimes H^2$ is spanned by π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$, the classes with $a_1 = a_2 = a_3$. At a CM point the Hodge group is a maximal torus of G (proof of Theorem 6.29), so the Hodge classes are the monomials of weight zero. The proof of Proposition 7.29(ii) counts them: 16 divisor classes $e_w^{(i)}e_{-w}^{(j)}$, 132 Hodge classes in degree four, of which the products of divisor classes span 100, and modulo that span $\omega_a \equiv \frac{1}{2}\big((a_2 - a_3)u_{12} + (a_1 - a_3)u_{13} + (a_1 - a_2)u_{23}\big)$ with u_{12}, u_{13}, u_{23} linearly independent. So the span meets that of the four π in exactly the classes with $a_1 = a_2 = a_3$. For a rational class these coordinates are the conjugates of α, which by Lemma 7.19 are equal only if $\alpha \in \mathbb{Q}$. The Chern character of a complex in the subring generated by divisor classes has its component in H^4 in the span of products of two divisor classes, which excludes $N\omega$. For X_c itself the monomials of weight zero in H^4 are the six products $e_w e_{-w} e_{w'} e_{-w'}$

over two opposite pairs, which are products of divisor classes, and the two tetrahedron monomials. ∎

At a CM point the Hodge classes of X_c in H^4 that are not products of divisor classes are among the classes that the proof of Theorem 6.29 pulls back along homomorphisms $X_c \times X_c \to X_c$; they are algebraic by Markman's theorem [Mar25a]. Li proves, independently and more generally, that every Hodge class on every power of every complex CM abelian fourfold is algebraic [Li26, Theorem 1.3], which gives Theorem 6.29 for every power X_c^n and not only for the square.

By Proposition 7.35, an object with $\mathrm{ch}(E) = N\omega$ cannot have its Chern character in the subring generated by divisor classes, so it cannot be assembled from line bundles alone. The next theorem shows that no sequence of the standard operations of the derived category, applied to line bundles on powers of X_t, gets any further.

Theorem 7.36 (Objects generated by line bundles)**.** *Let $t \in C$, $X = X_t$, and let $L(X) \subseteq \mathrm{Sp}(V,\psi)$ be the Lefschetz group of X, the centraliser of $\mathrm{End}^0(X)$ in $\mathrm{Sp}(V,\psi)$. Let $\mathcal{A}$ be the class of abelian varieties isogenous to a power of X; through an isogeny $X^n \to B$ the group $L(X)$ acts diagonally on $H^*(B,\mathbb{Q})$, and the action does not depend on the isogeny. Let $\mathcal{L}$ be the smallest class of objects of the categories $D^b(B)$, $B \in \mathcal{A}$, that contains the line bundles and is closed under shifts, cones, derived tensor products, derived duals, translations, the functors Lf^* and Rf_* for homomorphisms f between members of $\mathcal{A}$, and the Fourier–Mukai functors $\Phi_K \colon D^b(B) \to D^b(B')$ with kernel $K \in \mathcal{L}$ on $B \times B'$.*

(i) *Every object of $\mathcal{L}$ has an $L(X)$-invariant Chern character.*

(ii) *$L(X) = \mathrm{Sp}(V,\psi)$ if t is not a CM point, and $L(X)$ is a maximal torus of $\mathrm{Sp}(V,\psi)$ if it is. In both cases the $L(X)$-invariant classes on $X \times X$ form the subring generated by divisor classes, of dimensions $1,3,6,10,15,10,6,3,1$, respectively $1,16,100,304,454,304,100,16,1$, in the degrees $0,2,\dots,16$.*

(iii) *No rational exceptional class ω_t is $L(X)$-invariant. Hence for no object $E \in \mathcal{L}$ on $X \times X$ and no $N \neq 0$ is $\mathrm{ch}(E) - N\omega_t$ invariant under $L(X)$; in particular $\mathrm{ch}(E) \neq N\omega_t$.*

(iv) *The $\mathrm{Sp}(V,\psi)$-module generated by $\pi_{12} - \pi_{13}$ is irreducible, of highest weight $2\varpi_2$ and dimension 308, and it meets the span of the four π in the plane of the classes with $a_0 = 0$ and $a_1 + a_2 + a_3 = 0$.*

Proof. (i) Two isogenies $X^n \to B$ differ by an element of $\mathrm{End}^0(X^n) = M_n(\mathrm{End}^0(X))$, which commutes with $L(X)$ acting diagonally, so the action is well defined. For the same reason a homomorphism f between members of $\mathcal{A}$ is $L(X)$-equivariant on H^1, so f^* is equivariant on H^*, and so is f_*, which is f^* conjugated by Poincaré duality, since $L(X) \subseteq \mathrm{SL}(V)$ fixes the orientation classes. Translations act trivially on cohomology. The Chern character is additive on distinguished triangles and multiplicative on derived tensor products, $\mathrm{ch}(E^\vee) = \sum_i (-1)^i \mathrm{ch}_i(E)$, $\mathrm{ch}(Lf^*E) = f^*\mathrm{ch}(E)$, and $\mathrm{ch}(Rf_*E) = f_*\mathrm{ch}(E)$ by the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2], the Todd class of an abelian variety being 1; and $\Phi_K = Rp_{2*}(p_1^*(\,\cdot\,) \otimes^L K)$. So the objects with $L(X)$-invariant Chern character form a class with all the closure properties, and it remains to see that it contains the line bundles, that is, that $c_1(\mathcal{O}(D))$ is $L(X)$-invariant. Pulled back along an isogeny, which is injective on rational cohomology, c_1 becomes a Hodge class of degree two on some X^n, an element of the invariants of the Hodge group in $\bigwedge^2(V^\vee \otimes \mathbb{Q}^n) = \bigwedge^2 V^\vee \otimes S^2\mathbb{Q}^n \oplus S^2V^\vee \otimes \bigwedge^2\mathbb{Q}^n$. If t is not a CM point, the Hodge group is G, whose invariants in $\bigwedge^2 V$ are the multiples of ψ and in S^2V are zero, since every

summand of $S^2(V_1 \otimes V_2 \otimes V_3)$ contains a factor $S^2 V_i$; so the Hodge classes of degree two are $\mathrm{Sp}(V,\psi)$-invariant. If t is a CM point, a monomial of degree two in a weight basis has weight zero for the Hodge torus exactly when its two weights are opposite, and then it has weight zero for $L(X)$, by the description in (ii).

(ii) If t is not a CM point, $\mathrm{End}^0(X) = \mathbb{Q}$ and $L(X) = \mathrm{Sp}(V,\psi)$. At a CM point $\mathrm{End}^0(X) \otimes \mathbb{C}$ is the algebra of diagonal matrices in a weight basis (proof of Theorem 6.29), and ψ pairs each weight with its opposite only, so $L(X)_{\mathbb{C}}$ is the torus of diagonal matrices $\mathrm{diag}(\lambda_w)$ with $\lambda_w \lambda_{-w} = 1$, of rank four. For t not CM, the first fundamental theorem for $\mathrm{Sp}(V,\psi)$ [GW09, Chapter 5] shows that its invariants in $\bigwedge^*(V \otimes \mathbb{Q}^2)$ are spanned by the products of the three quadratic invariants, which are the divisor classes. Their dimensions follow from the skew Cauchy formula $\bigwedge^{2m}(V \otimes \mathbb{Q}^2) = \bigoplus_\lambda S_\lambda V \otimes S_{\lambda'}\mathbb{Q}^2$, over the partitions $\lambda = (2^r, 1^s)$ of $2m$ with at most two columns, and the rule that $S_\lambda V$ contains the trivial representation of $\mathrm{Sp}(V,\psi)$, then once, exactly when both columns of λ have even length at most 8 [GW09]: the invariants of degree $2m$ have dimension $\sum(s+1)$, over the even $r, s \geq 0$ with $2r + s = 2m$ and $r + s \leq 8$, which is $1, 3, 6, 10, 15, 10, 6, 3, 1$ for $m = 0, \ldots, 8$. At a CM point the invariants are spanned by the monomials of weight zero for $L(X)$; a set of weights $\pm e_j$ with sum zero is a union of opposite pairs, so each such monomial is, up to sign, a product of monomials of degree two and weight zero, which are divisor classes. For each opposite pair $\{w, -w\}$ the four vectors $e^{(1)}_{\pm w}$, $e^{(2)}_{\pm w}$ give one monomial of weight zero in degree 0, four in degree 2 and one in degree 4, so the numbers of these monomials are the coefficients of $(1 + 4y + y^2)^4$.

(iii) If t is not a CM point, the $\mathrm{Sp}(V,\psi)$-invariant combinations of the π are those with $a_1 = a_2 = a_3$ (Proposition 7.35), and the coefficients of ω_t are the distinct conjugates of α (Lemma 7.19). At a CM point use the map $x \mapsto E_x$ of the proof of Lemma 7.20, which is $\mathrm{Sp}(V,\psi)$-equivariant: it carries ω_t to an element (r, β) of $\mathbb{Q} \times F$ with $\beta \notin \mathbb{Q}$, acting on U_{ij} by the conjugate $\sigma_k(\beta)$ (Proposition 7.2(ii)), and ω_t is $L(X)$-invariant only if E_{ω_t} preserves the weight spaces of $L(X)$ in $\bigwedge^2 V$. Take weight vectors $v = x \otimes y \otimes z$ and $w = x \otimes y' \otimes z'$ of weights $(1,1,1)$ and $(1,-1,-1)$ for the Hodge torus. They are not opposite, so $v \wedge w$ spans a weight space of $L(X)$, every weight of $L(X)$ on V having multiplicity one. But $v \wedge w = (x \cdot x) \otimes (y \cdot y' \otimes z \wedge z' + y \wedge y' \otimes z \cdot z')$ has nonzero components in U_{12} and in U_{13}, on which E_{ω_t} acts by $\sigma_3(\beta) \neq \sigma_2(\beta)$. So the line is not preserved. The rest of (iii) follows from (i).

(iv) Write $H^2(X) = \mathbb{Q}\theta \oplus P$, with P the primitive part, the irreducible $\mathrm{Sp}(V,\psi)$-module of highest weight ϖ_2 and dimension 27. Through the form $\bigwedge^2 \psi$ the four π are self-adjoint endomorphisms of $H^2(X)$, and $\pi_{12} - \pi_{13}$ kills θ and is a trace-free self-adjoint endomorphism of P, an element of $S^2 P$ orthogonal to the invariant. Restricting $S^2(\bigwedge^2 V) = S_{(2,2)}V \oplus \bigwedge^4 V$ from $\mathrm{GL}(V)$ to $\mathrm{Sp}(V,\psi)$ gives, with dimensions $336 = 308 + 27 + 1$ and $70 = 42 + 27 + 1$, the decomposition $S^2 P = V(2\varpi_2) \oplus V(\varpi_4) \oplus V(\varpi_2) \oplus \mathbb{Q}$. The element $\pi_{12} - \pi_{13}$ is G-invariant. The summand $V(\varpi_2) \cong P$ has no G-invariant, since $(\bigwedge^2 V)^G = \mathbb{Q}\theta$, and neither has $V(\varpi_4)$, which lies in $\bigwedge^4 V$, whose G-invariants are the line of θ^2 (Proposition 6.74(i)), invariant under all of $\mathrm{Sp}(V,\psi)$. So $\pi_{12} - \pi_{13}$ is a nonzero vector of the irreducible module $V(2\varpi_2)$, which it generates; its dimension is 308 by Weyl's dimension formula. The same argument puts every trace-free combination of $\pi_{12}, \pi_{13}, \pi_{23}$ in $V(2\varpi_2)$, while π_0 and $\pi_{12} + \pi_{13} + \pi_{23}$ are $\mathrm{Sp}(V,\psi)$-invariant and independent modulo it; so adding the four π raises the dimension by two. ∎

Figure 38 compares the Hodge classes of $X_t \times X_t$ with the subring generated by divisor classes, degree by degree.

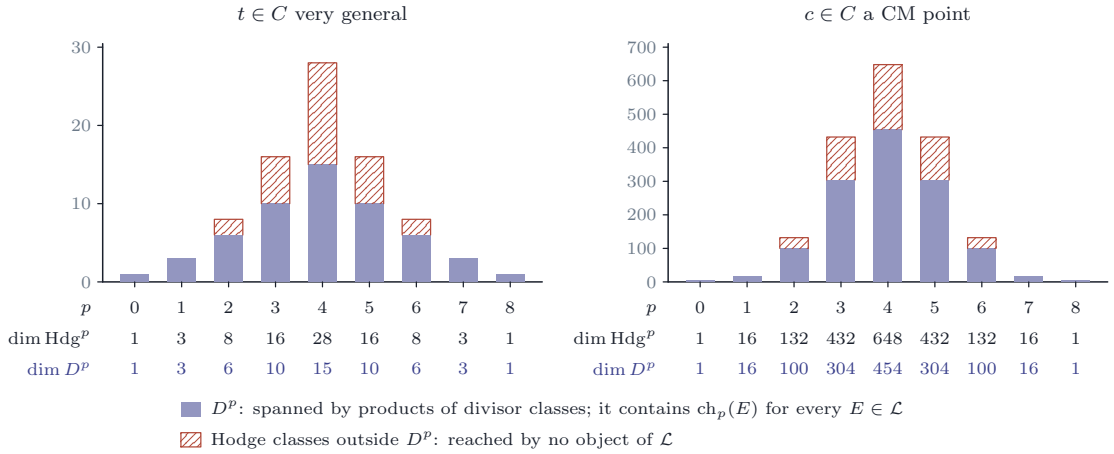


FIGURE 38. Theorem 7.36(ii). The Hodge classes of $X_t \times X_t$ degree by degree, against the subring D^p generated by divisor classes, which contains the Chern character of every object of $\mathcal{L}$: at a very general point of the Mumford curve (left) and at a CM point (right). The hatched classes, the exceptional ones among them, are reached by no object of $\mathcal{L}$.

Remark 7.37 (Where a construction has to leave $\mathcal{L}$). Theorem 7.36 extends Proposition 7.35 from sums of line bundles to everything the standard operations make out of them, at every point of the curve, including the CM points, where the exceptional classes are algebraic. There the cycles that represent them are not Chern characters of objects of $\mathcal{L}$, and neither are the two Hodge classes of X_c in degree four that are not products of divisor classes. At a point that is not CM, every Chern character of an object of $\mathcal{L}$ lies in the isotypic component of the trivial representation of $\mathrm{Sp}(V,\psi)$, while by Theorem 7.36(iv) an object with $\mathrm{ch}(E) = N\omega$ must have a nonzero component in the one of highest weight $2\varpi_2$. Every construction of such an object therefore contains a step that does not commute with the Lefschetz group on cohomology: passing to a direct summand, a subobject or a Jordan–Hölder factor, choosing a point of a moduli space, or deforming from a member with more endomorphisms. The last is what semiregularity does along the curve (Theorem 7.27).

Lemma 7.38 (The summands of a semiregular object)**.** *Let Z be a smooth projective variety and E_1, E_2 perfect complexes on Z. On $\mathrm{Ext}^2(E_1 \oplus E_2, E_1 \oplus E_2) = \bigoplus_{i,j} \mathrm{Ext}^2(E_j, E_i)$ the semiregularity map is $\sigma_{E_1\oplus E_2}(\zeta) = \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$. So $\sigma_{E_1\oplus E_2}$ is injective if and only if σ_{E_1} and σ_{E_2} are injective, their images meet only in zero, and $\mathrm{Ext}^2(E_1, E_2) = \mathrm{Ext}^2(E_2, E_1) = 0$.*

Proof. The Atiyah class is natural, so $\mathrm{At}(E_1 \oplus E_2)$ is the diagonal sum of $\mathrm{At}(E_1)$ and $\mathrm{At}(E_2)$. In $\sigma(\zeta) = \sum_q \frac{1}{q!} \mathrm{tr}(\zeta \circ \mathrm{At}^q)$ of Theorem 2.2 the composite $\zeta \circ \mathrm{At}^q$ has diagonal blocks $\zeta_{ii} \circ \mathrm{At}(E_i)^q$, and the trace of a block matrix of morphisms is the sum of the traces of its diagonal blocks. In particular the off-diagonal blocks ζ_{12} and ζ_{21} lie in the kernel, so injectivity forces them to vanish; on the diagonal blocks the map is $(\zeta_{11}, \zeta_{22}) \mapsto \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$, which is injective exactly when both summands are injective and their images meet only in zero. ∎

The condition on the images cannot be dropped, and the objects that show it are the simplest ones: the structure sheaves of abelian subvarieties of half dimension, each of which is semiregular on its own.

Lemma 7.39 (Abelian subvarieties of half dimension are semiregular)**.** *Let A be an abelian variety of dimension $2n$ and $B \subseteq A$ an abelian subvariety of dimension n. Then $\mathrm{ev}_{\mathcal{O}_B}\colon HT^2(A) \to \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$ is surjective, its kernel is $\mathrm{Ann}_{HT^2}([B])$, of dimension $6n^2 - n$, and $\sigma_{\mathcal{O}_B}$ is injective.*

Proof. The normal bundle of B is trivial, so $\mathrm{ch}(\mathcal{O}_B) = [B]$ by the Grothendieck–Riemann–Roch theorem for the embedding, and $\mathrm{Ext}^*(\mathcal{O}_B, \mathcal{O}_B) \cong \bigwedge^* \mathbb{C}^{2n}$ (proof of Corollary 3.17), so $\dim \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B) = \binom{2n}{2} = 2n^2 - n$, while $\dim HT^2(A) = 2\binom{2n}{2} + 4n^2 = 8n^2 - 2n$. Choose a complementary abelian subvariety B'; the isogeny $B \times B' \to A$ identifies $H^1(A, \mathbb{C})$ with $H^1(B) \oplus H^1(B')$ compatibly with the Hodge decompositions, identifies $HT^2(A)$ with $HT^2(B \times B')$ compatibly with contraction, and carries $[B]$ to a nonzero multiple of $1 \otimes [\mathrm{pt}_{B'}]$, where $[\mathrm{pt}_{B'}]$ spans $\bigwedge^{2n} H^1(B')$. Write $T = T_B \oplus T_{B'}$ for the tangent space and $H^{0,1}(A) = H^{0,1}(B) \oplus H^{0,1}(B')$. A class $\zeta \in H^{0,2}(A) = \bigwedge^2 H^{0,1}(A)$ has $\zeta \cup [B] = \zeta_{BB} \otimes [\mathrm{pt}_{B'}]$, with ζ_{BB} its component in $\bigwedge^2 H^{0,1}(B)$, because every factor from $H^{0,1}(B')$ is killed by the top class of B'; so the annihilator in $H^{0,2}$ has dimension $\binom{2n}{2} - \binom{n}{2}$. For $a \otimes v \in H^{0,1}(A) \otimes T = H^1(T_A)$ the contraction is $a \wedge (v \lrcorner [B]) = a_B \otimes (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$, since v_B kills $[\mathrm{pt}_{B'}]$ and $a_{B'} \wedge (v_{B'} \lrcorner [\mathrm{pt}_{B'}])$ would have type $(n-1, n+1)$ on the n-fold B'; the map $v \mapsto v_{B'} \lrcorner [\mathrm{pt}_{B'}]$ has rank n, as does $a \mapsto a_B$, so the contraction has rank n^2 on the space $H^{0,1}(A) \otimes T$ of dimension $4n^2$ and the annihilator there has dimension $3n^2$. For $\vartheta \in \bigwedge^2 T$ only the component in $\bigwedge^2 T_{B'}$ contracts $[\mathrm{pt}_{B'}]$ nontrivially, and injectively; so the annihilator in $\bigwedge^2 T$ has dimension $\binom{2n}{2} - \binom{n}{2}$. Altogether $\dim \mathrm{Ann}_{HT^2}([B]) = 2\binom{2n}{2} - 2\binom{n}{2} + 3n^2 = 6n^2 - n$. By the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ of Setup 4.54 the kernel of $\mathrm{ev}_{\mathcal{O}_B}$ lies in $\mathrm{Ann}_{HT^2}([B])$, so $\mathrm{ev}_{\mathcal{O}_B}$ has rank at least $(8n^2 - 2n) - (6n^2 - n) = 2n^2 - n = \dim \mathrm{Ext}^2(\mathcal{O}_B, \mathcal{O}_B)$: it is surjective and its kernel is the annihilator. If $\sigma_{\mathcal{O}_B}(y) = 0$, write $y = \mathrm{ev}(x)$; then $x \lrcorner [B] = 0$, so x lies in the annihilator, which is the kernel, and $y = 0$. ∎

Remark 7.40 (Sums of abelian subvarieties). Let $B_1, \dots, B_r \subseteq A$ be abelian subvarieties of dimension $n \geq 3$, translated so that any two of them meet in finitely many points or not at all, which generic translates achieve, and let $E = \bigoplus_j \mathcal{O}_{B_j}[k_j]$. The sheaves $\mathcal{E}xt^q(\mathcal{O}_{B_i}, \mathcal{O}_{B_j})$ for $i \neq j$ are supported on $B_i \cap B_j$ and vanish for $q \neq n$, by the Koszul resolution of $\mathcal{O}_{B_i}$ restricted to B_j, so $\mathrm{Ext}^2(\mathcal{O}_{B_i}[k_i], \mathcal{O}_{B_j}[k_j]) = H^{2+k_j-k_i-n}(B_i \cap B_j, \mathcal{E}xt^n)$ vanishes when all k_j are equal and $n \geq 3$, and when $|k_i - k_j| \leq 1$ and $n \geq 4$; and each $\sigma_{\mathcal{O}_{B_j}}$ is injective by Lemma 7.39. In these cases every hypothesis of Lemma 7.38, applied to one summand and the sum of the others, holds except the one on the images. Now let $v \in HT^2(A)$ satisfy $v \lrcorner \mathrm{ch}(E) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$ with $v \lrcorner [B_1] \neq 0$. Then $\zeta = \bigoplus_j \mathrm{ev}_{\mathcal{O}_{B_j}}(v)$ is nonzero, because v is not in the annihilator of $[B_1]$, and $\sigma_E(\zeta) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$: the images of the $\sigma_{\mathcal{O}_{B_j}}$ meet, and E is not semiregular. This is exactly the situation at a member of a Weil family on which a Weil class is a combination of classes of abelian subvarieties of dimension n, as on a product of elliptic curves: a tangent direction v of the family keeps $\mathrm{ch}(E)$ of Hodge type but moves any $[B_j]$ outside $\mathbb{Q}\theta^n \oplus W(A)$, and Corollary 3.17 excludes such sums for this reason. The sum is obstructed although its summands are not.

Corollary 7.41 (Hochschild deformations and direct summands)**.** *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and E a perfect complex on Y with $\mathrm{ch}(E) = N\omega$, $N \neq 0$.*

(i) *E extends to the first order deformation of $D^b(Y)$ given by $x \in HH^2(Y) = HT^2(Y)$ only if $x \in \mathbb{C}\kappa$. No B-field, no Poisson bivector and no commutative deformation off the curve carries it.*

(ii) *If σ_E is injective and $E = E_1 \oplus E_2$ with both summands nonzero, then $\operatorname{Ext}^2(E_1, E_2) = \operatorname{Ext}^2(E_2, E_1) = 0$, each σ_{E_i} is injective, each E_i extends to first order along the curve, and $\kappa \lrcorner \operatorname{ch}(E_i) = 0$. If moreover $\dim \operatorname{Ext}^2(E, E) = 119$, then $\dim \operatorname{Ext}^2(E_i, E_i) = 120 - \dim \operatorname{Ann}_{HT^2}(\operatorname{ch} E_i)$, and*

$$\operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) + \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = HT^2(Y), \qquad \operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) \cap \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = \mathbb{C}\kappa.$$

Proof. (i) The kernel of ev_E on $HH^2(Y)$ is the space of deformations of $D^b(Y)$ along which E deforms to first order [Tod09], [Mar25a, Section 8.3]. If $\mathrm{ev}_E(x) = 0$ then $N\, x \lrcorner \omega = \sigma_E(\mathrm{ev}_E(x)) = 0$, so $x \in \mathbb{C}\kappa$ by Theorem 7.21(ii). The three kinds of deformation named are the three summands of $HT^2(Y)$.

(ii) The first two assertions are Lemma 7.38. By Theorem 7.27(ii), $\mathrm{ev}_E(\kappa) = 0$, and ev_E is the diagonal sum of ev_{E_1} and ev_{E_2}, so each vanishes at κ; then $\kappa \lrcorner \operatorname{ch}(E_i) = \sigma_{E_i}(\mathrm{ev}_{E_i}(\kappa)) = 0$. Put $e(x) = 120 - \dim \operatorname{Ann}_{HT^2}(x)$, the rank of contraction into x. It is subadditive in x, and by Theorem 7.34(i) $\dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_i)$. The cross terms vanish, so

$$119 = \dim \operatorname{Ext}^2(E, E) = \sum_i \dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_1) + e(\operatorname{ch} E_2) \geq e(N\omega) = 119,$$

and every inequality is an equality. Hence $\dim \operatorname{Ann}(\operatorname{ch} E_1) + \dim \operatorname{Ann}(\operatorname{ch} E_2) = 121$. The intersection of the two annihilators lies in $\operatorname{Ann}(N\omega) = \mathbb{C}\kappa$ and contains κ, so it is $\mathbb{C}\kappa$, and the sum has dimension 120. ∎

Part (ii) plays for the Mumford classes the role of Theorem 6.32: a semiregular object attaining the bound either does not split, or splits into two pieces whose annihilators fill $HT^2(Y)$ and meet only in the direction of the curve. The two annihilators are the analogue of the two branches P and Q of Remark 2.10.

Proposition 7.42 (Tate classes at every closed point of every reduction)**.** *There are a finitely generated subring $R \subset \mathbb{C}$, a model $W_R \to C_R$ of $W \to C$ over R, with C_R smooth and projective over R with geometrically connected fibres and W_R an abelian scheme, projective over C_R, and a finite surjective morphism $C'_R \to C_R$ from a second curve of the same kind, with the following properties. Put $Y' = (W_R \times_{C_R} W_R) \times_{C_R} C'_R$, with projection $\pi' \colon Y' \to C'_R$, and let ω be a rational exceptional class. For every prime ℓ there is a global section ω_ℓ of the lisse sheaf $R^4\pi'_*\mathbb{Q}_\ell(2)$ on $C'_R[1/\ell]$ whose restriction to $C'_R \otimes_R \mathbb{C}$ is the pull-back of $\omega \otimes 1$; write $\omega_\ell(\bar{x})$ for its value at a geometric point over a point x.*

(i) *At a closed point x of $C'_R[1/\ell]$ the residue field $\kappa(x)$ is finite, Y'_x is the square of an abelian fourfold over it, and $\omega_\ell(\bar{x})$ is fixed by $\operatorname{Gal}(\bar{\kappa}/\kappa(x))$.*

(ii) *At such a point $\omega_\ell(\bar{x})$ is a $\mathbb{Q}_\ell$-linear combination of classes of algebraic cycles of codimension two on Y'_x, defined over $\kappa(x)$.*

(iii) *Let s be a closed point of $\operatorname{Spec} R[1/\ell]$ and η_s the generic point of the reduction C'_s. Then $\omega_\ell(\bar{\eta}_s)$ is fixed by the Galois group of the function field $\kappa(\eta_s)$, and it is not a $\mathbb{Q}_\ell$-linear combination of products of divisor classes of $Y'_{\bar{\eta}_s}$.*

Every closed point x of a reduction C_s of C is the image of a closed point x' of C'_s, and $Y'_{x'} = Y_x \otimes_{\kappa(x)} \kappa(x')$; so (i) and (ii) hold at every closed point of every reduction of C, after a finite extension of its residue field.

Proof. The models. The family $W \to C$ is defined over a subfield $k_1 \subset \mathbb{C}$ finitely generated over $\mathbb{Q}$, say as $W_1 \to C_1$. Let K_1 be the function field of C_1 and fix an embedding over k_1

of an algebraic closure $\bar{K}_1$ into $\mathbb{C}$; it sends the geometric generic point of C_1 to a point $t \in C$, and $Y_t = X_t \times X_t$ is the base change of $Y_{\bar{K}_1}$. The class ω_t is a Hodge class on an abelian variety, hence absolutely Hodge, so it comes from an absolutely Hodge class of $Y_{\bar{K}_1}$, and $\mathrm{Gal}(\bar{K}_1/K_1)$ acts on these through a finite quotient [Del82, Theorem 2.11 and Proposition 2.9]. Let $K' \subset \bar{K}_1$ be a finite extension of K_1 on which the action is trivial, k' the algebraic closure of k_1 in K', and C' the smooth projective curve over k' with function field K'; it is geometrically connected over k' and maps finitely onto $C_1 \otimes_{k_1} k'$. Spreading out $W_1 \to C_1$, C' and this map over a finitely generated subring of k', and inverting one element of it, gives a normal R and models with the stated properties.

The section. By smooth and proper base change $R^4\pi'_*\mathbb{Q}_\ell(2)$ is lisse on $C'_R[1/\ell]$. Its stalk at the geometric generic point $\operatorname{Spec}\bar{K}_1$ is $H^4(Y_{\bar{K}_1}, \mathbb{Q}_\ell(2)) = H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell$, and $\omega_t \otimes 1$ is the ℓ-adic component of an absolutely Hodge class, so it is fixed by $\mathrm{Gal}(\bar{K}_1/K')$. The scheme $C'_R[1/\ell]$ is normal and integral with function field K', so $\mathrm{Gal}(\bar{K}_1/K')$ maps onto its fundamental group [Stacks, Tag 0BQI], and the vector extends to a global section ω_ℓ. On the connected curve $C'_R \otimes_R \mathbb{C}$ it agrees with the pull-back of $\omega \otimes 1$, the two being global sections of the same local system that agree at a point over t.

(i) and (ii). A scheme of finite type over $\mathbb{Z}$ has finite residue fields at its closed points. A global section of a lisse sheaf is fixed by the fundamental group of every point, which is (i); and (ii) is Li's theorem [Li26, Theorem 1.1], with $n = r = 2$, for the abelian fourfold that is the fibre of W_R at the image of x, base changed to $\kappa(x)$.

(iii) The first assertion holds for the same reason as (i). For the second, let $\bar{\eta}$ be a geometric generic point of C'_R and $\bar{s}$ a geometric point over s. The morphism $C'_R \to \operatorname{Spec} R$ is flat and proper with geometrically connected and geometrically reduced fibres, so the specialisation map $\pi_1(C'_{\bar{\eta}}) \to \pi_1(C'_{\bar{s}})$ is surjective [Stacks, Tag 0C0P], and it is compatible with the actions of the two groups on the first cohomology of the fibres of W_R, identified by cospecialisation. The image of $\pi_1(C'_{\bar{\eta}})$ contains that of $\pi_1(C'_{\mathbb{C}})$, the closure of the image of the topological fundamental group of a finite cover of C, whose Zariski closure contains $G_{\mathbb{Q}_\ell}$ because the connected monodromy group of $W \to C$ is G (proof of Corollary 6.22). So the Zariski closure of the monodromy of W_R over $C'_{\bar{s}}$ contains $G_{\mathbb{Q}_\ell}$. A divisor class of $Y'_{\bar{\eta}_s}$ is defined over a finite extension of $\kappa(\eta_s)$, so it is fixed by an open subgroup of the Galois group, whose image in $\pi_1(C'_s)$ is open [Stacks, Tag 0BQI] and meets $\pi_1(C'_{\bar{s}})$ in an open subgroup; the Zariski closure of the image of an open subgroup has the same identity component, so the class is $G_{\mathbb{Q}_\ell}$-invariant. The G-invariants of $\bigwedge^2(V \oplus V)$ are the three divisor classes of Proposition 6.74(ii), which are $\mathrm{Sp}(V,\psi)$-invariant, so every product of divisor classes of $Y'_{\bar{\eta}_s}$ is $\mathrm{Sp}(V,\psi)$-invariant. The class $\omega_\ell(\bar{\eta}_s)$ corresponds to $\omega \otimes 1$, whose coefficients a_1, a_2, a_3 are the distinct conjugates of α (Lemma 7.19), and the $\mathrm{Sp}(V,\psi)$-invariant classes among the ω_a are those with $a_1 = a_2 = a_3$ (Proposition 7.35). The last sentence of the statement holds because $C'_R \to C_R$ is finite and surjective. ∎

At the generic point of each reduction the class is therefore a Tate class over a global function field that is not a combination of products of divisor classes, exactly as over $\mathbb{C}$, while at every closed point it is a combination of classes of cycles. The next theorem shows that a bound on those cycles is all that is missing, with no passage through p-adic Hodge theory.

Theorem 7.43 (The Mumford target is a degree bound in positive characteristic)**.** *Keep Proposition 7.42, fix a prime ℓ and a relatively ample line bundle on $Y' \to C'_R$, and for a finite set Φ of tuples of Hilbert polynomials say that ω is* Φ-represented *at a point x of*

$C'_R[1/\ell]$ if $\omega_\ell(\bar{x}) = \sum_i q_i \operatorname{cl}(Z_i)$ for some $q_i \in \mathbb{Q}_\ell$ and closed subschemes $Z_i \subset Y'_{\bar{x}}$ of pure codimension two whose Hilbert polynomials form a tuple in Φ. The following are equivalent.

(a) *ω_t is algebraic for every $t \in C$.*

(b) *For some finite Φ, the closed points of $C'_R[1/\ell]$ at which ω is Φ-represented are Zariski dense in C'_R.*

(c) *For some finite Φ and some Zariski dense set S of closed points of $\operatorname{Spec} R[1/\ell]$, the class ω is Φ-represented at infinitely many closed points of C'_s for every $s \in S$.*

For a single closed point s of $\operatorname{Spec} R[1/\ell]$, the class ω is Φ-represented at infinitely many closed points of C'_s, for some finite Φ, if and only if $\omega_\ell(\bar{\eta}_s)$ is a $\mathbb{Q}_\ell$-linear combination of classes of algebraic cycles on $Y'_{\bar{\eta}_s}$.

Proof. One standard property of the ℓ-adic cycle class is used beyond those over a field: if $\mathcal{O}$ is a discrete valuation ring in which ℓ is invertible, $\mathcal{Y} \to \operatorname{Spec} \mathcal{O}$ is smooth and projective, and $\mathcal{Z} \subset \mathcal{Y}$ is a closed subscheme flat over $\mathcal{O}$, then the specialisation isomorphism of ℓ-adic cohomology carries the cycle class of the geometric generic fibre of $\mathcal{Z}$ to that of its geometric special fibre [SGA $4\frac{1}{2}$, Cycle]. The specialisation isomorphism also carries the values of a global section of a lisse sheaf to each other, by its construction.

Step 1: the represented loci are closed. For a tuple $\underline{P} = (P_1, \dots, P_m)$ let $\mathcal{H}_{\underline{P}}$ be the fibre product over $C'_R[1/\ell]$ of the relative Hilbert schemes $\operatorname{Hilb}^{P_i}(Y'/C'_R[1/\ell])$. It is projective over $C'_R[1/\ell]$ [Gro61], so it has finitely many connected components. On a component T, with universal subschemes $\mathcal{Z}_i$, put $A(h) = \{q \in \mathbb{Q}_\ell^m : \sum_i q_i \operatorname{cl}(\mathcal{Z}_{i,\bar{h}}) = \omega_\ell(\bar{h})\}$ for $h \in T$. If h' is a specialisation of h, there is a discrete valuation ring mapping to T with generic point to h and closed point to h' [Stacks, Tag 054F], and the property above, applied to the pull-backs of Y' and of the $\mathcal{Z}_i$, gives $A(h) = A(h')$. Any two points of the connected noetherian scheme T are joined by a chain of specialisations and generisations, through the generic points of its irreducible components, so A is constant on T, say A_T. Let $\Sigma_{\underline{P}}$ be the union of the images in $C'_R[1/\ell]$ of the components T with $A_T \neq \emptyset$. It is closed, as a finite union of images of proper morphisms, and it is the set of points at which ω is $\{\underline{P}\}$-represented: at a point x of the image of such a T, a closed point of the fibre of T over x has residue field finite over $\kappa(x)$ and gives subschemes of $Y'_{\bar{x}}$. Put $\Sigma_\Phi = \bigcup_{\underline{P} \in \Phi} \Sigma_{\underline{P}}$.

Step 2: (b) implies (a). Σ_Φ is closed and contains a dense set, so it is $C'_R[1/\ell]$ and contains the generic point ξ of C'_R, whose residue field is K'. So $\omega_t \otimes 1 = \omega_\ell(\bar{\xi}) = \sum_i q_i \operatorname{cl}(Z_i)$ with $q \in \mathbb{Q}_\ell^m$ and subschemes Z_i of $Y_{\bar{K}_1}$, and so of Y_t. The cycle class maps are compatible with the comparison isomorphism $H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell \cong H^4(Y_t, \mathbb{Q}_\ell)$ [Del82, Section 1 and Example 2.1(a)], so the linear system $\omega_t = \sum_i q_i [Z_i]$, which has rational coefficients, is solvable over $\mathbb{Q}_\ell$, hence over $\mathbb{Q}$, and ω_t is algebraic. The Z_i are defined over a finite extension of K', so they spread out to subschemes, flat over a nonempty open subset [Stacks, Tag 052A], of the pull-back of $W \times_C W$ to a finite cover of C, and by the local constancy of the cycle class used in the proof of [I, Theorem 16.4], ω is algebraic at every point of a nonempty Zariski open subset of C, which is uncountable. Corollary 6.30 gives (a).

Step 3: (c) implies (b). Let D be the set of closed points at which ω is Φ-represented and $\bar{D}$ its closure. For $s \in S$ the set $D \cap C'_s$ is infinite, so its closure is the irreducible curve C'_s, and $C'_s \subseteq \bar{D}$. The points u of $\operatorname{Spec} R[1/\ell]$ over which the fibre of $\bar{D}$ has dimension at least one form a closed set, by the upper semicontinuity of the dimension of the fibres of a proper morphism [Stacks, Tag 0D4I]; it contains the dense set S, so it contains the generic

point. The generic fibre of $\bar{D}$ is then a closed subset of dimension one of the irreducible curve $C'_R \otimes_R k'$, so it contains ξ, and $\bar{D} = C'_R[1/\ell]$.

Step 4: (a) implies (c). By (a) and Step 1, applied over $\bar{K}_1$ with the Betti class in place of the ℓ-adic one, ω_t is the class of a rational combination of subschemes of $Y_{\bar{K}_1}$, defined over a finite extension L of K'. Enlarge R to a finitely generated subring of the algebraic closure of k' in L, let $C''_R \to C'_R$ be the normalisation in L, and spread the subschemes out over C''_R; by generic flatness [Stacks, Tag 052A] they are flat over a dense open subset U of $C''_R[1/\ell]$, with constant Hilbert polynomials. By Step 1 applied on U, ω is represented by their fibres at every point of U. The image of U in $C'_R[1/\ell]$ contains a dense open subset U', every closed point of U' is the image of a closed point of U, and U' meets C'_s in a nonempty open, hence infinite, subset for every s in a dense open subset of $\operatorname{Spec} R[1/\ell]$.

The single reduction. If ω is Φ-represented at infinitely many closed points of C'_s, then $\Sigma_\Phi \cap C'_s$ is closed and infinite, so it is C'_s and contains η_s. Conversely, cycles at $\bar{\eta}_s$ are defined over a finite extension of $\kappa(\eta_s)$, and spreading them out as in Step 4 over a finite cover of C'_s represents ω, with fixed Hilbert polynomials, at every closed point of a nonempty open subset of C'_s. ∎

Remark 7.44 (What is left in positive characteristic). By Proposition 7.42(ii) the class is represented at every closed point of $C'_R[1/\ell]$, so Theorem 7.43 turns the Mumford target into a statement of uniformity alone: the Hilbert data of some representation must stay in one finite set on a Zariski dense set of closed points, for instance at infinitely many closed points of infinitely many reductions. The passage to characteristic zero is then done by the specialisation of cycle classes along C'_R, and neither the p-adic variational Hodge conjecture nor a lifting of cycles is needed. Li's theorem gives no bound: where its proof carries cycles into a prescribed isogeny class, it does so by an isogeny whose degree it does not control [Li26, proof of Theorem 1.2, Step 3].

Nor do the natural representatives at the supersingular points supply the bound. At a closed point x whose fibre is supersingular, the Frobenius eigenvalues on $H^2(Y'_{\bar{x}}, \mathbb{Q}_\ell(1))$ are roots of unity, so by Tate's theorem [Tat66] this space is spanned by divisor classes, and $\omega_\ell(\bar{x}) \in \bigwedge^4 H^1$ is a combination of products of two of them. But for a closed point s of $\operatorname{Spec} R[1/\ell]$ and a finite set of Hilbert polynomials, only finitely many closed points of C'_s carry such a combination $\sum_i q_i \operatorname{cl}(D_i) \operatorname{cl}(D'_i)$ with effective divisors D_i, D'_i whose Hilbert polynomials lie in that set: otherwise Step 1 of the proof of Theorem 7.43, run on relative Hilbert schemes of divisors, would give one at $\bar{\eta}_s$, against Proposition 7.42(iii). This holds whether or not the supersingular points are dense. They are dense in C'_R when R has dimension one. Indeed, for every prime $p \geq 5$ not dividing the discriminants of F and D, which is all but finitely many, the supersingular points of the reduction at a prime over p of Kisin's integral model of the Shimura curve, with hyperspecial level at p, are its whole Newton jumping locus and number $(p^r - 1)(g - 1)$ on each component of genus g, r being the local degree of F [SZ11, Theorem 1.2 and Corollary 5.17]. For all but finitely many s the curve C'_s maps finitely onto such a component, of genus at least two, so it has at least $p - 1$ supersingular points, while a closed subset of C'_R of dimension one meets all but finitely many of the C'_s in boundedly many points.

Within one reduction the bound is equivalent to the class being algebraic at the generic point of C'_s (Theorem 7.43), a Tate class over a global function field that is not a combination of products of divisor classes (Proposition 7.42(iii)). This is the codimension two case of the variational Tate conjecture along C'_s, which Morrow proves for divisors in its crystalline form [Mor19]. From one reduction alone, the way back to characteristic zero would be the p-adic variational Hodge conjecture for $W \times_C W$ over the Witt vectors of $\kappa(s)$, with

rational rather than ℓ-adic coefficients and with the crystalline class of the cycle matching the de Rham class of ω. Bloch, Esnault and Kerz lift such a class in $K_0 \otimes \mathbb{Q}$ of the special fibre to the formal completion for $p > 15$, the dimension being nine, and the algebraisation of the formal class is the remaining step of their programme [BEK14, Theorem 1.3]. The route therefore runs through the uniform bound or through this conjecture.

Five of the routes of Remark 7.32 survive the rigidity theorems: *specialisation*, *reduction modulo p*, *semiregularity*, *Kuga–Satake* and *motivated classes*. The next corollary shows that they are not five separate obligations.

Corollary 7.45 (The five routes reduce to one statement)**.** *Let $W \to C$ be as in Corollary 6.22, let ω be a rational exceptional class, and let the* Mumford target *be the first case of* (F2)*: the Hodge conjecture for $X_t \times X_t$ for every $t \in C$.*

(i) *Each of the following is equivalent to the Mumford target: ω_t is algebraic for uncountably many $t \in C$; $B(W \times_C W)$ holds for $L_{W \times_C W}$; ω is represented with Hilbert data in one finite set at infinitely many points of C, for instance at infinitely many CM points; ω is represented with Hilbert data in one finite set on a Zariski dense set of closed points of the arithmetic model of Theorem 7.43.*

(ii) *Each of the following implies the Mumford target: a perfect complex E at one point of C with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, no other component, and $\dim \mathrm{Ext}^2(E, E) = 119$; the algebraicity, for uncountably many t, of the Kuga–Satake class of the surface S_λ of Theorem 7.4 attached to X_t; a rational map as in Proposition 7.25(iv), under its hypotheses, from $X_t \times X_t$ for uncountably many t.*

(iii) *There is a countable set $\Sigma_0 \subset C$ containing the CM points such that the Mumford target holds as soon as the Hodge conjecture holds for $X_c \times X_c$ at one point $c \notin \Sigma_0$.*

Proof. (i) The first two are Corollary 6.30, with Lemma 7.20 for one class in place of two, the third is Corollary 7.28 and the fourth is Theorem 7.43. (ii) The first is Theorem 7.27(iii). The second and the third give the Hodge conjecture for $X_t \times X_t$ at each of the uncountably many points, by Theorem 7.4(ii) and Proposition 7.25(iv), and then (i) applies. (iii) In the proof of Corollary 7.28 the set of points at which ω is algebraic is the union of the countably many closed sets $\Sigma_{\underline{P},\underline{q}}$, each finite or equal to C. Let Σ_0 be the union of the finite ones and of the CM points. If ω_c is algebraic at a point $c \notin \Sigma_0$, then c lies in some $\Sigma_{\underline{P},\underline{q}}$ that is not finite, which is C, and (i) applies. ∎

Remark 7.46 (Which routes are needed)*.* Of the five routes, specialisation, reduction modulo p and motivated classes are the Mumford target itself in three other forms, by Corollary 7.45(i) and, for the motivated classes, Corollary 6.30; semiregularity and Kuga–Satake are stronger inputs that imply it. So any one route suffices. Each of the three forms is the whole of the problem, a bound on Hilbert data or a case of (L) that is equivalent to the target, and the two stronger inputs ask for an object and for a cycle.

The target can also be read at a single member, through the motivic group of X rather than through cycles on the family; it then becomes the question of which of four groups this is.

Proposition 7.47 (Four possible motivic groups)**.** *In Setup 7.1, let M be the group of the Tannakian category of homological motives generated by $h(X)$, with the Betti realisation as fibre functor, let $M^1 = M \cap \mathrm{Sp}(V, \psi)$, and let N be the normaliser of G in $\mathrm{Sp}(V, \psi)$. Over $\bar{\mathbb{Q}}$ identify V_1, V_2 and V_3 with one plane, carrying ϵ_1, ϵ_2, ϵ_3 to one form, and let $\zeta \in \mathrm{Sp}(V, \psi)(\bar{\mathbb{Q}})$ be the cyclic permutation $v_1 \otimes v_2 \otimes v_3 \mapsto v_3 \otimes v_1 \otimes v_2$ of the three factors.*

It normalises G, and other identifications change it by an element of $G(\bar{\mathbb{Q}})$; so it acts through V on the Hodge classes of every power of X, independently of the choice.

(i) *N/G is a form of the symmetric group S_3, $M = \mathbb{G}_m \cdot M^1$, and M^1 is one of G, $G \cdot A_3$, N and $\mathrm{Sp}(V,\psi)$, where $G \cdot A_3$ is the inverse image in N of the subgroup of order three of N/G. The last three contain ζ.*

(ii) *The following are equivalent: $M^1 = G$; the Hodge conjecture holds for X^n for every $n \geq 1$; the two exceptional classes of $X \times X$ are algebraic; some algebraic class on some power of X is not fixed by ζ.*

(iii) *Let $\mathrm{Det} \in (V^{\otimes 4})^G \subset H^4(X^4, \mathbb{Q})$ be the Hodge class of Remark 7.8, and write r_1 also for the Hodge class $t \in (V^{\otimes 4})^G$ with $\hat{t} = r_1 = s_1 + s_2 + s_3$, in the notation of Steps 2 and 3 of the proof of Theorem 7.7. Both are fixed by N and not by $\mathrm{Sp}(V,\psi)$, and each of them is algebraic if and only if $M^1 \neq \mathrm{Sp}(V,\psi)$. So the algebraicity of either excludes the last case only; by this argument it does not give the exceptional classes, which are not algebraic when M^1 is $G \cdot A_3$ or N.*

(iv) *Let $W \to C$ be as in Corollary 6.22, and suppose that it is defined over a number field; this holds for the restriction of the universal family of a fine moduli space of polarised abelian fourfolds with level structure to the normalisation of one of the Shimura curves of* [Mum69] *in it, and for its pull-backs to finite étale covers. Then the set Σ_0 constructed in the proof of Corollary 7.45(iii) lies in $C(\bar{\mathbb{Q}})$, and the Mumford target holds as soon as, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, some algebraic cycle on some power of X_t has a class that is not fixed by ζ.*

Proof. The group. On an abelian variety homological and numerical equivalence coincide [Lie68], so the homological motives generated by $h(X)$ and the Lefschetz motive form a semisimple abelian category [Jan92]. The Künneth projectors are algebraic [Kle68, §2], so after the usual change of sign in the commutativity constraint this is a Tannakian category over $\mathbb{Q}$, with the Betti realisation as a fibre functor, faithful by the definition of homological equivalence. Its group M is reductive, the category being semisimple, and the M-invariants in the realisation of a motive P are the image of $\mathrm{Hom}(\mathbf{1}, P)$ [DM82, Theorem 2.11]; so for all m and p the M-invariants of $H^{2p}(X^m, \mathbb{Q})(p)$ are the classes of algebraic cycles. The group M acts faithfully on $V \oplus \mathbb{Q}(1)$ and fixes the polarisation class, so it lies in $\mathrm{GSp}(V,\psi)$ and acts on $\mathbb{Q}(1)$ through the multiplier. Through ψ and the Künneth projectors every tensor space of $V \oplus \mathbb{Q}(1)$ is a sum of direct summands of spaces $H^k(X^k, \mathbb{Q})(j)$ cut out by algebraic projectors, so the M-invariants there are algebraic classes, hence Hodge classes, fixed by $\mathrm{MT}(X) = \mathbb{G}_m \cdot G$; as a reductive group is the subgroup fixing its invariants in the tensor spaces [Del82, Proposition 3.1], $\mathrm{MT}(X) \subseteq M$ and $G \subseteq M^1$. An element of M with multiplier ν is a square root of ν times an element of M^1, so $M = \mathbb{G}_m \cdot M^1$. The homotheties act trivially on the spaces $H^{2p}(X^m, \mathbb{Q})(p)$, so there the algebraic classes are the M^1-invariants, as the Hodge classes are the G-invariants.

The groups between G and $\mathrm{Sp}(V,\psi)$. Over $\bar{\mathbb{Q}}$ the Lie algebra $\mathfrak{sp}(V,\psi) \cong S^2V$ is the sum of $\mathrm{Lie}\, G = T_1 \oplus T_2 \oplus T_3$, where $T_i \cong S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and of $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, which is an irreducible representation of G of dimension 27; so a G-stable subspace of $\mathfrak{sp}(V,\psi)$ containing $\mathrm{Lie}\, G$ is $\mathrm{Lie}\, G$ or $\mathfrak{sp}(V,\psi)$. If H is a closed subgroup with $G \subseteq H \subseteq \mathrm{Sp}(V,\psi)$, its Lie algebra is such a subspace. If it is $\mathrm{Lie}\, G$, then $H^\circ = G$, which H normalises, so $H \subseteq N$; if it is $\mathfrak{sp}(V,\psi)$, then $H^\circ = \mathrm{Sp}(V,\psi)$, which is connected, so $H = \mathrm{Sp}(V,\psi)$.

The normaliser. Since V is absolutely irreducible, the centraliser of G in $\mathrm{Sp}(V,\psi)$ is $\{\pm 1\}$, which lies in G as the image of $(-1,1,1)$, and $G(\bar{\mathbb{Q}})$ maps onto the group of inner automorphisms of $G_{\bar{\mathbb{Q}}}$. So $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}})$ embeds in the outer automorphism group

of $G_{\bar{\mathbb{Q}}}$, which permutes the three factors, SL_2 having no outer automorphism. The six permutations of the factors, after identifications as above, preserve $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ and normalise G, so $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$. The Galois group acts on it by conjugation through its action on the three factors, which is transitive, and the subgroups of S_3 stable under conjugation by a transitive subgroup are 1, A_3 and S_3. So the algebraic subgroups of N containing G are G, $G \cdot A_3$ and N, and with the previous paragraph this proves (i), ζ mapping to an element of order three of N/G.

(ii) If $M^1 = G$, then $M = \mathrm{MT}(X)$, and every Hodge class on every power of X is algebraic. The Hodge conjecture for $X \times X$ includes the exceptional classes. If these are algebraic, so is every rational exceptional class ω_a of Setup 7.18, and ω_a is not fixed by ζ: ζ permutes the three factors cyclically, so it fixes θ and carries $U_{ij} = S^2V_i \otimes S^2V_j \otimes \bigwedge^2 V_k$ onto $U_{\zeta(i)\zeta(j)}$; labelling the factors so that ζ sends the i-th to the $(i+1)$-th, it carries π_{12} to π_{23}, π_{23} to π_{13} and π_{13} to π_{12} and fixes π_0. So it fixes ω_a only if $a_1 = a_2 = a_3$, whereas these are the distinct conjugates of an irrational element of F (Lemma 7.19). Finally, if $M^1 \neq G$, then M^1 contains ζ by (i), and every algebraic class, being M^1-invariant, is fixed by ζ.

(iii) The class Det spans $(S^4V)^G \otimes \bigwedge^4 \mathbb{Q}^4$, where $(S^4V)^G$ is a line (Remark 7.8). As N normalises G, it preserves this line, and it fixes it, because $N(\bar{\mathbb{Q}})$ is generated by $G(\bar{\mathbb{Q}})$ and the permutations of the factors, which fix Cayley's hyperdeterminant, a symmetric function of the three factors, and the restriction of Det to the diagonal is a nonzero multiple of it (Remark 7.8). $\mathrm{Sp}(V,\psi)$ fixes no nonzero element of S^4V, which would be a nonzero quartic form constant on the single orbit $V \setminus \{0\}$. The permutations of the factors conjugate s_k into $s_{\sigma(k)}$, so they commute with r_1, as does G; and $t \mapsto \hat{t}$ is $\mathrm{Sp}(V,\psi)$-equivariant, since the pairing $\int_X v \cup w \cup \theta^3$ on V is G-invariant and alternating, hence a multiple of the form ψ. So the class r_1 is fixed by N. On $S^2V \subset V \otimes V$ the endomorphism r_1 acts by 3 on $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$ and by -1 on the T_i, since s_k is 1 on S^2V_k and -1 on $\bigwedge^2 V_k$; as S^2V is irreducible under $\mathrm{Sp}(V,\psi)$, the class r_1 is not $\mathrm{Sp}(V,\psi)$-invariant. If $M^1 \neq \mathrm{Sp}(V,\psi)$, then $M^1 \subseteq N$ by (i), and the two classes are M^1-invariant, hence algebraic; if $M^1 = \mathrm{Sp}(V,\psi)$, they are not. When M^1 is $G \cdot A_3$ or N it contains ζ, which fixes no exceptional class.

(iv) A fine moduli space as in the statement is defined over a number field, and the curve contains infinitely many CM points, which are algebraic, an abelian variety with complex multiplication being defined over a number field; so the curve is the Zariski closure of a set of algebraic points and is defined over $\bar{\mathbb{Q}}$, and so are its normalisation, the restricted family and their finite étale covers. Now let $W \to C$ be defined over $\bar{\mathbb{Q}}$. The moduli map of the family is a finite morphism defined over $\bar{\mathbb{Q}}$, so the CM points of C, whose images are algebraic, lie in $C(\bar{\mathbb{Q}})$. The other points of Σ_0 lie in the finite sets $\Sigma_{\underline{P},\underline{q}}$ of the proof of Corollary 7.28, images of unions of connected components of fibre products of relative Hilbert schemes of $W \times_C W \to C$. These are base changes of schemes over $\bar{\mathbb{Q}}$, and each connected component of the base change is the base change of a connected component over $\bar{\mathbb{Q}}$, which is algebraically closed; so a finite $\Sigma_{\underline{P},\underline{q}}$ consists of algebraic points. If $t \notin C(\bar{\mathbb{Q}})$, then $t \notin \Sigma_0$, the Hodge group of X_t is G, and (ii) for X_t gives the Hodge conjecture for $X_t \times X_t$; Corollary 7.45(iii) gives the target. ∎

The four cases differ already on X^4 and X^6 (Figure 39). Over $\bar{\mathbb{Q}}$, $(V^{\otimes 2m})^G$ is the tensor product of three copies of the space of SL_2-invariants of $(\bar{\mathbb{Q}}^2)^{\otimes 2m}$, of dimension $c = 2$ for $m = 2$ and $c = 5$ for $m = 3$, with a basis of non-crossing pairings that ζ and the transpositions of the factors permute. So the algebraic classes in the multilinear part $V^{\otimes 2m} \subset H^{2m}(X^{2m}, \mathbb{Q})$ have dimension c^3, $(c^3+2c)/3$, $(c^3+3c^2+2c)/6$ and $(2m-1)(2m-3)\cdots 1$ when M^1 is G,

$G \cdot A_3$, N and $\mathrm{Sp}(V,\psi)$, the last by the first fundamental theorem for $\mathrm{Sp}(V,\psi)$: 8, 4, 4, 3 on X^4 and 125, 45, 35, 15 on X^6. The middle two count the orbits of the basis under $\langle\zeta\rangle$ and under S_3: ζ fixes the c triples of equal pairings and a transposition fixes c^2 of them. If M were known to be connected, M^1 would be connected too, since $-1 \in G$ lies in its identity component; it would then be G or $\mathrm{Sp}(V,\psi)$, and the algebraicity of Det alone would give the Hodge conjecture for every power of X. The argument does not decide whether M is connected. It is the standard use of the motivic group as the stabiliser of the algebraic classes, and we claim no priority for Proposition 7.47. It proves no new case of the Hodge conjecture: it says what one new cycle on one power of one member would have to do.

The same invariance explains why the classes known or expected to be algebraic on abelian varieties stop short of the exceptional classes, on all abelian varieties at once.

Proposition 7.48 (The system generated by the known classes)**.** *Keep Setup 7.1, and ζ and $G \cdot A_3$ as in Proposition 7.47. For an abelian variety A let $A_X \subseteq A$ be the largest abelian subvariety isogenous to a power of X, and A' the identity component of the intersection of the kernels of the homomorphisms $A \to X$. Then $A_X \times A' \to A$ is an isogeny,* $\mathrm{Hom}(X,A') = 0$*, and*

$$H^1(A,\mathbb{Q}) = (V \otimes \mathrm{Hom}^0(A_X,X)) \oplus H^1(A',\mathbb{Q}),$$

the first summand embedded by $v \otimes f \mapsto f^*v$. Let $\mathrm{GL}(V)$ act on $H^*(A,\mathbb{Q}) = \bigwedge^* H^1(A,\mathbb{Q})$ through the factor V. Let $\mathcal{S}$ be the smallest family of subspaces $\mathcal{S}(A) \subseteq H^*(A,\mathbb{Q})$, over all complex abelian varieties A, that is closed under linear combinations, cup products, exterior products, and pull-back and push-forward along homomorphisms, and contains:

(a) *the divisor classes;*

(b) *for every subfield $E \subseteq \mathrm{End}^0(A)$, the rational classes whose extension to $\bar{\mathbb{Q}}$ lies in $\bigoplus_\sigma \bigwedge^d H^1(A,\bar{\mathbb{Q}})_\sigma$, the sum over the embeddings $\sigma\colon E \to \bar{\mathbb{Q}}$ of the top exterior powers of the eigenspaces, $d = \dim_E H^1(A,\mathbb{Q})$; among them are the Weil classes;*

(c) *the Hodge classes of the abelian varieties without isogeny factor X;*

(d) *the Hodge classes of the powers of X that are fixed by ζ, among them* Det *and r_1.*

Then every class of $\mathcal{S}$ is invariant under $G \cdot A_3$ acting through V, and the classes obtained from (a), (b) *and* (c) *alone are invariant under $\mathrm{Sp}(V,\psi)$. The Hodge classes in $\mathcal{S}(X^m)$ are the Hodge classes of X^m fixed by ζ, and $\mathcal{S}$ contains no exceptional class of $X \times X$.*

Proof. Up to isogeny $A = A_X \times A'$ by Poincaré reducibility, with A_X isogenous to some X^k and no isogeny factor X in A', so $\mathrm{Hom}(X,A') = \mathrm{Hom}(A',X) = 0$, X being simple. As $\mathrm{End}^0(X) = \mathbb{Q}$, $\mathrm{Hom}^0(A_X,X)$ has dimension k and the displayed map is an isomorphism. A homomorphism $h\colon A \to B$ is, up to isogeny, $h_X \times h'$ with $h_X\colon A_X \to B_X$ and $h'\colon A' \to B'$, and h^* acts on H^1 as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with $\mathrm{GL}(V)$; so pull-backs are equivariant, and so are cup and exterior products, $\mathrm{GL}(V)$ acting on $H^*(A \times B) = H^*(A) \otimes H^*(B)$ diagonally. The orientation class of A spans $\bigwedge^{\mathrm{top}} H^1(A) = (\det V)^{\otimes k} \otimes (\det \mathrm{Hom}^0(A_X,X))^{\otimes 8} \otimes \bigwedge^{\mathrm{top}} H^1(A')$, on which $\mathrm{SL}(V)$ acts trivially, so Poincaré duality and push-forwards are $\mathrm{SL}(V)$-equivariant. Since $G \cdot A_3 \subseteq \mathrm{Sp}(V,\psi) \subseteq \mathrm{SL}(V)$, the classes invariant under either group form a family closed under all the operations, and it remains to look at the generators.

(a) A divisor class of A is the sum of its Künneth components in $\bigwedge^2 H^1(A_X)$, $H^1(A_X) \otimes H^1(A')$ and $\bigwedge^2 H^1(A')$, which are Hodge classes. The middle one corresponds to a homomorphism from A' to the dual of A_X, which is zero. The first is a Hodge class of A_X, so a G-invariant of $\bigwedge^2(V \otimes \mathbb{Q}^k) = S^2V \otimes \bigwedge^2 \mathbb{Q}^k \oplus \bigwedge^2 V \otimes S^2\mathbb{Q}^k$; as $(S^2V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q}\psi$ (proof of Theorem 7.36(i)), it lies in $\psi \otimes S^2\mathbb{Q}^k$, which $\mathrm{Sp}(V,\psi)$ fixes.

$M = \mathbb{G}_m \cdot M^1$, $M^1 = M \cap \mathrm{Sp}(V,\psi)$	dim of the algebraic classes in $V^{\otimes 2m}$: X^4, $m=2$	X^6, $m=3$	formula, $c=2$ or 5	$\zeta \in M^1$	Det, r_1 algebraic	exceptional classes algebraic
$\mathrm{Sp}(V,\psi)$	3	15	$(2m-1)!!$	yes	no	no
dim: $9 < 36$						
N	4	35	$(c^3+3c^2+2c)/6$	yes	yes	no
index 2						
$G \cdot A_3$	4	45	$(c^3+2c)/3$	yes	yes	no
index 3						
$G = \mathrm{Hg}(X)$	8	125	c^3	no	yes	yes

$M^1 = G \Leftrightarrow$ the Hodge conjecture for every $X^n \Leftrightarrow$ the two exceptional classes of $X \times X$ are algebraic $\Leftrightarrow$ some algebraic class on some power of X is not fixed by ζ

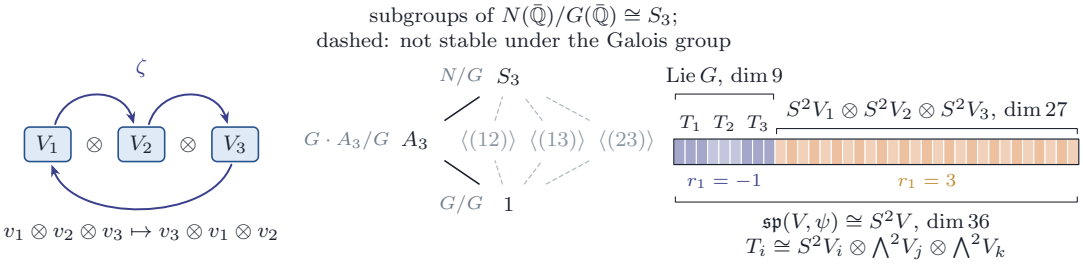


FIGURE 39. The four possible groups $M^1 = M \cap \mathrm{Sp}(V,\psi)$ of Proposition 7.47, where $M = \mathbb{G}_m \cdot M^1$, as the chain $G \subset G \cdot A_3 \subset N \subset \mathrm{Sp}(V,\psi)$: the first two inclusions have index 3 and 2, and N, whose identity component is G, has dimension 9 against 36. For each case the table gives the dimension of the algebraic classes in the multilinear part $V^{\otimes 2m}$ of $H^{2m}(X^{2m},\mathbb{Q})$ for $m=2$ and $m=3$, from the $c=2$, respectively $c=5$, non-crossing pairings; whether M^1 contains the cyclic permutation ζ of the three tensor factors; whether Det and r_1 are algebraic; and whether the two exceptional classes of $X \times X$ are. Only $M^1 = G$ makes the exceptional classes algebraic, and it is equivalent to the Hodge conjecture for every power of X (Proposition 7.47(ii)); the proposition does not decide which of the four cases occurs. Below, from left to right: ζ; the subgroups of $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$, of which only 1, A_3 and S_3 are stable under the Galois group, which acts transitively on the three factors; and $\mathfrak{sp}(V,\psi) = \mathrm{Lie}\, G \oplus S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, $36 = 9+27$, the second summand irreducible under G, with the eigenvalues -1 and 3 of r_1 on the two summands.

On the third $\mathrm{GL}(V)$ acts trivially. (b) The field E acts on $Q = \mathrm{Hom}^0(A_X, X)$ and on $H^1(A')$, and the σ-eigenspace of $H^1(A, \bar{\mathbb{Q}})$ is $V \otimes Q_\sigma \oplus H^1(A', \bar{\mathbb{Q}})_\sigma$. Its top exterior power is $(\det V)^{\otimes \dim Q_\sigma} \otimes (\det Q_\sigma)^{\otimes 8} \otimes \det H^1(A', \bar{\mathbb{Q}})_\sigma$, on which $\mathrm{SL}(V)$ acts trivially. (c) For such an abelian variety $A_X = 0$, and $\mathrm{GL}(V)$ acts trivially. (d) The Hodge classes of X^m are the invariants of G, which acts through V, and those fixed by ζ are invariant under the group $G \cdot A_3$ that G and ζ generate.

So every class of $\mathcal{S}$ is $G \cdot A_3$-invariant, and those obtained from (a), (b) and (c) are $\mathrm{Sp}(V, \psi)$-invariant. The Hodge classes of X^m in $\mathcal{S}$ contain those of (d) and are fixed by ζ, so they are exactly the Hodge classes fixed by ζ; and an exceptional class of $X \times X$ is not fixed by ζ (proof of Proposition 7.47(ii)). ∎

So no argument that uses only these inputs and these operations reaches the exceptional classes. The inputs include all Weil classes, whether or not they are known to be algebraic, and with them Markman's cycles [Mar25a]; the whole Hodge conjecture for the abelian varieties without isogeny factor X; and the algebraicity of Det and r_1. A proof needs a cycle outside this system: on a product of a power of X with an abelian variety, a class that is not invariant under $G \cdot A_3$ acting through V, or a cycle through a variety that is not an abelian variety, such as the Kuga–Satake class of S_λ would provide (Proposition 7.12). Proposition 7.48 extends the second limit of Remark 7.17; it is an obstruction to a method, not to the target, and we claim no priority for it.

The same invariance, used the other way round, shows that the statement (F2) of Corollary 6.71 is not smaller than the conjecture for abelian varieties. For an abelian variety Y let $R(Y) \subseteq H^*(Y, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients, as in Remark 6.72.

Proposition 7.49 (Statement (F2) is the conjecture for abelian varieties)**.** *Statement* (F2) *of Corollary 6.71 implies the Hodge conjecture for every abelian variety, and is therefore equivalent to it.*

Proof. The conjecture for abelian varieties contains (F2). For the converse, recall from Remark 6.72 that (F2) gives every Hodge class of degree $2q$ on Y as soon as one Hodge class of degree $2q$ on Y lies outside $R(Y)$. Let X be as in Setup 7.1, and let ζ act on the cohomology of every abelian variety through the factor V, as in Proposition 7.48. Every class of $R(Y)$ lies in $\mathcal{S}(Y)$, so it is fixed by ζ, for every Y; and an exceptional class $e_0 \in H^4(X \times X, \mathbb{Q})$ is not (Proposition 7.48). Let θ be an ample class on $X \times X$, which ζ fixes. If $\zeta e_0 = \lambda e_0$, then $\lambda \neq 1$ and $e = e_0 + \theta^2$ has $\zeta e = \lambda e_0 + \theta^2$, which is not a multiple of e; otherwise put $e = e_0$. In either case e is a Hodge class of degree 4 on $X \times X$ with $\zeta e \notin \bar{\mathbb{Q}}\, e$.

Let A be an abelian variety and $\beta \neq 0$ a Hodge class of degree $2p$ on A, and put $Y = A \times X \times X$. As ζ acts on $H^*(Y) = H^*(A) \otimes H^*(X \times X)$ diagonally, a class $\beta \otimes e$ in $R(Y)$ would satisfy $\zeta\beta \otimes \zeta e = \beta \otimes e$, which forces $\zeta e \in \bar{\mathbb{Q}}\, e$. So the Hodge class $\beta \otimes e$ lies outside $R(Y)$, and (F2) makes it algebraic. In the same way $e \cup \theta^4 \notin R(X \times X)$, since otherwise $(\zeta e - e) \cup \theta^4 = 0$ and the hard Lefschetz theorem would give $\zeta e = e$; so (F2) makes every Hodge class of degree 12 on $X \times X$ algebraic. By the Hodge–Riemann bilinear relations the intersection pairing between the Hodge classes of degrees 4 and 12 on $X \times X$ is nondegenerate, so there is a Hodge class e' of degree 12 with $\int_{X \times X} e \cup e' = 1$. Then

$$\beta = \mathrm{pr}_{A*}\big((\beta \otimes e) \cup \mathrm{pr}_{X \times X}^* e'\big),$$

with pr_A and $\mathrm{pr}_{X \times X}$ the projections of Y, and β is algebraic. ∎

Remark 7.50 (The rule set with Proposition 7.49). Theorem 6.70 is a statement about the rule set of Setup 6.69, and Proposition 7.49 is a rule that the set does not contain. Adjoined to it, the rule changes the fifth minimal set of Theorem 6.70(ii): the minimal sufficient sets become

$$\{\mathrm{F3}\},\quad \{\mathrm{L},\mathrm{M}\},\quad \{\mathrm{L},\mathrm{F3}'\},\quad \{\mathrm{V},\mathrm{F3}'\},\quad \{\mathrm{F2},\mathrm{F3}'\},$$

and $\mathrm{P2}_{\mathrm{split}}$ lies in none of them. Indeed the new rule has the premise (F2) and the conclusion **AB**, the conjecture for abelian varieties, so the sets C_1, C_2 and C_4 of the proof of Theorem 6.70 stay closed, and a sufficient set meets H_1, H_2 and H_4. The sets meeting these three that are minimal with this property are the five above, by the argument of that proof without H_3, and each is sufficient: $\{\mathrm{F2},\mathrm{F3}'\}$ gives **AB** by the new rule and then **HC** by Proposition 6.35(i). So the results of this paper on Weil classes do not shorten a route to the conjecture. What they do is reduce one case of it, the algebraicity of the Weil classes of every CM field, to the single statement $\mathrm{P2}_{\mathrm{split}}$, by a route that does not pass through (F2). Like (F3), which is the whole conjecture, (F2) is the whole conjecture for abelian varieties, carried by the classes that are hardest to reach; the exceptional classes of $X \times X$ are its smallest instance, and the proof above shows that the classes $\beta \otimes e$ built from them already carry all of it.

8. The main theorem in every family

[I, Theorem 1.1] describes the algebraic locus of the Weil classes in one family: it contains an explicit CM point, it is dense, it is a countable union of closed algebraic strata, and it is the whole family as soon as it has an interior point. The first two properties use the Weil classes; the last two hold for every flat class on every smooth projective family (Lemma 8.2). This section carries the theorem to the families that carry (F2) and (F3′), and records what each construction of the paper contributes there.

For (F2) the families are abelian schemes with a member of CM type. Every Hodge class of every abelian variety lies in such a family (Proposition 8.4), and (F2) is equivalent to the statement that in every one of them the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Theorem 8.5). The base points of [I, Theorem 17.5] are the CM seeds of the Weil families, and André's theorem makes the Weil classes of the split families seeds for every Hodge class of CM type. For (F3′) the cohomology of a variety splits into the part reached from abelian varieties by algebraic correspondences and a complement, and (F3′) is the statement that the complement carries no Hodge class (Proposition 8.7). On the first varieties where (F3′) is open, the squares of K3 surfaces with real multiplication, the algebraic locus of the real multiplication contains every CM point, is dense and stable under a group of rational Hodge isometries, and is either everything or meagre (Theorem 8.9). In both cases what is left is the interior point, and Theorem 8.10 lists what the constructions of the paper give towards it.

8.1. **Algebraic loci.**

Definition 8.1 (Algebraic locus)**.** Let S be a smooth connected complex quasi-projective variety, $f\colon \mathcal{X} \to S$ a smooth projective morphism, and ξ a global section of $R^{2p}f_*\mathbb{Q}$ over S^{an}.[43] The *algebraic locus* of ξ is

$$\Sigma(\xi) = \{\, s \in S :\ \xi_s \text{ is the class of an algebraic cycle with rational coefficients on } \mathcal{X}_s \,\}.$$

[43]By Ehresmann's theorem f is a locally trivial fibration of differentiable manifolds, so $R^{2p}f_*\mathbb{Q}$ is a local system with fibre $H^{2p}(\mathcal{X}_s,\mathbb{Q})$ at s. A global section is a class ξ_s on every fibre that is locally constant under the identifications of nearby fibres, equivalently a class on one fibre fixed by the monodromy. The

Lemma 8.2 (The dichotomy in every family)**.** *In Definition 8.1 the following hold.*

(i) $\Sigma(\xi)$ *is the union of countably many Zariski closed subsets of* S*.*

(ii) *Either* $\Sigma(\xi) = S$*, or* $\Sigma(\xi)$ *is meagre in* S^{an}*.*

(iii) *The following are equivalent:* $\Sigma(\xi) = S$*;* $\Sigma(\xi)$ *has an interior point;* $\Sigma(\xi)$ *is not meagre;* ξ_s *is algebraic for every* s *outside a countable union of proper closed analytic subsets of* S*. If* $\Sigma(\xi)$ *is dense, they are also equivalent to* $\Sigma(\xi)$ *being closed.*

Proof. (i) The proof of [I, Theorem 16.4] applies with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by $(\mathcal{X}, S, \xi)$, as in Lemma 6.27: for a tuple $\underline{P}$ of Hilbert polynomials the fibre product of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(\mathcal{X}/S)$ is projective over S [Gro61], the class of a flat family of cycles is a flat section, two flat sections over a connected base that agree at one point agree everywhere, and the image of a proper morphism is closed. So the set of points at which $\xi_s = \sum_i q_i[Z_i]$ with Z_i of Hilbert polynomial P_i is Zariski closed, and $\Sigma(\xi)$ is the union of these strata over the countably many pairs $(\underline{P}, \underline{q})$.

(ii) and (iii). A Zariski closed subset of the irreducible variety S that has an interior point in S^{an} is all of S, and a proper one is nowhere dense. If no stratum is S, then $\Sigma(\xi)$ is a countable union of nowhere dense closed sets, hence meagre, and has no interior point, since S^{an} is a Baire space. If one stratum is S, then $\Sigma(\xi) = S$. This gives (ii) and the equivalence of the first three conditions of (iii). The complement of a countable union of proper closed analytic subsets is not meagre, by the Baire category theorem, so the fourth condition implies the third, and it is implied by the first. Finally, a dense set that is closed is S. ∎

So the dichotomy of [I, Theorem 1.1(iii)] and the equivalences of [I, Theorem 1.1(iv)] hold for every flat class on every smooth projective family; what [I, Theorem 1.1] adds for the Weil classes is a point of $\Sigma(\xi)$ and the density, and neither is automatic, since $\Sigma(\xi)$ may be empty.

8.2. **(F2) is propagation from CM points.**

Lemma 8.3 (Points of CM type are dense)**.** *Let* G *be a connected reductive group over* $\mathbb{Q}$*, let* $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$ *be a homomorphism, and let* $\mathcal{D}$ *be the orbit of* u_0 *under conjugation by* $G(\mathbb{R})^+$*, with the topology of* $G(\mathbb{R})^+/\mathrm{Stab}(u_0)$*. Then the points* $u \in \mathcal{D}$ *for which* $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$ *for some maximal torus* T *of* G *defined over* $\mathbb{Q}$ *are dense in* $\mathcal{D}$*.*

Proof. Let $u_1 \in \mathcal{D}$ and let W be a neighbourhood of u_1. The centraliser of the torus $u_1(\mathrm{U}(1))$ in $G_{\mathbb{R}}$ is connected and reductive, and a maximal torus of it defined over $\mathbb{R}$ is a maximal torus T_1 of $G_{\mathbb{R}}$ containing $u_1(\mathrm{U}(1))$. Write $\mathfrak{g}$ for the Lie algebra of G and $\mathfrak{t}_1 \subseteq \mathfrak{g}_{\mathbb{R}}$ for that of T_1, and choose a regular element $Y_1 \in \mathfrak{t}_1$. The map $\varphi\colon G(\mathbb{R})^+ \times \mathfrak{t}_1 \to \mathfrak{g}_{\mathbb{R}}$, $(g, Y) \mapsto \mathrm{Ad}(g)Y$, has differential $(Z, Y') \mapsto [Z, Y_1] + Y'$ at $(1, Y_1)$, which is onto because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$ for a regular semisimple Y_1. So for neighbourhoods N_1 of 1 in $G(\mathbb{R})^+$ with $N_1 u_1 N_1^{-1} \subseteq W$, and N_2 of Y_1 in $\mathfrak{t}_1$ consisting of regular elements, the image $\varphi(N_1 \times N_2)$ contains a nonempty open subset of $\mathfrak{g}_{\mathbb{R}}$. The rational points $\mathfrak{g}(\mathbb{Q})$ are dense in $\mathfrak{g}_{\mathbb{R}}$, so that open subset contains some $Y \in \mathfrak{g}(\mathbb{Q})$, and $Y = \mathrm{Ad}(g)Y'$ with $g \in N_1$ and $Y' \in N_2$. As Y is rational and regular semisimple, the identity component T of its centraliser in G is a maximal torus defined over $\mathbb{Q}$, and $T_{\mathbb{R}} = gT_1g^{-1}$ because the centraliser of Y' has identity component T_1. Then $u = gu_1g^{-1}$ lies in W and $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$. ∎

set of points s at which ξ_s is of type (p, p) is in general a proper subset of S, and it is algebraic by [I, Theorem B.6].

For an abelian variety the lemma is used with G its Hodge group, the smallest algebraic $\mathbb{Q}$-subgroup of $\mathrm{GL}(H^1(A,\mathbb{Q}))$ whose real points contain the image of the homomorphism $\mathrm{U}(1) \to \mathrm{GL}(H^1(A,\mathbb{R}))$ that defines the Hodge structure. If $\mathrm{Hg}(A)$ lies in a $\mathbb{Q}$-torus T, then A is of CM type.[44] Indeed T lies in a maximal $\mathbb{Q}$-torus T' of $\mathrm{GL}(H^1(A,\mathbb{Q}))$, which is the group of units of a commutative semisimple $\mathbb{Q}$-subalgebra $L \subseteq \mathrm{End}(H^1(A,\mathbb{Q}))$ of dimension $2\dim A$; the algebra L commutes with $T' \supseteq \mathrm{Hg}(A)$, so it lies in the commutant of $\mathrm{Hg}(A)$, which is $\mathrm{End}^0(A)$.

Proposition 8.4 (A family with a CM member through every Hodge class)**.** *Let A_0 be a complex abelian variety and γ a Hodge class of degree $2p$ on A_0. There are a smooth connected complex quasi-projective variety S, a polarised abelian scheme $\pi\colon \mathcal{A} \to S$, a global section ξ of $R^{2p}\pi_*\mathbb{Q}$ that is a Hodge class at every point, and points s_0, $c \in S$, such that $\mathcal{A}_{s_0} \cong A_0$ with $\xi_{s_0} = \gamma$ under this isomorphism and $\mathcal{A}_c$ is of CM type.*

Proof. Let $\ell \geq 1$ be an integer with $\ell\gamma$ integral. We construct a section with value $\ell\gamma$ at s_0 and divide it by ℓ at the end; until then γ stands for $\ell\gamma$. Fix a polarisation of A_0, put $V = H_1(A_0,\mathbb{Q})$, and fix a level $N \geq 3$. As in [I, Setup 16.1], the quotient $\mathcal{S}_g$ of the Siegel space $\mathcal{H}$ of complex structures on $V_{\mathbb{R}}$ compatible with the polarisation, by the principal congruence subgroup of level N of the stabiliser of $H_1(A_0,\mathbb{Z})$, is a smooth quasi-projective variety carrying a universal family $f\colon \mathcal{A}_g \to \mathcal{S}_g$, polarised by the fixed polarisation [BB66]; let $a_0 \in \mathcal{S}_g$ be the point of A_0 and $J_0 \in \mathcal{H}$ a lift, through which the fibre of $R^{2p}f_*\mathbb{Z}$ at every point of the image of $\mathcal{H}$ is identified with $H^{2p}(A_0,\mathbb{Z}) = \bigwedge^{2p} H_1(A_0,\mathbb{Z})^\vee$ by parallel transport in the simply connected space $\mathcal{H}$. Put $\mathbb{V} = R^{2p}f_*\mathbb{Z}$, a polarised variation of $\mathbb{Z}$-Hodge structure. By [I, Theorem B.6] the connected component $\mathcal{C}$ of its locus of Hodge classes that contains (a_0,γ) is a closed algebraic subvariety of the total space of $\mathbb{V}$; it has finitely many irreducible components.

Let $G = \mathrm{Hg}(A_0)$, which lies in the symplectic group of the polarisation and fixes γ [Del82, §3], let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$, $u_0(e^{i\theta}) = \cos\theta + J_0 \sin\theta$, be the homomorphism defining the Hodge structure, and let $\mathcal{D} = \{gJ_0g^{-1} : g \in G(\mathbb{R})^+\}$. For $g \in G(\mathbb{R})^+$ the complex structure $J = gJ_0g^{-1}$ is compatible with the polarisation and positive for it, since g preserves the polarisation, so $\mathcal{D} \subseteq \mathcal{H}$; and $\mathcal{D}$ is the orbit of Lemma 8.3 when J is identified with $u_J = gu_0g^{-1}$. For $J \in \mathcal{D}$ the group $u_J(\mathrm{U}(1))$ lies in $G(\mathbb{R})$, so $\mathrm{Hg}(A_J) \subseteq G$ fixes γ, which is therefore a Hodge class on A_J. The map sending $J \in \mathcal{D}$ to the image of J in $\mathcal{S}_g$ together with γ is thus a continuous map from the connected space $\mathcal{D}$ to the locus of Hodge classes of $\mathbb{V}$, so it takes values in $\mathcal{C}$. By Lemma 8.3 there are points $J_k \in \mathcal{D}$ converging to J_0 with $u_{J_k}(\mathrm{U}(1))$ in the real points of a $\mathbb{Q}$-torus of G; then $\mathrm{Hg}(A_{J_k})$ lies in that torus and A_{J_k} is of CM type. One of the finitely many irreducible components of $\mathcal{C}$, say $\mathcal{C}_0$, contains the images of infinitely many J_k, and, being closed, also their limit (a_0,γ).

Let $\rho\colon S \to \mathcal{C}_0$ be a resolution of singularities [Hir64]: S is smooth, connected and quasi-projective, since $\mathcal{C}_0$ is irreducible and quasi-projective, and ρ is projective and surjective. Let $\pi\colon \mathcal{A} \to S$ be the pull-back of $\mathcal{A}_g$ along $S \to \mathcal{C}_0 \to \mathcal{S}_g$, a polarised abelian scheme. A point of $\mathcal{C}_0$ is a pair (a,v) with $v \in \mathbb{V}_a$ of Hodge type, and $(a,v) \mapsto v$ is a flat section of the pull-back of $\mathbb{V}$ to $\mathcal{C}_0$, because the total space of a local system is a covering space of the base; its pull-back to S, divided by ℓ, is a flat section ξ of $R^{2p}\pi_*\mathbb{Q}$ of Hodge type at every point. Take s_0 over (a_0,γ) and c over the image of some J_k in $\mathcal{C}_0$. ∎

[44] For an elliptic curve $G = \mathrm{SL}_2$ and $\mathcal{D}$ is the upper half-plane. The compact torus $u_\tau(\mathrm{U}(1))$ fixing τ lies in the real points of a maximal $\mathbb{Q}$-torus of SL_2 exactly when τ is imaginary quadratic, and the torus is then the group of elements of norm one of $\mathbb{Q}(\tau)$. So the points of Lemma 8.3 are the curves with complex multiplication, and the lemma is the familiar density of the imaginary quadratic points in the upper half-plane.

Theorem 8.5 ((F2) is propagation from CM points)**.** *The following are equivalent.*

(i) (F2).

(ii) *The Hodge conjecture holds for every complex abelian variety.*

(iii) *For every polarised abelian scheme $\pi\colon \mathcal{A} \to S$ over a smooth connected complex quasi-projective variety and every global section ξ of $R^{2p}\pi_*\mathbb{Q}$, if ξ_s is algebraic at one point $s \in S$, then $\Sigma(\xi) = S$.*

(iv) *The same, for the points s at which $\mathcal{A}_s$ is of CM type: if ξ_c is algebraic at one point c with $\mathcal{A}_c$ of CM type, then $\Sigma(\xi) = S$.*

(v) *For every π and ξ as in* (iii) *with ξ_c algebraic at a point c with $\mathcal{A}_c$ of CM type, $\Sigma(\xi)$ has an interior point.*

Proof. (i) and (ii) are equivalent by Proposition 7.49. If (ii) holds and ξ_s is algebraic, then ξ_t is a Hodge class at every t by the first part of the proof of Proposition 6.17, which uses only the theorem of the fixed part and the semisimplicity of polarisable Hodge structures, and it is algebraic by (ii); so (ii) implies (iii). (iv) is the case of (iii) at points of CM type, and (iv) and (v) are equivalent by Lemma 8.2(iii).

Assume (iv). Let F be a CM field and $n \geq 2$, and let δ_0 be the trivial class, the discriminant of a hermitian space that is a sum of n hyperbolic planes, each of determinant -1. The family of (F, n, δ_0)-Weil type over its Shimura variety with a level structure, as in [I, Setup 17.1] and [I, Setup 16.1], is a polarised abelian scheme over a smooth connected quasi-projective base, on which the Weil classes are global sections ([I, Proposition 17.3(i)]). By [I, Theorem 1.1(i)] they are algebraic at a member isogenous to a product of abelian varieties with complex multiplication by F, which is of CM type, and by (iv) they are algebraic at every member. Let A be an abelian variety of split Weil type for F in the sense of [Mil20, §2.1]: F acts on it with $\dim_F H^1(A, \mathbb{Q}) = 2n$ and $\dim V_\sigma^{1,0} = n$ for every embedding σ, and for some polarisation the hermitian form of [I, Remark 17.2] has a totally isotropic subspace of dimension n, so that it is a sum of n hyperbolic planes and A is of (F, n, δ_0)-Weil type. By [I, Proposition 17.3(iv)] the Hodge structure $H^1(A, \mathbb{Q})$ is isomorphic, compatibly with F, to that of a member A_s of the family, so there is an isogeny $A \to A_s$ commuting with F; its pull-back maps $W_F(A_s)$ onto $W_F(A)$ and algebraic classes to algebraic classes. So every abelian variety of split Weil type has algebraic Weil classes, for $n \geq 2$ by this argument and for $n = 1$ by the theorem of Lefschetz on $(1,1)$ classes. By André's theorem [Mil20, Theorem 1 and its proof], every Hodge class on an abelian variety B of CM type is a sum $\sum f_\Delta^*(t_\Delta)$ of pull-backs along homomorphisms $f_\Delta\colon B \to A_\Delta$ of Weil classes t_Δ on abelian varieties A_Δ of split Weil type for CM fields F_Δ; so the Hodge conjecture holds for every abelian variety of CM type. Now let A_0 be any abelian variety and γ a Hodge class on it, and take S, π, ξ, s_0, c from Proposition 8.4. Then ξ_c is algebraic, so by (iv) $\xi_{s_0} = \gamma$ is algebraic. This is (ii). ∎

Figure 40 draws the theorem in one family.

The implication from (iii) to (ii) is the argument of Deligne and André as Milne records it [Mil20, §5, Theorem 3 and Remark 2].[45] In that argument the families can be taken to be polarised abelian schemes and the starting points to be of CM type, which is what (iv) records, and the converse holds. The seeds are explicit: for the Weil classes they are the

[45] Deligne's proof that Hodge classes on abelian varieties are absolutely Hodge uses families of the same kind [Del82]. By his Principle B [Del82, §2] the property of being absolutely Hodge passes from one member of a family to every member, so a family through a given Hodge class with a member of CM type reduces the question to that member. Algebraicity is not known to pass from one member to all, and (iii) is precisely that property.

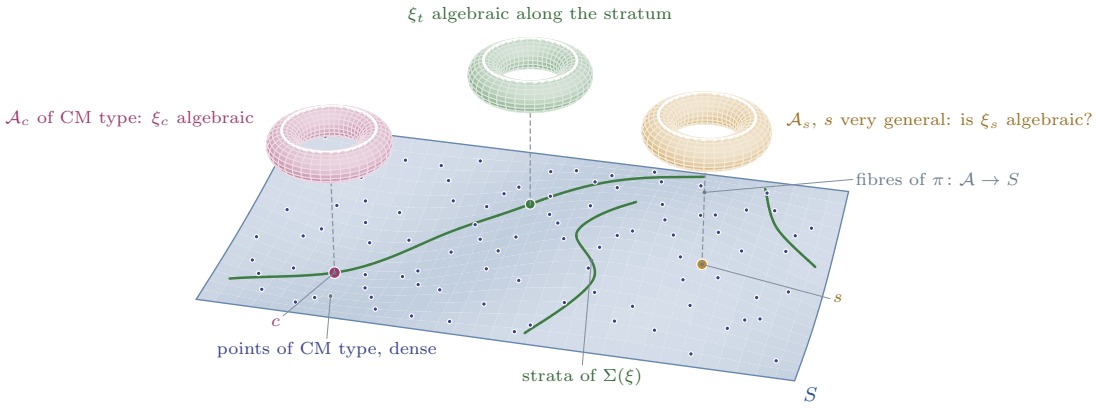


FIGURE 40. Theorem 8.5 in one family. The sheet is the base S of a polarised abelian scheme $\pi\colon \mathcal{A} \to S$ through a given Hodge class, as in Proposition 8.4; the small dots are its points of CM type, dense by Lemma 8.3, and the green curves are strata of the algebraic locus $\Sigma(\xi)$ of the flat class ξ (Lemma 8.2). Above three points the fibres are drawn as tori, and the loop on each stands for ξ, carried from fibre to fibre by parallel transport. At the CM member $\mathcal{A}_c$ the class is algebraic, and it stays algebraic along the stratum through c. At a very general member $\mathcal{A}_s$ it is algebraic exactly when $\Sigma(\xi)$ has an interior point, and (F2) is the statement that this happens in every such family.

base points of [I, Theorem 17.5], and for every other Hodge class they are the CM members supplied by Proposition 8.4, at which the class is algebraic once the Weil classes of the split families are. For the Weil classes of the split family of a CM field, whose base point is of CM type, statement (v) is the condition of [I, Theorem 1.1(iv)]. So the condition of [I, Theorem 1.1(iv)], required in every polarised abelian scheme with a member of CM type at which the class is algebraic, is exactly (F2); it is not a reduction of (F2) to something smaller, because the interior point it asks for is the conclusion of (iii).

8.3. **(F3′) and the part reached from abelian varieties.**

Definition 8.6 (The abelian part)**.** For a smooth complex projective variety X and $p \geq 0$ let $H^{2p}_{\mathrm{ab}}(X) \subseteq H^{2p}(X, \mathbb{Q})$ be the sum of the images of the maps $\Gamma_*\colon H^k(B, \mathbb{Q}) \to H^{2p}(X, \mathbb{Q})$, $y \mapsto \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^* y)$, over all complex abelian varieties B, the point included, all k, and all algebraic cycles Γ on $B \times X$ with rational coefficients and of the codimension that lands in degree $2p$.

Proposition 8.7 ((F3′) is a statement about the complement of the abelian part)**.** *Let X be a smooth complex projective variety and $p \geq 0$.*

(i) *$H^{2p}_{\mathrm{ab}}(X)$ is a sub-Hodge structure of $H^{2p}(X, \mathbb{Q})$, spanned by the images of finitely many of the maps Γ_*.*

(ii) $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$.

(iii) *Let $T^{2p}_{\mathrm{ab}}(X)$ be a complement of $H^{2p}_{\mathrm{ab}}(X)$ in $H^{2p}(X, \mathbb{Q})$ as a Hodge structure. Then* (F3′) *holds for X in degree $2p$ if and only if $T^{2p}_{\mathrm{ab}}(X)$ contains no nonzero Hodge class, in particular whenever the Hodge group of $T^{2p}_{\mathrm{ab}}(X)$ fixes no nonzero vector.*

(iv) *X lies in the class $\mathcal{A}$ of Proposition 6.35(iii) if and only if $T^{2p}_{\mathrm{ab}}(X) = 0$ for every p and the odd cohomology is reached in the same way.*

Proof. (i) An algebraic cycle Γ of codimension e on $B \times X$, with $\dim B = b$, induces a morphism of Hodge structures $H^k(B,\mathbb{Q}) \to H^{k+2e-2b}(X,\mathbb{Q})(e-b)$, so its image is a sub-Hodge structure; a sum of sub-Hodge structures is one, and a finite-dimensional space is spanned by finitely many of them.

(ii) Every $\Gamma_*\beta$ with β a Hodge class lies in both sets, so $\mathrm{Ab}^p(X)$ is contained in the intersection. Conversely take (B_j, Γ_j, k_j), $1 \le j \le r$, as in (i), and let

$$\Phi = \sum_j \Gamma_{j*} \colon \bigoplus_j H^{k_j}(B_j,\mathbb{Q})(t_j) \longrightarrow H^{2p}_{\mathrm{ab}}(X),$$

with the Tate twists t_j that make it a morphism of pure Hodge structures of weight $2p$; it is surjective. Polarisable Hodge structures form a semisimple category, so Φ has a section σ that is a morphism of Hodge structures. For a Hodge class $\gamma \in H^{2p}_{\mathrm{ab}}(X)$ the class $\sigma(\gamma)$ is of type (p,p), so each component $\beta_j \in H^{k_j}(B_j,\mathbb{Q})$ is a Hodge class, k_j being even, and $\gamma = \sum_j \Gamma_{j*}\beta_j \in \mathrm{Ab}^p(X)$.

(iii) A complement exists by semisimplicity, and the Hodge classes of $H^{2p}(X,\mathbb{Q})$ are the sums of those of the two summands; by (ii) the Hodge classes of the first summand are $\mathrm{Ab}^p(X)$. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ exactly when the second has none, and a Hodge class of $T^{2p}_{\mathrm{ab}}(X)$ is a vector fixed by its Hodge group [Del82, §3]. (iv) is the definition of $\mathcal{A}$ read degree by degree. ∎

So (F3′) asks nothing of the abelian part, where by (ii) and Proposition 7.49 the Hodge conjecture reduces to (F2), and it asks of the complement that it carry no Hodge class. The abelian part contains every algebraic class, taking B a point, so under the Hodge conjecture the complement carries none. To prove (F3′) for X in degree $2p$ without it, one has to show that every Hodge class of X is a sum of classes $\Gamma_*\beta$, that is, to construct algebraic cycles Γ on products $B \times X$ that reach it, which is the step Remark 6.36 isolates. Proposition 6.35(iii), Proposition 6.42 and Theorem 6.52 are cases in which the complement is zero, and Proposition 6.39 and Proposition 6.40 cases in which the conjecture itself makes its Hodge classes algebraic.

8.4. **The first open case of (F3′) in families.** The frontier of (F3′) in Remark 6.68 begins with $S \times S$ and $S^{[2]}$ for a K3 surface S with real multiplication. These surfaces come in families over Hermitian symmetric domains, and the argument of [I, Theorem 1.1] applies to them with rational Hodge isometries in place of isogenies.

Setup 8.8 (A family with real multiplication)**.** Let E be a totally real field of degree $r \ge 2$ and T a finite-dimensional $\mathbb{Q}$-vector space with an action of E and a nondegenerate symmetric form q for which E acts by self-adjoint maps, so that $q = \mathrm{Tr}_{E/\mathbb{Q}}\, q_E$ for a unique E-bilinear form q_E. Put $m = \dim_E T \ge 3$, and suppose that q_E has signature $(2, m-2)$ at one real place σ_1 of E and is negative definite at the others, and that (T,q) embeds isometrically in $\Lambda_{\mathbb{Q}}$, where Λ is the K3 lattice; write $N = T^{\perp} \cap \Lambda$. Let $T_1 = T \otimes_{E,\sigma_1} \mathbb{R}$ and

$$\mathcal{D}_E = \{\, [\omega] \in \mathbb{P}(T_1 \otimes_{\mathbb{R}} \mathbb{C}) :\ q(\omega,\omega) = 0,\ q(\omega,\bar\omega) > 0 \,\}^{+},$$

one of its two components, a bounded symmetric domain of dimension $m-2$. For $\omega \in \mathcal{D}_E$ let S_ω be a K3 surface with a marking $H^2(S_\omega,\mathbb{Z}) \cong \Lambda$ carrying $H^{2,0}(S_\omega)$ to $\mathbb{C}\omega$, which exists by the surjectivity of the period map [Huy16, Chs. 6–7]; it is projective, since N has signature $(1, 21-rm)$ and so contains a class of positive square, which is of type $(1,1)$ [Huy16, Ch. 8]. Each $e \in E$ acts on $T_1 \otimes \mathbb{C}$ by the scalar $\sigma_1(e)$ and is self-adjoint, so it

preserves the Hodge decomposition of T, and $T \subseteq H^2(S_\omega, \mathbb{Q})$ is a sub-Hodge structure containing $T(S_\omega)_\mathbb{Q}$. Write $\tilde{e}_\omega \in H^4(S_\omega \times S_\omega, \mathbb{Q})$ for the class of the endomorphism $e \oplus 0$ of $H^2(S_\omega, \mathbb{Q}) = T \oplus N_\mathbb{Q}$, a Hodge class, and let

$$\Sigma_E = \{\, \omega \in \mathcal{D}_E : \ \tilde{e}_\omega \text{ is algebraic for every } e \in E \,\}$$

be the algebraic locus of the real multiplication; as $e \mapsto \tilde{e}_\omega$ is $\mathbb{Q}$-linear, it suffices to ask this for e in a $\mathbb{Q}$-basis of E. The set Σ_E does not depend on the choice of the surfaces S_ω. If (S, φ) and (S', φ') are two marked choices for ω, then $\psi = \varphi'^{-1}\varphi$ is a rational Hodge isometry $H^2(S, \mathbb{Q}) \to H^2(S', \mathbb{Q})$, induced by an algebraic class Z on $S \times S'$ by Buskin's theorem [Bus19], and Z^t induces the adjoint ψ^{-1}; the composite $Z \circ \tilde{e} \circ Z^t$ acts on $H^2(S', \mathbb{Q})$ as $\psi(e \oplus 0)\psi^{-1}$, which is $e \oplus 0$ for the marking φ', and by zero on H^0 and H^4, so it is the class $\tilde{e}'$ of S', and $\tilde{e}$ is algebraic if and only if $\tilde{e}'$ is. Let $G = \mathrm{Res}_{E/\mathbb{Q}} \mathrm{SO}(T, q_E)$, so that $G(\mathbb{R}) = \mathrm{SO}(T_1) \times \prod_{\sigma \neq \sigma_1} \mathrm{SO}(T_\sigma)$; write $G(\mathbb{R})^+$ for the identity component of this Lie group, which acts transitively on $\mathcal{D}_E$ through its first factor, and $\Gamma_\mathbb{Q} = G(\mathbb{Q}) \cap G(\mathbb{R})^+$.

The surfaces with real multiplication of [vG08, vGS25] lie in families of this kind,[46] and so do the surfaces S_λ of Theorem 7.4, for which E is cubic, $m = 3$ and $\mathcal{D}_E$ is a disc.

Theorem 8.9 (The algebraic locus of a real multiplication)**.** *Keep Setup 8.8.*

(i) *The points ω at which S_ω has complex multiplication, that is, at which the Hodge group of $T(S_\omega)_\mathbb{Q}$ is commutative, are dense in $\mathcal{D}_E$, and they lie in Σ_E.*

(ii) *Σ_E is stable under $\Gamma_\mathbb{Q}$, and $\Gamma_\mathbb{Q}$ is dense in $G(\mathbb{R})^+$.*

(iii) *Σ_E is dense in $\mathcal{D}_E$, and either $\Sigma_E = \mathcal{D}_E$ or Σ_E is meagre.*

(iv) *The following are equivalent: Σ_E is closed; Σ_E has an interior point; Σ_E is not meagre; $\Sigma_E = \mathcal{D}_E$; the Hodge conjecture holds for $S_\omega \times S_\omega$ for every ω outside a countable union of proper closed analytic subsets of $\mathcal{D}_E$. When they hold, the Hodge conjecture and* (F3′) *hold for $S_\omega \times S_\omega$ and $S_\omega^{[2]}$ at every ω at which $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_\mathbb{Q})$ is E or a CM field.*

Proof. We first record two facts about the very general point. The Hodge structure of S_ω on T is given by $u_\omega \colon \mathrm{U}(1) \to \mathrm{SO}(T_\mathbb{R})$, acting by z^2 on $\mathbb{C}\omega$, by z^{-2} on $\mathbb{C}\bar{\omega}$ and trivially on their orthogonal complement; it commutes with E, lies in $G(\mathbb{R})^+$, and $u_{g\omega} = g u_\omega g^{-1}$ for $g \in G(\mathbb{R})^+$.

(a) *No nonzero $t \in T$ is orthogonal to every $\omega \in \mathcal{D}_E$.* The lines $\mathbb{C}\omega$, $\omega \in \mathcal{D}_E$, span $T_1 \otimes \mathbb{C}$, since $\mathcal{D}_E$ is a nonempty open subset of the nondegenerate quadric $q(\omega, \omega) = 0$ in $\mathbb{P}(T_1 \otimes \mathbb{C})$, so such a t would have zero image in T_1; but $T \to T_1$ is injective, since T is a free E-module and σ_1 is injective.

(b) *An endomorphism f of T that is an endomorphism of Hodge structures at every $\omega \in \mathcal{D}_E$ is multiplication by an element of E.* Such an f commutes with every u_ω, hence with the elements X_ω of the Lie algebra of $G(\mathbb{R})$ that span the Lie algebras of the groups $u_\omega(\mathrm{U}(1))$. They lie in $\mathfrak{so}(T_1) \oplus 0$, which acts on $T' = \bigoplus_{\sigma \neq \sigma_1} T_\sigma$ by zero, and their span is stable under $G(\mathbb{R})^+$, since $u_{g\omega} = g u_\omega g^{-1}$, so it is an ideal of $\mathfrak{so}(T_1)$. That ideal has nonzero projection to every simple ideal, as the eigenvalues $2i, -2i, 0, \dots, 0$ of X_ω on $T_1 \otimes \mathbb{C}$ show also for $m = 4$, where $\mathfrak{so}(T_1) \otimes \mathbb{C} \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and $T_1 \otimes \mathbb{C}$ is the tensor product of

[46] For a projective K3 surface S the transcendental part $T(S)_\mathbb{Q}$ is an irreducible Hodge structure whose algebra of Hodge endomorphisms is a field, totally real or CM [Zar83]; S has real multiplication when this field is totally real and not $\mathbb{Q}$. The space $T(S)_\mathbb{Q}$ is then a vector space over the field, so its degree divides $\dim T(S)_\mathbb{Q}$, and a very general member of a family of Setup 8.8 has $T(S_\omega)_\mathbb{Q} = T$ and real multiplication by exactly E, as the proof of Theorem 8.9 shows.

the two standard representations; so it is all of $\mathfrak{so}(T_1)$, and f commutes with $\mathfrak{so}(T_1)$. For $m \geq 3$ the representation of $\mathfrak{so}(T_1)$ on $T_1 \otimes \mathbb{C}$ is irreducible and nontrivial, so f preserves $T_1 = \mathfrak{so}(T_1) \cdot T_{\mathbb{R}}$ and T' and acts on $T_1 \otimes \mathbb{C} = T \otimes_{E,\sigma_1} \mathbb{C}$ by a scalar λ. An automorphism τ of $\mathbb{C}$, acting on $T \otimes_{\mathbb{Q}} \mathbb{C}$ through the second factor, commutes with f and maps $T \otimes_{E,\sigma_1} \mathbb{C}$ onto $T \otimes_{E,\tau\sigma_1} \mathbb{C}$, so f acts on the latter by $\tau(\lambda)$; and every embedding of E is of the form $\tau\sigma_1$. So $f \otimes 1$ is multiplication by an element e of $E \otimes_{\mathbb{Q}} \mathbb{C}$ on $T \otimes_{\mathbb{Q}} \mathbb{C} = T \otimes_E (E \otimes_{\mathbb{Q}} \mathbb{C})$, and e is fixed by $\mathrm{Aut}(\mathbb{C})$ because f is defined over $\mathbb{Q}$; hence $e \in E$.

For $t \in T$ the set of ω with $t \perp \omega$ is a closed analytic subset of $\mathcal{D}_E$, and for $f \in \mathrm{End}_{\mathbb{Q}}(T)$ so is the set where f is an endomorphism of Hodge structures, namely where $f(\omega) \in \mathbb{C}\omega$ and $f^*(\omega) \in \mathbb{C}\omega$, f^* the adjoint. By (a) and (b) these sets are proper for $t \neq 0$ and $f \notin E$. There are countably many of them, and outside their union $\mathrm{NS}(S_\omega)_{\mathbb{Q}} \cap T = 0$, so $T(S_\omega)_{\mathbb{Q}} = T$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_{\mathbb{Q}}) = E$.

(i) The Hodge group of T acts trivially on $T \cap \mathrm{NS}(S_\omega)_{\mathbb{Q}}$, whose orthogonal complement in T is $P = T(S_\omega)_{\mathbb{Q}}$, so it is commutative exactly when the Hodge group H of P is. The group G is connected and reductive, and $\omega \mapsto u_\omega$ identifies $\mathcal{D}_E$ with the orbit of Lemma 8.3, since $g \in G(\mathbb{R})^+$ fixes u_ω exactly when it preserves the line $\mathbb{C}\omega$, the eigenspace of $u_\omega(z)$ for z^2. By Lemma 8.3 the points at which $u_\omega(\mathrm{U}(1))$ lies in the real points of a maximal $\mathbb{Q}$-torus of G are dense; at them the Hodge group of T, the smallest algebraic $\mathbb{Q}$-subgroup whose real points contain $u_\omega(\mathrm{U}(1))$, lies in that torus, so these points have complex multiplication. Now let ω be any point at which H is commutative. The Hodge structure P is irreducible [Zar83], so P is an irreducible representation of the torus H, whose rational points are Zariski dense in it. The $\mathbb{Q}$-subalgebra L of $\mathrm{End}(P)$ that $H(\mathbb{Q})$ generates is therefore commutative, and P is a simple faithful L-module; so L is a field with $\dim_L P = 1$, and its commutant $E' = \mathrm{End}_{\mathrm{Hdg}}(P)$ is L. By [Zar83] E' is totally real or CM. If it were totally real, the embedding τ through which it acts on the line $H^{2,0}(S_\omega)$ would be real, and that line would lie in $P \otimes_{E',\tau} \mathbb{C}$, which is one-dimensional and stable under complex conjugation, so it would also contain the different line $H^{0,2}(S_\omega)$, which is impossible. So E' is a CM field, the Hodge conjecture holds for $S_\omega \times S_\omega$ by Proposition 6.39(i), and the Hodge classes $\tilde{e}_\omega$ are algebraic.

(ii) Let $g \in \Gamma_{\mathbb{Q}}$ and $\omega \in \Sigma_E$. Then $g\omega \in \mathcal{D}_E$, and $\phi = \mathrm{id}_N \oplus g$ is a rational isometry of $\Lambda_{\mathbb{Q}} = N_{\mathbb{Q}} \oplus T$ carrying $\mathbb{C}\omega$ to $\mathbb{C}g\omega$, so through the markings it is a rational Hodge isometry $H^2(S_\omega, \mathbb{Q}) \to H^2(S_{g\omega}, \mathbb{Q})$, induced by an algebraic class $Z \in H^4(S_\omega \times S_{g\omega}, \mathbb{Q})$ [Bus19], whose transpose induces ϕ^{-1}. As g is E-linear, $\phi \circ (e \oplus 0) \circ \phi^{-1} = e \oplus 0$, so, as in Setup 8.8, $\tilde{e}_{g\omega} = Z \circ \tilde{e}_\omega \circ Z^t$ is algebraic.

For the density of $\Gamma_{\mathbb{Q}}$, let Y be an $E \otimes \mathbb{R}$-linear map of $T \otimes \mathbb{R}$ that is skew-adjoint for q and has $1 - Y$ invertible. Its Cayley transform $c(Y) = (1 + Y)(1 - Y)^{-1}$ is $E \otimes \mathbb{R}$-linear, has $c(Y)^* c(Y) = 1$ because $1 - Y$ is the adjoint of $1 + Y$, and has determinant one for the same reason, so $c(Y) \in G(\mathbb{R})$; and c is a homeomorphism onto the open neighbourhood $\{g \in G(\mathbb{R}) : 1 + g \text{ invertible}\}$ of 1, with inverse $g \mapsto (g - 1)(g + 1)^{-1}$. The E-linear skew-adjoint maps of T are dense among these Y, and c maps them into $G(\mathbb{Q})$. So $\Gamma_{\mathbb{Q}}$ is dense in a connected open neighbourhood of 1, its closure is an open subgroup of $G(\mathbb{R})^+$, and it is all of $G(\mathbb{R})^+$.

(iii) and (iv). Density follows from (i). If Σ_E contains a nonempty open set U_0, then $\Sigma_E = \mathcal{D}_E$: for $\omega \in \mathcal{D}_E$ choose $\omega' \in U_0$ and $g_1 \in G(\mathbb{R})^+$ with $g_1\omega' = \omega$; the set of $g \in G(\mathbb{R})^+$ with $g^{-1}\omega \in U_0$ is open and contains g_1, so by (ii) it contains some $g \in \Gamma_{\mathbb{Q}}$, and $\omega = g(g^{-1}\omega) \in \Sigma_E$ by (ii).

It remains to show that Σ_E has an interior point as soon as it is not meagre. Fix $h \in N$ with $h^2 > 0$ and $h \cdot \delta \neq 0$ for every $\delta \in N$ with $\delta^2 = -2$, which exists because the

hyperplanes $\delta^\perp$ are locally finite in the positive cone of $N_\mathbb{R}$ [Huy16, Ch. 8]. Let $\mathcal{D}^\circ$ be the set of $\omega \in \mathcal{D}_E$ that are orthogonal to no $\delta \in \Lambda$ with $\delta^2 = -2$ and $h \cdot \delta = 0$. Writing such a δ as $n + t$ with $n \in N_\mathbb{Q}$ and $t \in T$, we have $t \neq 0$, because $N_\mathbb{Q} \cap \Lambda = N$, and $\delta \perp \omega$ exactly when $t \perp \omega$; so by (a) the complement of $\mathcal{D}^\circ$ lies in a countable union of proper closed analytic subsets and is meagre.

Let $\omega \in \mathcal{D}^\circ$. No class of $\mathrm{NS}(S_\omega)$ of square -2 is orthogonal to h, so after composing the marking with an isometry $\pm w$, w a product of reflections in such classes, which fixes ω and so changes only the choice of S_ω, the class h is that of an ample line bundle L on S_ω [Huy16, Ch. 8]. The deformations of the pair (S_ω, L) form a smooth family over a base that the local period map identifies with a neighbourhood of ω in the domain of periods orthogonal to h [Huy16, Ch. 6], and that domain contains $\mathcal{D}_E$, since $h \in N$. Restricting to $\mathcal{D}_E$, we obtain an open neighbourhood U of ω in $\mathcal{D}_E$ and a smooth family $\mathcal{X}_U \to U$ of K3 surfaces with markings, whose fibre over u is a choice of S_u, carrying a line bundle that restricts to L on S_ω. On S_ω the bundle $L^{\otimes 3}$ is very ample and has no higher cohomology [Huy16, Ch. 2]; by semicontinuity and base change, after shrinking U, its direct image is free and its sections embed $\mathcal{X}_U$ in $\mathbb{P}^M \times U$ over U. The classes $\tilde{e}$ are flat sections of the fourth cohomology of $\mathcal{X}_U \times_U \mathcal{X}_U$, which is projective over U. As in the proof of Lemma 8.2(i), with the Hilbert schemes of $\mathbb{P}^M \times \mathbb{P}^M$ [Gro61], which are projective, for each e in a $\mathbb{Q}$-basis of E the set of $u \in U$ at which $\tilde{e}_u = \sum_i q_i [Z_i]$ with rational q_i and Z_i of Hilbert polynomial P_i is the image, under a proper map, of a union of connected components of a fibre product of relative Hilbert spaces over U, and so it is closed in U. Taking the union over the countably many $(\underline{P}, \underline{q})$ and the intersection over the basis, $\Sigma_E \cap U$ is a countable union of subsets of U that are closed in U.

The sets U constructed for the points of $\mathcal{D}^\circ$ cover it, and countably many of them, U_i, suffice, since $\mathcal{D}_E$ has a countable basis. If Σ_E is not meagre, neither is $\Sigma_E \cap \mathcal{D}^\circ$, so some $\Sigma_E \cap U_i$ is not meagre, and by the Baire category theorem in U_i one of the countably many closed subsets of U_i that make it up has an interior point. This proves (iii), and in (iv) the equivalence of the first four conditions, since a dense closed set is everything. The fifth condition implies the third, because at each point where the Hodge conjecture holds for $S_\omega \times S_\omega$ the Hodge classes $\tilde{e}_\omega$ are algebraic, and the complement of a countable union of proper closed analytic subsets is not meagre.

If $\Sigma_E = \mathcal{D}_E$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_\mathbb{Q}) = E$, then E preserves $T(S_\omega)_\mathbb{Q}$, the smallest sub-Hodge structure of T whose complexification contains $H^{2,0}(S_\omega)$, since each $e \neq 0$ maps it onto a sub-Hodge structure of the same dimension that contains $H^{2,0}(S_\omega)$; and each $e \in E$ acts on it through the algebraic class $\pi_T \circ \tilde{e}_\omega \circ \pi_T$, where π_T is the algebraic projector onto $T(S_\omega)_\mathbb{Q}$ of the proof of Proposition 6.39; so Proposition 6.39(iii) gives the Hodge conjecture for $S_\omega \times S_\omega$ and $S_\omega^{[2]}$, and with it (F3′). If the field is CM, Proposition 6.39(i) gives the same. Outside the countable union found at the beginning of the proof the field is E, so the fourth condition implies the fifth. ∎

On the families of van Geemen and Schütt with real multiplication by $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ the real multiplication is induced by explicit cycles [vGS25, Theorem 1.2 and Section 4.8], and on the double planes branched along six lines that have real multiplication, necessarily by a quadratic field here, by Schlickewei's cycles [Sch10, vG08]. Whenever the members on which such cycles are known contain a nonempty open subset of $\mathcal{D}_E$, (iv) gives $\Sigma_E = \mathcal{D}_E$. For the family of the surfaces S_λ, Theorem 8.9 gives the real multiplication at every CM point and on a dense set stable under $\Gamma_\mathbb{Q}$, and Theorem 7.4(ii) gives it at S_λ if the Kuga–Satake class of S_λ is algebraic; whether Σ_E is closed there is open. Since Σ_E

is stable under the group $\Gamma_{\mathbb{Q}}$, every $\Gamma_{\mathbb{Q}}$-orbit lies either in Σ_E or in its complement: the Hodge isometries of (ii) move the real multiplication along an orbit and never bring it to an orbit where it is not yet algebraic, and by Remark 7.5 the composites of Hodge similarities that return to $T(S_\lambda)$ span only the maximal multiquadratic subfield of E, which is $\mathbb{Q}$ for a cubic field.

8.5. **What the constructions reach.**

Theorem 8.10 (Where (F2) and (F3′) stand)**.**

(i) What carries over. *In every polarised abelian scheme of Theorem 8.5 the algebraic locus of a flat class is a countable union of closed algebraic strata and is either everything or meagre (Lemma 8.2); in the Weil families it is moreover dense and contains an explicit CM point (*[I, Theorem 1.1]*); in every family of Setup 8.8 the algebraic locus of the real multiplication is dense, contains every CM point, is stable under $\Gamma_{\mathbb{Q}}$ and is either everything or meagre (Theorem 8.9); and every Hodge class of every abelian variety lies in a polarised abelian scheme with a CM member (Proposition 8.4).*

(ii) (F2). *It holds if and only if the algebraic locus of every flat Hodge class algebraic at a CM member of a polarised abelian scheme has an interior point (Theorem 8.5). In the Weil families the constructions of the paper that supply an interior point are: an object with injective semiregularity map, or satisfying the flatness condition, at one point (Theorem 2.3,* [I, Theorem 17.7]*, Proposition 3.15); a bound on the Hilbert data of representatives along the Hecke orbit of a base point (*[I, Theorem 16.9]*), which is equivalent to its conclusion (*[I, Theorem 16.17]*); and the Lefschetz standard conjecture for the total space over a curve through a base point, which implies it and, when the monodromy is Zariski dense, follows from it (Corollary 6.31). For the squares of Mumford fourfolds the last two take the forms of Theorem 7.43 and Corollary 6.30. Markman's sheaves are objects of the first kind, and with them he proves the algebraicity of the Weil classes on every abelian fourfold of Weil type and on the split sixfolds of every imaginary quadratic field* [Mar25a, Mar25b]*; with* [MZ99] *this gives the Hodge conjecture for abelian varieties of dimension at most five. The open cases named in Section 6 include the non-split Weil sixfolds of imaginary quadratic fields, the Weil eightfolds of quartic CM fields, where an object of the first kind needs* $\dim \mathrm{Ext}^2(E,E) \geq 88$ *(Corollary 5.15), and the squares of Mumford fourfolds, where the classes are algebraic at every CM point (Theorem 6.29) and on every member once one Kuga–Satake class is algebraic (Theorem 7.4), or once one complex at one CM point has* Ext^2 *of dimension* 119 *(Theorem 7.27); on the square of a Mumford fourfold the two exceptional classes are algebraic exactly when one class on a product with an abelian sixfold is (the sixfold link of* [BB26]*).*

(iii) (F3′). *It holds if and only if, for every variety and every degree, the complement of the abelian part carries no Hodge class (Proposition 8.7). The complement is zero on the class $\mathcal{A}$, which contains the Fermat and Delsarte hypersurfaces of every dimension and the cyclic covers branched along them (Proposition 6.42, Corollary 6.53), and it carries no Hodge class on the varieties of Proposition 6.43, Proposition 6.44 and Proposition 6.46, where* (F3′) *is proved, and of Proposition 6.39 and Proposition 6.40, where the conjecture itself is. On the first open varieties, $S \times S$ and $S^{[2]}$ for a K3 surface with real multiplication, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 8.9); the real multiplication is algebraic on the members of the families of* [vGS25] *and*

on the quadratic double planes of [Sch10]*, and whether the locus is closed is open for the family of S_λ, except for the two cyclic cubic fields of conductor* 7 *and* 9 *with the algebra split at one real place, where the dihedral correspondences of* [BB26] *make the locus the whole family. Beyond the varieties of abelian type, a Hodge class off the abelian part is removed only by a new algebraic cycle on a product $B \times X$ (Remark 6.36).*

Neither (F2) nor (F3′) *is proved here, and by Theorem 9.1 together they are the Hodge conjecture.*

Proof. Each statement is the result cited with it. For (ii), Markman's theorems are recalled in the introduction and their reach in Section 6, where the non-split sixfolds and the fields of degree at least four are among the open entries, and Theorem 5.14 and Corollary 5.15 give the number 88. For (iii), the known cases of the real multiplication are those recalled in Remark 7.5. ∎

9. The closure theorem

We collect the results of the paper in one theorem. Its first clause is [I, Theorem 1.1]; the second and third give the base of that theorem in its sharpest form; the fourth and fifth say what closing the algebraic locus requires and which constructions cannot supply it; and the last places the Weil classes inside the Hodge conjecture and states what [I, Theorem 1.1] becomes when it is carried to every family. The theorems we quote from the literature are used as theorems: the semiregularity theorem of Buchweitz and Flenner [BF03], in the form for perfect complexes that Pridham proves [Pri24]; the theorem of Cattani, Deligne and Kaplan [CDK95]; the theorem of Baily and Borel [BB66]; the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21]; the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99]; Markman's theorems on abelian varieties of Weil type [Mar25a, Mar25b]; the theorem of Bando and Siu on stable reflexive sheaves [BS94], Kuranishi's theorem in the form of Goldman and Millson [Kur65, GM88], Grivaux's Chern classes of coherent sheaves [Gri10] and Voisin's theorem on Weil tori [Voi02b]; and the theorems of Deligne, André and Milne recorded in Section 6.6 [Del71, And96b, Mil20].

Theorem 9.1 (Closure)**.** *Let F be a CM field of degree $2m$, let $n \geq 2$, and let δ be a discriminant. Write $K = \mathbb{Q}(\sqrt{-d})$ when $m = 1$.*

(i) The algebraic locus. *The algebraic locus Σ_F of the Weil classes contains an explicit CM point at which they are polynomials with rational coefficients in divisor classes, it is dense, it is a countable union of closed algebraic strata, and it is either the whole domain or meagre. It is closed if and only if the Hodge conjecture holds for the very general member of the family, and then the Weil classes are algebraic on every member (*[I, Theorem 1.1]*).*

(ii) One class on one connected domain. *For an abelian $2n$-fold A of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ one has $\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = 1$ for $k \neq n$ and 3 for $k = n$, and the Hodge conjecture for A in every degree is the algebraicity of the single class ω_1 of* [I, (11)]*, written out with 2^{2n-1} integer coefficients. For the family of the triple (K, n, δ) the conjecture is equivalent to the algebraicity of the absolutely Hodge class ω_s at every point s of the connected period domain $\mathcal{D}_{n,n}$ of dimension n^2 (*[I, Theorem 4.14]*,* [I, Corollary 4.15]*,* [I, Theorem 15.14]*,* [I, Corollary 15.35]*).*

(iii) A base point in every family. *Every family of every CM field contains a member isogenous to a product of CM abelian m-folds at which the Weil classes are polynomials in divisor classes (*[I, Theorem 17.5]*). For an imaginary quadratic field every family also contains a member with quaternionic multiplication extending K, at which the Weil classes are the real and imaginary parts of the n-th power of one explicit $(1,1)$ class, and every product of n abelian surfaces of Weil type is a member whose Weil classes are an explicit integral polynomial in $2n$ divisor classes (*[I, Theorem 15.25], [I, Theorem 15.27], [I, Theorem 15.10]*).*

(iv) Closing the locus. *If at every point of the orbit $G_F(\mathbb{Q})\cdot s_0$ the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then $\Sigma_F = \mathcal{D}_F$; this hypothesis is equivalent to its conclusion, and the transported cycle does not satisfy it whenever a component of the base cycle has finite stabiliser (*[I, Theorem 16.9], [I, Theorem 16.17], [I, Theorem 16.15]*). The bound is needed only along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit (*[I, Theorem 17.14]*). The locus of members with smaller Hodge group, which contains every member at which this paper exhibits algebraic Weil classes, contains the base point, is dense, stable under $G_F(\mathbb{Q})$ and a countable union of closed algebraic strata, and is meagre, so closing the locus requires the algebraicity of the Weil classes at a member with full Hodge group (*[I, Proposition 17.12], [I, Corollary 17.13]*). If at a base point some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the divisor classes of the family has injective semiregularity map, or satisfies the flatness condition of Proposition 3.15, then $\Sigma_F = \mathcal{D}_F$ (Theorem 2.3, [I, Theorem 17.7]). When the Chern character is exactly a multiple of a Weil class, no complex at any member of any family, for any CM field, has injective semiregularity map (Theorem 3.5, Theorem 5.4). In the polarised form, for an imaginary quadratic field and $n \geq 3$, every object has $\dim \mathrm{Ext}^2(E,E) \geq (4+\rho)n^2 - 2n$, with $\rho \leq 3$ the rank of the Hankel matrix of the coefficients, and an object attaining the bound meets the criterion (Theorem 3.9); at a quartic CM field and $n = 2$ the number takes fifteen values, an object meeting the criterion has $\dim \mathrm{Ext}^2(E,E) \geq 88$, and the characters of rank 88 whose Hodge locus leaves the polarised family are exactly those of θ^4 shape, which the Weil tori exclude (Theorem 5.14, Corollary 5.15, Proposition 5.17, Corollary 5.18). The statement is needed only for the split families of the CM fields of degree at least four, which carry the Weil classes of every CM field by descent and scalar extension (Proposition 6.13), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point (Corollary 6.31).*

(v) Constructions that do not reach a Weil class. *Each of the following fails to produce a Weil class on a general member, in the sense and under the hypotheses of the theorem cited, which Section 6 delimits:*

(a) *an object restricted from a projective space to the dual and transported by the Poincaré kernel (*[I, Theorem 7.3]*);*

(b) *equivariant for the field with the norm character (*[I, Theorem 5.6]*);*

(c) *an object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully, whether through semiregularity or through the weakened criterion of* [Mar25b, Question 11.4] *(*[I, Theorem 9.7], [I, Corollary 9.18], [I, Theorem 9.19]*);*

(d) *an object with a line bundle summand off the polarisation line, in particular a sum of line bundles (*[I, Theorem 9.20]*, Corollary 4.21);*
(e) *an object whose Chern character is effective at a base point (Theorem 4.5);*
(f) *induction through hyperplane sections, in either direction (*[I, Theorem 10.3], [I, Theorem 10.8]*);*
(g) *Kuga–Satake from a weight two sub-structure of* $H^2(A, \mathbb{Q})$ *(*[I, Proposition 11.11], [I, Corollary 11.12]*);*
(h) *a divisorial argument at a general member of a family of nontrivial discriminant (*[I, Proposition 15.18]*);*
(i) *a secant object whose support has two smooth branches of codimension two meeting transversally at a point, which includes the candidate of* [Mar25b, Section 12] *(Theorem 4.61, Corollary 4.63, Corollary 4.77);*
(j) *a secant object of rank one whose support is a complete intersection of two divisors (Theorem 4.81), or a secant object quasi-isomorphic to a two-term complex of sums of powers of the polarisation or of simple semi-homogeneous bundles (Theorem 4.89, Theorem 4.92); on the fourfold, a secant sheaf* $\mathcal{I}_S(3\Theta)$ *with smooth support unless* $d \leq 9$*, that is* $d \in \{1, 3, 5, 7\}$ *in Setup 4.51, and one whose ideal sheaf is presented by two bundles with classes in the subring generated by line bundles unless* $d \equiv 15, 23 \pmod{24}$ *(Corollary 4.68, Corollary 4.84);*
(k) *a representative of the Weil* K*-theory class that is a locally free sheaf on a smooth germ of codimension at least two of its support, or whose support has codimension at least* n *(Theorem 2.5, Theorem 2.13);*
(l) *at a quartic CM field, a complex with secant character that is locally free on a smooth curve at some point, in particular the explicit pair of* [Mar25c, Example 11.2.7] *(Theorem 5.28, Corollary 5.30);*
(m) *a convolution of fewer than* $2n$ *line bundles with nondegenerate differences (Theorem 4.26); on* E_0^6*, every convolution of the Weil pieces with three multiples of the polarisation, and on* E_0^8 *every one whose pieces have their shifts within six consecutive values (Theorem 4.35, Corollary 4.48).*

(vi) The Weil classes inside the conjecture. *The Hodge conjecture is equivalent to the conjunction of* (F2)*, the algebraicity of the Hodge classes of abelian varieties outside the subring generated by divisor classes and Weil classes of quotients, and* (F3′)*, the algebraicity of every Hodge class modulo the images of Hodge classes of abelian varieties under algebraic correspondences; and* (F2) *alone is equivalent to the Hodge conjecture for abelian varieties (Proposition 7.49, Proposition 6.35). The statement* (F3′) *holds on every variety whose cohomology is reached from abelian varieties by correspondences, on every variety of dimension at most three, on every fourfold and fivefold whose zero-cycles are supported in dimension at most three or that is dominated by a variety of the first kind, on every smooth ample divisor of odd dimension in a variety of the first kind, and on every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, and on every complete intersection of diagonal hypersurfaces of Vandermonde type, where the Hodge conjecture follows for two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$ *and for the very general member of degree* 2 *in every dimension, of degree* 3 *or* 4 *up to dimension seven and of degree* 6 *up to dimension five (Proposition 6.35, Proposition 6.42, Proposition 6.43, Proposition 6.44, Proposition 6.46, Proposition 6.47, Corollary 6.53, Theorem 6.55, Corollary 6.56, Theorem 6.66). The first*

instance of (F2)*, the two exceptional classes on the square of a Mumford fourfold, is algebraic at every CM point of the family, is algebraic on every member once one Kuga–Satake class is algebraic, and is algebraic on every member exactly when the Lefschetz standard conjecture holds for one ninefold (Theorem 6.29, Theorem 7.4, Corollary 6.30). Over the rule set formed by the implications of the paper and of the literature, together with Proposition 7.49, the minimal sets of statements from which the conjecture follows are* $\{\mathrm{F3}\}$, $\{\mathrm{L},\mathrm{M}\}$, $\{\mathrm{L},\mathrm{F3}'\}$, $\{\mathrm{V},\mathrm{F3}'\}$ *and* $\{\mathrm{F2},\mathrm{F3}'\}$*, the first two being each equivalent to it (Theorem 6.70, Remark 7.50). Carried to every family,* [I, Theorem 1.1] *gives exactly these two statements back:* (F2) *is equivalent to the statement that in every polarised abelian scheme with a member of CM type the algebraic locus of a flat Hodge class that is algebraic at such a member has an interior point (Theorem 8.5);* (F3′) *is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 8.7); and on the squares of K3 surfaces with real multiplication, the first varieties on which* (F3′) *is open, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 8.9), and it is the whole family for real multiplication by* $\sqrt{2}$*,* $\sqrt{3}$ *and* $\sqrt{6}$ *in families, for complex multiplication by* $\mathbb{Q}(\sqrt{-5})$ *and* $\mathbb{Q}(\sqrt{-7})$ *on the Fricke families, and for the cubic fields* $\mathbb{Q}(\zeta_7)^+$ *and* $\mathbb{Q}(\zeta_9)^+$ *on the surfaces of the Mumford fourfolds of the algebra split at one real place* [BB26].[47]

Proof. Each clause is the theorem or theorems cited with it, and no clause is used in the proof of another. For (i), [I, Theorem 1.1] is proved at the end of [I, Section 17.3]. For (ii), [I, Theorem 4.14] counts the Hodge classes, [I, Corollary 4.15] reduces every degree to ω_1, [I, Theorem 4.9(i)] shows that ω_1 has integer coefficients and exactly 2^{2n-1} nonzero terms, [I, Theorem 15.14(a)] and (b) are the two implications for the family, and [I, Proposition B.8] gives the connectedness and the dimension. For (iii), [I, Theorem 17.5] gives the CM point, [I, Theorem 15.25(v)] the cycle at a quaternionic point, [I, Theorem 15.27] places such a point in every family by the discriminant computation of [I, Theorem 15.25(iv)] and [I, Lemma 15.26], and [I, Theorem 15.10] treats products.

For (iv), the bounded criterion is [I, Theorem 16.9] with [I, Theorem 16.17] and [I, Theorem 16.15], and its form for a CM field is [I, Theorem 17.7(iv)]; the escaping sequence is [I, Theorem 17.14], and the special locus is [I, Proposition 17.12] with [I, Corollary 17.13]. The semiregularity criterion is Theorem 2.3 with Remark 3.6, carried to every CM field by [I, Theorem 17.7(v)]; the flatness form is Proposition 3.15. The pure form is excluded by Theorem 3.5, which rests on Lemma 3.2, Lemma 3.3 and Proposition 3.4, and for every CM field by Theorem 5.4. The number of the polarised form is Theorem 3.9(i), (iv) and (v); the quartic rank and the Hodge loci are Theorem 5.14, Corollary 5.15, Proposition 5.17 and Corollary 5.18. The reduction to the split families is Proposition 6.13, and the equivalence with the Lefschetz standard conjecture is Corollary 6.31. For (v), each item is the statement of the theorem cited, under its own hypotheses, which Section 6 delimits item by item.

For (vi), Proposition 6.35(i) gives the Hodge conjecture from (F3′) and the conjecture for abelian varieties, and Proposition 7.49 gives the latter from (F2); the converse implications hold because (F2) is a case of the conjecture and (F3′) is weaker than it. The cases of (F3′)

[47]These are the statements (L), (M) and (V) of Definition 6.14: $B(X)$ of [I, Definition 10.6] for every smooth projective X; every Hodge class motivated in the sense of André [And96b, §2.1]; and, for every smooth projective morphism f to a smooth connected base, a flat section of $R^{2p}f_*\mathbb{Q}$ that is algebraic at one point is algebraic at every point. The statement (F3) is the Hodge conjecture for the varieties that are not abelian, which is equivalent to the whole conjecture (Proposition 6.33).

are Proposition 6.35(iii) and (iv), Proposition 6.42, Proposition 6.43, Proposition 6.44, Proposition 6.46, Corollary 6.53, Theorem 6.55, Corollary 6.56 and Theorem 6.66. The statements on the Mumford classes are Theorem 6.29, Theorem 7.4(ii) and Corollary 6.30. The statements on the families are Theorem 8.5, Proposition 8.7 and Theorem 8.9; the families on which the real multiplication is algebraic are treated in [BB26]. The minimal sets are Theorem 6.70(ii) with the rule of Proposition 7.49 adjoined, as computed in Remark 7.50, and Theorem 6.70(vi) shows that {F3} and {L, M} are each equivalent to the conjecture. ■

The theorem arranges the paper around [I, Theorem 1.1]. Clause (ii) is the problem in its smallest form, one explicitly written class on one connected domain, and clause (iii) shows that the base case is never the difficulty: every family has a base point, for every field, every dimension and every discriminant, with a cycle written down rather than an existence statement. Clause (iv) reduces the closedness of the locus to one object at one point and computes the number that object must attain, and clause (v) shows that the constructions that fail do so for identifiable structural reasons: a divisor-generated Chern character cannot leave the divisor subring, a norm eigenclass sits in the wrong summand of the character grading, a Chern character on the Weil line has an annihilator exactly as large as the family, a support with two transversal branches of codimension two through a point carries a local obstruction that the compensating scalar cannot meet, and a curve along which an object is locally free is tangent to at most one of the two eigenplanes of the real multiplication. Clause (vi) shows where the Weil classes stand in the whole conjecture: the conjecture for abelian varieties is carried by the classes beyond the Weil lines, and the rest is algebraicity modulo abelian varieties, which holds on every variety whose cohomology comes from abelian varieties, on the Delsarte hypersurfaces of every dimension and on the diagonal complete intersections of Vandermonde type (Figure 41). Section 8 shows that the structure [I, Theorem 1.1] finds in the Weil families, a CM seed, density and the dichotomy, is present in every family that carries (F2) or (F3′), and that in each of them the one missing property is the interior point that closes the locus (Theorem 8.10).

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Declarations

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Ethics approval and consent. This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere except in the earlier paper named on the title page, which this paper and its companion replace, and the work of others is credited by citation throughout.

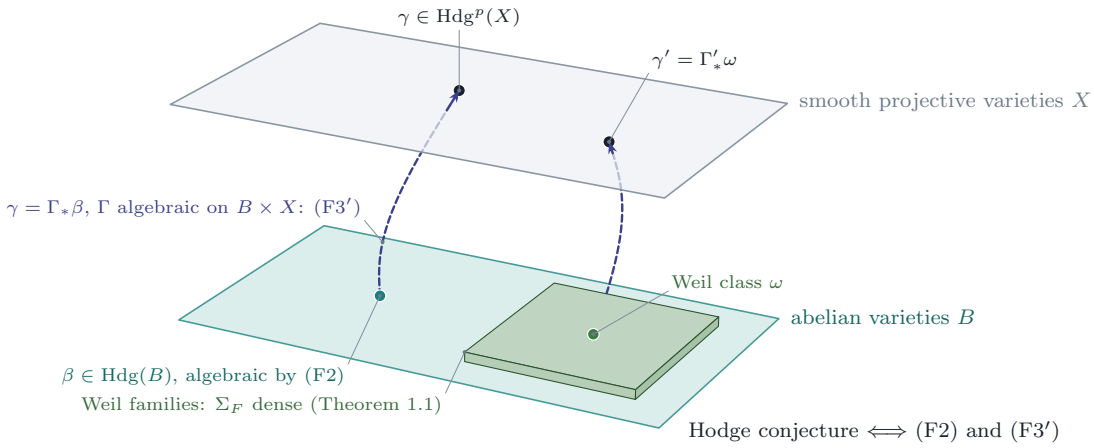


FIGURE 41. The two halves of the conjecture in Theorem 9.1(vi). A Hodge class γ on a smooth projective variety X is the image $\Gamma_*\beta$ of a Hodge class β on an abelian variety B under an algebraic correspondence Γ on $B \times X$, which is (F3′); the class β is algebraic, which is (F2), equivalently the conjecture for abelian varieties (Proposition 7.49). The raised plate on the lower sheet stands for the families of Weil type, on which [I, Theorem 1.1] makes the algebraic locus of the Weil class ω dense; the dashed part of each correspondence is the part behind the upper sheet.

Data availability. The archive [BMB26] is deposited on Zenodo under the DOI 10.5281/zenodo.23225483. It holds programs that repeat much of the finite arithmetic of this paper and its companion [I] as an independent check, a Lean 4 certificate of part of that arithmetic, the Macaulay2 computations of local Ext algebras with their output, and the generators and TikZ sources of the figures; its programs perform 2218 checks, all of which pass, and the Lean kernel checks four hundred and nine theorems, each depending on no axiom or on propositional extensionality alone. Deep Bhattacharjee computed all the calculations, in C, Python, Julia, Macaulay2, Lean 4 and shell scripts, with the assistance of Claude Code, an AI model developed by Anthropic. Every result the programs certify can be reproduced from the archive with the programs and the Lean kernel, and the authors are responsible for the mathematical statements and their proofs. No other data were generated or analysed.

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